

GE Healthcare

Signa Pre-EXCITE to HDx 1.5T Series System Upgrade

Technical Publication



5166008
Rev 11

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
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Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルには英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1 . . .	Mar 6, 2006 . .	Initial Release.
2 . . .	Jul 17, 2006 . .	Section 2: Added CAUTION note to upgrade flowchart that refers to the IIS Key and SaveINFO operation. Revised s/w load instructions and added LCD monitor adjustments (MRIhc15805, MRIhc16251, MRIhc16325). Added blocks for restoreINFO, ICW, and update of eLicense (MRIhc15873, MRIhc15892, MRIhc15894, MRIhc16209). Section 4: Added CAUTION note to not install O-ring and grease when connecting RF Body cables to penetration panel. Section 7, 8, & 9: Updated the type of fastener information used to attach the 16 Channel Switch mounting plate for a welded or bolted rear pedestal. Section 14-4-10: Added note informing the installer not to connect additional monitors to host computer prior to operating system software load (MRIhc15870).
3 . .	Oct 26, 2006 . .	Section 7: Updated PAC replacement instructions with information on removing the PAC Control Housing (MRIhc17607).
4 . .	Feb 22, 2007 . .	Section 2: Reversed flowchart procedures <i>Software Option Install w/eLicense</i> and <i>Install and Calibration Wizard</i> . Section 7: Removed PAC2B part number. Section 14: Added note that Humidity iButton can be installed in either direction.
5 . .	May 24, 2007 . .	Section 5: Revised de-installation process by adding a Warning and illustration for removing mounting frame from magnet room side because frame may be made of magnetic material. Section 14: Installation instructions for OW table were revised due to introduction of new OW table. Updated iButton positioning and added block diagram in Environmental Sensor installation.
6 . .	Aug 10, 2007 . .	Sections 6, 7, 8 and 9: Updated the Hybrid Splitter upgrade by adding the Netcom Hybrid Splitter which has different Isolation Filter mounting brackets than the existing Teledyne Hybrid Splitter.
7 . .	Sep 28, 2007 . .	Section 1: Added sub-section 1-3 for non-English label installation instructions. Section 2: Removed reference to performing IIS key re-generation at beginning of upgrade flowchart (PQR 13100244). Added LCC magnet with Horizon enclosure to flowchart in sub-section 2-3. Section 4: Sub-section 4-2: Revised information on Run 421 in cable table 4-1 for Cx and LCC magnets that need a longer Run 421. Also updated cable maps to reflect this information. Revised Run 421 cable connection to SRI-3 instructions in sub-section 7-1-3-C (PQR 13094054). Added information for connection of Mobile system Heat Exchanger power cable, Run 046, to PDU. Updated cable maps to reflect this information. Added heat Exchanger power cable terminal connection in PDU information to table in sub-section 4-7-1-C (PSR 13120695). Section 14: Added 1-wire Hub power cord connection options to J4, J5, or J6 (PQR 13097707).
8 . .	May 28, 2008 . .	Revised title page to <i>Signa Pre-EXCITE to HDx 1.5T Series System Upgrade</i> . The change reflects the addition of HDxT, HDi, and Vibrant products. Section 2: Added True Mobile flowchart (MRIhc37270). Section 4: Updated cable map to show the cabling of the existing and the new GOC. Updated cable maps to show new Temperature and humidity Sensor installation and cables. Corrected the J3 and J4 connection designations at the hybrid splitter (PQR 13172263). Section 5: Added notes pertaining to penetration panel installation differences for mobile vans. Section 6: Added instructions for installing gradient lead board and rear pedestal stabilization. Section 7: Added instructions for installing gradient lead board (PQR13162916). Revised illustrations to show new gradient lead board. Sections 8 and 9: Added instructions and revised illustration for installing gradient lead board. Section 10: Revised views of rear of magnet by removing obsolete terminal block. Added information in section 10-4 if site received a new front end bell. Section 11: Added instructions that if the cradle upgrade wasn't completed by mechanical installer or mobile van partner, the GE FE was to be notified. Section 12: Revised views of rear of magnet by removing obsolete terminal block. Section 13: Added information for referring to IIQ Kit for installing radial supports in section 13-6. Added gradient lead board to views of rear of magnet. Section 14: Revised entire section to add OW upgrade for systems that are shipped after July 1, 2008 with new GOC. In Sub-section 14-5, revised temperature/humidity sensor power cord connection to J6 on power strip (PQR 13166741) and added power draw to Max Amp table (PQR 13166207). Section 15: Corrected table reference from 4-1 to 4-2. Added Heat Exchanger PDU connections to table in section 15-4. Section 17: Corrected Section reference from 1-8 to 1-9.
9 . .	Sep 22, 2008 . .	Section 1: Updated Introduction section to reflect the addition of HDxT, Release 15.x. Section 4: (ECO 2054786) Sub-section 4-4, updated cable maps by adding Run 1382. Sub-section 4-7-1, added Runs 1381 and 1382 to PDU connection table. Section 14: Sub-Section 14-4-7, corrected power and ground cable run numbers connected to OW. Sub-section 14-4-12, Added note about SCSI Tower being an option for HDxT and 15.x.
10 . .	Jun 12, 2009 . .	Section 1: Updated HDx to HDxT where applicable. Sections 6, 7, 8, and 9: Removed installation of plates on back of bridge (PQR 13246367) and supplied additional information on applying laser labels on WideOpen enclosures (PQR 13246299). Section 14: Removed the word DASM from Table of Contents (PQR 13237400). Section 18: Revised catalog information flowcharts and illustrations (PQR 13117708).
11 . .	Oct 14, 2010 . .	Section 1: Added 200 gauss zone illustration and notice about safety training (CAPA 1859293). Section 2: Updated eLicense model type to MR from Signa HDx (per Direction 2397845). Section 4: Added Section 4-4, Extra Mobile Upgrade Cables (MRIhc48625). Updated cable maps for removal of OW components and installation of Z400 GOC (ECO 2096124). Sections 6, 8, & 9: Updated ferrous warnings for moving Rear Pedestal which contains longitudinal drive motor. Section 14: Updated instructions for removal of obsolete components and GOC. Added instructions for installing Z400 GOC (ECO 2096124). Revised Ferrous Material Handling warnings for Dock, Rear Pedestal and Blower Box throughout manual. Added 200G plot for moving restrictions of blower box. Section 17-1: Added check for proper blower box anchoring. (CAPA 1859293).

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SECTION 1 – GETTING STARTED

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1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 1.5T pre-EXCITE system to Signa® 1.5T HDx (14.x), HDxt (15.x), HDi (14.x & 15.x), or Vibrant (14.x & 15.x) system. Refer to illustration 1-1 for pictorial overview of upgraded system.

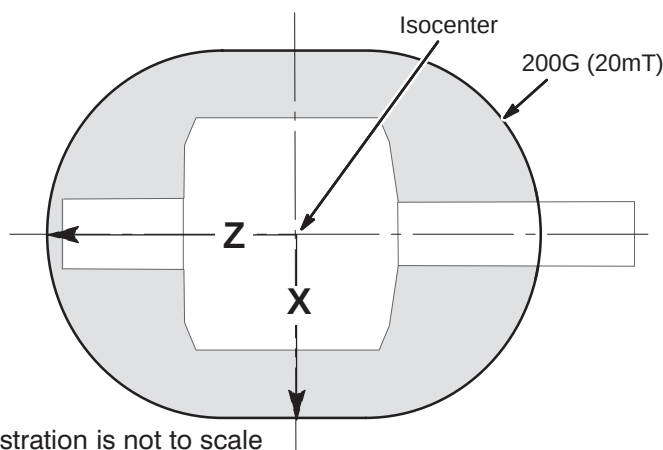
NOTICE:

During this hardware upgrade, the magnet may be ramped to full field strength. Individuals working in the magnet room must be MR safety trained.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE. THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.



Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
Unshielded Magnet	Measure ²	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa HDxt 1.5T system are listed in the *Signa HDx 1.5T Upgrade Pre-Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-3 NON-ENGLISH LABEL INSTALLATION

Components and accessories delivered to this site may have English labels attached to them. For non-English sites, optional multi-language labels have also been supplied.

If required, non-English labels should be attached over the English labels.

- For Surface Coils and Coil Cables – Apply the appropriate label over the English label. The label is included with the coil component.
- For all other labels, follow the label update procedures in the manual or on the applicable multi-language label sheet.

Verify at the end of the installation that the original English labels are covered by new labels representing the native language of the customer.

1-4 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 1.5T HDxt Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-5 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. The IIS key will need to be re-generated. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-6 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-7 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-8 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-8-1 MATERIAL POLICY OVERVIEW

1-8-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-8-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-8-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the HDx upgrade to GoldSeal which includes:

- All removed cabinets
- All removed cables
- Penetration Panel
- SRI and PAC modules
- All removed Magnet Enclosure parts
- All removed Rear Pedestal parts
- LPCC, Carriage, and/or removed Bridge and related parts
- Octane or Indigo or Operator Console, and Computer cabinet with all associated cables and modules

1-8-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-8-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-8-5 REGION SPECIFIC RETURN INFORMATION

1-8-5-1 Sites in Americas

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

1-8-5-1 Sites in Americas (Continued)

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)-747-6997 or Dial Comm 8*579-6997
Fax: (414)-747-6855
Field ASPEN: 1-800-525-1516, Box #26041

1-8-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS-Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-8-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS-E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1-60-91-57-35 or (+33) 1 64 97 55 41
Fax: (+33) 1-60-86-12-31

For non-Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1-8-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1-9.

1-9 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-10 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-11 UPGRADE SYSTEM CONFIGURATION

Illustrations 1-1 and 1-2 show the major equipment installed for a Signa HDxt 1.5T Upgrade.

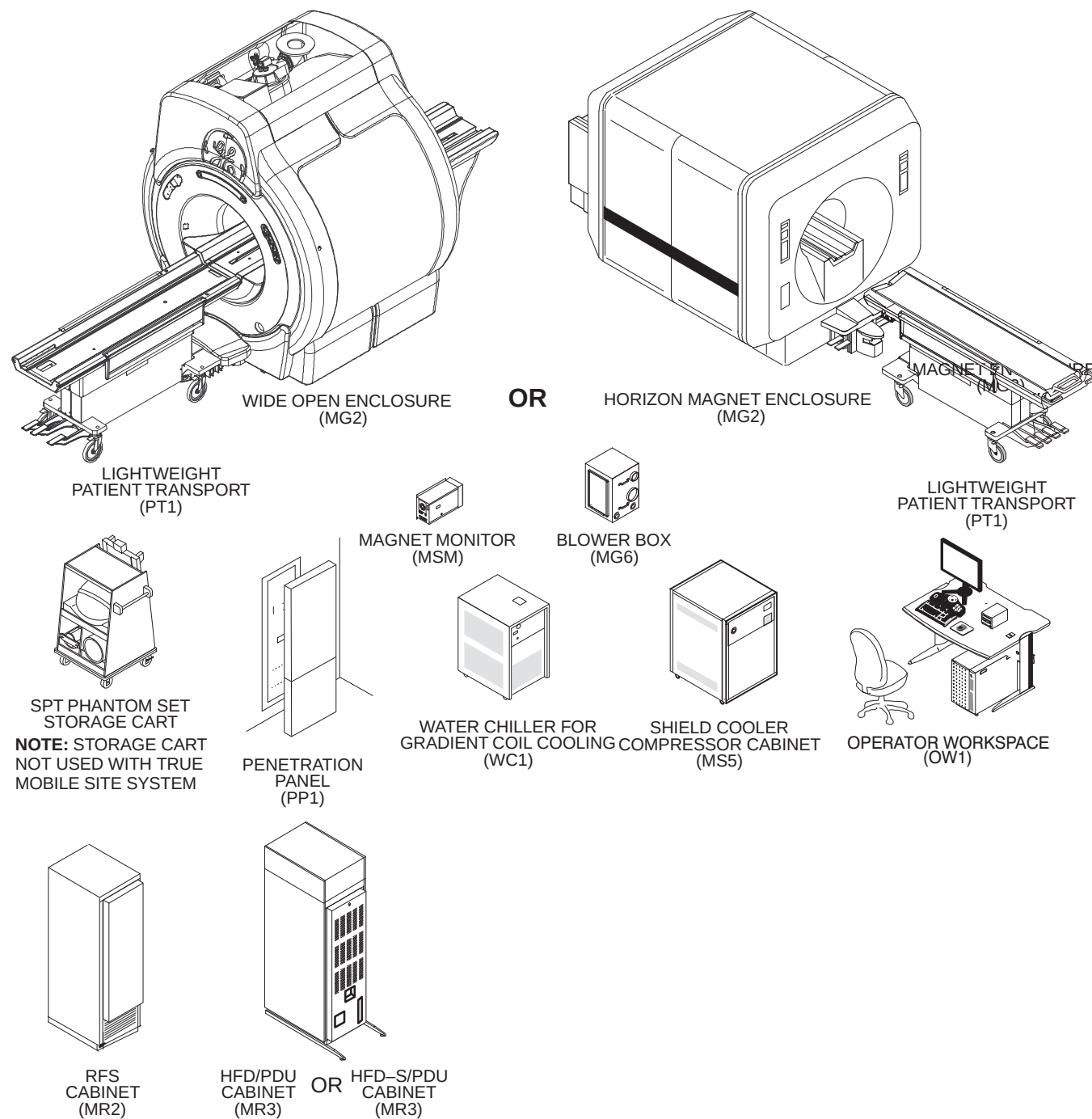


ILLUSTRATION 1-2
PRE-EXCITE UPGRADED TO EXCITE HDxt NON-TWINSPEED SYSTEM MAJOR EQUIPMENT

1-12 UPGRADE SYSTEM CONFIGURATION (Continued)

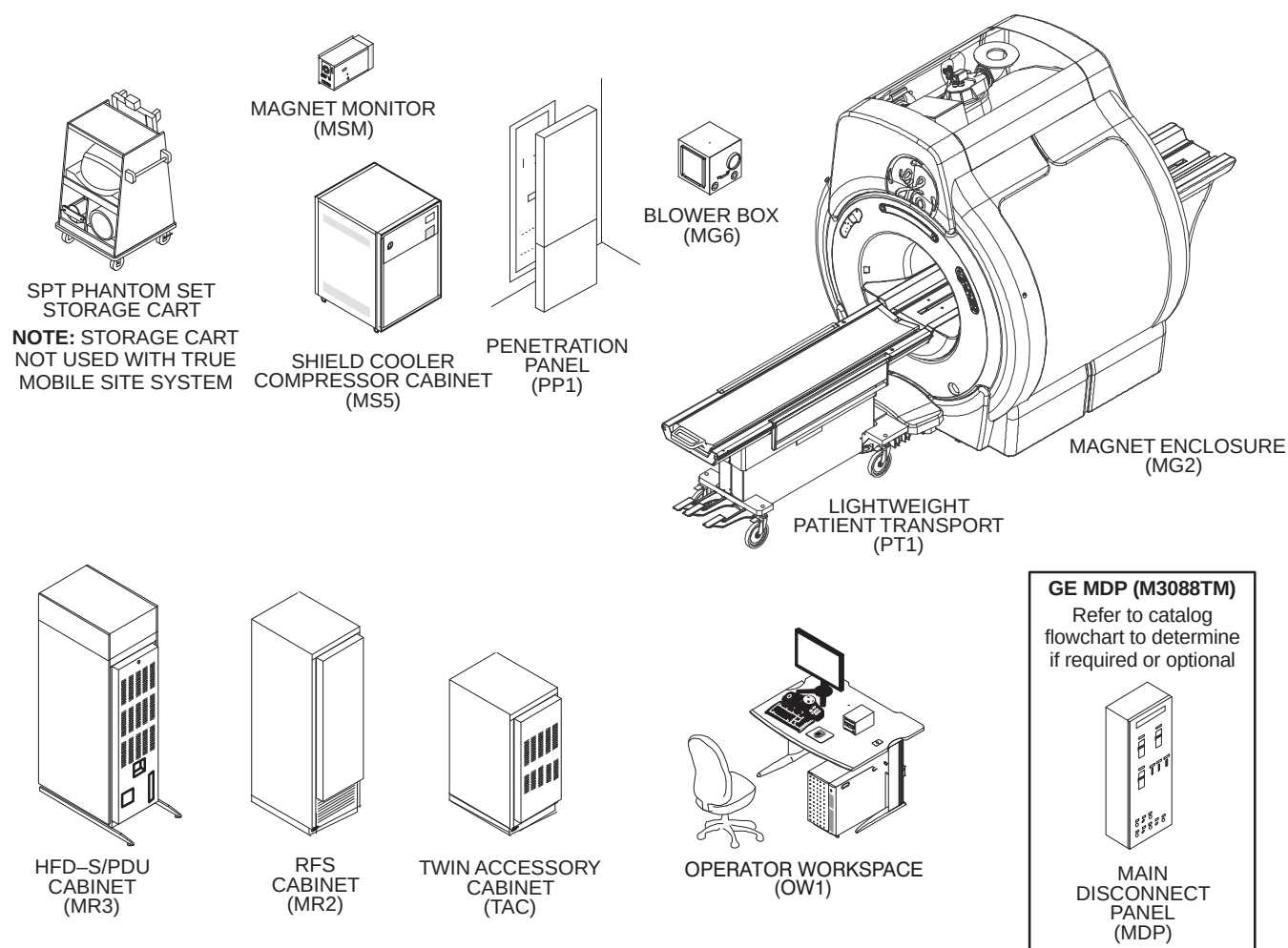


ILLUSTRATION 1-3
PRE-EXCITE UPGRADED TO Signa 1.5T EXCITE HDxt TwinSpeed System

SECTION 2 – UPGRADE INSTALLATION PROCESS

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2-1 FIXED-SITE UPGRADE PROCESS

2-1-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

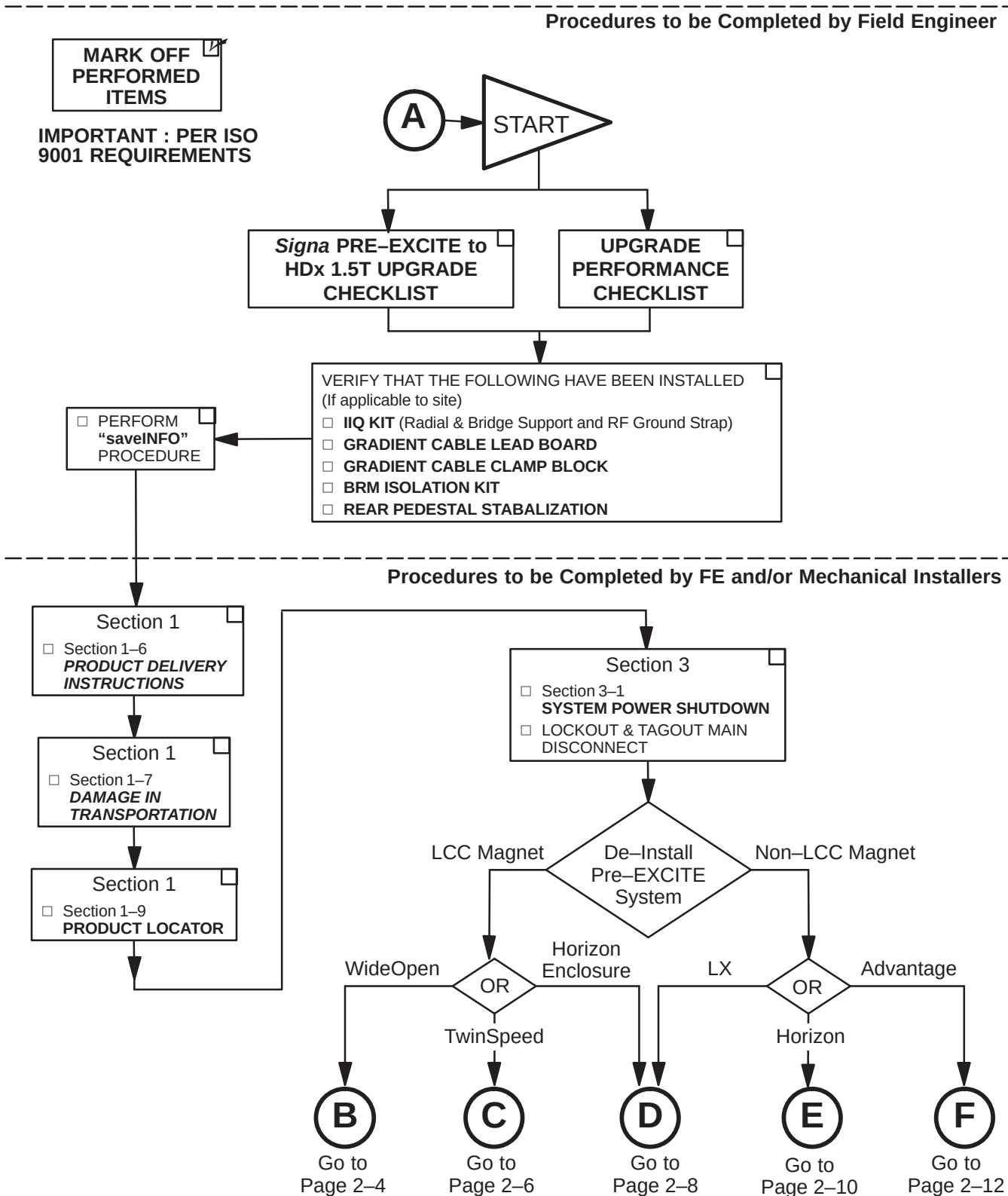
2-1-2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to Signa HDx 1.5T, Release 14.x.

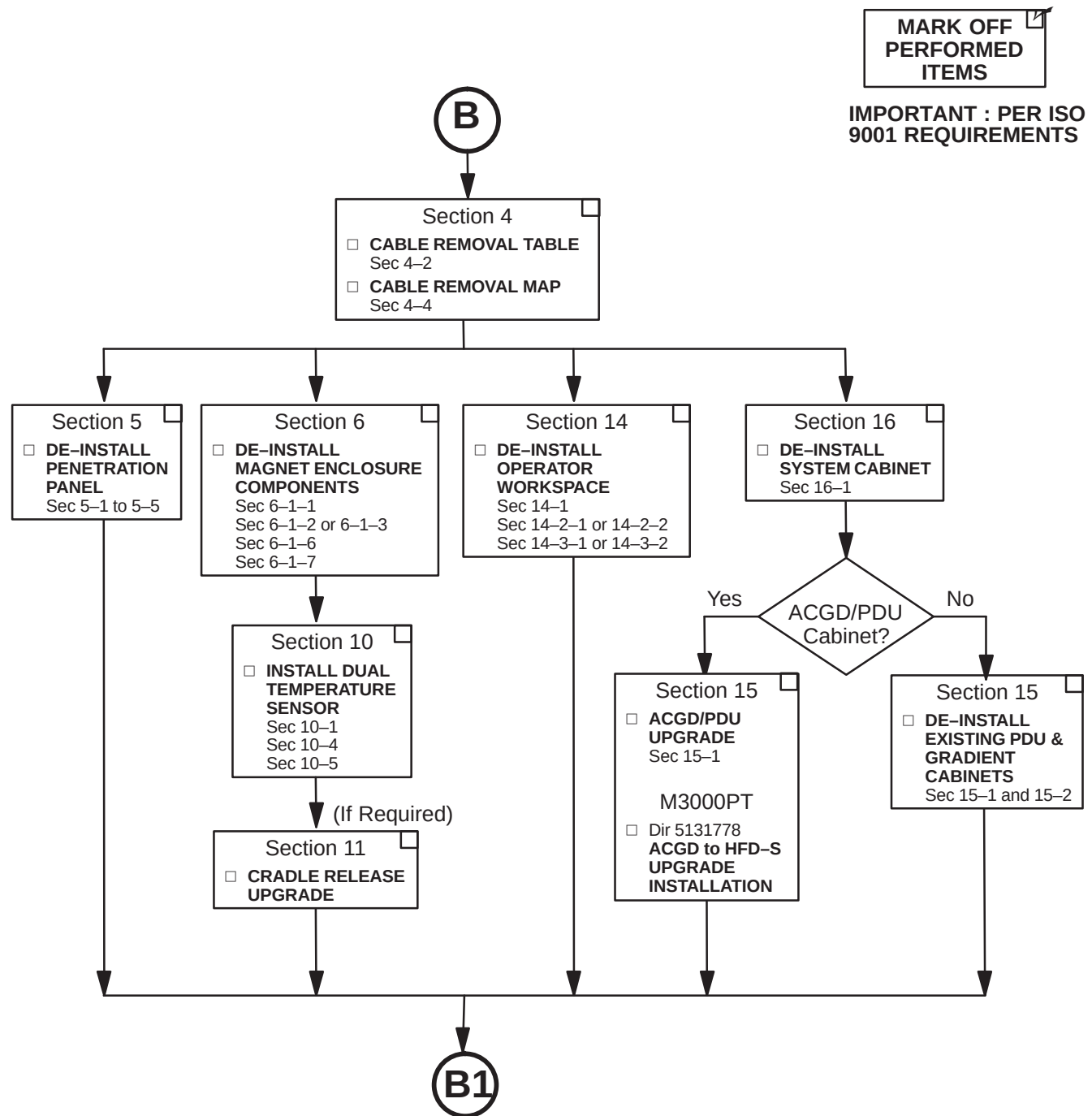
The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa HDx equipment has been delivered together. **Make sure that every part required is available before starting.**

2-1-3 SYSTEM MECHANICAL UPGRADE START FLOWCHART

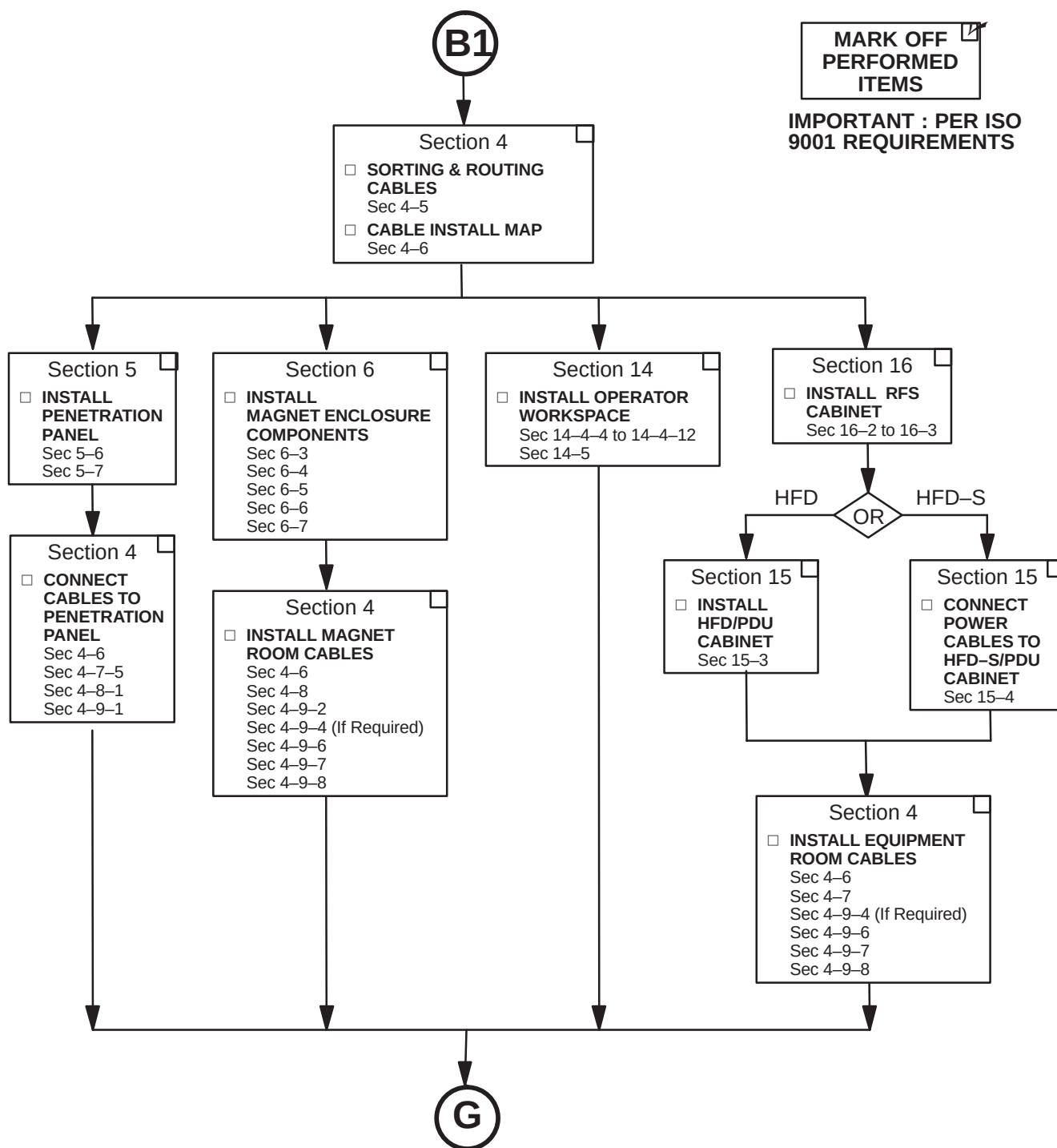


2-1-3-A SYSTEM MECHANICAL UPGRADE FLOWCHART FOR WideOpen

 De-Installation Procedures to be Completed by FE and/or Mechanical Installers


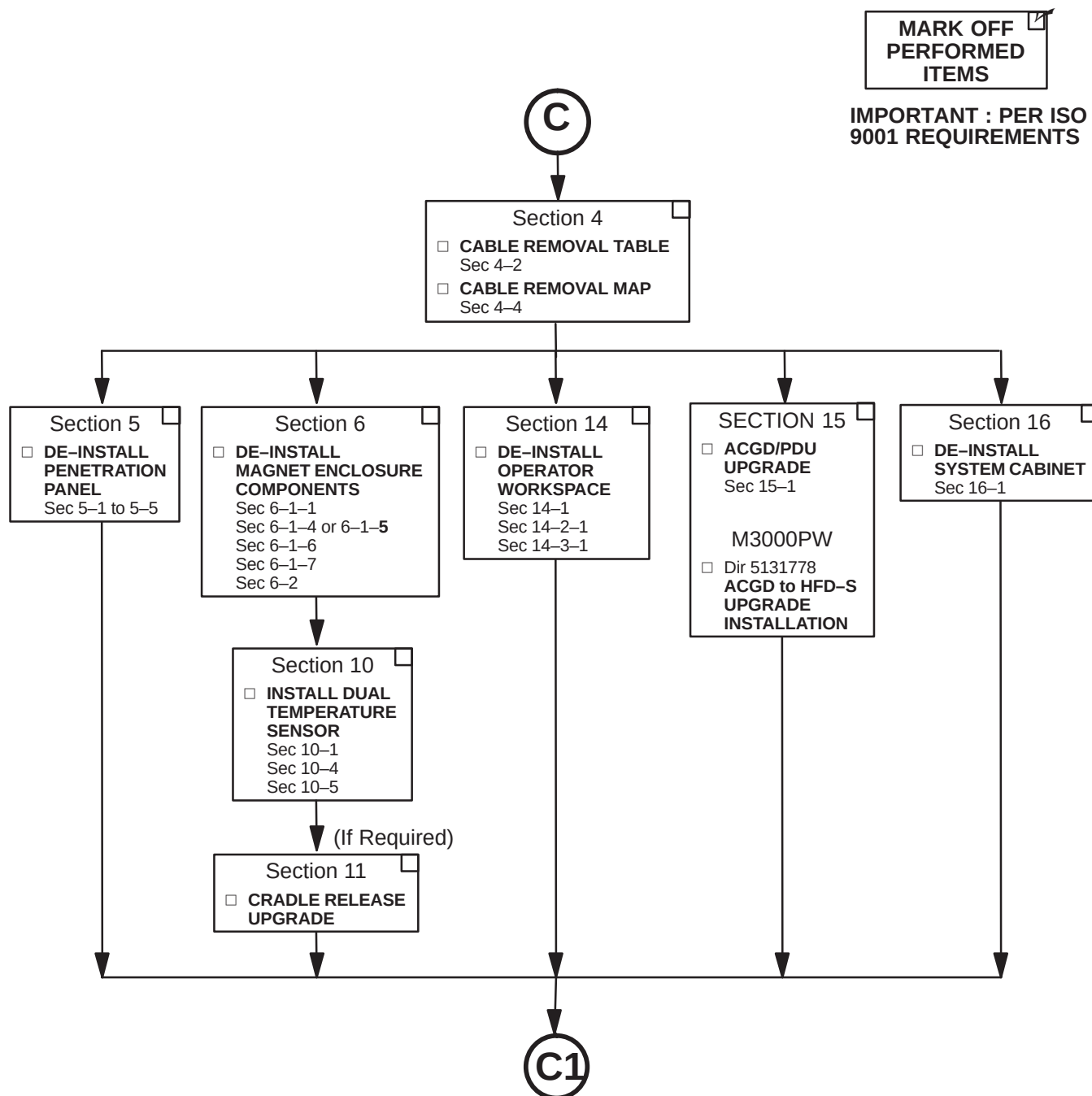
2-1-3-A SYSTEM MECHANICAL UPGRADE FLOWCHART FOR WideOpen (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



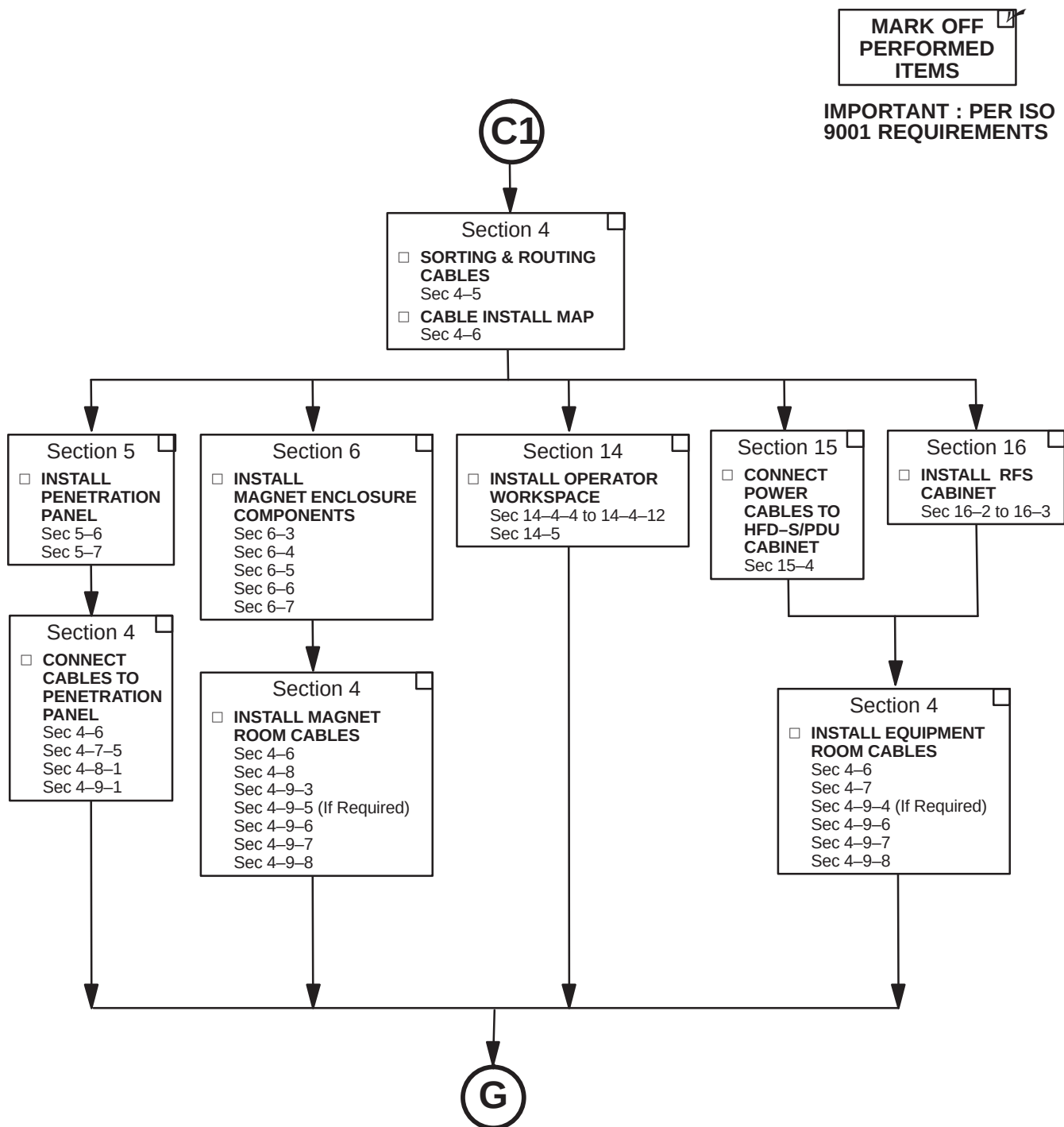
2-1-3-B SYSTEM MECHANICAL UPGRADE FLOWCHART FOR TwinSpeed

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



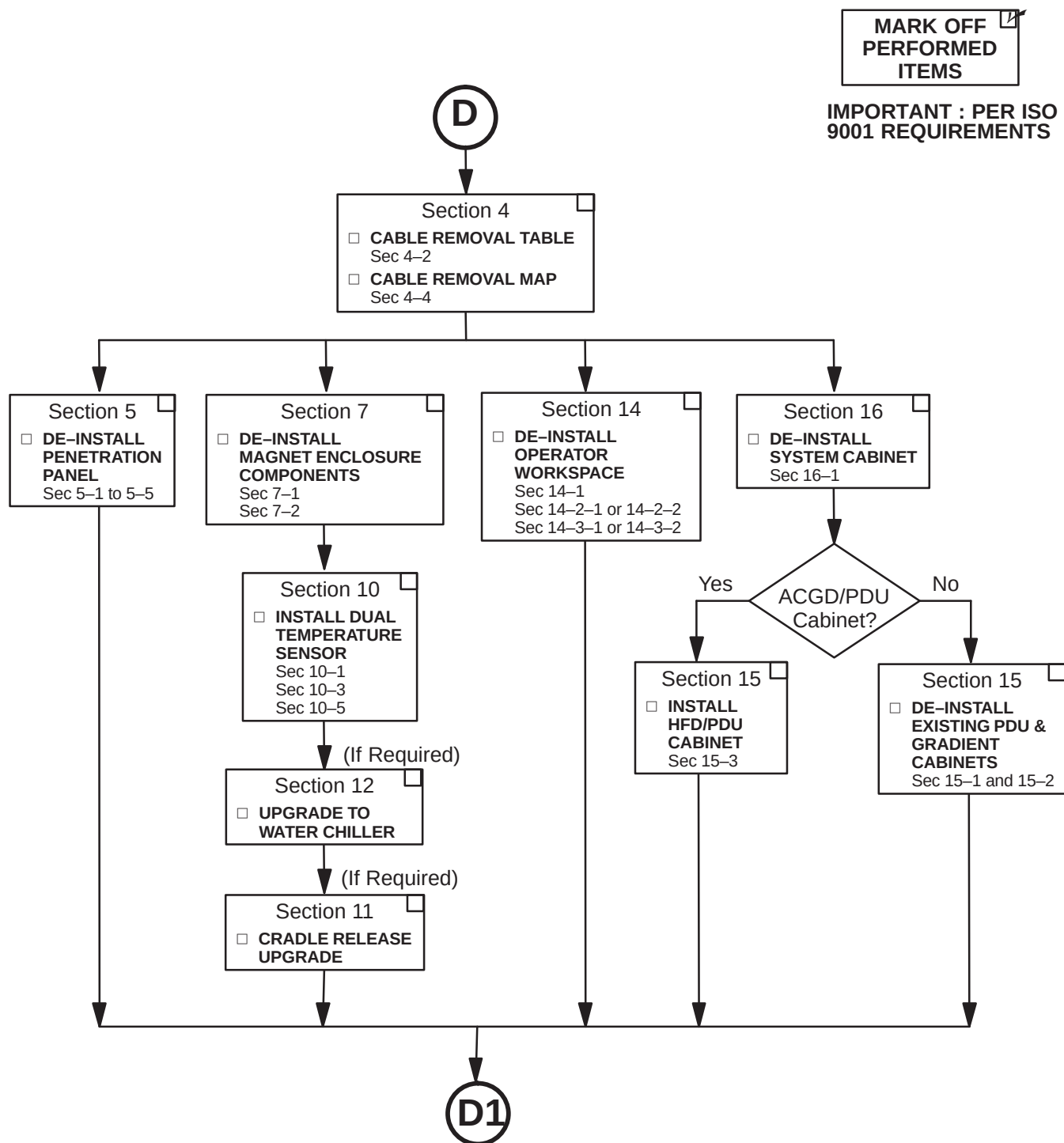
2-1-3-B SYSTEM MECHANICAL UPGRADE FLOWCHART FOR TwinSpeed (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



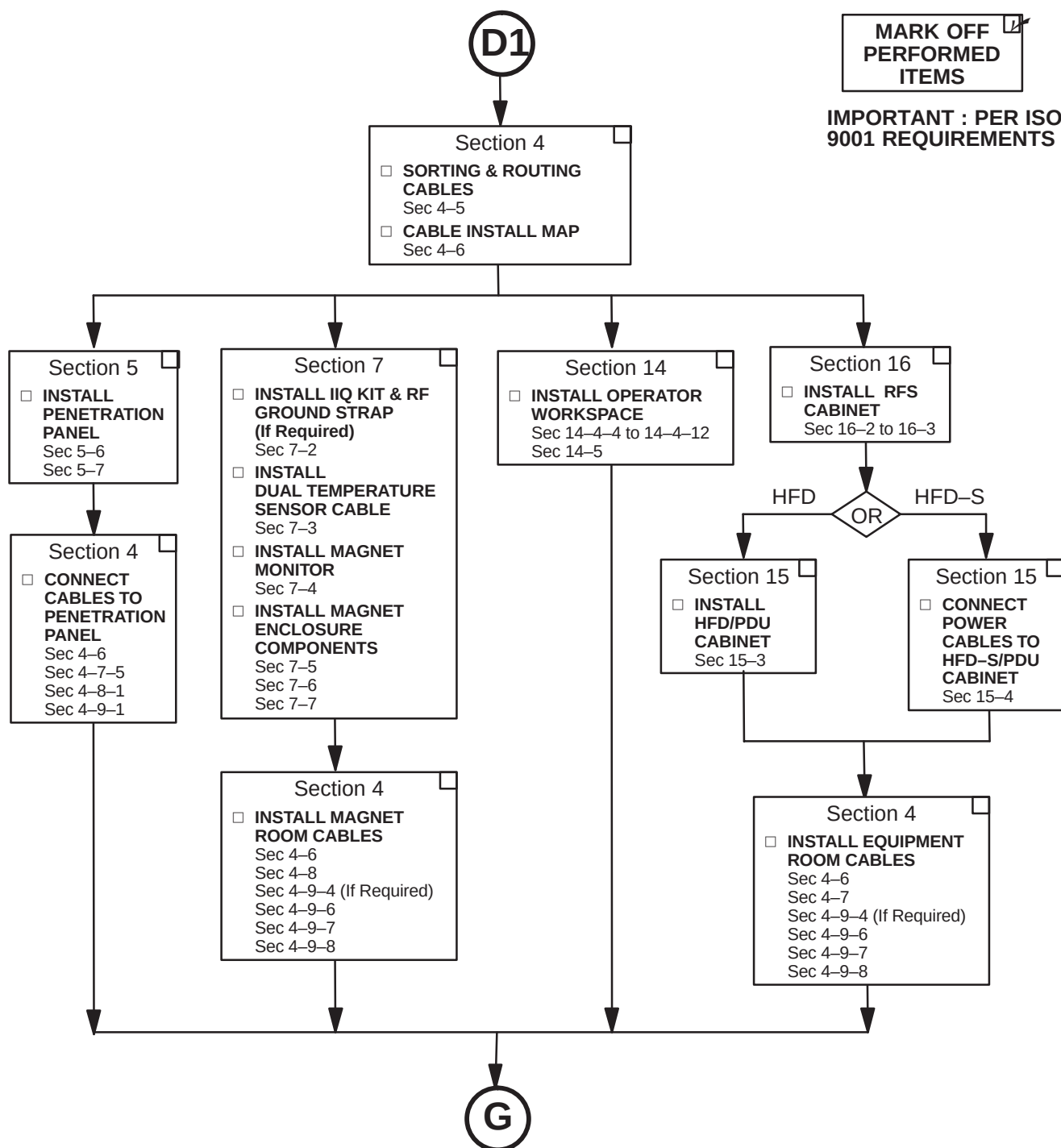
2-1-3-C SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC OR LCC WITH HORIZON ENCLOSURE LX

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



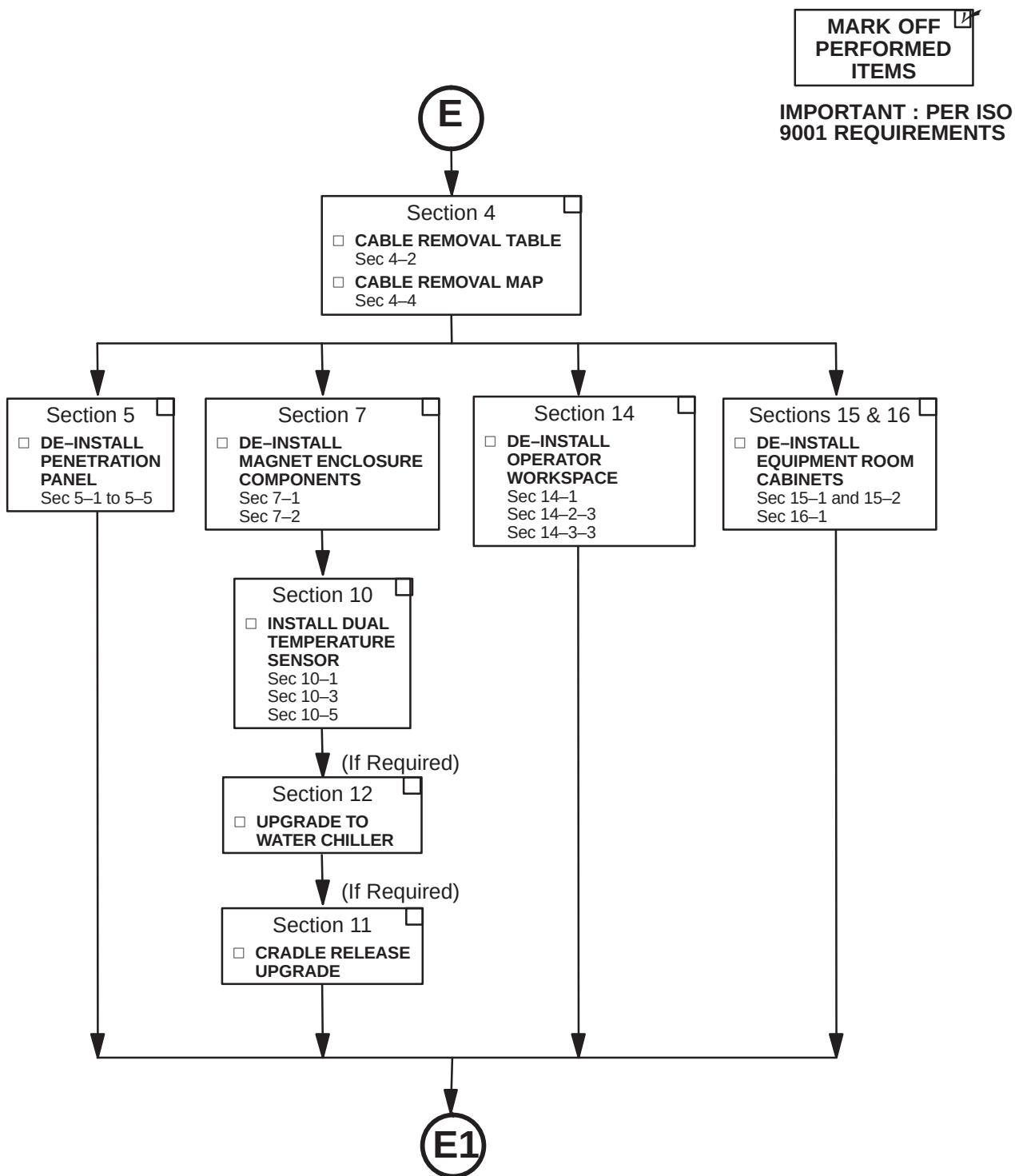
2-1-3-C SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC OR LCC WITH HORIZON ENCLOSURE LX (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



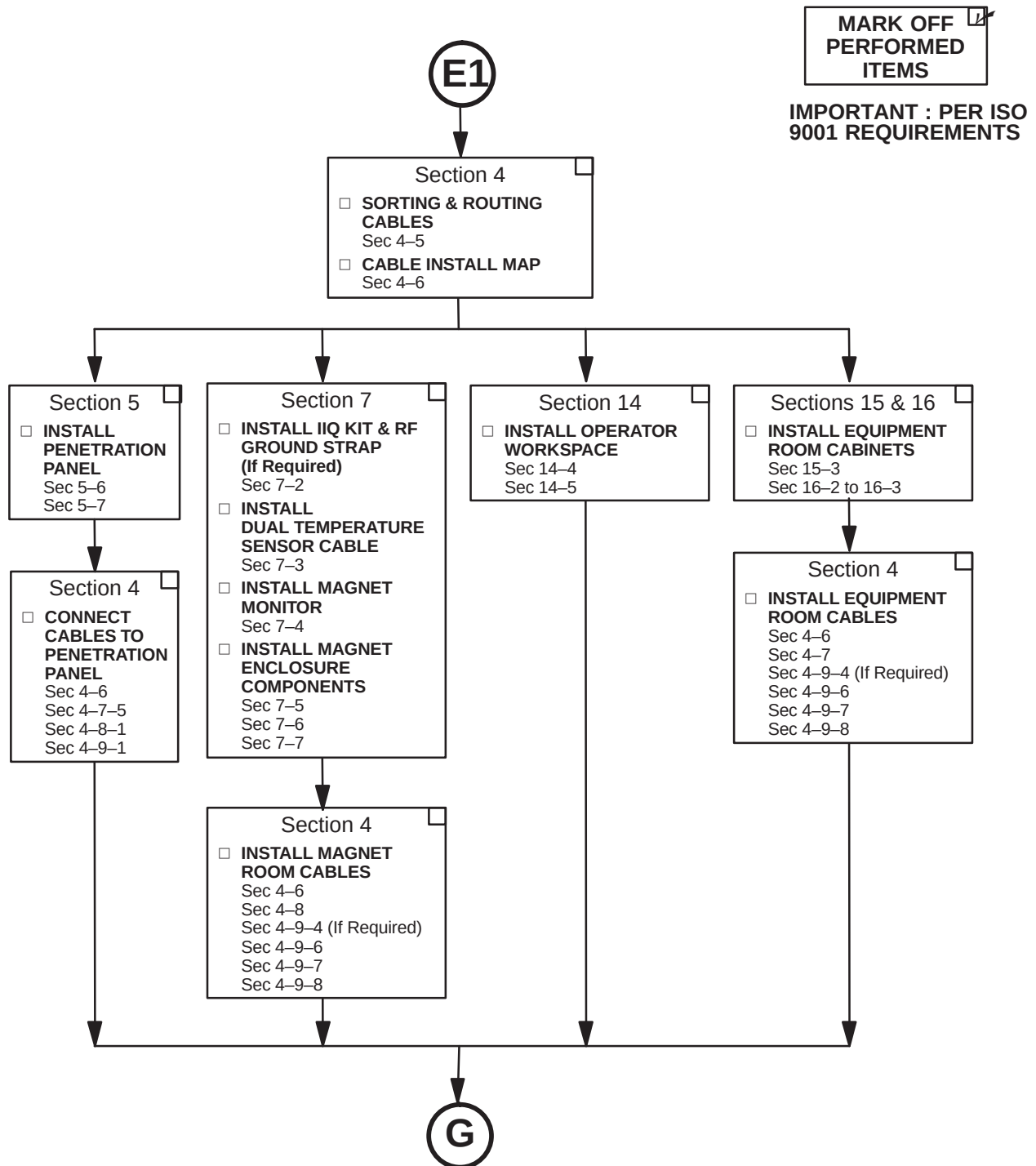
2-1-3-D SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC HORIZON

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



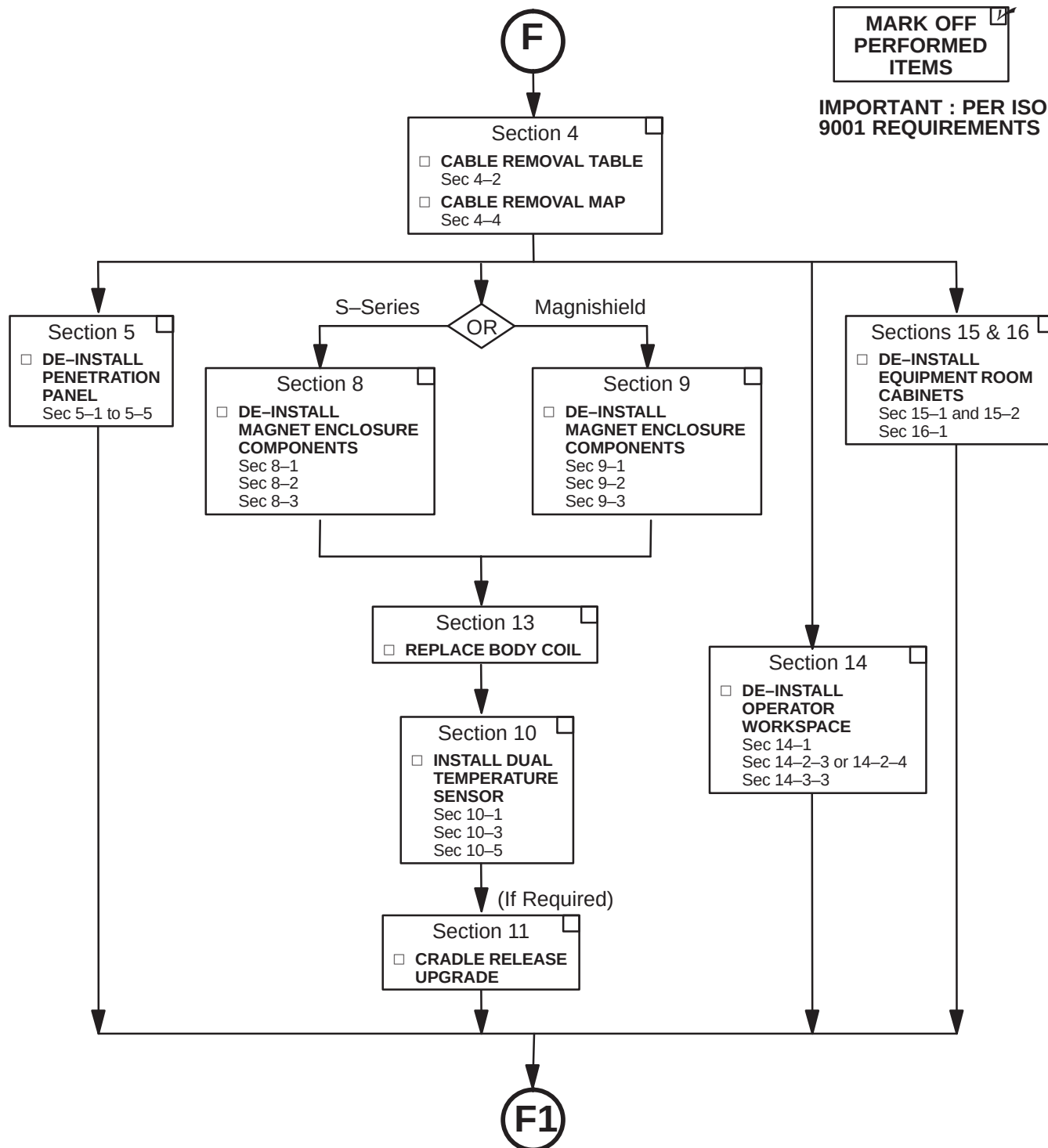
2-1-3-D SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC HORIZON (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



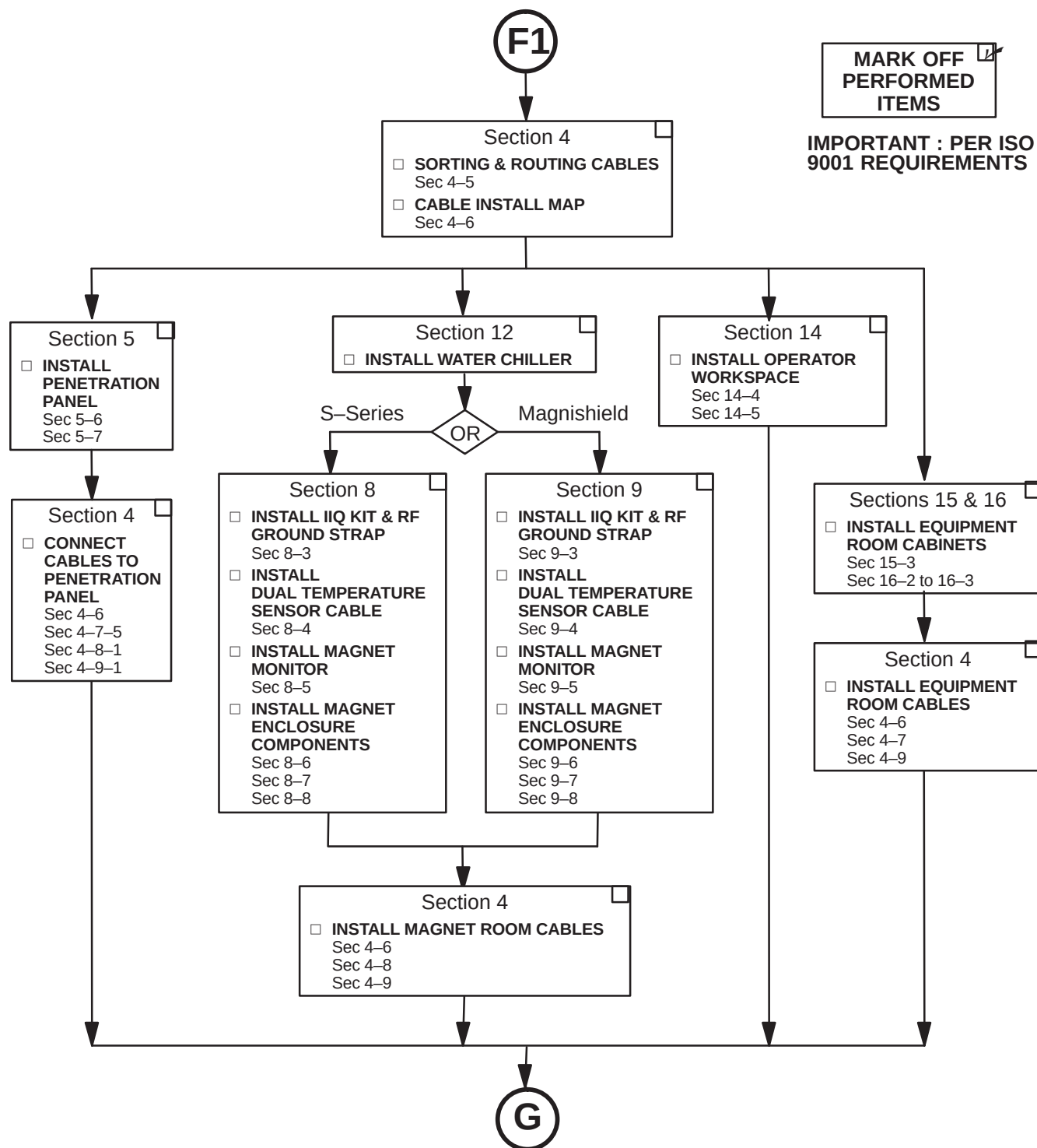
2-1-3-E SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC ADVANTAGE

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



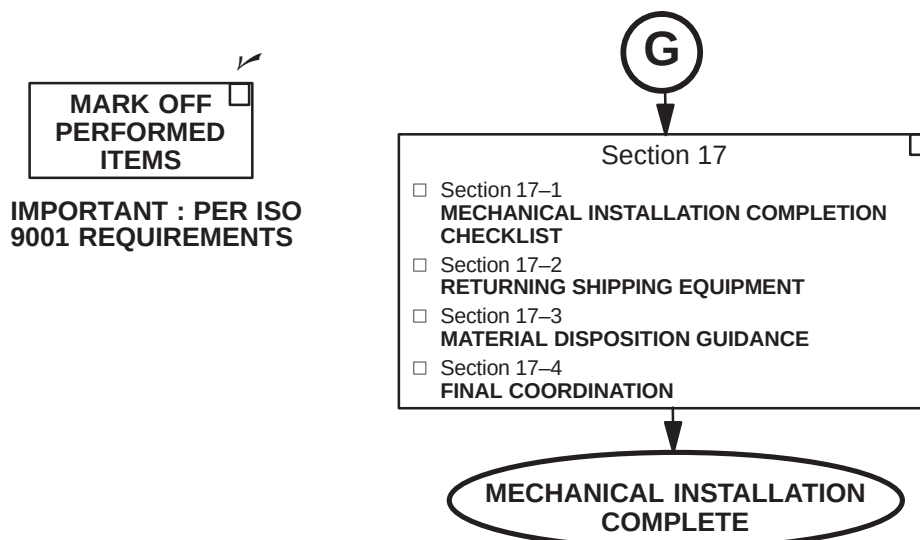
2-1-3-E SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC ADVANTAGE (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers

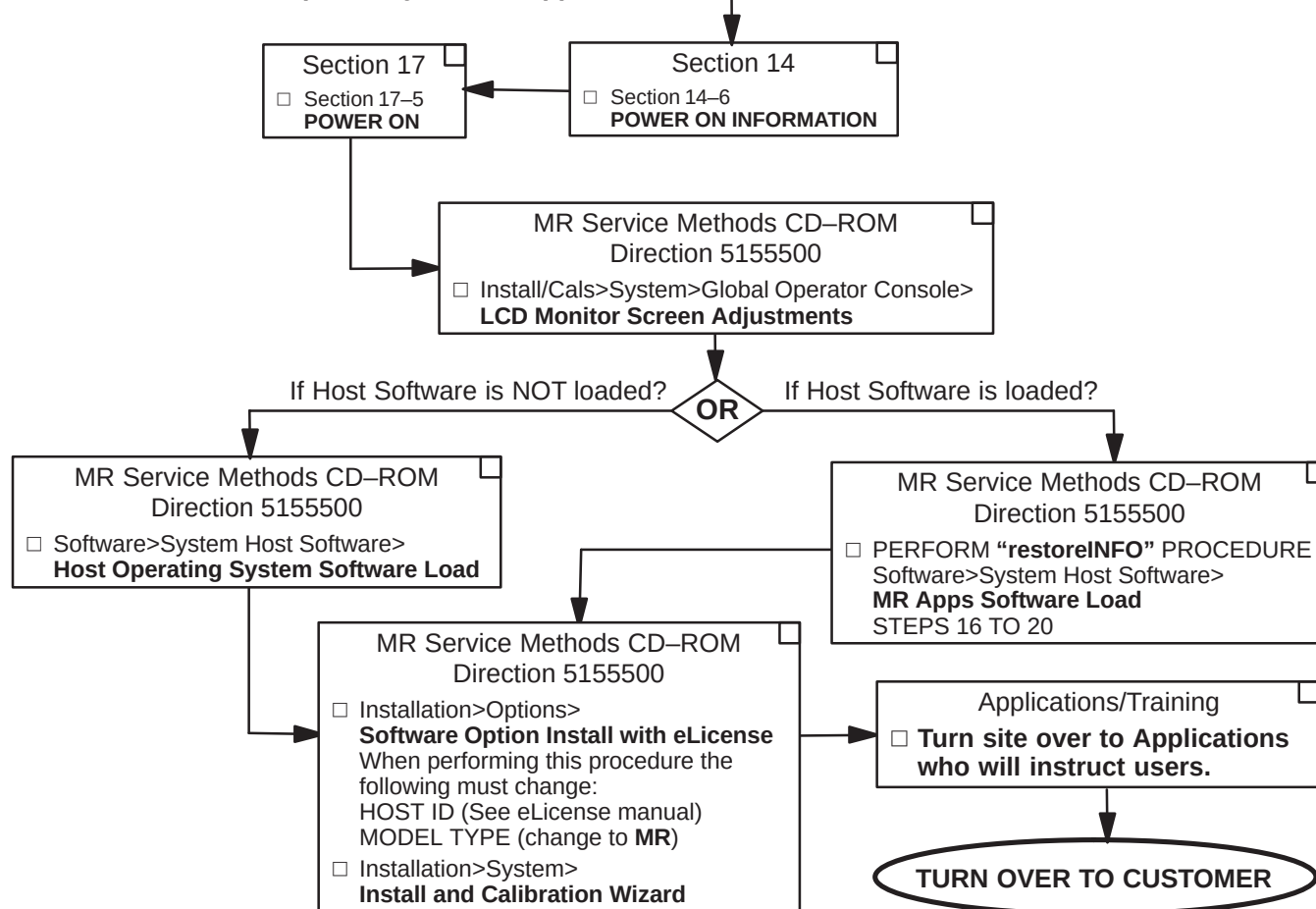


2-1-4 UPGRADE COMPLETION CHECKS AND POWER ON FLOWCHART

Procedures to be Completed by FE and/or Mechanical Installers



Procedures to be Completed by FE and Applications



2-2 TRUE MOBILE UPGRADE PROCESS

2-2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR AWAY FROM THE MAGNET AS POSSIBLE.

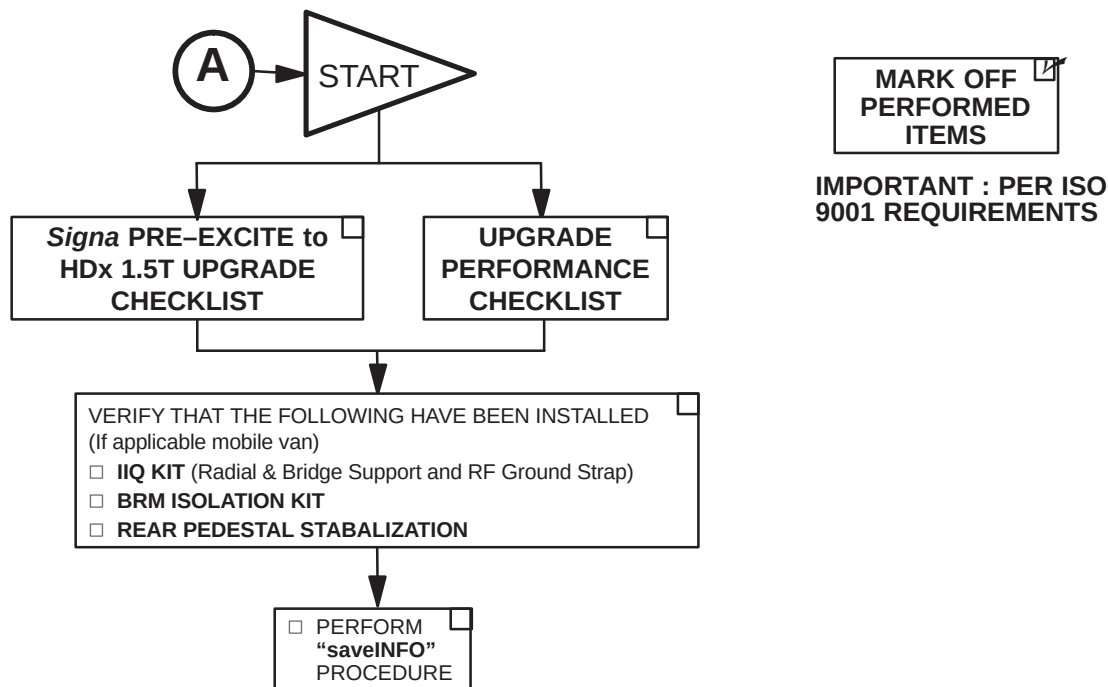
2-2-2 UPGRADE PROCEDURE

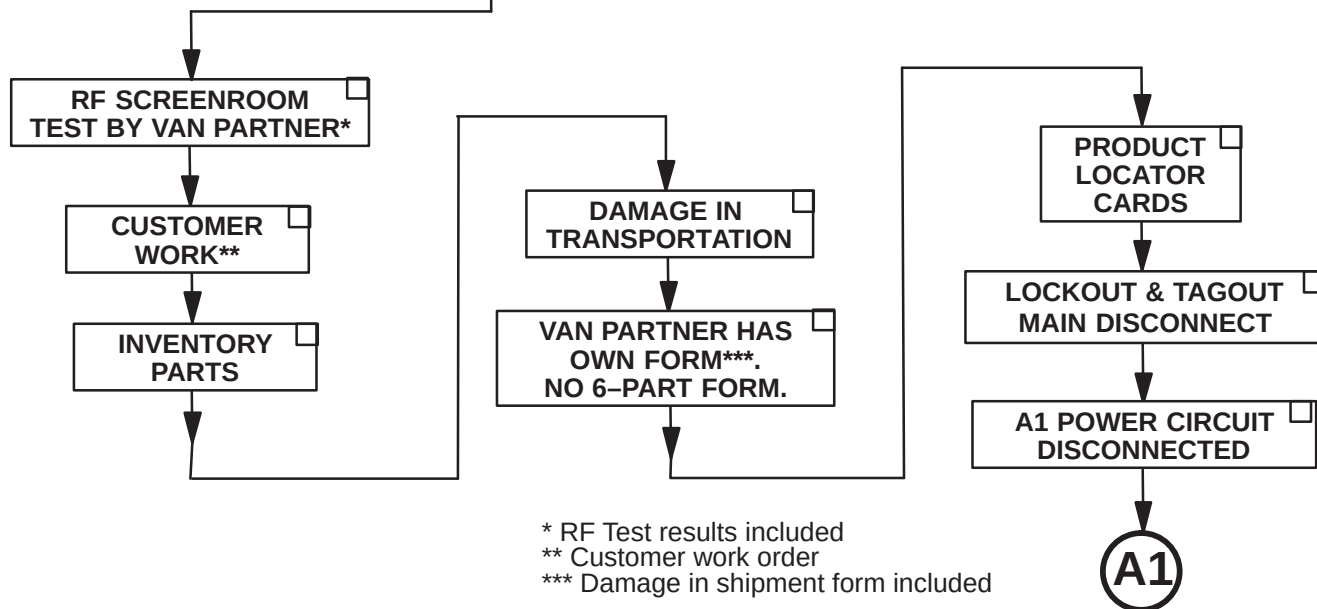
The following upgrade flow charts contain information for upgrading to Signa HDx 1.5T Release 14.x.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa HDx equipment has been delivered together. **Make sure that every part required is available before starting.**

2-2-3 TRUE MOBILE MECHANICAL SYSTEM UPGRADE START FLOWCHART

 Procedures to be Completed by Field Engineer


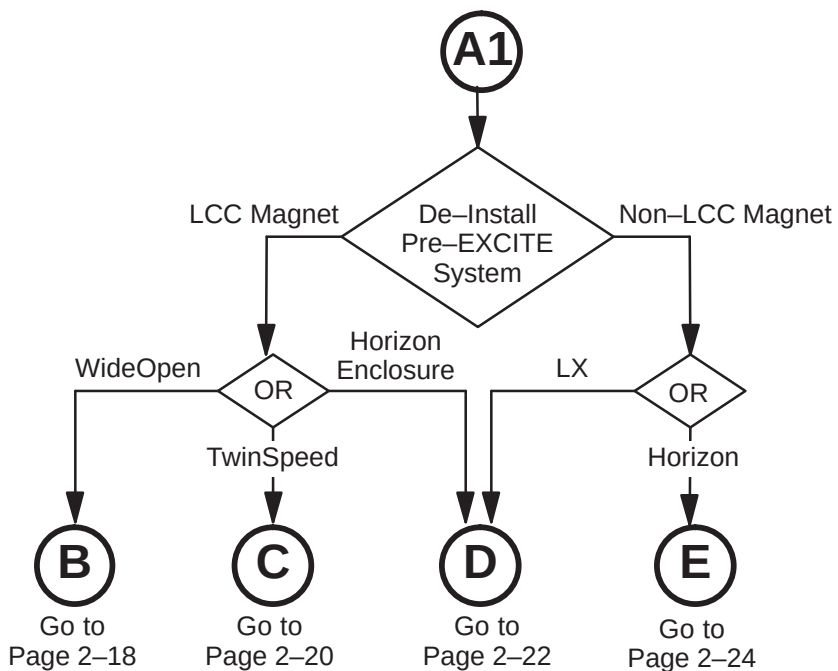
 Procedures to be Completed by FE and/or Van Partner


2-2-3 TRUE MOBILE MECHANICAL SYSTEM UPGRADE START FLOWCHART (Continued)

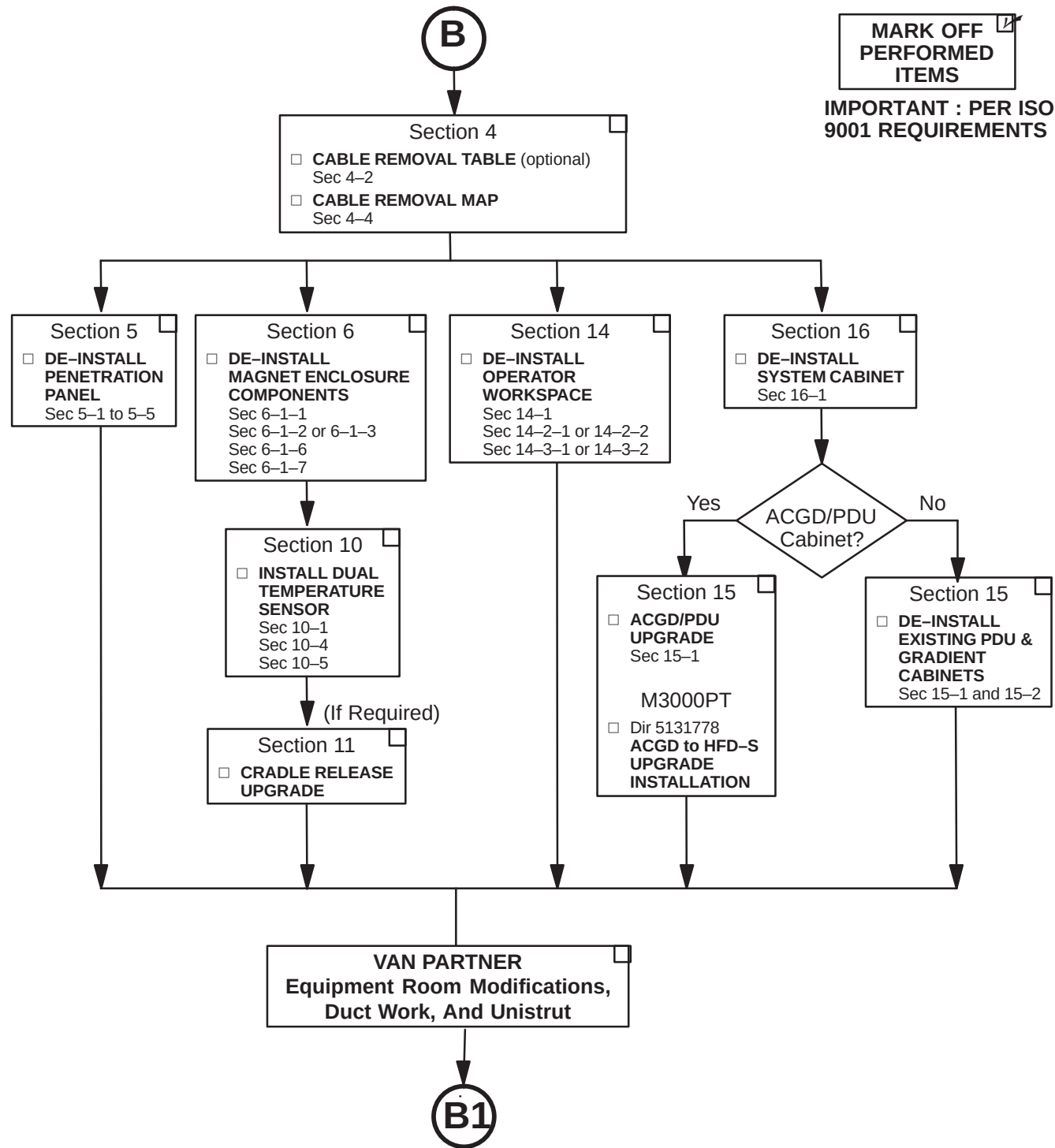
Procedures to be Completed by FE and/or Van Partner

MARK OFF PERFORMED ITEMS

**IMPORTANT : PER ISO
9001 REQUIREMENTS**

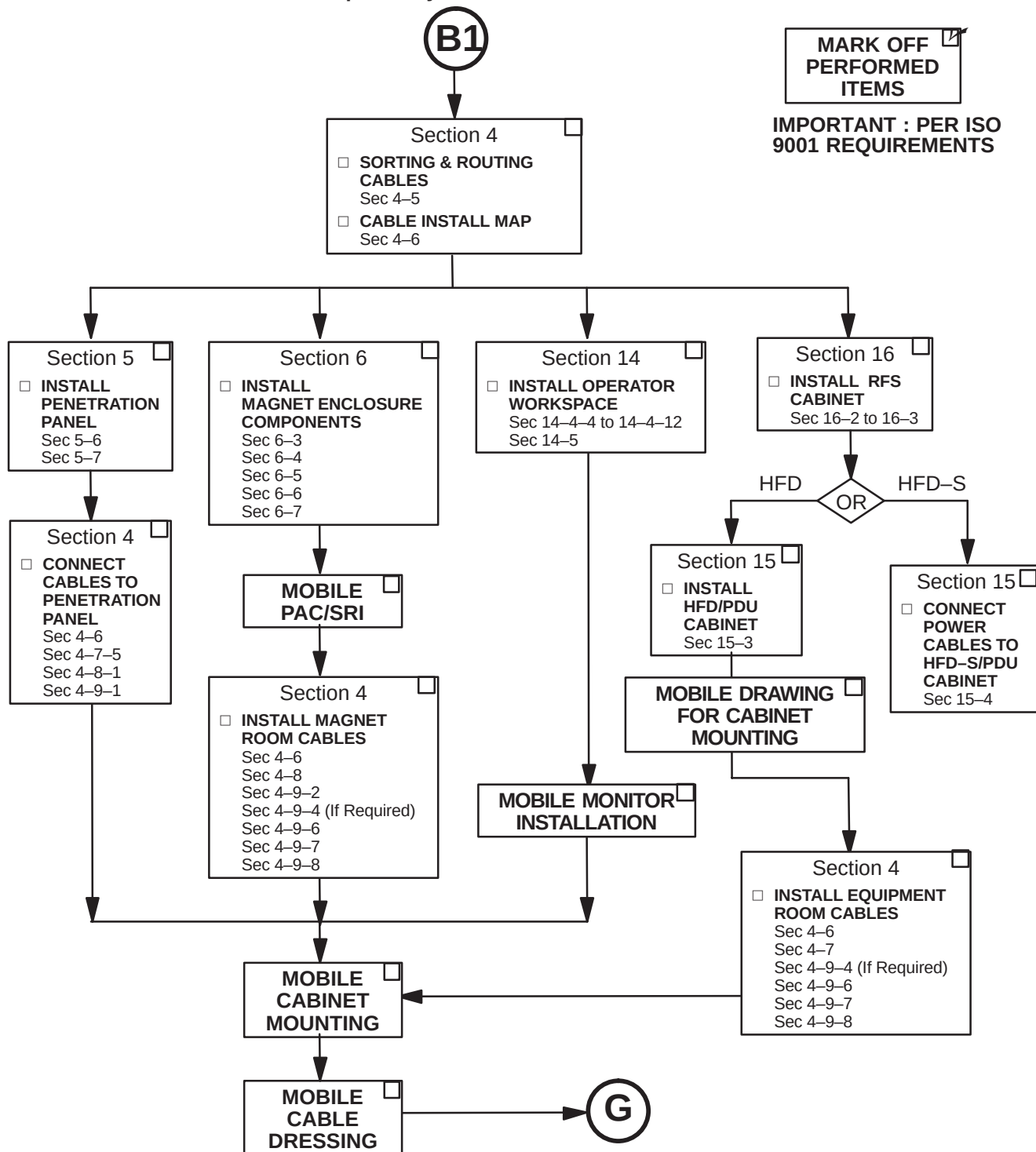


2-2-3-A SYSTEM MECHANICAL UPGRADE FLOWCHART FOR WideOpen

 De-Installation Procedures to be Completed by FE and/or Van Partner


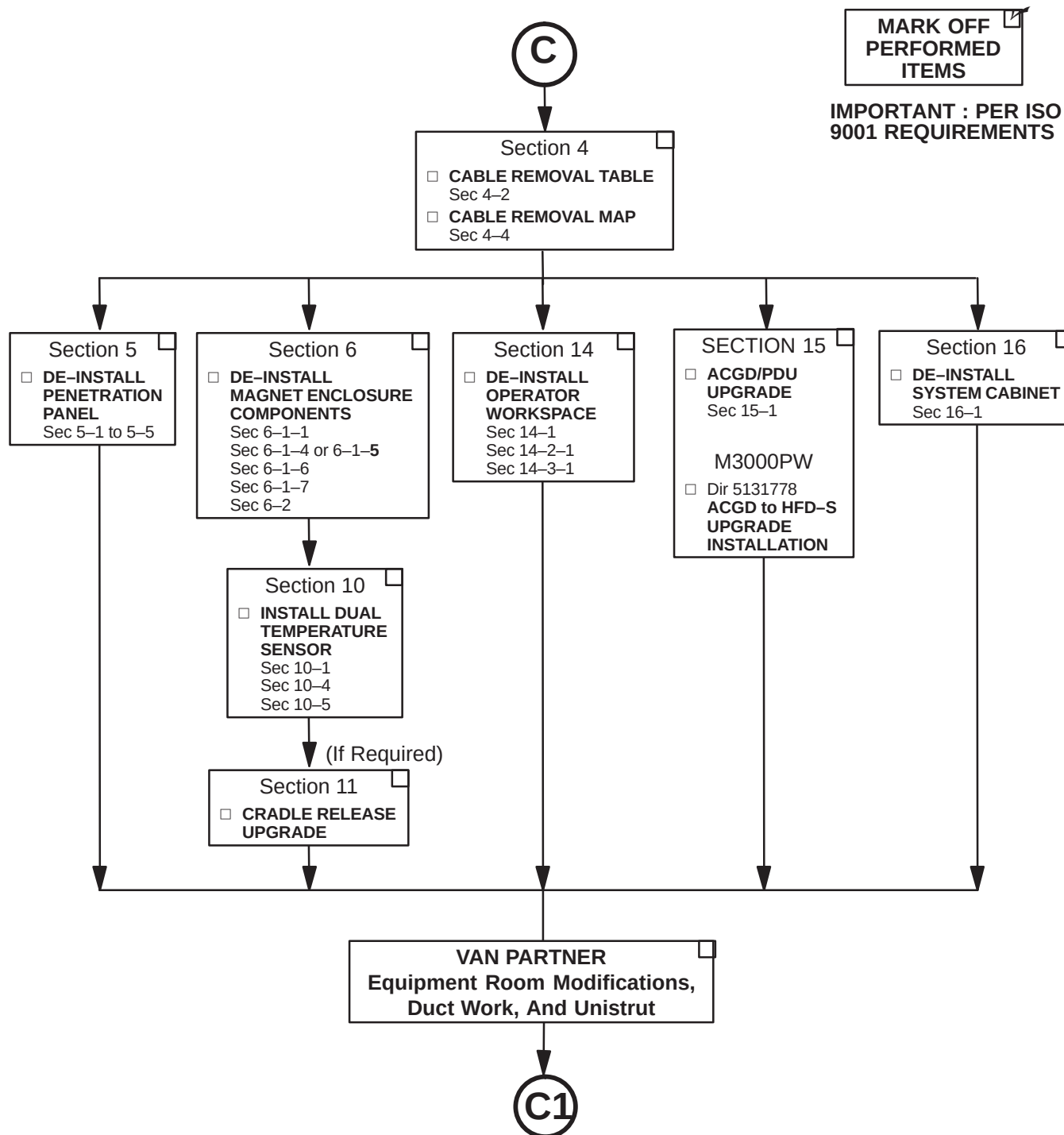
2-2-3-A SYSTEM MECHANICAL UPGRADE FLOWCHART FOR WideOpen (Continued)

Installation Procedures to be Completed by FE and/or Van Partner



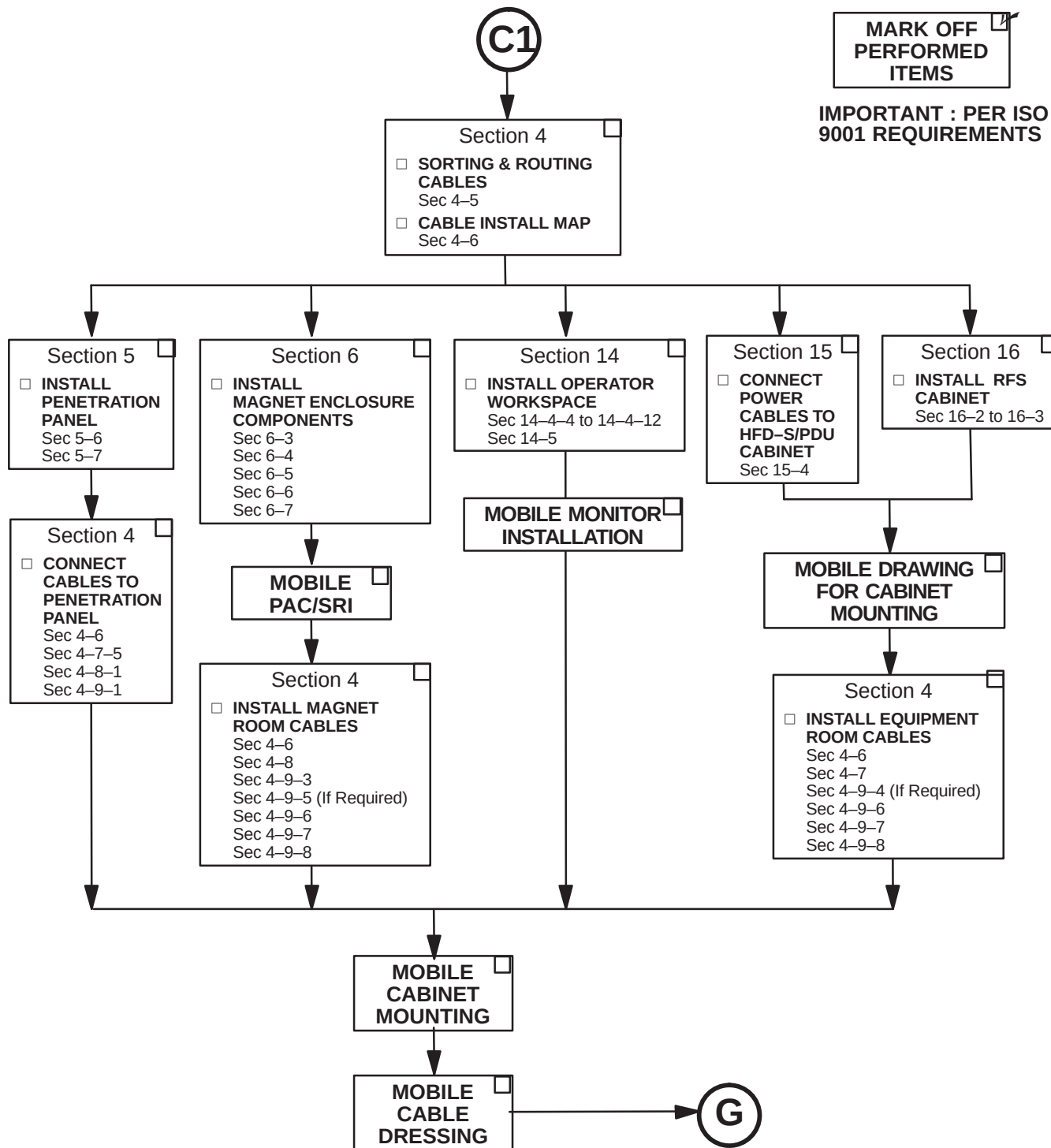
2-2-3-B SYSTEM MECHANICAL UPGRADE FLOWCHART FOR TwinSpeed

De-Installation Procedures to be Completed by FE and/or Van Partner



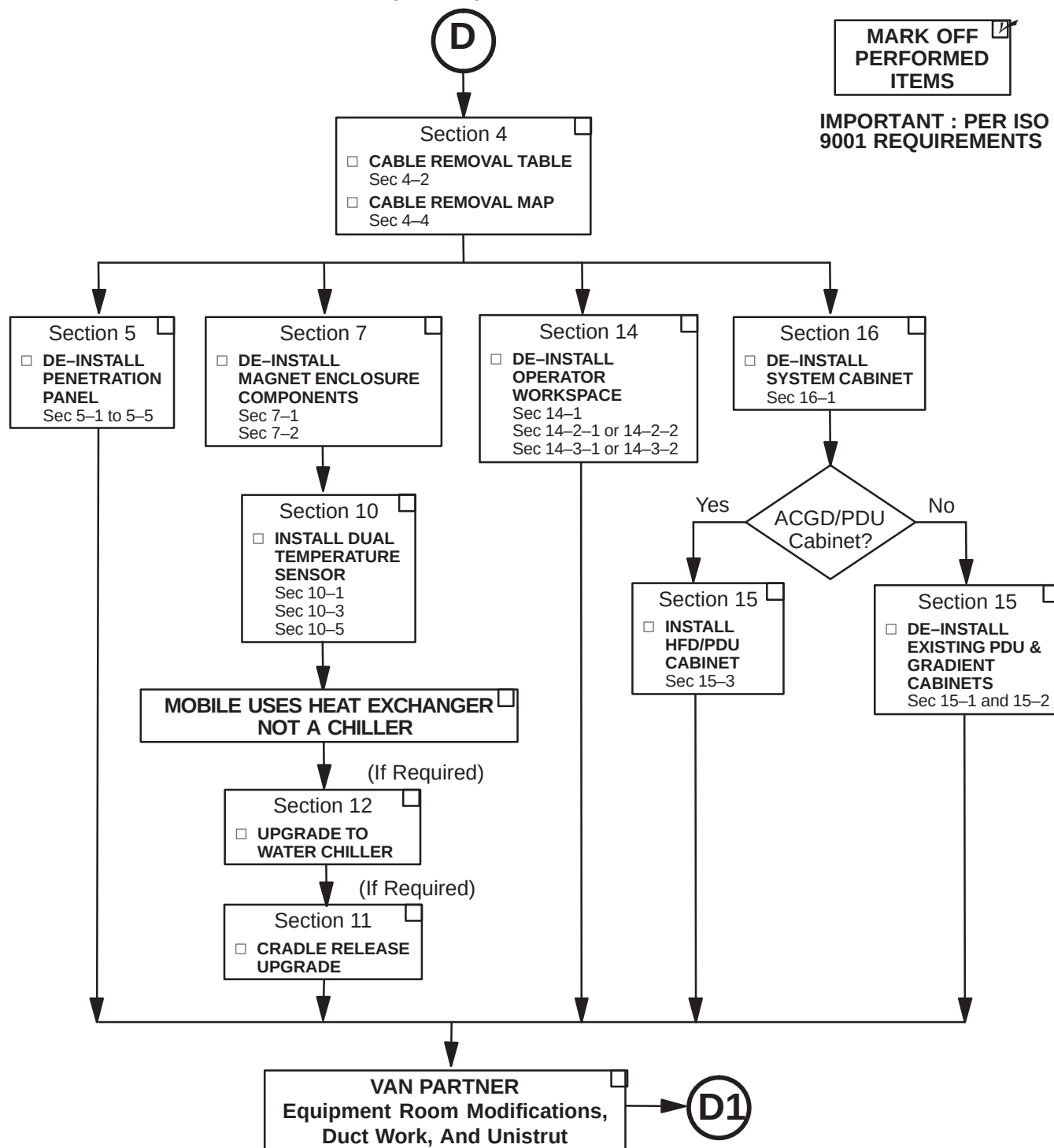
2-2-3-B SYSTEM MECHANICAL UPGRADE FLOWCHART FOR TwinSpeed (Continued)

Installation Procedures to be Completed by FE and/or Van Partner



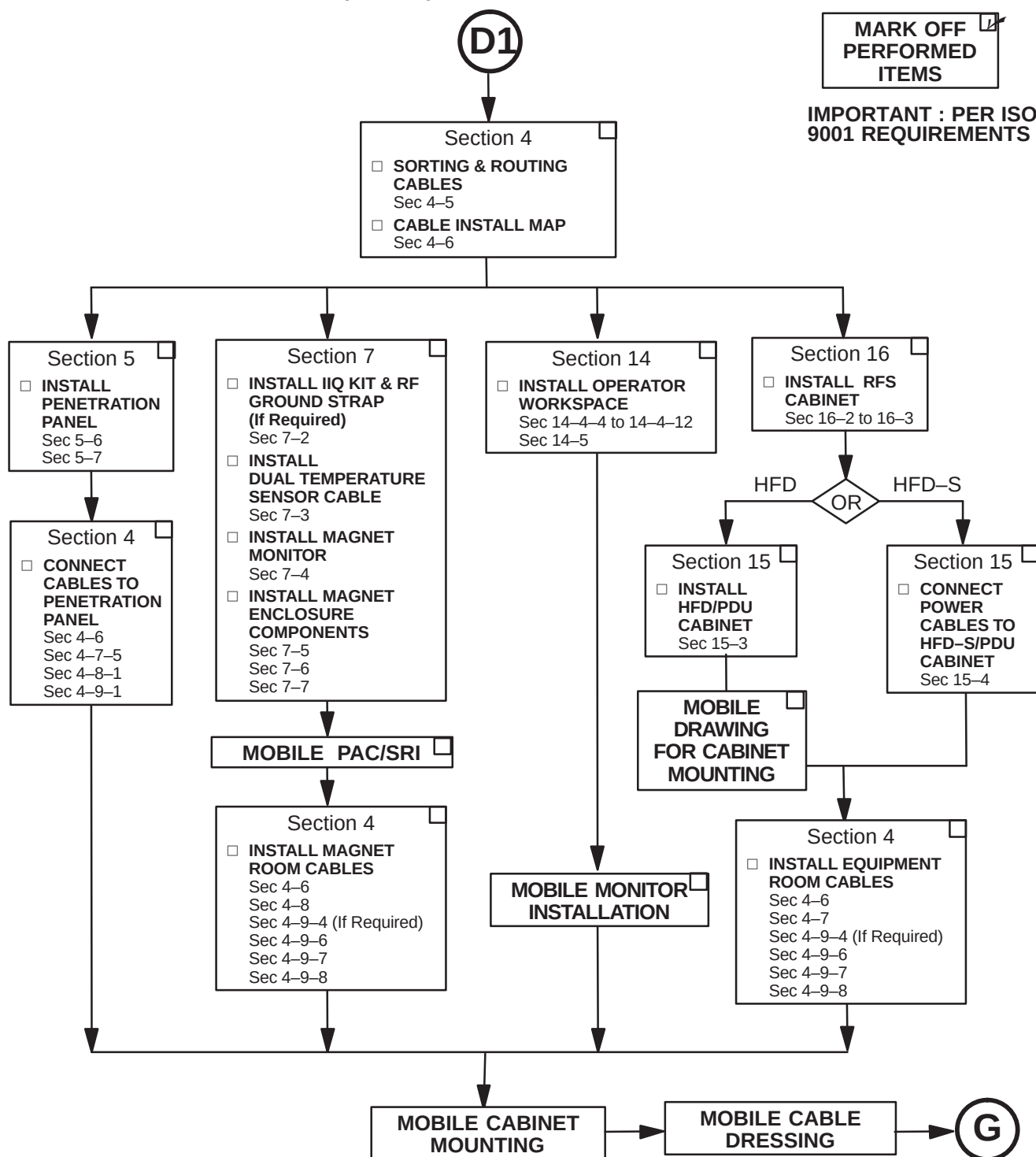
2-2-3-C SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC OR LCC WITH HORIZON ENCLOSURE LX

De-Installation Procedures to be Completed by FE and/or Van Partner



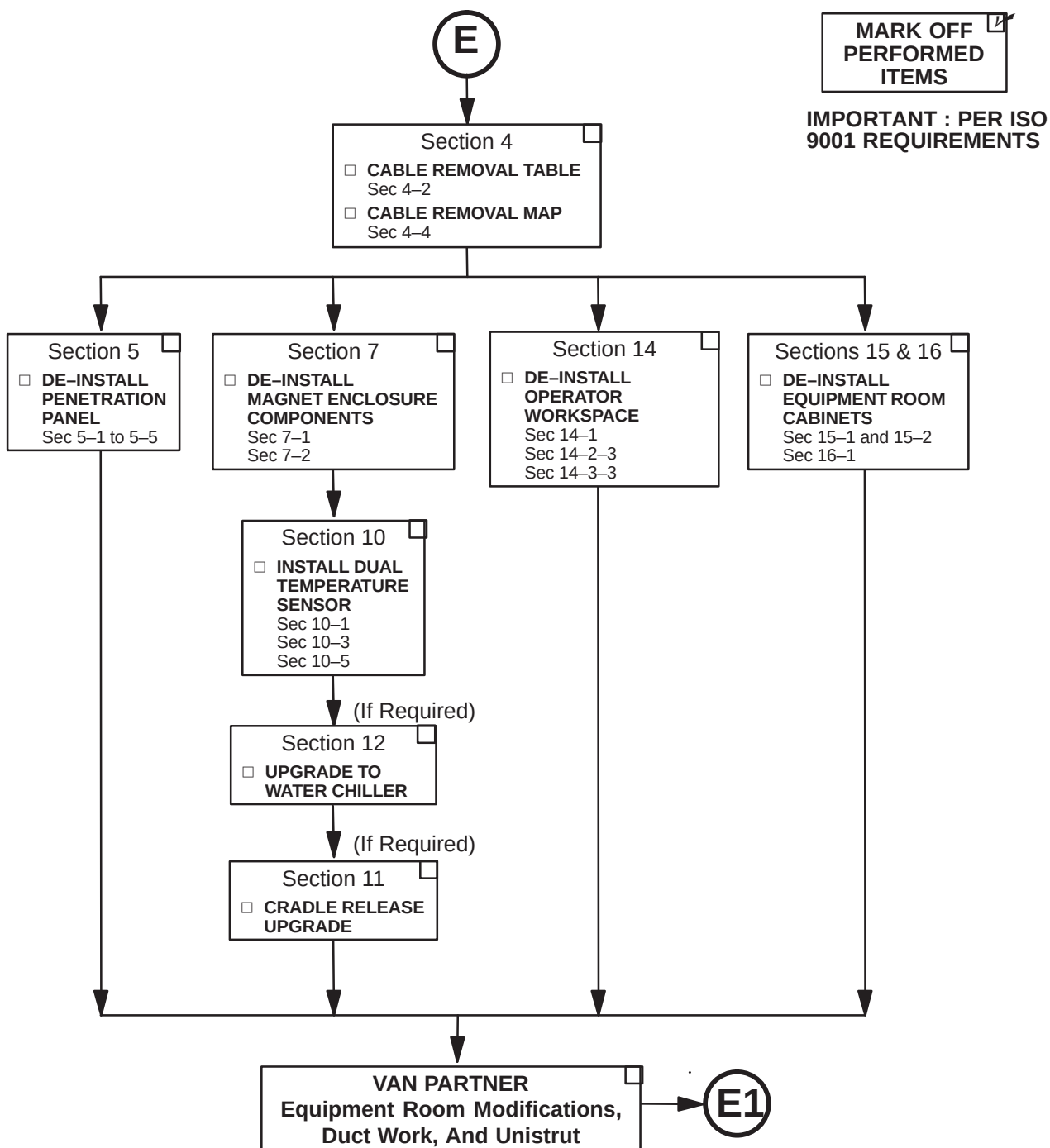
2-2-3-C SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC OR LCC WITH HORIZON ENCLOSURE LX (Continued)

Installation Procedures to be Completed by FE and/or Van Partner



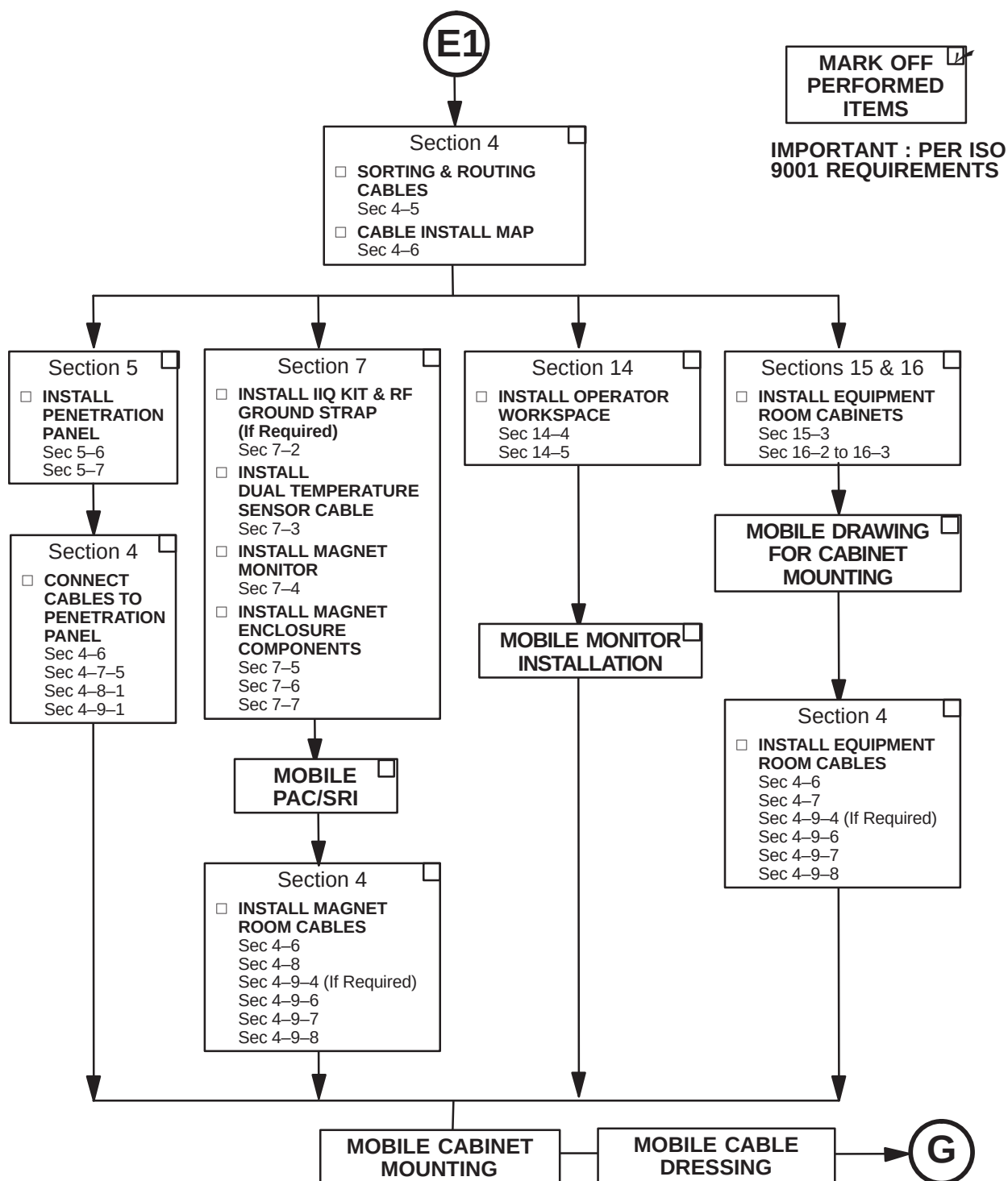
2-2-3-D SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC HORIZON

De-Installation Procedures to be Completed by FE and/or Van Partner



2-2-3-D SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON-LCC HORIZON (Continued)

Installation Procedures to be Completed by FE and/or Van Partner

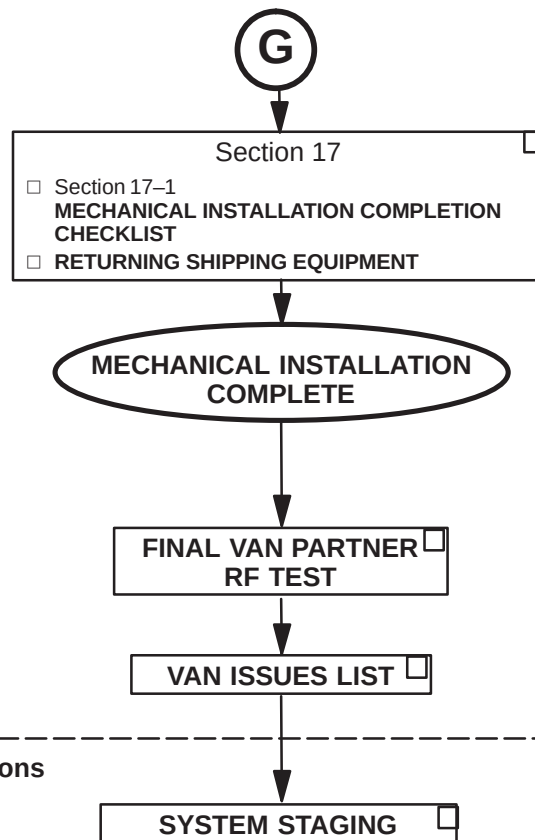


2-2-4 UPGRADE COMPLETION CHECKS AND POWER ON FLOWCHART

Procedures to be Completed by FE and/or Van Partner

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



Procedures to be Completed by FE and Applications

SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
2. Make sure that site customer has archived all patient information. Execute "Save Info".
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – SYSTEM CABLES

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LCC Pre-WideOpen with Octane, RF/PDU, SGD, and Magnet Monitor	
Pre-LCC LX with Octane, SGD, RF/PEN, PDU, w/o Magnet Monitor	
Pre-LCC LX with Indigo, SGD, RF/PEN, PDU, w/o Magnet Monitor	
Pre-LCC LX with Octane, Grad/GRAM, RF/PEN, PDU, w/o Magnet Monitor	
Pre-LCC LX with Indigo, Grad/GRAM, RF/PEN, PDU, w/o Magnet Monitor	
Horizon 5.x w/Operator Console/Computer, Grad/GRAM, RF/PEN, PDU, w/o Mag Mon	
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4-1 OVERVIEW

This section provides information on removing and re-labeling Pre-EXCITE system cables, and installing new cables for upgrading to Signa 1.5T HDx.

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx

Table 4-1 lists all cables to be removed from various early system configurations for upgrading to a Signa 1.5T HDx system.

Signa Advantage 4.x systems that are being upgraded to Signa HDx 1.5T require many more cables to be removed or re-labeled than indicated on the cable removal maps on pages 4-9 thru 4-12. Table 4-1 lists all cables removed from various early system configurations, including Advantage 5.4 and Horizon 5.x, for upgrading to Signa HDx.

On the following table 4-1, the letters at the top of each column on the right side of the table are defined by the following cable removal explanations. An "X" in the box indicates that the cable should be removed.

For upgrading to HDx 1.5T, remove these cables if existing system is:

- A** – Signa LCC WideOpen LX with Octane Computer, ACGD and Magnet Monitor
- B**– Signa LCC TwinSpeed LX with Octane Computer, ACGD and Magnet Monitor
- C** – Signa Pre-LCC LX with Octane Computer and ACGD, and without Magnet Monitor
- D** – Signa LCC LX with Octane Computer, RF/PDU, SGD, and Magnet Monitor
- E** – Signa Pre-LCC LX with Octane Computer, RF/PEN, PDU, SGD (or Grad/GRAM), and w/o Magnet Monitor
- F** – Signa Pre-LCC LX with Indigo Computer, RF/PEN, PDU, SGD (or Grad/GRAM), and w/o Magnet Monitor
- G** – Signa Horizon 5.x with Operator Console/Computer, RF/PEN, PDU, Grad/GRAM and w/o Magnet Monitor
- H**– Signa Advantage 5.x with Operator Console/Computer, RF, PEN, PDU, Gradient and w/o Magnet Monitor
- I** – Signa Advantage 4.5 or later but pre-5.4 with Operator Console, Computer, RF, PEN, PDU, and Gradient
- J** – Signa Advantage 4.x but pre-4.5 with Operator Console, Computer, RF, PEN, PDU, and Gradient

Note

Discard means to remove and return to GE for recycling

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

TABLE 4-1
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T HDx

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
002	PD1-A2-TS1-32/NEUT/GND	MR2-A12-SERVICE									X	X	
003	PD1-A2-TS1-1,2,3/NEUT/GND	MR3-A10-AC IN								X	X	X	
007	PD1-FEEDER 15	PP2-A6-AC IN										X	
008	PD1-FEEDER 16	PP2-A6-SERVICE										X	
009	PD1-A2-TS1-10,11,12/NEUT/GND	MR1-A7-J1					X	X	X	X	X	X	
011	PD1-A2-TS1-7,8,9	RESISTIVE SHIM CABINET											SEE NOTE 1
012	PD1-A2-TS1-21/NEUT/GND	OC1-A8-J3 (AC IN)									X	X	
013	PD1-A2-TS1-33/NEUT/GND	OC1-A8-J6 (SERVICE)									X	X	
014	PD1-A2-TS1-34/NEUT/GND	CC1-A20-SERVICE									X	X	
015	PD1-A2-TS1-13,14,15/NEUT/GND	CC1-A20-AC IN									X	X	
016	PD1-A2-TS1-16,17,18/NEUT/GND	CC2-A20-AC IN										X	
017	PD1-A2-TS1-34/NEUT/GND	CC2-A20-SERVICE										X	
020	PD1-A2-TS1-28,29,30	RESISTIVE SHIM CABINET											SEE NOTE 1
021	PD1-A2-GND	CC1-A20-GND									X	X	
022	PD1-A2-GND	CC2-A20-GND										X	
023	PD1-GND BUS	OC1-GND							X	X	X	X	
027	PD1-A2-TS1-32/NEUT/GND	PP2-A15-J1-SERVICE								X	X	X	
028	PD1-A2-TS1-31/NEUT/GND	PP2-A15-J2-AC IN								X	X	X	
030	PD1-A2-TS1-16,17,18,NEUTGND	MR2-A12-J1 AC IN				X	X	X	X	X	X	X	
035	PD1-A2-TS1-22,23,24/NEUT/GND	CC2-A20-AC IN										X	
036	PD1-FEEDER 5-TS1-4,5,6	MR6										X	
037	PD1-GND BUS	MR2-A11 GND	X	X	X	X	X	X	X	X	X	X	
038	PD1-GND BUS	MR1-A7 GND	X	X	X		X	X	X	X	X	X	
041	PD1-TS1-1,2,3,GND	MR3-CABI/FJ6				X	X	X	X				
042	PD1-TS1-4,5,6,GND	MR3-CABI/FJ7				X	X	X	X				
043*	MR7-A4-J7	PD1-A2					X	X	X				
044	PP1-GND	MR1-A7-GND	X	X	X	X	X	X	X				
045	PP1-GND	MG6-GND	X	X	X	X	X	X	X				
046	PD1-TS1-20,NEUT,GND	HE1					X	X	X				SEE NOTE 13
047	PD1-TS1-X,X,NEUT,GND	OW1-A2-A5-J1						X					
048	PD1-GND BUS	OW1-A3-A5-GND	X	X	X	X	X	X					
049	OW1 A1 A1	OW1 A2 A5 J2	X	X	X	X	X	X					
050	OC1-AC-IN	PD1-TS1-13,14,15,N,GND							X	X			
064	MR2-A12-J1-AC OUT	CC1							X	X			
065	PD1-TS1-26,27,GND	WC1											SEE NOTE 13
200	MR2-A11-J19	MR3-A9-J1									X	X	
201	MR2-A11-J20	MR3-A9-J2									X	X	
202	MR2-A11-J21	MR3-A9-J3									X	X	
203	MR2-A11-J22	MR3-A9-J4									X	X	
207	MR2-A11-J6	RF DOOR SWITCH											SEE NOTE 2
223	OC1-A8-J13	MR2-A1-J36									X	X	
224	CC1-A21-J19	MR1-A7-J8									X	X	

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T HDx

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
226	MR1-A7-J5	MR2-A11-J14									X	X	
227	PD1-A1-A2-J2	MR2-A11-J16									X	X	
228	CC1-A21-J10	MR2-A11-J17									X	X	
229	MR2-A11-J1	MR1-A7-J3	X	X	X	X	X	X	X	X	X	X	
231	MR2-A11-J2	PP1-J8	X	X	X	X	X	X	X	X	X	X	
233	MR2-A1-J5	PP1-J7									X	X	
235	MR1-A7-J13	PP1-J44	X	X	X	X	X	X	X	X	X	X	
236	MR1-A7-J12	PP1-J4	X	X	X	X	X	X	X	X	X	X	
238	CC1-A21-J14	OC1-A8-J8									X	X	
239	CC1-A21-J15	OC1-A8-J10									X	X	
244	CC1-A21-J13	BTI-HOST										X	
248	CC2-A21-J1	OC1-A9-J1										X	
249	CC2-A21-J22	OC1-A8-J12										X	
256	OC1-A8-J11	PP1-J12									X	X	
257	MG3-A3-J1	PP1-J44	X	X	X	X	X	X	X	X	X	X	
258	MG3-A17-J1	PP1-J4	X	X	X	X	X	X	X	X	X	X	
260	PP1-J7	MG3-A17-J2									X	X	
261	PP1-J6	MG2-A11-A1-J1										X	
262	PP1-J8	MG3-A3-A5-OUTPUT	X	X	X	X	X	X	X	X	X	X	
274	CC1-A21-J20	MR1-A7-J7									X	X	
280	PP2-A1-J2	PP1-J15										X	
281	PP2-A1-J3	PP1-J11										X	
282	MG3-A2-J8	PP1-J12	X	X	X	X	X	X	X	X	X	X	
287	CC2-A21-J15	CC1-A21-J2										X	
296	PP1-J18	EO2											SEE NOTE 8
297	PP1-J18	EO1											SEE NOTE 8
300	PP1-J15	MG2-A15											
301	CC2-A21-J20	CC1-A21-J28										X	
302	CC2-A21-J21	CC1-A21-J29										X	
318	PP1-J11	MG3-A7-LONG DRIVE	X	X	X	X	X	X	X	X	X	X	
322	MR2-A11-J11	PP2-A6-J18										X	
325	CC1-A21-J27	CC2-A21-J19										X	
327	CC1-A21-J25	CC2-A21-J17										X	
346	MR2-A11-J41	CM1-FRONT									X	X	
347	MR2-A11-J42	CM1-BACK									X	X	
354	CM1	ST1									X	X	
370	MR1-A7-J19	PP1-A9-J46										X	
384	PP2-A15-J17	PP1-J20								X	X	X	
385	PP1-J20	MG2-A30-TS1,2,3											
386	PP2-A15-J21	PP1-J47								X	X	X	
387	MG2-A29-P1	PP1-J47	X	X	X	X	X	X	X	X	X	X	
388	PP2-A15-J22	PP1-J11								X	X	X	

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T HDx

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
389	PP2-A15-J20	PP1-J15								X	X	X	
404	MG3 A17 J1	MG2 A16 A5 J6	X	X	X	X	X	X	X	X	X	X	
405	MR3-A9-TS1-1/2	PP1-A7-1,2								X	X	X	
405	MG3-TS1-1,2	PP1-A7-1,2								X	X	X	
406	MR3-A9-TS1-5/6	PP1-A7-3,4								X	X	X	
406	MG3-TS1-3,4	PP1-A7-3,4								X	X	X	
407	MR3-A9-TS1-5/6	PP1-A7-5,6								X	X	X	
407	MG3-TS1-5,6	PP1-A7-5,6								X	X	X	
408	PP1-J16	PD1-A14-A1-J1								X	X	X	
409	PP1-J16	MG2-A12-A1-J7								X	X	X	
410	PP1-J14	PP2-A15-J23										X	
411	PP1-J14	MG3-A13-J1										X	
412	PP1-J51	MG3-A32-J1								X	X	X	
413	PP1-J51	PP2-A15-J28(A16-J9)								X	X	X	
414	MG2-A35-MIC	MG3-A2-J9											
415	PP1-A8-J45	MG3-A33-J7										X	
416	MR1-A7-J21	PP1-A8-J45										X	
417	PP2-A15-J19	MR2-A11-J32									X	X	
418	MR2-A11-J39	PP2-A15-J										X	
419	MR2-A11-J26	PP1-A10-J1									X	X	
420	MG3-A33-J1	PP1-A10-J2									X	X	
421	MG2-A33-J3	MG2-A31-J1											SEE NOTE 14
422	MG2-A33-J2	MG3-A1-J4											
424	PP1-A10-J5/J6	MG2-A33-J11/J12									X	X	
425	MR2-A11-J12	PP1-J6										X	
426	MR1-A7-J9	MR3-A9-J5									X	X	
427	MR2-A11-J17	CC1-A21-J13									X	X	
428	MR2-A11-J44	CC1-A21-J35									X	X	
429	BT1	CC1-A21-J25									X	X	
430	MR2-A11-J24	CC1-A21-J26									X	X	
431	MR2-A11-J25	CC1-A21-J27									X	X	
433	OC1-A9-J1	CC1-A21-J10									X	X	
434	CC1-A8-J12	CC1-A21-J29									X	X	
435	MR2-A11-J13	PP1-A10-J3									X	X	
436	MG2-A12-A1-J1	MG3-A33-J3									X	X	
437	MG2-A12-A1-J2	MG3-A33-J4									X	X	
439	MG3-A3-J4	MG3-A33-J1									X	X	
440	MG3-A3-J3	MG3-A34-J2									X	X	
441	MG3-A33-J7	MG3-A34-J8									X	X	
446	PP1-A9-J46	MG2-A12-A13-J5										X	
447	PP1-A9-J46	MG2-A12-A13-J3										X	
448	MR2-A11-J44	CC1-A21-J1										X	

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T HDx

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
449	MR2-A11-J24	CC1-A21-J34										X	
450	MR2-A11-J25	CC1-A21-J35										X	
453	PP1-J50	MG2-A11-A1-J4									X	X	
454	MR2-A11-J34	PP1-J50									X	X	
457	OM1	PP1-J17											SEE NOTE 3
458	OM3	PP1-J17											SEE NOTE 3
470	MR1-A7-J17	PP1-A11-J71								X	X	X	
474	PP1-A11-J72	MG3-A12-A1-J3								X	X	X	
475	PP1-A11-J73	MG3-A12-A1-J4								X	X	X	
476	PP1-A11-J74	MG3-A33-J7								X	X	X	
477	PP1-A11-J75	MG3-A12-A1-J5								X	X	X	
478	PP1-A11-J76	MG3-A12-A1-J6								X	X	X	
485	MG3 A17 J2	MG2 A16 A3 OUT	X	X	X	X	X	X	X	X	X	X	
486	PP1-A14-J85	MR1-A7-J22	X	X	X	X	X	X	X	X	X	X	
487	PP1-A15-J78	MR1-A7-J20	X	X	X	X	X	X	X	X	X	X	
488	PP1-J80	MR2-A11-J8	X	X	X	X	X	X	X	X	X	X	
489	PP1-J81	MR2-A11-J9	X	X	X	X	X	X	X	X	X	X	
490	PP1-J82	MR2-A11-J10	X	X	X	X	X	X	X	X	X	X	
491	PP1-J80	MG3-A36-J7	X	X	X	X	X	X	X	X	X	X	
492	PP1-J81	MG3-A36-J8	X	X	X	X	X	X	X	X	X	X	
493	PP1-J82	MG3-A36-J9	X	X	X	X	X	X	X	X	X	X	
494	PP1-A14-J87	MG3-A36-J6	X	X	X	X	X	X	X	X	X	X	
495	PP1-A15-J78	MG3-A36-J10	X	X	X	X	X	X	X	X	X	X	
499	PP1-A14-J84	MR2-A11-J5	X	X	X	X	X	X	X	X			
605	MR2-A11-J11	PP1-J10			X		X	X	X	X	X	X	
605	MS1-A3-A1-P403	PP1-J10			X		X	X	X	X	X	X	
611	PP1-SHIM	TAC-J22											
612	PP1-SHIM	MG2-A12-A1											SEE NOTE 9
700	PP1 A14 J86	MG2/MG3 A17 J2	X	X	X	X	X	X	X	X	X	X	
701	MR2-J11-J15	RF DOOR SWITCH											SEE NOTE 2
702	MR2-J11-J18	MR1-A7-J5	X	X	X	X	X	X	X	X	X	X	
703	MR2 A11 J23	PD1 A14 A1 J2	X		X	X	X	X	X	X			
704	PP1-J12	OC1-A10-J9							X	X			
706	MR2 A11 J14	PD1 A14 A1 J3 / MR1 PD1 J3	X		X	X	X	X	X	X			
707	MR2-A11-J26-ESTOP	OC1-J8							X	X			
708	MR2-A11-J16	MR1-A7-J27	X	X	X	X	X	X	X	X			
709	MR1-A7-J26	MR3-A9-J1	X	X	X	X	X	X	X	X			
710	MR2-A11-J17	MR3-A9-J2	X	X	X	X	X	X	X	X			
711	PP1-A17-J91	MG2-A33								X	X	X	
712	PP1-A17-J91	MG2-A11-A1								X	X	X	
711/712	PP1-A17-J91	MR2-A11-J13	X	X	X	X	X	X	X				
711/712	PP1-A17-J91	MG2-A33-J11,J12,J13 MG2-A11-J9,J10											

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T HDx

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
715	PP1-J14	MG2-A33-J1											
716	PP1-J89	MG2-A11-J2											
718	MR2-A11-J29	OC1-A10-J4							X	X			
719	PP1-J6	MR2-A11-J4								X			
726	PP1-A16-J77	MR1-A7-J18	X	X	X	X	X	X	X	X	X	X	
727	PP1-A16-J77	MG3-A36-J5	X	X	X	X	X	X	X	X	X	X	
728	MG3 A36 J11	MG2 A16 OUT J1	X	X	X	X	X	X	X	X	X	X	
729	MG3 A36 J12	MG2 A16 OUT J2	X	X	X	X	X	X	X	X	X	X	
730	MG3 A36 J13	MG2 A16 OUT J3	X	X	X	X	X	X	X	X	X	X	
731	MG3 A36 J14	MG2 A16 OUT J4	X	X	X	X	X	X	X	X	X	X	
732	MG3 A36 J15	MG2 A16 OUT J5	X	X	X	X	X	X	X	X	X	X	
733	MG3 A36 J16	MG2 A16 OUT J6	X	X	X	X	X	X	X	X	X	X	
734	MG3 A36 J17	MG2 A16 OUT J7	X	X	X	X	X	X	X	X	X	X	
735	MG3 A36 J18	MG2 A16 OUT J8	X	X	X	X	X	X	X	X	X	X	
736	MG3 A36 J19	MG2 A16 A8-J9	X	X	X	X	X	X	X	X	X	X	
743	MG2-A12-A2-J2	MG3-A3-J3								X	X	X	
744	MG2-A12-A2-J1	MG3-A3-J4								X	X	X	
749	MG2-A33-J10	MG2-A36	X	X	X	X	X	X	X	X	X	X	
753*	MR7-A4-J1	MR3-A11-J5					X	X	X				SEE NOTE 4
754*	MR7-A4-J2	MR3-A11-J4					X	X	X				SEE NOTE 4
755*	MR7-A4-J3	MR3-A11-J3					X	X	X				SEE NOTE 4
756*	MR7-A4-J4	MR3-A11-J8					X	X	X				SEE NOTE 4
757*	MR7-A4-J5	MR3-A11-J7					X	X	X				SEE NOTE 4
758*	MR7-A4-J6	MR3-A11-J6					X	X	X				SEE NOTE 4
759*	MR3-11-J12	MR7-A4-J8					X	X	X				SEE NOTE 4
760*	MR3-11-J11	MR7-A4-J9					X	X	X				SEE NOTE 4
761*	MR3-11-J10	MR7-A4-J10					X	X	X				SEE NOTE 4
762	MR3CABI/FJ3	PP1-A7-1/2				X	X	X	X				SEE NOTE 6
762	MG2-A12-A1-1/2	PP1-A7-1/2											SEE NOTE 5
763	MR3CABI/FJ4	PP1-A7-3/4				X	X	X	X				SEE NOTE 6
763	MG2-A12-A1-3/4	PP1-A7-3/4											SEE NOTE 5
764	MR3CABI/FJ5	PP1-A7-5/6				X	X	X	X				SEE NOTE 6
764	MG2-A12-A1-5/6	PP1-A7-5/6											SEE NOTE 5
767	MR1-A7-J37	PP1-J51								X			
768	MR1-A7-J31	PP1-J20	X	X	X	X	X	X	X				
769	MR1-A7-J34	PP1-J15	X	X	X	X	X	X	X				
770	MR1-A7-J41	PP1-J11	X	X	X	X	X	X	X				
771	MR1-A7-J36	PP1-J47	X	X	X	X	X	X	X				
772	MR1-A7-J40	PP1-J14	X	X	X	X	X	X	X				
773	MR1-A7-J35	PP1-A17-J1	X	X	X	X	X	X	X				
774	MR1-A7-J32	MR2-A11-J25	X	X	X	X	X	X	X				
775	MR1-A7-J42	PP1-A11-J92	X	X	X	X	X	X	X				

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T HDx

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
776	MR1-A7-J44	PP1-A11-J93	X	X	X	X	X	X	X				
777	MR1-A7-J43	PP1-A11-J94	X	X	X	X	X	X	X				
778	MG2-A12-A2-J3	PP1-A11-J72	X	X	X	X	X	X	X				
779	MG2-A12-A2-J4	PP1-A11-J73	X	X	X	X	X	X	X				
780	MG3-A3-J6	PP1-A11-J74	X	X	X	X	X	X	X				
781	MG2-A12-A2-J5	PP1-A11-J75	X	X	X	X	X	X	X				
782	MG2-A12-A2-J6	PP1-A11-J76	X	X	X	X	X	X	X				
783	MR1-A7-J37	PP1-F5,F6	X	X	X	X	X	X	X				
784	MG6-J1	PP1-F5,F6	X	X	X	X	X	X	X				
785	MR1-A7-J47	PP1-F7,F8	X		X	X	X	X	X				
786	MG6-J2	PP1-F7,F8	X		X	X	X	X	X				
787	MR2-A11-J30	CC1-A1-A1-SERIAL A									X	X	
788	PP1-J12	OW1-A1-A4-J7	X	X	X	X	X	X					
789	MR2-A11-J26	OW1-A1-A2-J18	X	X	X	X	X	X					
790	OW1-A2-A6-XO,RO,X1,R1	MR2-A11-J21						X					
791	MR2-A11-J30	OW1-A2-A4-EXP1	X	X	X	X	X	X					
792	OW1-A2-A1-MIC/AUDIO	OW1-A1-A4-J11,J13				X	X	X					
793	OW1-A2-A1-COM 1	OW1-A1-A4-J9						X					
794	OW1-A2-A3-MIC/AUDIO	OW1-A1-A4-J12,J14						X					
795	OW1-A2-A3-PWR	OW1-A1-A4-J17						X					
796	OW1-A2-A3-KBD/MS	OW1-A1-A4-J8						X					
797	OW1 A2 A1 PC	OW1 A3 LCD	X	X	X	X	X	X					
799	OW1-A15-MONITOR (OR 820)	OW1-A4 (B/W) (OR 820)				X	X	X					
800	OC1-J1	CC1-A2-J1 (MUX)							X	X			
801	OC1-J10	CC1-A2-J2 (SECUR. KEY)							X	X			
802	OC1-J12	CC1-A1-J9 (PLASMA)							X	X			
803	OC1-J13	CC1-A1-A10 (VIDEO)							X	X			
804	OC1-J3	CC1-A1-A1 (BODY)							X	X			
805	OW1-A1-A4-J6	OW1-A7 (NON-SCIM KYBRD)	X	X	X	X	X	X					
806	OW1-A2-A3-SER 1	OW1-A13 (MODEM)						X					
807	OW1-A16	OW1-A5 (DASM)	X	X	X	X	X	X					
809	OW1-A15-MIC	OW1-A1-A4-J12,J14	X		X	X	X						
810	OW1-A15-MD,KYBD	OW1-A1-A4-J8	X	X	X	X	X						
811	OW1-A2-A4-PORT2	OW1-A15-ETHERNET	X	X	X	X	X						
812	OW1-A15-COM1	OW1-A13 (MODEM)	X	X	X	X	X						
813	OW1-A2-A1-COM2	OW1-A15-COM2	X	X	X	X	X						
814	OW1-A1-A4-J9	OW1-A2-A1-KBD,MOUSE	X	X	X	X	X						
815	OW1-A16-SCSI	OW1-A15-SCSI (PCI)	X	X	X	X	X						
816	OW1-A2-A6-J1	OW1-A15-BIT3	X	X	X	X	X						
817	OW1-A1-A2-J17	OW1-A2-A5-J3	X	X	X	X	X						
818	MR2-A11-J31	OW1-A16-PORT6,7,8	X	X	X	X	X						

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T HDx

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
819	OW1-A2-A7-SCSI	OW1-A15-SCSI	X	X	X	X	X						
820	OW1-A15-MONITOR	OW1-A4 (B/W)				X	X	X					
836	MR2-A11-J21	OW1-A15-BIT3	X	X	X	X	X						
848	MG2-A4-J5	MG2-A11-J4											
861	MG2-A4-J4	MG2-A33-J5,6,7,8,9	X	X	X	X	X	X	X				
862	MG2-A4-J1	MG2-A5-J1											
863	MG2-A4-J2	MG2-A5-J2											
897	MG2-A4-J3	MG2-A33-J4											
939	MDS-UPS	MSM4											
941	MSM1-J11	SCC-J1											
942	MSM1-J5	OW1-A2-A4-PORT 7											
966	MR1-A7-J1	MR3-PD1-7,8,9,GND	X	X	X								
967	MR1-A7-CB1	MR3-PD1-10,11,12,GND	X	X	X								
968	MR2-A11-J1	MR3-PD1-4,5,6,GND	X	X	X								
969	OW1-A2-A5-GND	MR3-PD1-18,19,GND,N	X	X	X	X	X						
970	WC1	MR3-PD1-16,17,GND	X		X	X							SEE NOTE 13
971	MR2-A11-J23	MR3-PD1-J2	X	X	X	X	X						
972	MR2-A11-J14	MR3-PD1-J3	X	X	X	X	X						
974	MR3-PD1-J5	TAC-J12											
976	MR3-CAB I/F-J3	TAC +/- X											SEE NOTE 10
977	MR3-CAB I/F-J4	TAC +/- Y											SEE NOTE 10
978	MR3-CAB I/F-J5	TAC +/- Y											SEE NOTE 10
979	TAC-X+,WB-,ZM-	PP1-XWB-,ZM-											SEE NOTE 10
980	TAC-Y+,WB-,ZM-	PP1-YWB-,ZM-											SEE NOTE 10
981	TAC-Z+,WB-,ZM-	PP1-ZWB-,ZM-											SEE NOTE 10
982	PP1-ZM+,ZM-	MG2-A12-A1-7/8											SEE NOTE 10
983	PP1-YZM+,ZM-	MG2-A12-A1-9/10											SEE NOTE 10
984	PP1-XZM+,ZM-	MG2-A12-A1-11/12											SEE NOTE 10
985	PP1-ZWB+,WB-	MG2-A12-A1-1/2											SEE NOTE 10
986	PP1-YWB+,WB-	MG2-A12-A1-3/4											SEE NOTE 10
987	PP1-XWB+,WB-	MG2-A12-A1-5/6											SEE NOTE 10
988	PP1-A7-GND	TAC-GND											SEE NOTE 10
989	MR3-PD1-16,17,N,GND	TAC-J8											
990	PP1-J52	VACUUM SENSOR											SEE NOTE 11
991	PP1-J52	TAC-J9											SEE NOTE 11
992	TSCC-J3	TAC-J10											
1004	OW1-A16-PORT 2	TAC-J27											SEE NOTE 12
1152	PP1-SHIM	MG2-A12-A1											SEE NOTE 9
	MG3-A2-J7	MG2-A11-A1-MIC	X	X	X	X	X	X	X				SEE NOTE 7

4-2 CABLE REMOVAL FOR UPGRADING TO SIGNA HDx (continued)

Table 4-1 Notes:

1. If part of existing system, the shim and magnet power supply cables, Runs 011 and 020, need to be reused for the new HFD/PDU Cabinet. Refer to Section 15-2-2 for instructions on removing and reterminating the cables.
2. A new RF Door Switch cable, Run 701, is provided for upgrade. If existing cable is routed within wall, it may be reused. If existing cable is labeled as Run 207, replace with new Run 701 or relabel per instructions in Section 4.
3. If optional Oxygen Monitor is installed on site, retain Runs 457 and 458 for connection to new Penetration Panel.
4. These cable Runs are removed only if GRAM Cabinet is part of existing site being deinstalled.
5. Magnet Room side of Gradient cables, Runs 762, 763, and 764 may be reused. Butt Splices and Shrink Wrap Tubing are not provided with upgrade for reterminating Gradient cables at rear of magnet. Disconnect only at Penetration Panel and reconnect to new HDx Penetration Panel.
Note: Remove and transfer to new HDx Pen Panel the magnet room side gradient cable bracket if site being upgraded has #1/0 gradient cables in the magnet room which are being retained for this upgrade. The U-bolts on the transferred bracket are larger than the U-bolts on the new Pen Panel – which are designed for #2 size gradient cables.
6. Equipment Room side of Gradient cables, Runs 762, 763, and 764:
For sites with ACGD/PDU cabinet:
 - Cables that are connected to ACGD/PDU cabinet will be reused. Disconnect only at Penetration Panel and reconnect to new HDx Penetration Panel.For sites with SGD cabinet:
 - Early production SGD cabinets were installed using the larger #1/0 gradient cables with braided mesh clamped in the interface panel for grounding. These cables must be removed and replaced with the new #2 gradient cables provided with upgrades.
 - If SGD cabinet was installed using the #2 cables, then these may be reused providing that the length is usable for connection to the new HFD cabinet. If needed, install the new #2 gradient cables provided with this upgrade.For sites with Gradient/GRAM cabinets:
 - De-install existing #1/0 cables and replace with new #2 cables to be routed and connected to HFD cabinet.
7. Existing microphone cable will be replaced with upgrade.
8. Existing Emergency Off Switch cables, Runs 296 and 297, may be retained and reused.
9. Early system configurations installed Run 612, later configurations installed Run 1152 for Shim option.
10. Existing cable Runs 982 to 987 in Magnet Room will not be replaced. They will be reused and connected to new Penetration Cabinet. Runs 976 to 981 in Equipment Room can also be reused with upgraded HFD-S/PDU Cabinet and new HDx Penetration Panel. The TAC Cabinet will not be replaced, so the cables will not be disconnected from the cabinet.
11. Runs 991 and 992 may not be present on all systems. Later versions of TwinSpeed moved the vacuum sensor into the TAC cabinet.
12. Run 1004 will reconnect to cable 2382771 inside the GOC cabinet.
13. For Mobile systems, the Heat Exchanger power cord will be retained and connected to TS-20, NEU, and GND on the PDU.
14. For Cx or LCC magnet with Horizon enclosure. Run 421 will be replaced by a longer Run 421, 46-287908G3. This cable has to be ordered separately.

4-3 RE-LABELING OF EXISTING CABLES

The Signa Advantage 4.x RF Door Switch cable, Run 207, may be reused but must be re-labeled. Refer to Table 4-2.

TABLE 4-2
CABLES TO BE RE-LABELED

OLD RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
207	4.x	RUN #207 MR2-A11-J6 46-243775G411	RUN #701 MR2-A11-J15 46-243775G807	RUN #207 DOOR SWITCH 46-243775G411	RUN #701 RF DOOR SWITCH 46-243775G807

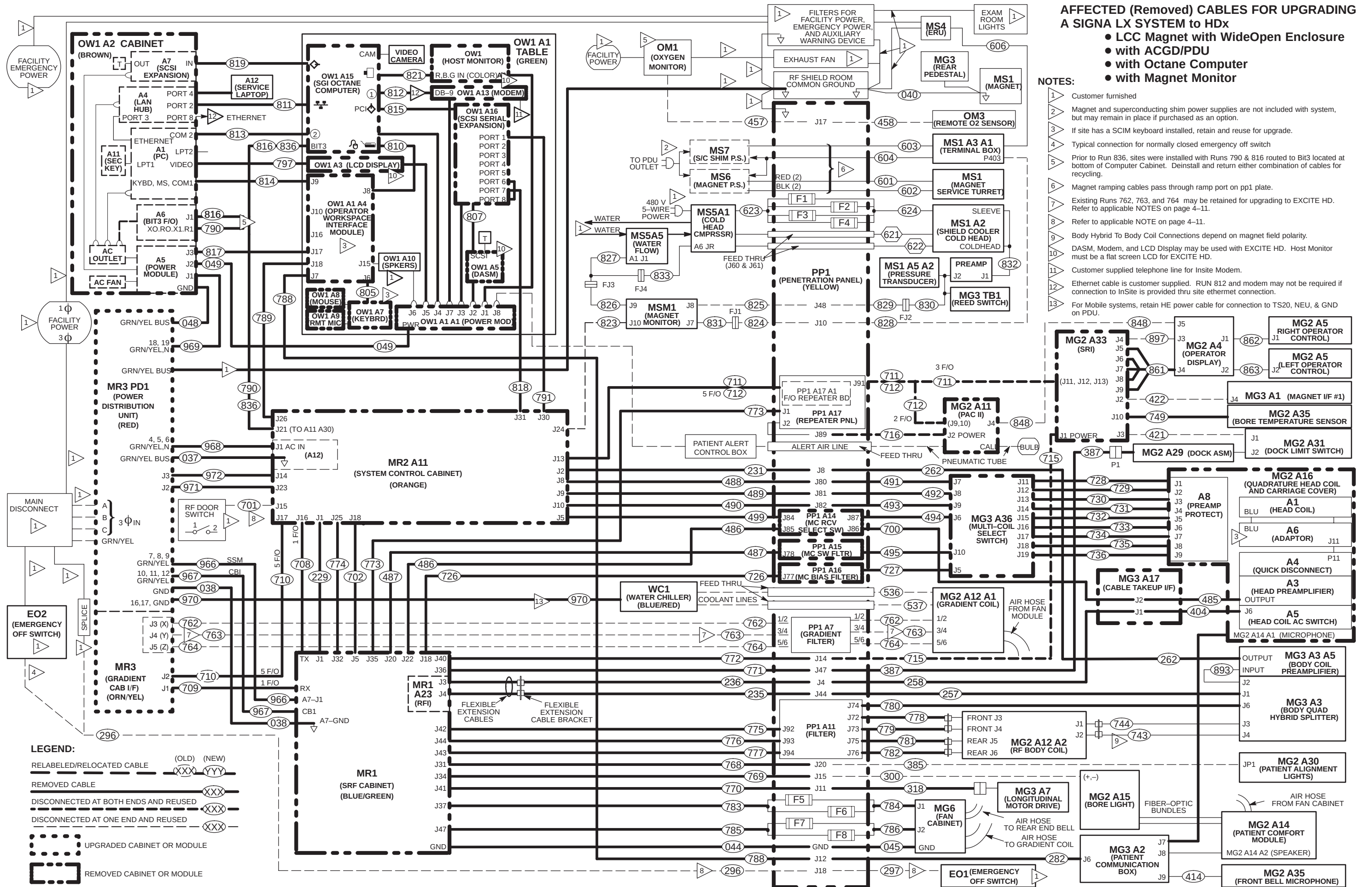
4-4 EXTRA MOBILE UPGRADE CABLES

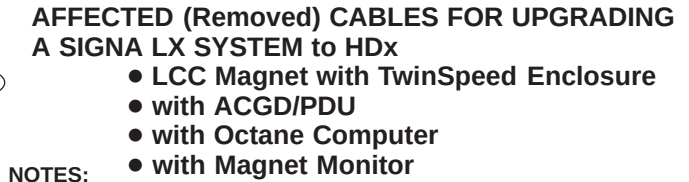
For Mobile upgrades, Runs 1189, 1190, and 1082 are shipped to site, but not installed. Discard Runs 1189, 1190 and 1082 for Mobile upgrades.

4-5 CABLE REMOVAL INTERCONNECT MAPS

The following "Cable Removal" interconnect maps refer only to cables removed, relabeled, and/or relocated when upgrading:

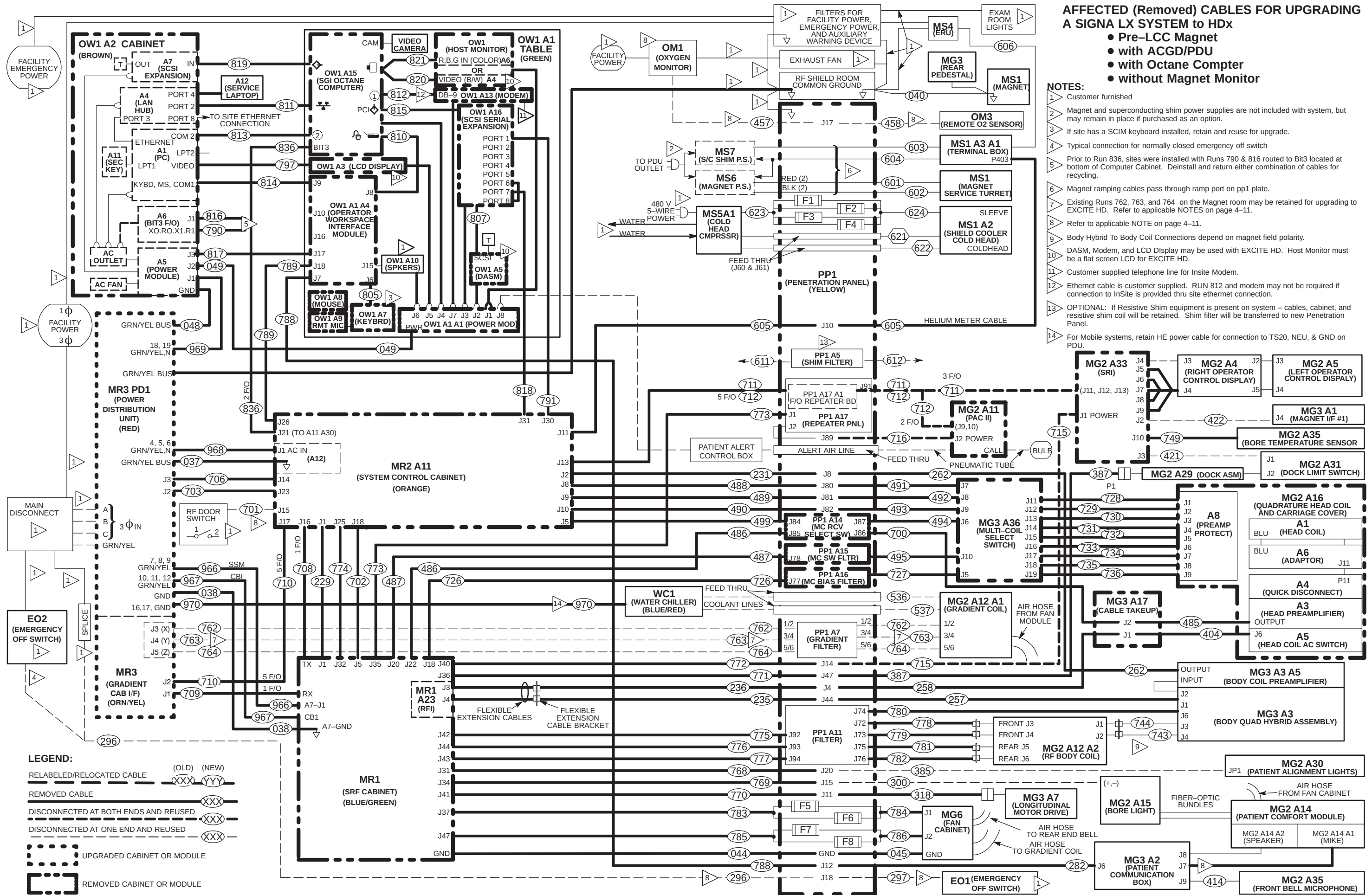
- Signa LCC WideOpen with Octane, ACGD, and Magnet Monitor Page 4-13
- Signa LCC TwinSpeed with Octane, ACGD, and Magnet Monitor Page 4-14
- Signa pre-LCC LX with Octane, ACGD and without Magnet Monitor Page 4-15
- Signa LCC WideOpen with Octane, RF/PDU, SGD, and Magnet Monitor Page 4-16
- Signa LCC Pre-WideOpen with Octane, RF/PDU, SGD, and Magnet Monitor Page 4-17
- Signa pre-LCC LX with Octane, SGD, RF/PEN, PDU and w/o Magnet Monitor Page 4-18
- Signa pre-LCC LX with Indigo, SGD, RF/PEN, PDU and w/o Magnet Monitor Page 4-19
- Signa pre-LCC LX with Octane, Grad/GRAM, RF/PEN, PDU and w/o Magnet Monitor Page 4-20
- Signa pre-LCC LX with Indigo, Grad/GRAM, RF/PEN, PDU and w/o Magnet Monitor Page 4-21
- Signa Horizon 5.x with Operator Console/Computer, Grad/GRAM, RF/PEN, PDU, w/o Mag Mon . Page 4-22
- Signa Advantage 5.x with Oper. Console/Computer, Grad/GRAM, RF, PEN, PDU, w/o Mag Mon . Page 4-23
- Signa Advantage 4.x with Operator Console. computer, RF, REN, PDU, and Gradient –
Cable removal map is not provided. Refer to Table 4-1, Columns H and I, for cable removal information on Advantage 4.x systems

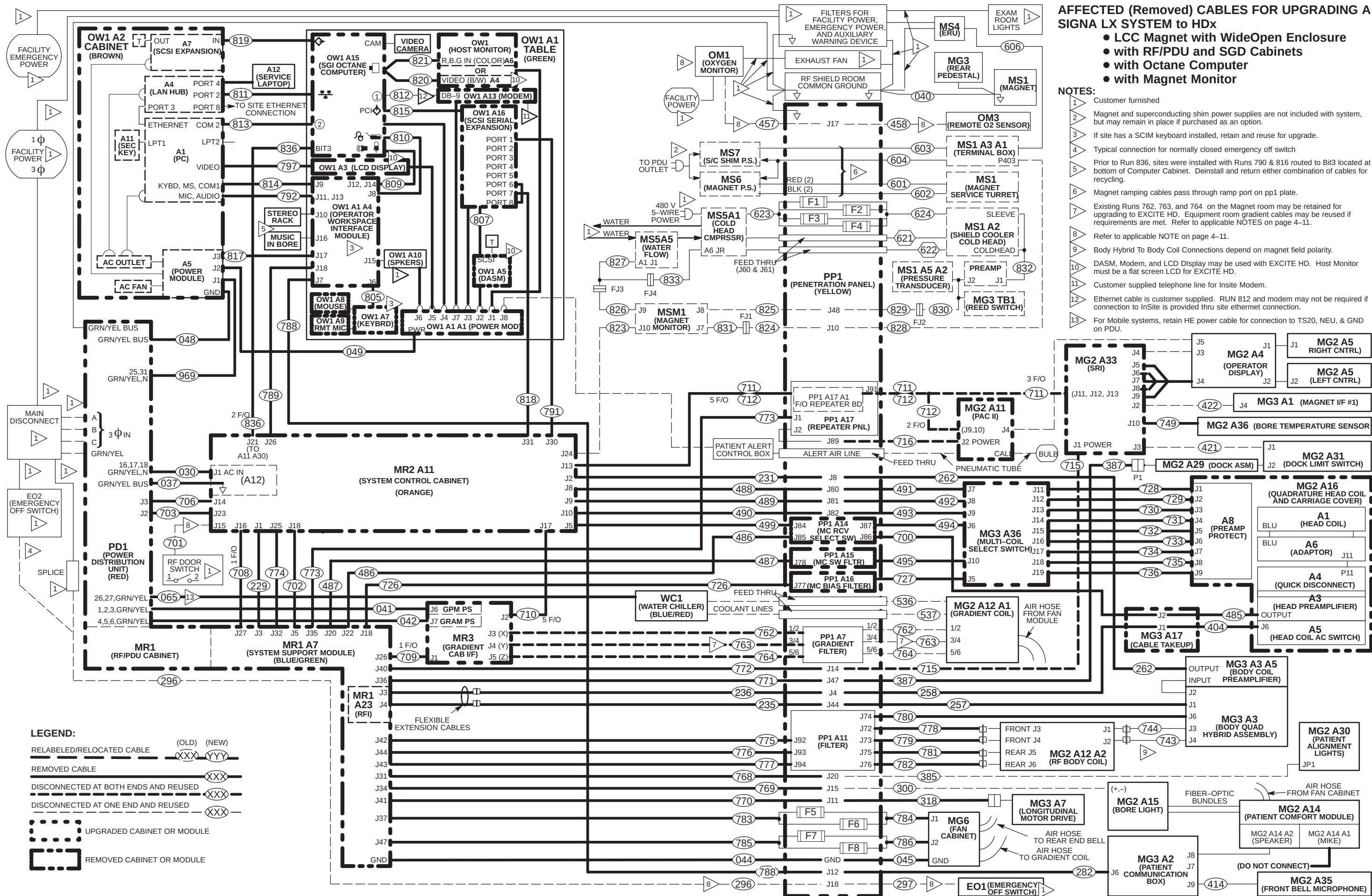


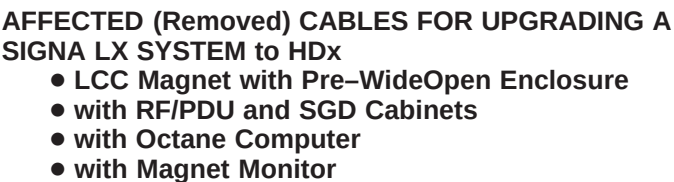


NOTES: • With Magnet Monitor

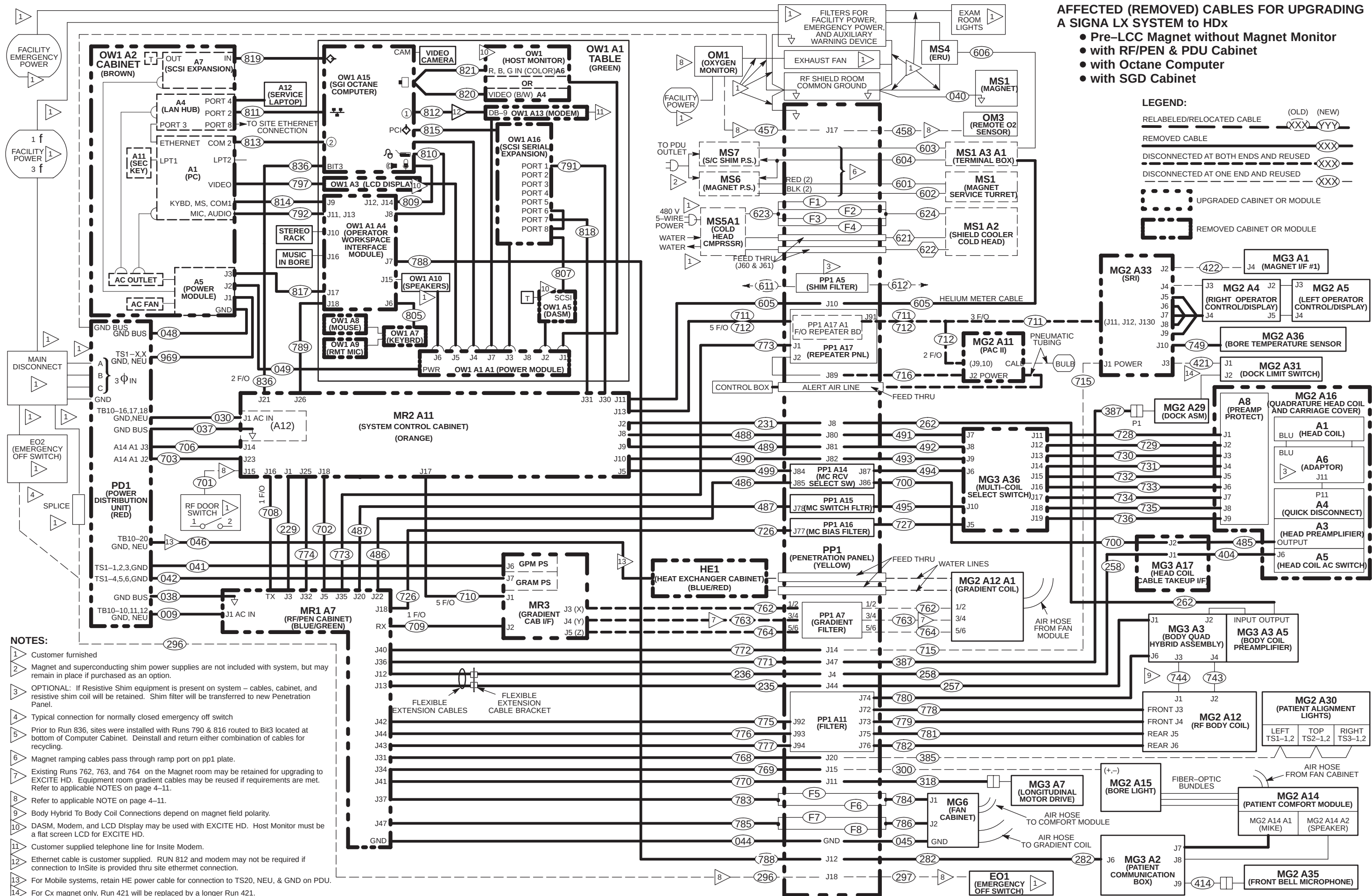
- 1 CUSTOMER FURNISHED
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 If site has a SCIM keyboard installed, retain and reuse for upgrade.
- 4 Typical connection for normally closed emergency off switch
- 5 Optional: May not be present on all systems.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 Existing gradient cable Runs 982 thru 987 in the Magnet Room should be retained and not upgraded. Runs 976 thru 981 and 988 in Equipment room may be retained for upgrading to EXCITE HD. Refer to applicable NOTES on page 4-11.
- 8 Optional: Used only if Shim Filter is installed.
- 9 Refer to applicable NOTE on page 4-11.
- 10 Customer supplied telephone line for Insite Modem.
- 11 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 12 Water Flow meter is used only for sites that have the TGWC cooling system installed.
- 13 DASM, Modem, and LCD Display may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
- 14 Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.







- 1 Customer furnished
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 If site has a SCIM keyboard installed, retain and reuse for upgrade.
- 4 Typical connection for normally closed emergency off switch
- 5 Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Equipment room gradient cables may be reused if requirements are met. Refer to applicable NOTES on page 4-11.
- 8 Refer to applicable NOTE on page 4-11.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 DASM, Modem, and LCD Display may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
- 11 Customer supplied telephone line for Insite Modem.
- 12 Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
- 13 For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.



**AFFECTED (REMOVED) CABLES FOR UPGRADING
A SIGNA LX SYSTEM to HDx**

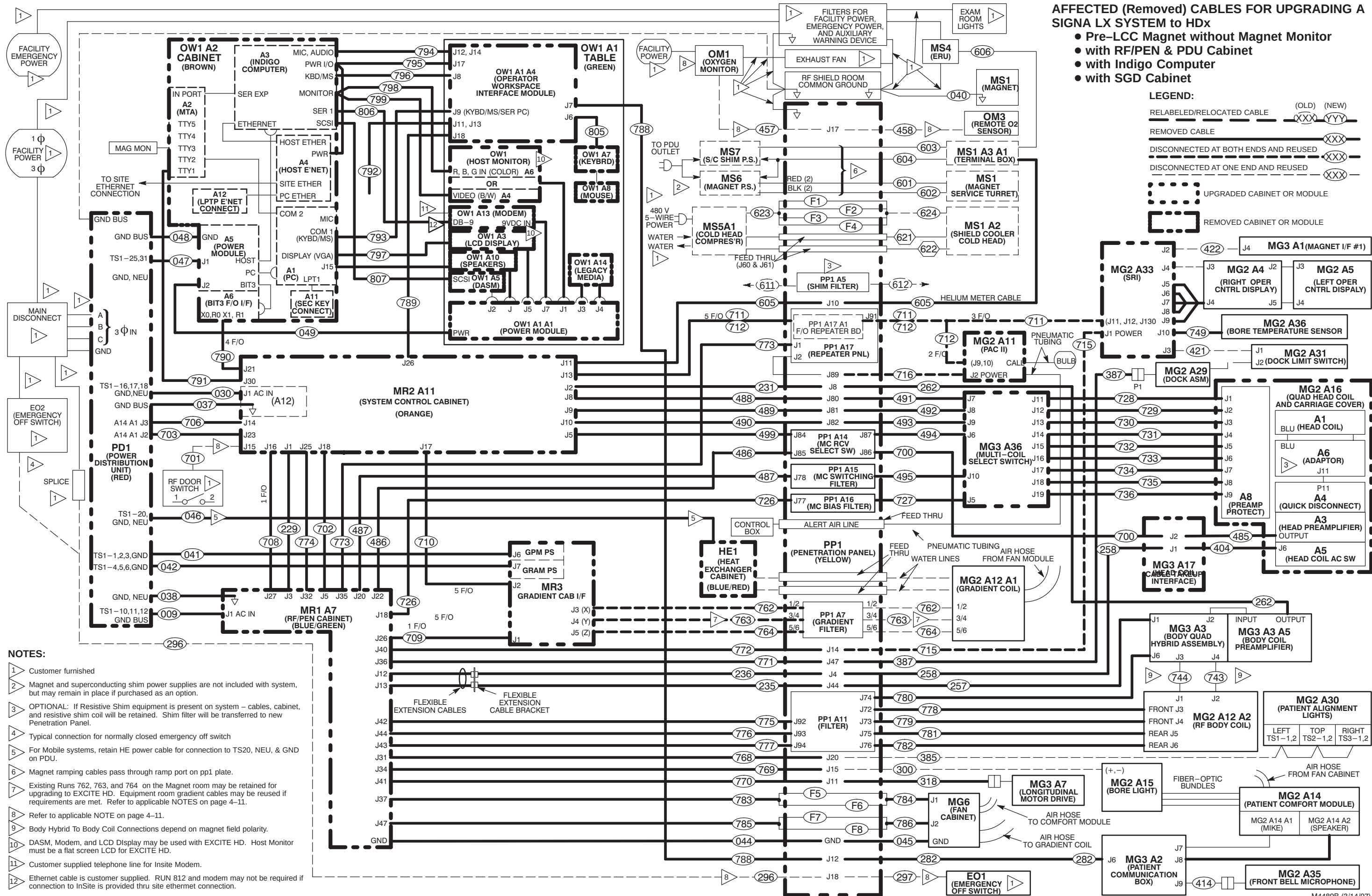
- Pre-LCC Magnet without Magnet Monitor
- with RF/PEN & PDU Cabinet
- with Octane Computer
- with SGD Cabinet

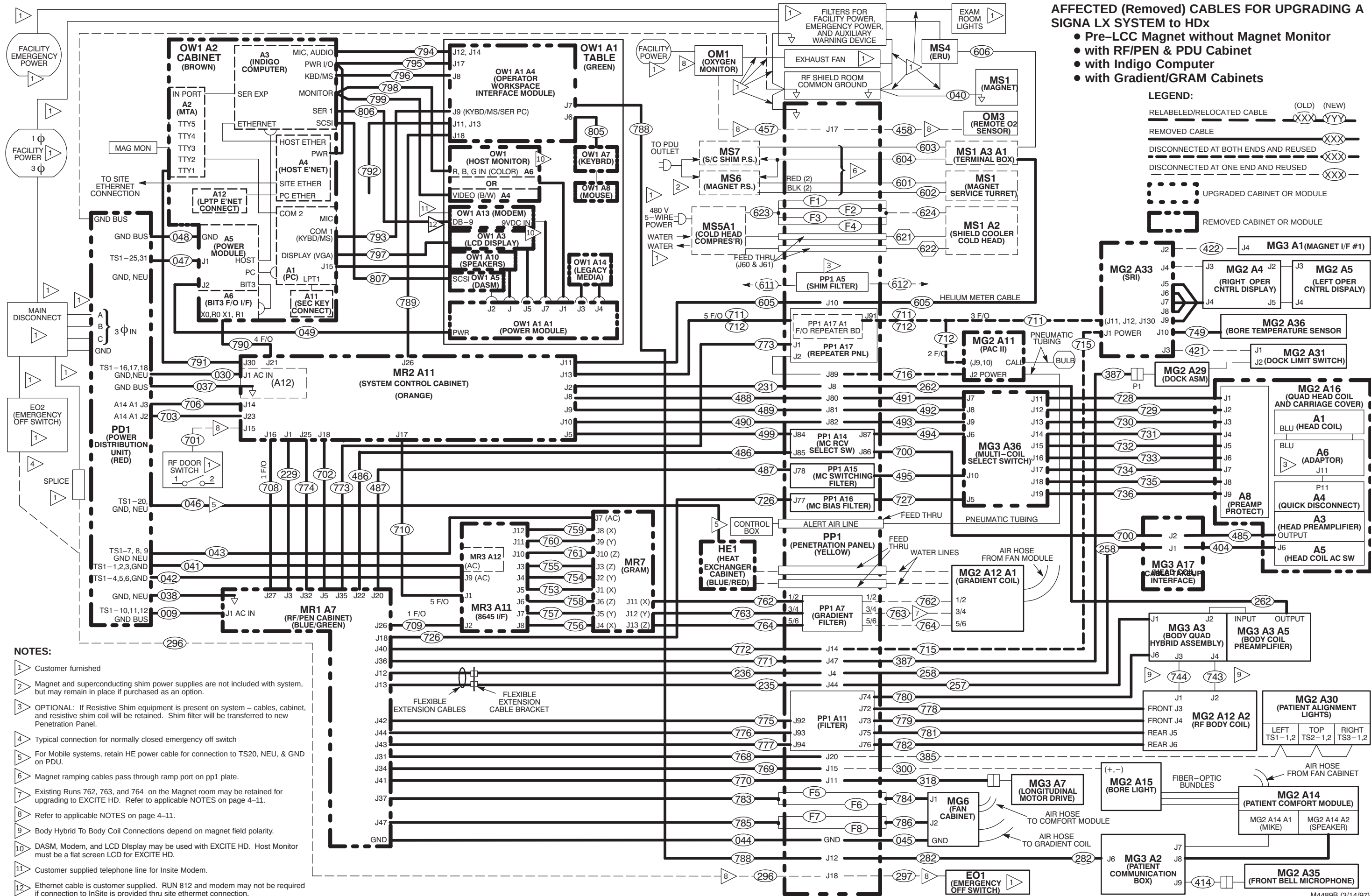
LEGEND:

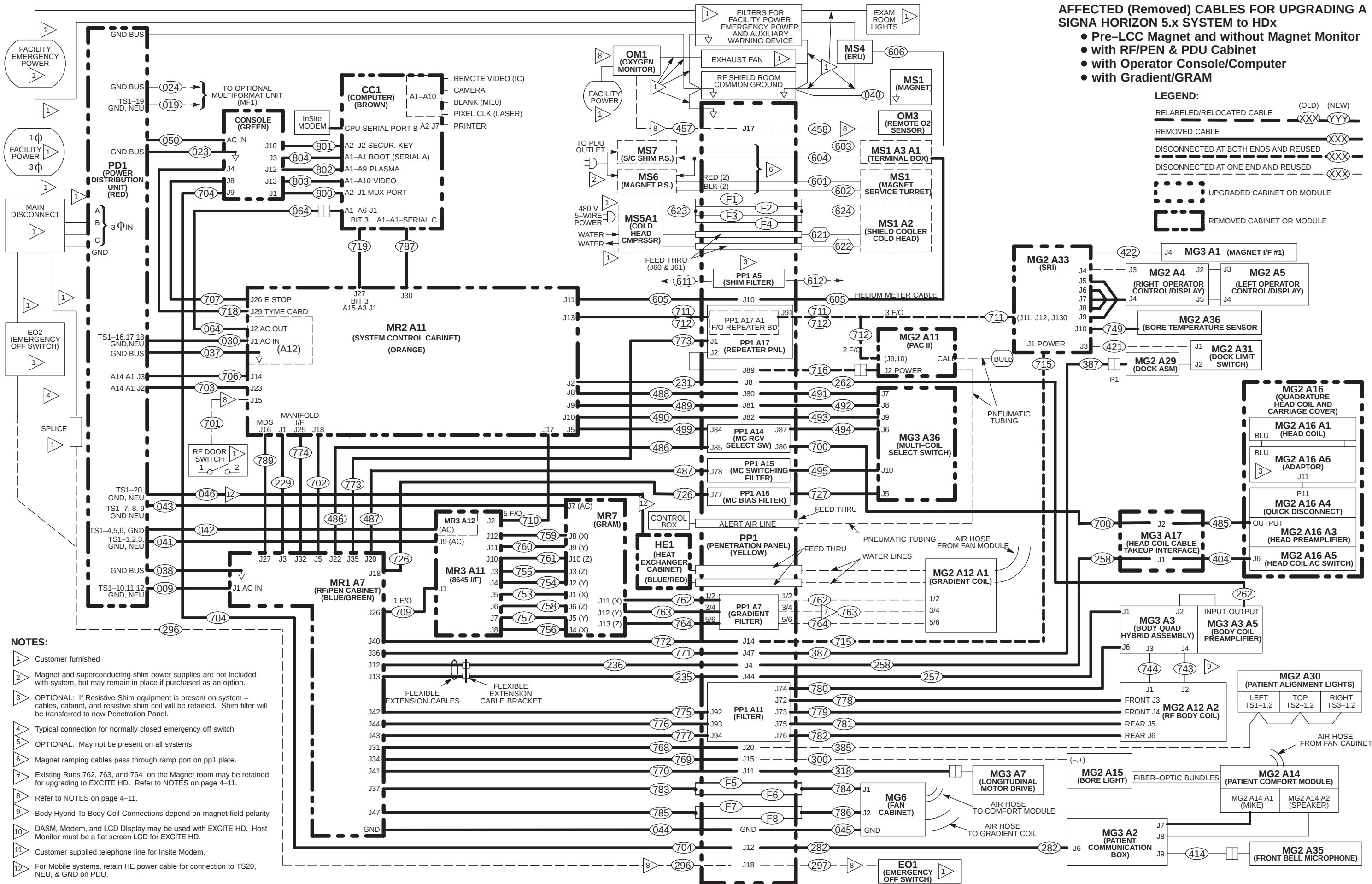
RELABELLED/RELOCATED CABLE (OLD) (NEW)
REMOVED CABLE
DISCONNECTED AT BOTH ENDS AND REUSED
DISCONNECTED AT ONE END AND REUSED

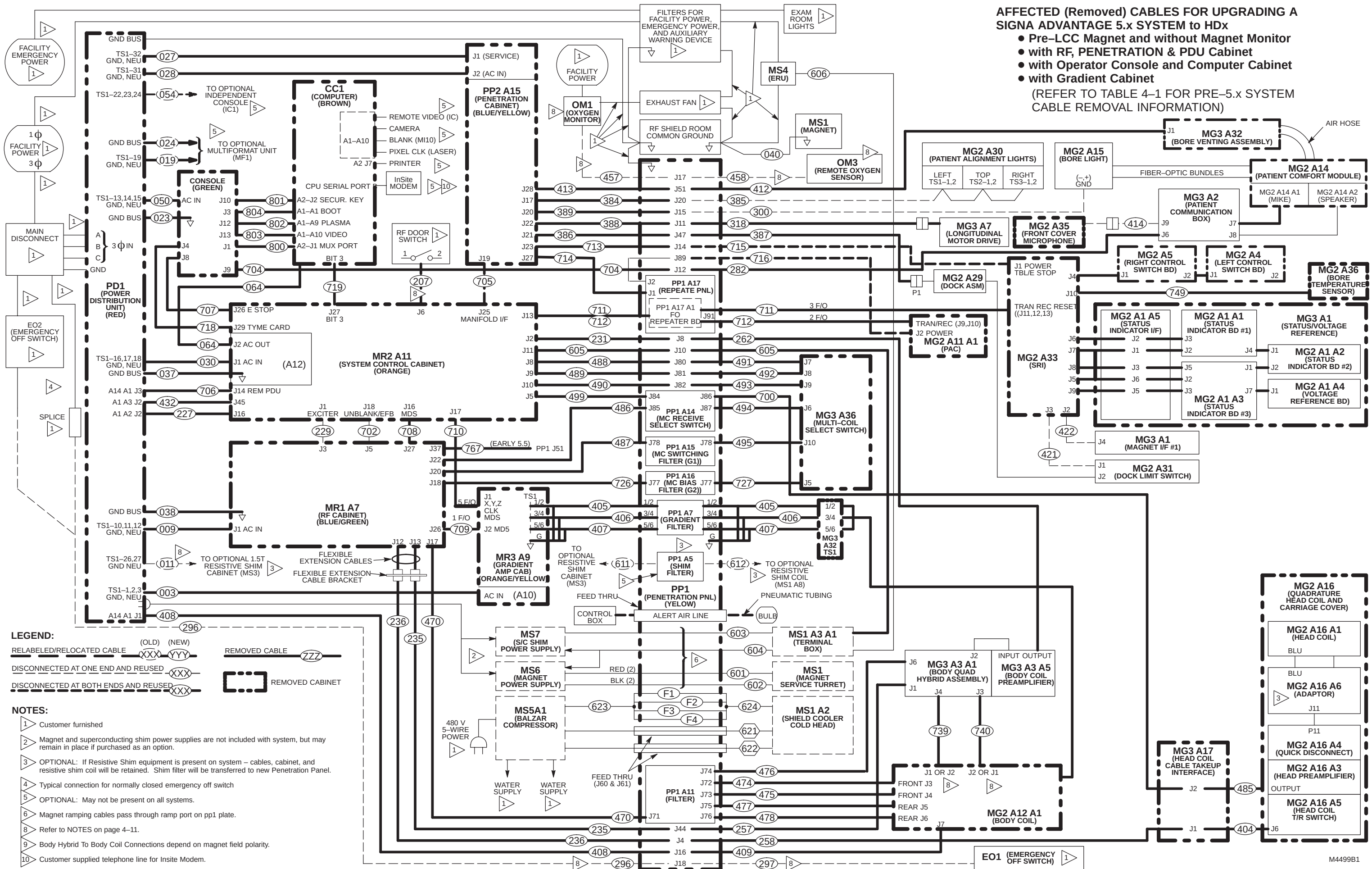
UPGRADED CABINET OR MODULE
REMOVED CABINET OR MODULE

- NOTES:**
- 1 Customer furnished
 - 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
 - 3 OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
 - 4 Typical connection for normally closed emergency off switch
 - 5 Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
 - 6 Magnet ramping cables pass through ramp port on pp1 plate.
 - 7 Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Equipment room gradient cables may be reused if requirements are met. Refer to applicable NOTES on page 4-11.
 - 8 Refer to applicable NOTE on page 4-11.
 - 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
 - 10 DASM, Modem, and LCD Display may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
 - 11 Customer supplied telephone line for Insite Modem.
 - 12 Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
 - 13 For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.
 - 14 For Cx magnet only, Run 421 will be replaced by a longer Run 421.









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4-6 SORTING AND ROUTING CABLES



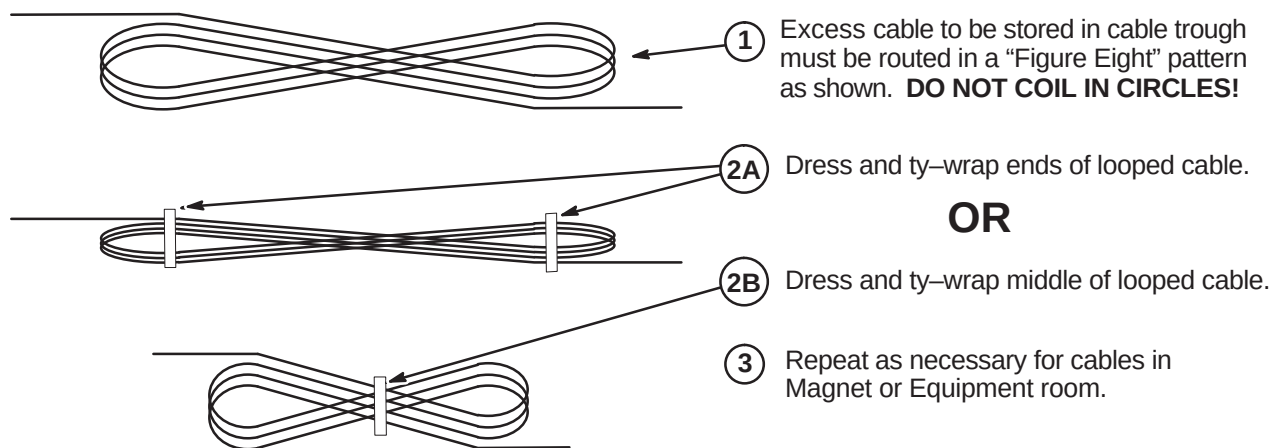
Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2-1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 2-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

4-7 HDx CABLE INTERCONNECT MAPS

The following cable maps illustrate the Signa HDx 1.5T system interconnect for all supplied cables and interface with customer supplied wiring.

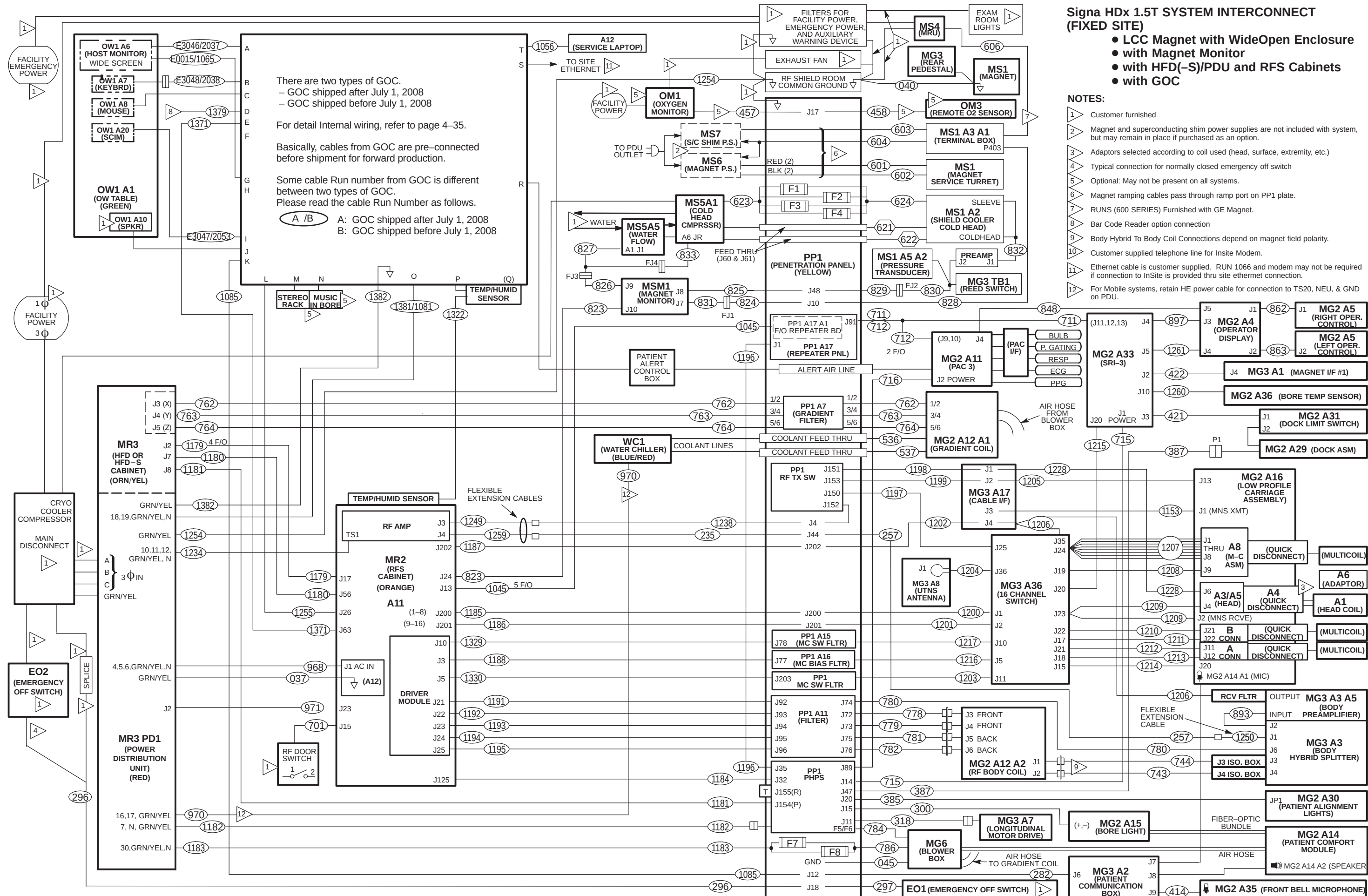
- Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.
 - Instructions for connections for optional laser cameras are supplied with option.
 - Connection procedures for other options are covered in the installation manual shipped with the option.
 - Spectroscopy option cabling can be found on page 4-35.
- Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
- Interconnect details for magnet subsystem are covered in the applicable magnet subsystem manual.

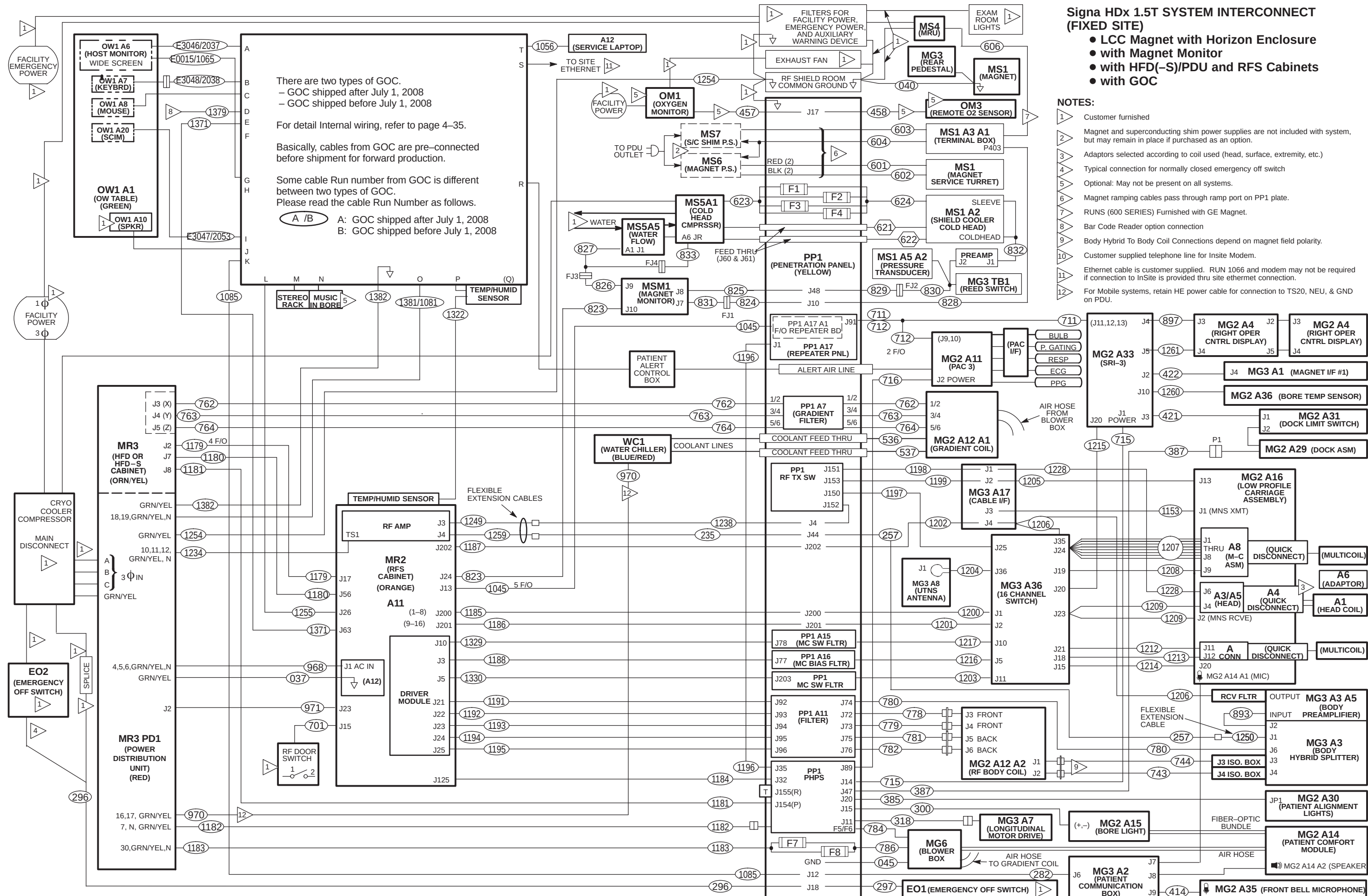
For the convenience of the mechanical installers, a duplicate System Interconnect is provided. This will allow the install team to post one interconnect map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation when marking off cables when their connections are complete.

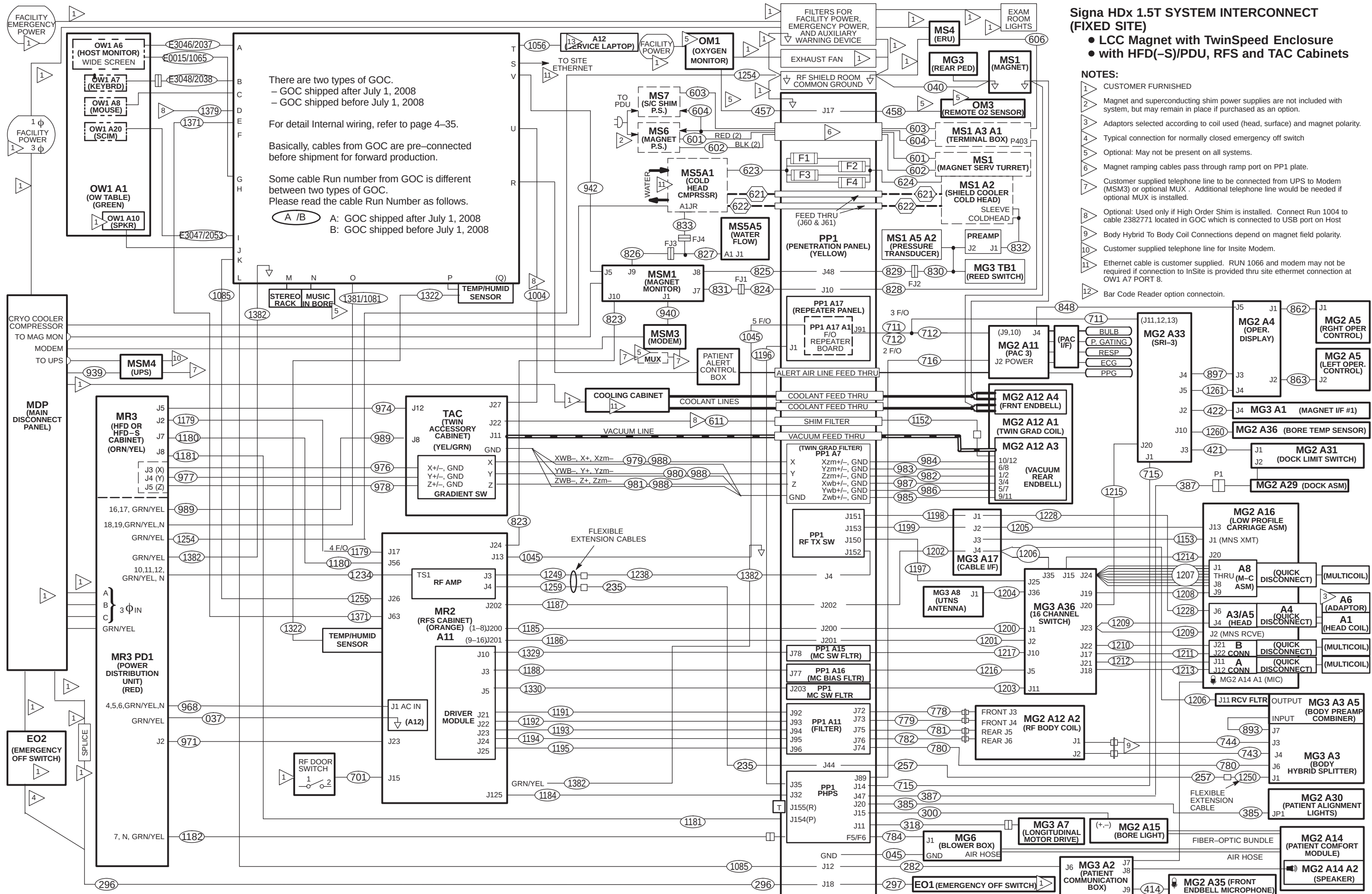
Fold-out HDx 1.5T System Interconnect Cable Maps are provided on the following pages:

Description	Page
– LCC Magnet with WideOpen Enclosure	6-27
– LCC Magnet with Horizon Style Enclosure	6-28
– LCC Magnet with TwinSpeed Enclosure	6-29
– Non-LCC Magnet with Horizon Style Enclosure	6-30
– LCC Magnet with WideOpen Enclosure (DUPLICATE)	6-31
– LCC Magnet with Horizon Style Enclosure (DUPLICATE)	6-32
– LCC Magnet with TwinSpeed Enclosure (DUPLICATE)	6-33
– Non-LCC Magnet with Horizon Style Enclosure (DUPLICATE)	6-34
– Signa EXCITE HD 1.5T System Interconnect for 2KW MNS Option	6-35
– MRCC and GWHX Cooling System Interconnects	6-36

Tape maps in a convenient location in the Magnet Room and the Equipment/Operator Room to check off routed cables. Additional information on each cable is located in the Appedndix. Upon completion of connecting cables, return fold-out maps to the binder for future reference.

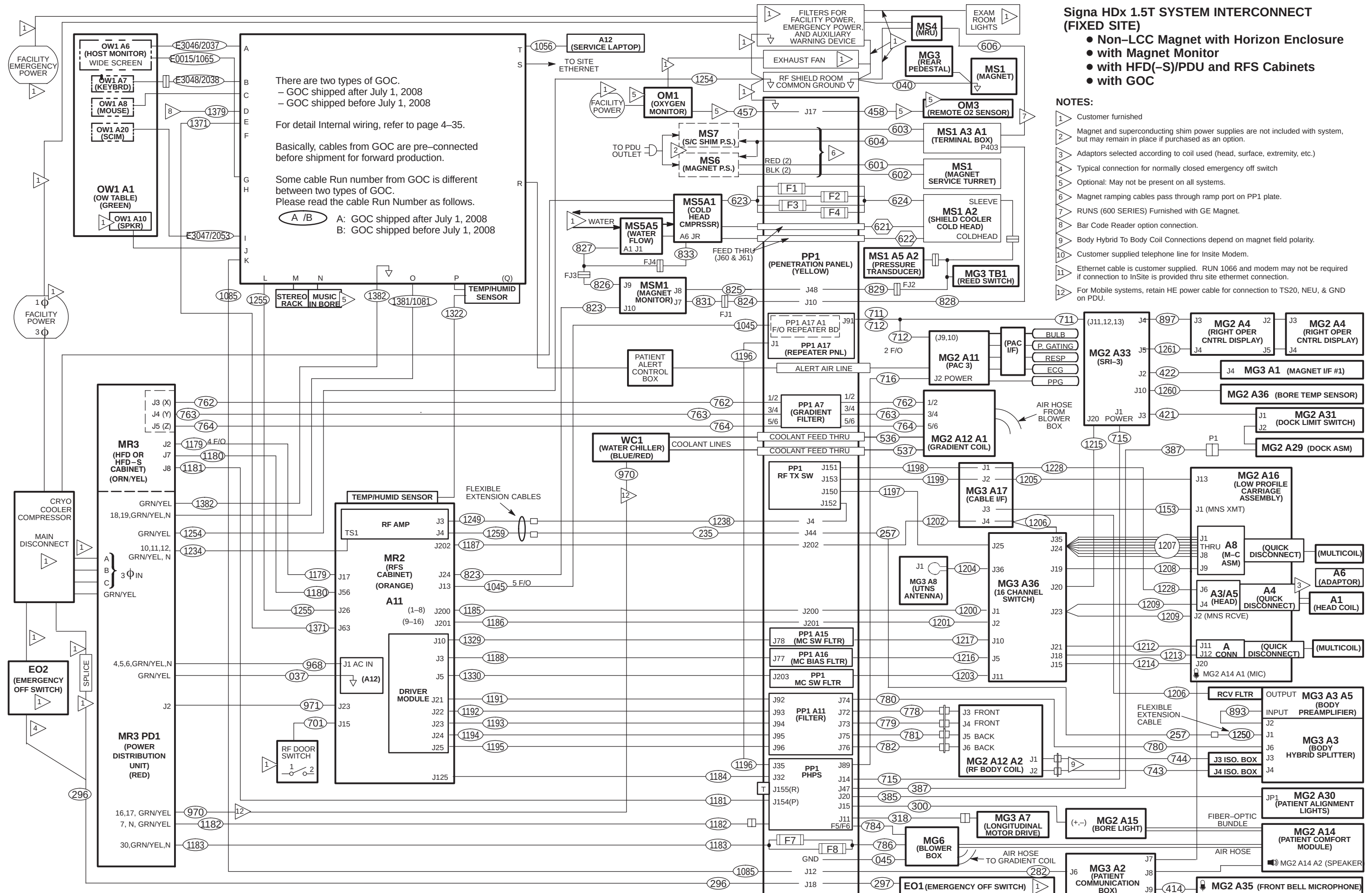


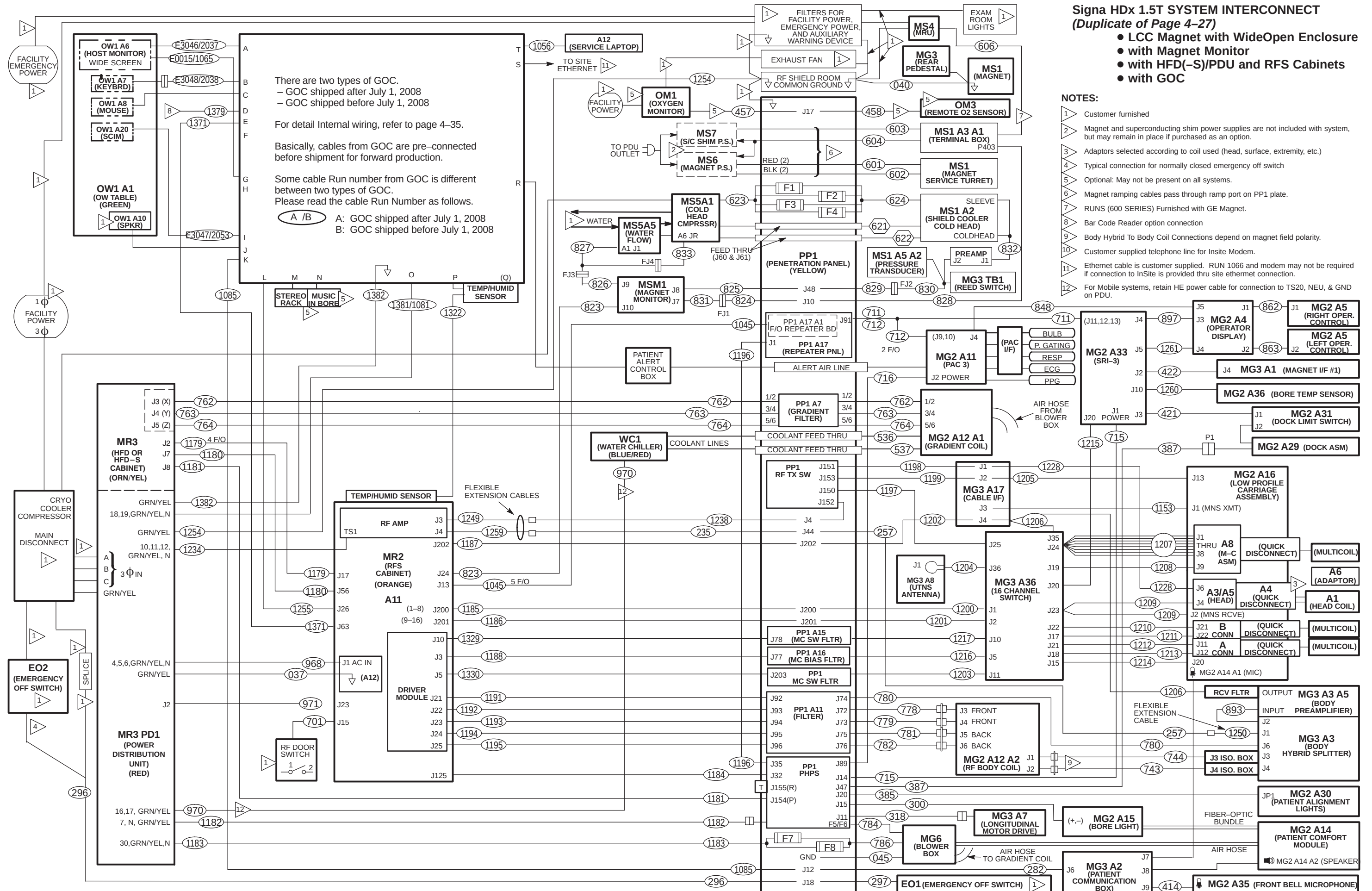


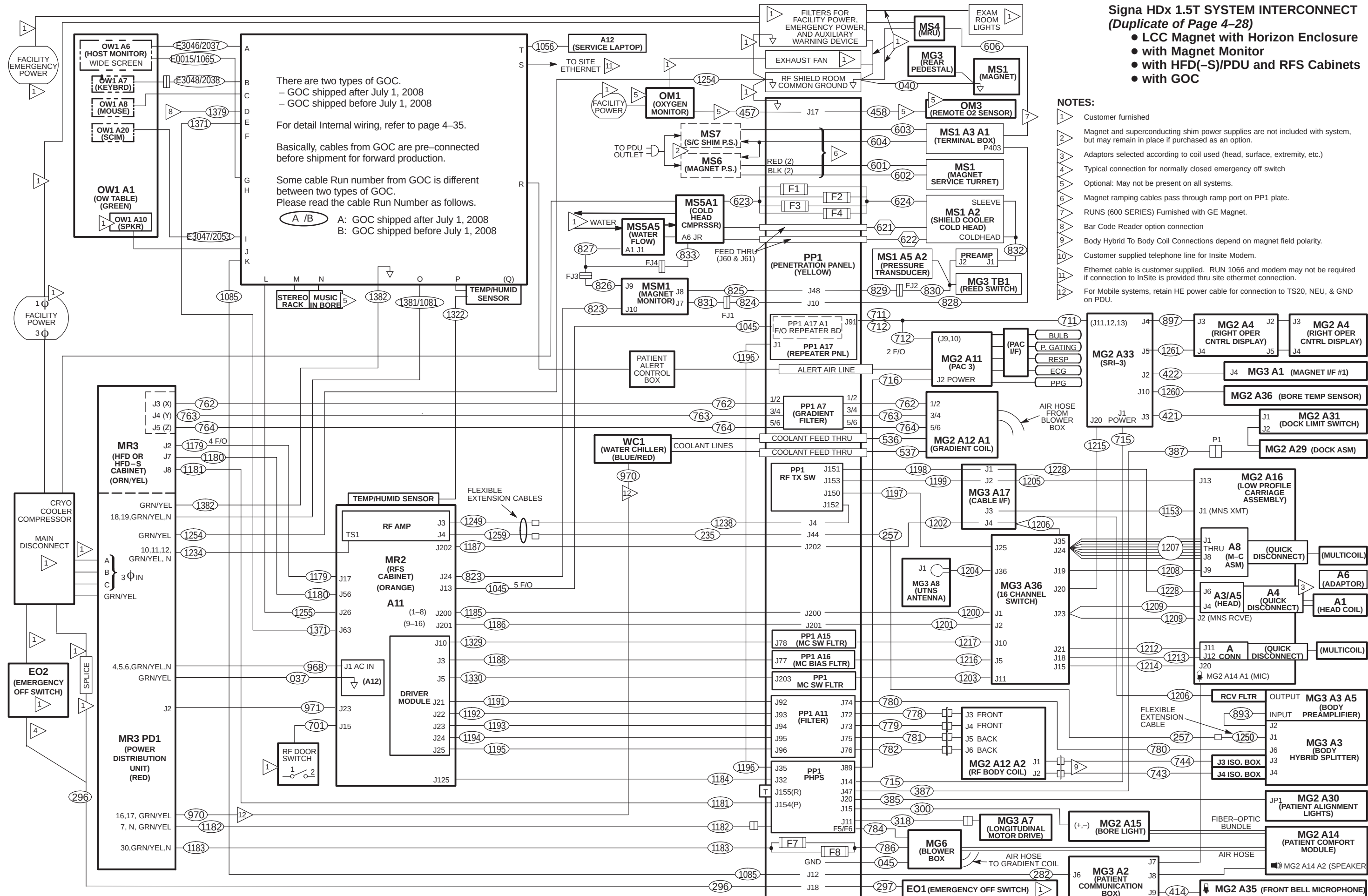


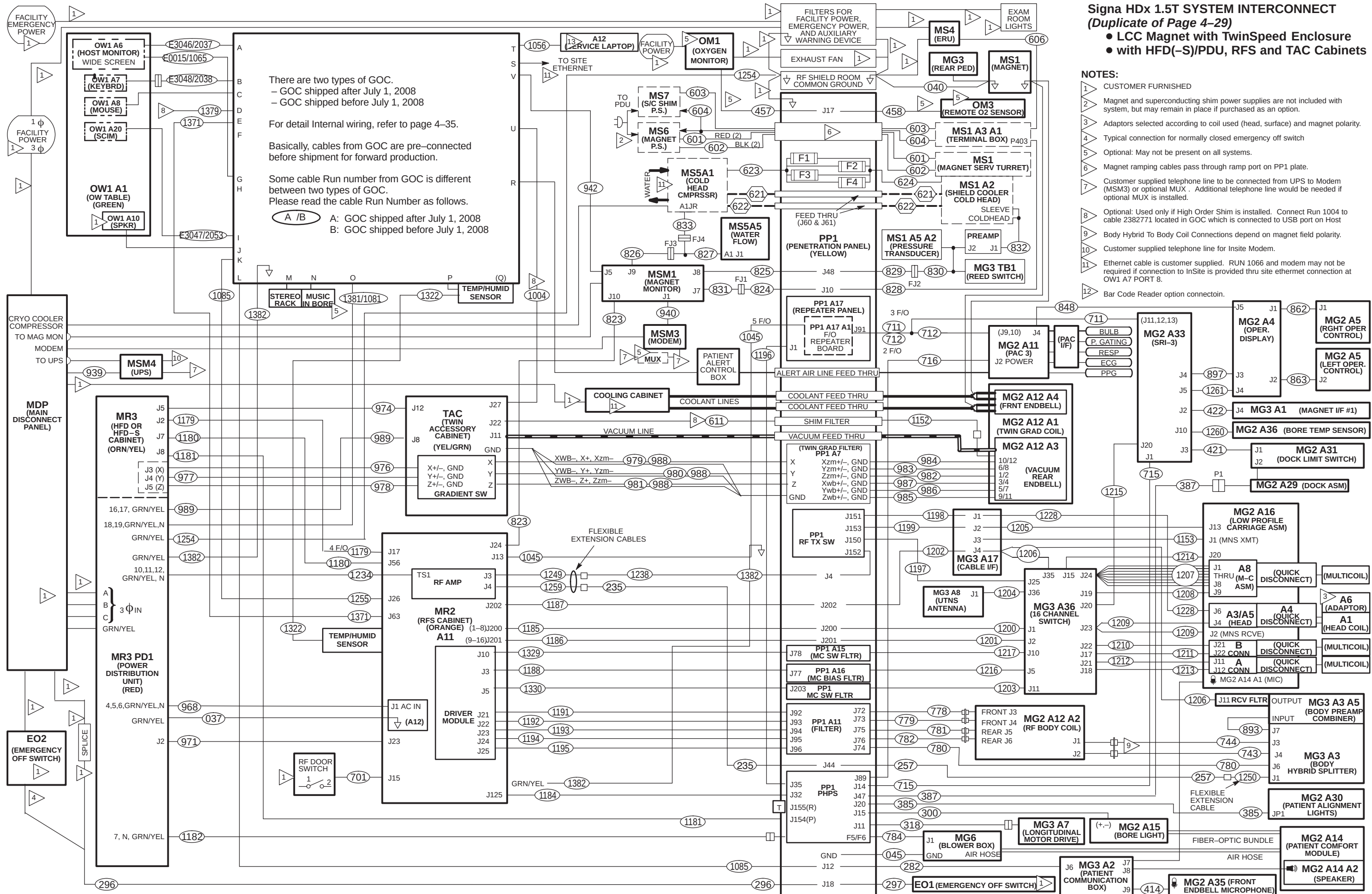
Signa HDx 1.5T SYSTEM INTERCONNECT (FIXED SITE) - LCC Magnet with TwinSpeed Enclosure - with HFD(–S)/PDU, RFS and TAC Cabinets

- NOTES:**
- CUSTOMER FURNISHED
 - Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
 - Adaptors selected according to coil used (head, surface) and magnet polarity.
 - Typical connection for normally closed emergency off switch
 - Optional: May not be present on all systems.
 - Magnet ramping cables pass through ramp port on PP1 plate.
 - Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
 - Optional: Used only if High Order Shim is installed. Connect Run 1004 to cable 2382771 located in GOC which is connected to USB port on Host
 - Body Hybrid To Body Coil Connections depend on magnet field polarity.
 - Customer supplied telephone line for Insite Modem.
 - Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8.
 - Bar Code Reader option connectoin.





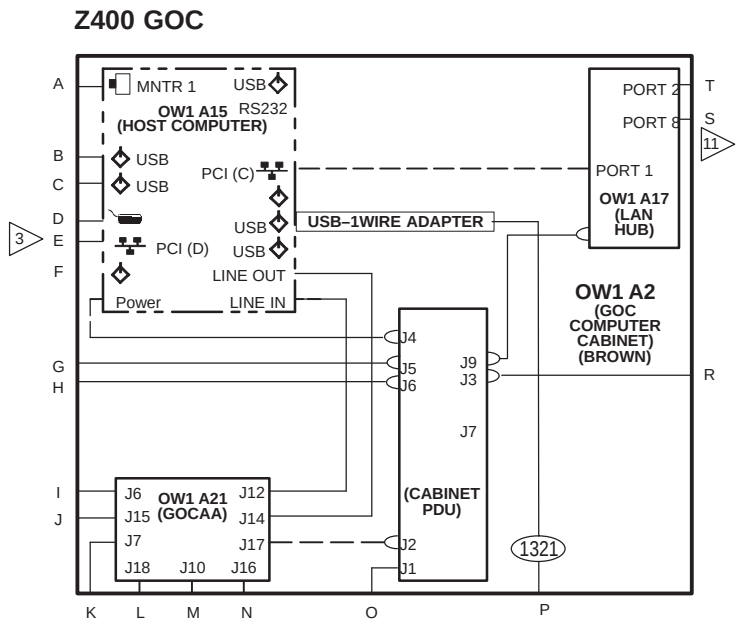




Signa HDx 1.5T SYSTEM INTERCONNECT **(Duplicate of Page 4–29)** • LCC Magnet with TwinSpeed Enclosure • with HFD(–S)/PDU, RFS and TAC Cabinets

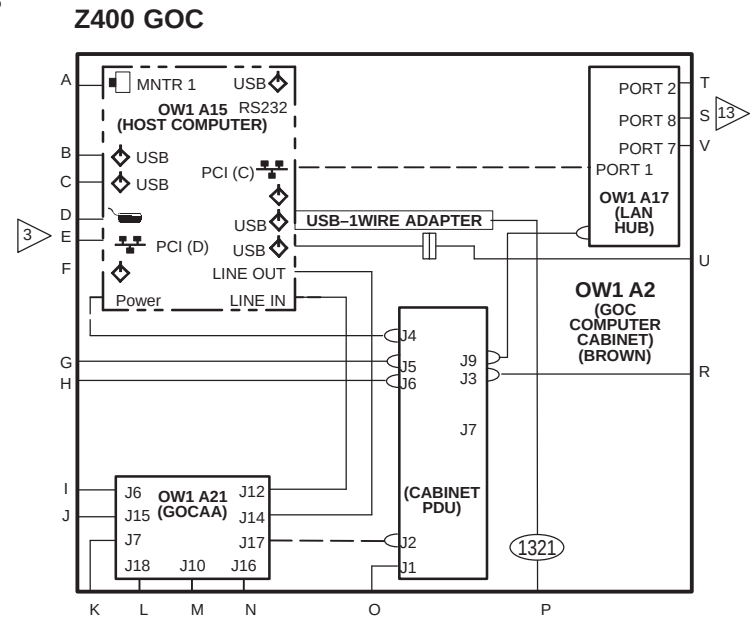
- NOTES:**
- CUSTOMER FURNISHED
 - Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
 - Adaptors selected according to coil used (head, surface) and magnet polarity.
 - Typical connection for normally closed emergency off switch
 - Optional: May not be present on all systems.
 - Magnet ramping cables pass through ramp port on PP1 plate.
 - Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
 - Optional: Used only if High Order Shim is installed. Connect Run 1004 to cable 2382771 located in GOC which is connected to USB port on Host
 - Body Hybrid To Body Coil Connections depend on magnet field polarity.
 - Customer supplied telephone line for Insite Modem.
 - Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8.
 - Bar Code Reader option connectoin.

1.5T HDx Series
GOC Configurations



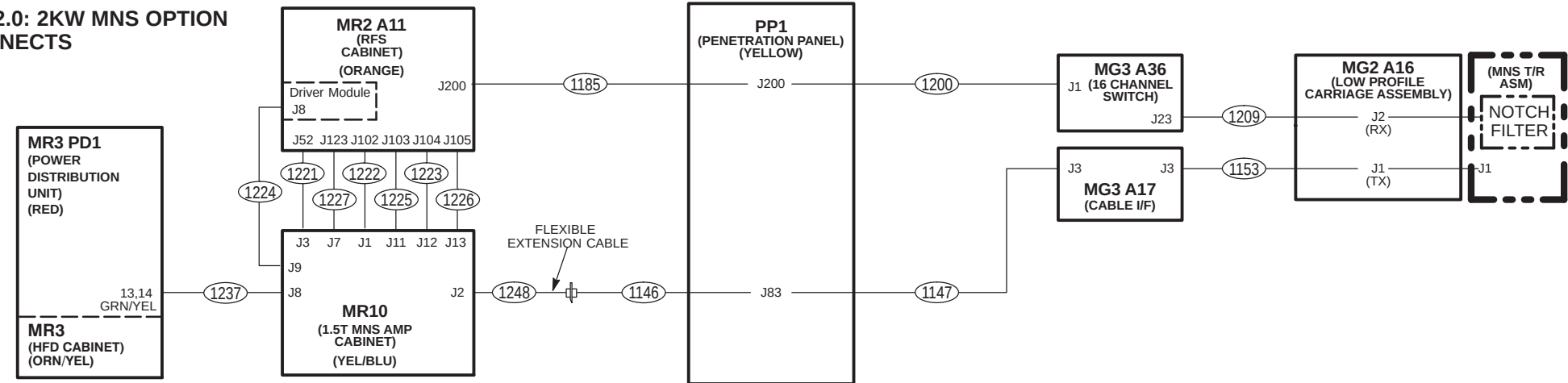
- NOTES:**
- 3 Bar Code Reader Option connection.
 - 10 Customer supplied telephone line for Insite Modem.
 - 11 Ethernet cable is customer supplied. RUN E3045/1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

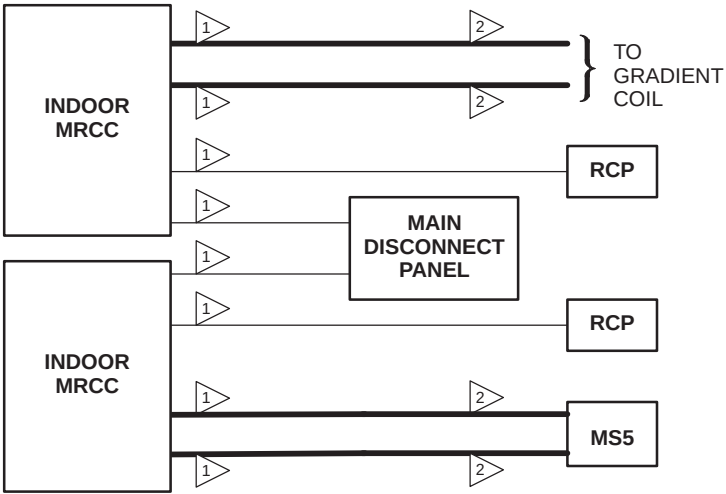
1.5T TwinSpeed HDx Series
GOC Configurations



- NOTES:**
- 3 Bar Code Reader Option connection.
 - 10 Customer supplied telephone line for Insite Modem.
 - 11 Ethernet cable is customer supplied. RUN E3045/1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

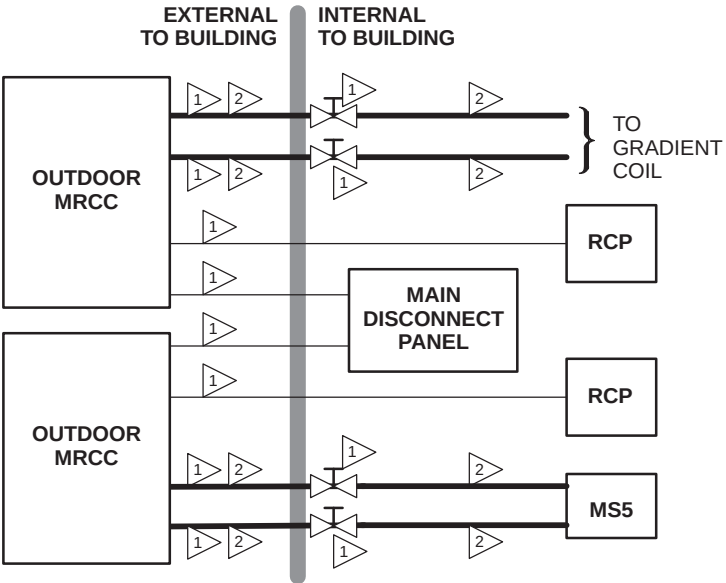
Signa 1.5T EXCITE HD, Release 12.0: 2KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS





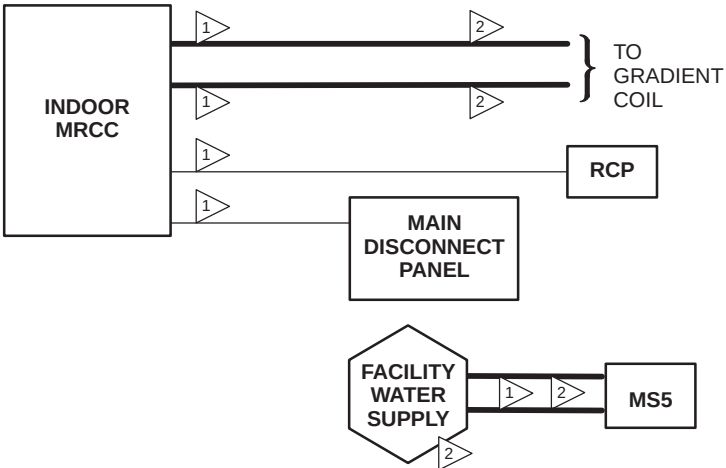
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



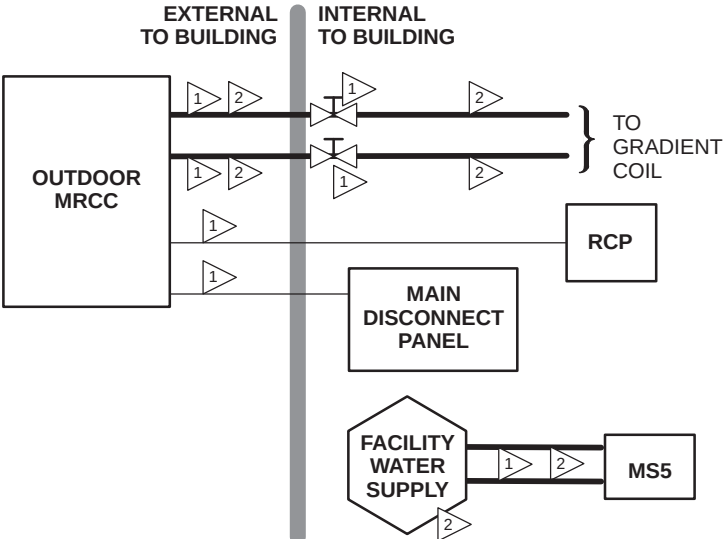
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



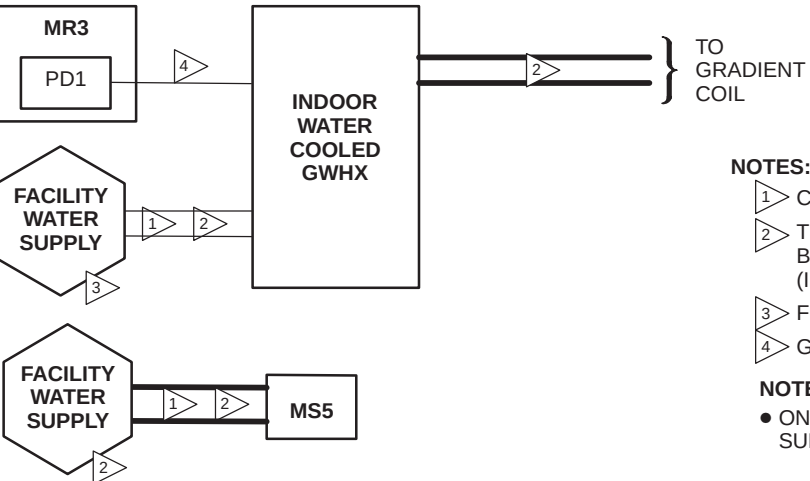
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
 - 3 FOR INDOOR WATER COOLED GWHX ONLY.
 - 4 GWHX PROVIDED CABLE.
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO GWHX SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR GWHX SUBSYSTEM GROUP INTERCONNECT DIAGRAM

4-8 EQUIPMENT ROOM CABLE INSTALLATION

The equipment room cables and cabinets can be installed simultaneously. Cable locations depend on where cabinets are installed.

4-8-1 HFD CABINET CABLE CONNECTIONS

Note: If site has an ACGD/PDU cabinet and is being upgraded to HFD-S/PDU, refer to Section 15-4 for connection of cables to PDU module.



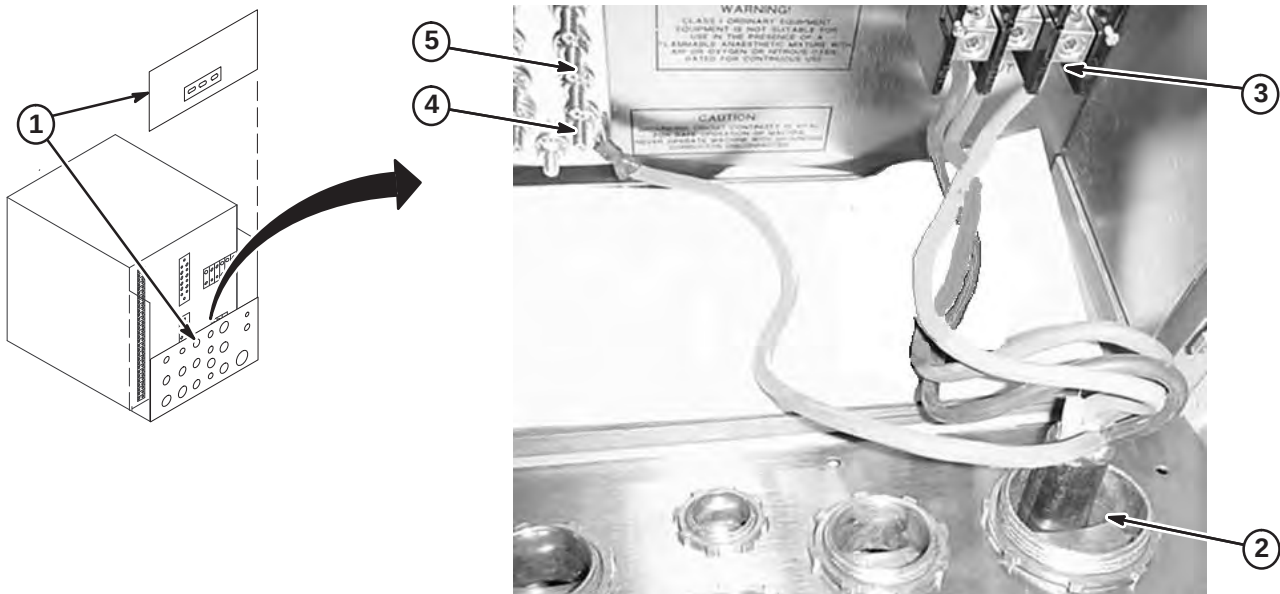
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE HFD CABINET.

4-8-1-A HFD Cabinet Power and Ground Cable Installation

1. Remove rear PDU module covers.
2. Route facility input power cable and ground wire thru interface panel.
3. Connect L1-L3 conductors to terminal block.
4. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.
5. Connect RF screen room ground conductor to Ground Bus Bar.

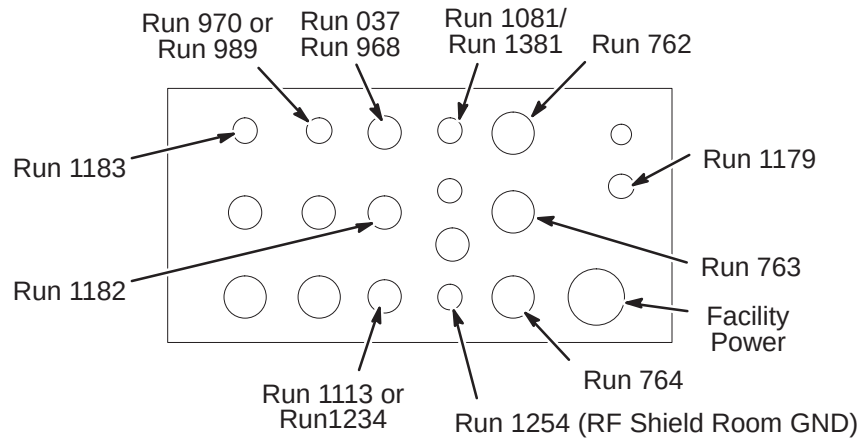


4-8-1-B Route Power Cables Thru Interface Panel

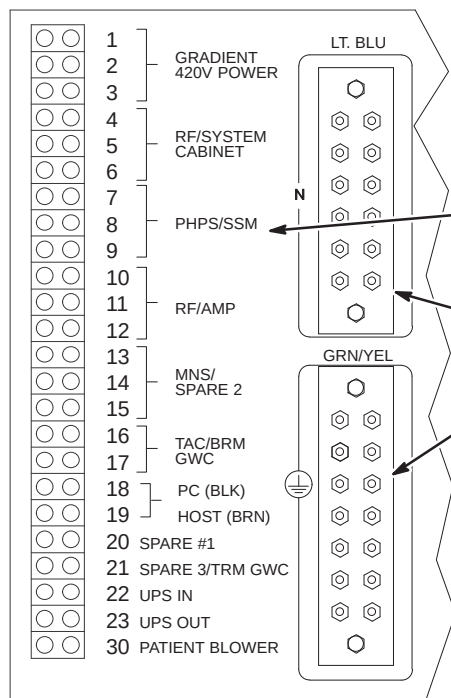
1. Route Runs thru interface panel as marked.

Note: The GRN/YEL ground wire, Run 037, **MUST** be routed with Run 968 power cable thru the same port.

2. Route any option Runs as marked.



4-8-1-C Connect Power Cables to PDU Module

**WARNING!**

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

① Route/connect power cables to pressure connectors and studs on back of PDU module as shown in applicable Table below

② Connect neutral wire cables to Neutral bus connectors.

③ Connect ground wire cables to Ground bus connectors. Trim and reterminate if necessary.

④ Tighten clamps of interface panel to secure cables in place.

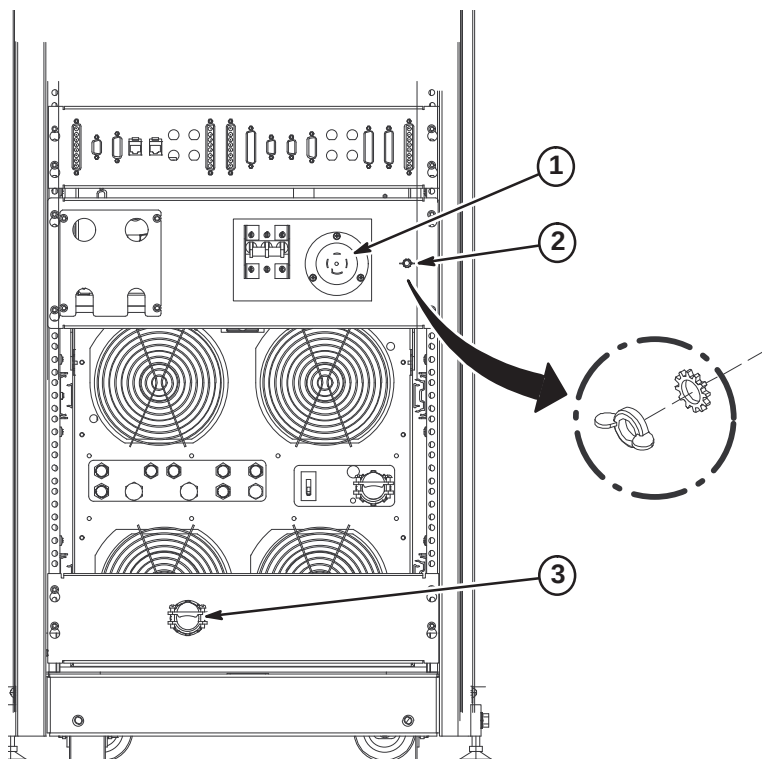
	CONNECT TO	RUN	PHASE	DESTINATION
NON-TwinSpeed	4,5,6	968	φ A, B, C	RFS Cabinet
	7	1182	φ A	PHPS Module
	10,11,12	1234	φ A, B, C	RF Amp in RFS Cabinet
	16,17	970	φ A,B	Water Chiller
	18,19	1081	φ C,A	Operator Workspace (Old GOC)
	19	1381	φ A	Operator Workspace (New GOC)
	20	046/973	φ A	Heat Exchanger (Mobile)
	30	1183	φ A	Blower Box
	GRN/YEL	1382		GOC Ground Wire (New GOC)
	GRN/YEL	037		System Cabinet Ground Wire
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)
TwinSpeed	4,5,6	968	φ A, B, C	RFS Cabinet
	7	1182	φ A	PHPS Module
	10,11,12	1113	φ A, B, C	NB RF Amp Cabinet
	16,17	989	φ A,C	TAC Cabinet
	18,19	1081	φ C,A	Operator Workspace
	19	1381	φ A	Operator Workspace (New GOC)
	(MDP)* or		φ A or	AirSys Water Cooled Cabinet
	30 (PDU)		φ A	AirSys Heat Exchanger
	GRN/YEL	1382		GOC Ground Wire (New GOC)
	GRN/YEL	037		System Cabinet Ground Wire
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)
COLOR CODE: for remainder of cables: 3φ = Black, Red, Orange Neutral = Light blue Single φ = Brown, Black Gnd = Grn/Yel * Connection to Main Disconnect Panel to be completed by customer electrician.				

4-8-2 EQUIPMENT ROOM CABINET POWER CABLE CONNECTIONS

CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

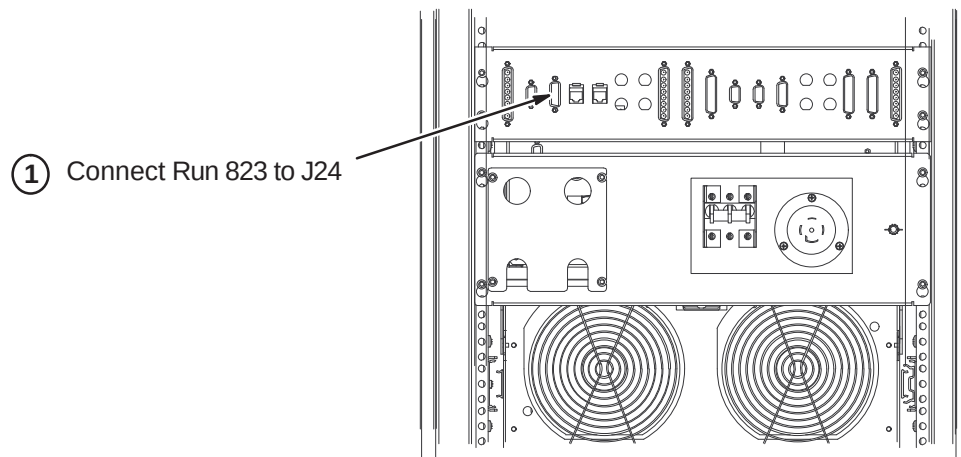
4-8-2-A RF/System Cabinet Power and Ground Cable Connections

1. Connect Run 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Route Run 1234 thru lower I/F panel and clamp in place. Connect as marked to RF Amp module.



4-8-3 INSTALLATION OF MAGNET MONITORING CABLES

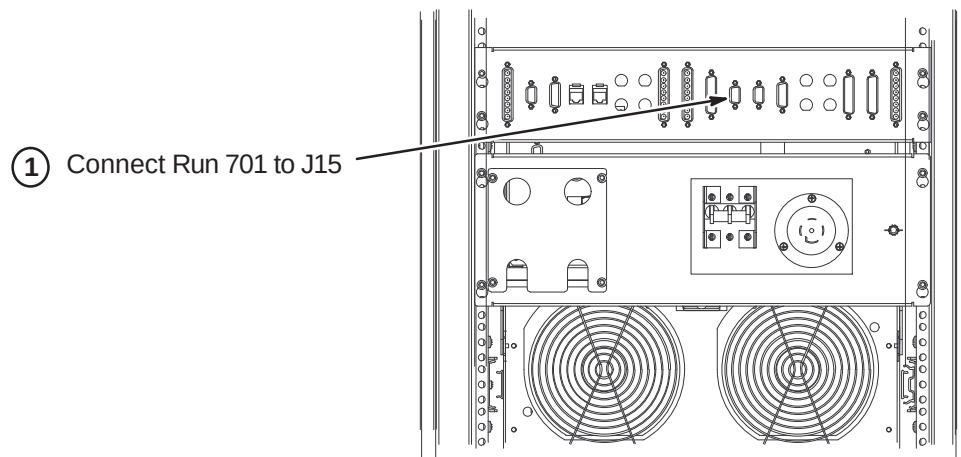
Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual*, for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



4-8-4 INSTALL RUN 701 – RF DOOR SWITCH (If Required)

Run 701 is a 100 ft cable with a 9-pin subminiature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminiature D plug.

1. Connect Run #701 to J15 on System Cabinet and route to RF Door Switch. Refer to site engineering drawings for cable routing path.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

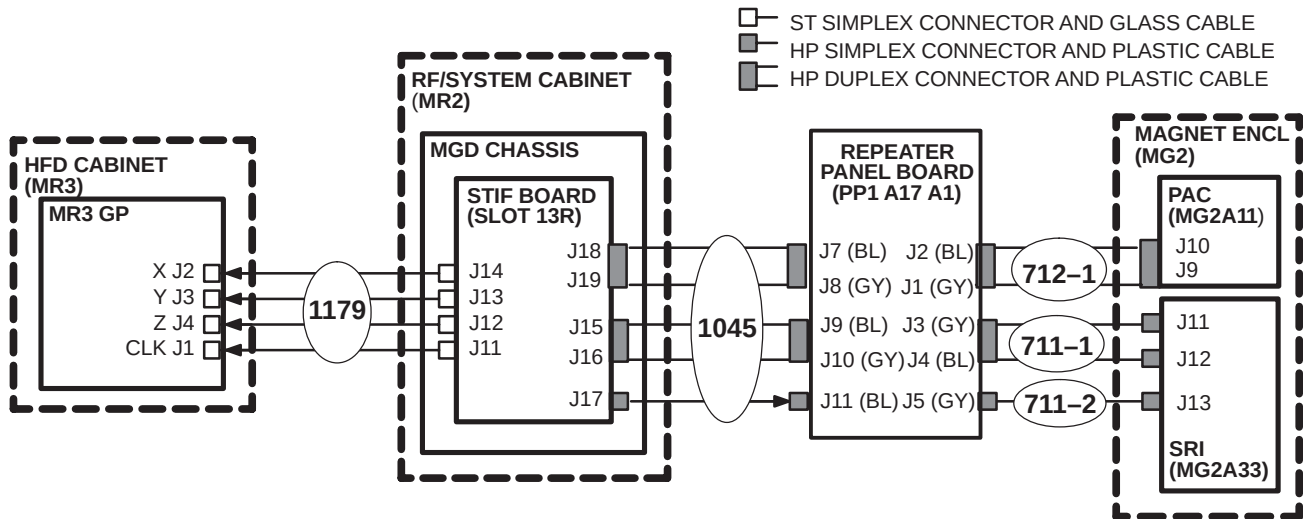


4-8-5 INSTALLATION OF FIBER OPTIC CABLES

Plan for storage of excess Run 1179 conduit, as it cannot be reterminated. For most Horizon amd LX configurations, Runs 711/712 in the magnet room will be retained for the upgrade to EXCITE HD.

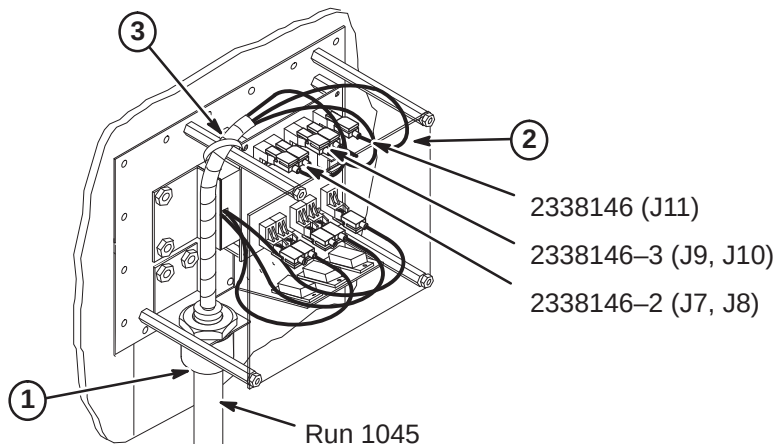
4-8-5-A Routing of Fiber Optic Cables

1. Unroll coiled fiber optic conduits for Runs 711/712, 1045, and 1179 along intended routes.
2. Connection of Runs is shown in diagram below.



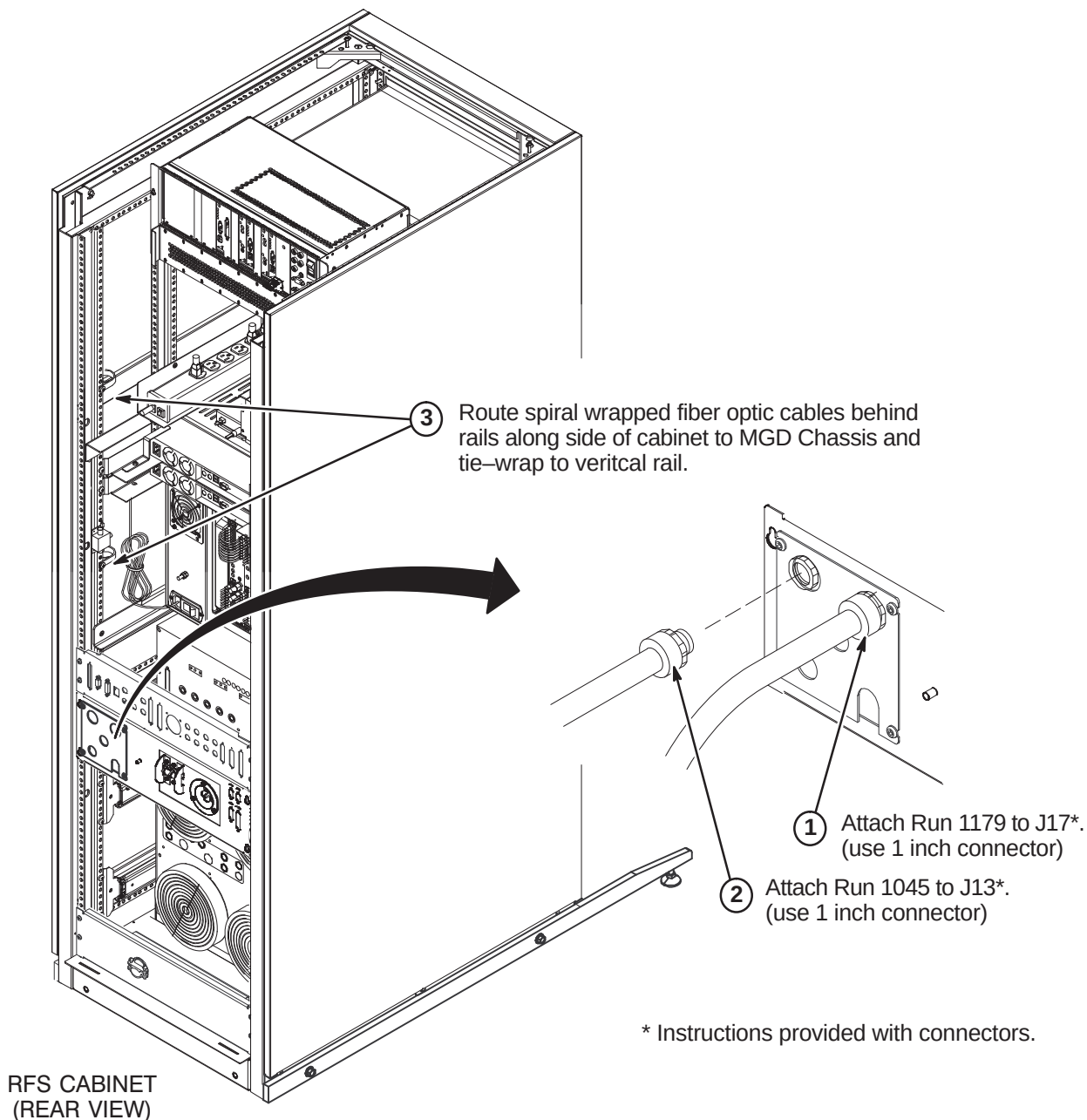
4-8-5-B Connect Run 1045 to Penetration Panel

1. Move Run 1045 to the Equipment Room side and attach "PP1-A17" end to access bracket on Penetration Panel. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
2. Adjust route of individual fiber optic runs to maintain large radius and connect Run 1045 as shown.
3. Adjust position of protective spiral wrap and secure with tie-wrap.
4. Route other end of Run 1045 to System Cabinet.



4-8-5-C Route Fiber Optic Runs 1045, 1083, and 1179 in RF/System Cabinet**CAUTION**

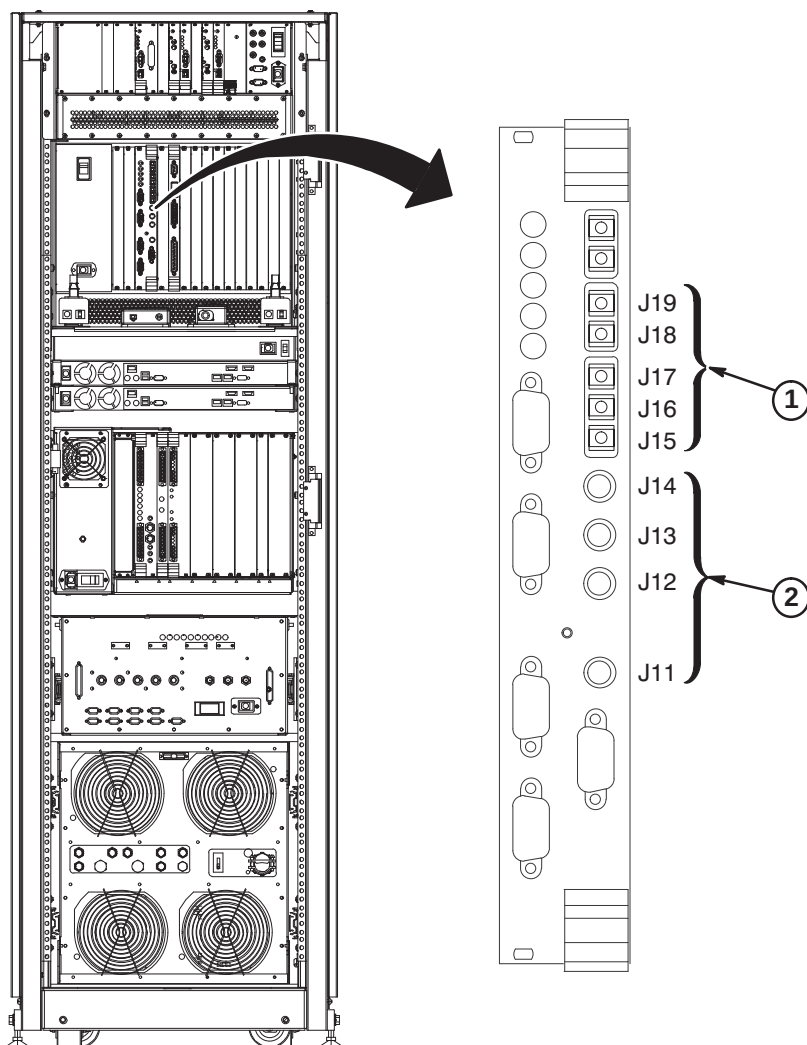
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



4-8-5-D Route/Connect Fiber Optic Runs 1179 and 1045 to STIF Board**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
2. Route/Connect Run 1179 fiber optic cables to J11, J12, J13, and J14 on the STIF Board.



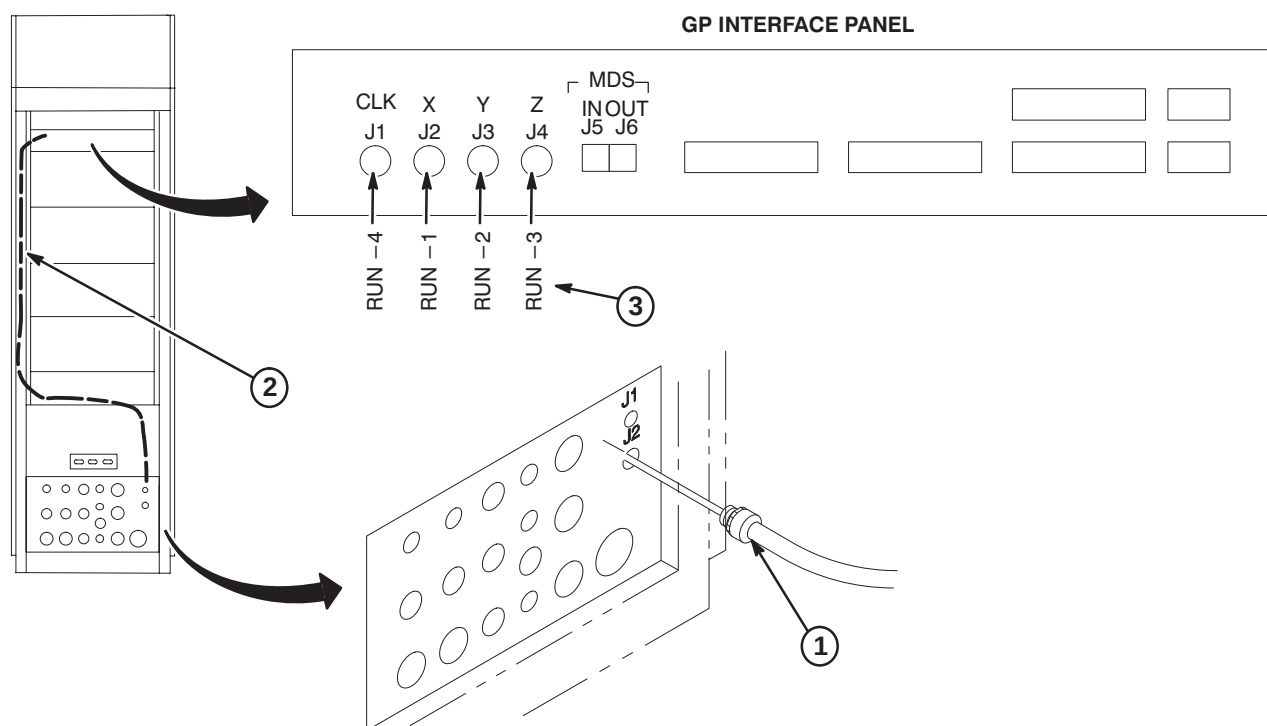
Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

4-8-5-E Route/Connect Fiber Optic Run 1179 to GP3 Module in HFD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.

1. Attach Run 1179 to I/F J2.
2. Route fiber optic cables to GP3 Module. Leave enough slack for extending GP module forward.
3. Connect fiber optics to GP3 module according to table.

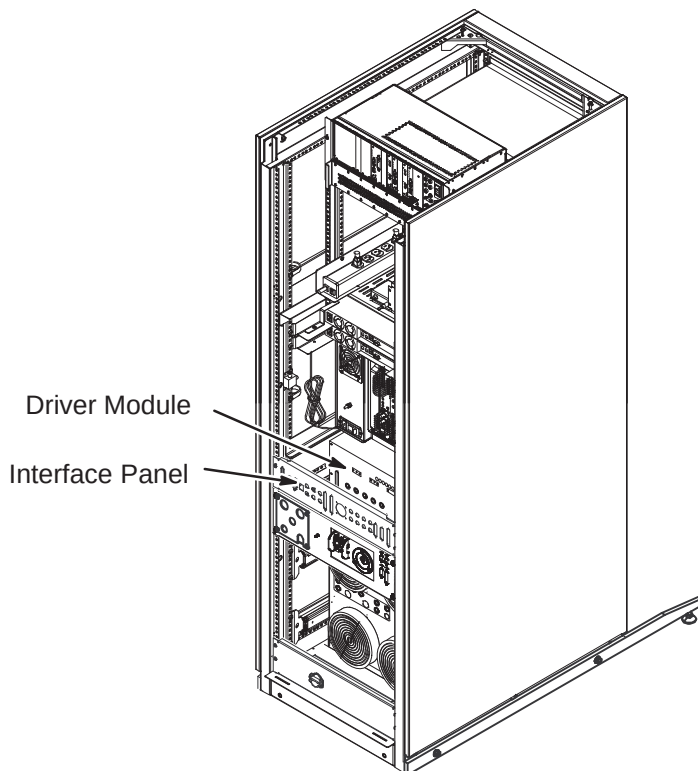


4-8-6 INSTALLATION OF REMAINING EQUIPMENT ROOM CABLES

Connect remaining Runs within equipment room. Dress and secure all cables after connections are made. Reference the applicable interconnect map in Section 4 and Cable Table in Appendix for to/from designator information and Run number reference.

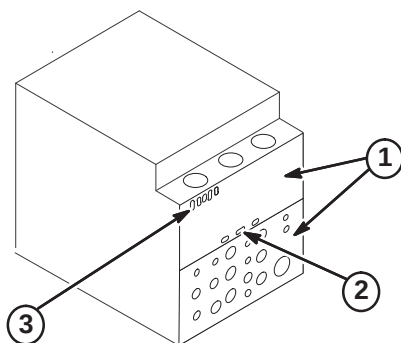
4-8-6-A RF/System Cabinet Connections

1. Connect routed cables to Interface Panels at bottom and middle of cabinet and to the Driver Module as marked. Refer to cable map.
2. Dress and secure cables. Adjust as necessary.



4-8-6-B HFD Cabinet PDU Module Connections

1. Attach rear PDU module covers removed earlier.
2. Connect Run 1179 to J2.
3. Connect Runs 974, 1180, and 1181 to J5, J7, and J8.



4-9 MAGNET ROOM CABLE INSTALLATION

The magnet enclosure and magnet room cables must be installed simultaneously.

Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

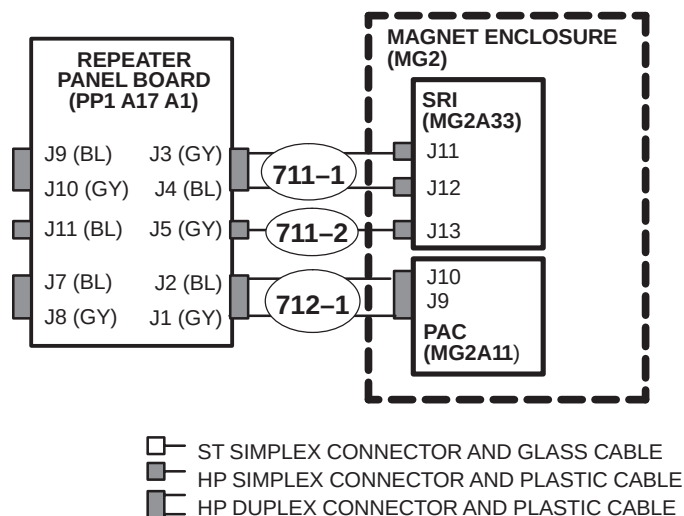


FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.

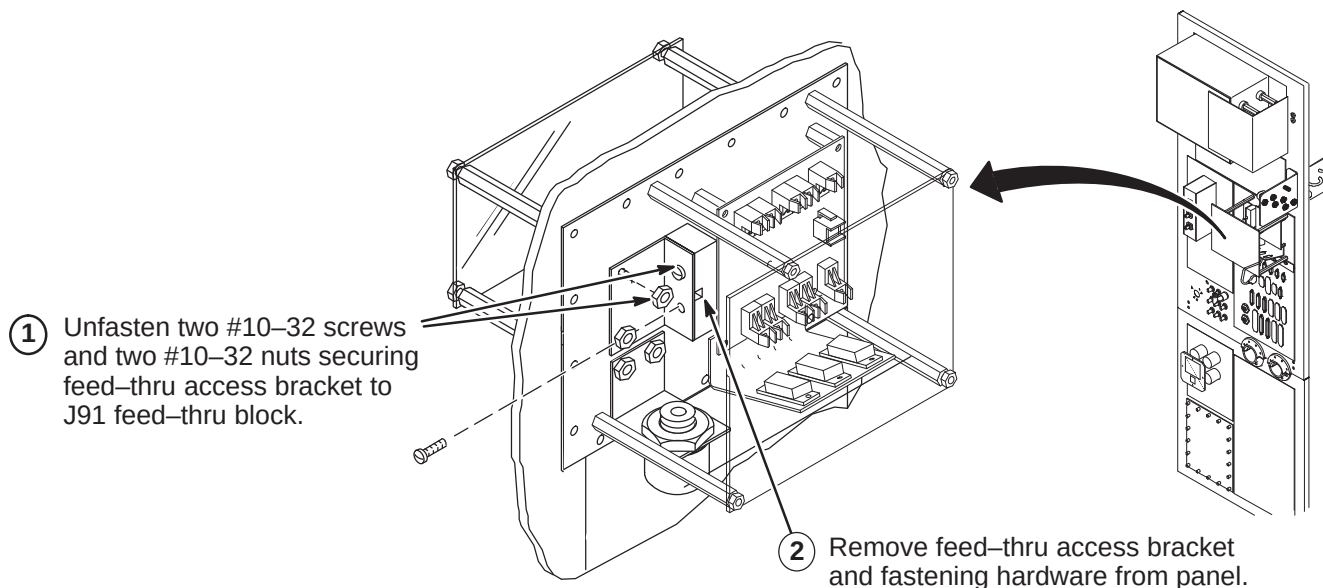
4-9-1 MAGNET ROOM FIBER OPTIC PAC/SRI INSTALLATION

For Horizon and LX configurations, existing Run 711/712 will be retained and connected to the new HDx Penetration Panel.

Shown below are the connections for the fiber optics cables routed in the magnet room from the Penetration Panel, thru the cable trough to the Rear Pedestal, and under the magnet to the PAC/SRI platform.

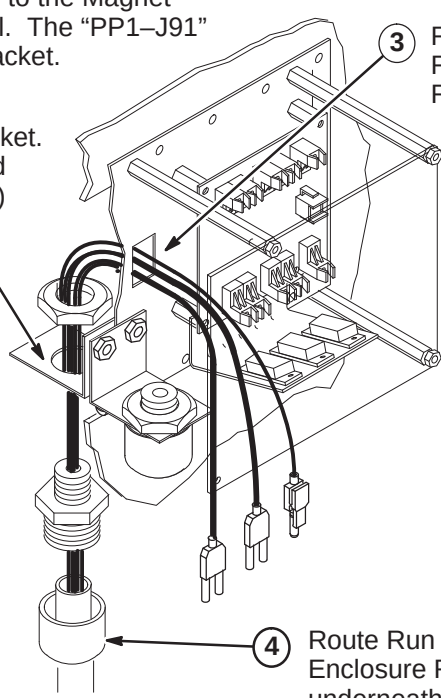


4-9-1-A Repeater Panel Feed-thru Bracket Removal



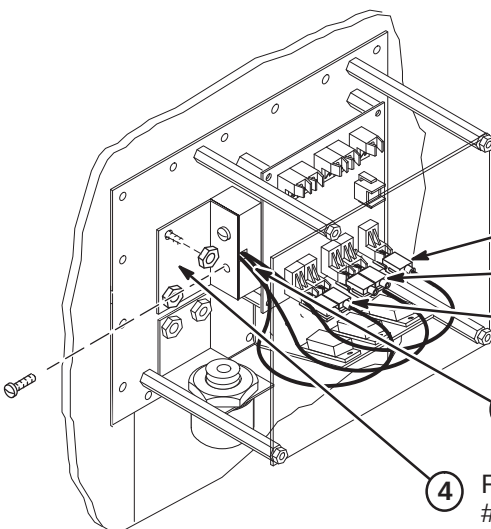
4-9-1-B Routing Run 711/712 in Magnet Room

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cut and ends terminated with the Fiber Optic Connector Repair Kit. Refer to Appendix for instructions. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- ① Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The "PP1-J91" end will be attached to interface bracket.
 - ② Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
 - ③ Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
 - ④ Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.
- 

4-9-1-C Connecting Fiber Optic Run 711/712 to Repeater Panel

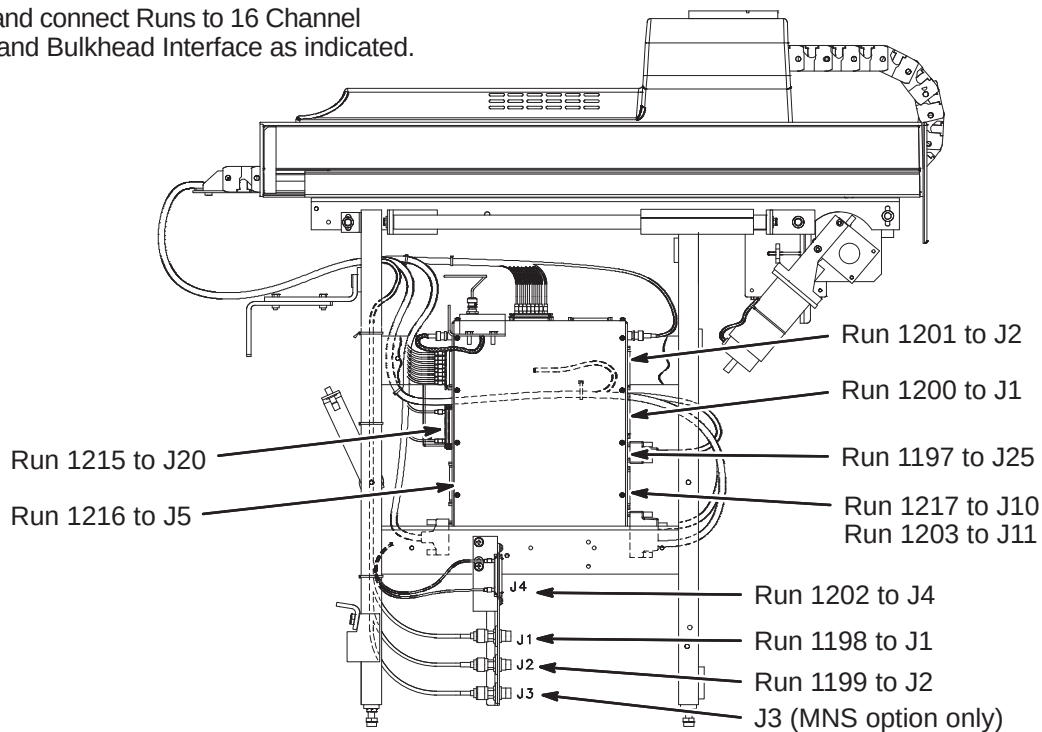
Note: Take care to maintain the minimum 2 inch bend radius.

- ① Adjust route of individual fiber optic runs to maintain large radius.
 - ② Connect Runs 712-1, 711-1, and 711-2 as shown.
 - ③ Gather fiber optic runs and route through wave guide notch in feed-thru block.
 - ④ Replace feed-thru access bracket and fasten with two #10-32 screws and two #10-32 nuts removed in step A1.
- 
- Run 711-2 (J5)
- Run 711-1 (J3, J4)
- Run 712-1 (J1, J2)

4-9-2 CONNECT REMAINING CABLES TO REAR PEDESTAL

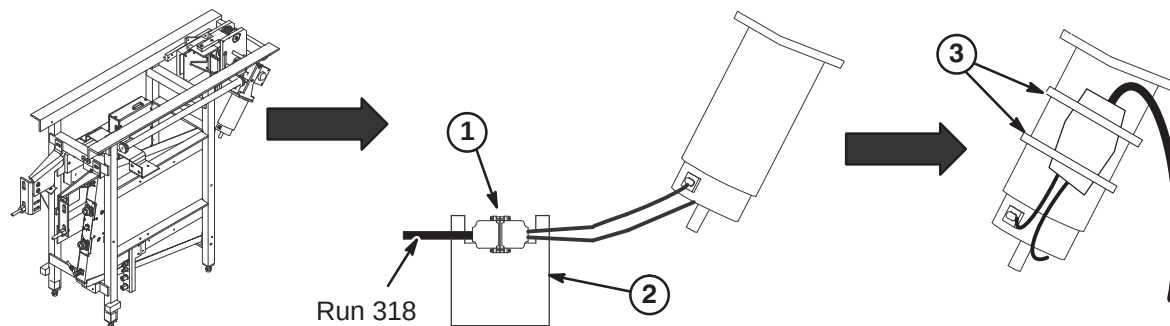
4-9-2-A Connect Cables to 16 Channel Switch and Bulhead Interface

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



4-9-2-B Route/Connect Run 318 to Longitudinal Drive Motor

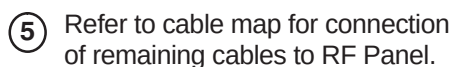
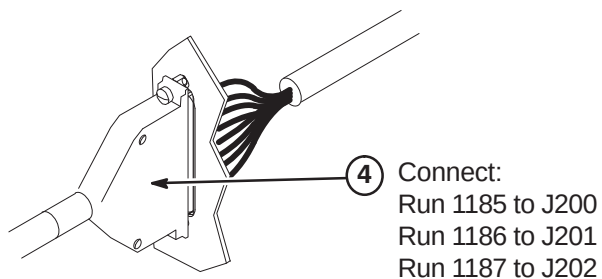
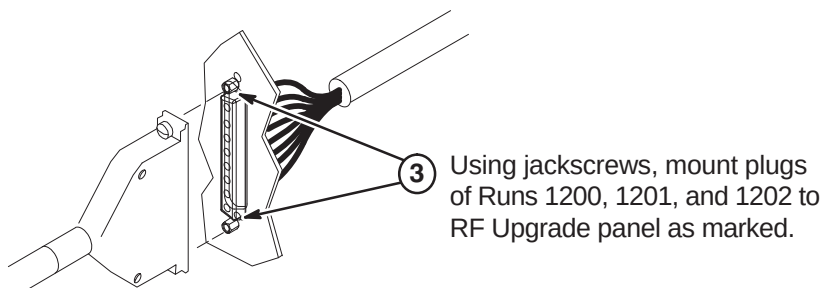
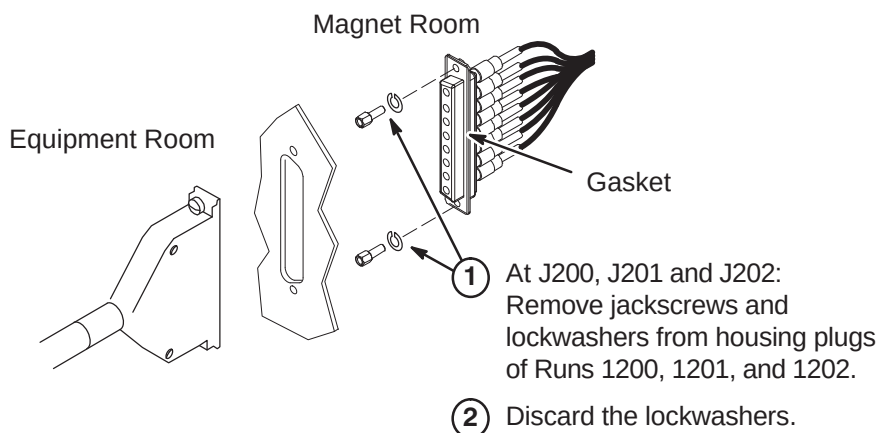
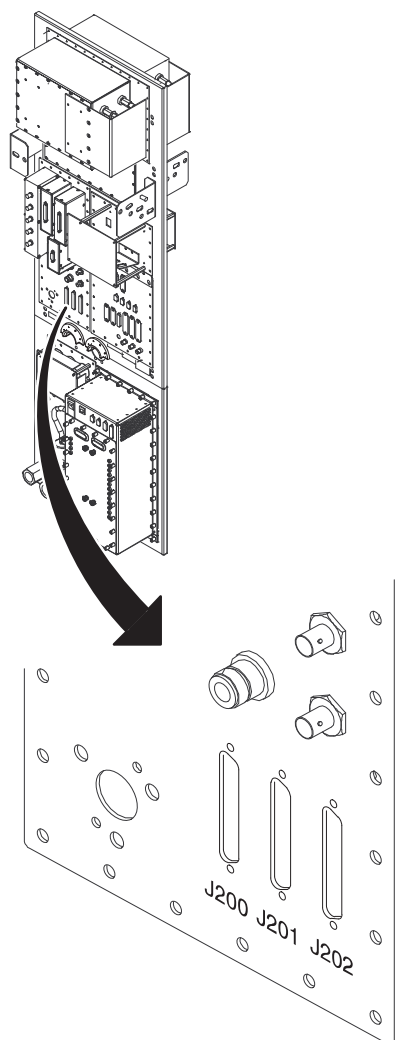
1. Route Run 318 from J11 on Penetration Panel and connect to Longitudinal Drive Motor cable.
2. Wrap both connectors with 177 x 200 Connector Wrap (2236130-2).
3. Ty-wrap cable to motor to prevent movement.



4-10 INSTALLATION OF CABLES THAT RUN BETWEEN EQUIPMENT ROOM AND MAGNET ROOM

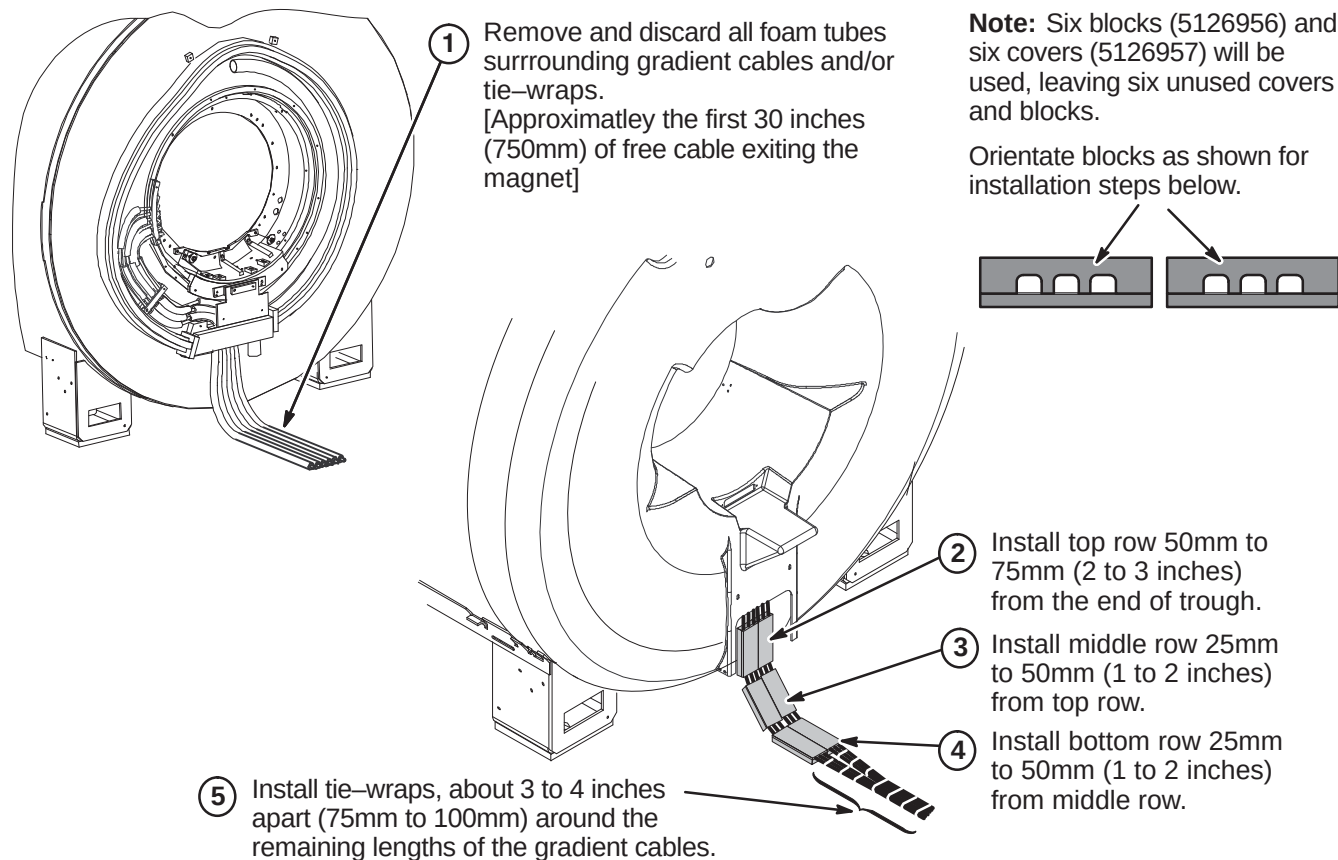
4-10-1 CONNECTION OF CABLES AT PENETRATION PANEL J200, J201, and J202

Note: Two people are required, one on each side of the Penetration panel, to install the cables in Step 1.



4-10-2 INSTALL GRADIENT CABLE RESTRAINT BLOCKS FOR WideOpen

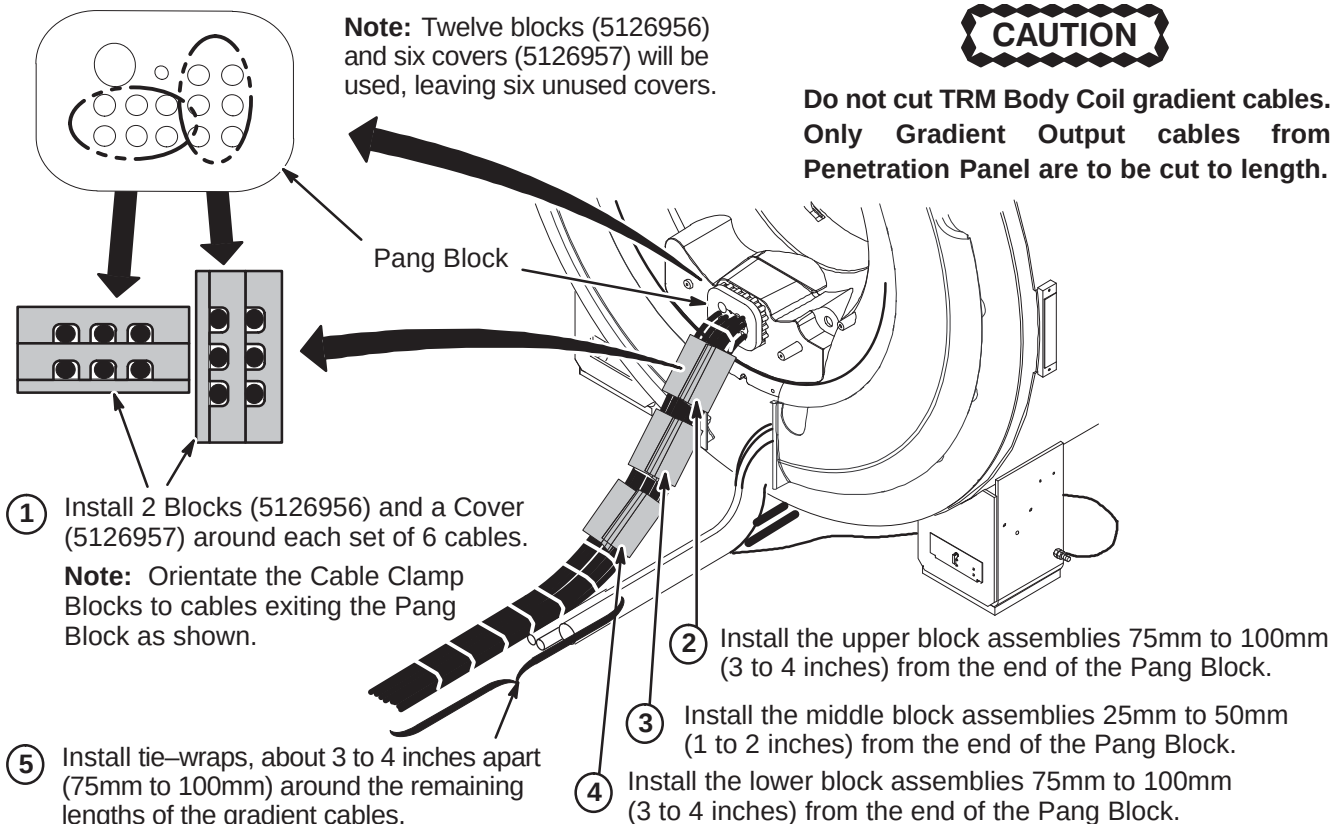
The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.



4-10-3 INSTALL GRADIENT CABLE RESTRAINT BLOCKS FOR TwinSpeed

Any existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.

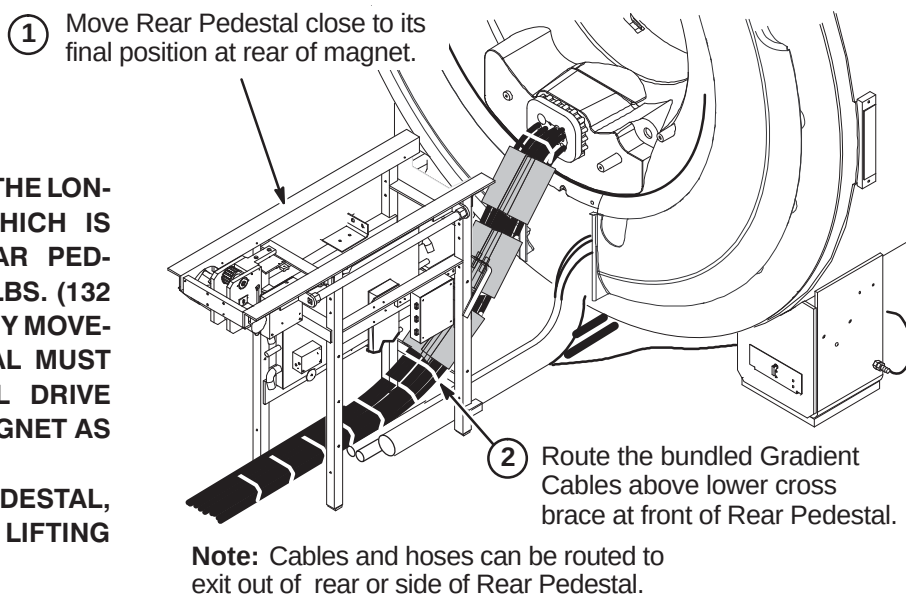
4-10-3-A Attach Gradient Cable Blocks



4-10-3-B Route Body Coil Gradient Cables Thru Rear Pedestal

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! THE REAR PEDESTAL CAN WEIGH UP TO 310 LBS. (132 KG). IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE. WHEN MOVING THE REAR PEDESTAL, USE PROPER SAFETY AND LIFTING TECHNIQUES.



4-10-4 GRADIENT CABLE INSTALLATION FOR NON-TwinSpeed

NOTE: Sites that already have a BRM-D Body Coil installed with leads crimped to gradient cables routed from Penetration Panel WILL NOT replace the existing cables with new gradient cables in Magnet Room. Sections **4-10-4-E** thru **4-10-4-H** will not be completed for these sites.

Three cables have been delivered to site. All have three lead wires with terminals at each end. The leads on this cable end are connected to the Equipment room and Magnet room sides of the Gradient Filter on the Penetration Panel. This procedure cuts the cables to length and connects them to the HFD cabinet and the cable leads from the Gradient Coil.

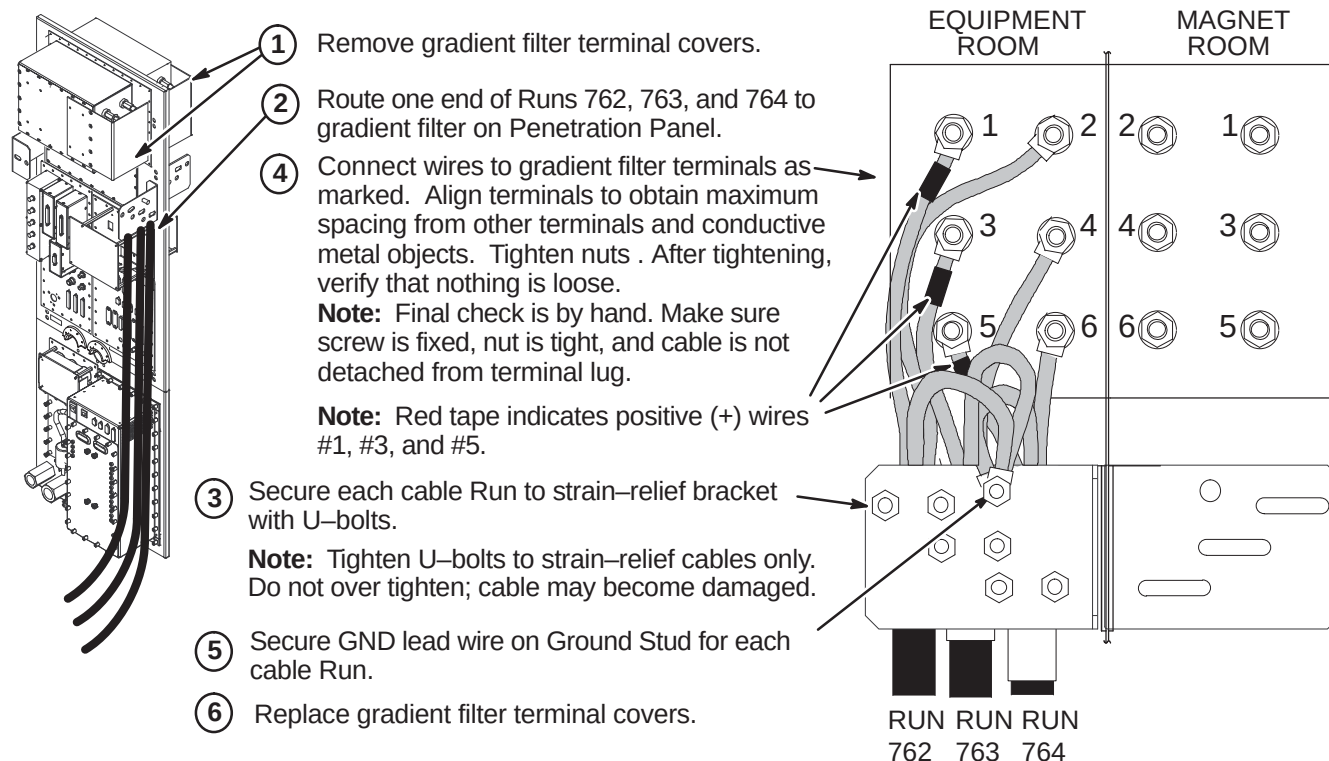
Verify that the length of supplied Run 762, 763, and 764 cable is sufficient to route from HFD Cabinet to Penetration Panel and, if required, Penetration Panel to Body Coil Gradient Cables at rear of magnet. **Each cable has identical terminations at each end for connection to both sides of the Penetration Panel Gradient Filter.**

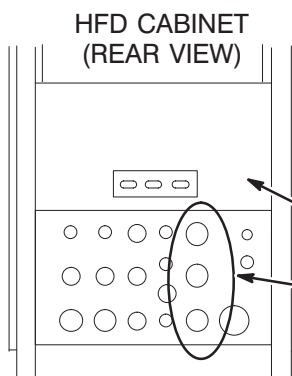
WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

4-10-4-A Connect Gradient Cables to Equipment Room Side of Penetration Panel



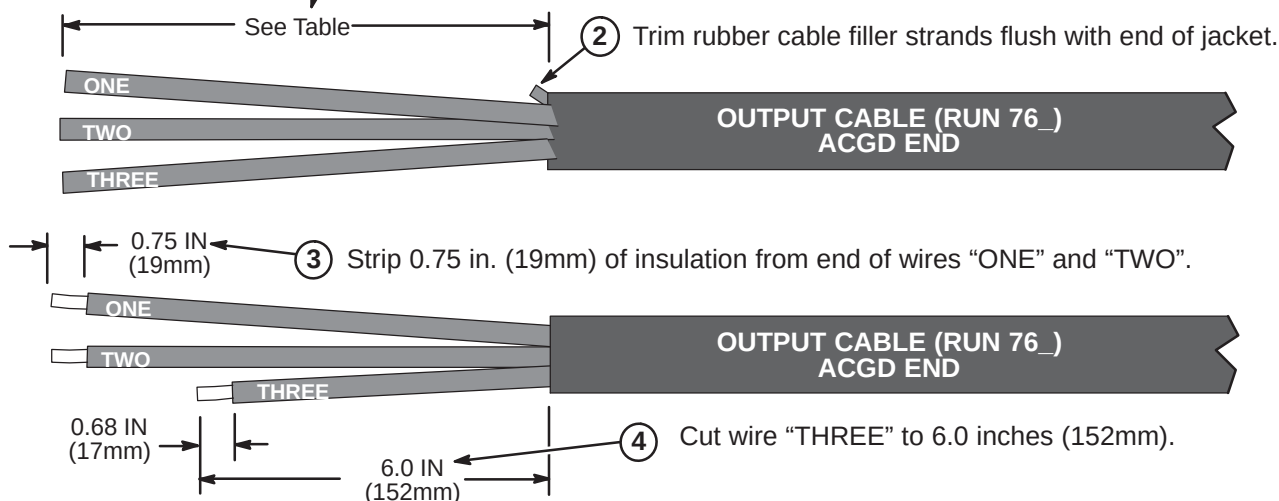
4-10-4-B Route Runs 762, 763, and 764 to HFD Cabinet

- ① Unroll and route each cable from equipment room side of Penetration Panel to HFD Cabinet. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet.
- ② Mark location for cutting.
- ③ Cut cables to length.
- ④ Remove Rear Cover Plate by removing seven screws that hold it in place.
- ⑤ Loosen screws on clamps for gradient output cables. After termination of cables in Procedure 4-10-4-C, insert ends of cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

4-10-4-C Terminate Runs 762, 763, and 764 for HFD Cabinet

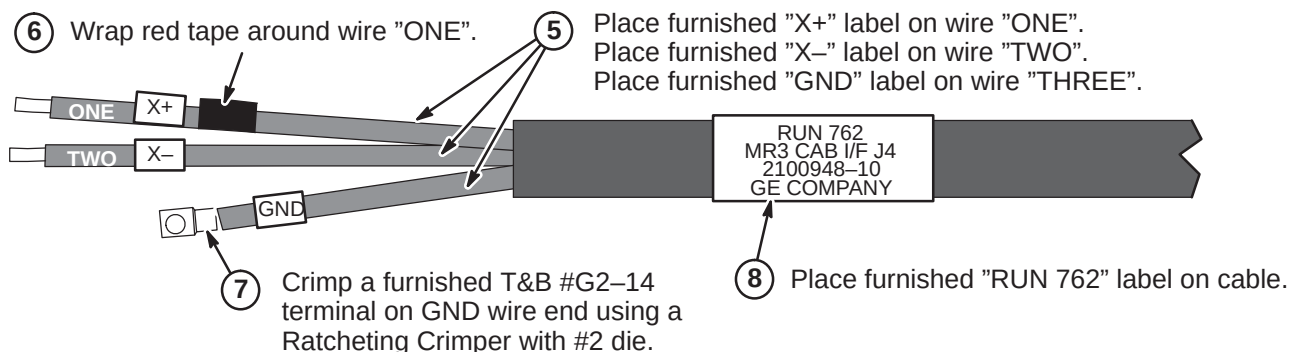
- ① Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 762 (X), 763 (Y), and 764 (Z).

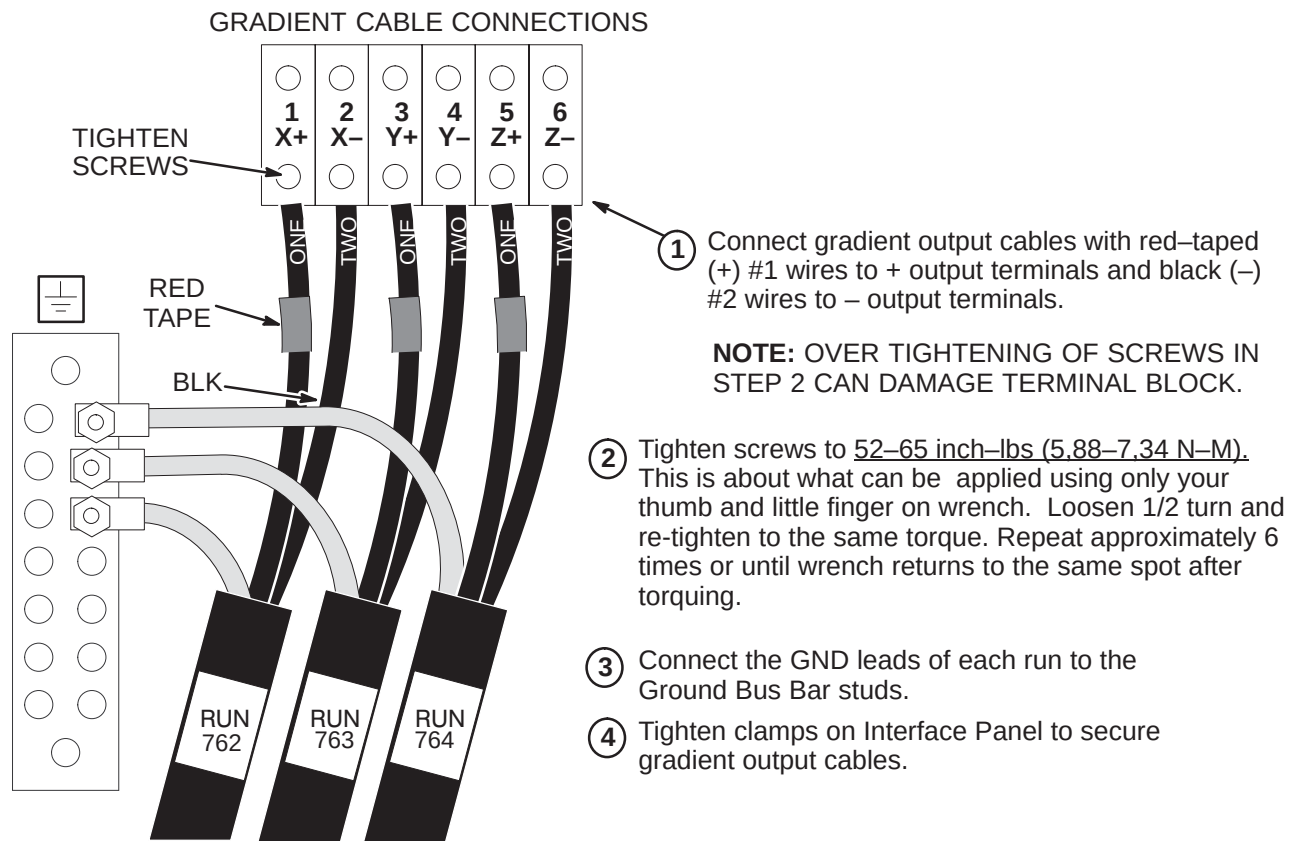
	Run 762	Run 763	Run 764
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

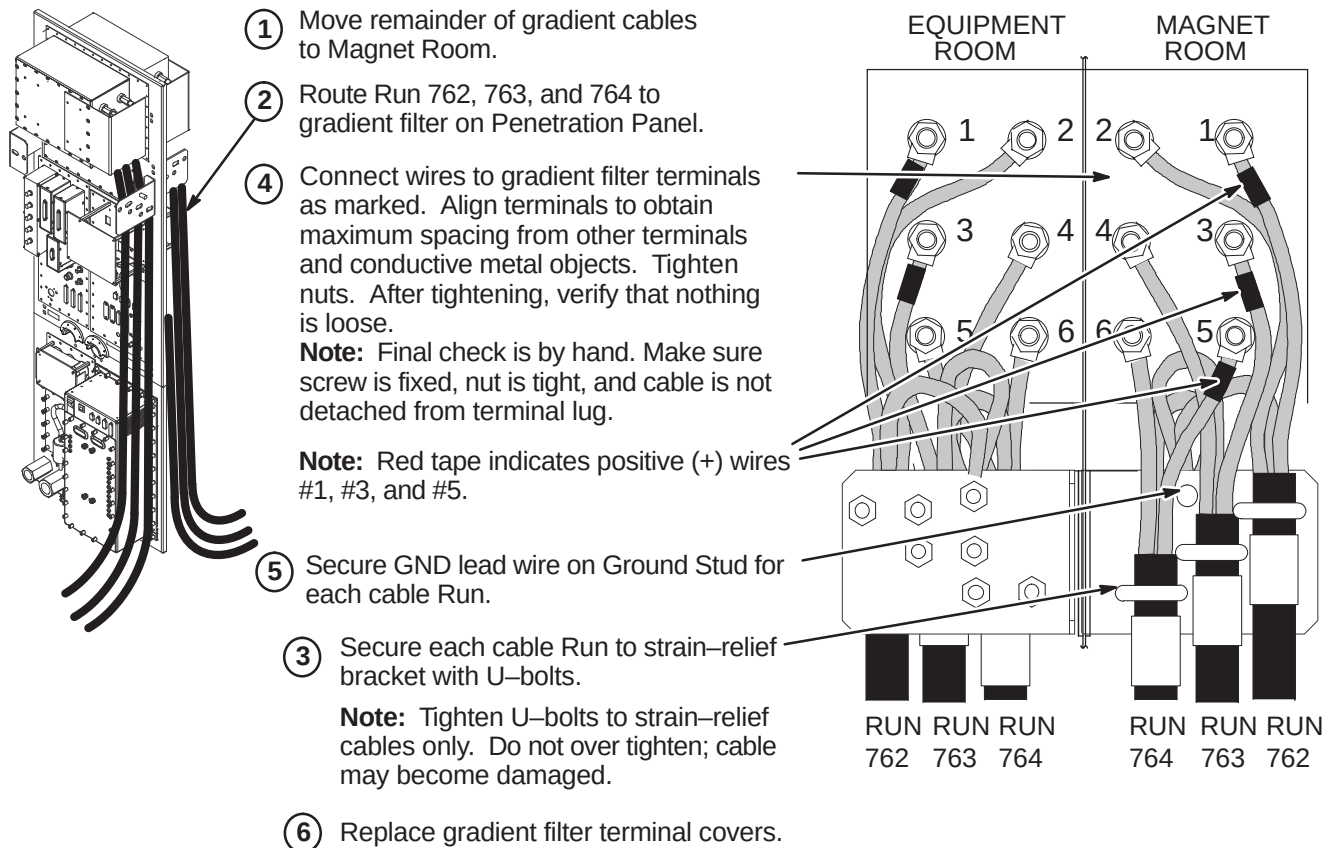
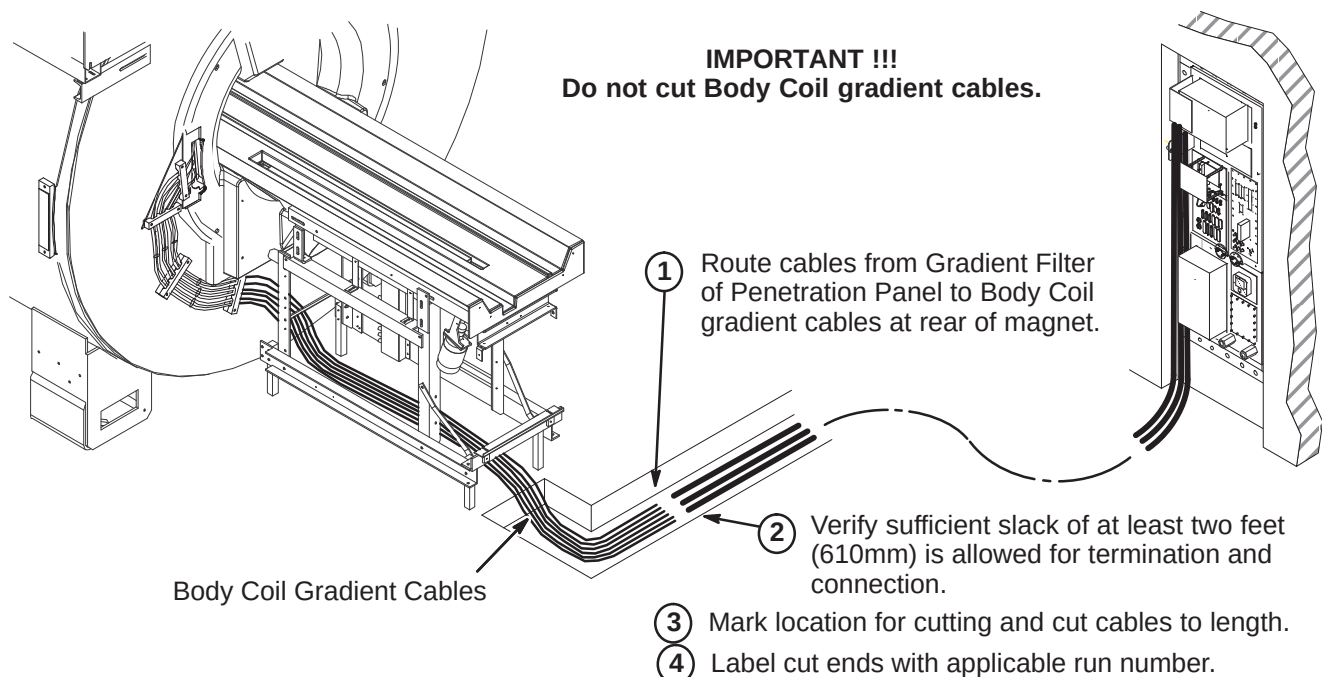
**Note:**

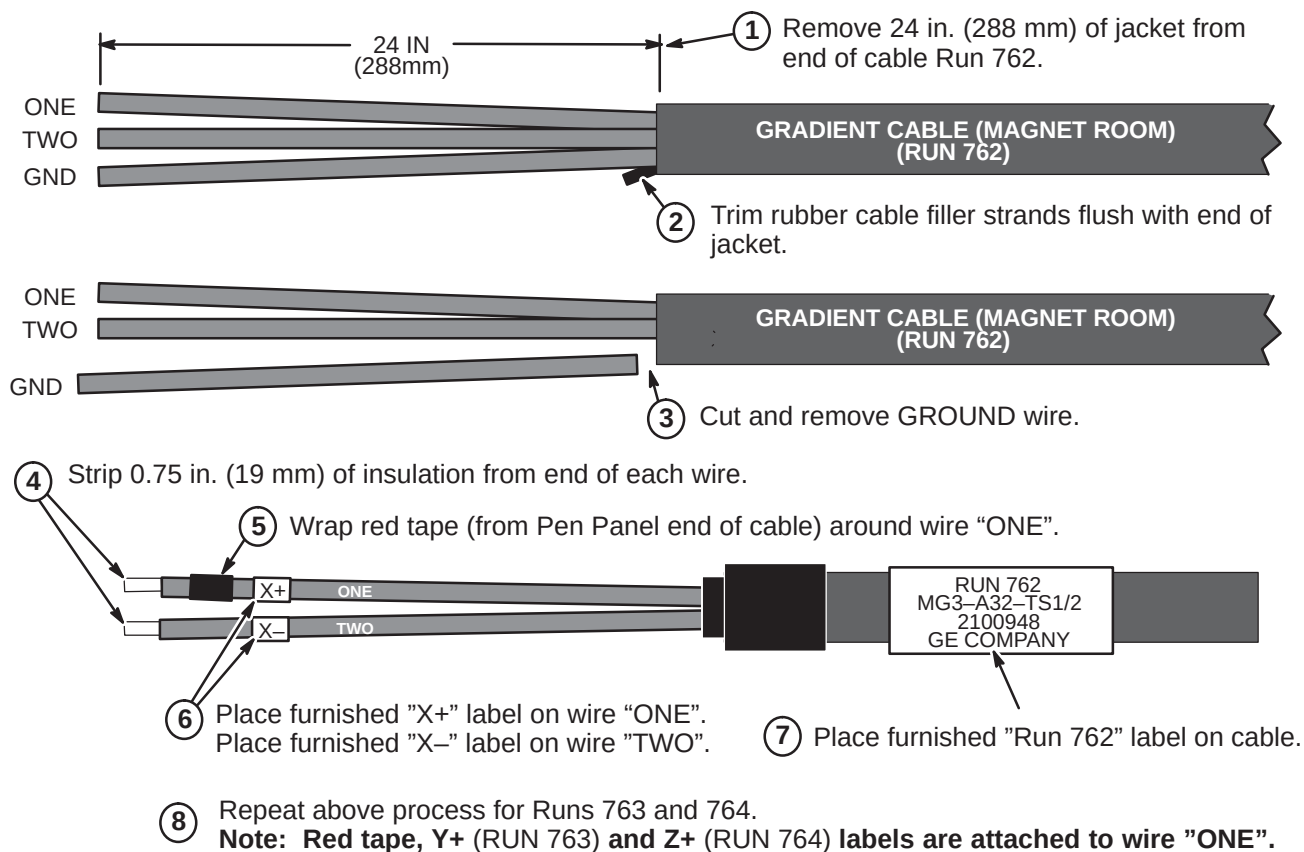
Steps below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y-) and 764 (Z+, Z-).

Red tape, Y+ (RUN 763) and Z+ (RUN 764) labels are attached to wire "ONE".



4-10-4-D Connect Runs 762, 763, and 764 at HFD Cabinet

4-10-4-E Connect Gradient Cables to Magnet Room Side of Penetration Panel**4-10-4-F Route Runs 762, 763, and 764 to Rear Pedestal Area**

4-10-4-G Terminate Runs 762, 763, and 764 for Magnet Room**WARNING!**

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

4-10-4-H Connection of Body Coil Gradient Cables to runs 762, 763, and 764

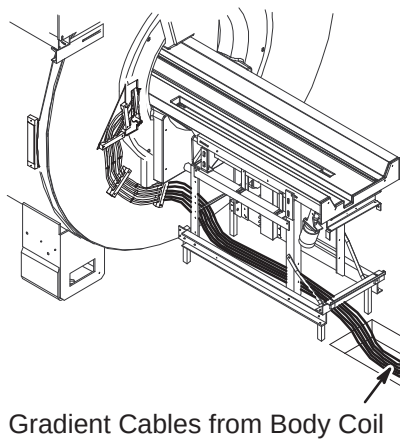
**Note:**

Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

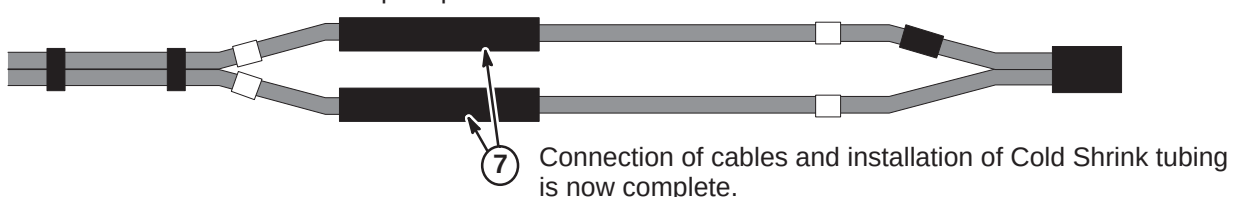
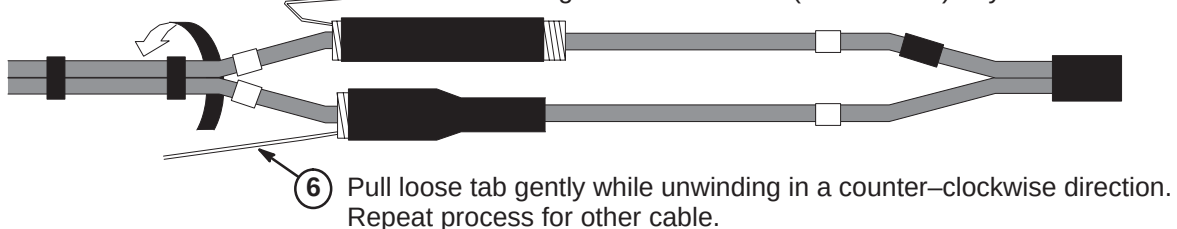
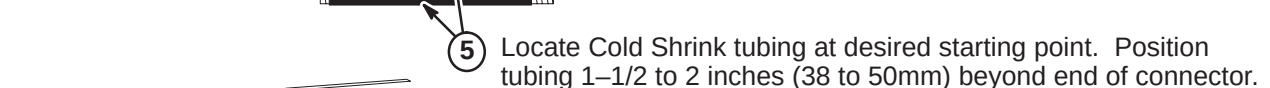
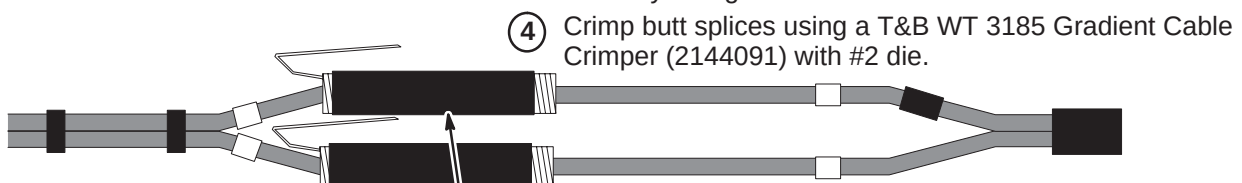
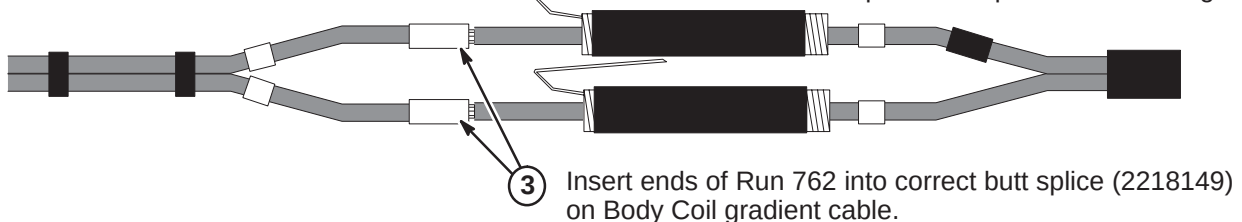
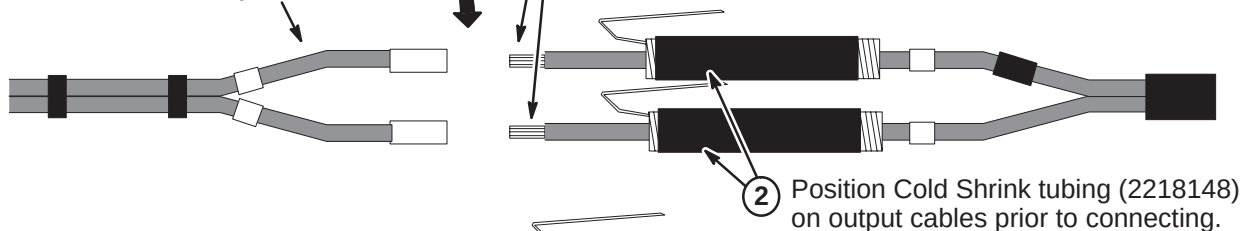
Note:

Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet

Gradient Output Cables from Penetration Panel

- ① Cable connection process shown below is for Run 762. Repeat process for Runs 763 and 764.

Note: Red tape, X+ (RUN 762), Y+ (RUN 763) and Z+ (RUN 764) labels are attached to wire "ONE".



4-10-5 GRADIENT CABLE INSTALLATION FOR TwinSpeed

NOTE: Sites that already have a TRM Body Coil installed with leads crimped to gradient cables routed from Penetration Panel WILL NOT replace the existing cables with new gradient cables in Magnet Room. Sections 4-10-5-L thru 4-10-5-Q will not be completed for these sites.

Also, the gradient cables in the Equipment Room do not have to be replaced. Existing gradient cables can be reused.

The following TwinSpeed gradient cables installation procedures are provided for those sites that have to install gradient cables.



FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

Nine cables have been delivered to site. Nine of the labeled pre-terminated leads will be connected to the Twin Gradient Filter on the Penetration Panel. The other three labeled pre-terminated ends will be connected to the TAC Cabinet. The other ends of all the cables will be cut to length, labeled, and terminated.

- Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. **Do not cut Body Coil gradient cables.**
- Locate furnished bag of parts that contain the terminals and labels required to be attached to the cables.

The nine cables with supplied labels are as follows:

RUN	LABEL	SUPPLIED LABELS
976/979	PP1-A7-X	(1) RUN 976, (1) RUN 979, (2) X+, (1) X-, (1) GND, (1) XZM-, (1) XWB-
977/980	PP1-A7-Y	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y-, (1) GND, (1) YZM- (1) YWB-
978/981	PP1-A7-Z	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z-, (1) GND, (1) ZZM- (1) ZWB-
982	PP1-A7-Z _{ZM}	(1) RUN 982, (1) ZZM- (1) ZZM+
983	PP1-A7-Y _{ZM}	(1) RUN 983, (1) YZM- (1) YZM+
984	PP1-A7-X _{ZM}	(1) RUN 984, (1) XZM- (1) XZM+
985	PP1-A7-Z _{WB}	(1) RUN 985, (1) ZWB- (1) ZWB+
986	PP1-A7-Y _{WB}	(1) RUN 986, (1) YWB- (1) YWB+
987	PP1-A7-X _{WB}	(1) RUN 987, (1) XWB- (1) XWB+

4-10-5-A General Steps for Cutting Cables to Length and Routing in Equipment Room

1. Move Run 979, 980, and 981 cables to equipment room. The end of cable with terminals will be connected to Penetration Panel.
2. Route cables from Gradient Filter of Penetration Panel to right side of interface panel at rear of Twin Accessory Cabinet (TAC).
3. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at TAC Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
4. Refer to Procedure **4-10-5-E** to terminate and label TAC end cable Runs 979, 980, and 981
5. Refer to Procedure **4-10-5-F** for connection of Runs 979, 980, and 981 to TAC cabinet.
6. Route remainder of cables from rear of TAC to rear of HFD cabinet. Make sure enough slack is provided for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
7. Refer to Procedure **4-10-5-H** for connection of Runs 976, 977, and 978 to TAC cabinet.
8. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
9. Refer to Procedure **4-10-4-H** for connection of Runs 976, 977, and 978 to HFD cabinet.

4-10-5-B Connect Runs 979, 980, and 981 to Penetration Panel

1. Remove gradient filter terminal covers.
2. Route Run 981 (Z) to left side and Runs 979 (X) and 980 (Y) to right side of gradient filter on Penetration Panel.

Note: The three X and three Y terminals are routed to the right side and three Z terminals to the left side.

3. Connect +, ZM- (cable marked as TWO), & WB- (cable marked as THREE) wires to gradient filter terminals as marked on each side of gradient filter.

Note: Red tape indicates positive (+) wires.

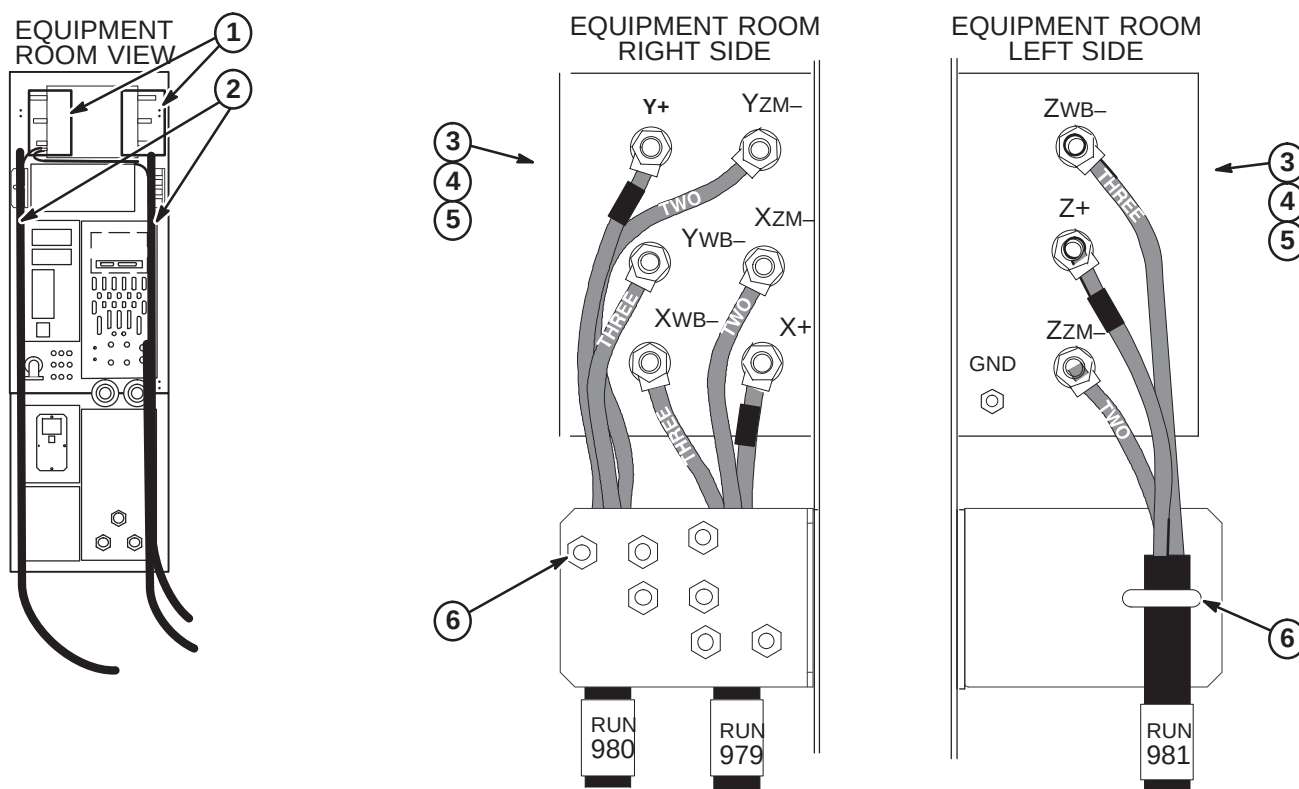
4. Align cables to obtain maximum spacing from other cables and conductive metal objects.

Note: When tightening nuts, it is important to hold screw body.

5. Tighten nuts. After tightening, verify that nothing is loose.

Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.

6. Route and secure each cable Run to strain-relief bracket with U-bolts.

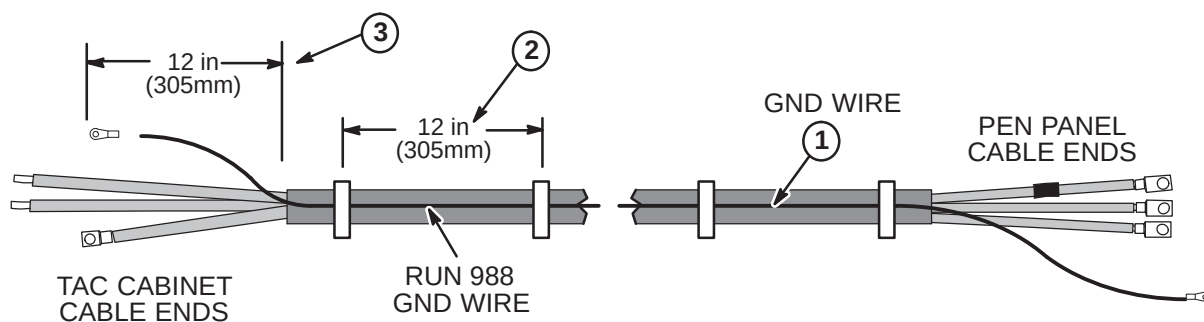


4-10-5-C Attach Ground Cables to Runs 979, 980, and 981

For each gradient cable routed between the Penetration Panel and TAC Cabinet, a ground wire, Run 988 (2283297), has to be ty-wrapped to each cable.

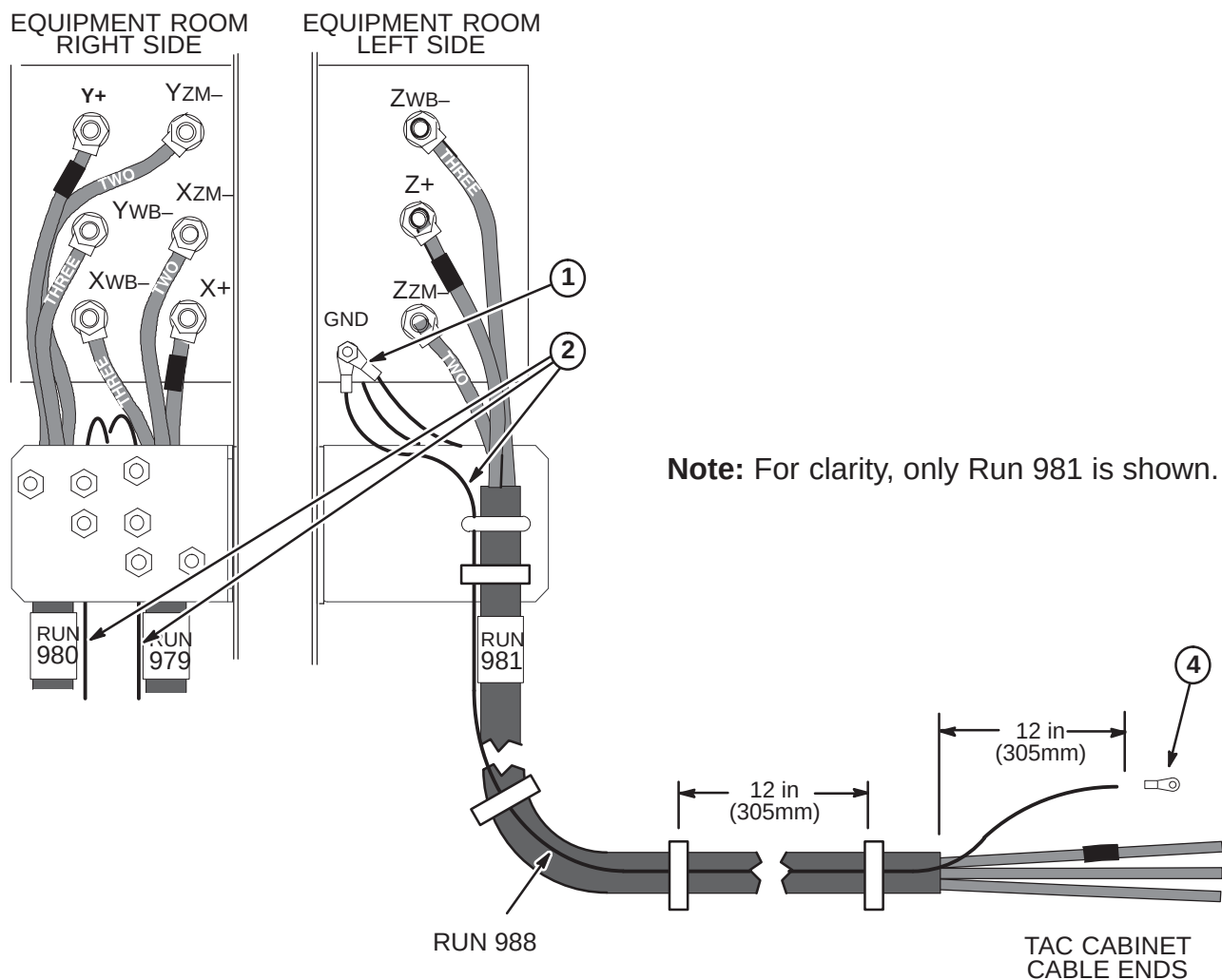
1. One Ground Wire will be routed along each of the Gradient Cables, Runs 979, 980, and 981.
2. Each Run 988 must be ty-wrapped about every 12 inches along entire length of each gradient cable.
Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. End of Run 988 at TAC cabinet should extend about 12 inches (305mm) past end of outer jacket after trimming.
Note: Refer to Procedure **4-10-5-E** for length of gradient cable jacket material to be removed.

Excess ground cable can not be coiled. It must be cut to length and reterminated.



4-10-5-D Connect Run 988 Cables Ground Wires to Gradient Filter

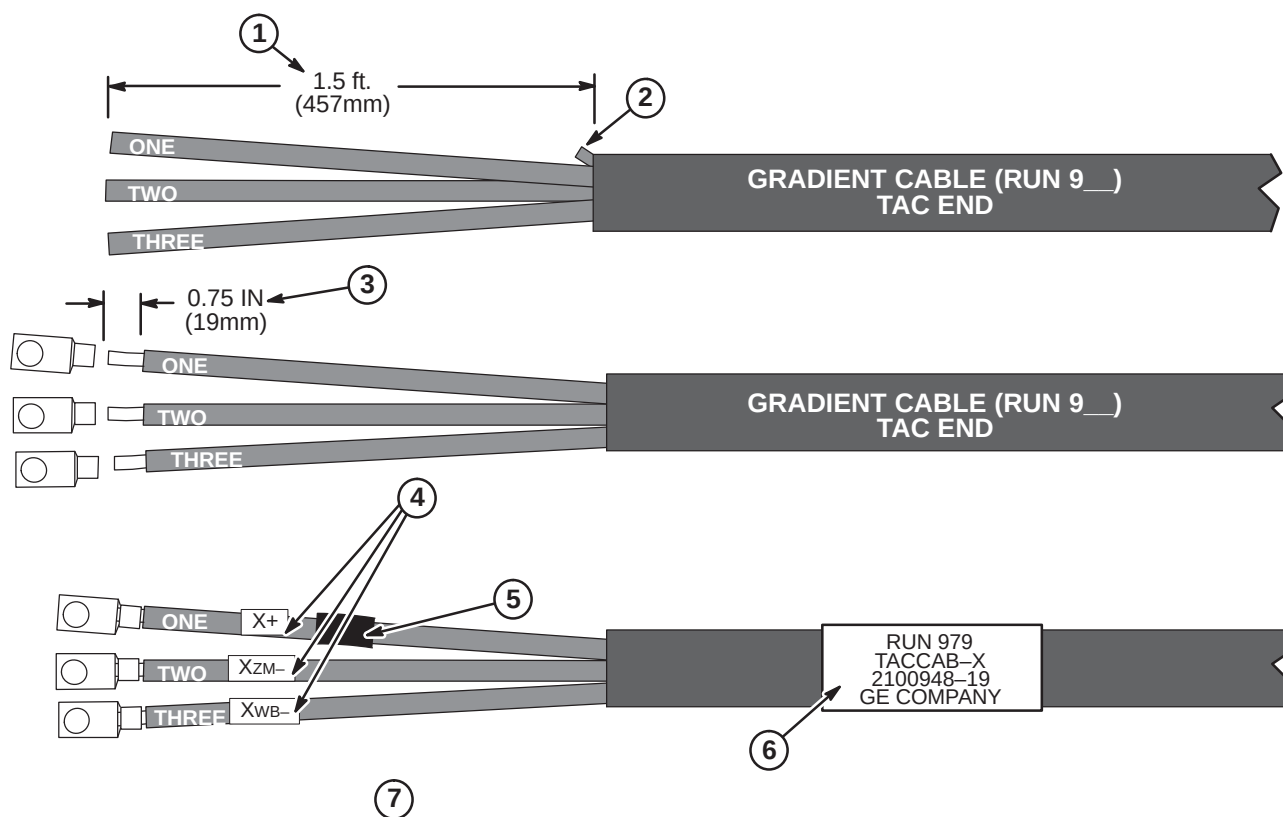
1. Connect the three Run 988 Gound Wires to GND on left side of Gradient Filter.
2. Each Run 988 must be ty-wrapped about every 12 inches along entire length of each gradient cable.
Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. Re-install gradient filter terminal covers.
4. Cut each Run 988 to length, reterminate with 3/8 ring terminal for connection to GND stud on TAC Cabinet.



4-10-5-E Terminate Runs 979, 980, and 981 for TAC Cabinet Connection

The following steps are required to terminate the end of the cable that was cut in Procedure **4-10-4-A**. The other end of the cable is connected to the Penetration Panel.

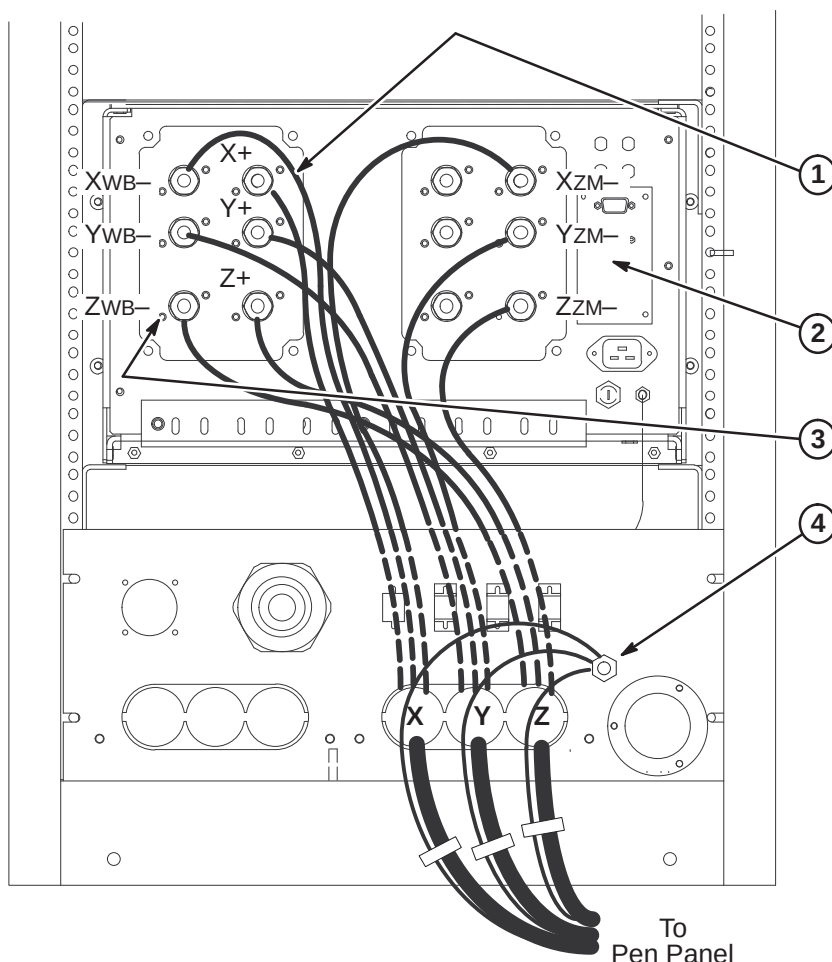
1. Remove 18 inches of jacket from end of cable Runs 979 (X), 980 (Y), and 981 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires "ONE", "TWO" and "THREE" and crimp a furnished T & B 1/2" (G2-3-12) terminal to each using Ratcheting Crimper with #2 die.
4. Place furnished "X+" label on wire "ONE".
Place furnished "XZM-" label on wire "TWO".
Place furnished "XWB-" label on wire "THREE".
5. Wrap red tape around wire "ONE".
6. Place furnished "RUN 979" label on cable.
7. Steps above show completion of Run 979. Repeat same process for Runs 980 (Y+, YZM-, YWB-) and 981 (Z+, ZZM-, ZWB-).



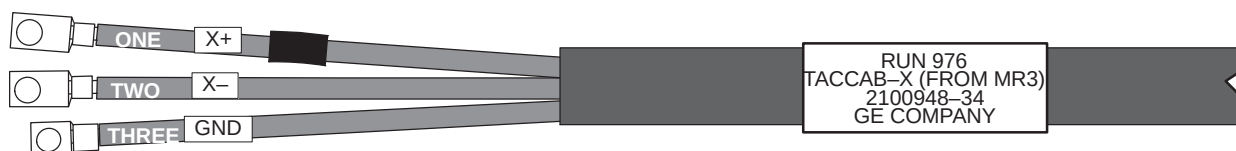
4-10-5-F Connect Runs 979, 980, and 981 to TAC Cabinet

To make installation easier, route and connect Run 981 (Z terminals) first, then Run 980 (Y), and last Run 979 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect XZM-, YZM-, and ZZM- cables as marked.
3. Connect XWB-, YWB-, and ZWB- cables as marked.
4. Route and connect the three Run 988 Ground wires to GND stud on interface panel.

**4-10-5-G Pre-terminated cable ends for Runs 976, 977, and 978**

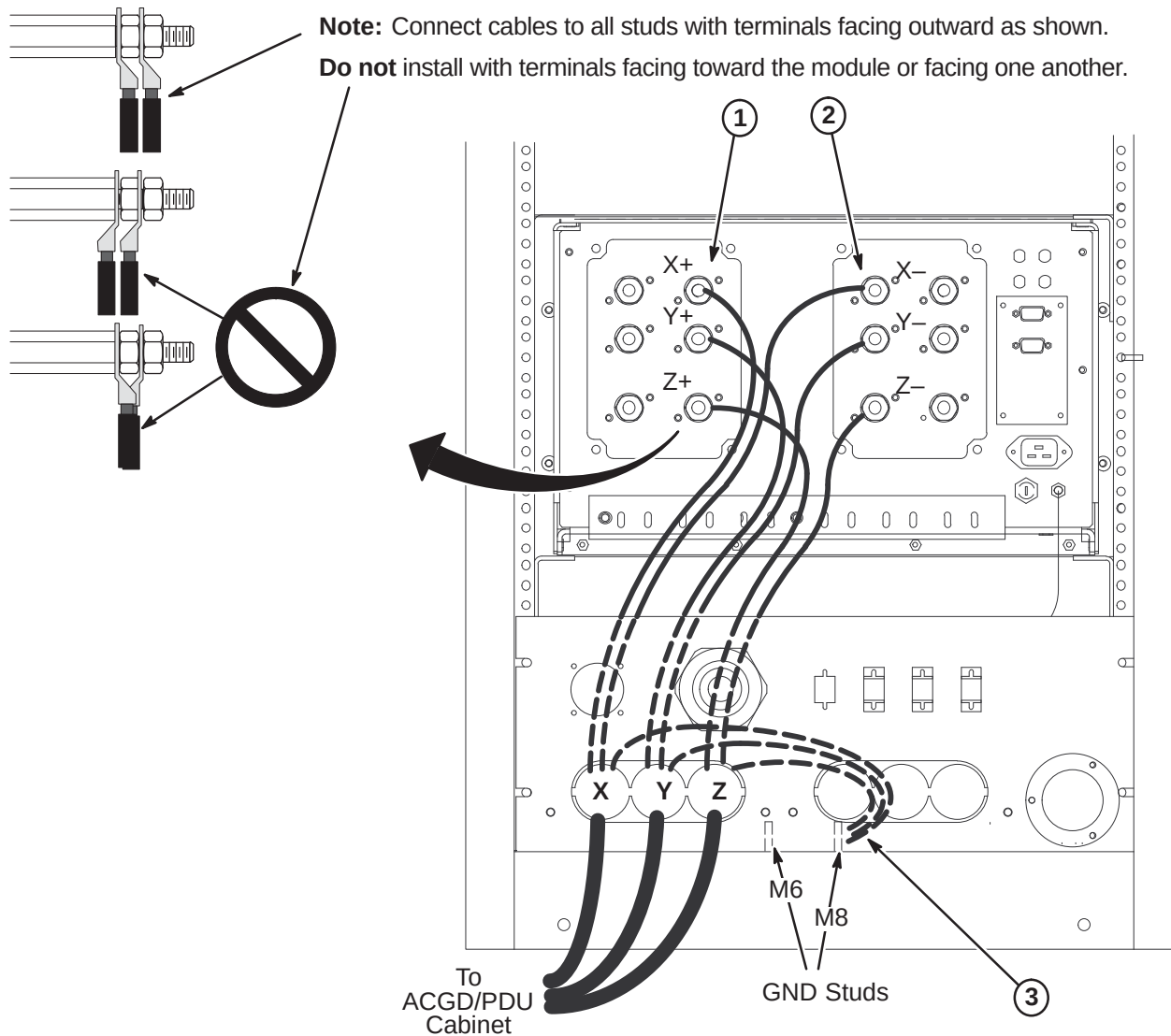
Cables delivered to site will be pre-terminated for the TAC Cabinet connection of Runs 976, 977, and 978. Refer to Procedure **4-10-4-A** for cutting cables to length for connection to HFD Cabinet. A typical pre-terminated end for each cable is shown below.



4-10-5-H Connect Runs 976, 977, and 978 to TAC Cabinet

To make installation easier, route and connect Run 978 (Z terminals) first, then Run 977 (Y), and last Run 976 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect X-, Y-, and Z- cables as marked.
3. Connect Ground leads from each cable to M8 GND stud located at bottom of cabinet.



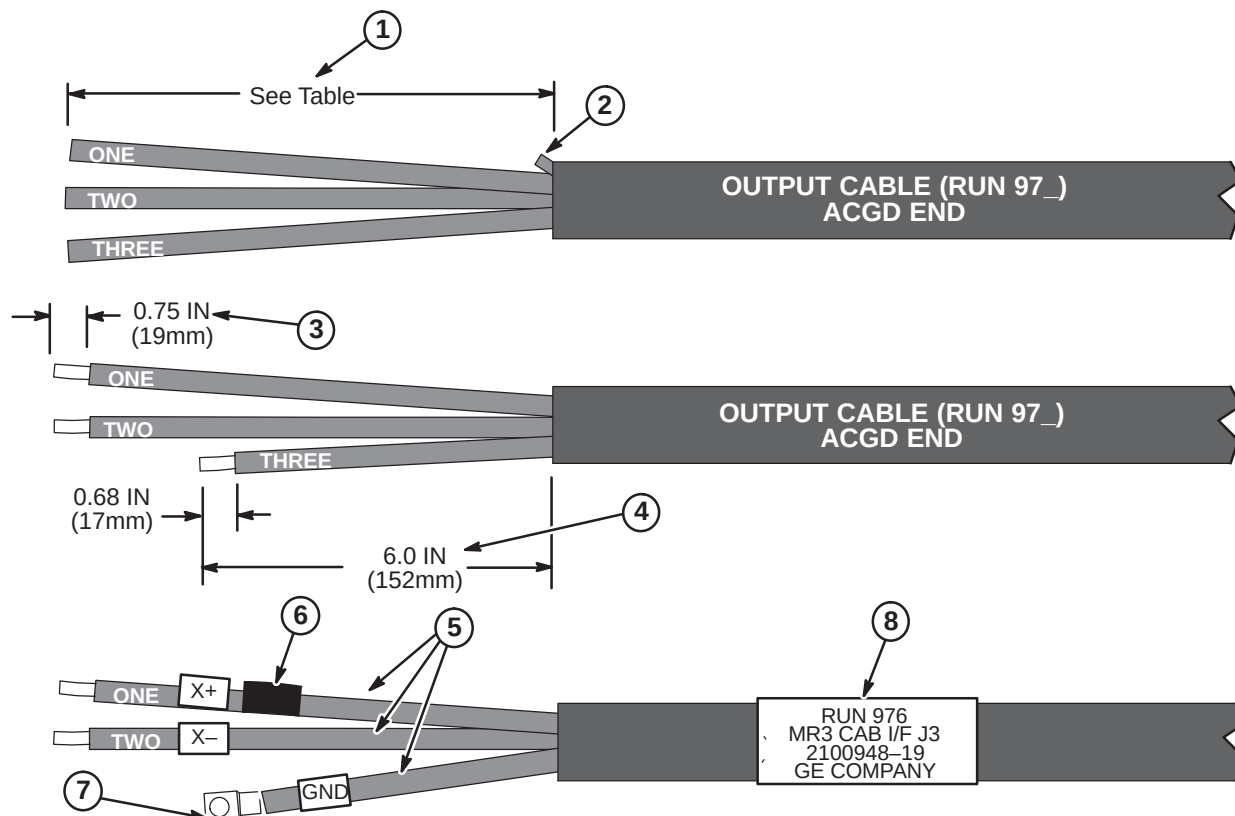
4-10-5-I Terminate Runs 976, 977, and 978 for Connection to HFD Cabinet

The following steps are required to terminate the end of the cable that is to be connected to the HFD Cabinet. The other end of the cable is connected to the TAC Cabinet.

	Run 976	Run 977	Run 978
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

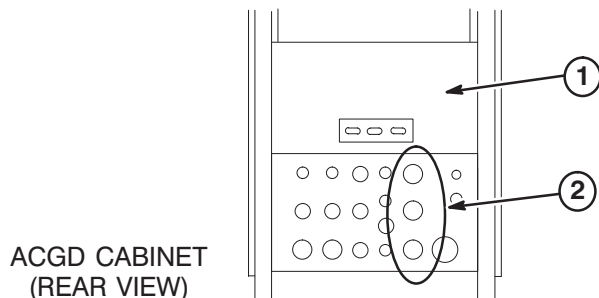
1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 976 (X), 977 (Y), and 978 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires "ONE" and "TWO".
4. Cut wire "THREE" to 6.0 inches (152mm).
5. Place furnished "X+" label on wire "ONE".
Place furnished "X-" label on wire "TWO".
Place furnished "GND" label on wire "THREE".
6. Wrap red tape around wire "ONE".
7. Crimp a furnished T&B #G2-38 terminal on GND wire end using a Ratcheting Crimper with #2 die.
8. Place furnished "RUN 976" label on cable.

Steps in illustration below show completion of Run 976. Repeat same process for Runs 977 (Y+, Y-) and 978 (Z+, Z-). **Red tape, Y+ (RUN 977) and Z+ (RUN 978) labels are attached to wire "ONE".**

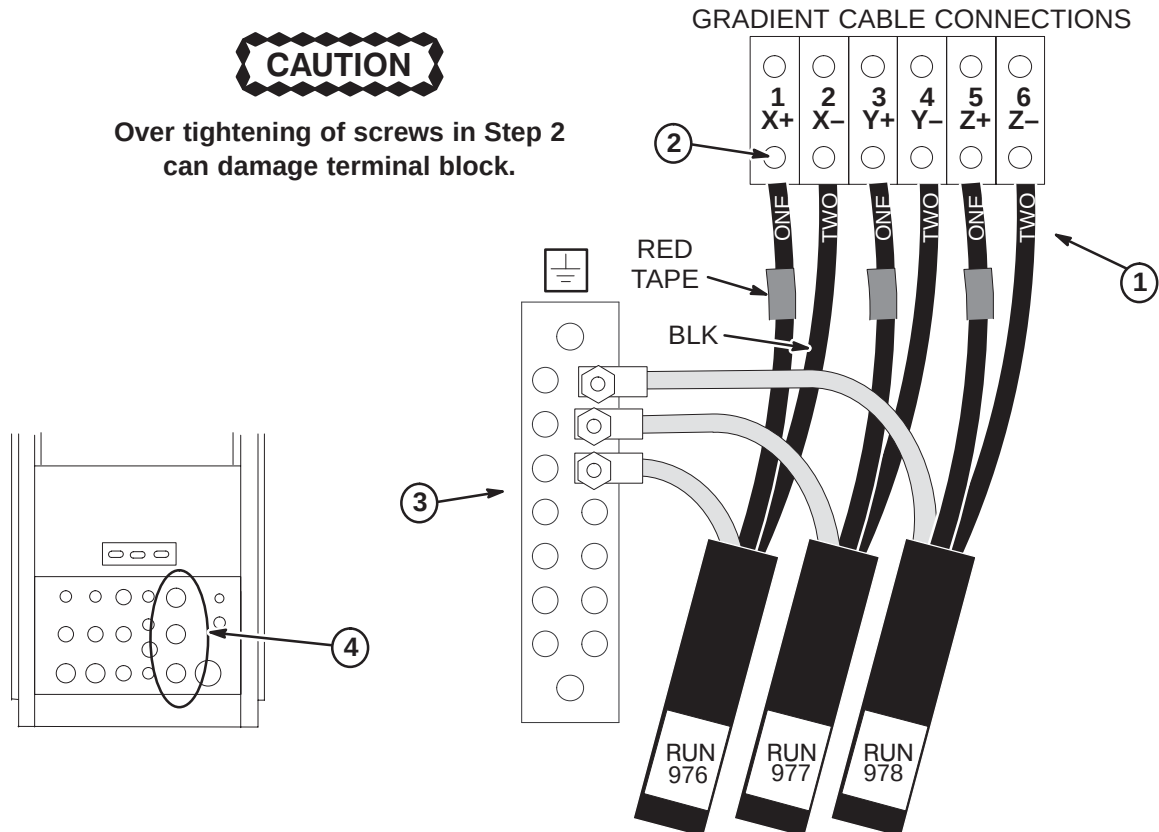


4-10-5-J Gradient Output Cable Interface Panel Access at HFD Cabinet

1. Remove Rear Cover Plate by removing seven screws that hold it in place.
2. Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 976 into J3, Run 977 into J4, and Run 978 into J5)

**4-10-5-K Gradient Output Cable Routing and Connection in HFD Cabinet**

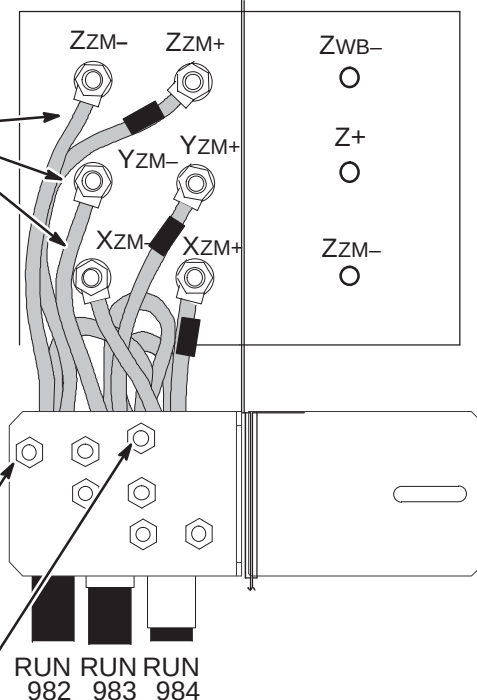
1. Connect gradient output cables with red-taped (+) #1 wires to + output terminals and black (-) #2 wires to - output terminals.
2. Tighten six screws to 52-65 inch-lbs (5.88-7.34 N-M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
3. Connect the GND leads of each run to the Ground Bus Bar studs.
4. Tighten clamps on Interface Panel to secure gradient output cables.



4-10-5-L Connect ZM (Zoom) Cables to Penetration PanelMAGNET ROOM
VIEW

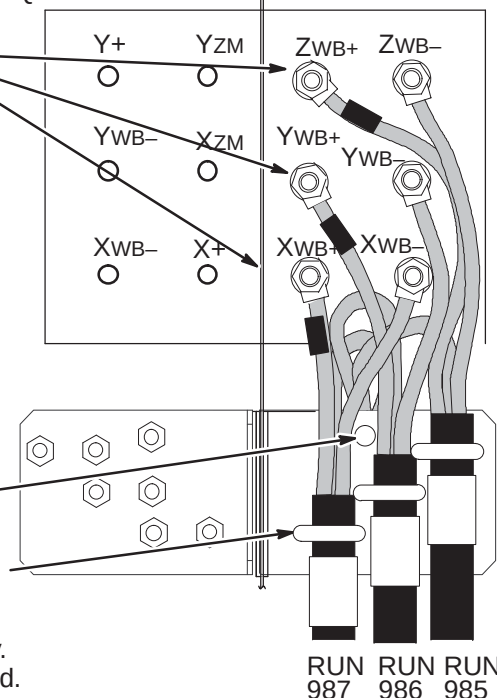
- ① Remove gradient filter terminal covers.
- ② Route Runs 982, 983, and 984 to right side of gradient filter on Penetration Panel.
- ③ Connect six ZM- and ZM+ wires to gradient filter terminals as marked.
Note: Red tape indicates positive (+) wires.
- ④ Align cables to obtain maximum spacing from other cables and conductive metal objects.
Note: When tightening nuts, it is important to hold screw body.
- ⑤ Tighten nuts to a torque of 35 ft-lbs (67-74Nm) or after tightening, verify that nothing is loose.
Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
- ⑥ Route and secure each cable Run to strain-relief bracket with U-bolts.
Note: Tighten U-bolts to strain-relief cables only. Do not over tighten; cables may become damaged.
- ⑦ Secure GND lead wire on Ground Stud for each cable Run.

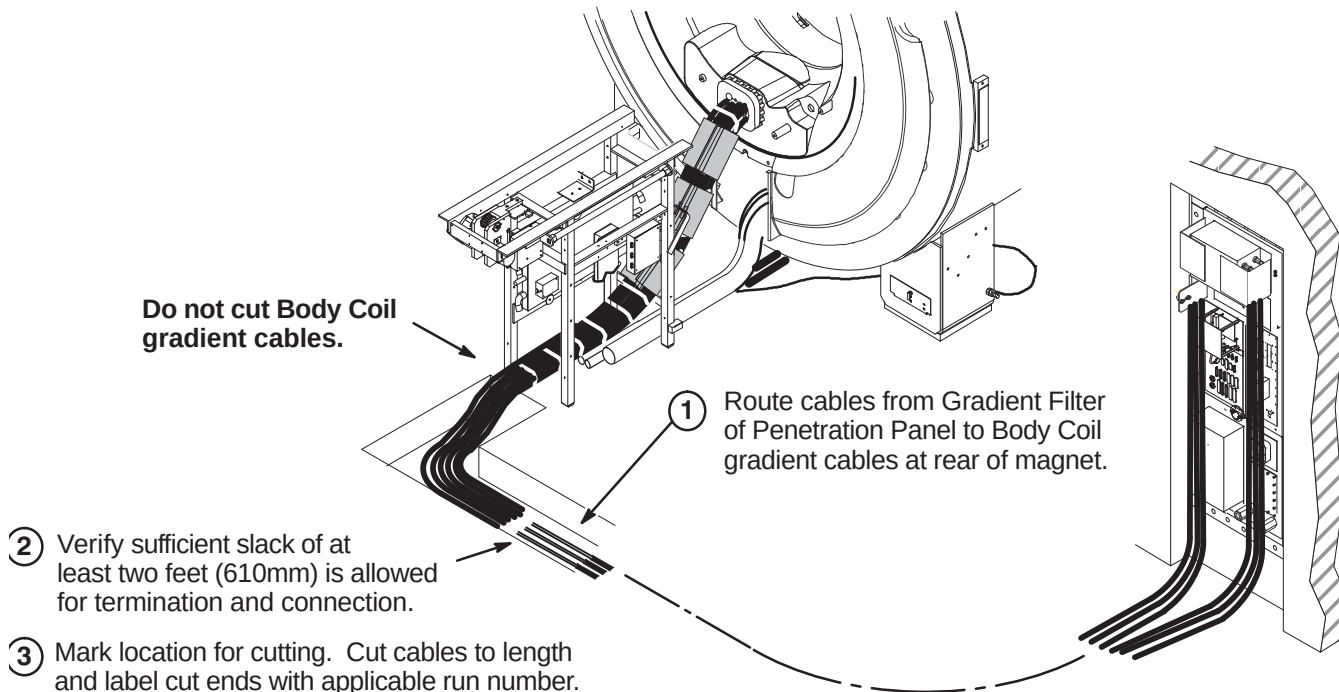
MAGNET ROOMEQUIPMENT ROOM

**4-10-5-M Connect WB (Whole Body) Cables to Penetration Panel**MAGNET ROOM
VIEW

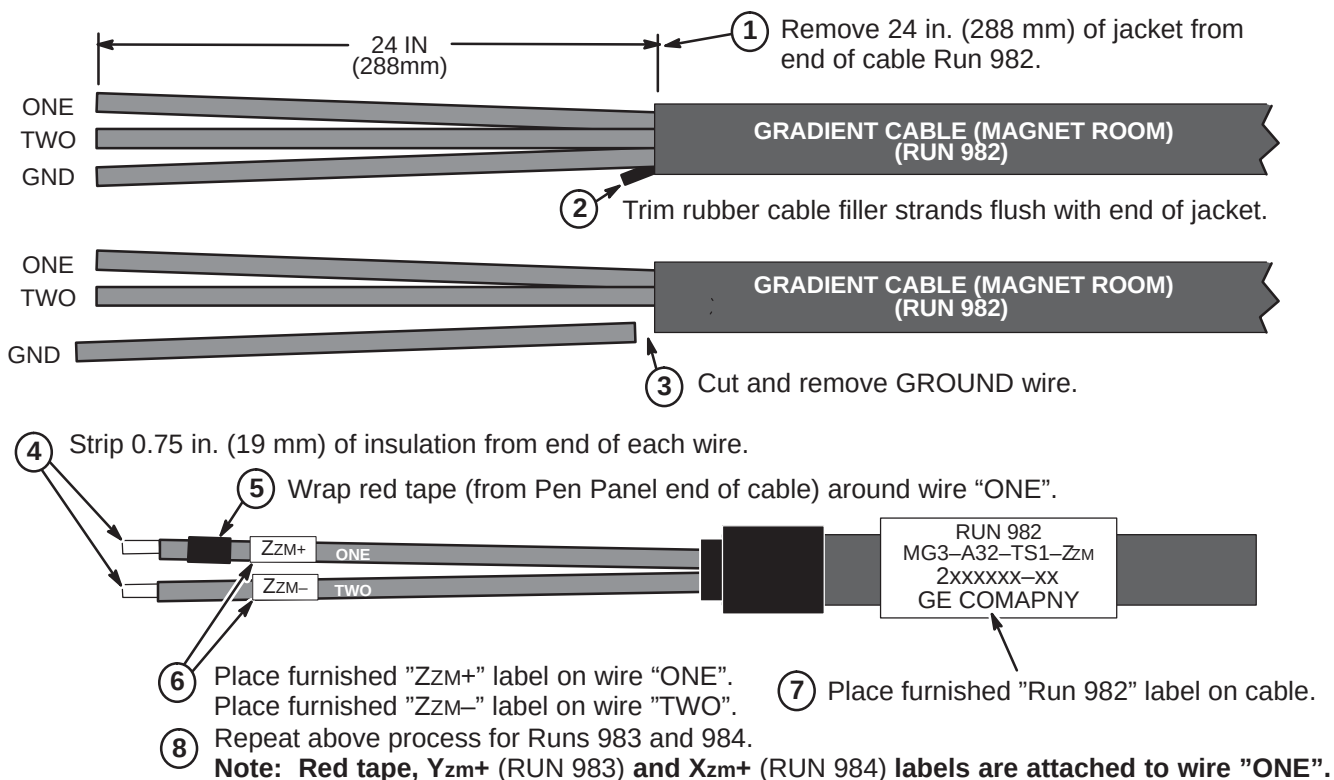
- ① Route Runs 985, 986, and 987 to left side of gradient filter on Penetration Panel.
- ② Connect six WB- and WB+ wires to gradient filter terminals as marked.
Note: Red tape indicates positive (+) wires.
- ③ Align cables to obtain maximum spacing from other cables and conductive metal objects.
Note: When tightening nuts, it is important to hold screw body.
- ④ Tighten nuts to a torque of 35 ft-lbs (67-74Nm) or after tightening, verify that nothing is loose.
Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
- ⑤ Secure GND lead wire on Ground Stud for each cable Run.
- ⑥ Route and secure each cable Run to strain-relief bracket with U-bolts.
Note: Tighten U-bolts to strain-relief cables only. Do not over tighten; cables may become damaged.
- ⑦ Replace gradient filter terminal covers.

EQUIPMENT ROOM MAGNET ROOM



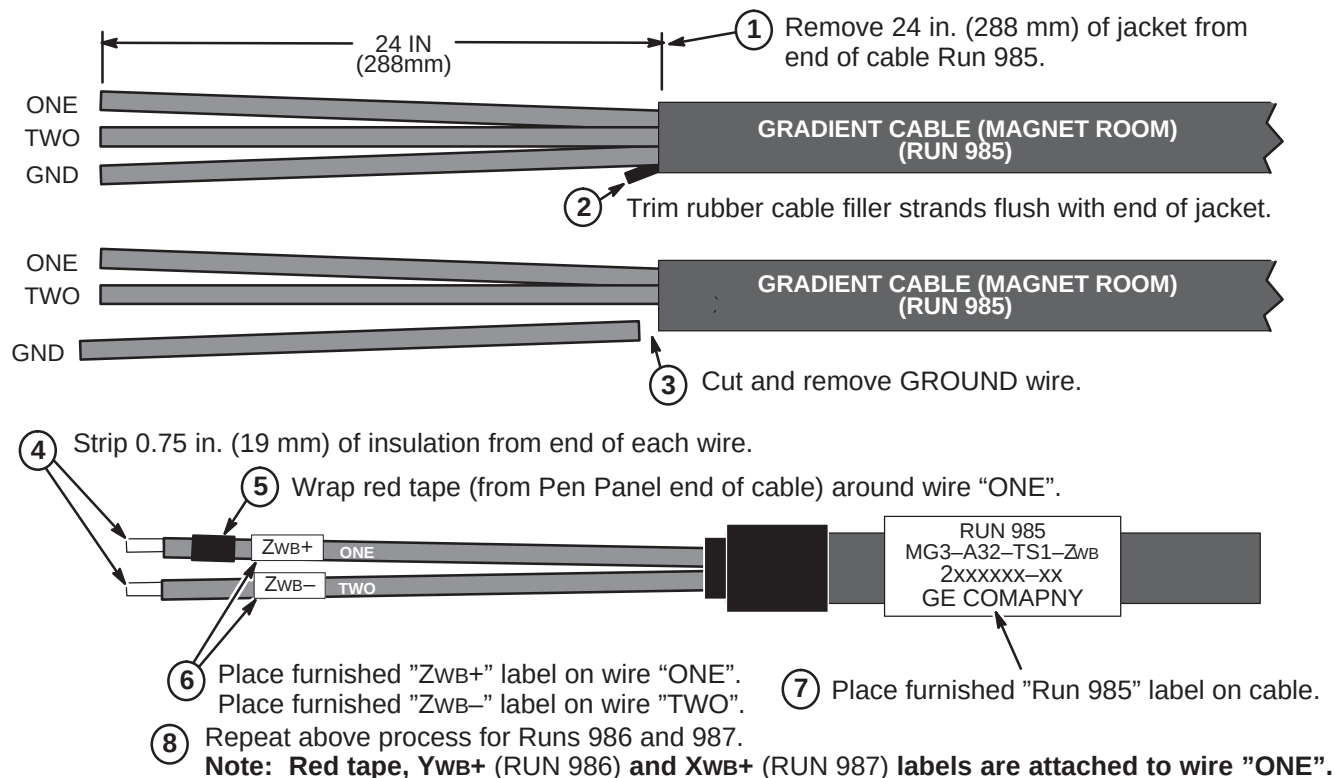
4-10-5-N Route Gradient Cables to Rear of Magnet**4-10-5-O Terminate Runs 982, 983, And 984 for Magnet Room**

Note: The following steps are required to terminate the end of the cable that was cut in Section 4-10-5-N. The other end of the cable is connected to the Penetration Panel.



4-10-5-P Terminate Runs 985, 986, And 987 for Magnet Room

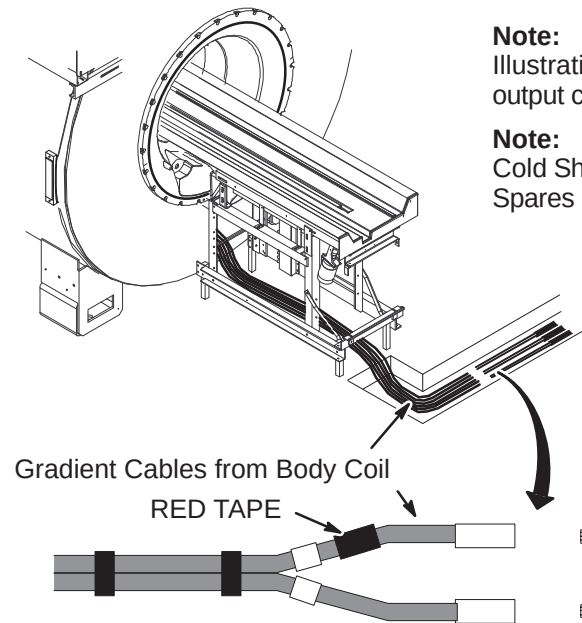
Note: The following steps are required to terminate the end of the cable that was cut in Section 4-10-5-N. The other end of the cable is connected to the Penetration Panel.

**WARNING!**

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

4-10-5-Q Connect Magnet Room Gradient Cables



Note:

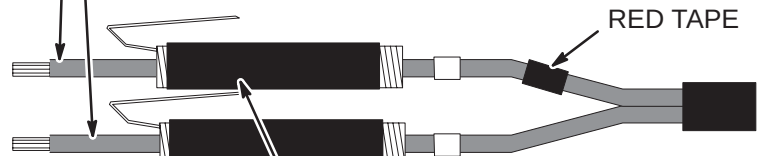
Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

Note:

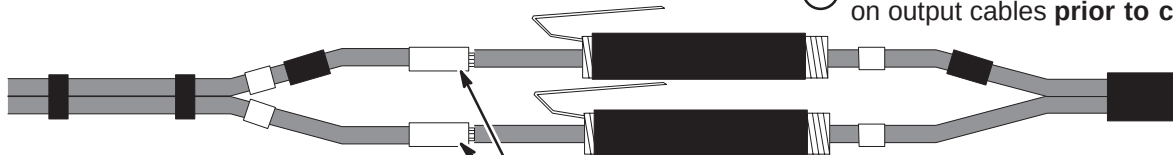
Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet

Gradient Output Cables from Penetration Panel

- ① Cable connection process shown below is for Run 987. Repeat process for Runs 982, 983, 984, 985 and 986.
Note: Red tape, XWB+ (RUN 987), YWB+ (RUN 986), ZWB+ (RUN 985), XzM+ (RUN 984), YzM+ (RUN 983), and ZzM+ (RUN 982) labels are attached to wire "ONE".

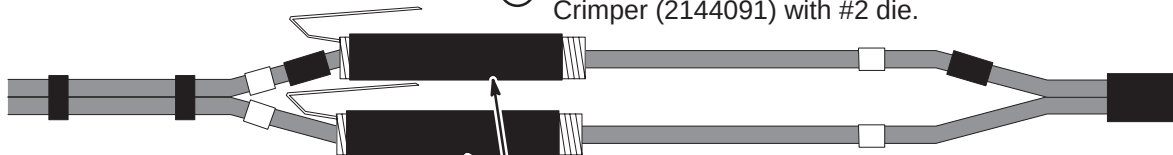


- ② Position Cold Shrink tubing (2218148) on output cables **prior to connecting**.

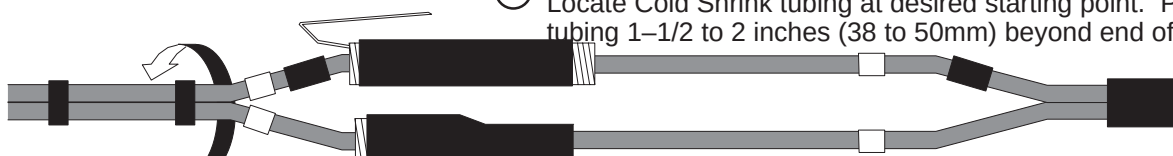


- ③ Insert ends of Run 987 into correct butt splice (2218149) on Body Coil gradient cable.

- ④ Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die.



- ⑤ Locate Cold Shrink tubing at desired starting point. Position tubing 1-1/2 to 2 inches (38 to 50mm) beyond end of connector.



- ⑥ Pull loose tab gently while unwinding in a counter-clockwise direction. Repeat process for other cable.



- ⑦ Connection of cables and installation of Cold Shrink tubing is now complete.

4-10-6 TIMES MICROWAVE RF CABLES INSTALLATION

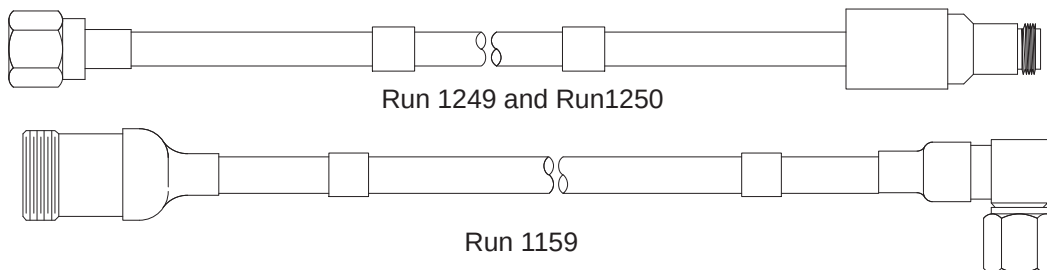
4-10-6-A Cable Information

Five cables, three extension cables, and other associated parts to be installed are provided for this procedure. Also provided are two Times Microwave stripping tools for LMR 1200 (2384858) and the LMR 600 (2352193), and the Times Microwave cutting tool (5111565).

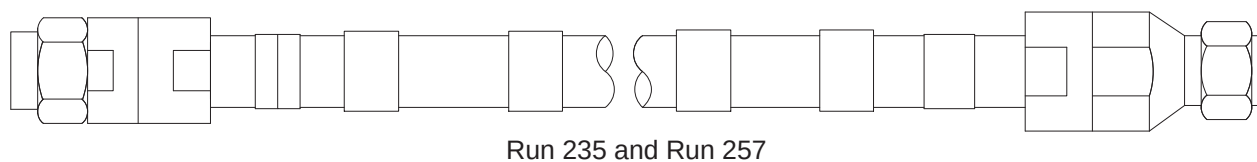


Do not subject LMR-600FR cable (Head) to a bend radius less than 3 inches (75mm). Do not subject LMR-1200FR cable (Body) to a bend radius less than 6.5 inches (165mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

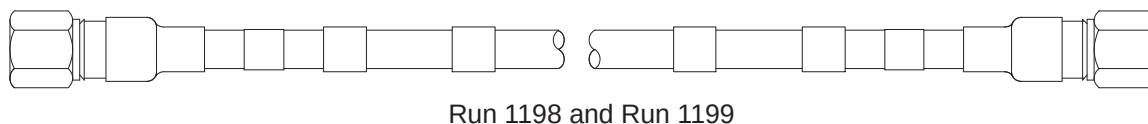
Two extension cables, 5116213-2, Run 1249 (Head) and 5121269, Run 1159 (Body) are connected to the RF Amplifier module in the RF/System Cabinet. The other extension cable, Run 1250, 5121245, is connected to the Body Hybrid Splitter in the Rear Pedestal.



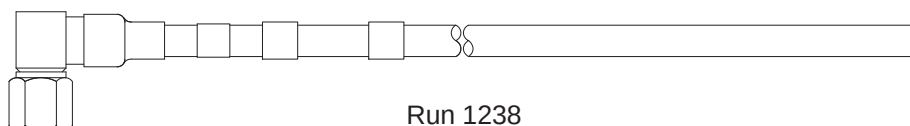
Cable 2316584-2 (Body, Runs 235 & 257), will be cut to length to make two cables and terminated for routing in the equipment and scan rooms.



Cable 5115915 (Head, Runs 1198 & 1199), will be cut to length to make two cables and terminated for routing in the scan room.



Cable 5115916 (Head, Run 1238), will be cut to length and terminated for routing in the equipment room.

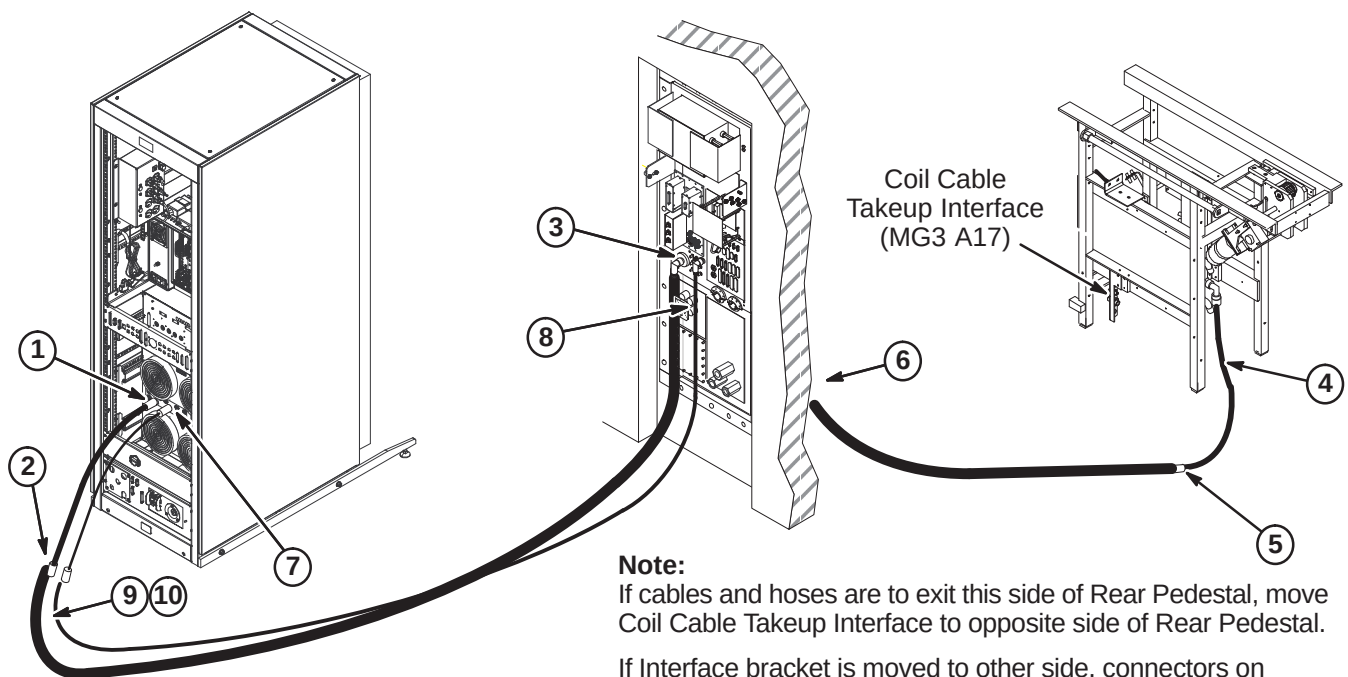


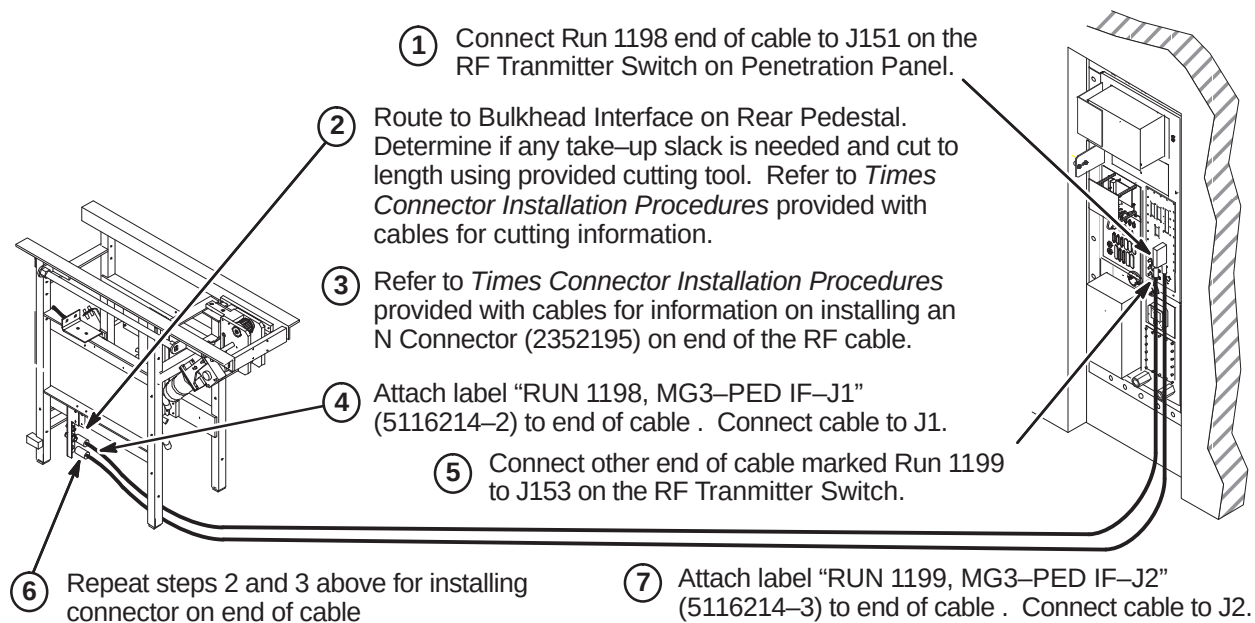
The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

4-10-6-B Install Runs 235, 257, 1238, 1249, 1250, and 1259

RF cables will be routed and connected as indicated in the following steps.

1. Connect extension cable Run 1259 to J4 Body Output at rear of RF Amp in RFS cabinet.
2. Connect end of Times RF Body cable marked "(RUN 235) FJ5" to Run 1259, Body extension cable. Attach label "RUN 235, Connect to Run 1259" (5121244) over existing label.
3. Route cable to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4-10-6-D** for termination procedure of cut end.
4. Connect Run 1250 to J1 on Hybrid Splitter.
5. Move remainder of RF Body cable to Scan Room and connect other end of cable marked "(RUN 257) MG3A3-J1" to Run 1250 at Rear Pedestal as shown below. Attach label "RUN 257, Connect to Run 1250" (5121244-2) over existing label.
6. Route remainder of cable to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4-10-6-D** for termination instructions of cut end.
7. Connect extension cable Run 1249 to J3 Head Output at rear of RF Amp IN RFS cabinet.
8. Connect end of Times RF Head cable marked "(RUN 1238) PP1-J4" to J4 at Penetration Panel. Route to RF Head extension cable at rear of SRF cabinet. Determine if any take-up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information.
9. Attach label "RUN 1238, CONNECT TO RUN 1249" (5116214) to end of cable after termination.
10. Connect cable to Run 1249.



4-10-6-C Install Runs 1198 and 1199

4-10-6-D Terminate and Connect RF Body Cables at Penetration Panel

Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2384859) on each end of the RF cables. Use provided cutting tool (5111565) and the LMR900 Stripping Tool (2384858).

Note

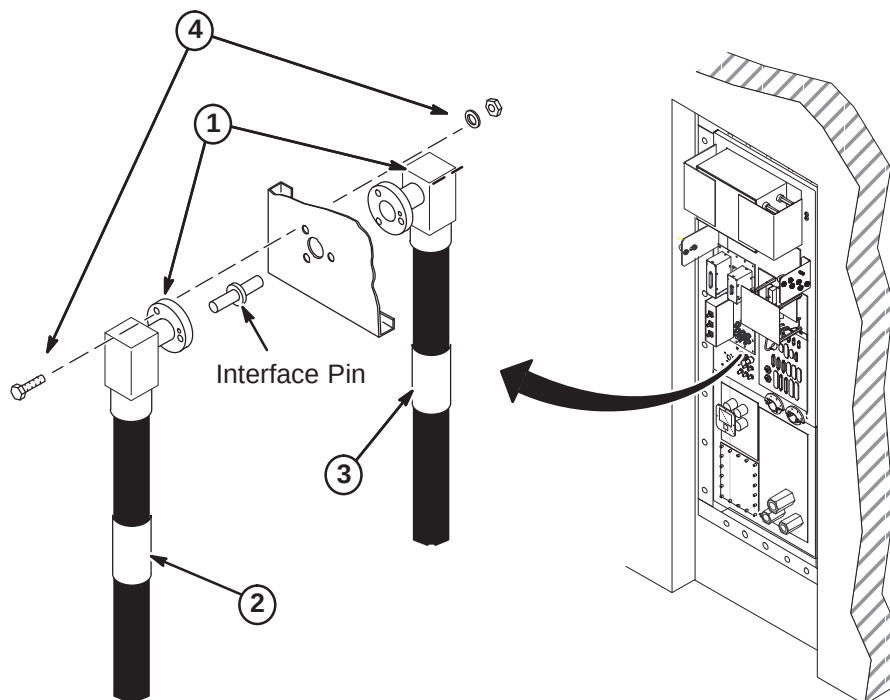
Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

2. Attach label "RUN 235, PP1-J44" (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label "RUN 257, PP1-J44" (2357230-2) to cable as shown on Scan room side of Penetration Panel.

CAUTION

Do NOT install O-Ring or Grease that may be provided with Cable Termination Kit. Installation of these parts will prevent the ground from making contact resulting in RF leaks.

4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
 - 1/4-20 x 1.25 Hex Head Bolt (brass)
 - 1/4 Screw Size Washer (phos bronze)
 - 1/4-20, 7/16" Hex x 3/16" thk Nut (brass)



4-10-7 PNEUMATIC PATIENT ALERT INSTALLATION

For most existing Horizon and LX configurations, the Patient Alert system has been installed prior to the eXCITE HD upgrade. For these upgrades, route and connect existing tubes and associated parts as needed to complete system.

For new installation of Patient Alert system, the customer and users should be involved in the following decisions:

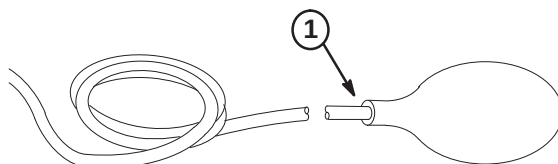
- Routing of Pneumatic Tubing
- Location of Control Box



Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

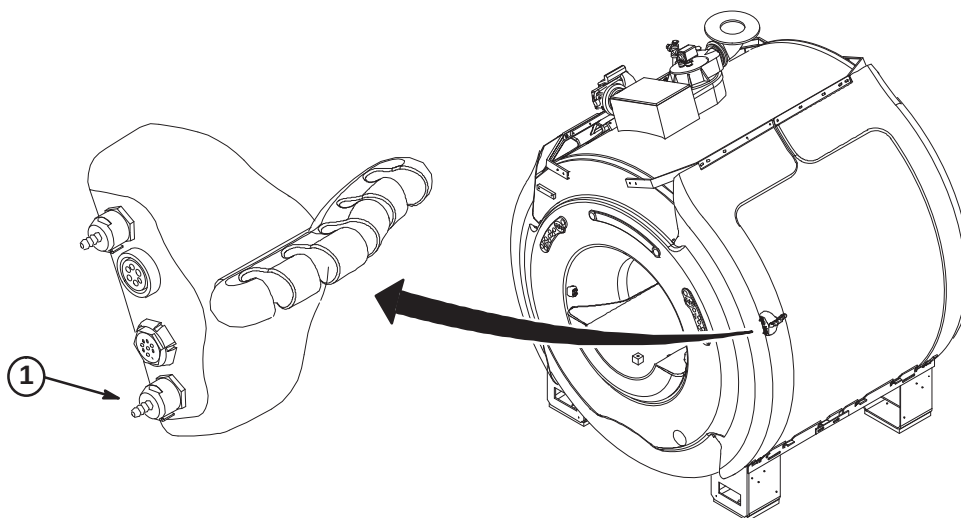
4-10-7-A SQUEEZE BULB INSTALLATION

1. Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
2. Before cutting pneumatic tubing from roll to length required, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the "CALL" connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.



4-10-7-B CONNECT TUBE TO REMOTE PAC INTERFACE ASSEMBLY

1. Connect end of tubing to "CALL" connection on Remote PAC Interface Assembly.



4-10-7-C MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION**CAUTION**

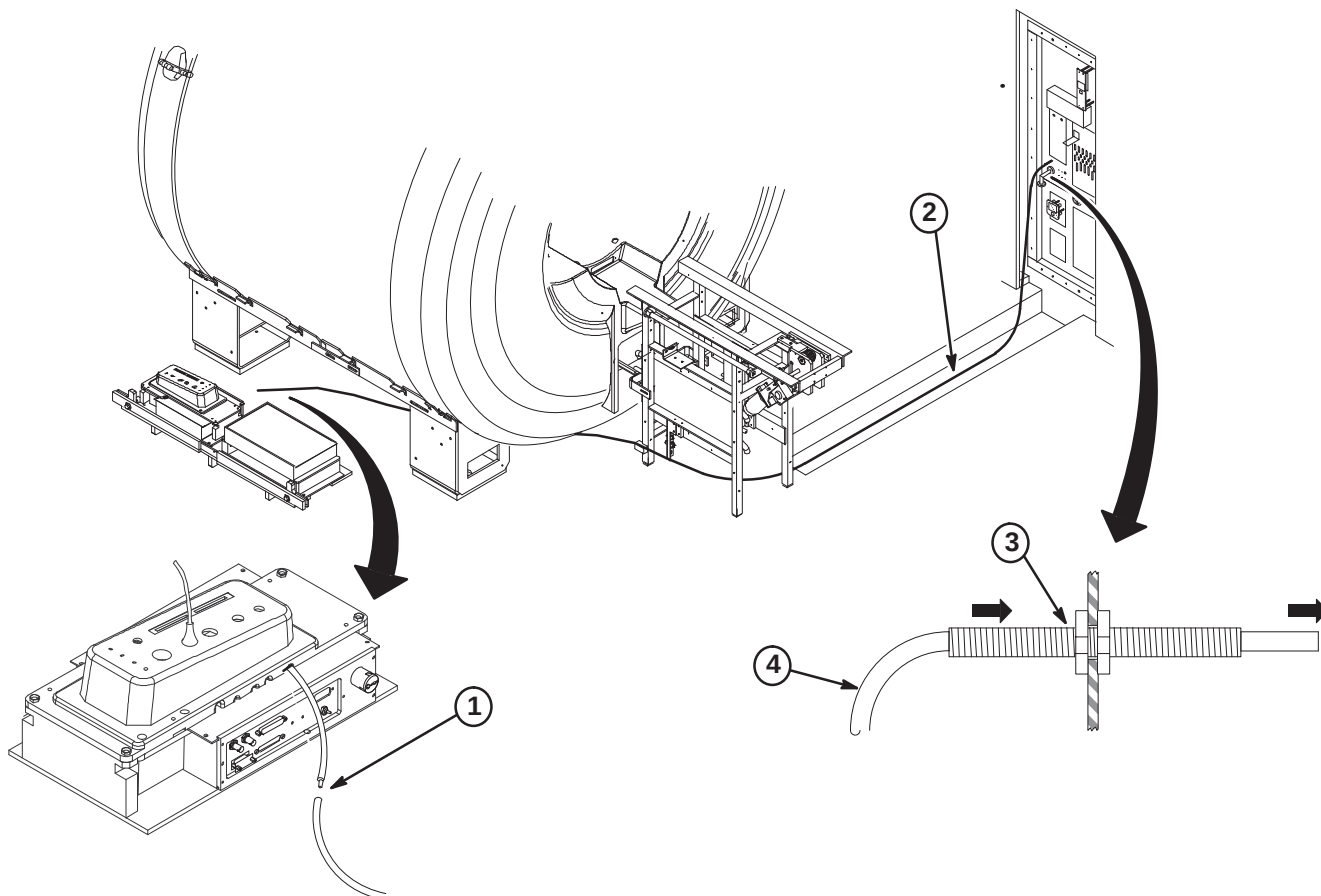
Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

1. Push end of pneumatic tubing onto PAC-II tubing connector.

Note: If the pneumatic tubing is to be pulled thru fiber-optic conduit, coat with baby powder, not grease, to allow easier pulling.

2. Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.
3. Insert Wave Guide thru "ALERT AIR LINE" hole and fasten to Penetration Panel with nut on each side.
4. Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.



4-10-7-D DETERMINING NEED OF EXTENDER BOX

If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator's Console), Extender Kit, 46-317758P2, must be ordered.

- **If the pneumatic tubing is long enough to reach control box:**
The extender is not needed. Proceed to Procedure 4-10-7-F, CONTROL BOX INSTALLATION.
- **If the pneumatic tubing is not long enough to reach control box:**
The extender is needed. Proceed to Procedure 4-10-7-E, INSTALLATION OF EXTENDER KIT.

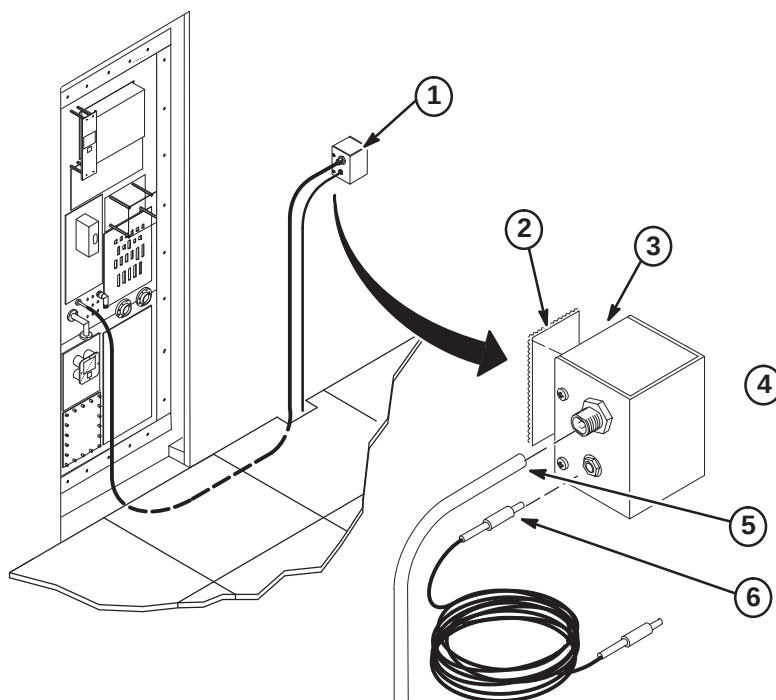
4-10-7-E INSTALLATION OF EXTENDER KIT (46-317758P2)

CAUTION

Do not mount Extender Box in Magnet Room because the box contains ferrous parts.

Install Extender Box

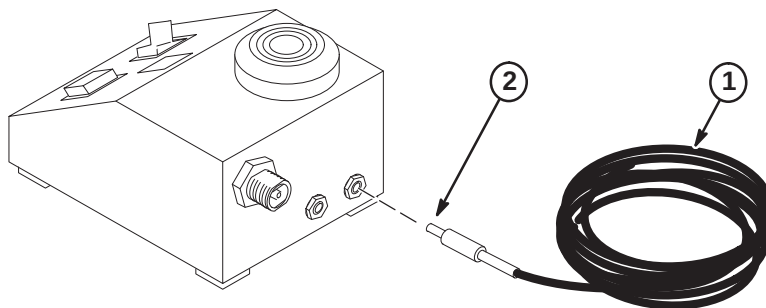
1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed-thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.



4-10-7-E Installation of Extender Kit (46-317758P2) (Continued)

Connection of Extender Wire to Control Box

1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.



4-10-7-F CONTROL BOX INSTALLATION OPTIONS

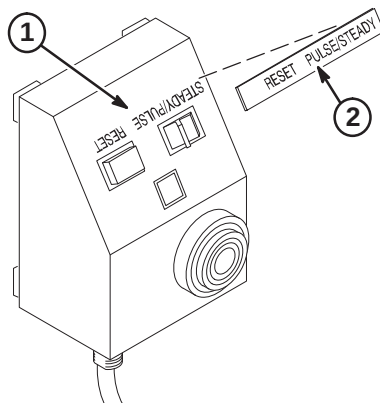
Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

- Mount control box on a wall or other vertical surface per Procedure **4-10-7-G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Mount control box under shelf per Procedure **4-10-7-G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Proceed to Procedure **4-10-7-K**, FINAL CONNECTIONS TO CONTROL BOX.

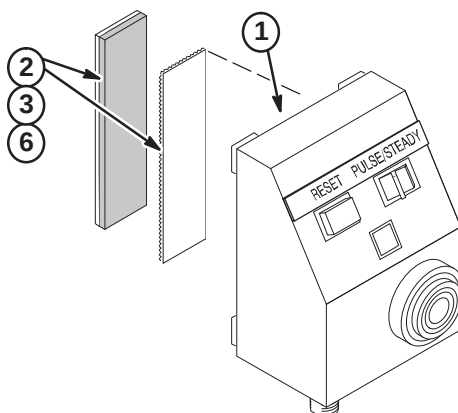
Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

4-10-7-G CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF

1. Clean Control Box surface.
2. Remove adhesive backing from metalized nameplate and apply over silk screened lettering as shown.
3. Select one of the following for installation of Control Box:
 - If control box will be mounted to WALL or SHELF with Velcro, go to Procedure **4-10-7-H**
 - If control box will be mounted to WALL with screws, go to Procedure **4-10-7-I**
 - If control box will be mounted to SHELF with screws, go to Procedure **4-10-7-J**

**4-10-7-H Control Box Mounted With Velcro**

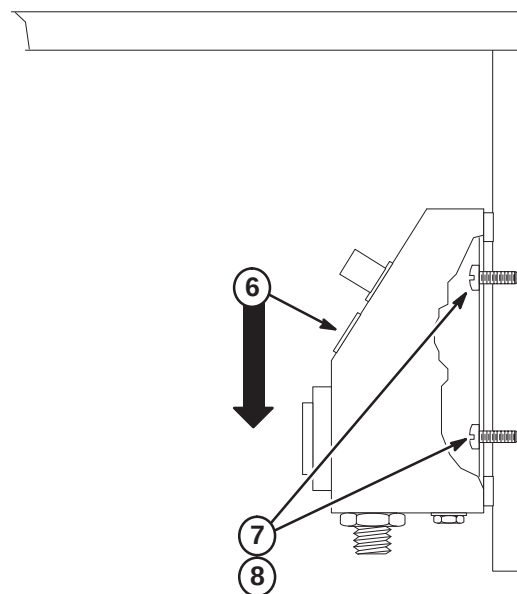
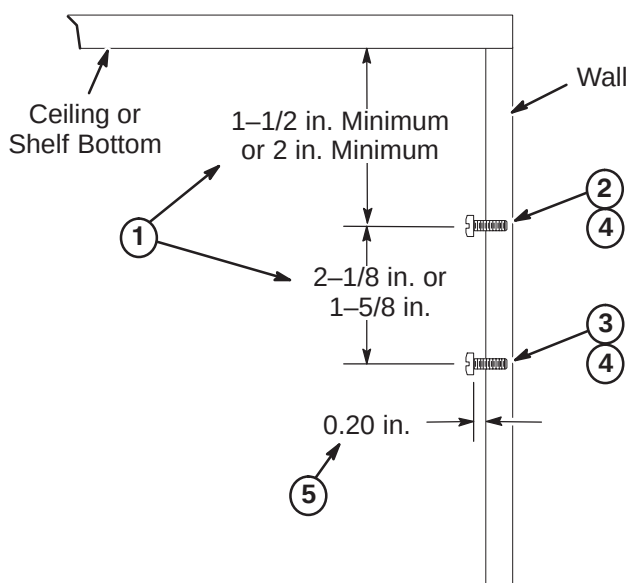
1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. Mount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.
8. Proceed to Procedure **4-10-7-K**, FINAL CONNECTIONS TO CONTROL BOX.



4-10-7-I Installing Control Box with Mounting Screws on Vertical Surface

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

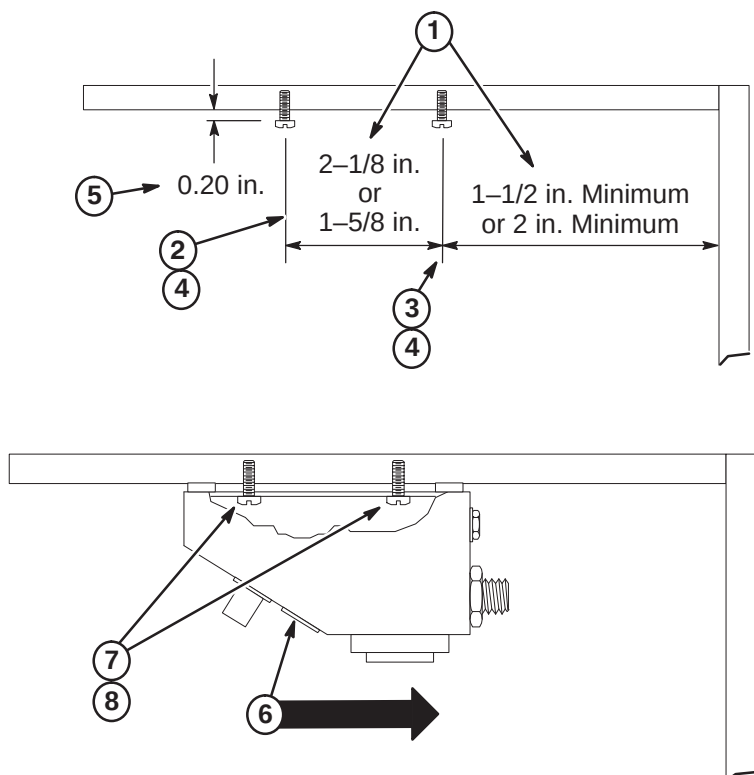
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark top mounting hole position on wall.
3. Mark bottom mounting hole position on wall.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4-10-7-K, FINAL CONNECTIONS TO CONTROL BOX**.



4-10-7-J Installing Control Box with Mounting Screws Under Shelf

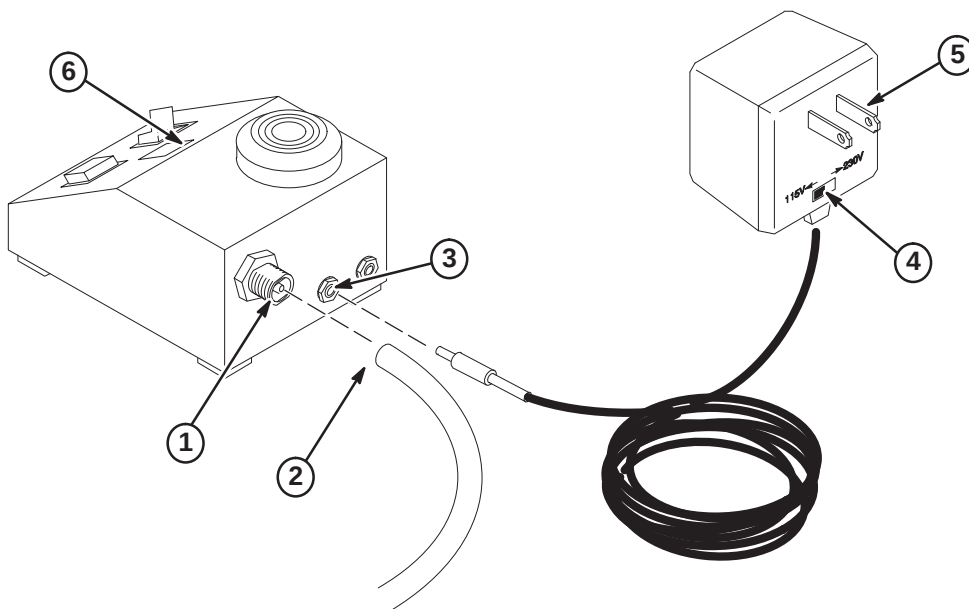
Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark front mounting hole position on underside of shelf.
3. Mark rear mounting hole position on underside of shelf.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push inward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4-10-7-K, FINAL CONNECTIONS TO CONTROL BOX**.



4-10-7-K FINAL CONNECTIONS TO CONTROL BOX

1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection.
2. Cut tubing and push fully onto feed-thru connector on back of control box.
3. Plug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is on.



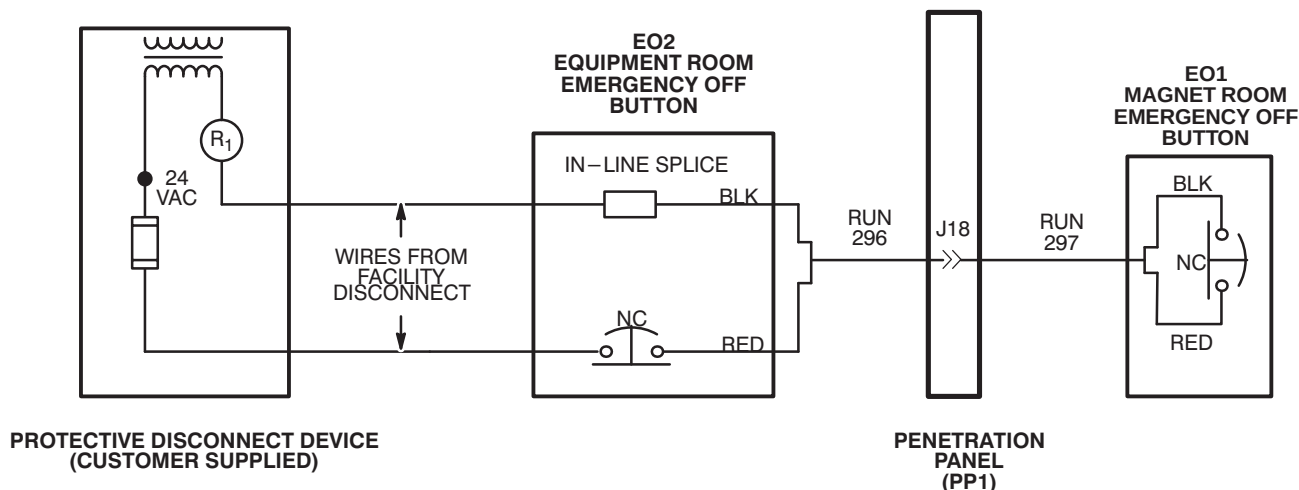
4-10-8 EMERGENCY OFF CONNECTIONS

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to *Direction 2355598, Signa EXCITE 3.0T Pre-Installation*, Section 5, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

Note: In Illustration below, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

2. At equipment room room "Emergency Off" (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2).
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2).
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push-on terminal.
5. Terminate black wire at end of Run 296 with local supplied "push-on" terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in-line splice.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1).

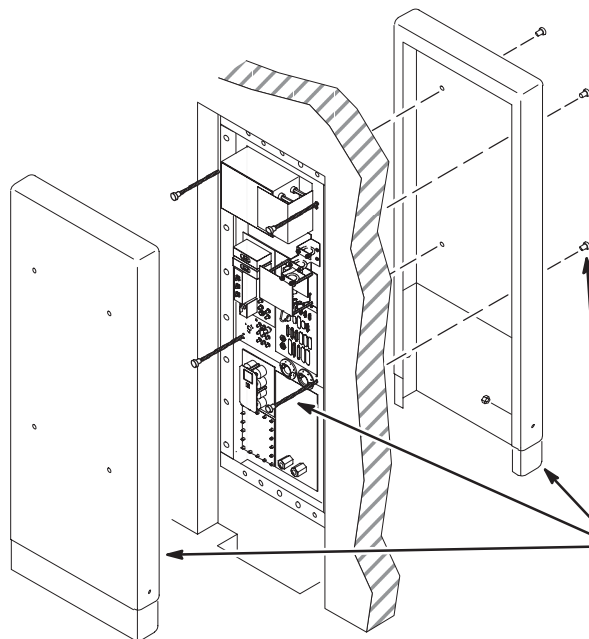


SECTION 5 – PENETRATION PANEL

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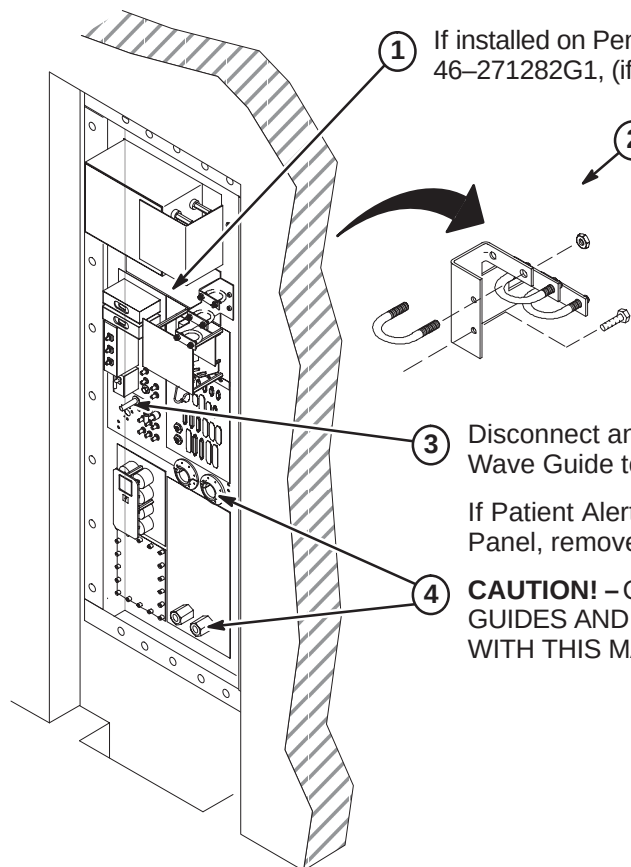
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5-1 REMOVE PENETRATION PANEL COVERS



- ① If installed, unfasten and save Penetration Panel covers, stand-offs, and mounting hardware.
- ② Move above items to a safe storage area. They will be reinstalled at end of upgrade.

5-2 REMOVE PARTS FOR TRANSFER TO NEW PENETRATION PANEL



- ① If installed on Penetration Panel, transfer the Shim Filter, only if it's a 46-271282G1, (if not this model, order one) to new Penetration Panel.
- ② Remove and transfer to new HDx Pen Panel the magnet room side gradient cable bracket (2133758), U-bolts (2113244), and Brass Screws (46-208560P82) if site being upgraded has #1/0 gradient cables in the magnet room which are being retained for this upgrade.
Note: The U-Clamps are larger than the clamps on the new Pen Panel which are designed for #2 size gradient cables.
- ③ Disconnect and remove Patient Alert tubing. Transfer Patient Alert Wave Guide to new Penetration Panel re-using mounting hardware.
If Patient Alert Extender Box is mounted on Penetration Panel, remove and re-install on new Penetration Panel.
- ④ **CAUTION!** – OLDER SYSTEMS USED BERYLIUM PASTE ON WAVE GUIDES AND GAS LINES. WASH HANDS IF YOU COME IN CONTACT WITH THIS MATERIAL.

5-3 DISCONNECT COOLANT HOSES

CAUTION

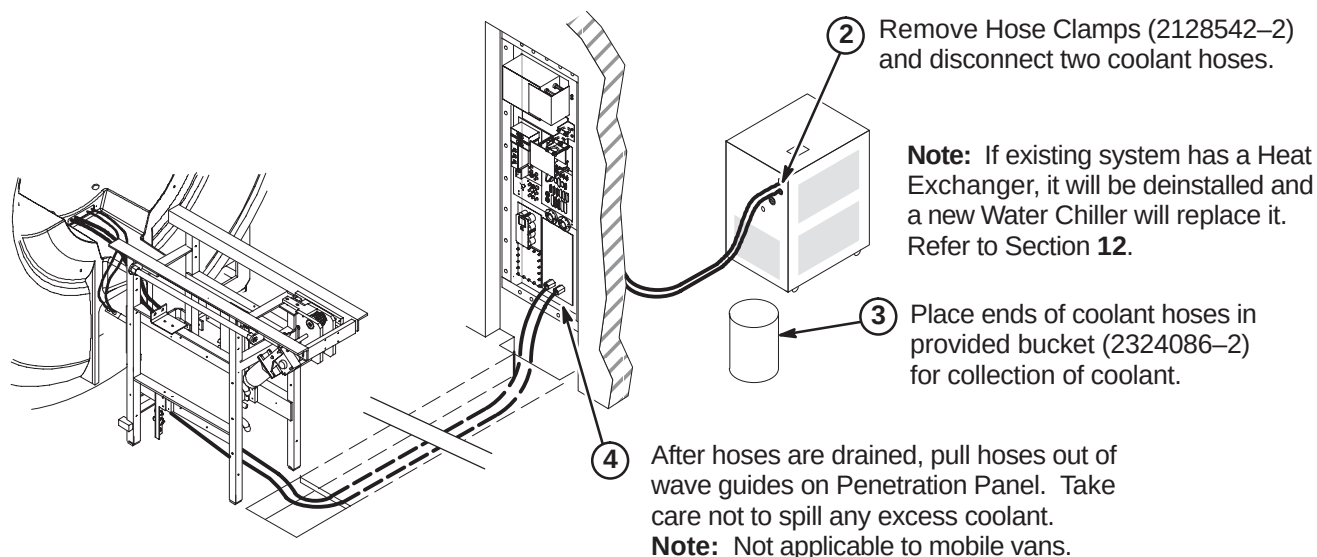
Coolant spillage possibility. The gradient coil cooling system is pressurized. Turn off the heat exchanger or water chiller before disconnecting the hoses. This takes pressure off the system and helps avoid coolant from being sprayed all over the immediate area.

WARNING!

POSSIBLE EYE INJURY! DRAINING OF COOLANT (ANTI-FREEZE) MAKES SPLASHING POSSIBLE. THIS COULD CAUSE EYE INJURY. BE SURE TO WEAR LATEX GLOVES AND SAFETY GLASSES.

Note: Four hands are better than two – While it is possible for one person to disconnect the coolant hoses, it is easier if two people do it. There would be less chance of spilling the coolant.

- ① Prior to disconnection of coolant hoses, calculate the amount of coolant that will be drained from system to determine if the 5 gallon bucket provided with upgrade will be sufficient for collection of coolant to be drained. Approximately two gallons of coolant will remain in gradient coil after disconnection. If more containers are needed for collection of drained coolant, they must be obtained locally.

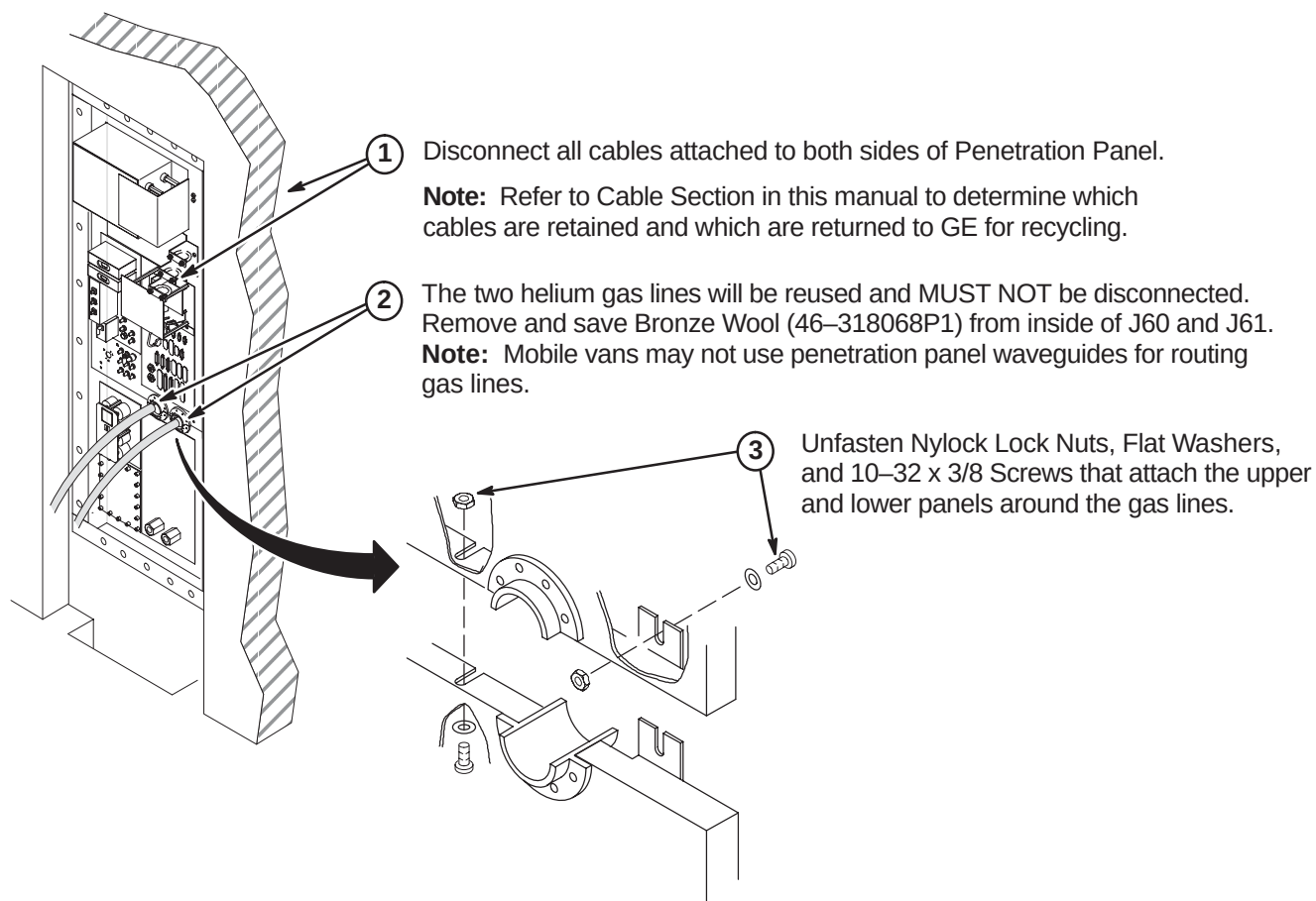


5-4 DISCONNECT PENETRATION PANEL CABLES AND UNFASTEN UPPER & LOWER PANELS**WARNING!****FERROUS MATERIAL HAZARD!**

THE MATERIAL USED TO SECURE THE PENETRATION PANEL TO THE THE SCREEN ROOM MAY BE MAGNETIC. FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

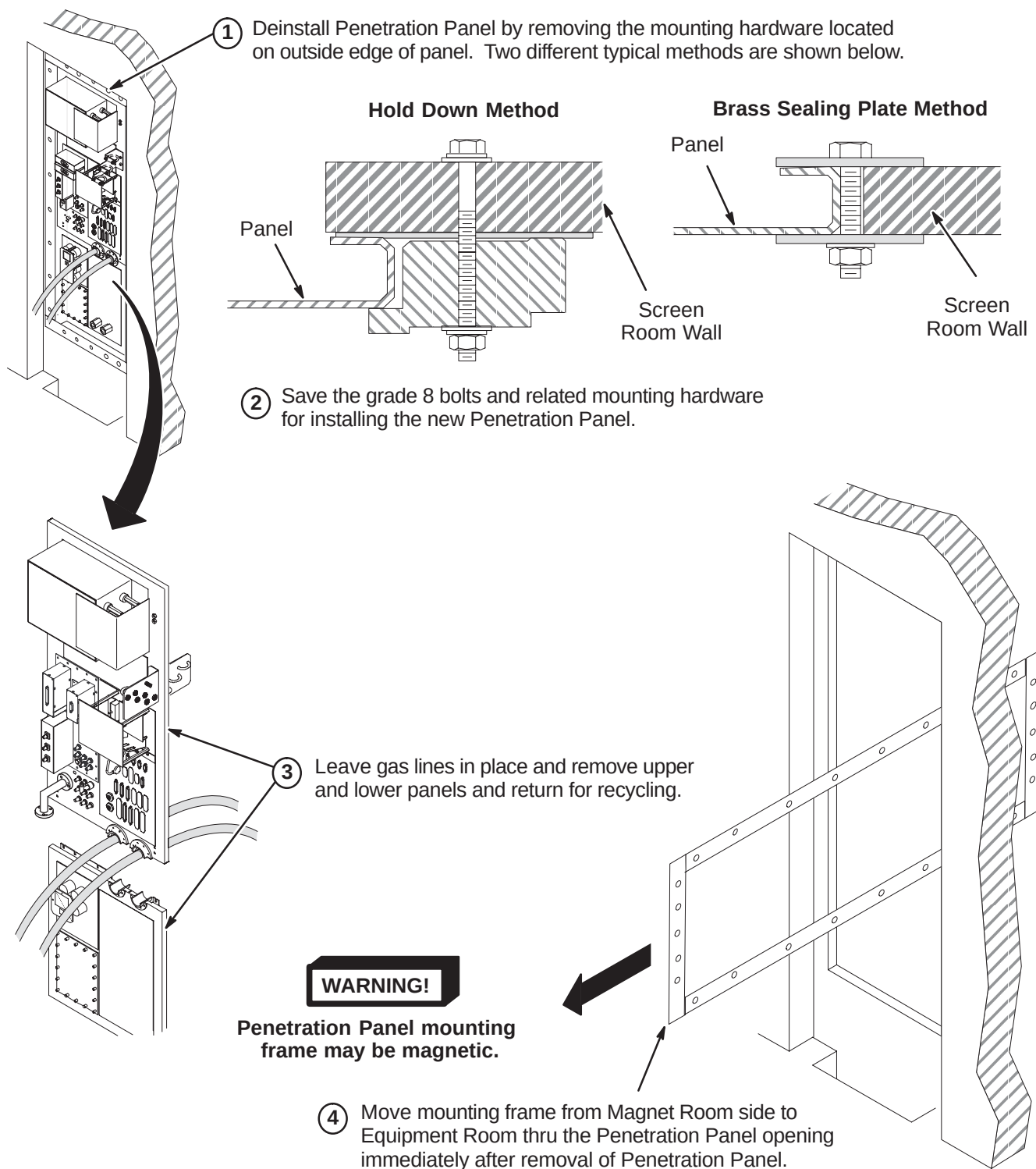
ALL MOUNTING MATERIALS OR RECTANGULAR FRAME THAT SECURES THE PENTRATION PANEL TO THE SCREEN ROOM ON THE MAGNET SIDE MUST BE PASSED FROM THE MAGNET ROOM TO THE EQUIPMENT ROOM THROUGH THE PENETRATION PANEL OPENING IMMEDIATELY AFTER REMOVAL OF PENETRATION PANEL.

REFER TO SECTION 5-5.

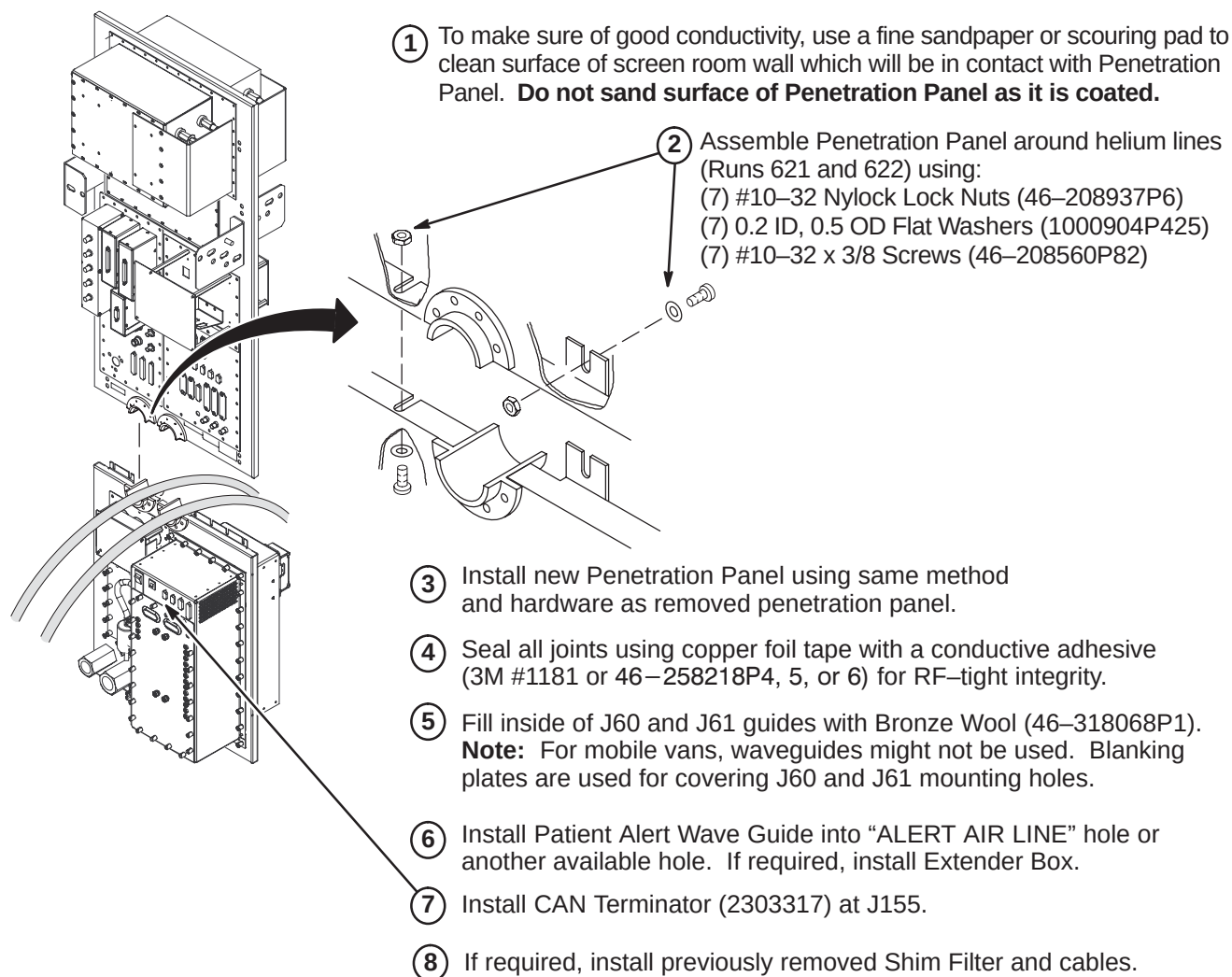


5-5 DE-INSTALL PENETRATION PANEL

At least two people are required to deinstall the Penetration Panel.



5-6 INSTALL EXCITE HD PENETRATION PANEL



5-7 CONNECT CABLES TO PENETRATION PANEL

For information and instructions on cable and hose connections at the Penetration Panel, refer to applicable:

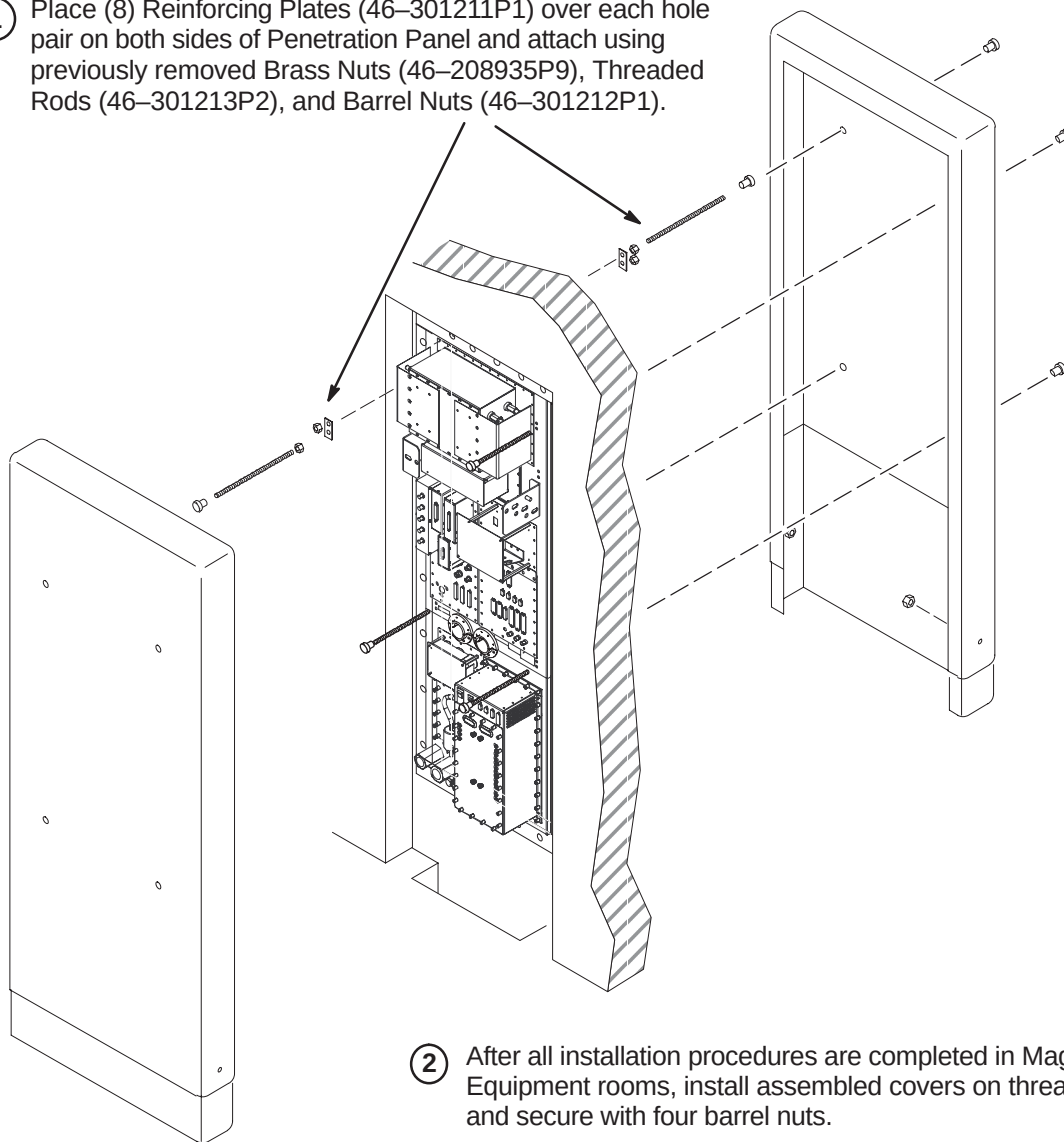
- Installation Flowchart in Section 2
- Coolant Hose installation in Section 12
- Cable Interconnect Map in Section 4
- Cable and System Installation in Section 7 .

Make sure to reinstall all customer added option connections.

5-8 INSTALL PENETRATION PANEL COVERS (If Previously Removed)**Note**

The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

- ① Place (8) Reinforcing Plates (46-301211P1) over each hole pair on both sides of Penetration Panel and attach using previously removed Brass Nuts (46-208935P9), Threaded Rods (46-301213P2), and Barrel Nuts (46-301212P1).

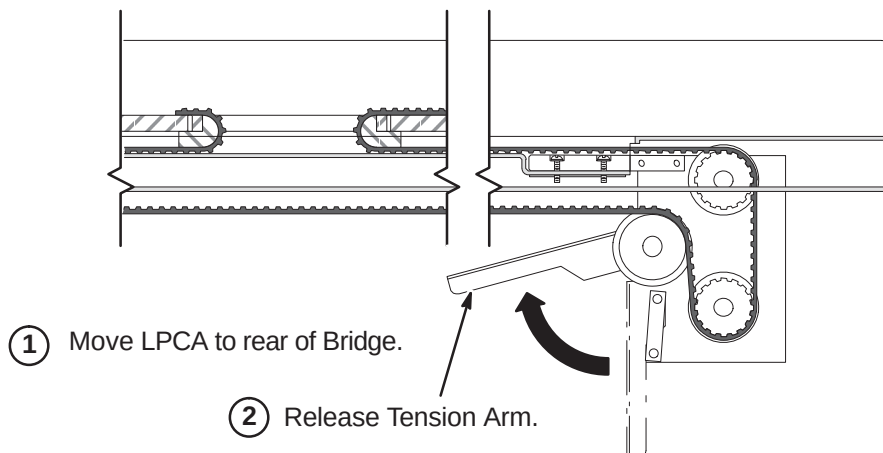
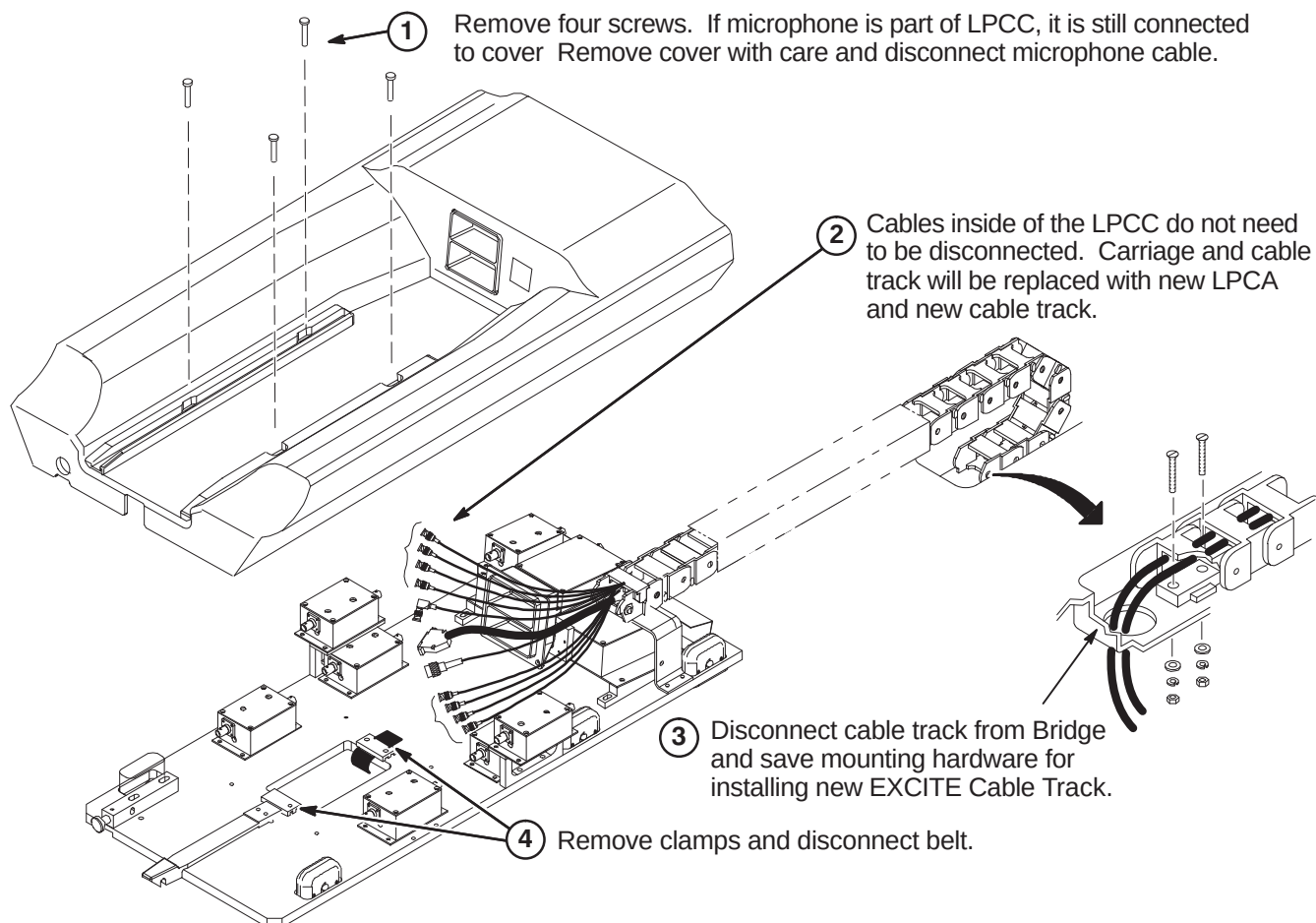


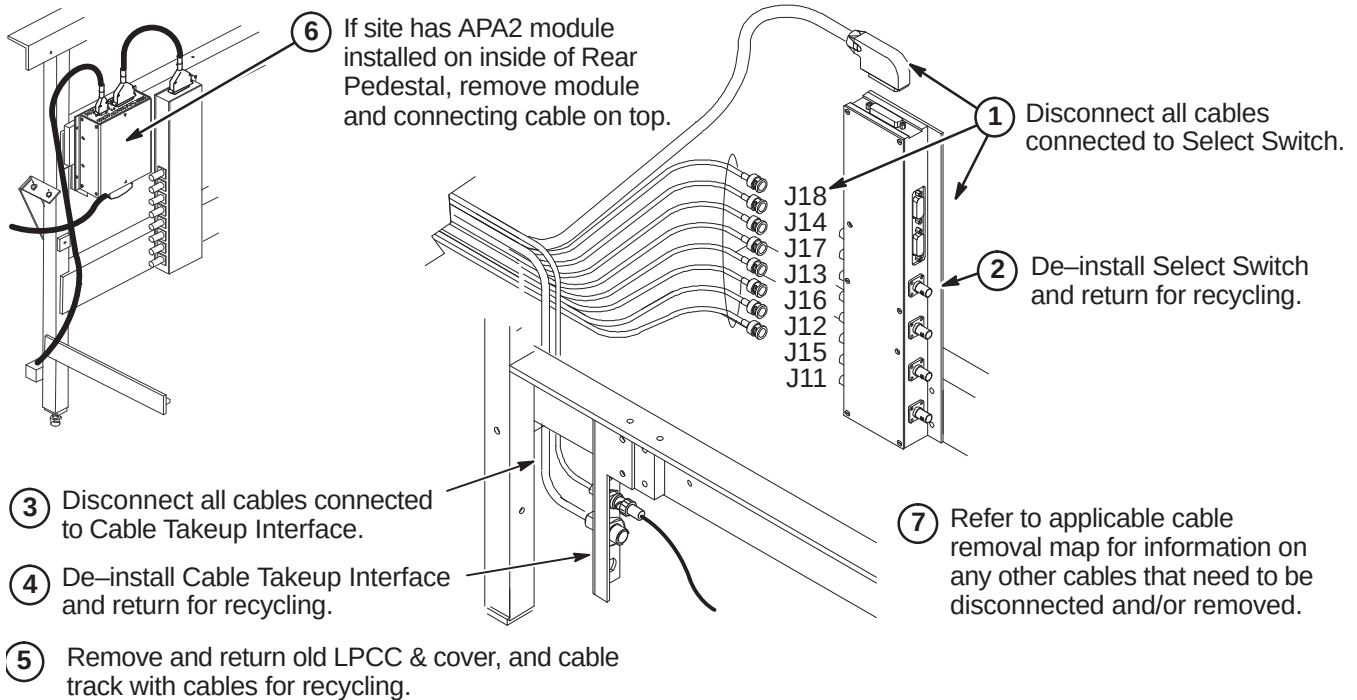
- ② After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

SECTION 6 – UPGRADE FROM WideOpen & TwinSpeed LX

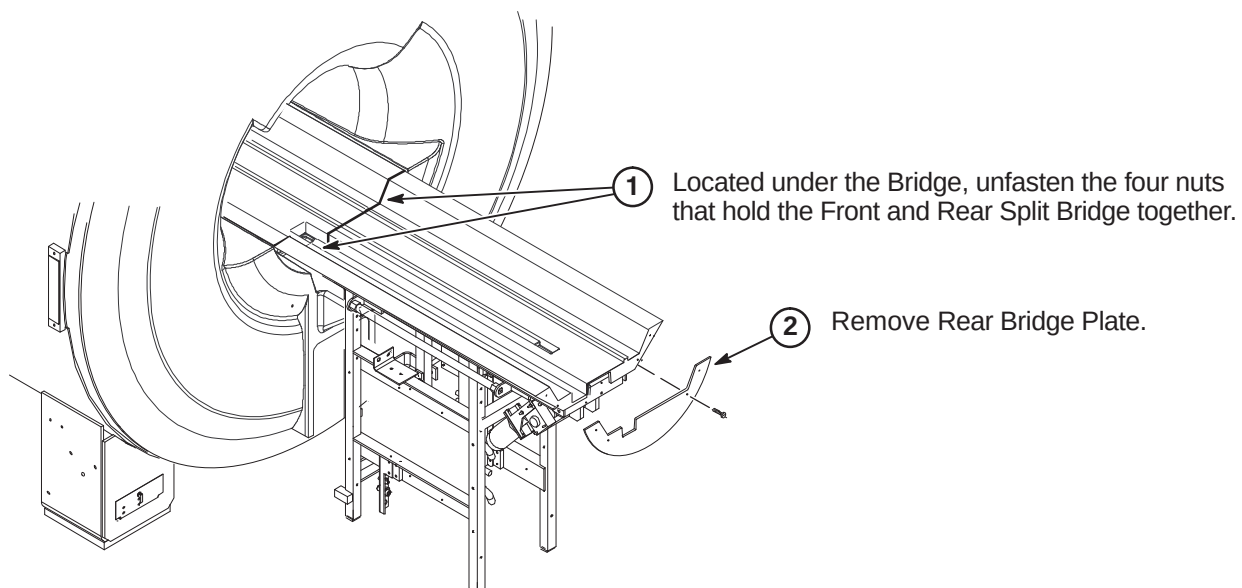
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6-1 DE-INSTALL COMPONENTS ON WideOpen OR WideOpen ENCLOSURE**6-1-1 DE-INSTALL EXISTING LPCA & CABLE TRACK****6-1-1-A Release Tension Arm****6-1-1-B Disconnect Low Profile Carriage Cover & Cable Track**

6-1-1-C Disconnect Cables In Rear Pedestal**6-1-2 DEINSTALL SPLIT BRIDGE FOR WideOpen**

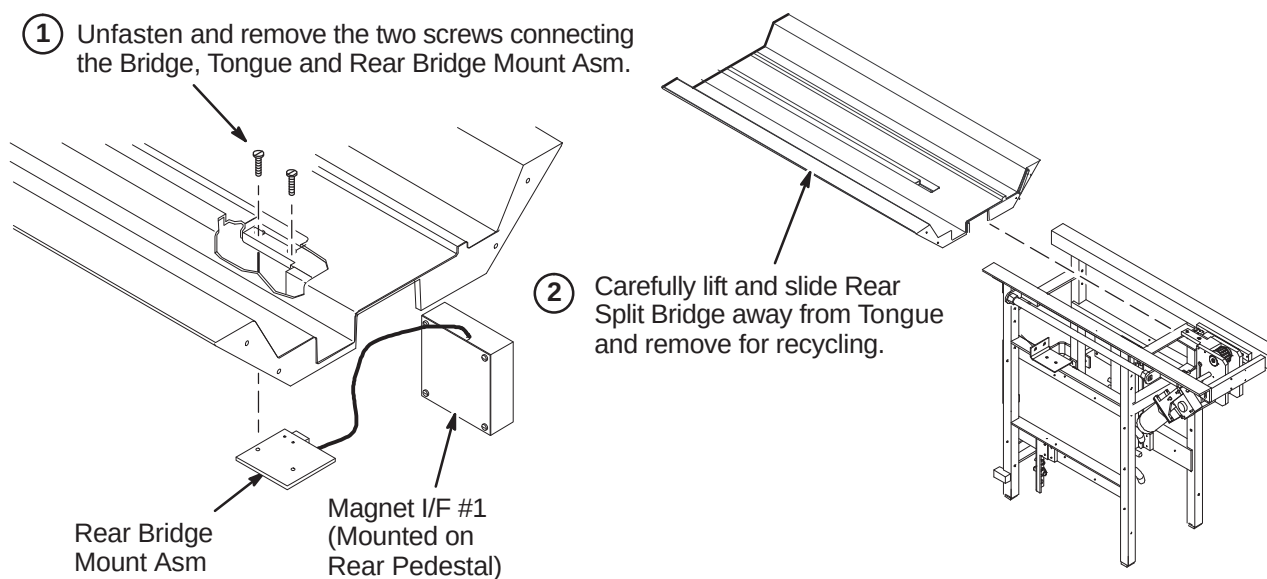
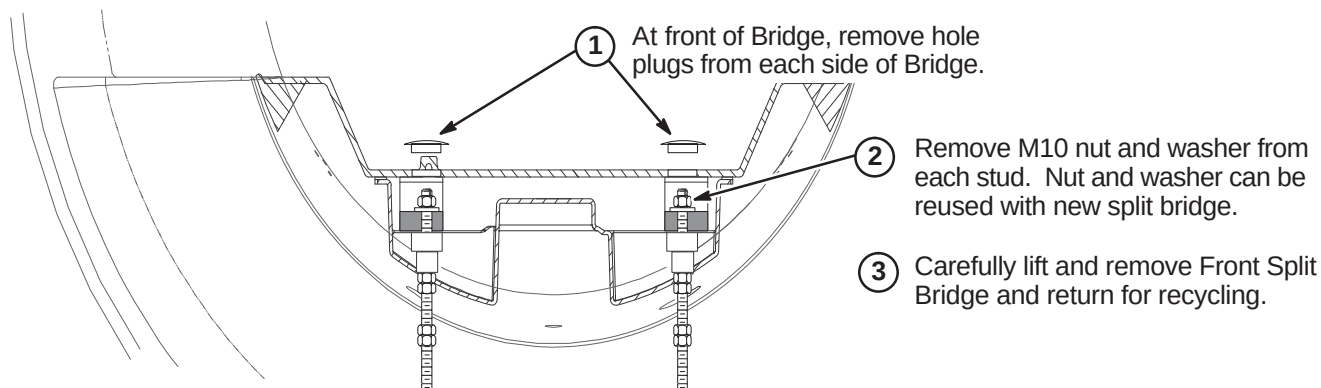
Upgrading to EXCITE HD requires a new Front and Rear Split Bridge if the site has an LCC magnet with a WideOpen or TwinSpeed enclosure. The following procedures are for removing either the original split bridge designed for a single cable track or the older style single piece long bridge. The WideOpen enclosure with a long bridge will also have the front end bell removed and replaced.

6-1-2-A Unfasten Rear Split Bridge

6-1-2-B Remove Rear Split Bridge**WARNING!****FERROUS MATERIAL HAZARD!**

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

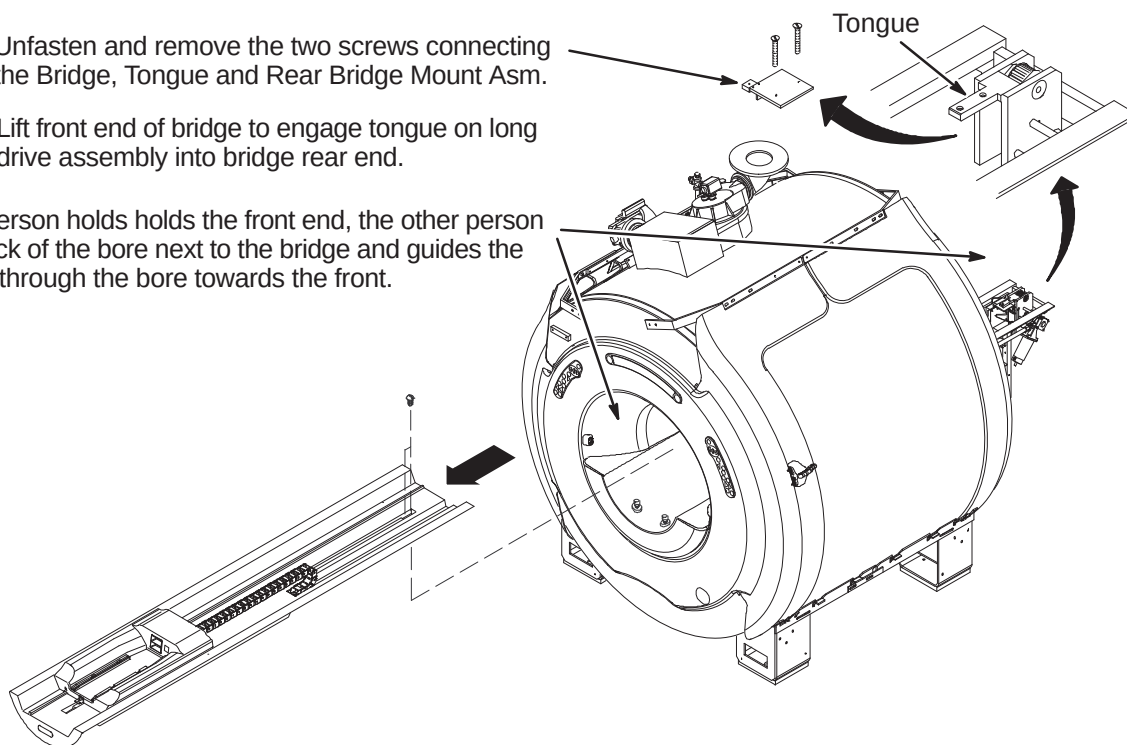
WHEN MOVING REAR PEDESTAL, USE PROPER SAFETY AND LIFTING TECHNIQUES.

**6-1-2-C Remove Front Split Bridge**

6-1-3 DE-INSTALL LONG BRIDGE FOR WideOpen

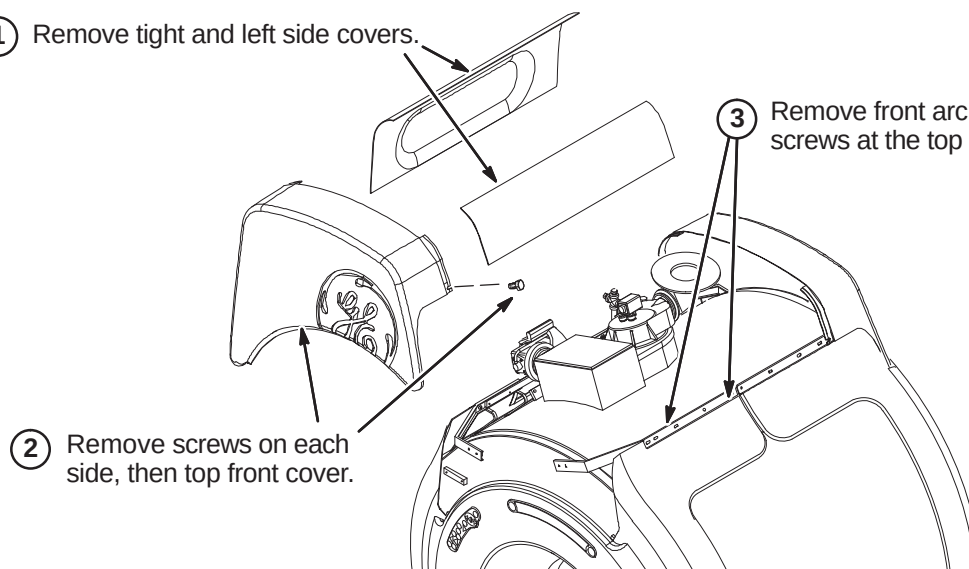
6-1-3-A Remove Long Bridge

- ① Unfasten and remove the two screws connecting the Bridge, Tongue and Rear Bridge Mount Asm.
- ② Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.
- ③ While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front.



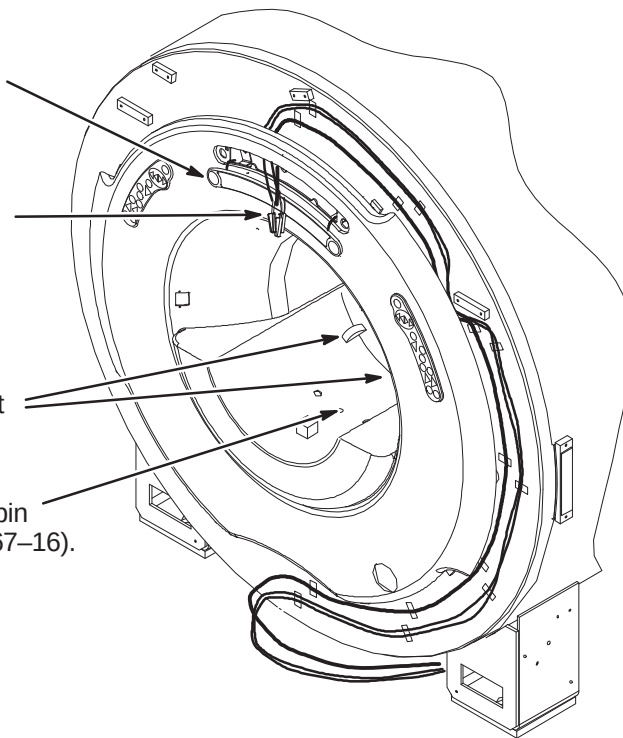
6-1-3-B Remove WideOpen Enclosure Covers

- ① Remove right and left side covers.
- ② Remove screws on each side, then top front cover.
- ③ Remove front arc covers by removing two screws at the top and sliding cover to the side.

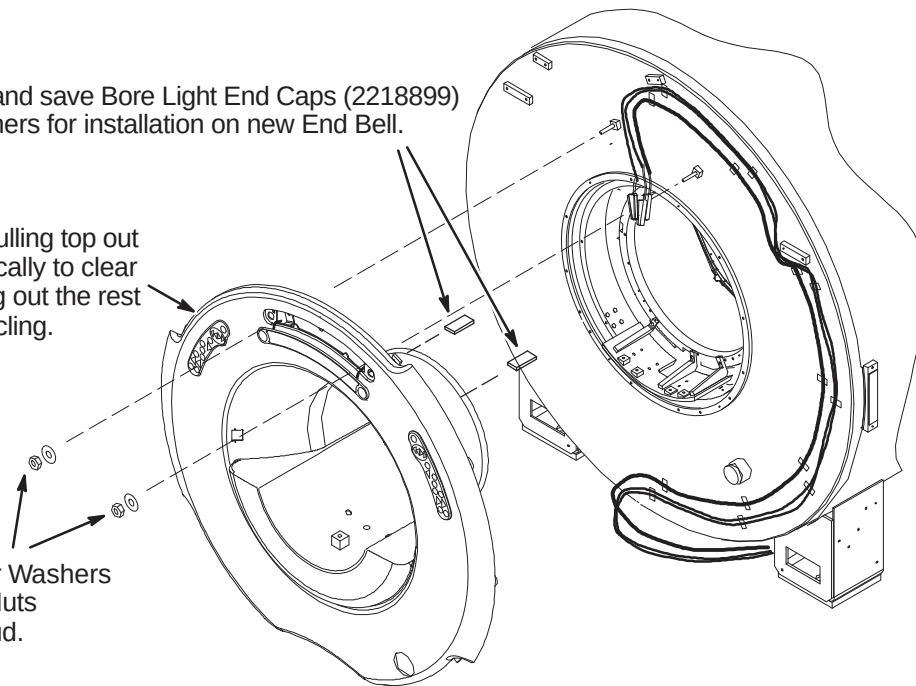


6-1-3-C Disconnect Operator Control Panel and Unfasten Front End Bell

- ① Insert 3mm allen wrench into hole in Control Panel and turn counter clockwise 1/2 turn.
- ② Carefully, using a small screwdriver, pry the Operator Display away from the front cover.
- ③ Disconnect the three cables attached to the Operator Display.
- ④ Lift open the two latches connecting the front end bell to the RF Coil.
- ⑤ Remove and discard the RF Tube Location pin and M6 x 30mm Hex Socket screw (2109867-16).

**6-1-3-D Remove Front End Bell**

- ③ Remove and save Bore Light End Caps (2218899) and fasteners for installation on new End Bell.
- ② Remove front end bell by pulling top out past studs, then lifting vertically to clear end bell inserts, then pulling out the rest of the way. Return for recycling.
- ① Unfasten and save Rubber Washers (2167321) and M10 Lock Nuts (2109877-3) from each stud.

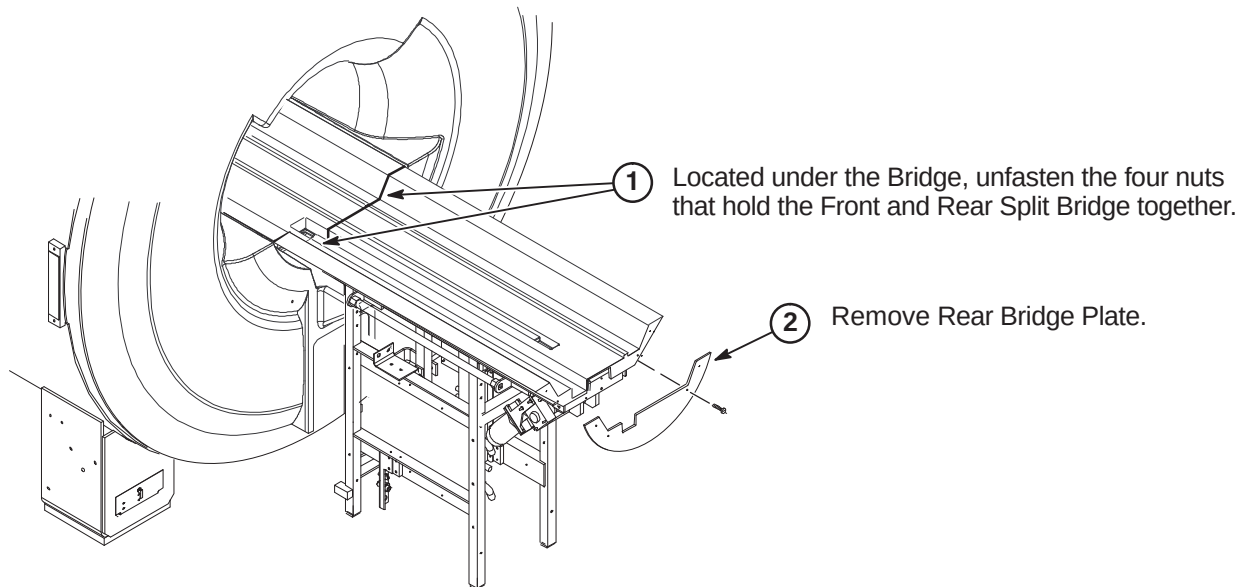


6-1-4 DE-INSTALL SPLIT BRIDGE FOR TwinSpeed

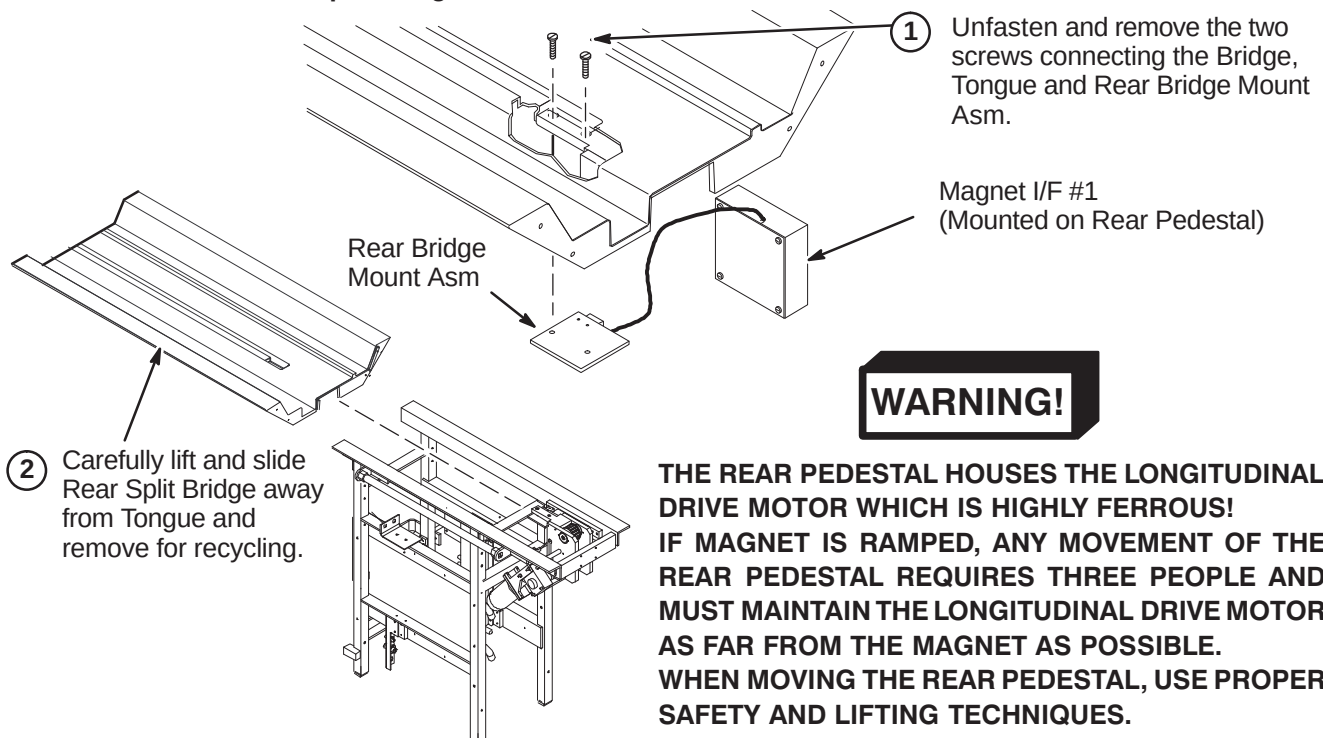
Upgrading to EXCITE HDTwinSpeed requires a new Front and Rear Split Bridge. The method of attachment at the front of the Bridge is different between a bridge that was installed for a forward production (new) installation and a Long bridge that was upgraded from a non-TwinSpeed WideOpen enclosure to TwinSpeed.

- Section 6-1-4 shows the deinstallation for a forward production Split Bridge.
- Section 6-1-5 shows the deinstallation of the upgraded Long Bridge.

6-1-4-A Unfasten Rear Split Bridge

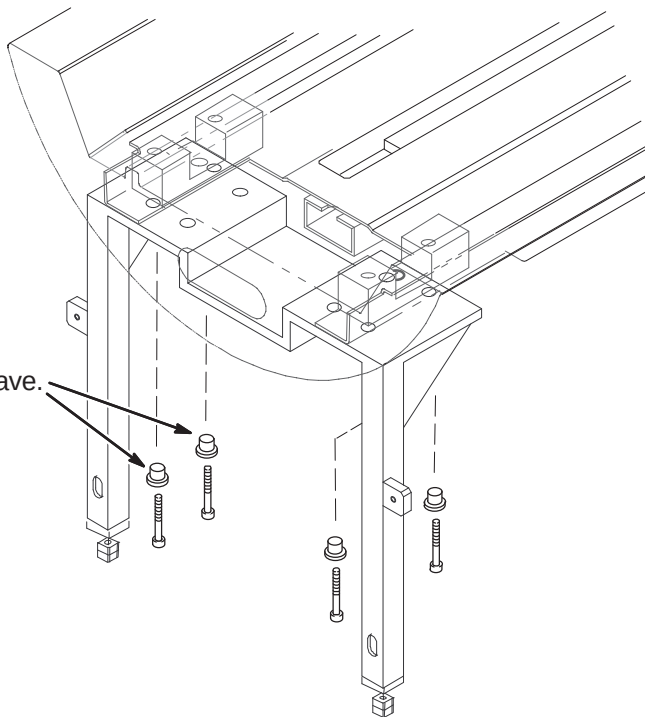


6-1-4-B Remove Rear Split Bridge

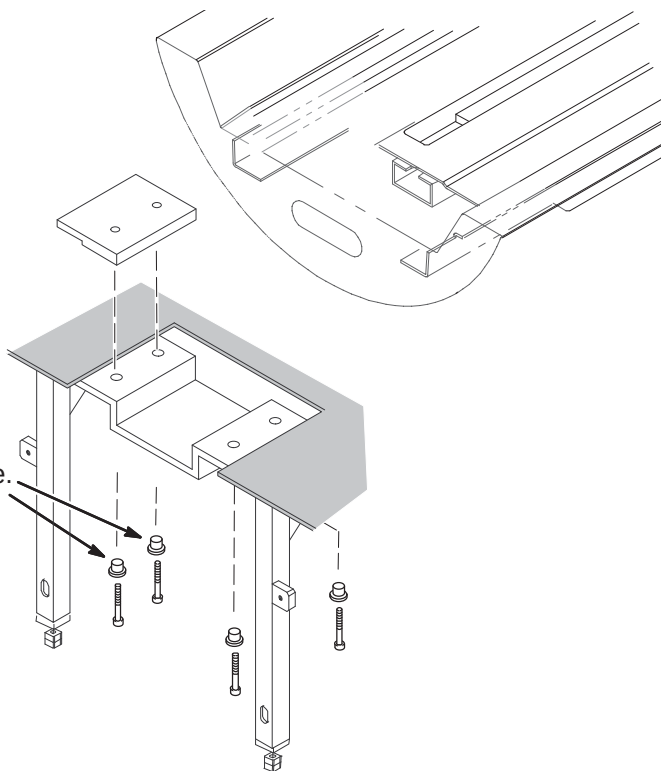


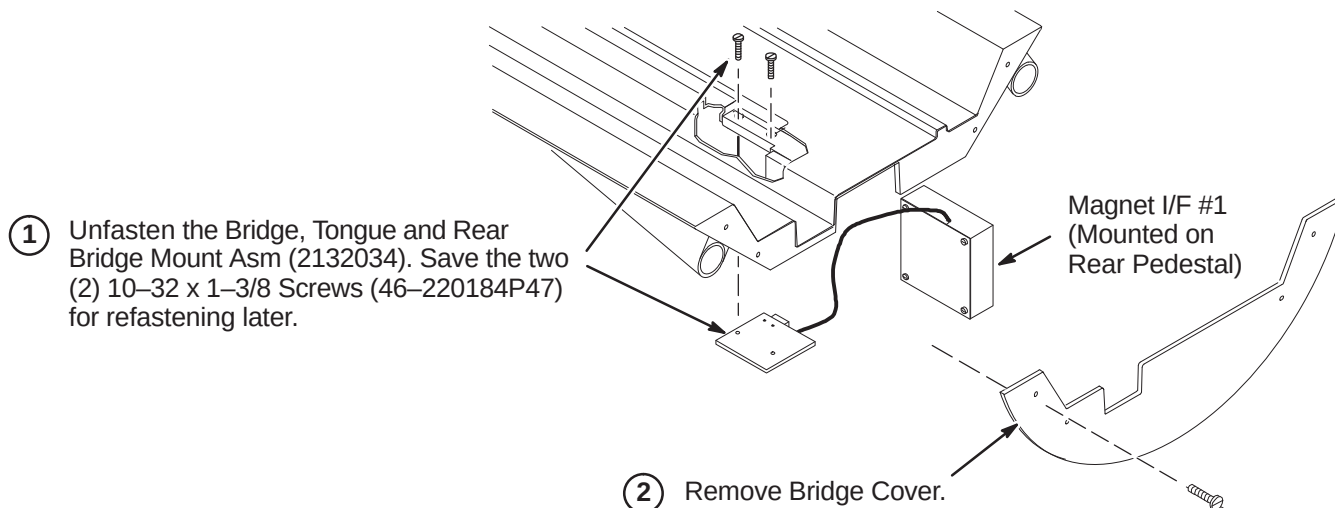
6-1-4-C Remove Forward Production Installed Front Split Bridge

- ① Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings and save. These will be reused for installing the new Bridge.
- ② Carefully lift and remove Front Split Bridge and return for recycling.

**6-1-5 DE-INSTALL LONG BRIDGE FOR TwinSpeed****6-1-5-A Unfasten Front of Long Bridge**

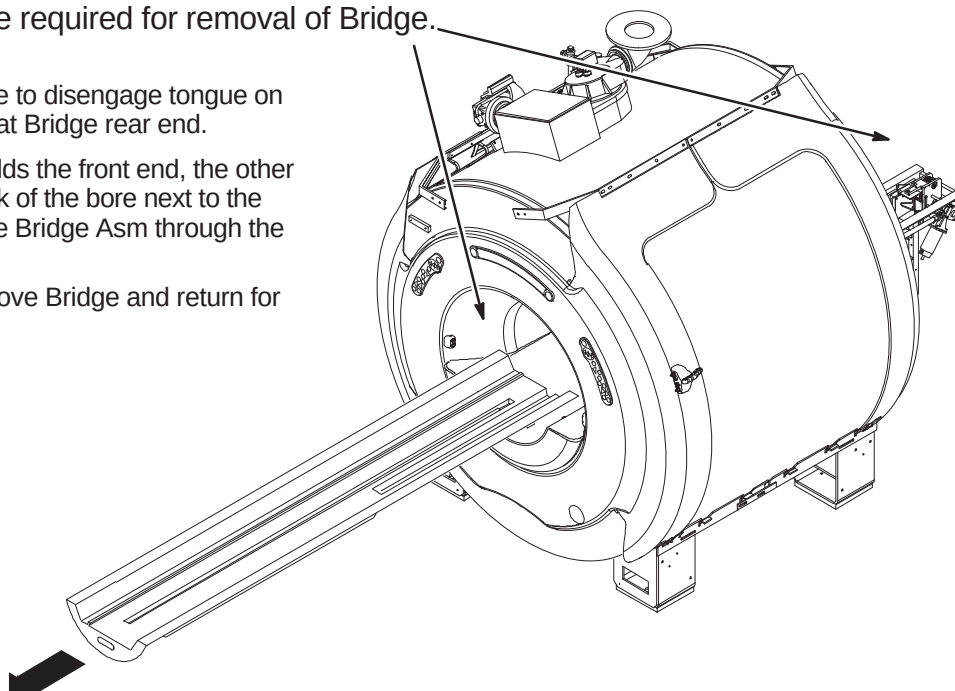
- ① Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings and save. These will be reused for installing the new Bridge.



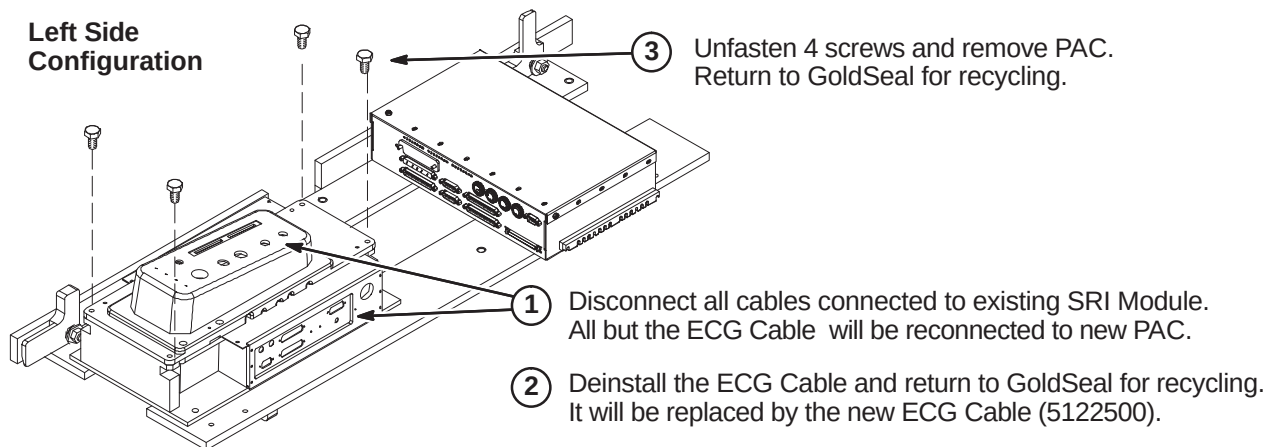
6-1-5-B Disconnect Bridge Mount Asm**6-1-5-C Remove Upgrade Long Bridge**

NOTE: Two people are required for removal of Bridge.

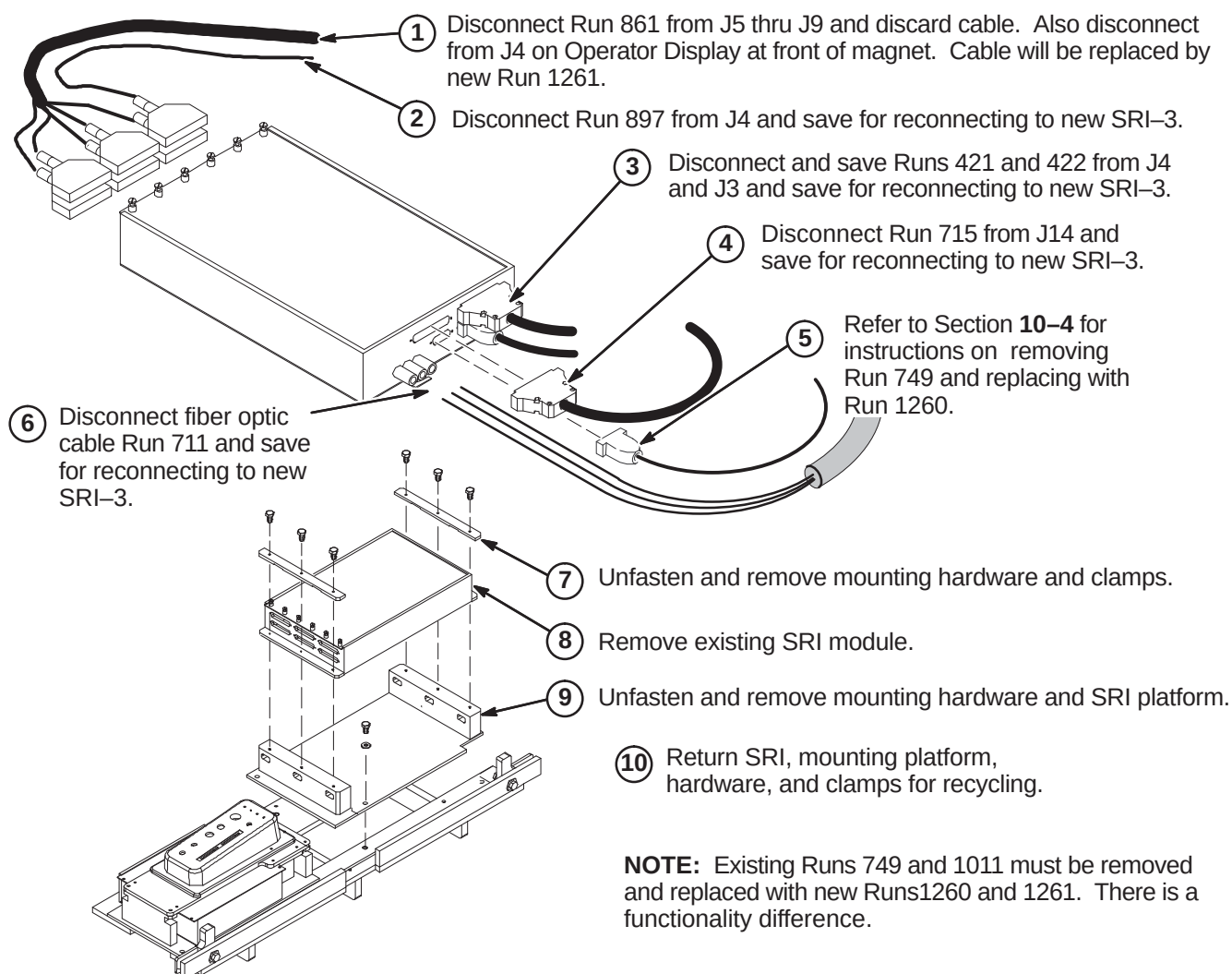
- ① Lift front end of bridge to disengage tongue on long drive assembly at Bridge rear end.
- ② While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore to the front end.
- ③ Carefully lift and remove Bridge and return for recycling.



6-1-6 DE-INSTALL PAC MODULE



6-1-7 DE-INSTALL SRI MODULE



6-2 INSTALLATION OF TwinSpeed SPECIFIC PARTS (TRM ONLY)

This upgrade section is performed only on a TwinSpeed system which uses a TRM coil. The Twin RF Body Coil will be replaced and the Manifold Protector Plate will be installed.

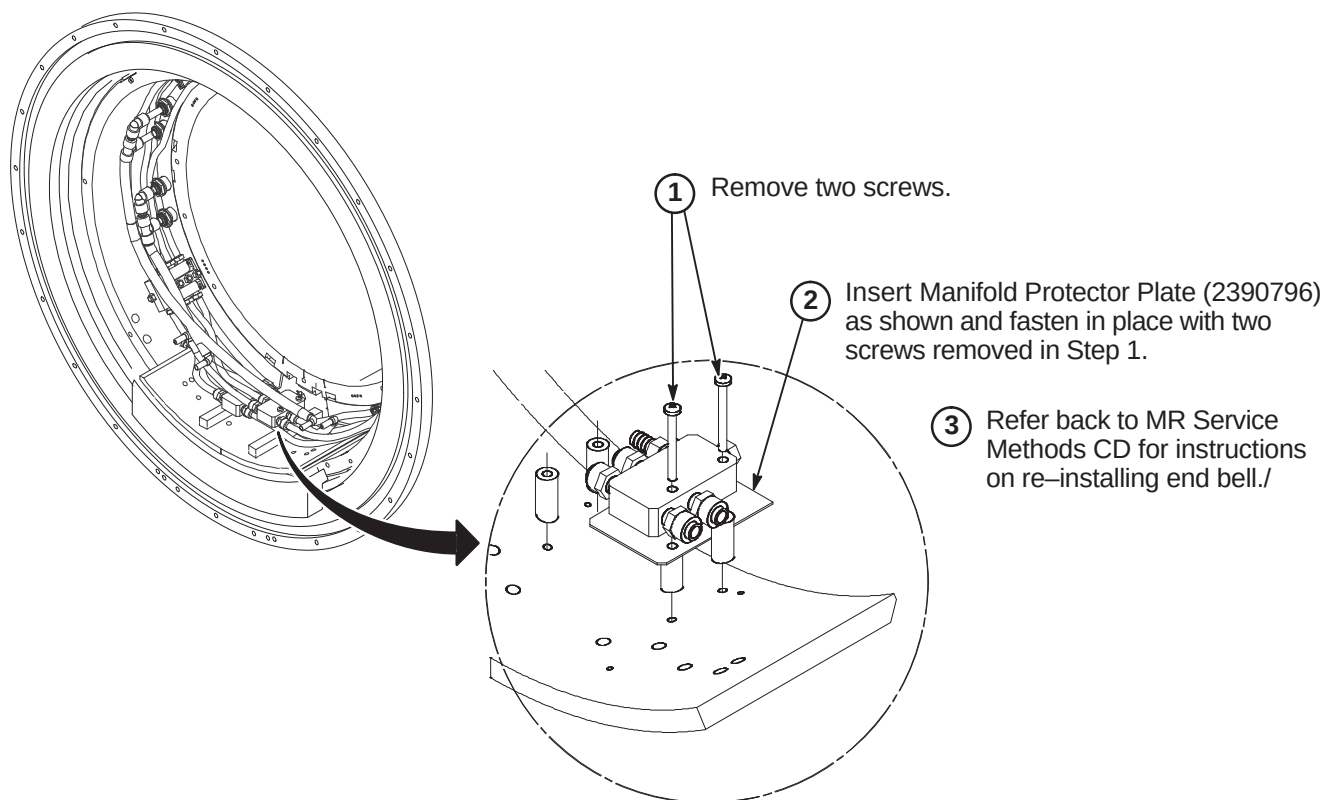
6-2-1 1.5T TwinSpeed RF BODY COIL REPLACEMENT

For replacement procedure, refer to applicable MR Service Methods CD-ROM and go to:

Replacements/1.5T Twin RF Body Coil Replacement Procedure.

When front End Bell is removed, perform Procedure 6-2-2 for installing the Manifold Protector plate.

6-2-2 INSTALL MANIFOLD PROTECTOR PLATE



6-3 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It would be required for earlier versions of the WideOpen style enclosure. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the HDx upgrade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2318581, *Signa Cx/K4 Magnet WideOpen Enclosure Bridge Support* and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.

Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade*, for instructions on installing this kit.

Note: If this procedure is not completed by Mechanical Installer or Van Partner, make sure that the GE Field Engineer has been notified that this upgrade procedure needs to be completed. For Van Partner, refer to the Design Review Checklist to report status of completion.

6-4 INSTALL NEW COMPONENTS AND CABLES

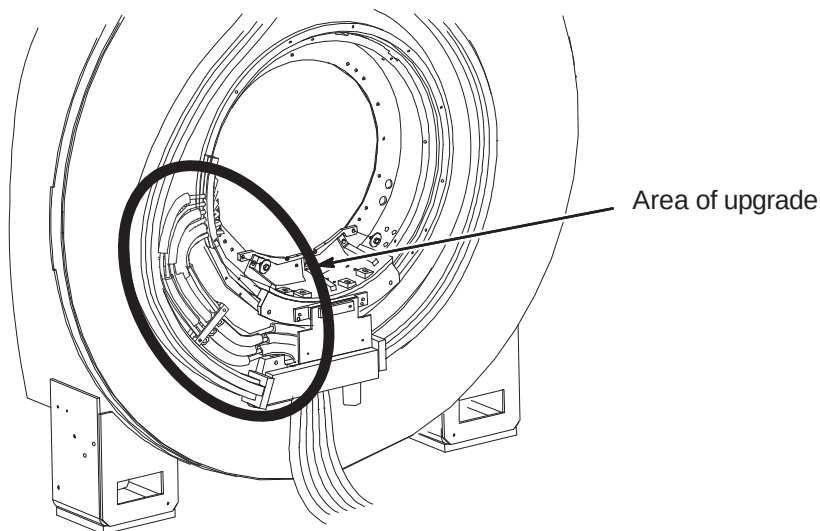
6-4-1 INSTALL BRM GRADIENT CABLE LEAD BOARD (MR/i or CV/i Fixed Site Systems Only)

Note

For a system being upgraded, whether it is a fixed-site or a mobile, if the gradient lead restraint is using the 90 degree mobile van bracket or the IIQ "S" bracket assembly, do not modify or upgrade it. If the system being upgraded does not contain either of these, then install the IIQ "S" bracket assembly.

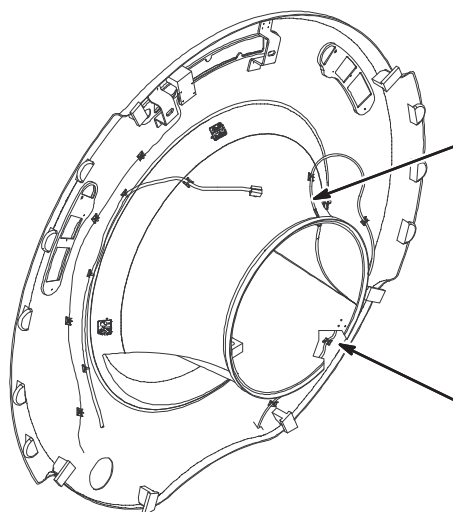
For fixed sites only, the mounting of the gradient cables at the rear of the magnet must use the Gradient Cable lead Board, also known as the "S" bracket assembly. If the cables are routed as shown in the illustration below, then the upgrade must be installed. This upgrade is part of M3335BD, *LCC WideOpen IIQ Upgrade*. Refer to Direction 2404642, provided with Gradient Cable Lead Board collector (5110336), for instructions on how to install the new design.

M3335BD also include parts and instructions for installing the Rear Pedestal Stabilization kit (5198961). Refer to Section 6-4-10.



6-4-2 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL

This procedure must be completed if existing or new Front End Bell does not have Run 1260 installed on it. To verify cable installed, check cable attached to J10 of SRI Module. It must be Run 1260.

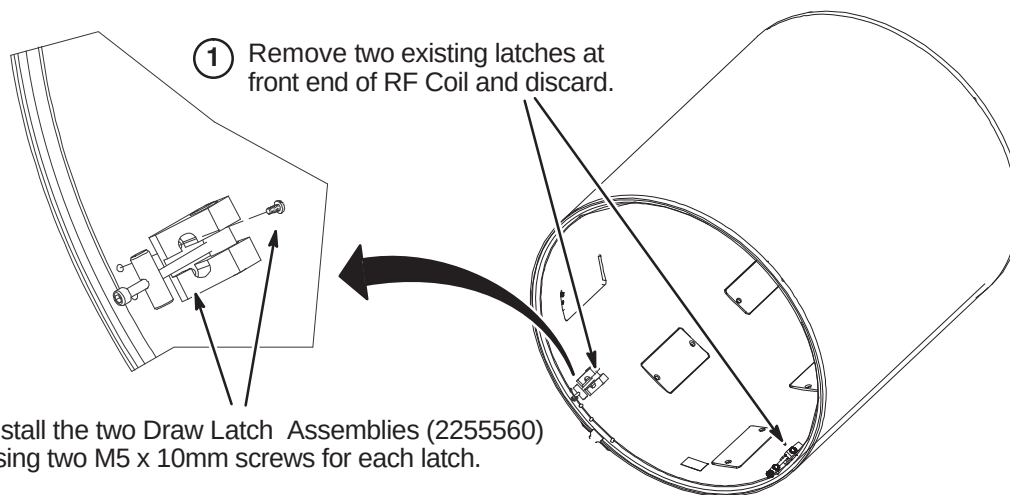


- ① EXCITE HD requires a new Dual Temperature Sensor cable, Run 1260 (5123812), to be installed.

Refer to Section **10-4** for instructions on removal of existing cable and installing the new cable.

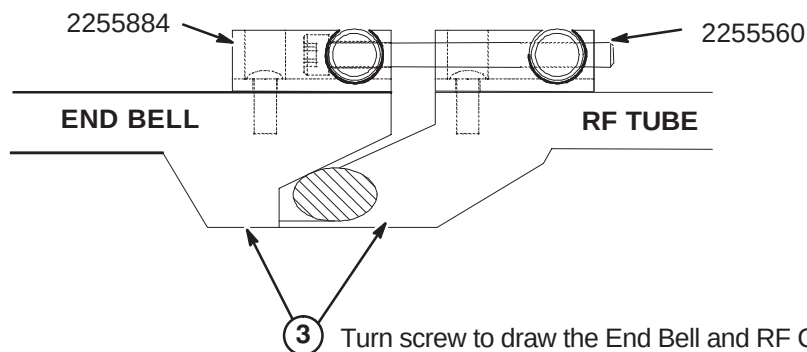
- ② Run 861 should also be removed at this time and replace with Run 1261.

Note: The new Front End Bell has the new style Latch Keeper Assembly (225584) installed for attaching to RF Coil in magnet bore. Refer to Section **6-4-3** for instructions on attaching the mating Draw Latch Assembly (225560) to the RF Coil.

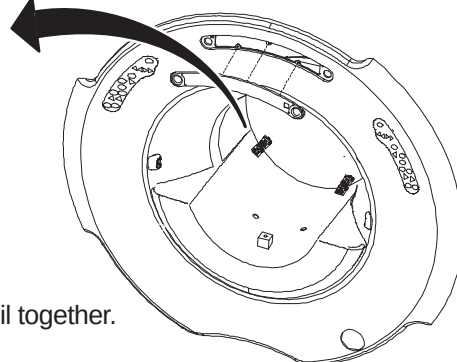
6-4-3 INSTALL LATCH KEEPER ASSEMBLY ON RF COIL

- ① Remove two existing latches at front end of RF Coil and discard.

- ② Install the two Draw Latch Assemblies (225560) using two M5 x 10mm screws for each latch.

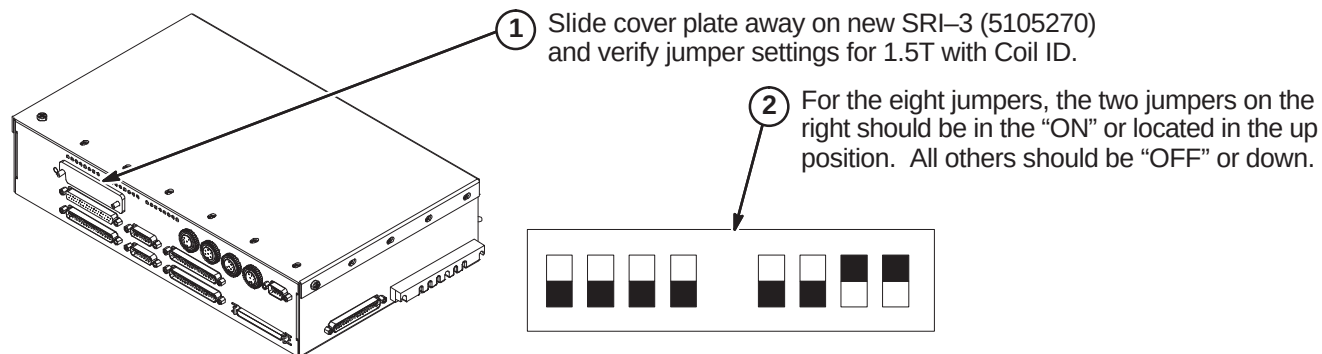


- ③ Turn screw to draw the End Bell and RF Coil together.

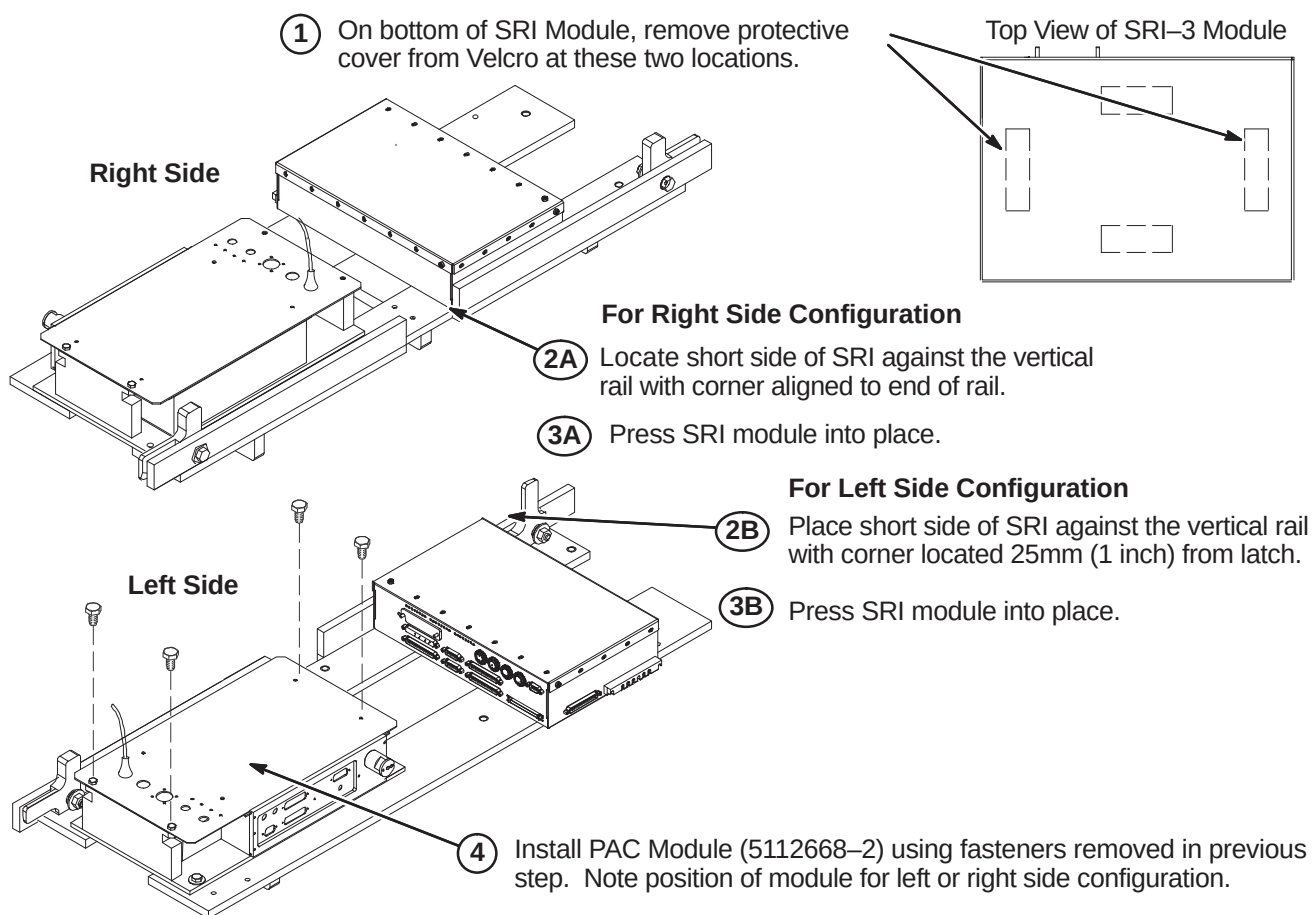


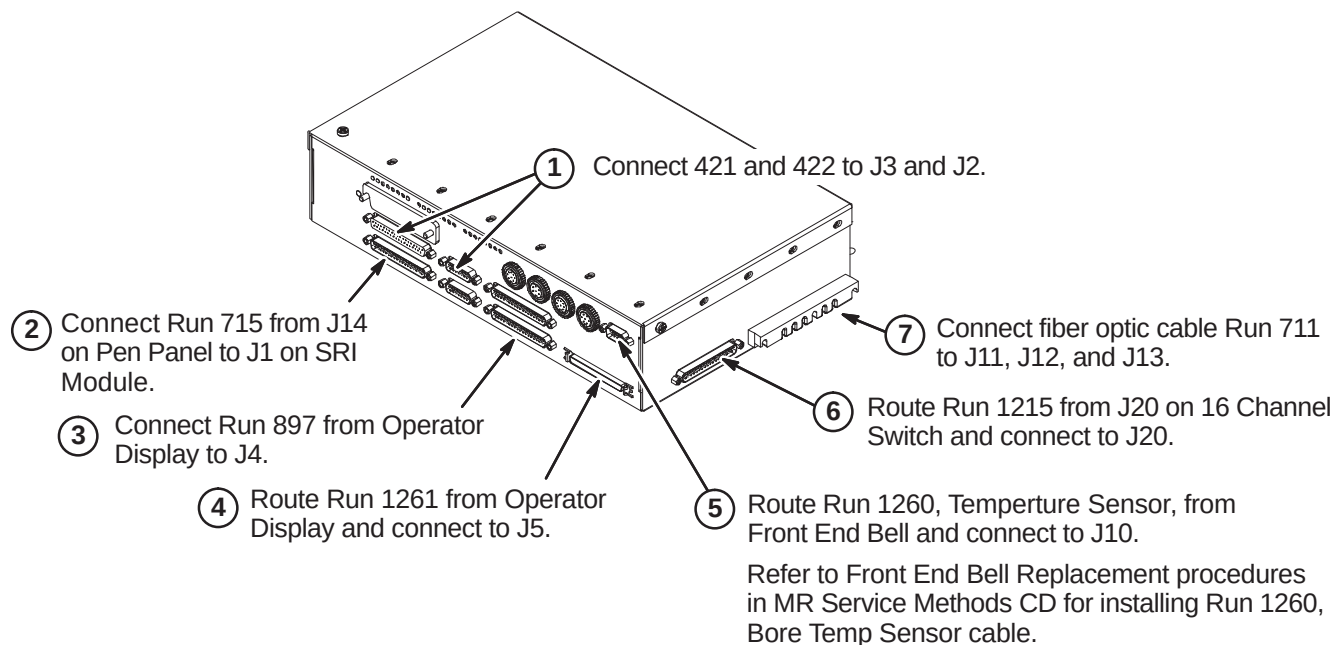
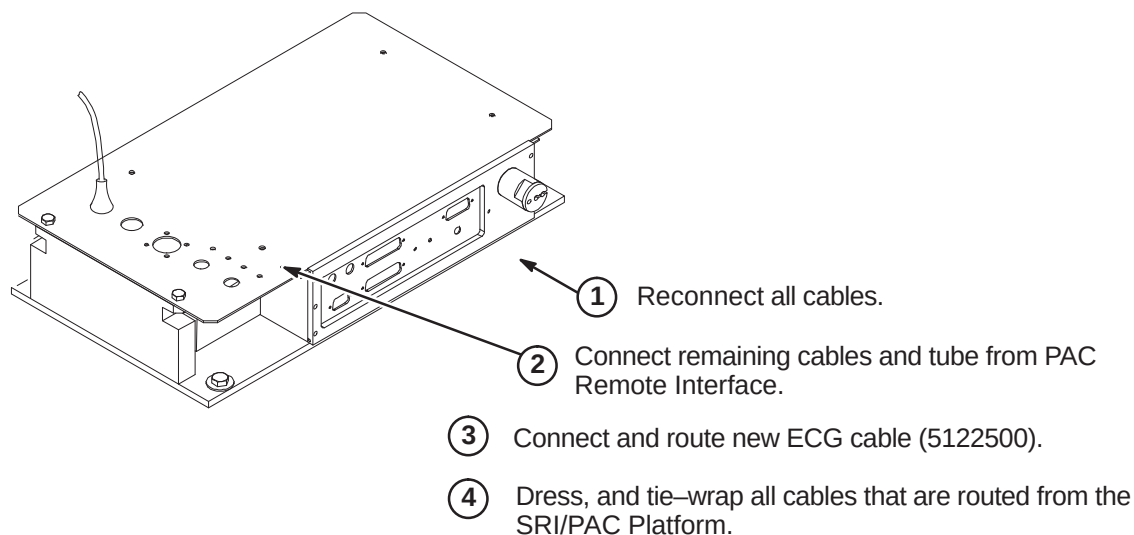
6-4-4 INSTALL PAC AND SRI MODULES

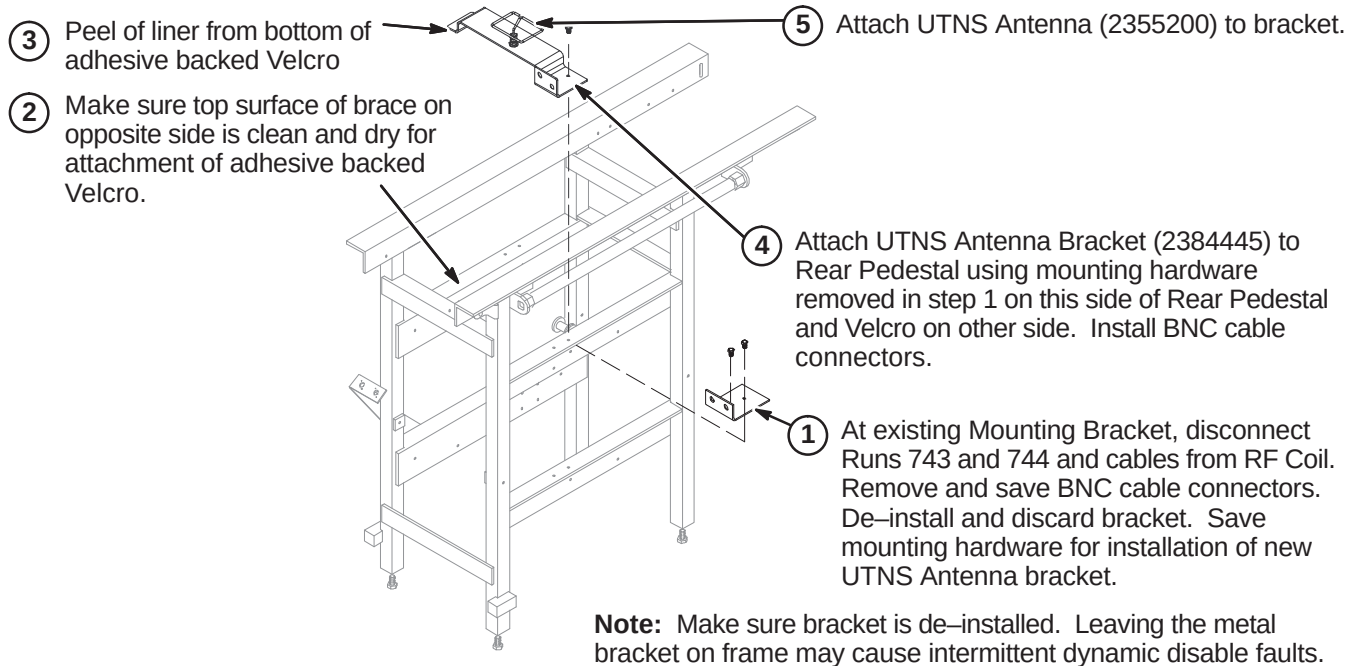
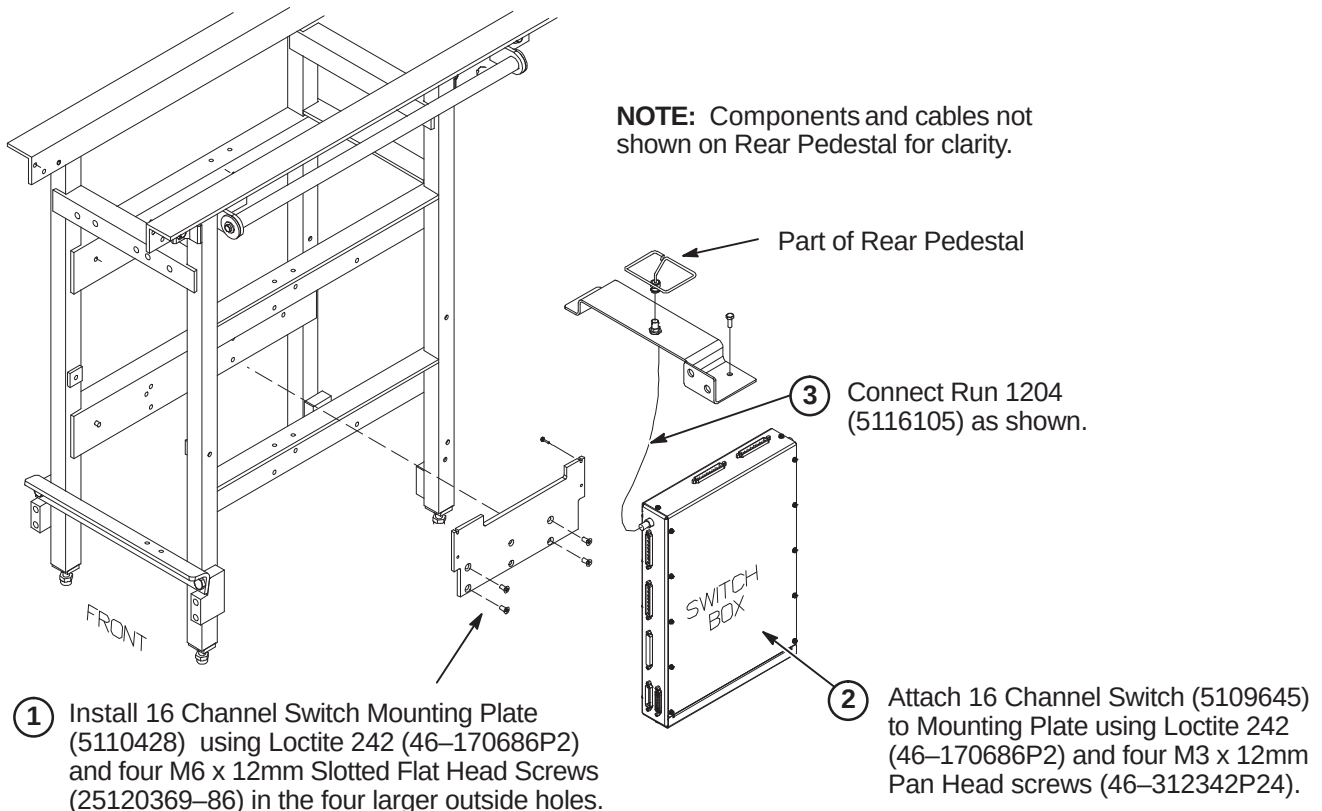
6-4-4-A Verify Jumper Positions for SRI

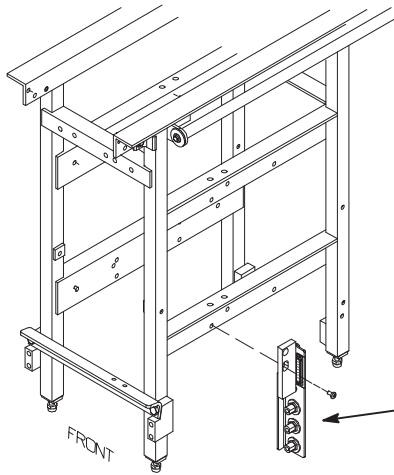


6-4-4-B Install New PAC and SRI On Right or Left Side of Enclosure



6-4-4-C Connect Cables to SRI-3 Module**6-4-4-D Connect Cables to PAC Module**

6-4-5 INSTALL NEW COMPONENTS ON REAR PEDESTAL**6-4-5-A UTNS Antenna Installation****6-4-5-B Install 16 Channel Switch**

6-4-5-C Install New Bulkhead Interface

NOTE: Components and cables not shown on Rear Pedestal for clarity.

- ① Install new Bulkhead Interface (5113250) using Loctite 242 (46-170686P2) and two M6 x 12mm Pan Head Screws (2109873-24).

6-4-6 HYBRID SPLITTER UPGRADE

Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

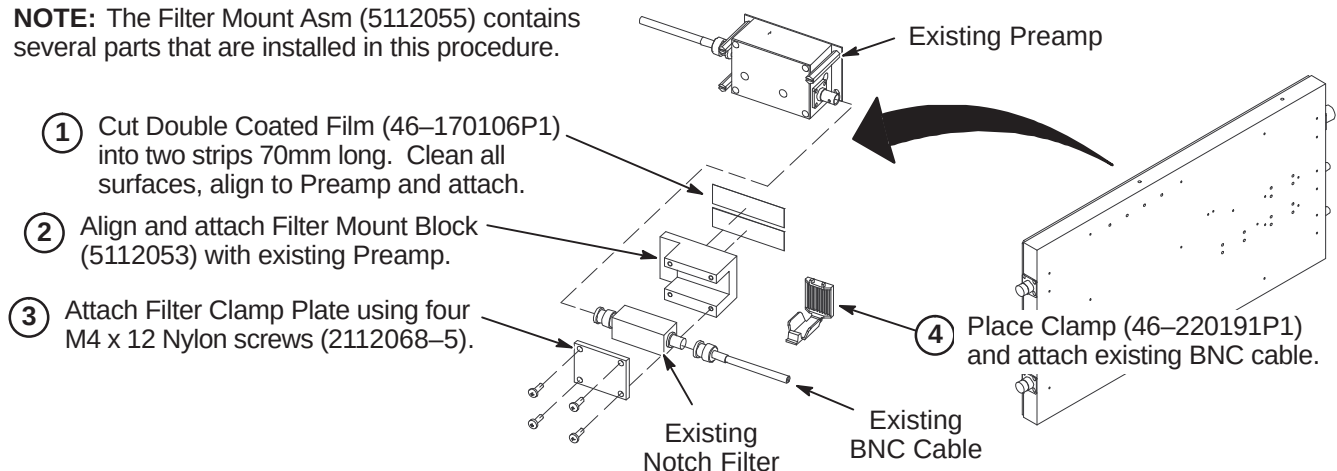
- Refer to Section 6-4-6-A for systems that do not have mount assembly for the Notch Filter
- Refer to Section 6-4-6-B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 6-4-6-C for instructions on installing Isolation Filters on systems with Teledyne Splitter

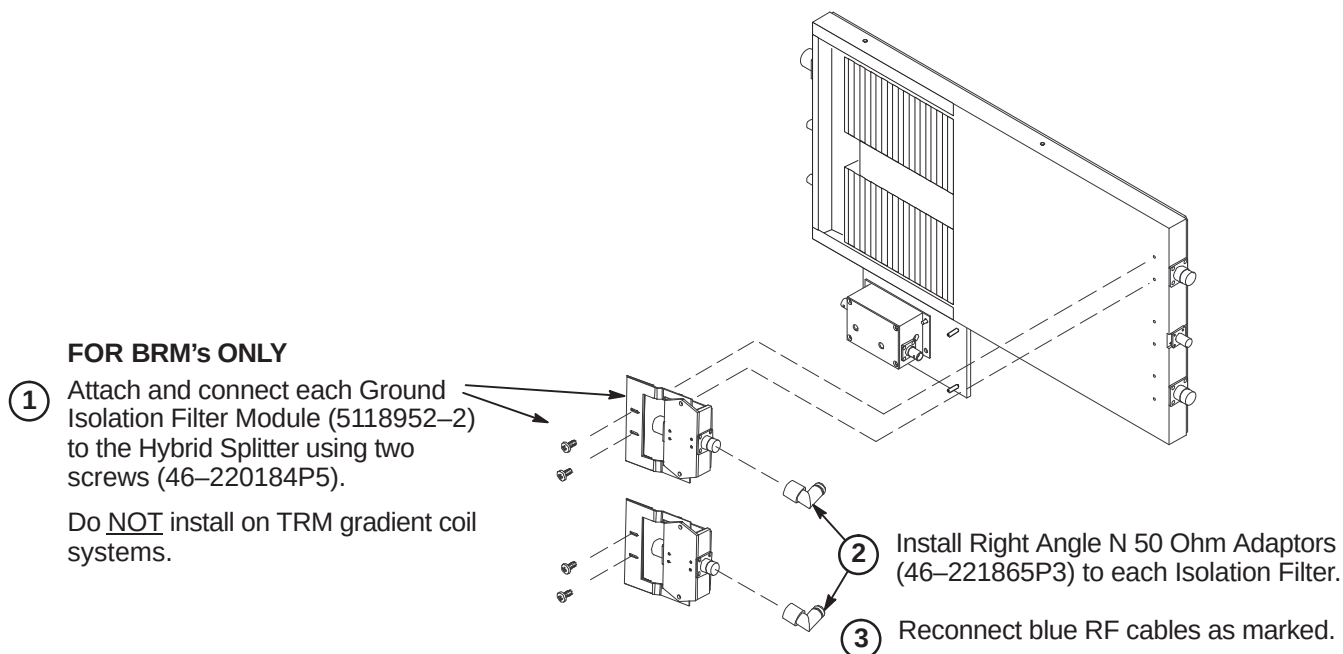
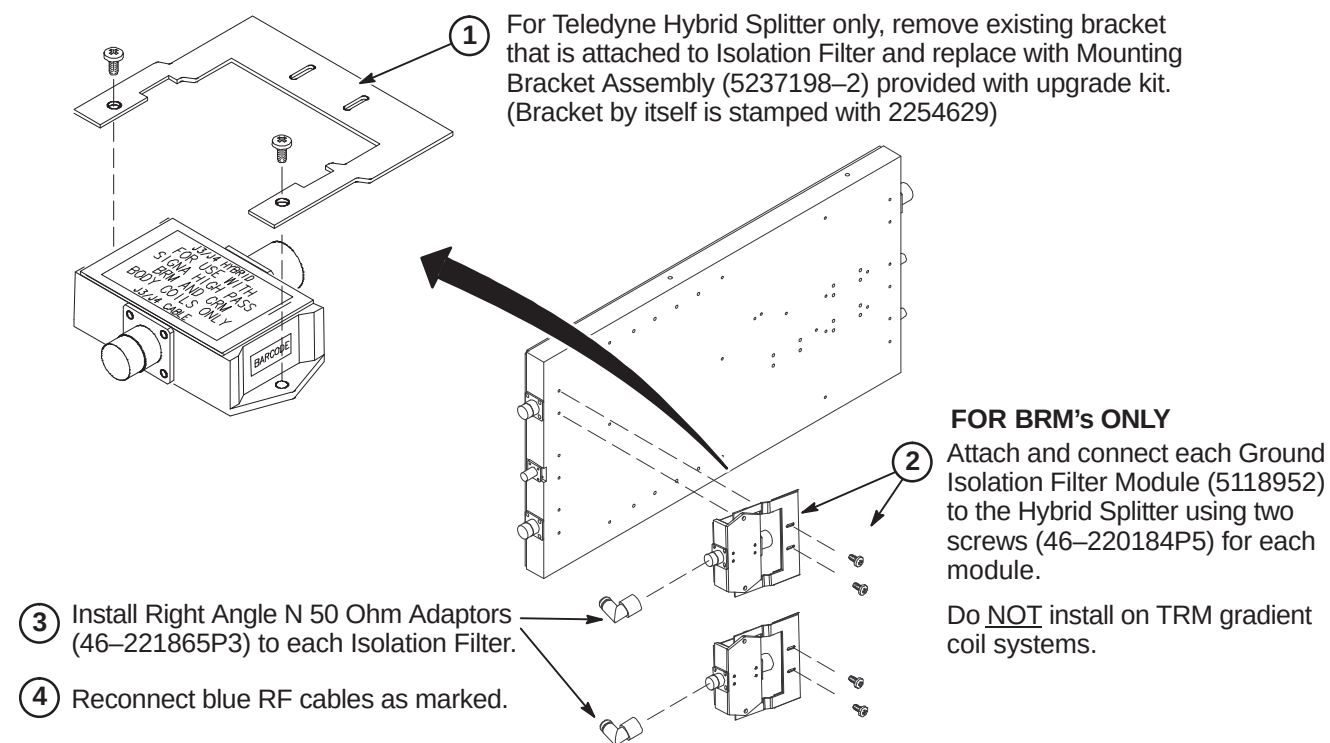
The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils. Do NOT install on systems with TRM gradient coils (TwinSpeed).

6-4-6-A Install Filter Mount Asm to Body Hybrid Splitter

This Filter Mount assembly may not be required on newer Netcom Hybrid Splitter as it may already have a Notch Filter mounting clamp of a different style installed.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.

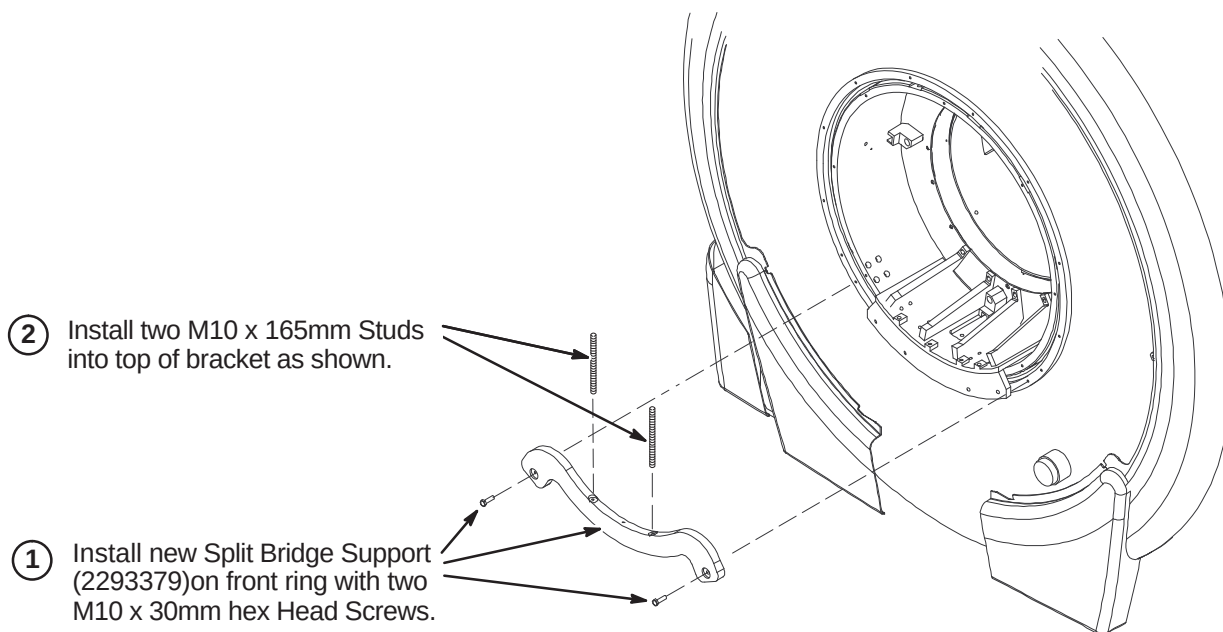


6-4-6-B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)**6-4-6-C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)**

6-4-7 INSTALL FRONT END BELL

The following procedures must be completed if new Front End Bell is installed on WideOpen enclosure that had an existing Long Bridge removed for this upgrade.

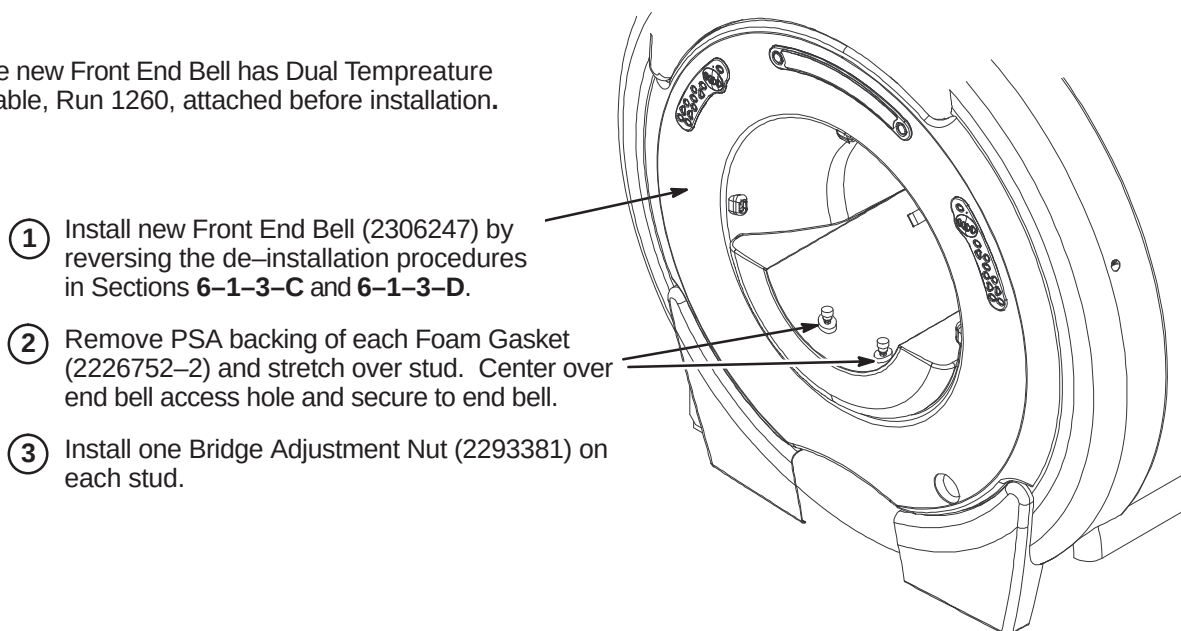
6-4-7-A Install Split Bridge Support

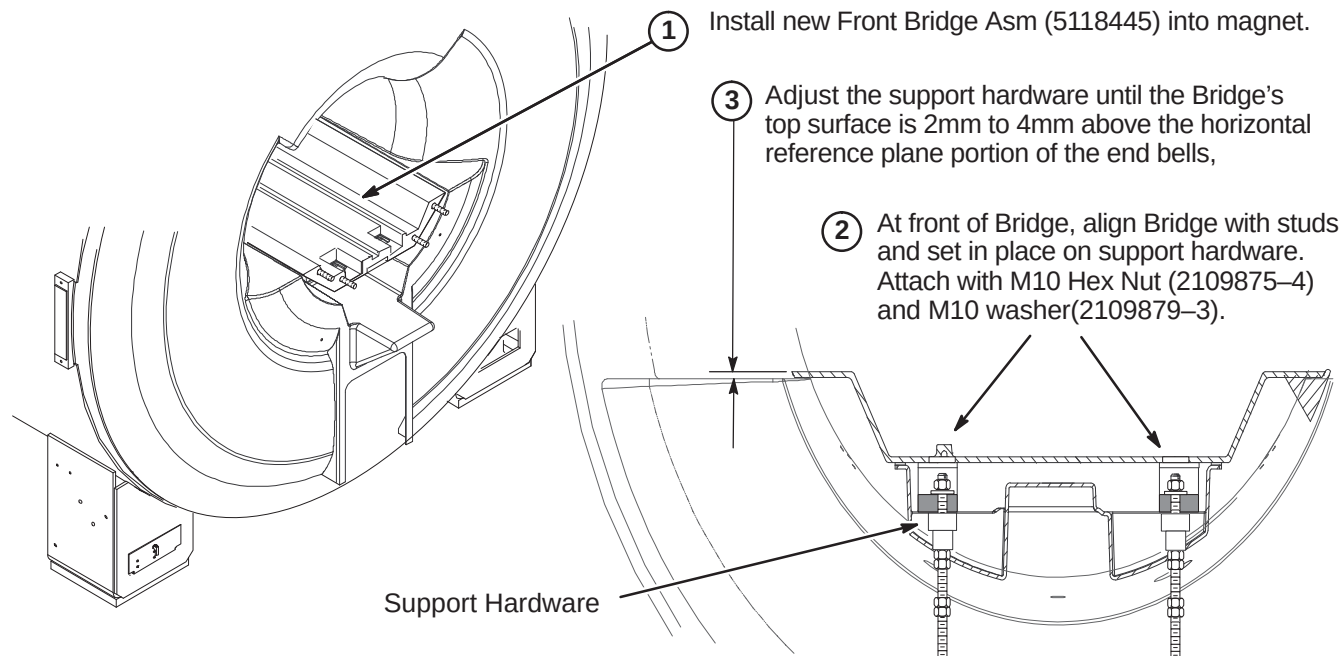
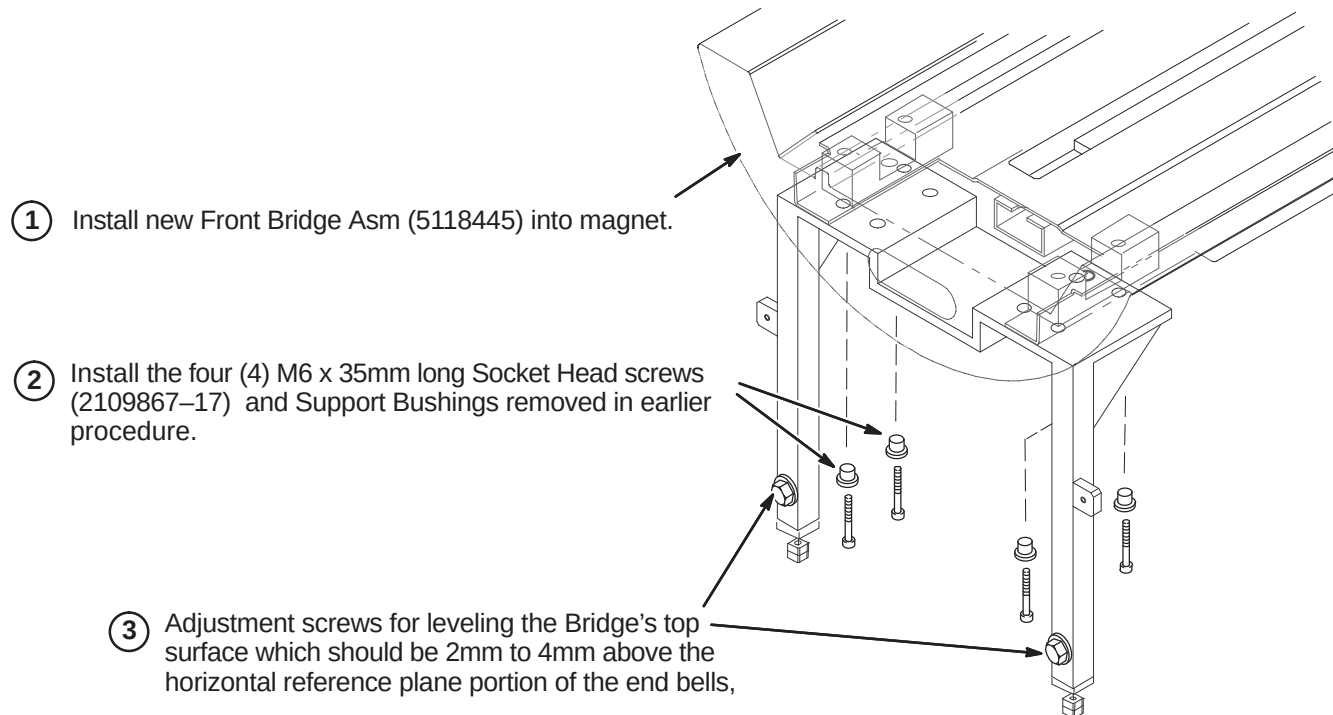


6-4-7-B Install Front End Bell

NOTE:

Make sure new Front End Bell has Dual Temperature Sensor cable, Run 1260, attached before installation.



6-4-8 INSTALL FRONT SPLIT BRIDGE FOR WideOpen**6-4-9 INSTALL FRONT SPLIT BRIDGE FOR TwinSpeed**

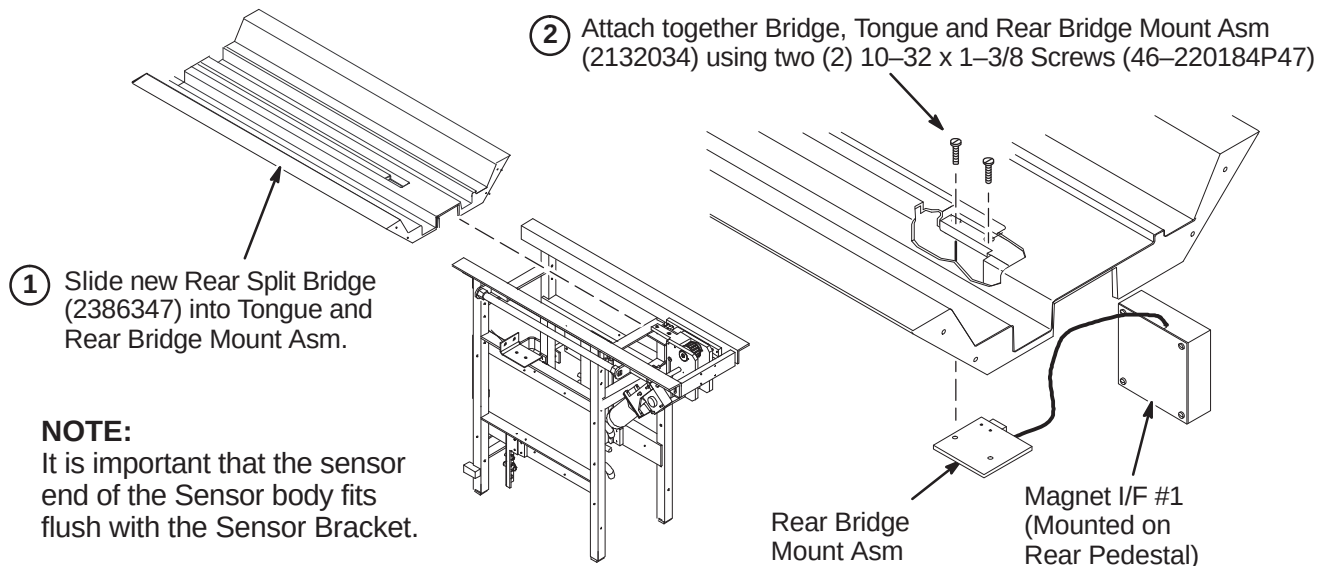
6-4-10 INSTALL REAR PEDESTAL

Prior to this upgrade, the site should have been audited to determine if the Rear Pedestal Stabilization Kit (5198961) has been installed. This kit includes parts and instructions for the installation of the kit. It would be required for earlier versions of the WideOpen style enclosure. If this upgrade has not been performed on the existing enclosure, then the applicable catalog should have been ordered and it must be installed at this point of the HDx upgrade.

Refer to steering Direction 5244044, included in the kit, for guidance.

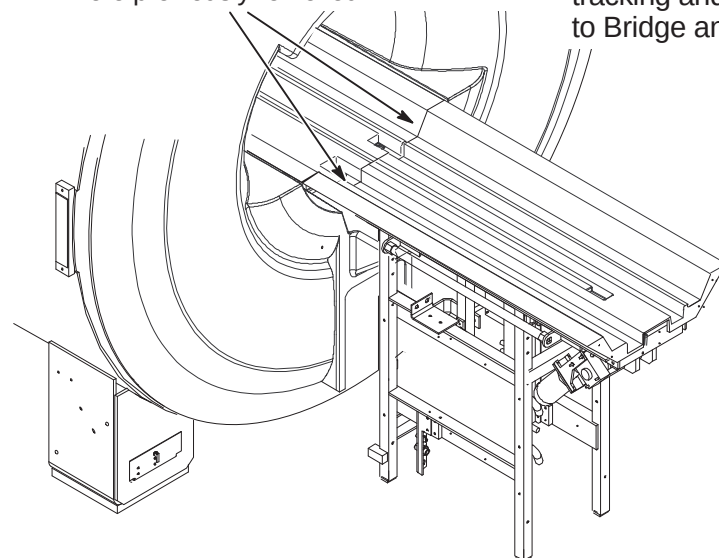
Note: If this procedure is not completed by Mechanical Installer or Van Partner, make sure that the GE Field Engineer has been notified that this upgrade procedure needs to be completed. For Van Partner, refer to the Design Review Checklist to report status of completion.

6-4-10-A Install Rear Split Bridge on Rear Pedestal



6-4-10-B Connect Rear Split Bridge to Front Split Bridge

- ① Move Bridge into position and fasten with four M10 nuts and washers that were previously removed.
- NOTE:**
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.



WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! THE REAR PEDESTAL CAN WEIGH UP TO 310 LBS. (132 KG). IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE. WHEN MOVING THE REAR PEDESTAL, USE PROPER SAFETY AND LIFTING TECHNIQUES.

6-4-11 LEVEL BRIDGE**NOTE:**

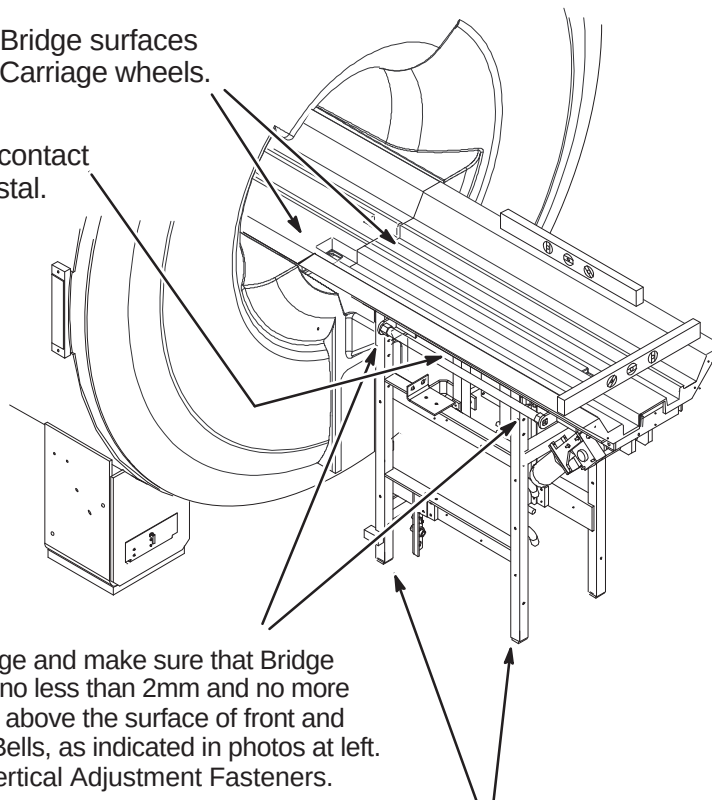
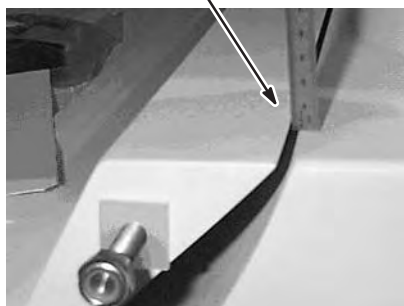
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:

Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.



2mm to 4mm above End Bell surface.



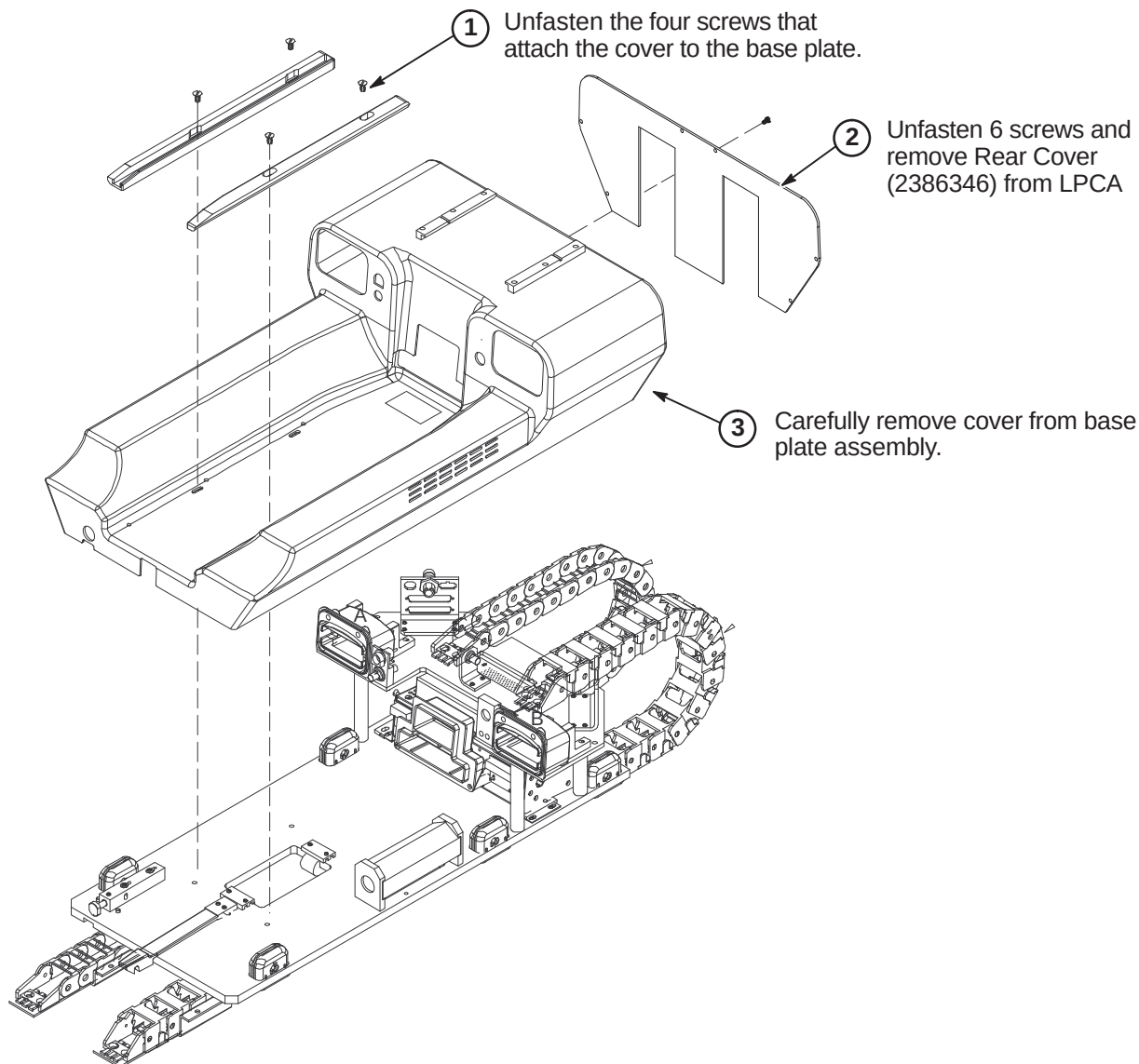
- ① Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of front and rear End Bells, as indicated in photos at left. Tighten Vertical Adjustment Fasteners.

- ② Adjust leg screws on both sides as needed to level Rear Pedestal.

6-4-12 INSTALL LPCA AND CABLE TRACKS

The WideOpen and TwinSpeed style enclosures will be upgraded to an LPCA with a dual cable track. The S-Series and Cx magnet enclosures will install a new LPCA with only one cable track.

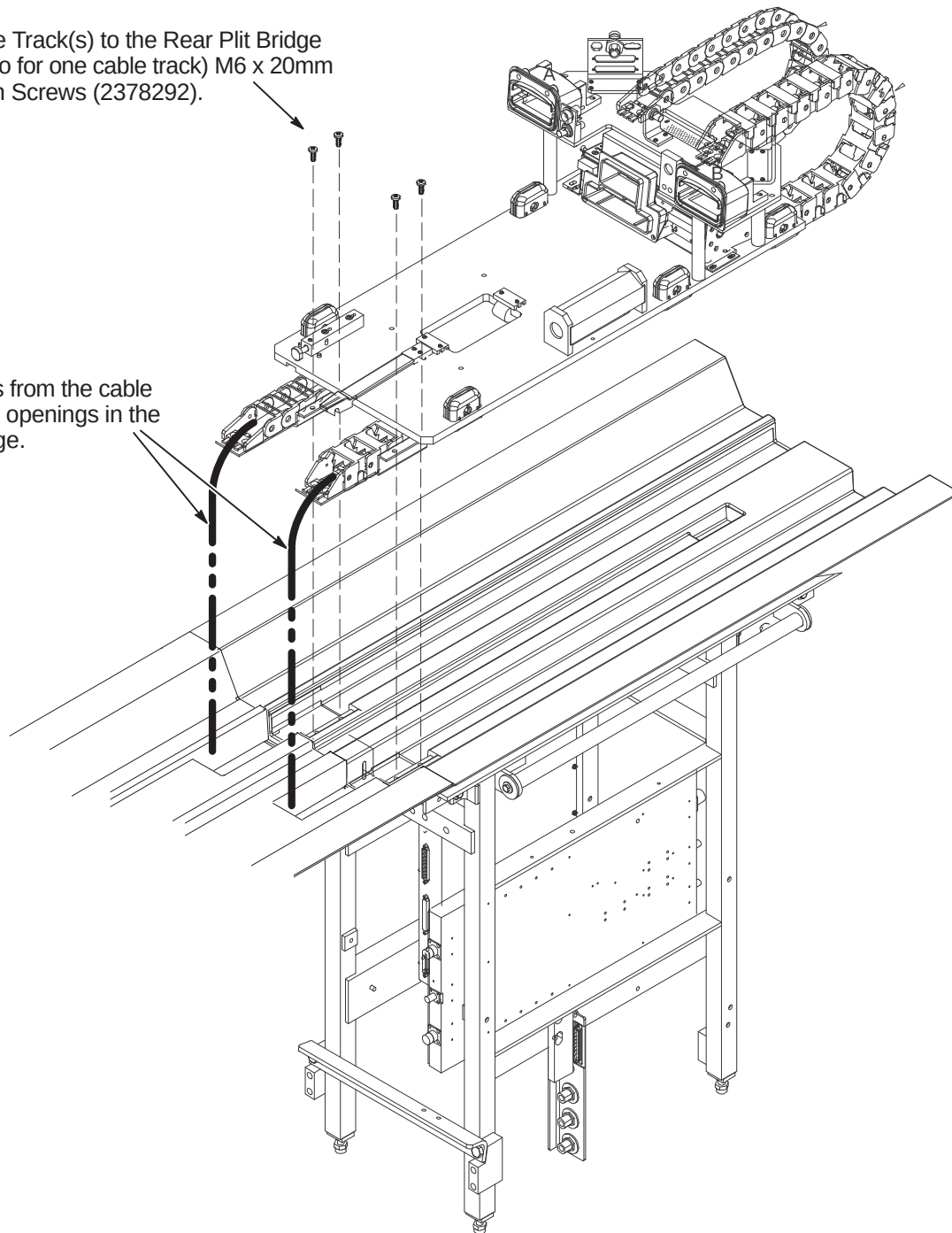
6-4-12-A Remove LPCA Cover

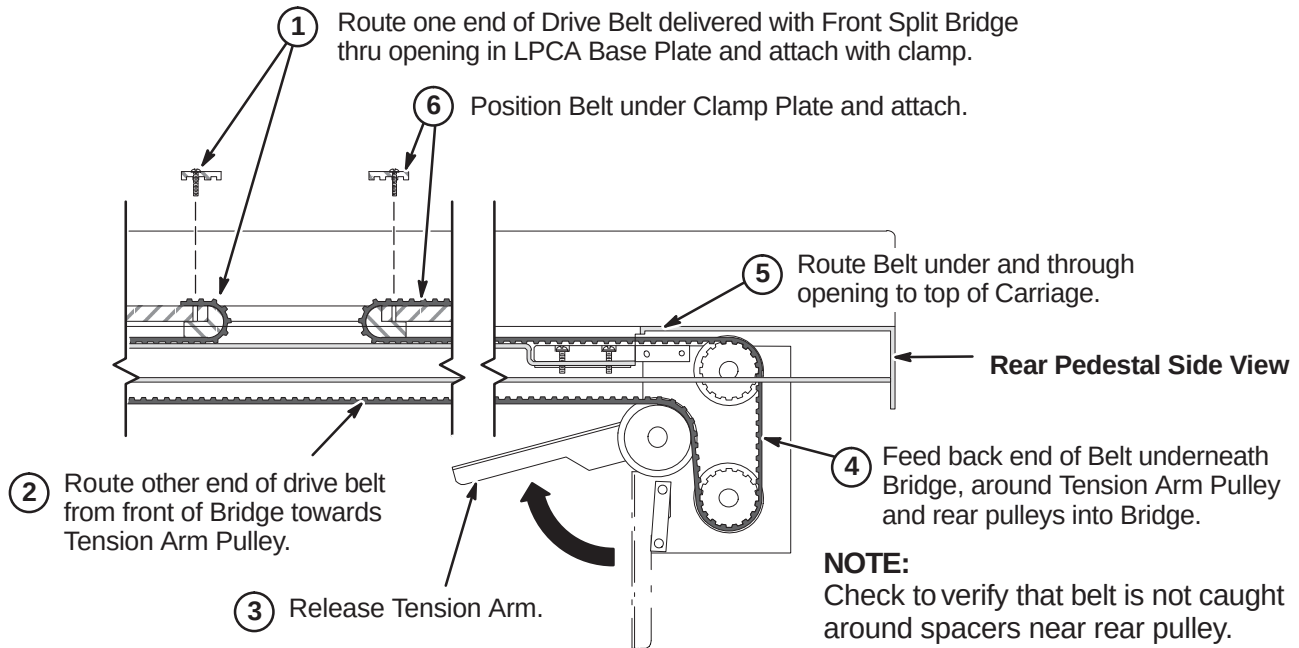
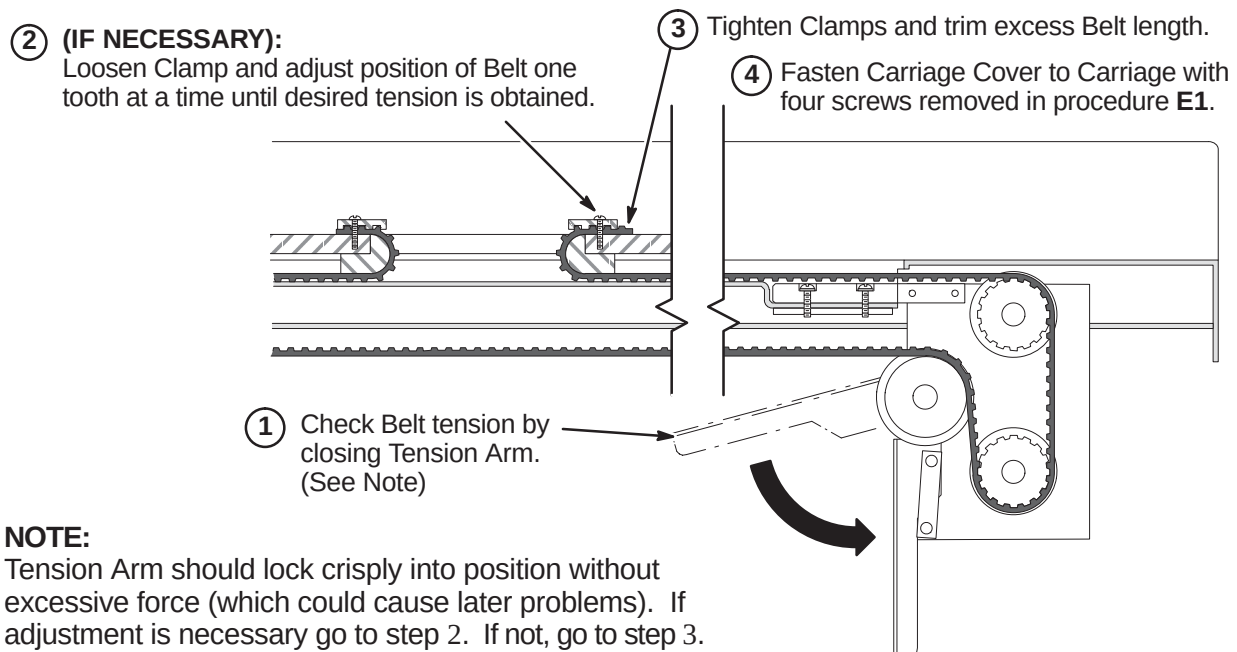


6-4-12-B Attach Cable Tracks to Bridge

- ② Attach the Cable Track(s) to the Rear Plit Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

- ① Feed the cables from the cable track(s) thru the openings in the Front Split Bridge.



6-4-12-C Route and Attach Drive Belt**6-4-12-D Final Adjustment of Drive Belt**

6-4-12-E Installation of Sensor Flag

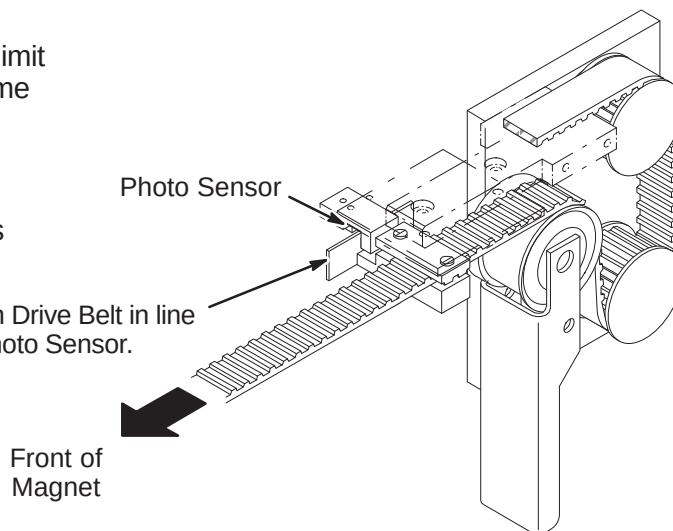
NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:

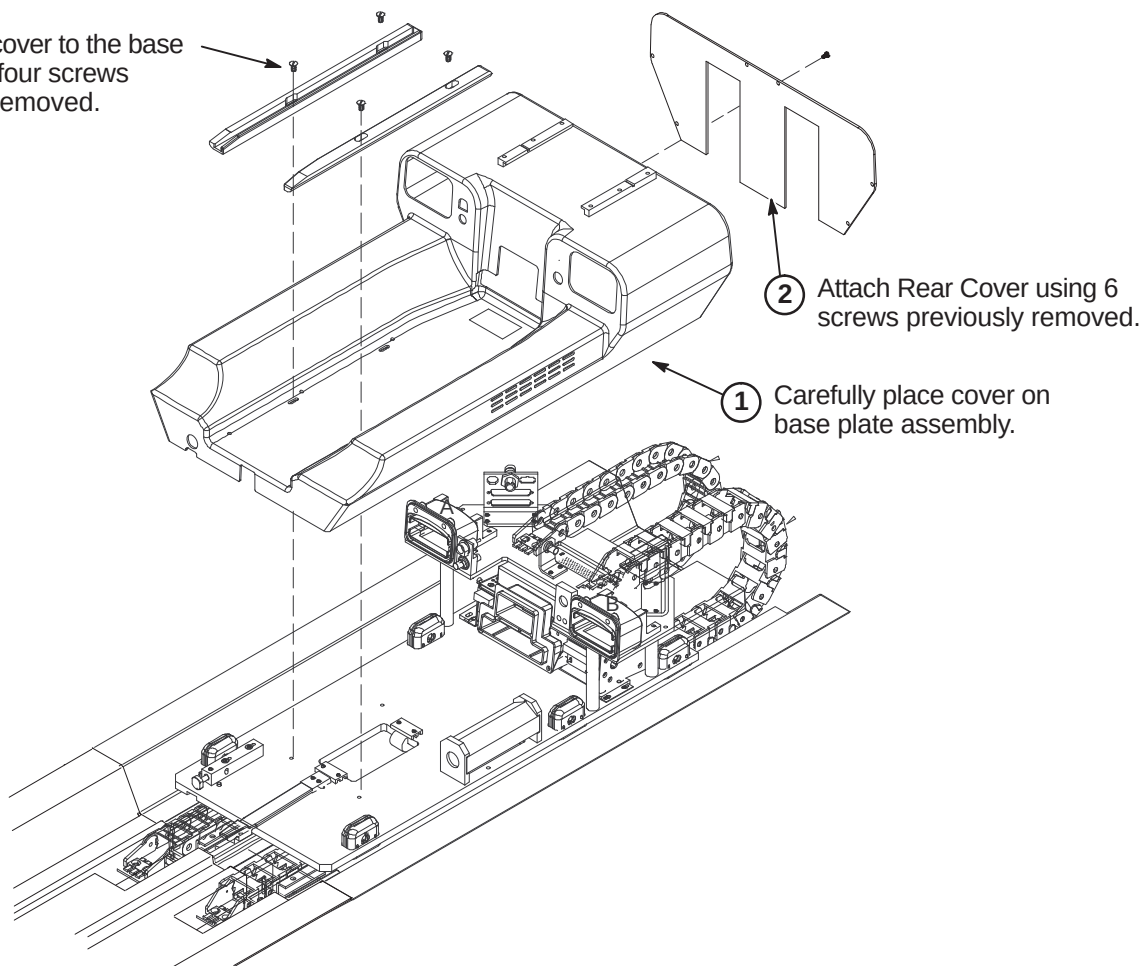
The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



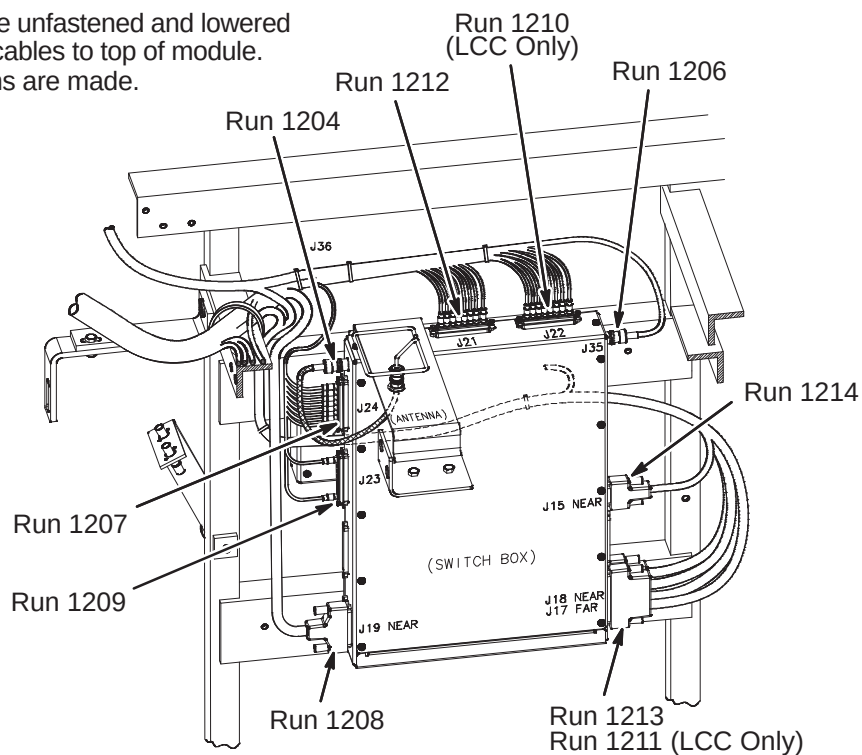
6-4-12-F Installation of LPCA Cover

- ③ Attach the cover to the base plate using four screws previously removed.

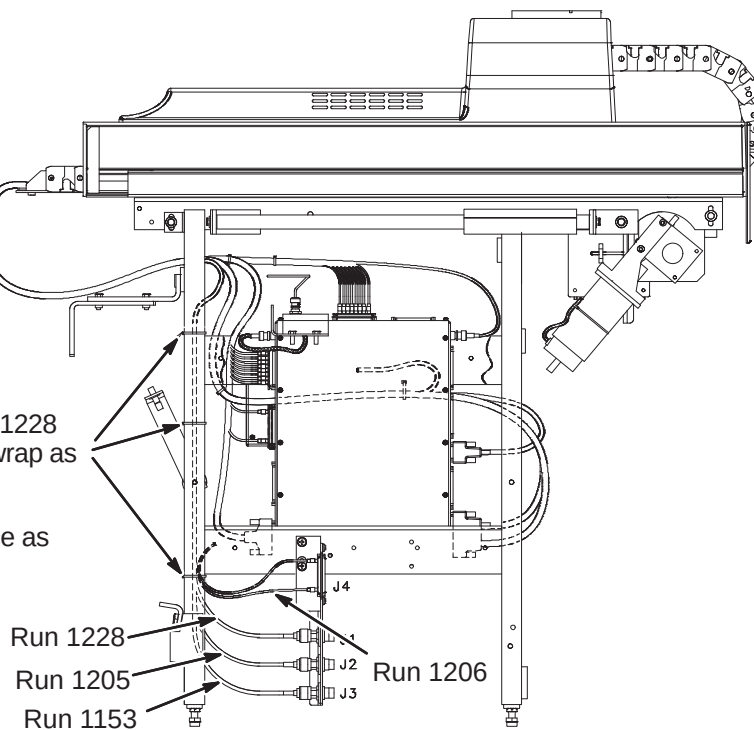


6-4-12-G Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

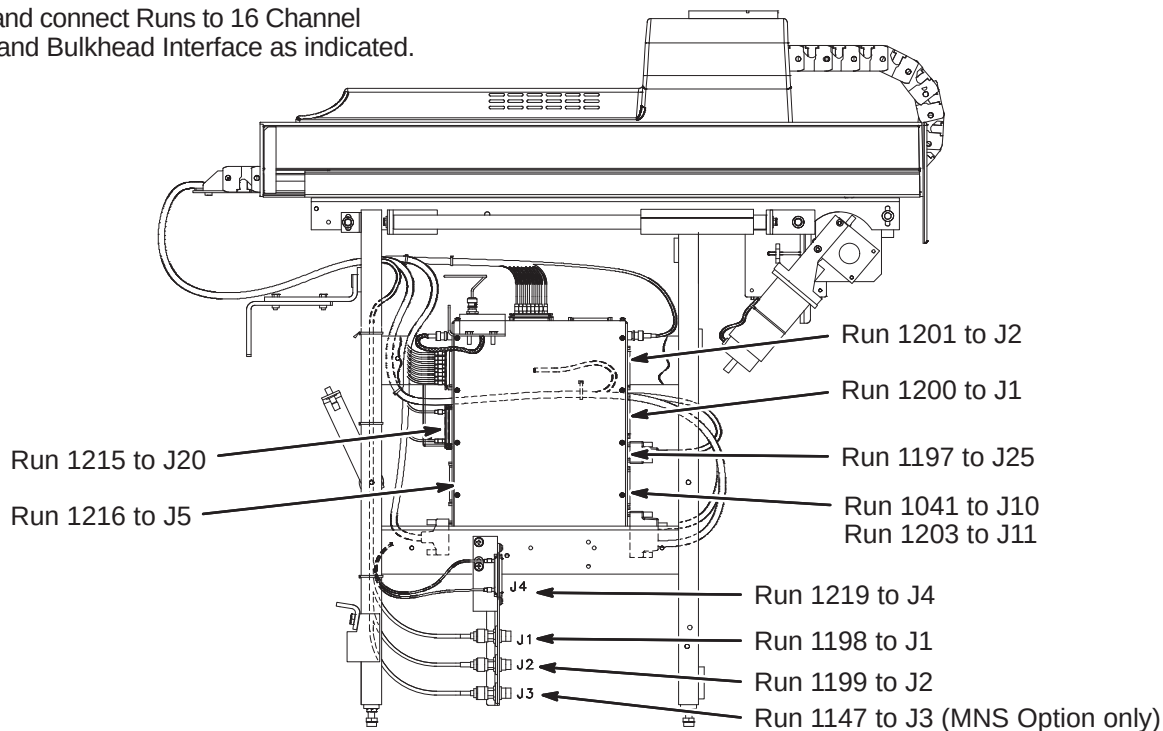
**6-4-12-H Connect Cables from Cable Track to Bulkhead Interface**

- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.
- ② Connect Runs to Bulkhead Interface as indicated.

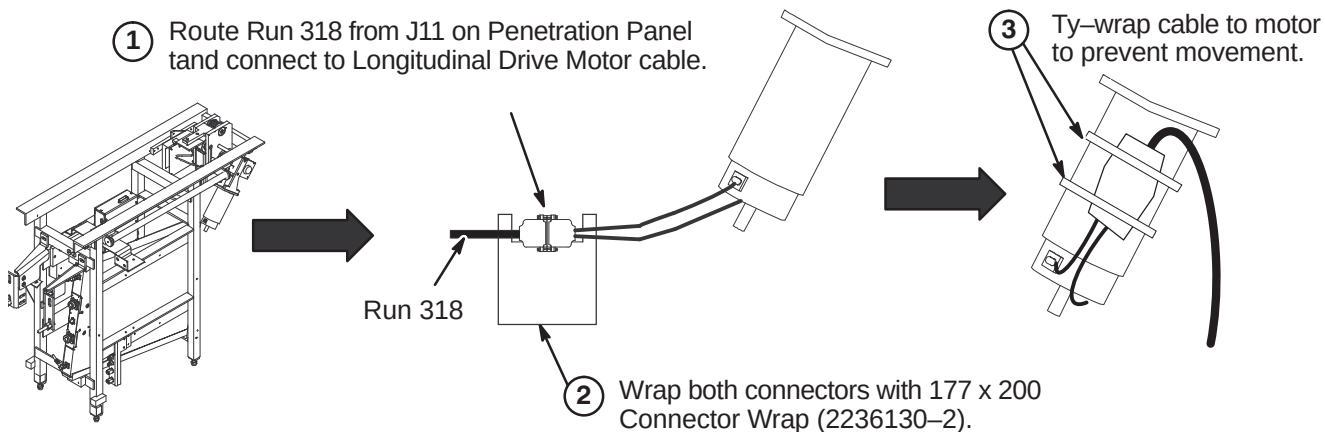


6-4-13 CONNECT REMAINING CABLES TO REAR PEDESTAL**6-4-13-A Cable Connections to 16 Channel Switch and Bulkhead Interface**

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

**6-4-13-B Route/Connect Run 318 to Longitudinal Drive Motor**

- ① Route Run 318 from J11 on Penetration Panel and connect to Longitudinal Drive Motor cable.

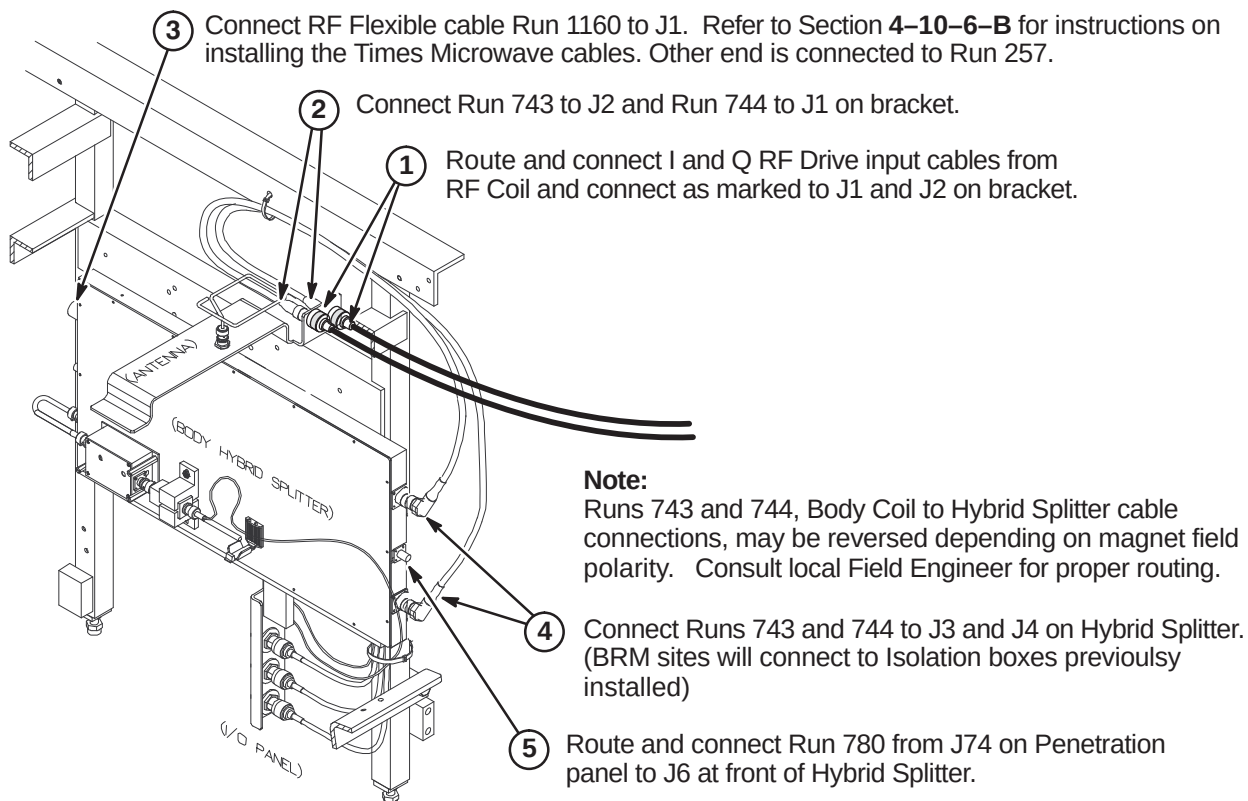


6-4-13-C Connect Cables to Hybrid Splitter

CAUTION

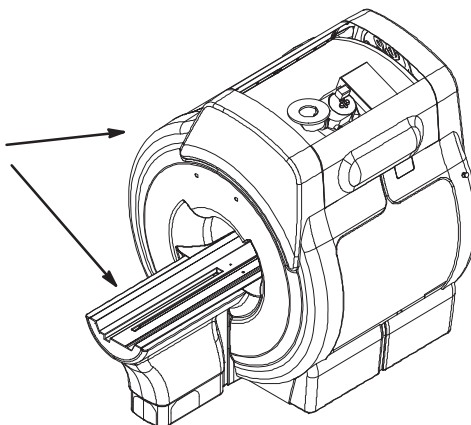
The RF Drive Input cables utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cables to the RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



6-5 INSTALL COVERS

- 1 Replace all enclosure and rear pedestal covers



6-6 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

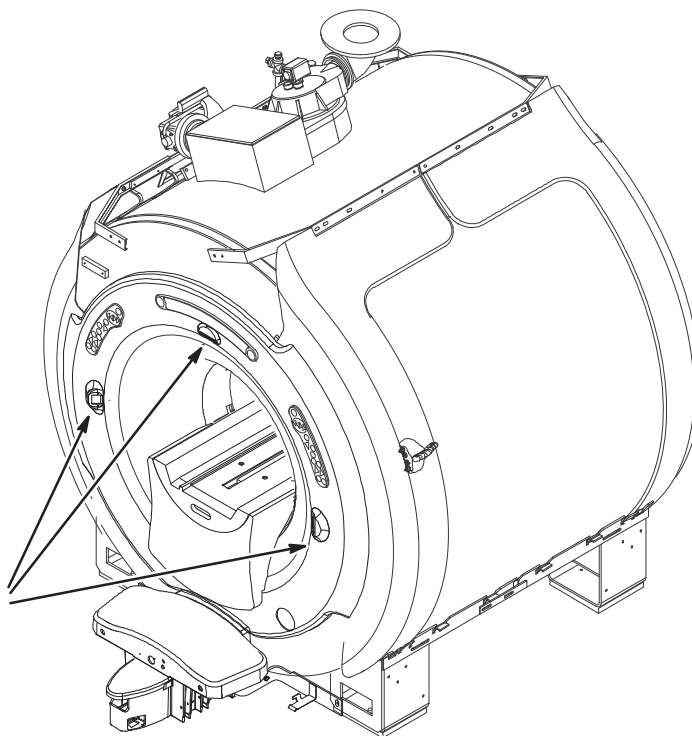
A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.



English

- ① For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



Spanish



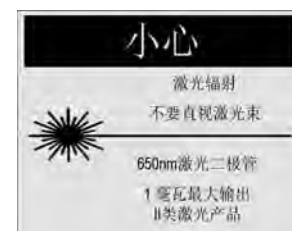
Italian



Swedish

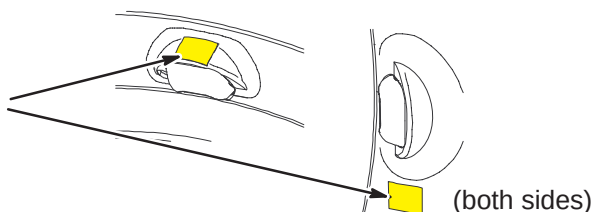


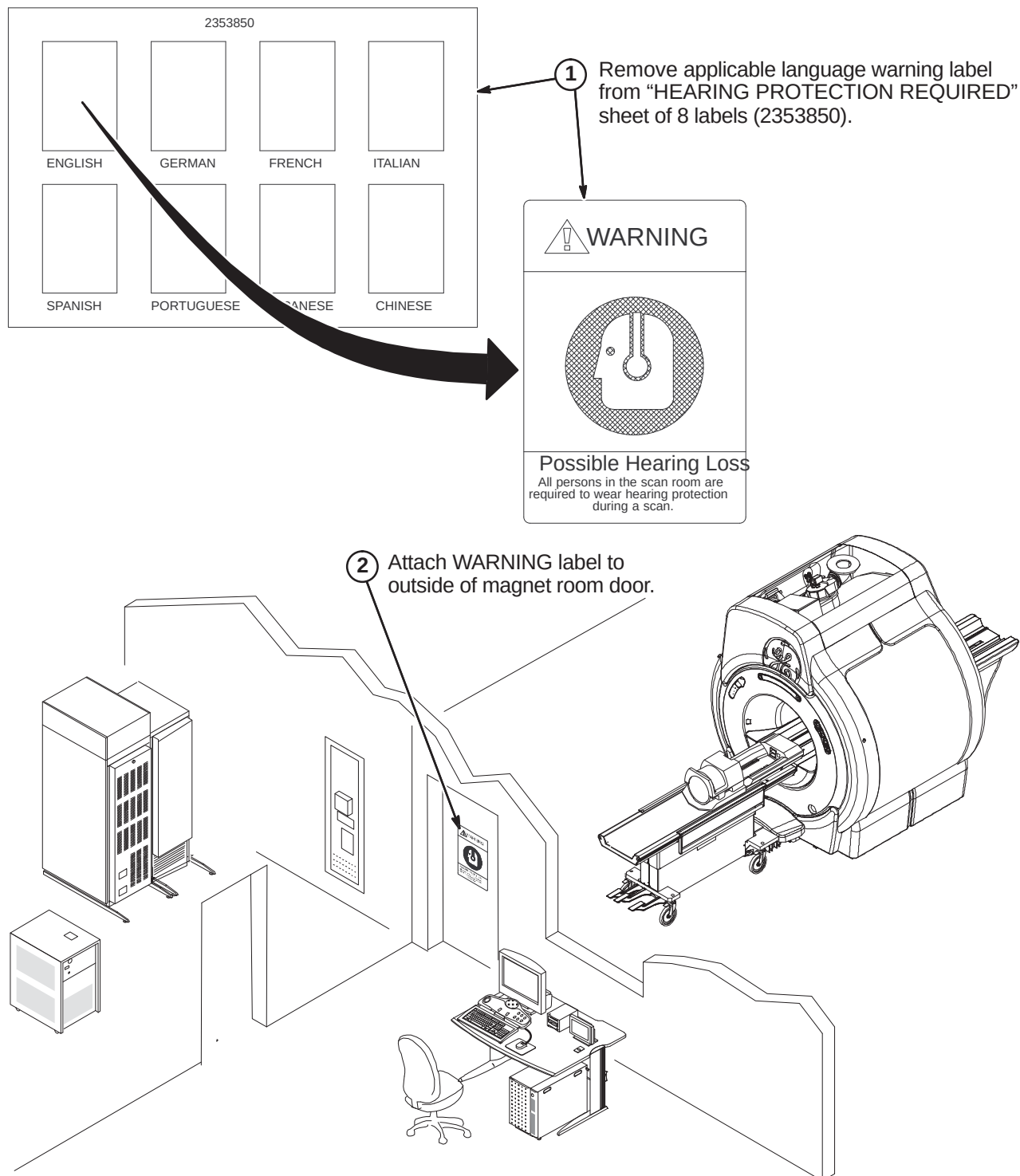
Japanese



Chinese

- ② If enclosure does not have labels attached, attach new labels as indicated.



6-7 INSTALL HEARING LOSS WARNING SIGN

SECTION 7 – S-SERIES/Cx/LCC HORIZON OR LX UPGRADE

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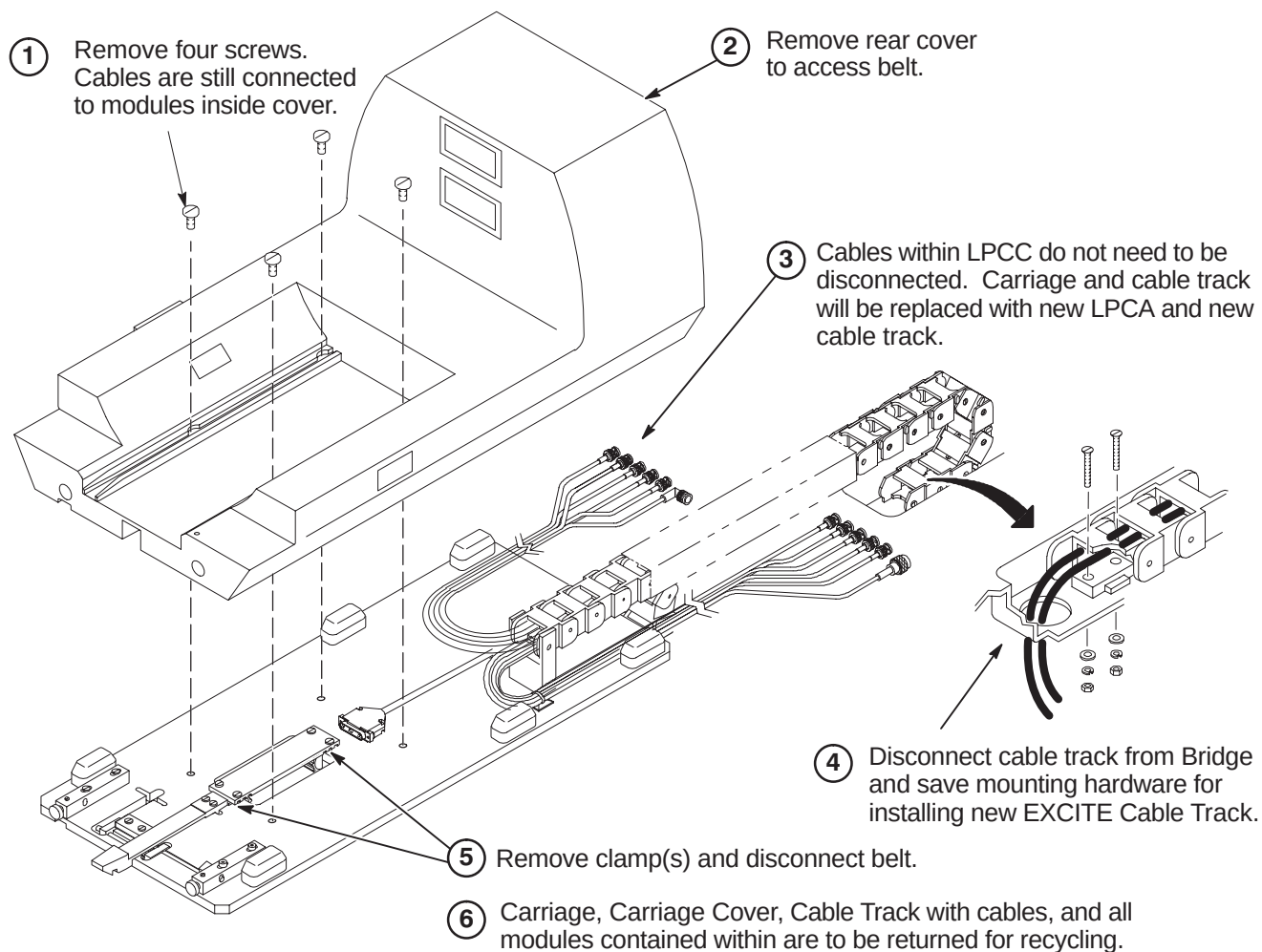
7-1 DE-INSTALL COMPONENTS ON HORIZON OR LX MAGNET ENCLOSURE

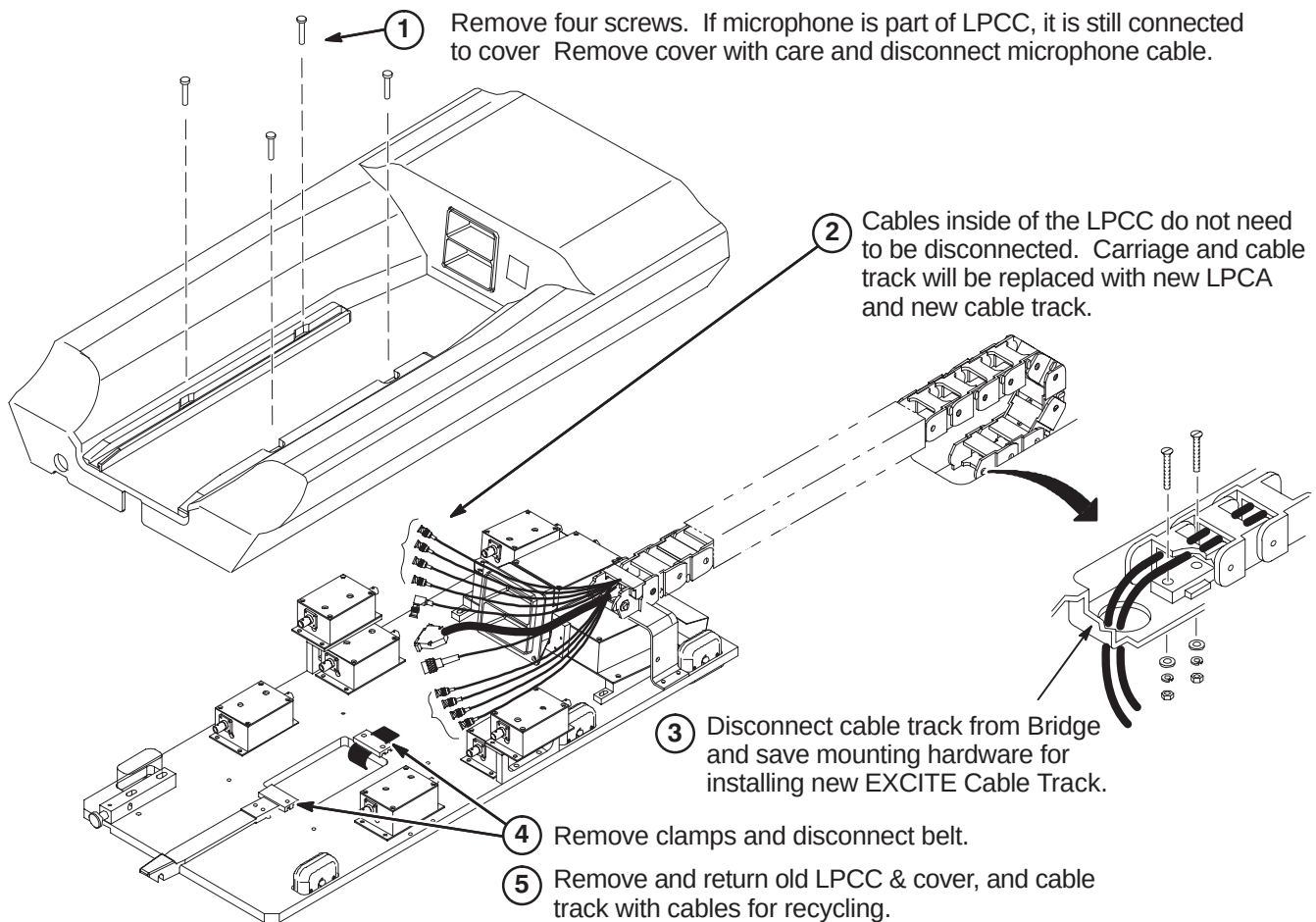
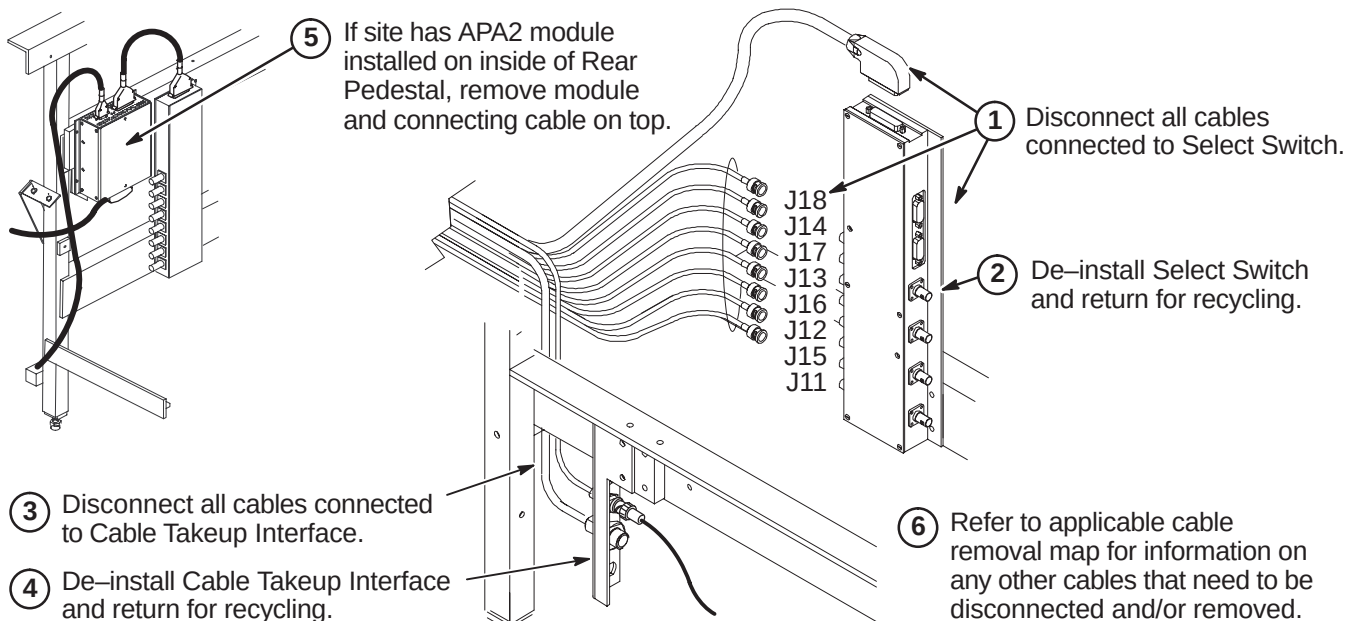
7-1-1 REMOVE EXISTING CARRIAGE COVER & CABLE TRACK

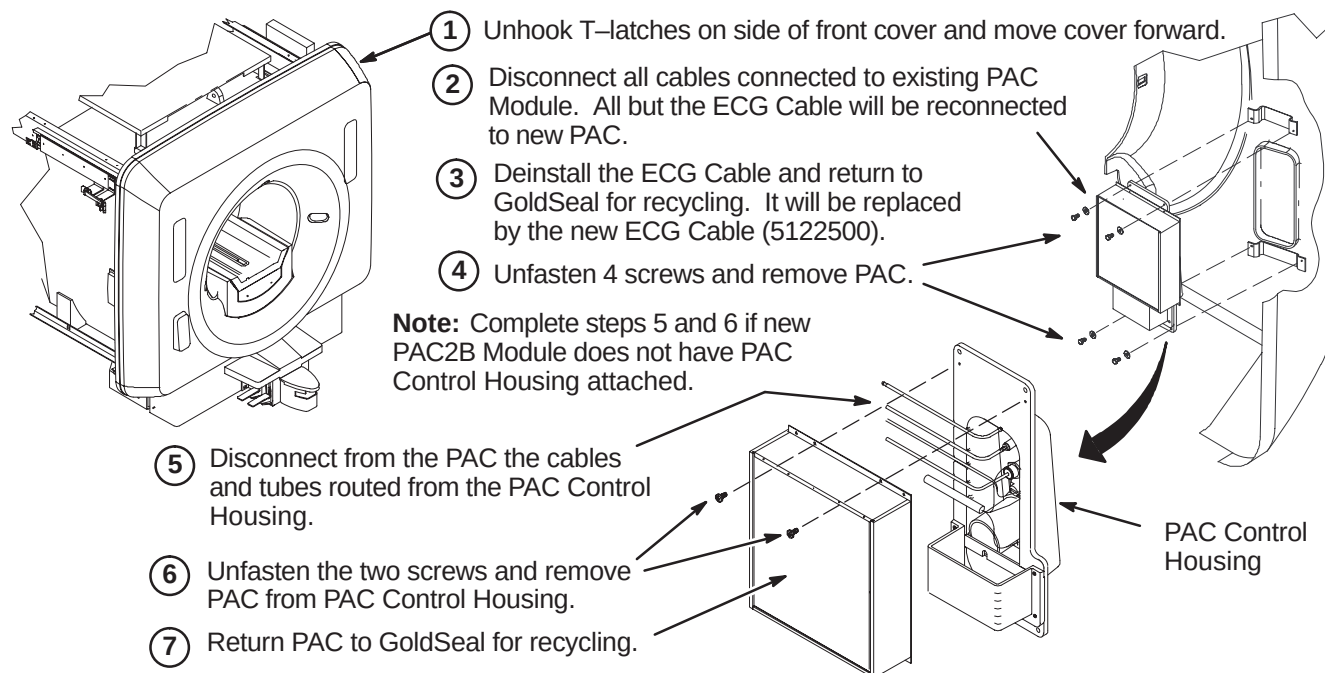
The site being upgraded may have either a High Profile Carriage Cover or a Low Profile Carriage Cover that will need to be replaced with the new LPCA and cable track.

Refer to sections 7-1-1-A or 7-1-1-B for the applicable removal procedure.

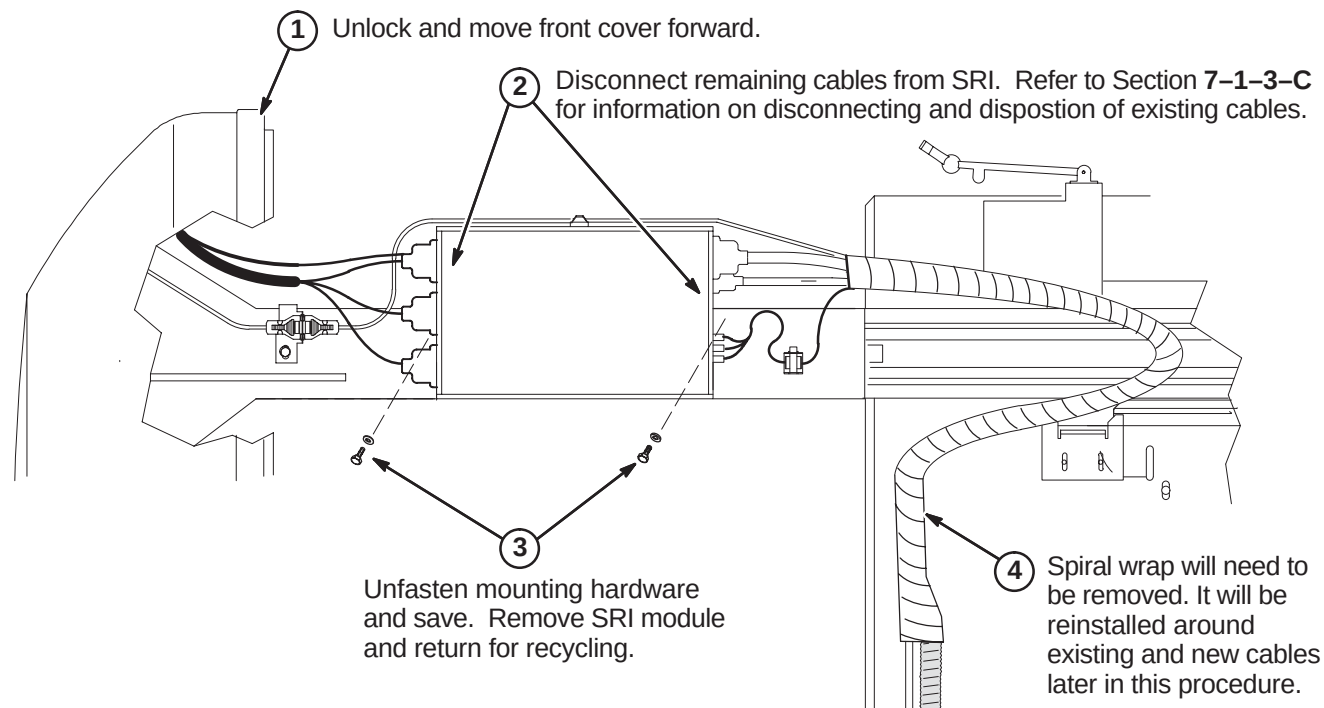
7-1-1-A Remove Existing High Profile Carriage Cover & Cable Track

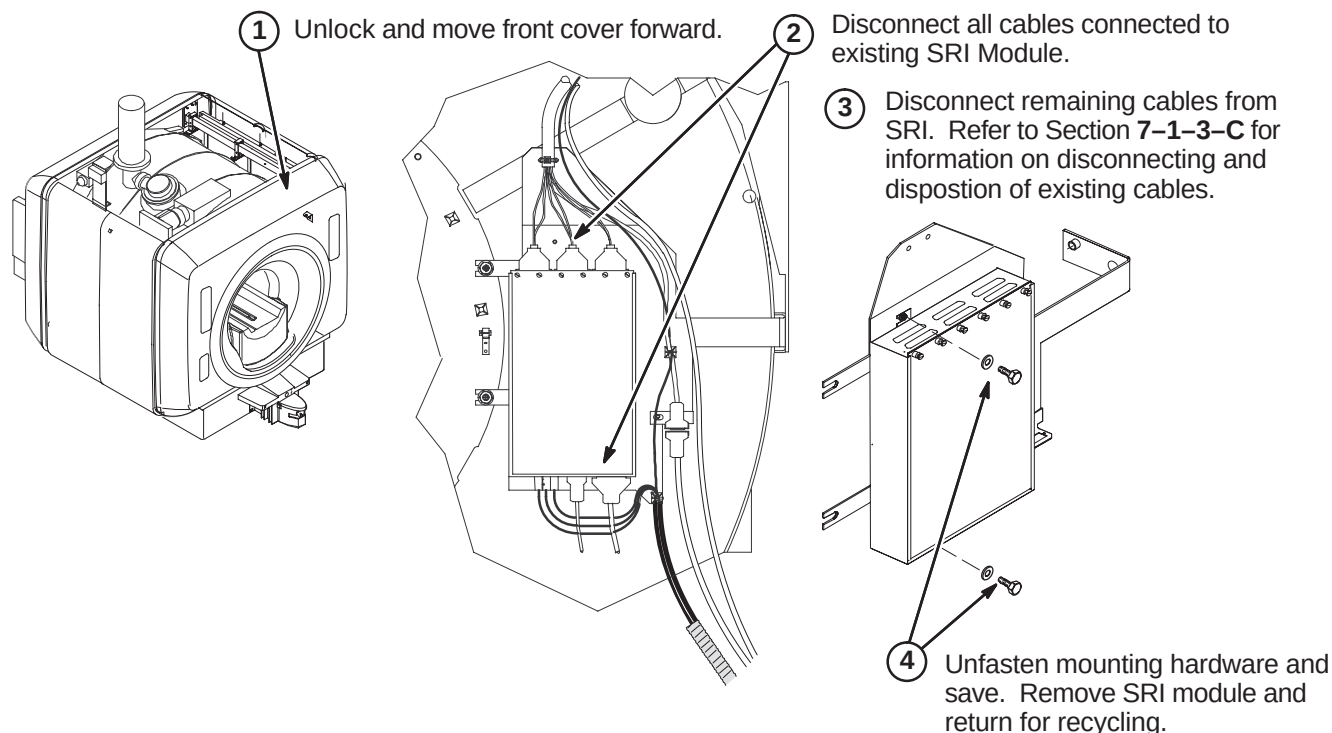
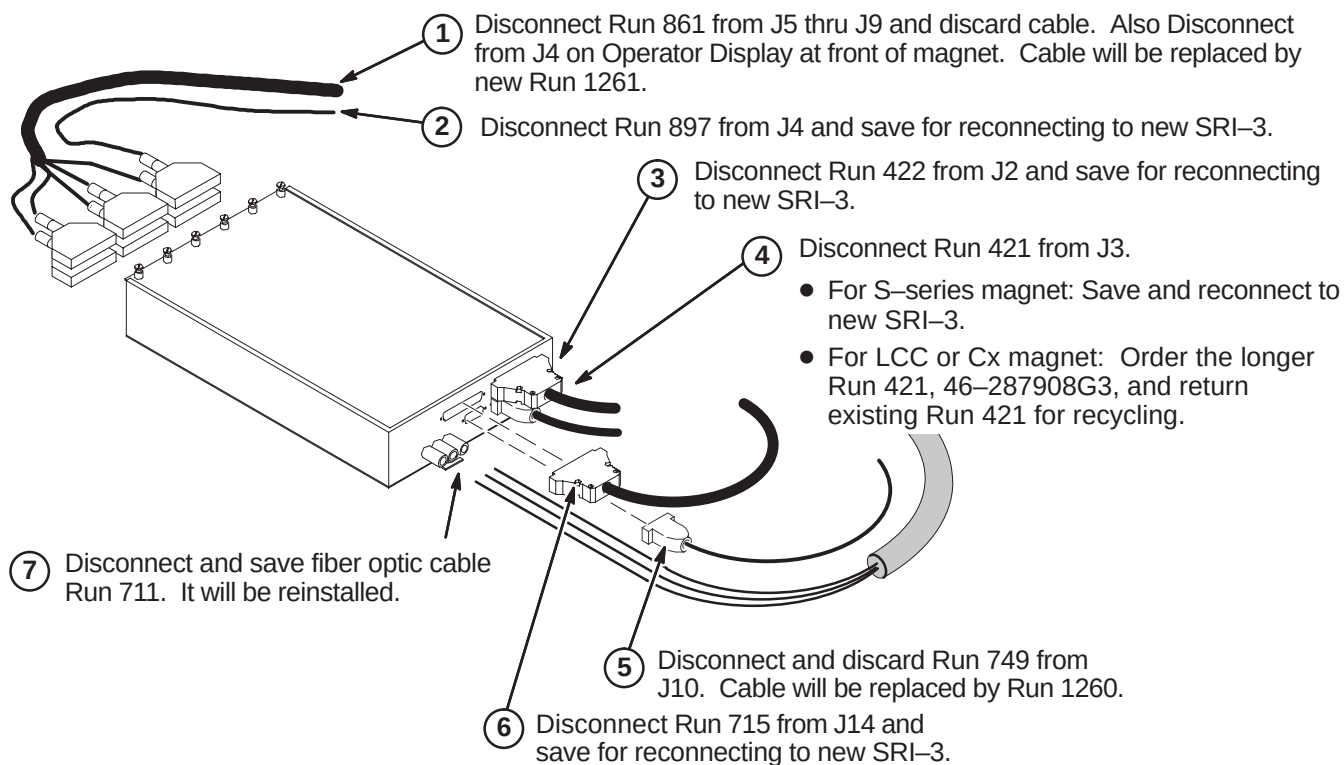


7-1-1-B Remove Existing Low Profile Carriage Cover & Cable Track**7-1-1-C Disconnect Cables In Rear Pedestal**

7-1-2 REMOVE PAC MODULE**7-1-3 REMOVE SRI MODULE**

Section 7-1-3-A contains instructions for removing the SRI module on an S-Series magnet. Section 7-1-3-B contains instructions for removing the SRI module on an Cx magnet.

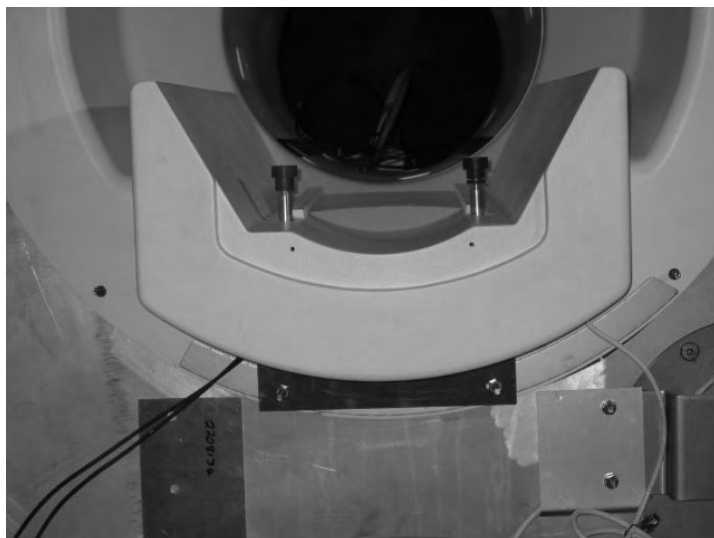
7-1-3-A De-Install SRI Module on S-Series Enclosure

7-1-3-B Remove SRI Module for Cx or LCC Enclosure**7-1-3-C SRI Cable Disposition**

7-2 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for all magnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the HDx upgrade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S-Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade*, for instructions on installing this kit.

For fixed sites only, the mounting of the gradient cables at the rear of the magnet must be upgraded. This upgrade is part of M3335BD, *LCC WideOpen IIQ Upgrade*. Refer to Direction 2404642, provided with Gradient Cable Lead Board collector (5110336), for instructions on how to install the new design.

7-3 INSTALL GRADIENT LEAD BOARD AND RESTRAINT BLOCKS (If Required for Fixed-Sites Only)

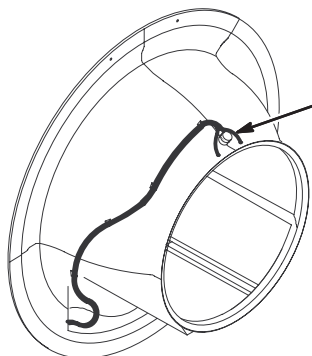
Prior to this upgrade, the site should have been audited to determine if the Gradient Lead Board Kit has been installed. This kit includes parts and instructions for the installation of the Gradient Lead Board at the rear of the magnet.

Also, the gradient cable Restraint Blocks Kit (5186281-2) should have been installed.

These kits are required for all magnets with Horizon style enclosures. If these upgrades has not been performed on the existing enclosure, then the following should have been ordered and it must be installed at this point of the HDx upgrade.

- LCC Early WideOpen – M3335BC
- LCC Horizon – M3335BB
- Cx – M3335BB
- S-Series – M3335BA

7-4 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL



EXCITE HD requires a new Dual Temperature Sensor cable, Run 1260 (5123812), to be installed.

Refer to Sections **10-3-7-B**, **10-3-8**, and **10-3-9** for instructions on removal of existing cable and installing the new cable.

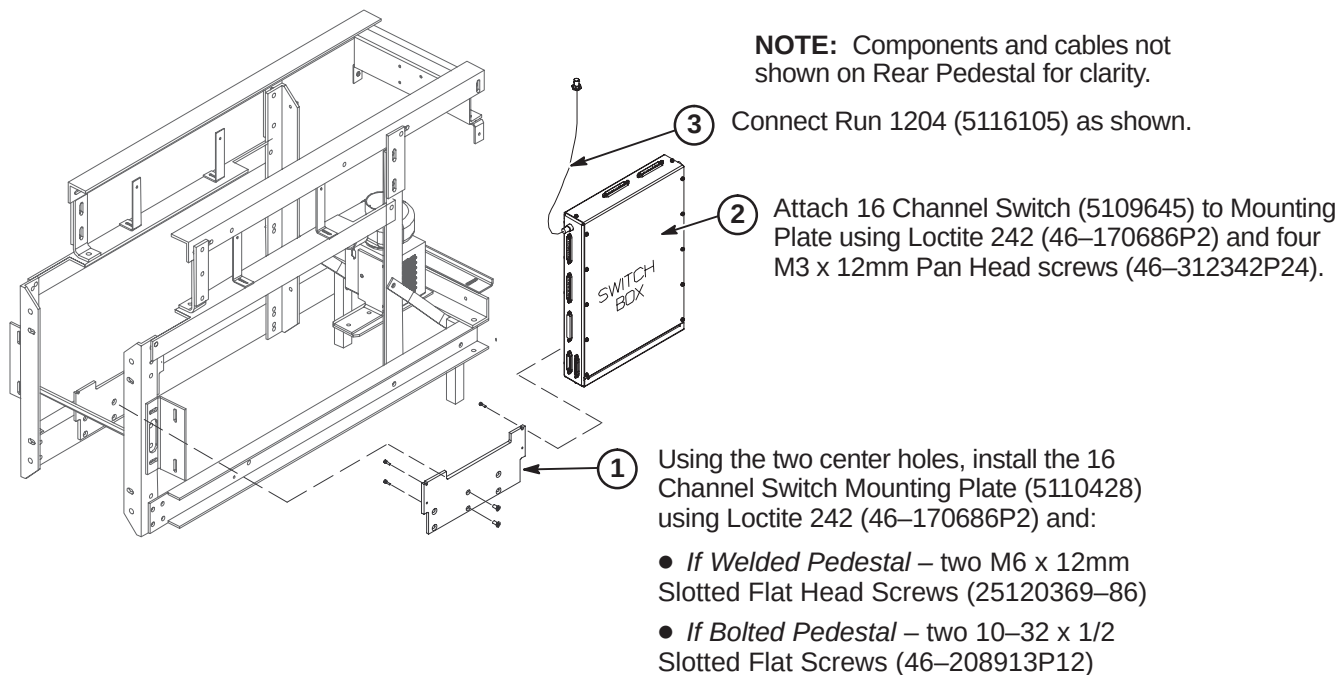
7-5 INSTALL MAGNET MONITOR

EXCITE HD requires Magnet Monitor on all systems. For this upgrade, all the parts and instructions are included in parts collector 5127838, Magnet Monitor Kit for Non-LCC Systems. Refer to Direction 2230681, included with collector, for instructions on installing magnet monitor on this magnet.

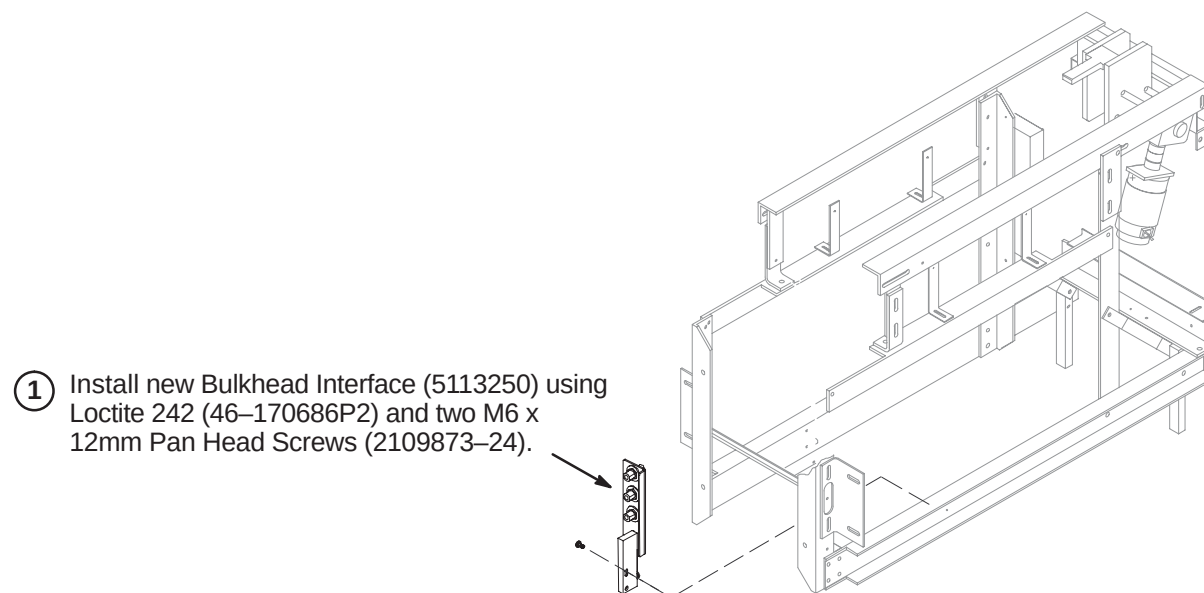
7-6 INSTALL NEW COMPONENTS AND CABLES

7-6-1 INSTALL NEW COMPONENTS ON REAR PEDESTAL

7-6-1-A Install 16 Channel Switch

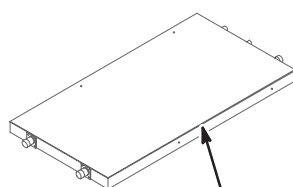


7-6-1-B Install Bulkhead Interface

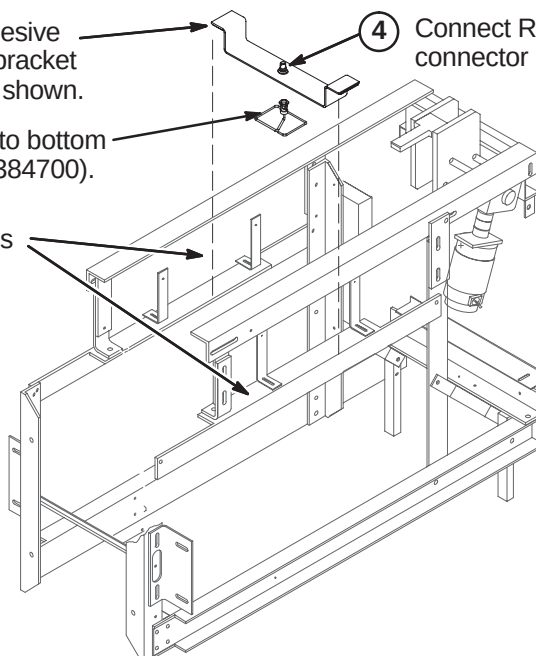


7-6-1-C Install UTNS Antenna

- ③ Peel of liner from bottom of adhesive backed Velcro on each side of bracket and attach to Rear Pedestal as shown.
- ② Attach UTNS Antenna (2355200) to bottom side of UTNS Upgrade bracket (2384700).
- ① Make sure top surfaces of side braces are clean and dry for attachment of adhesive backed Velcro.
- ④ Connect Run 1204 to UTNS Antenna connector at top of bracket.



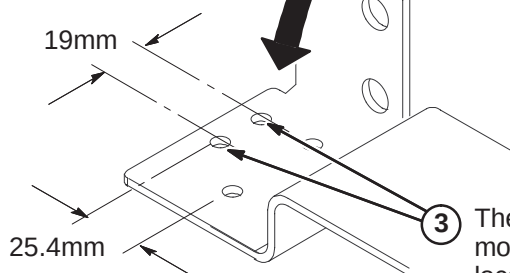
NOTE:
For clarity, Hybrid Splitter not shown in installed position.

**7-6-1-D Install RF Drive Cable Mounting Bracket for LCC and Cx Magnets Only**

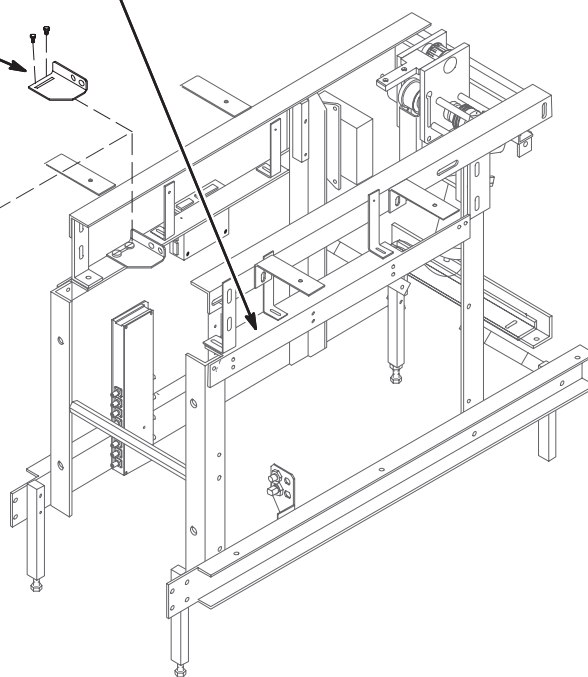
- ① Make sure top surfaces of side braces are clean and dry for attachment of adhesive backed Velcro.
- ② At existing Mounting Bracket, disconnect Runs 743 and 744 and RF Drive Input cables from Coil. Remove and save BNC cable connectors. De-install and discard bracket. Save mounting hardware for installation of new plastic bracket.

Note: Make sure above bracket is de-installed. Leaving the metal bracket on frame may cause intermittent dynamic disable faults.

- ④ Attach Plastic Bracket to Rear Pedestal using mounting hardware removed in Step 5. Attach with velcro on other side of bracket. Install BNC connectors.
- ⑤ Reconnect Runs 743 & 744 and RF Drive cables.



- ③ The new plastic bracket (2384445) requires two new mounting holes. Drill two 7mm diameter holes at locations dimensioned from the existing holes.



7-6-2 HYBRID SPLITTER UPGRADE

Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

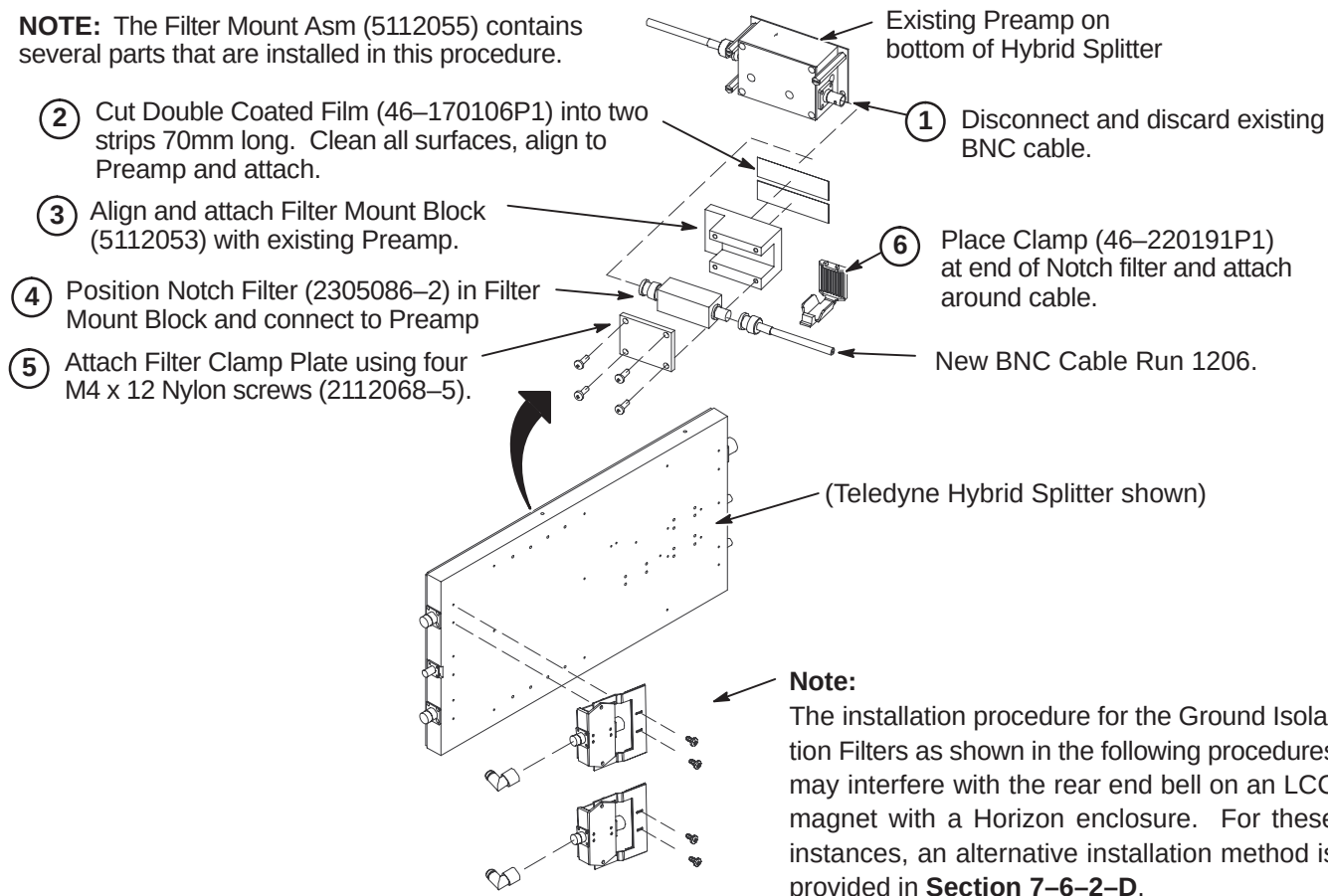
- Refer to Section 7-6-2-A for systems that do not have mount assembly for the Notch Filter
- Refer to Section 7-6-2-B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 7-6-2-C for instructions on installing Isolation Filters on systems with Teledyne Splitter

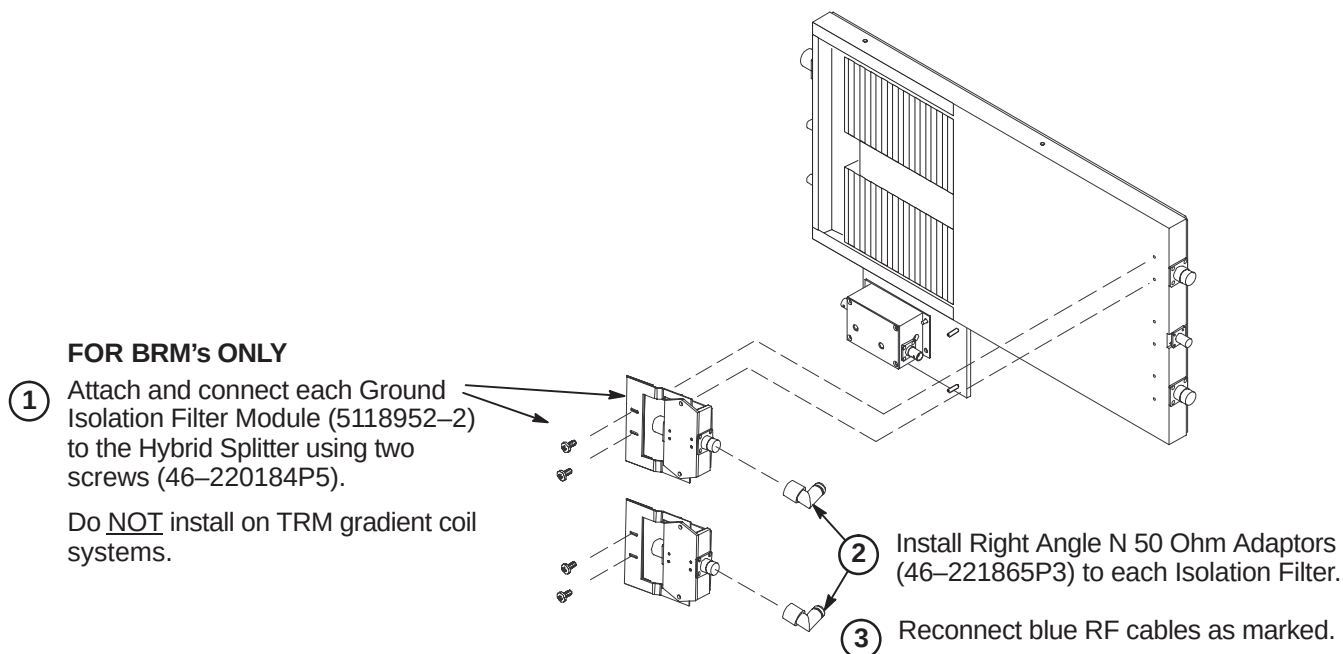
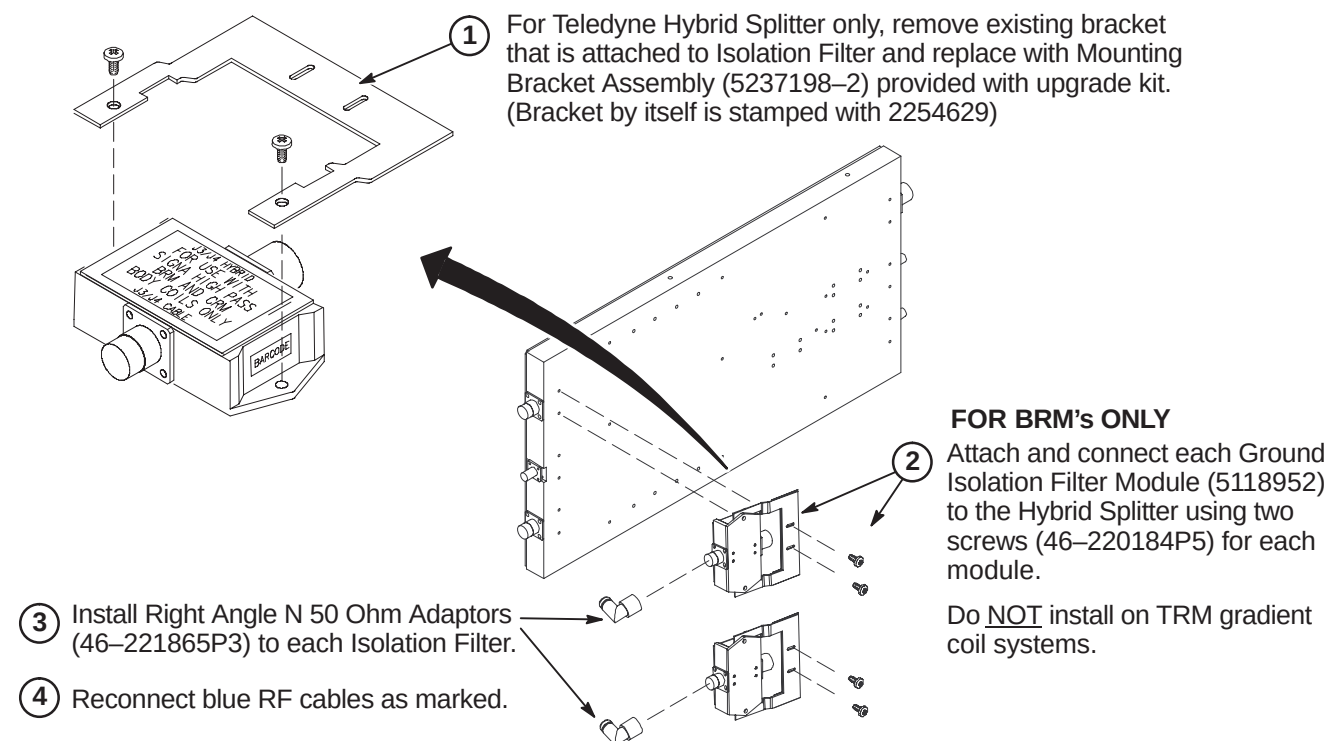
The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils.

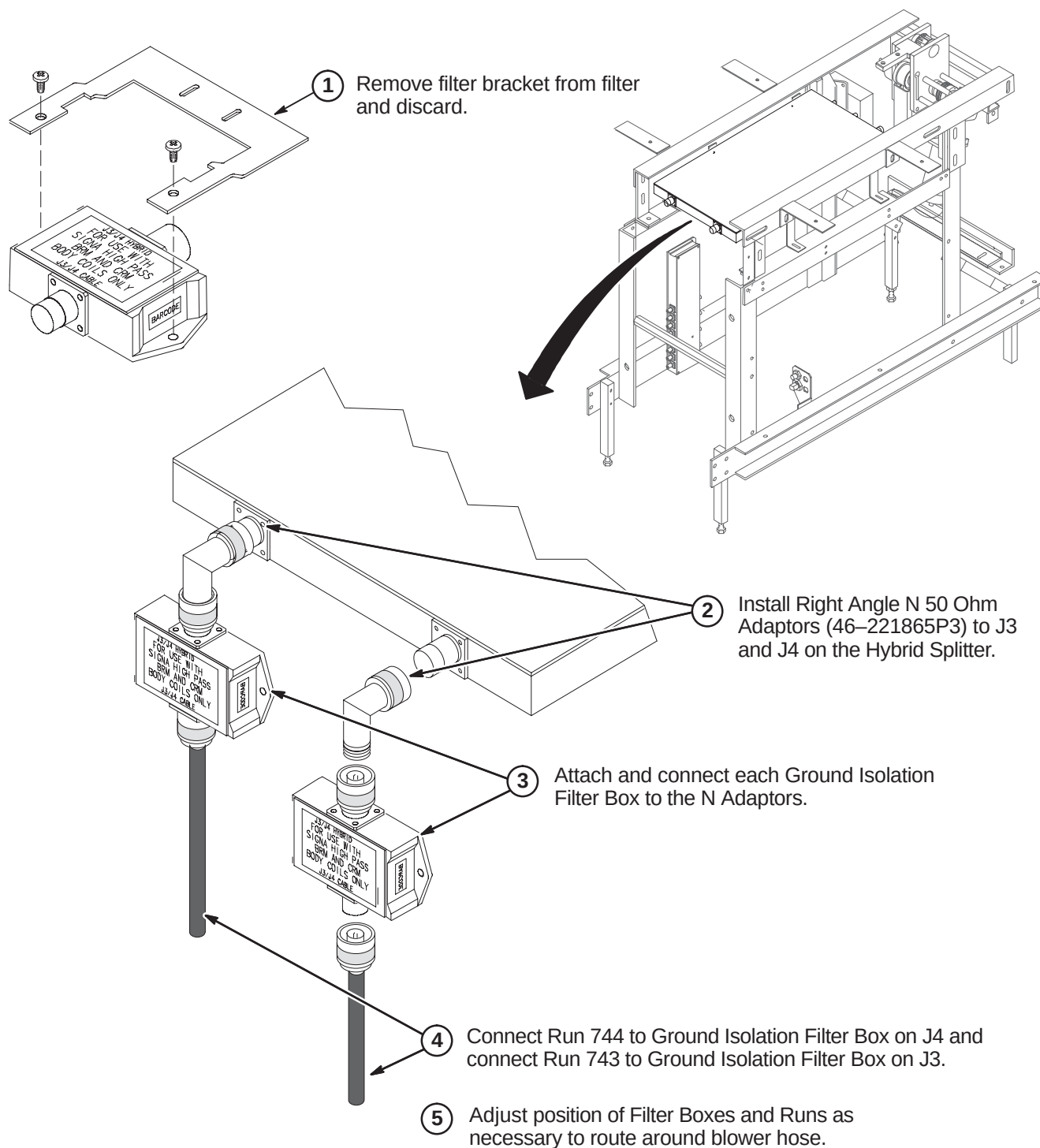
7-6-2-A Install Notch Filter and Filter Mount Block Asm

This Filter Mount assembly may not be required on newer Netcom Hybrid Splitter as it may already have a Notch Filter mounting clamp of a different style installed.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.

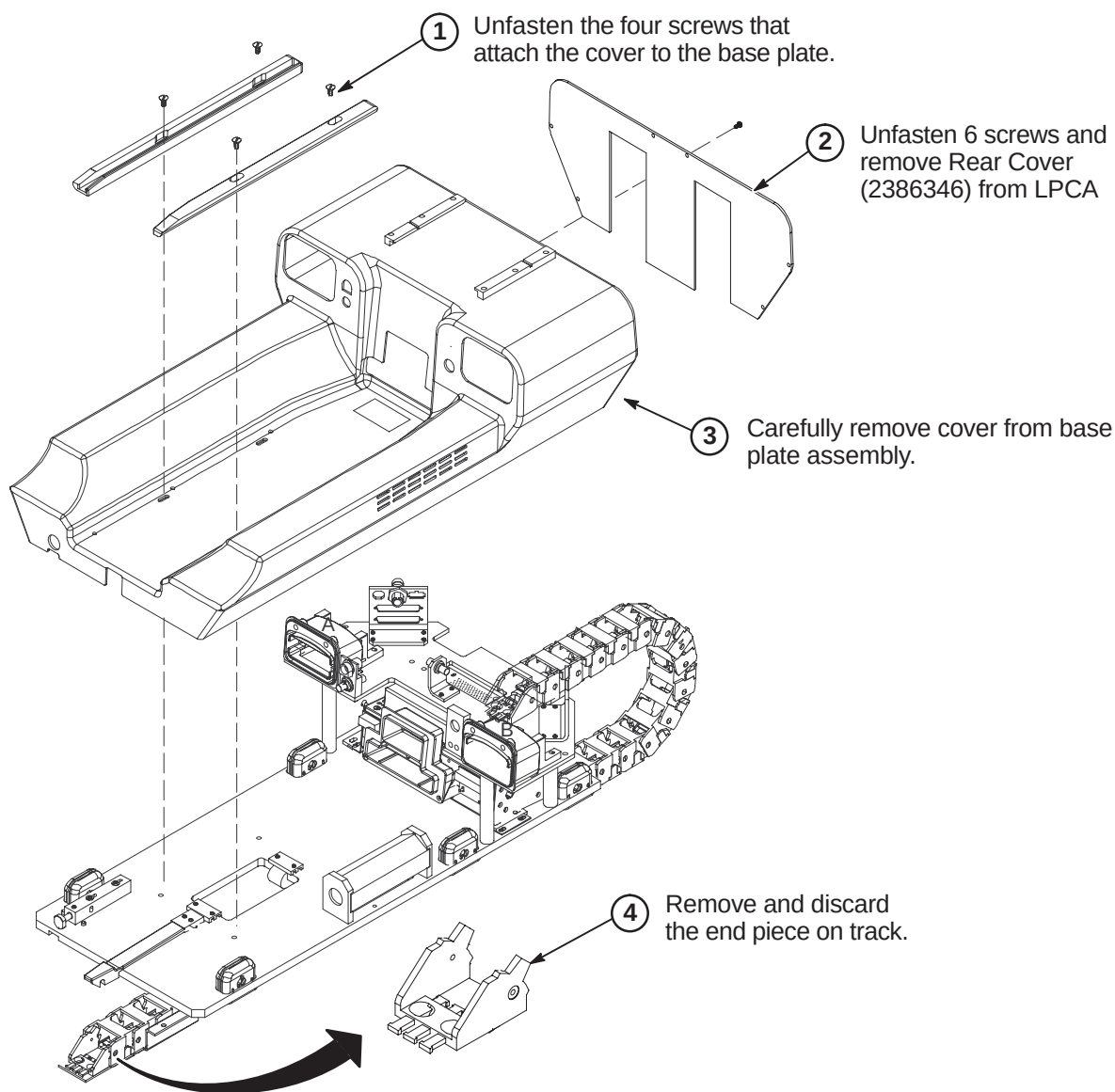


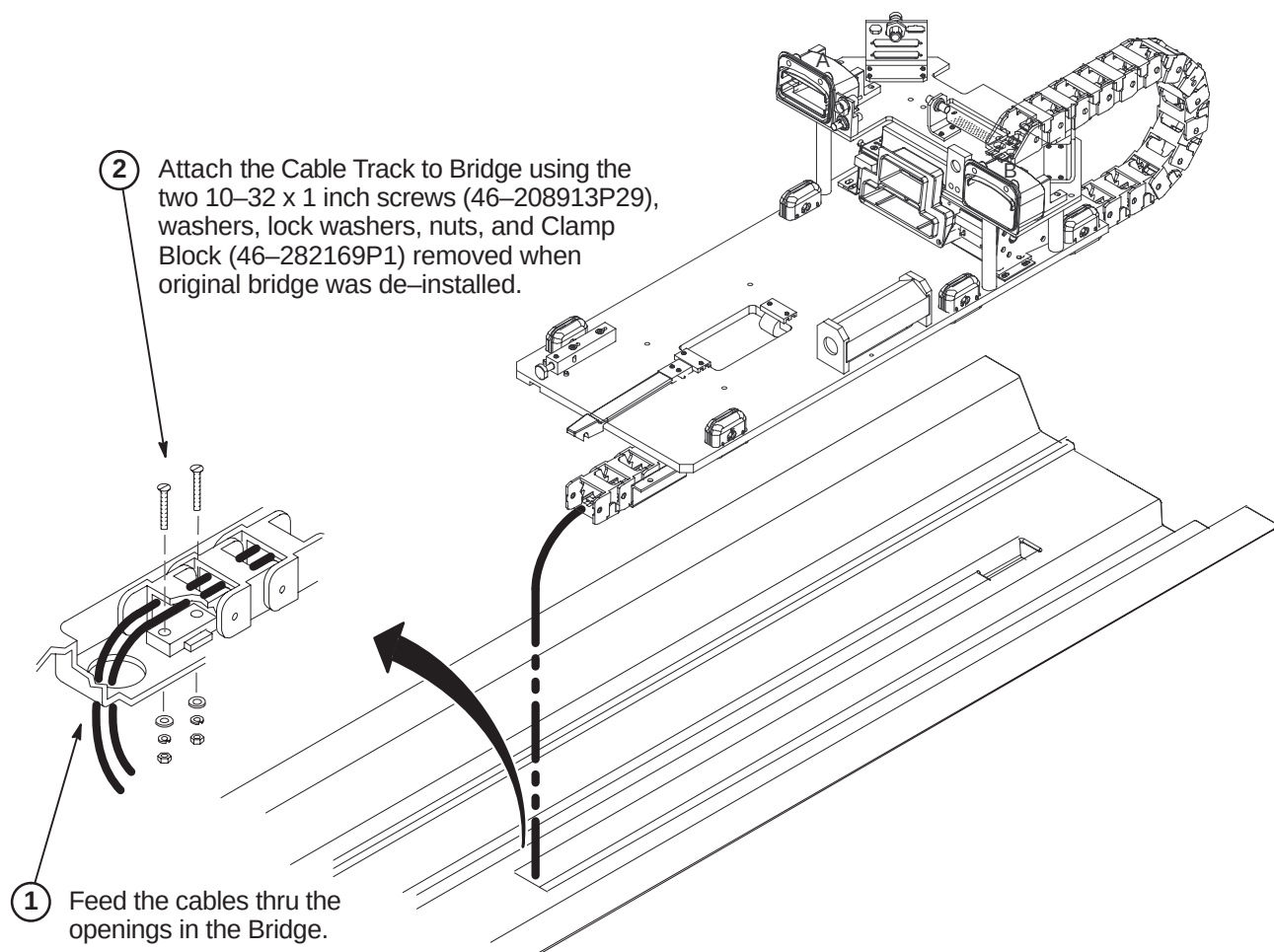
7-6-2-B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)**7-6-2-C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)**

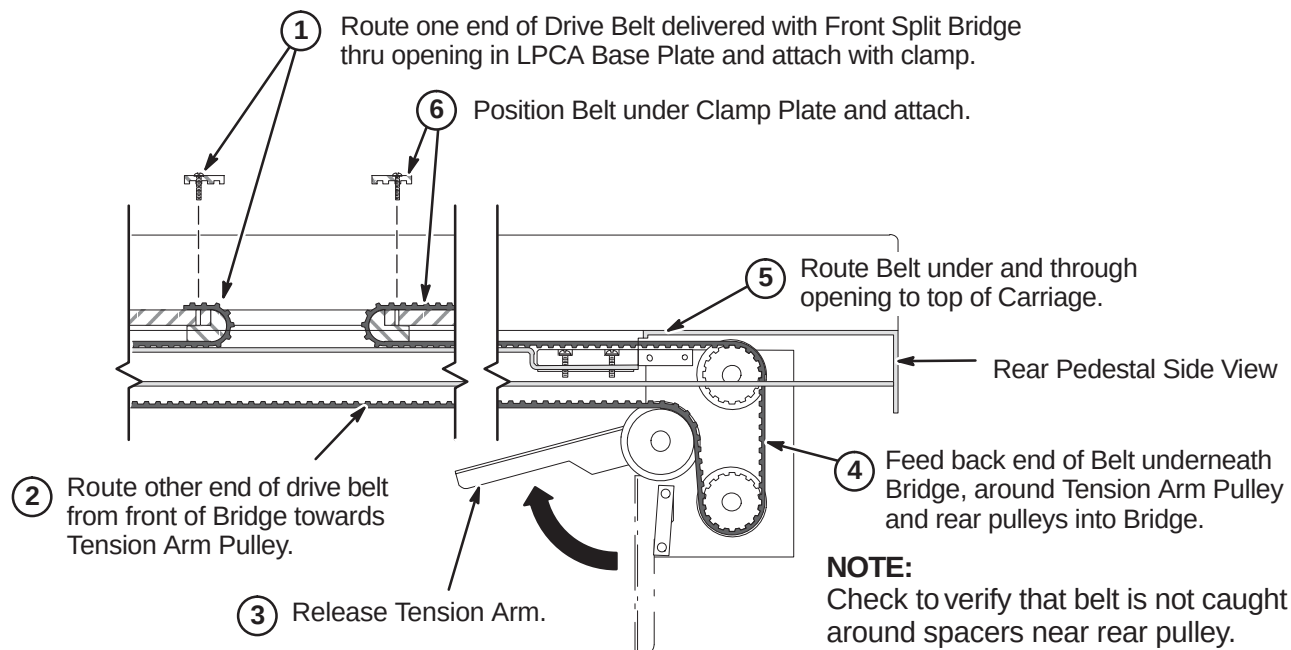
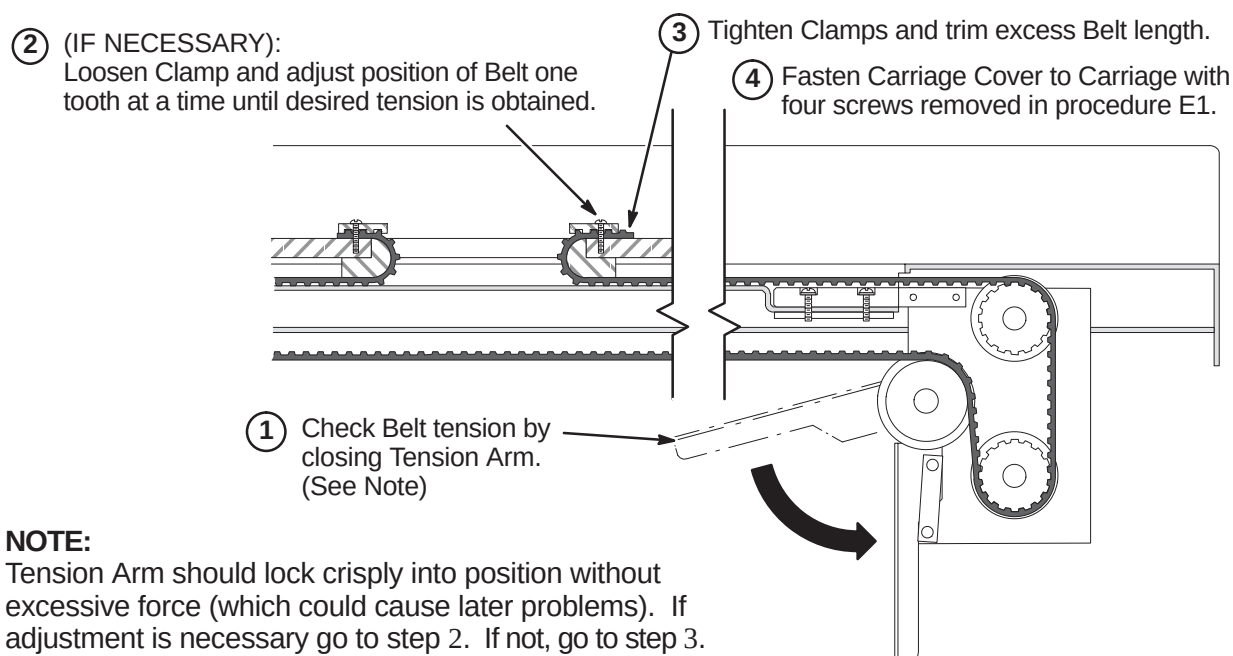
7-6-2-D Alternative Ground Isolation Filter Installation for LCC Magnet with Horizon Enclosure

7-6-3 INSTALL LPCA AND CABLE TRACK

7-6-3-A Remove LPCA Cover



7-6-3-B Attach Cable Track to Bridge

7-6-3-C Route and Attach Drive Belt**7-6-3-D Final Adjustment of Drive Belt**

7-6-3-E Installation of Sensor Flag

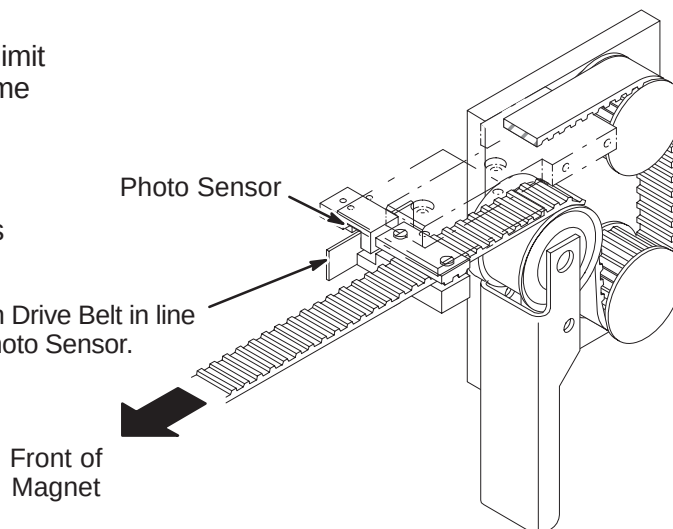
NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:

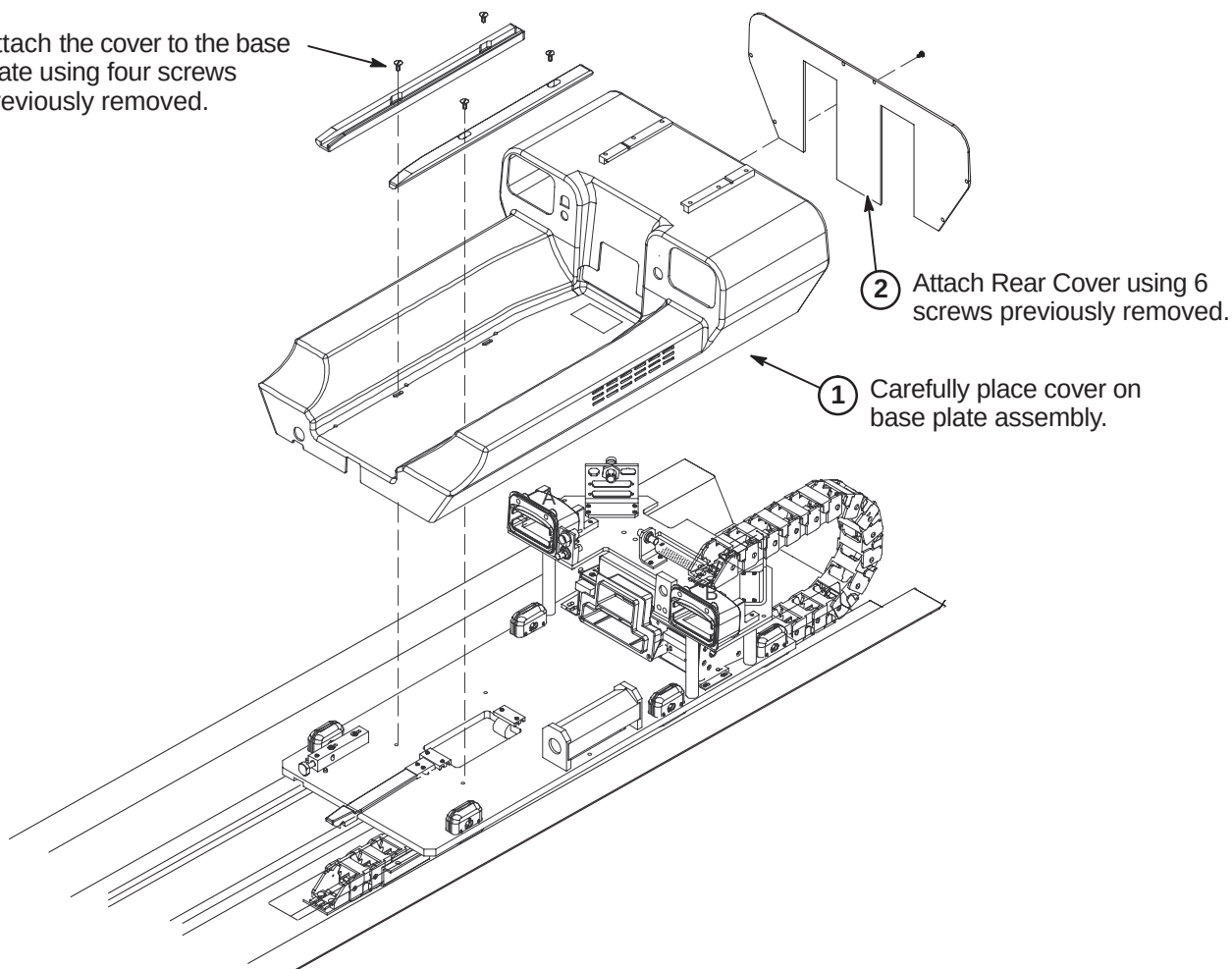
The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



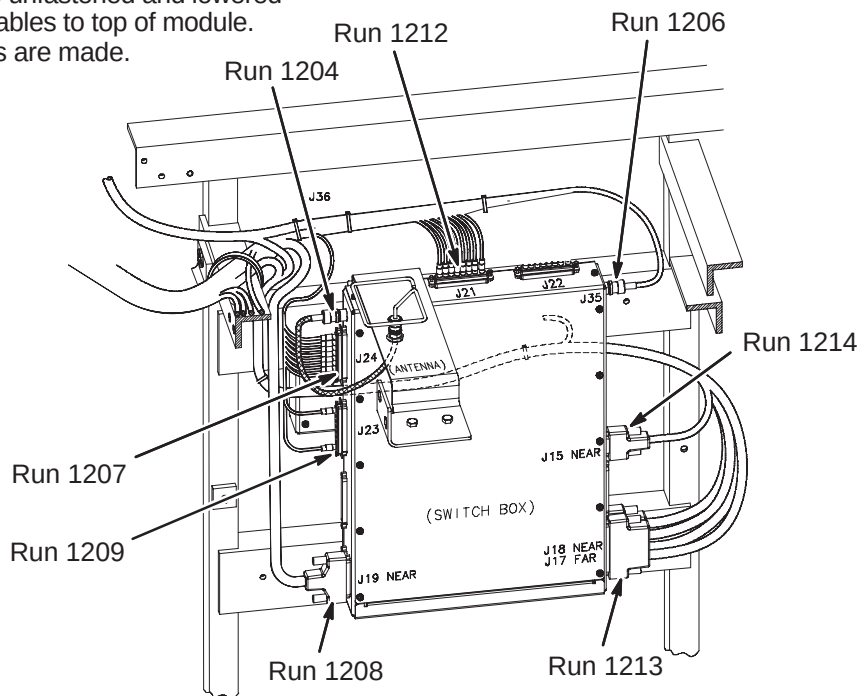
7-6-3-F Installation of LPCA Cover

- ③ Attach the cover to the base plate using four screws previously removed.



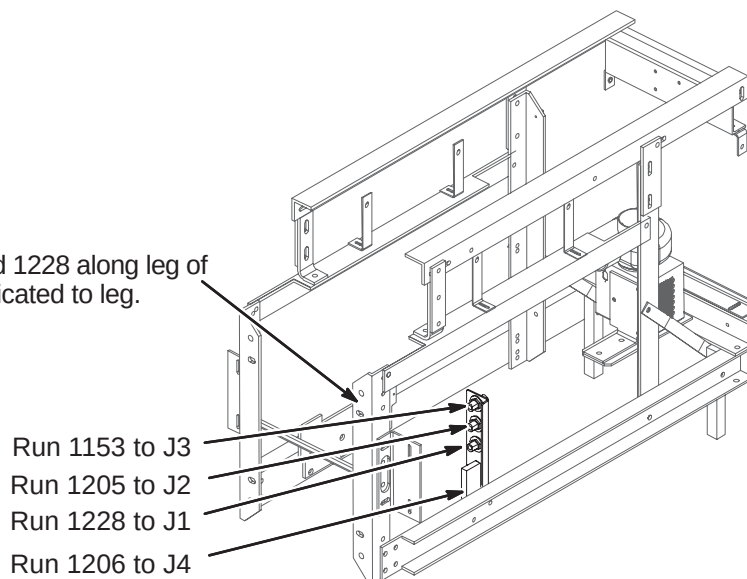
7-6-3-G Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**7-6-3-H Connect Cables from Cable Track to Bulkhead Interface**

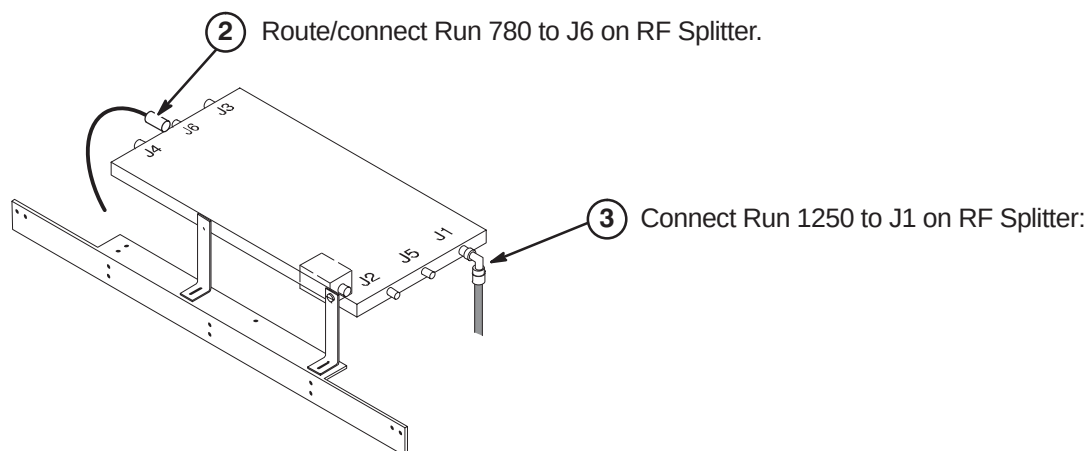
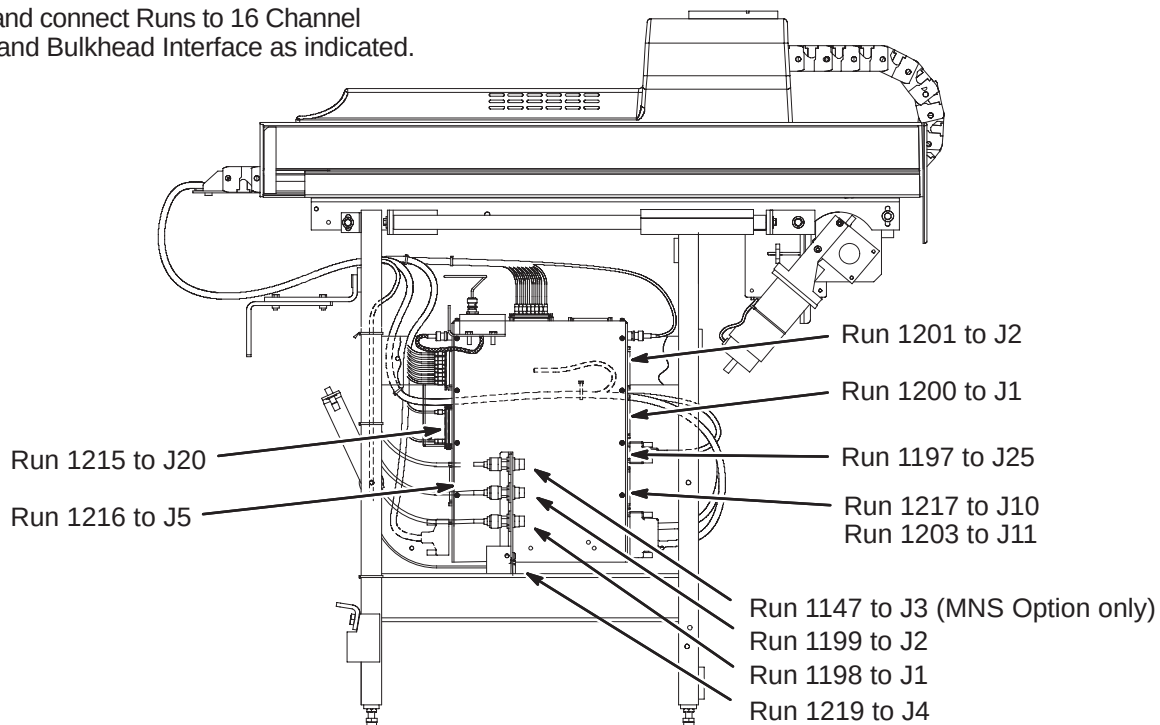
- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.

- ② Connect Runs to Bulkhead Interface as indicated.



7-6-4 CONNECT REMAINING CABLES TO REAR PEDESTAL

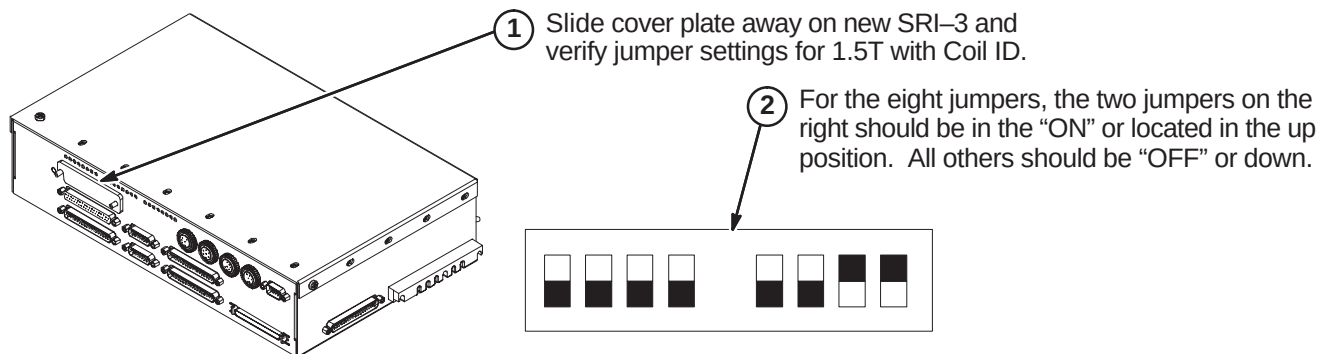
- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



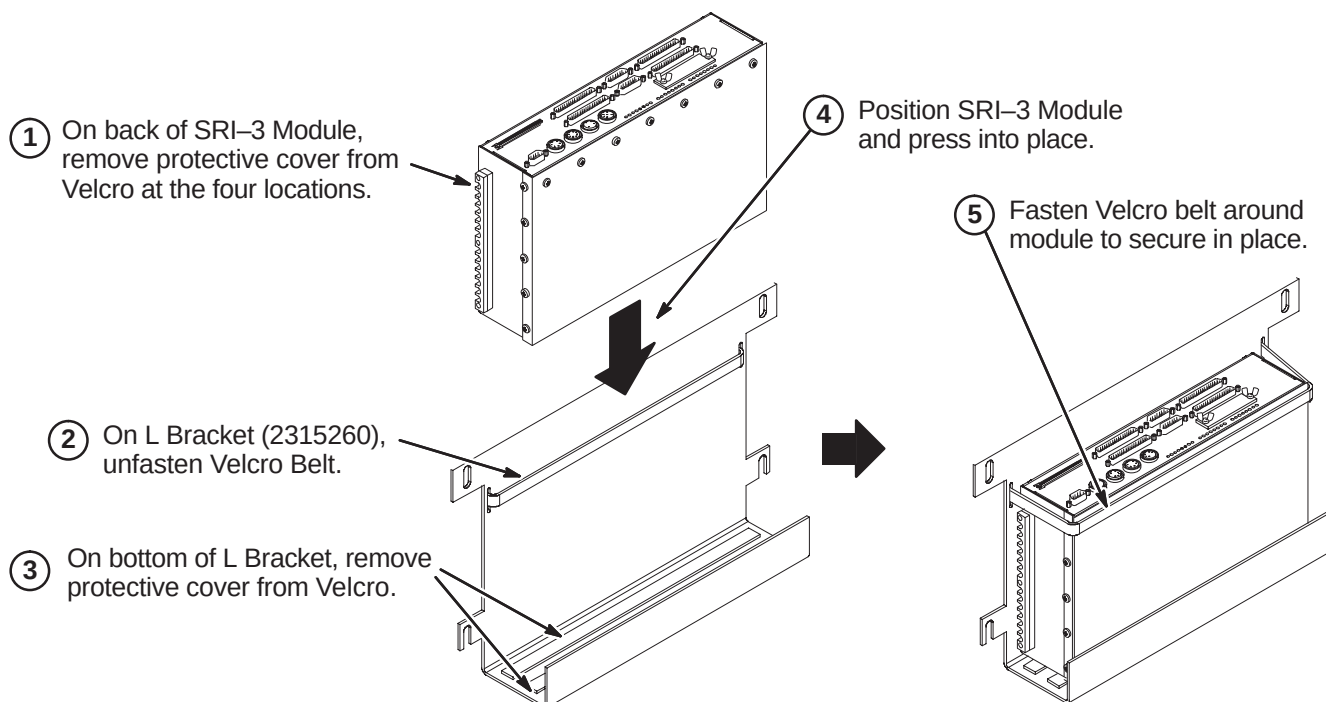
7-6-5 INSTALL SRI-3 MODULE

Sections 7-6-5-C contains instructions for installing the SRI module on an S-Series magnet. Section 7-6-5-D contains instructions for installing the SRI module on an Cx magnet. The remaining procedures in this section are applicable to either magnet type.

7-6-5-A Verify SRI-3 Jumper Settings

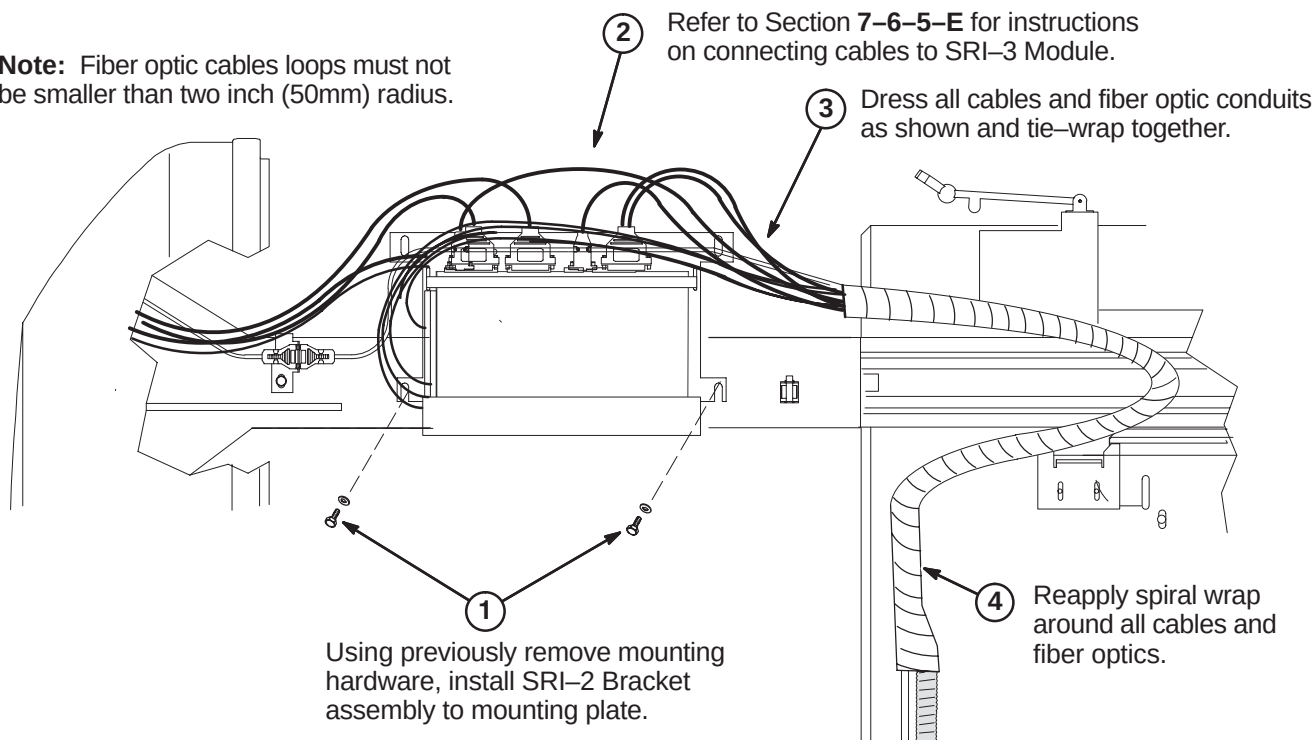


7-6-5-B Position SRI-3 in L Bracket

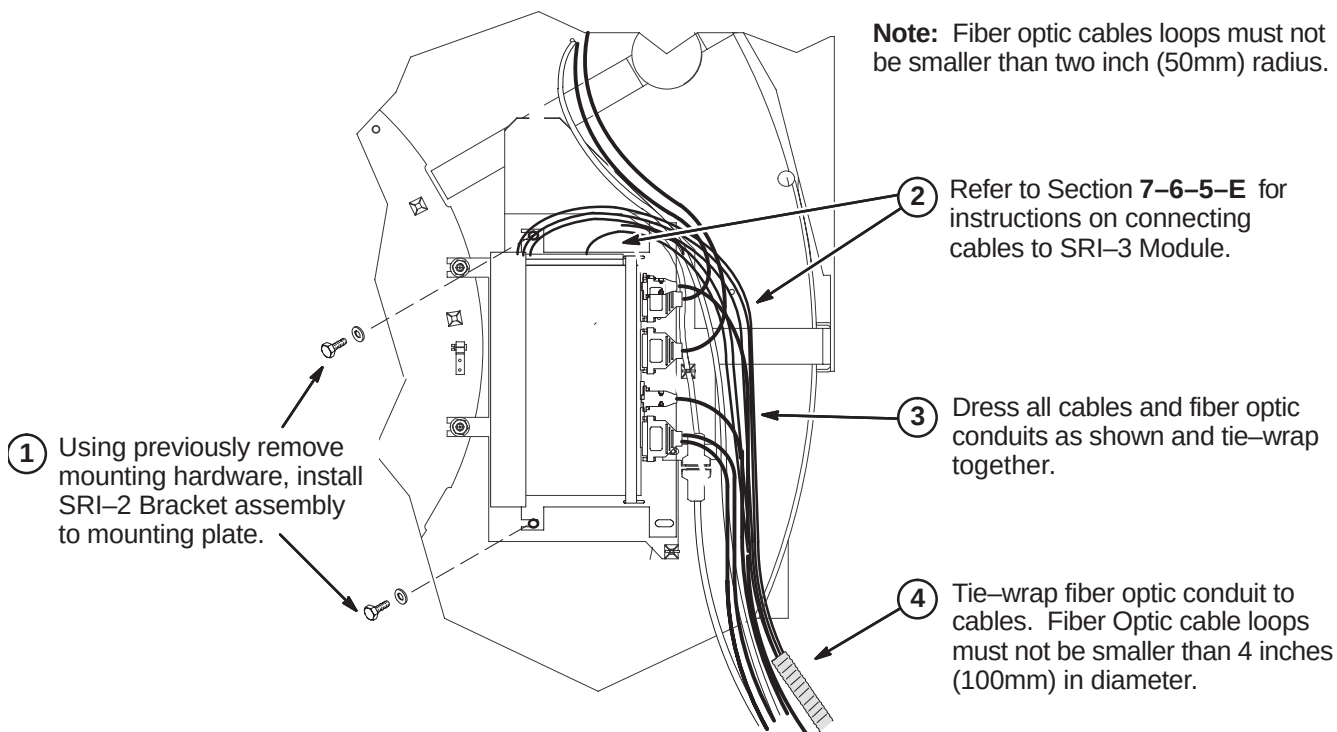


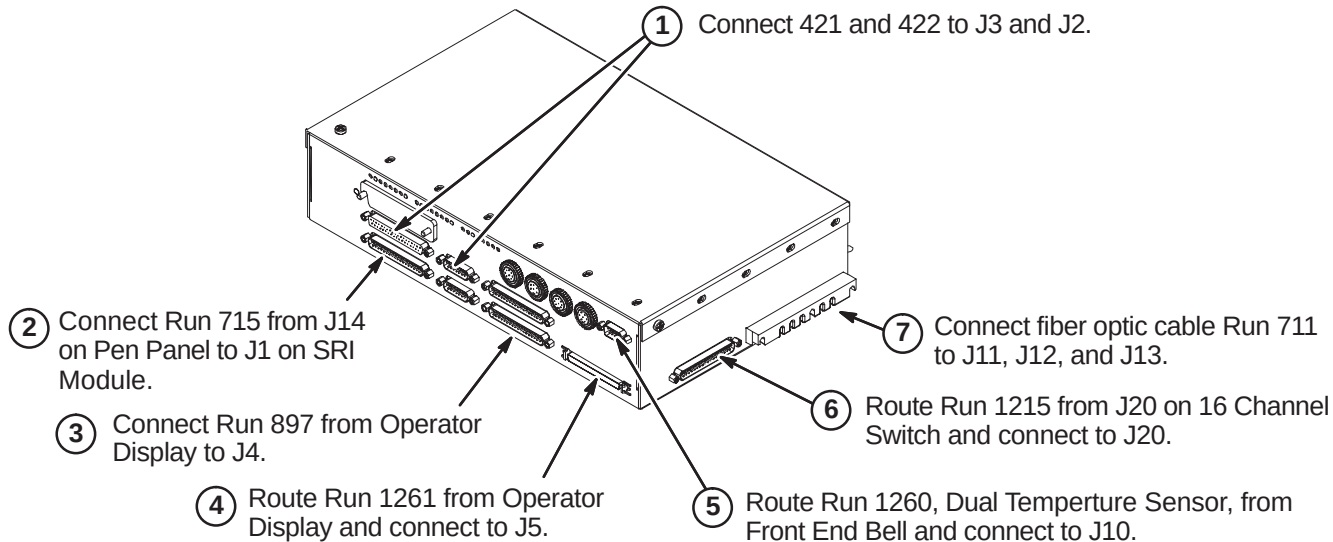
7-6-5-C Install SRI-3 on S-Series Magnet

Note: Fiber optic cables loops must not be smaller than two inch (50mm) radius.

**7-6-5-D Install SRI-3 on Cx or LCC Magnet**

Note: Fiber optic cables loops must not be smaller than two inch (50mm) radius.

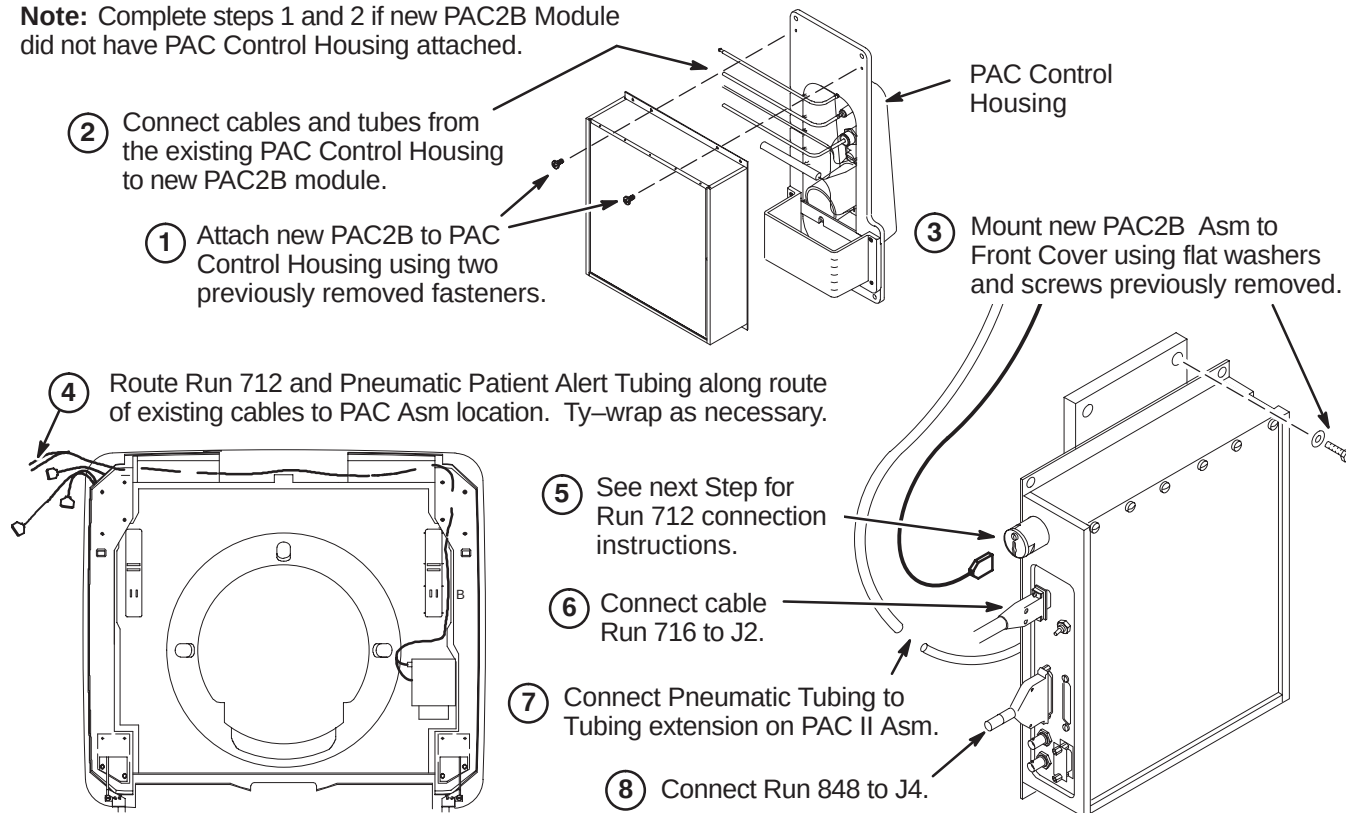


7-6-5-E Connect Cables at SRI-3 Module Location

7-6-6 INSTALL PAC MODULE

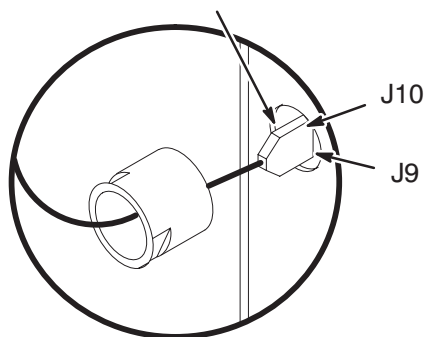
7-6-6-A Install PAC2B and Connect Run 712, 716, & Pneumatic Patient Alert Tubing

Note: Complete steps 1 and 2 if new PAC2B Module did not have PAC Control Housing attached.



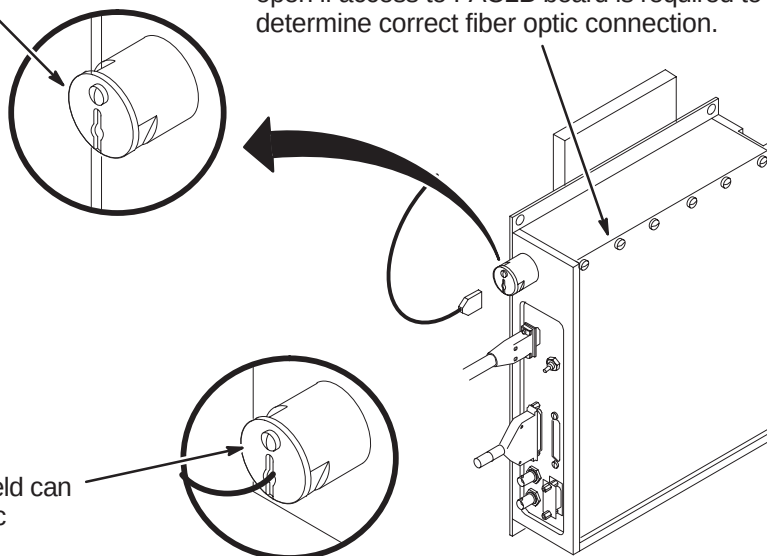
7-6-6-B Connect Run 712

- 1 Remove shield cover by removing one screw and unscrew shield from module.
- 2 Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.

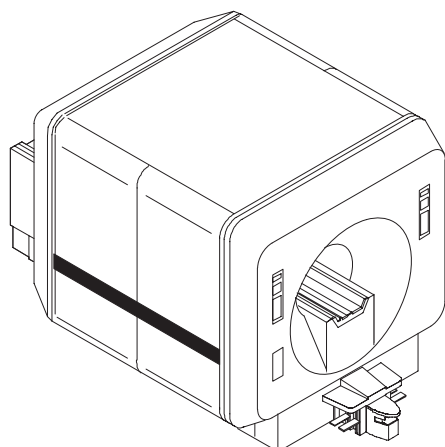


- 3 Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

Note: Loosen screws and slide module cover open if access to PAC2B board is required to determine correct fiber optic connection.

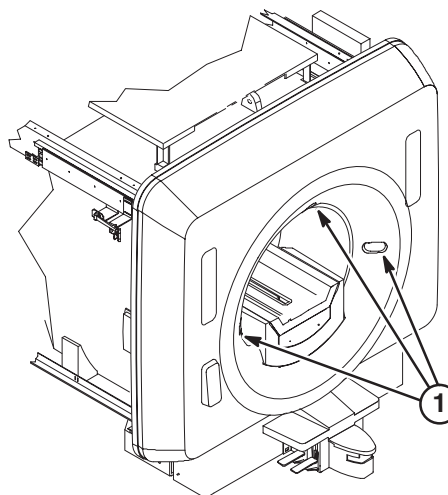


7-7 INSTALL COVERS AND RATING PLATES



① Replace all covers on enclosure and rear pedestal.

7-8 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS



A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

① For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



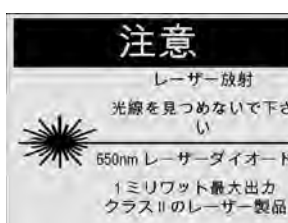
Spanish



Italian



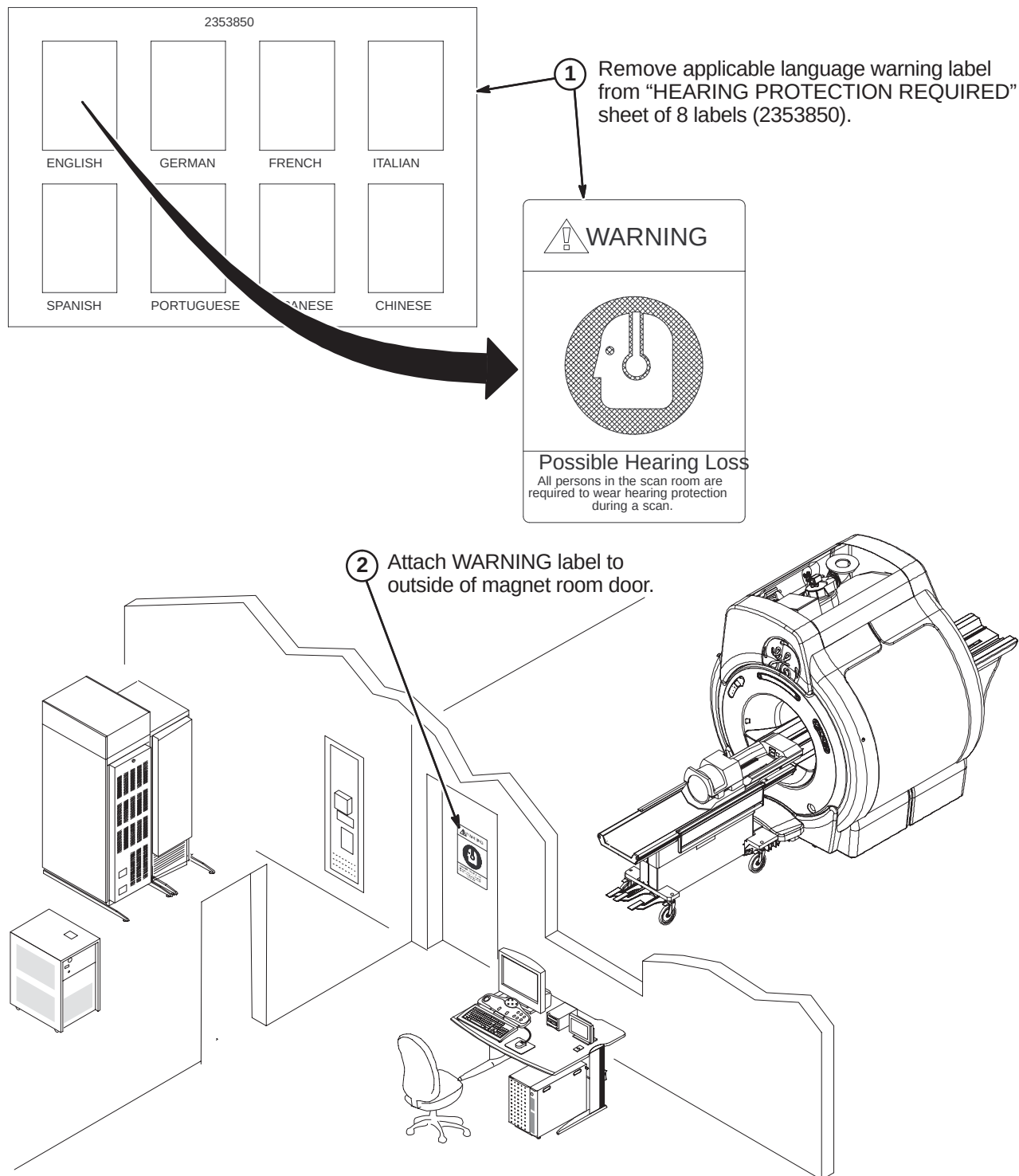
Swedish



Japanese



Chinese

7-9 INSTALL HEARING LOSS WARNING SIGN

SECTION 8 – S-SERIES ADVANTAGE UPGRADE

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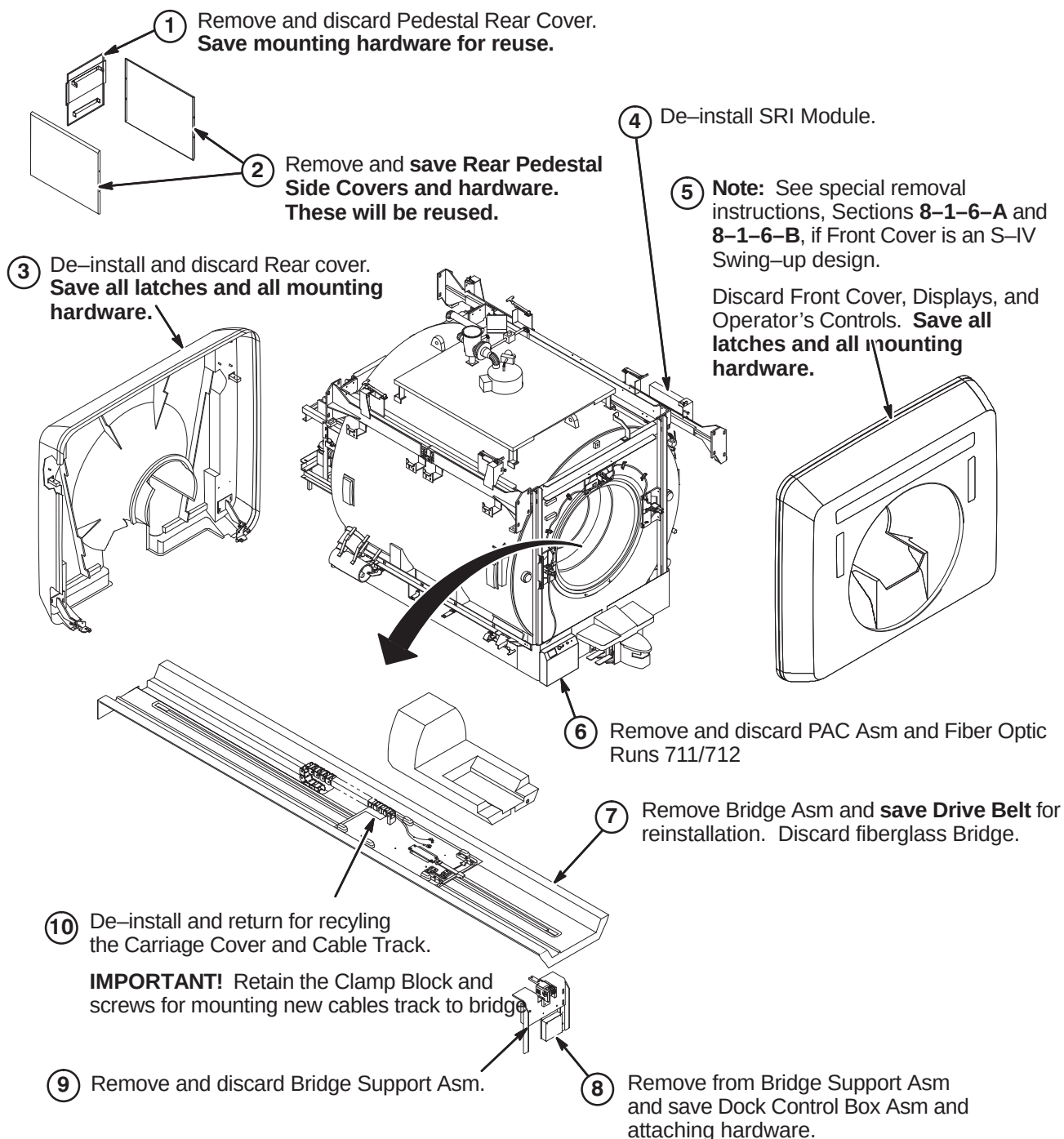
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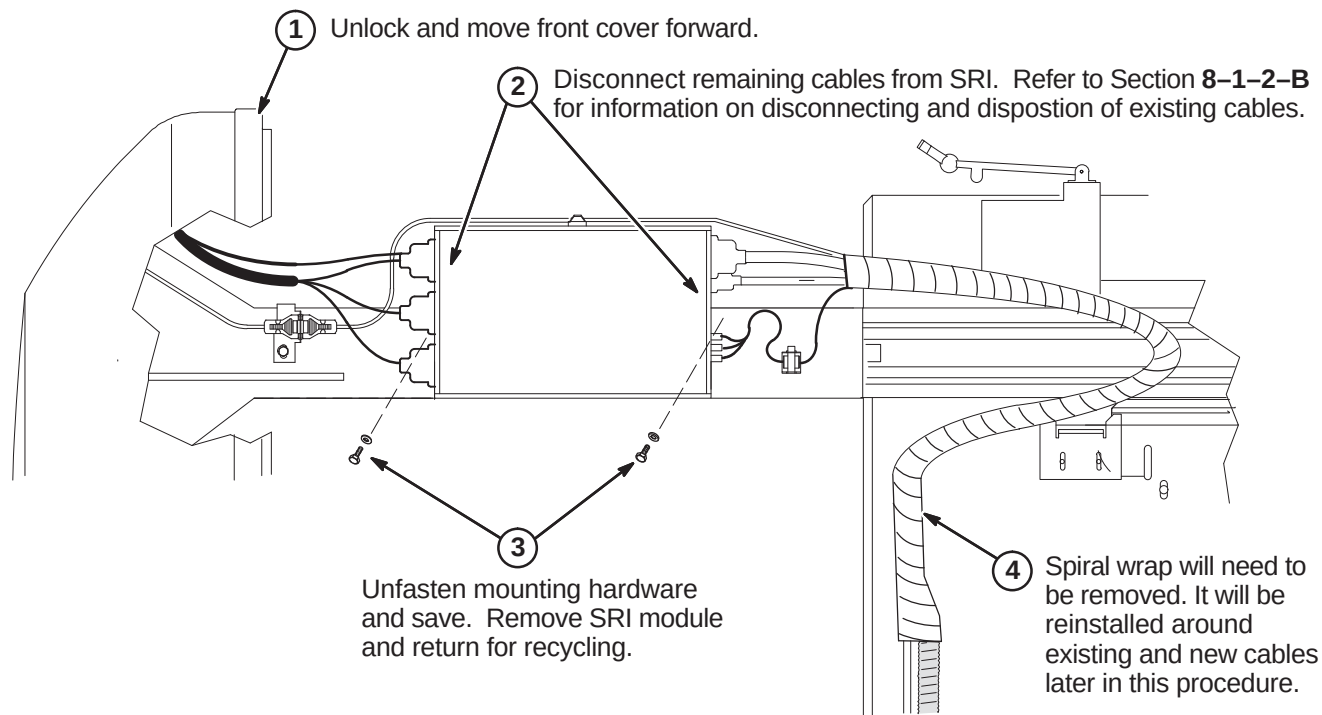
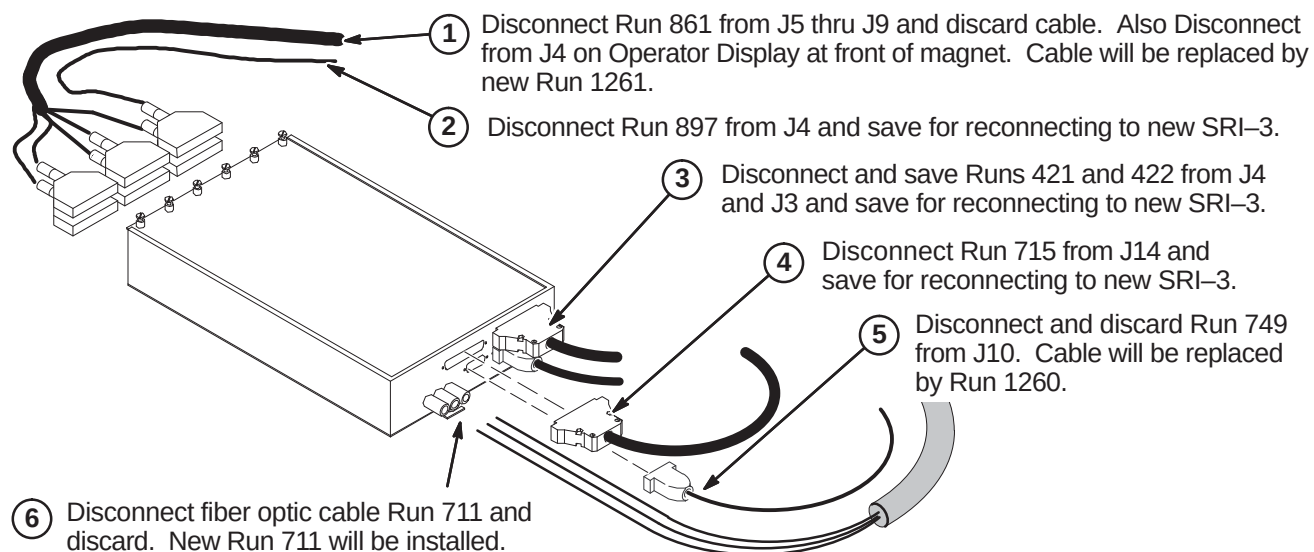
8-1 DE-INSTALL ADVANTAGE S-SERIES MAGNET ENCLOSURE

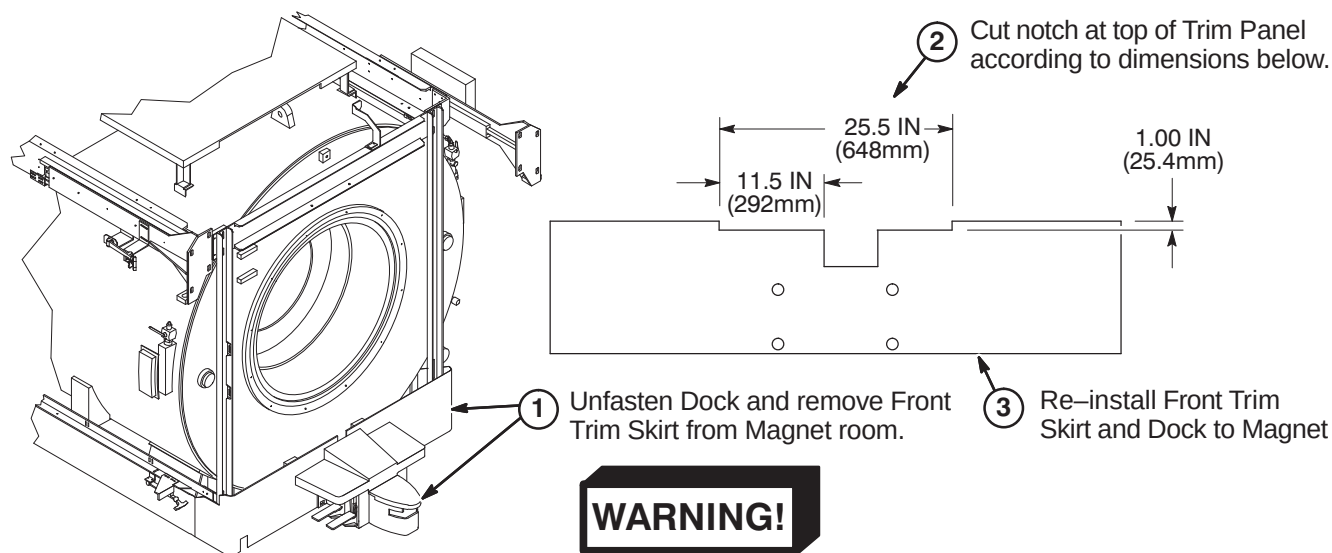
Refer to Section 4, Table 4-1, for information on cable removal. Also refer to applicable cable map for additional information on cable removal.

Discarded Material Return Policy – Refer to Section for procedures on how to handle all de-installed material that is designated as “discard” in the following procedures.

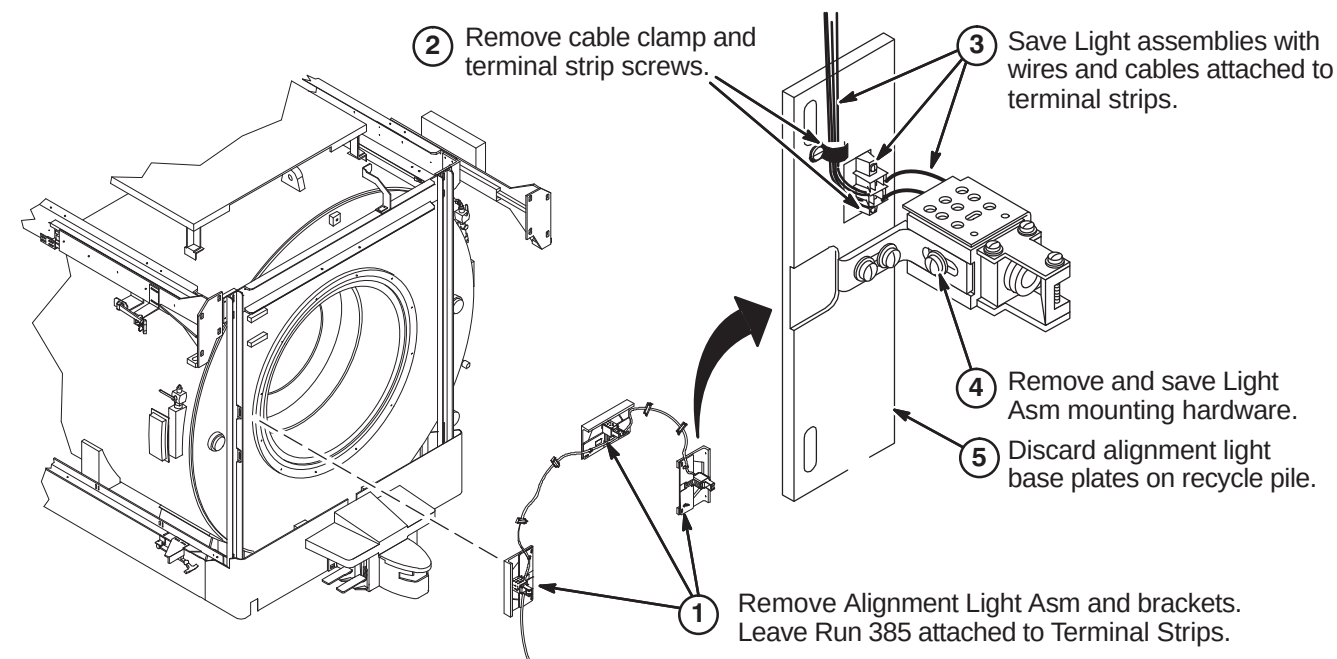
8-1-1 DE-INSTALL COVERS, BRIDGE, BRIDGE SUPPORT, DSIM/SRI AND PAC ASM



8-1-2 REMOVE EXISTING SRI MODULE**8-1-2-A De-Install SRI Module****8-1-2-B SRI Cable Disposition**

8-1-3 REMOVE AND NOTCH FRONT TRIM COVER**FERROUS MATERIAL HAZARD!**

Dock Assembly contains ferrous materials. When dock is not fastened to the magnet, two people must apply downward pressure to the top of the dock at all times. Push the dock slowly along the floor away from the magnet. When the distance from the magnet is roughly where the end of the patient table would be, the dock may be carefully lifted and removed from the magnet room.

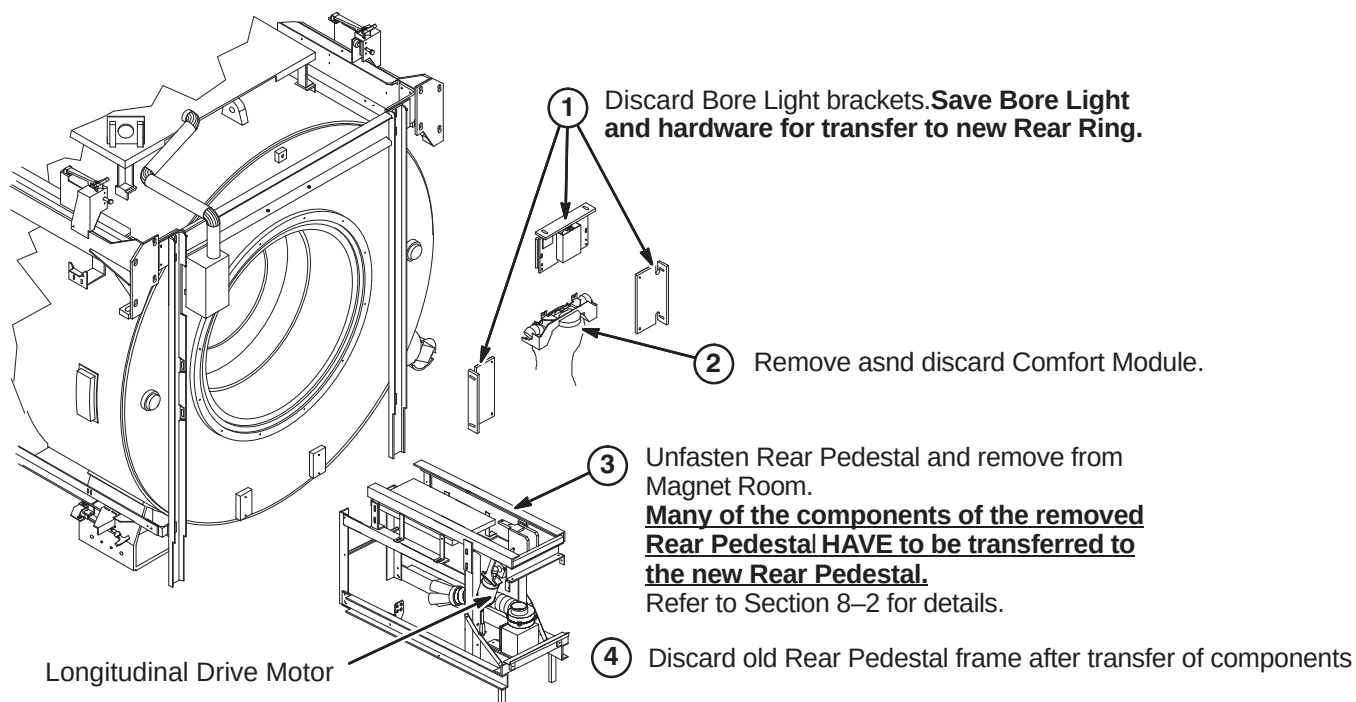
8-1-4 REMOVE ALIGNMENT LIGHTS

8-1-5 REMOVE REAR PEDESTAL, BORE LIGHT, & COMFORT MODULE

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

WHEN REMOVING REAR PEDESTAL, IT MUST BE MOVED DIRECTLY BACK FROM THE MAGNET AND NOT ROTATED. USE PROPER SAFETY AND LIFTING TECHNIQUES.

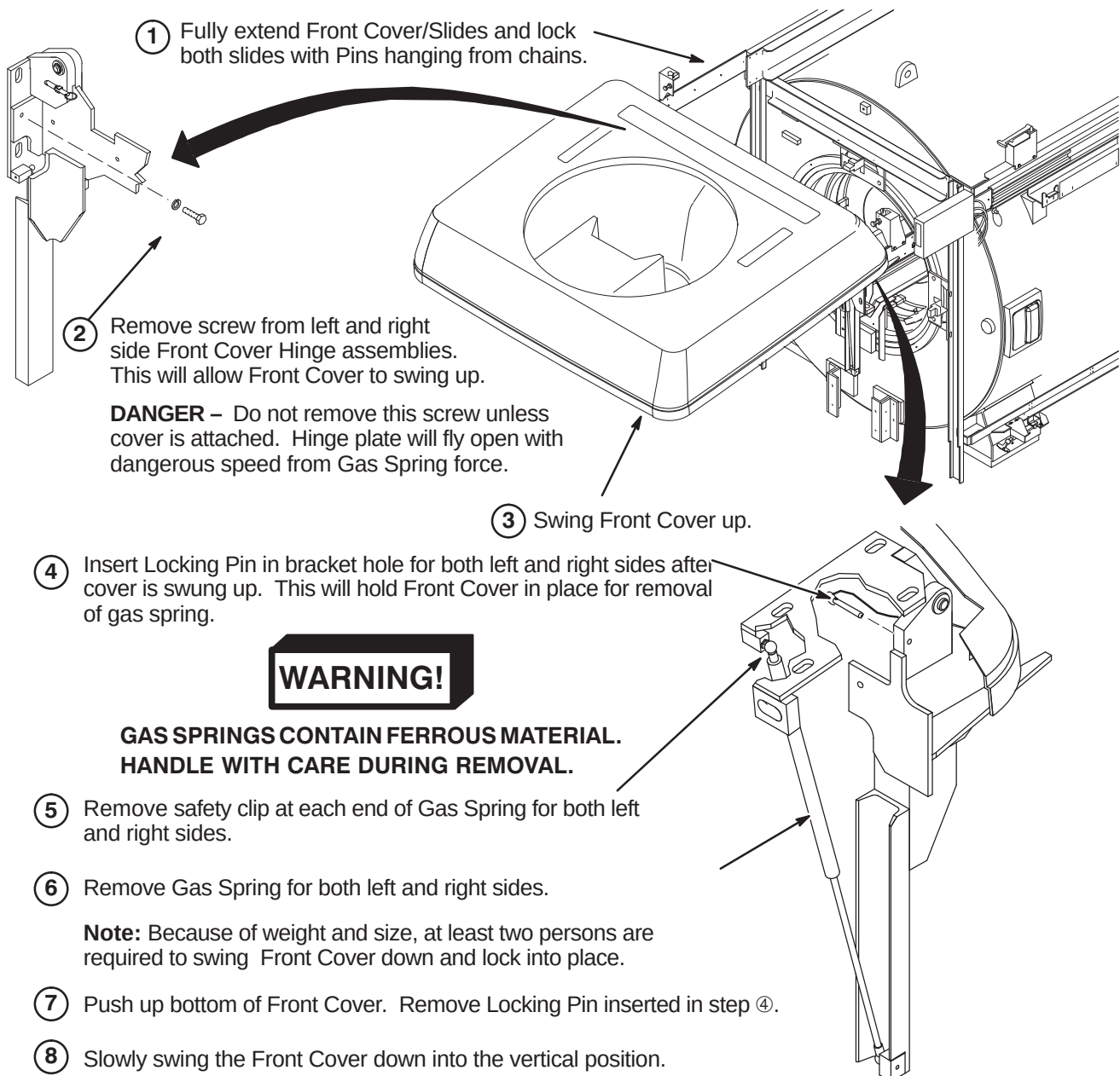


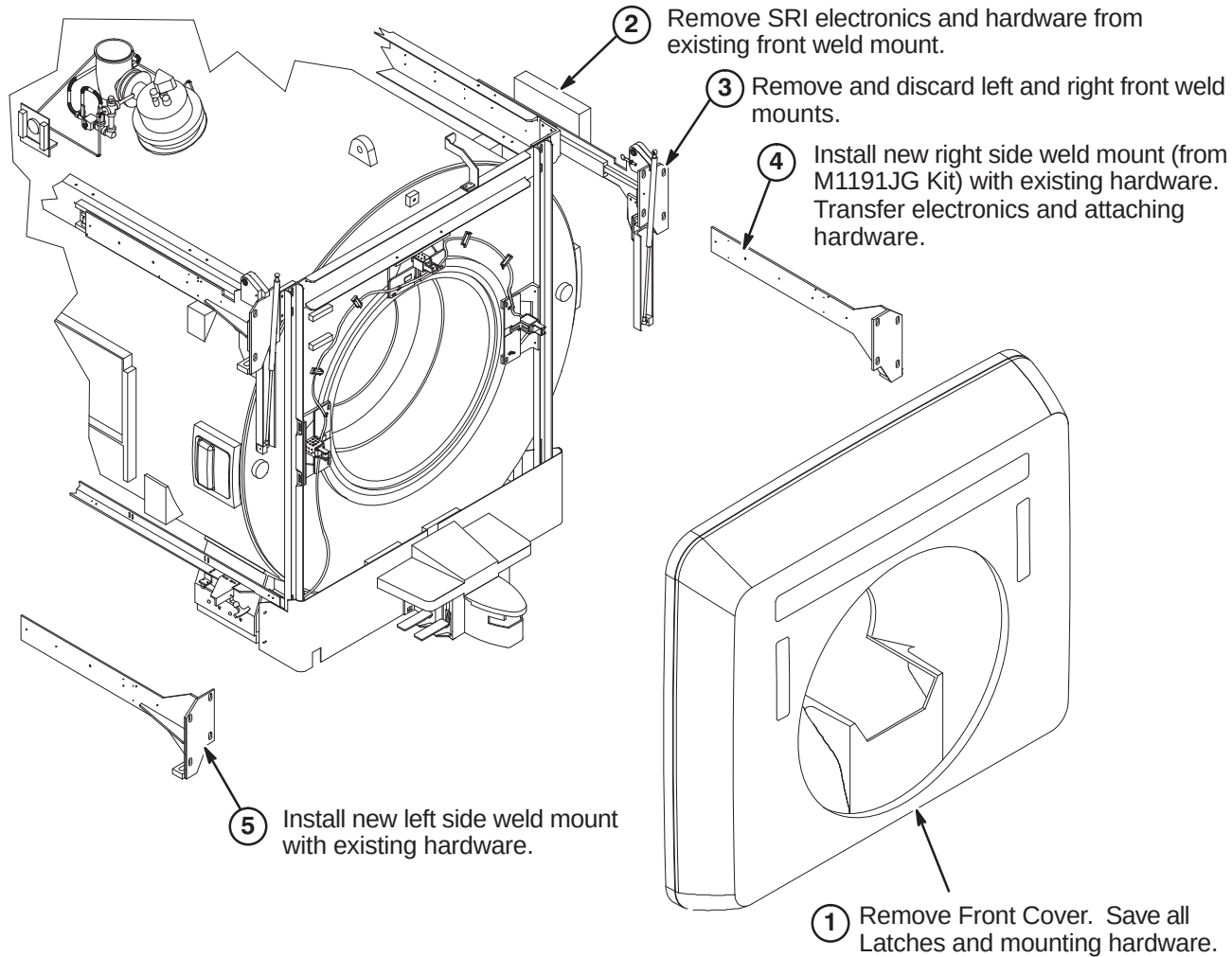
8-1-6 DE-INSTALL FRONT COVER GAS SPRING (S-IV & S-X Magnets)

8-1-6-A Preliminary Front Cover Removal Instructions

CAUTION

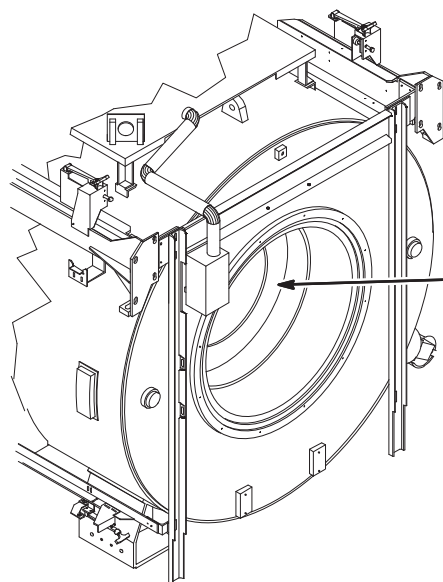
MAGNET MATERIAL HAZARD! The cover dolly was not designed for cover installation with magnet at field. The caster assemblies and fastening hardware are magnetic. If magnet is at field, three installers are required to remove front cover from dolly and carry cover into position at front of magnet.



8-1-6-B Remove Front Cover, Install New Weld Mounts (S-IV & S-X Magnets)

8-1-7 BODY COIL REPLACEMENT

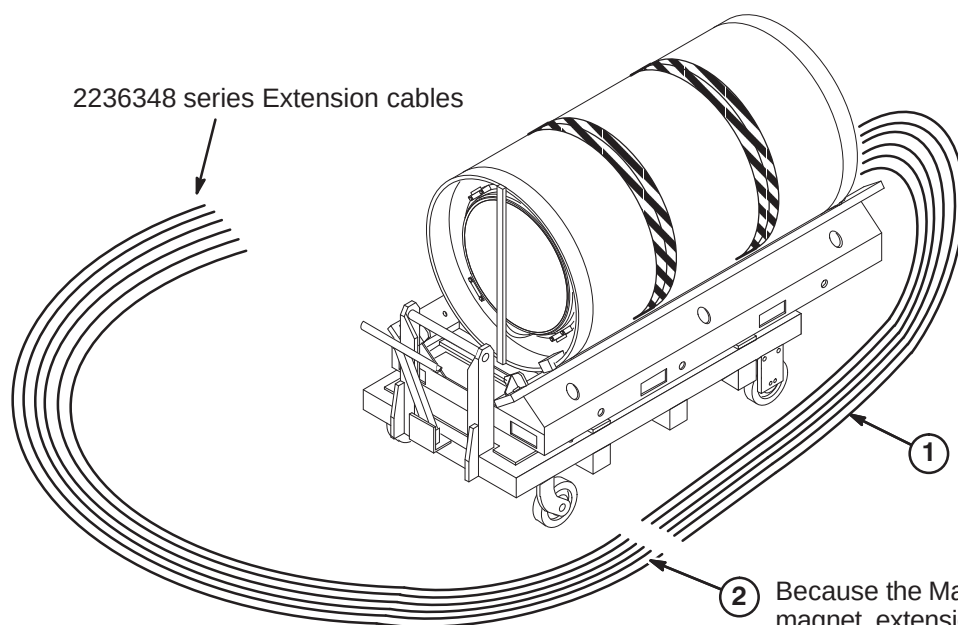
8-1-7-A Coil Removal Instructions



① Remove Gradient Body Coil.

Note: Refer to procedures in Section 13 and *Signa EXCITE HD 1.5T Service Methods CD-ROM*, Direction 5133315 Replacement>Coil Replacements>Gradient Coil Replacement for Non-Cx Magnets, for instructions on de-installing the existing Body Coil.

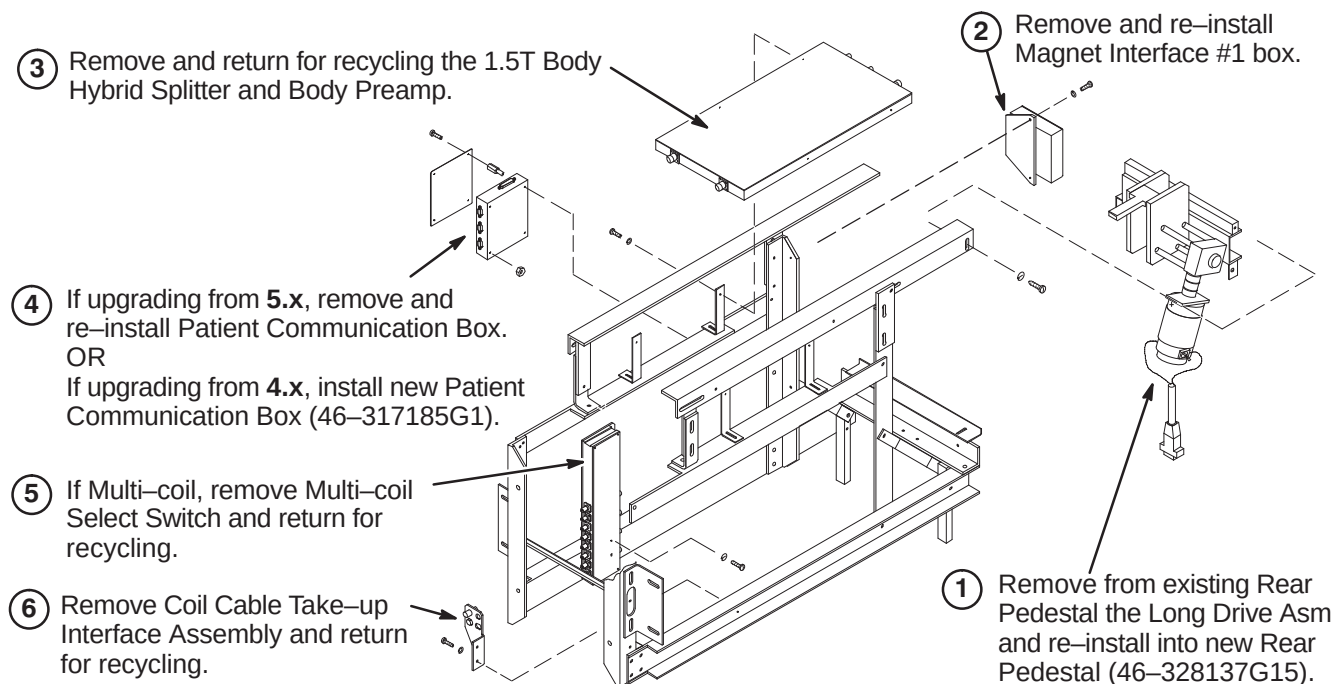
8-1-7-B BRM-D Gradient Coil Pre-Installation Instructions



2236348 series Extension cables

① The new BRM-D coil will have long gradient cable leads.

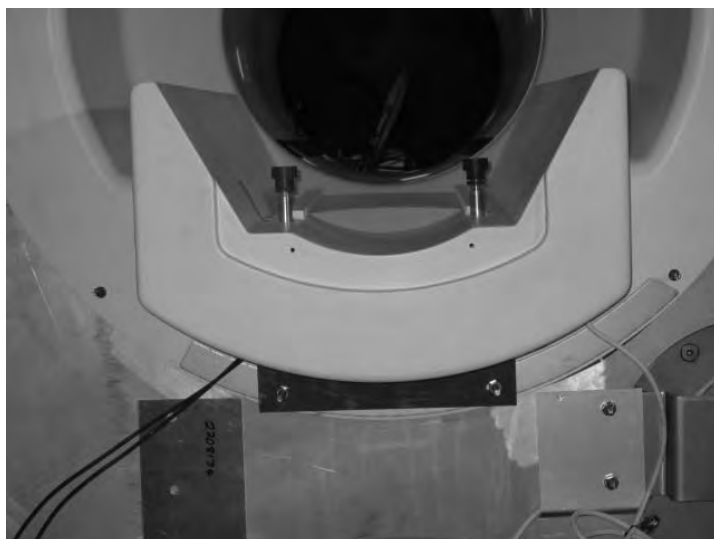
② Because the Magnishield is not an Active-Shield magnet, extension gradient leads have to be added. The procedure for attaching the extension leads is located in Section 13-2 must be completed outside of the magnet room.

8-2 TRANSFER EXISTING COMPONENTS TO NEW REAR PEDESTAL FRAME

8-3 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for all magnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the HDx upgrade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S-Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade*, for instructions on installing this kit.

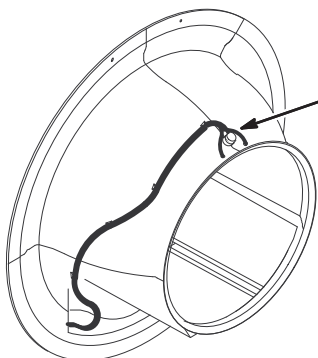
8-4 INSTALL GRADIENT LEAD BOARD AND RESTRAINT BLOCKS (If Required for Fixed-Sites Only)

Prior to this upgrade, the site should have been audited to determine if the Gradient Lead Board Kit (5198802) has been installed. This kit includes parts and instructions for the installation of the Gradient Lead Board at the rear of the magnet.

Also, the gradient cable Restraint Blocks Kit (5186281-2) should have been installed.

These kits are required for all magnets with Horizon style enclosures. If these upgrades has not been performed on the existing enclosure, then M33335BA should have been ordered and it must be installed at this point of the HDx upgrade.

8-5 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL



The new Front End Bell is equipped with a Dual Temperature Sensor cable, Run 749, that has to be replaced with a new cable, Run 1260 (5123812).

EXCITE HD requires Run 1260 to be installed.

Refer to Sections **10-3-7-B**, **10-3-8**, and **10-3-9** for instructions on removal of existing cable and installing the new cable.

8-6 INSTALL MAGNET MONITOR

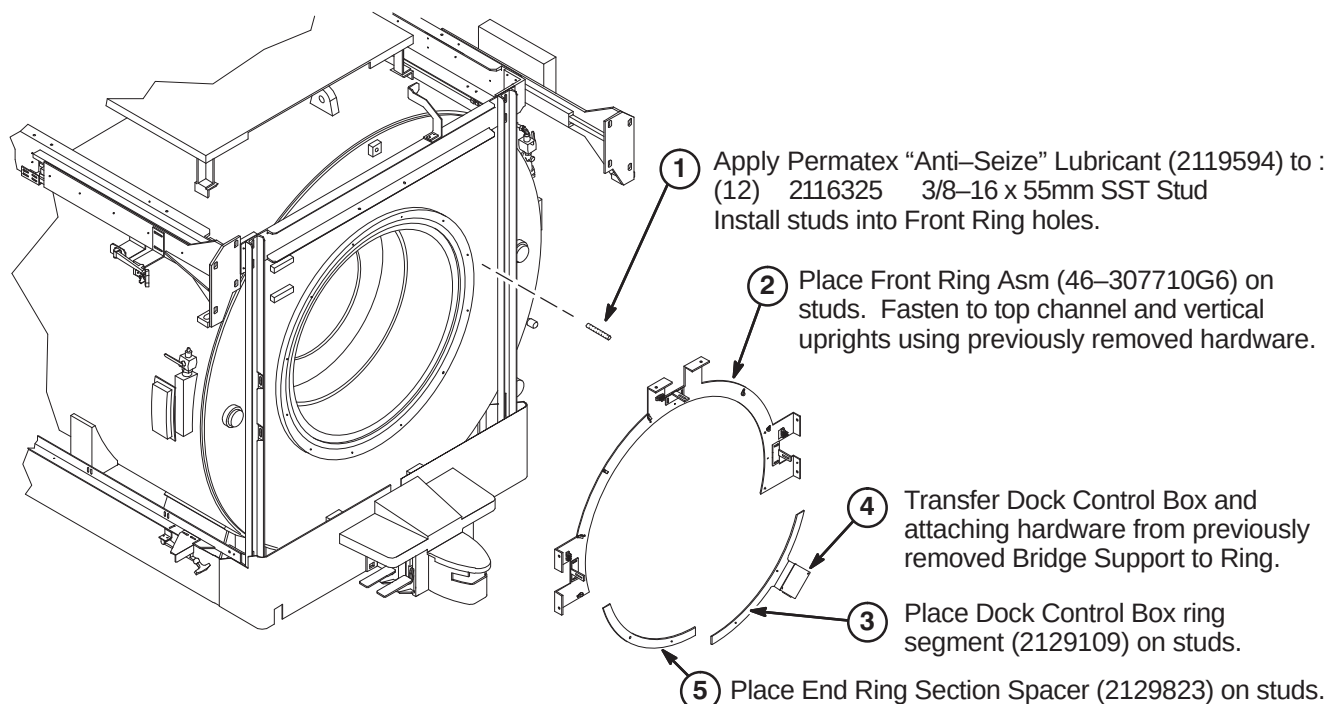
EXCITE HD requires Magnet Monitor on all systems. For this upgrade, all the parts and instructions are included in parts collector 5127838, Magnet Monitor Kit for Non-LCC Systems. Refer to Direction 2230681, included with collector, for instructions on installing magnet monitor on this magnet.

8-7 INSTALL NEW COVERS, COMPONENTS, AND CABLES

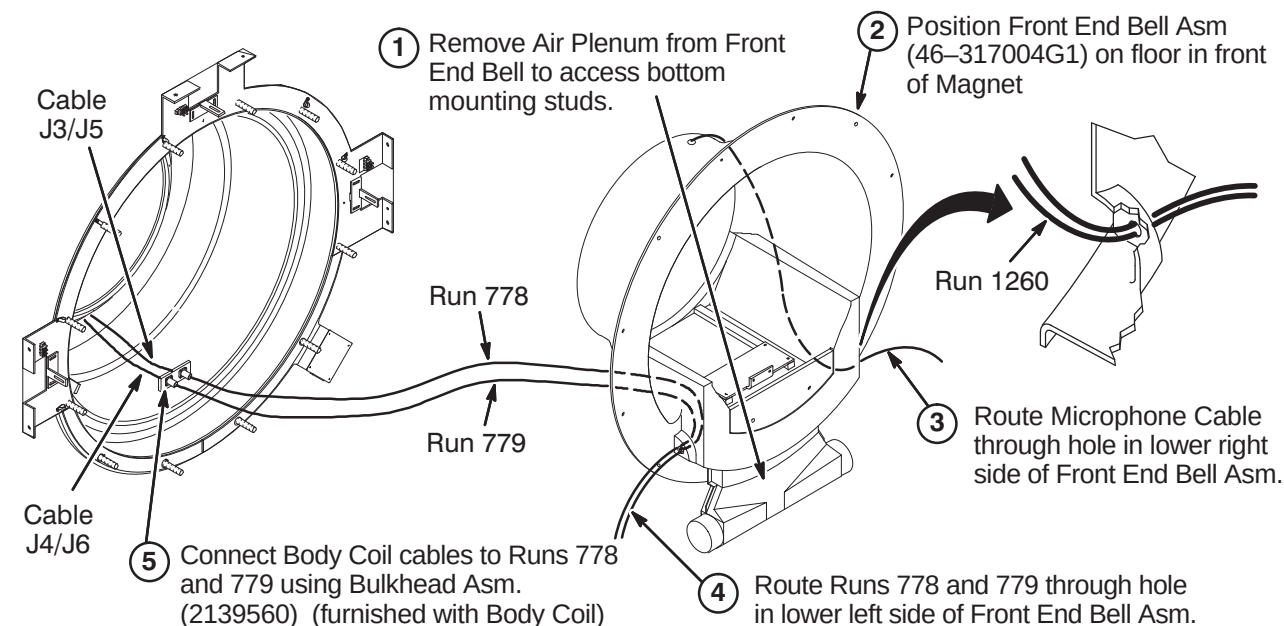
The new Gradient Coil must be installed before starting the installation of new parts referred to in the following steps. Refer to Section for Gradient Coil installation instructions.

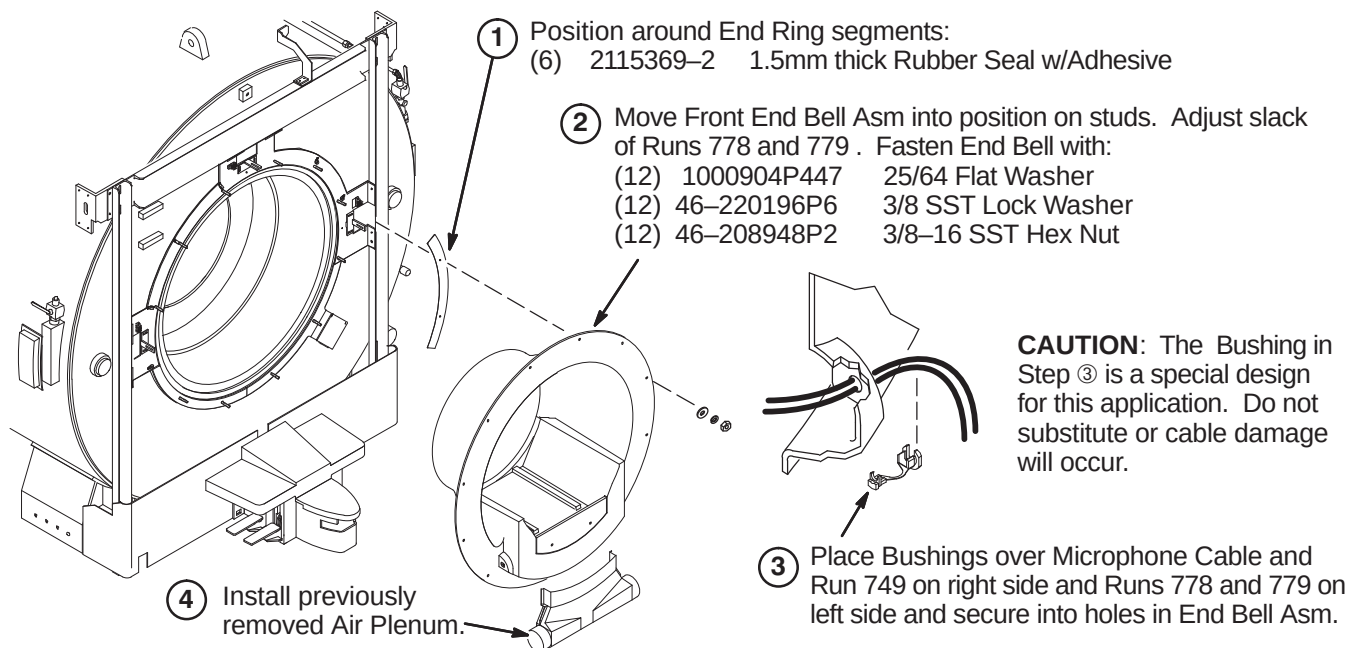
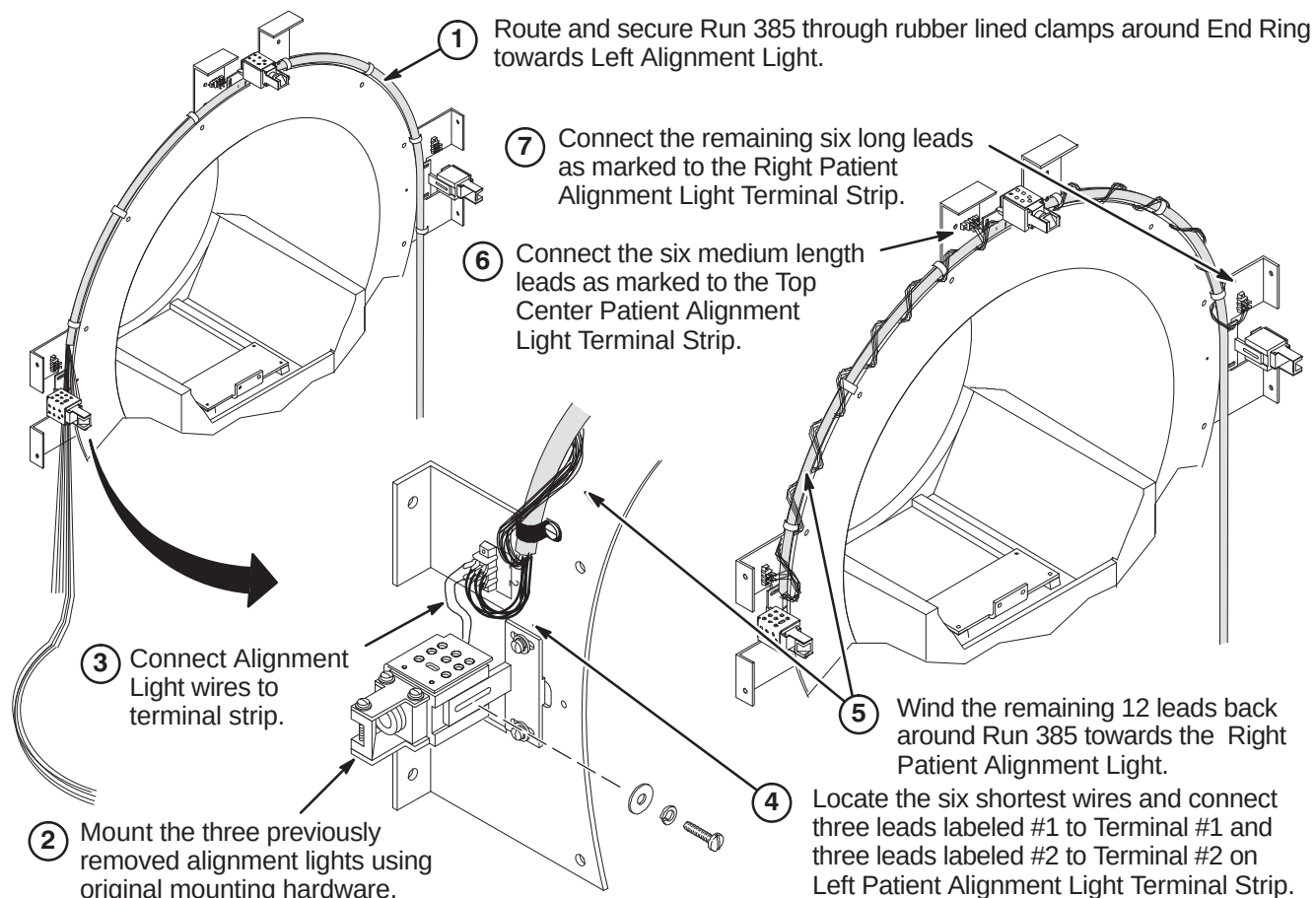
8-7-1 INSTALL COMPONENTS AT FRONT OF MAGNET

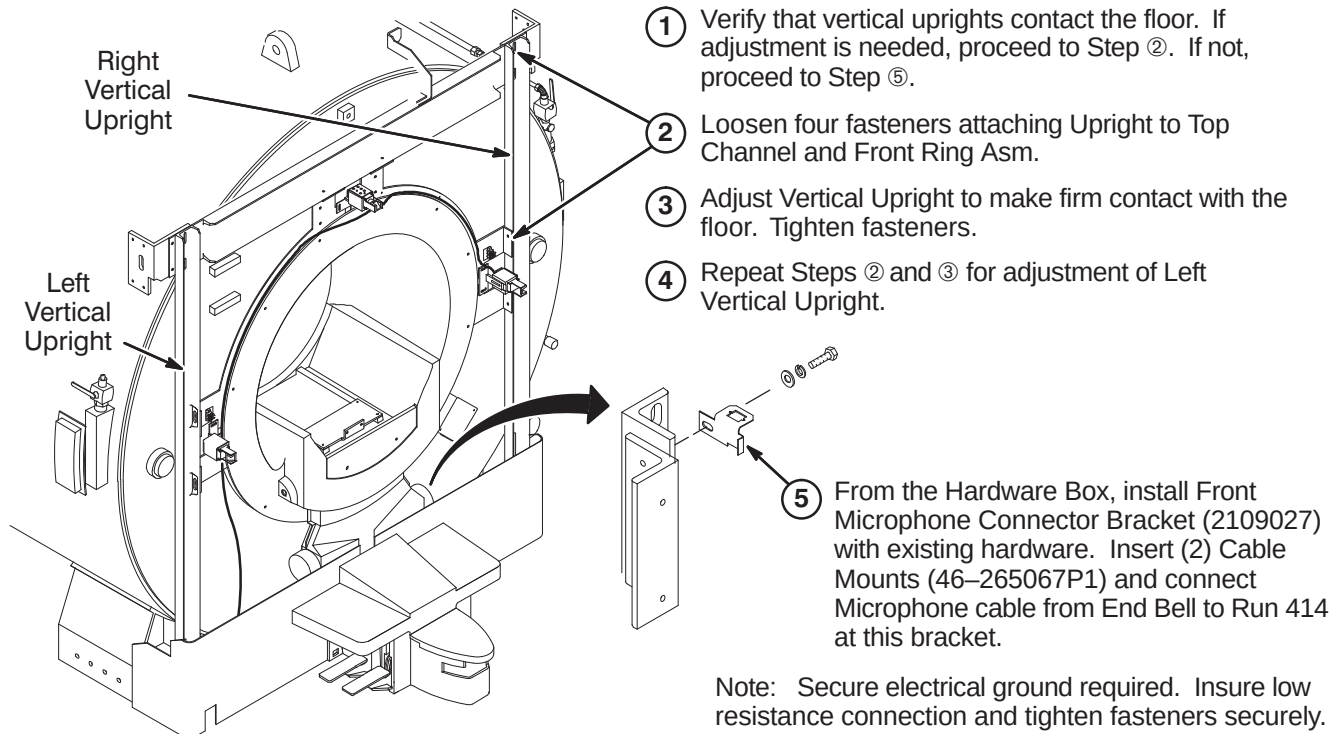
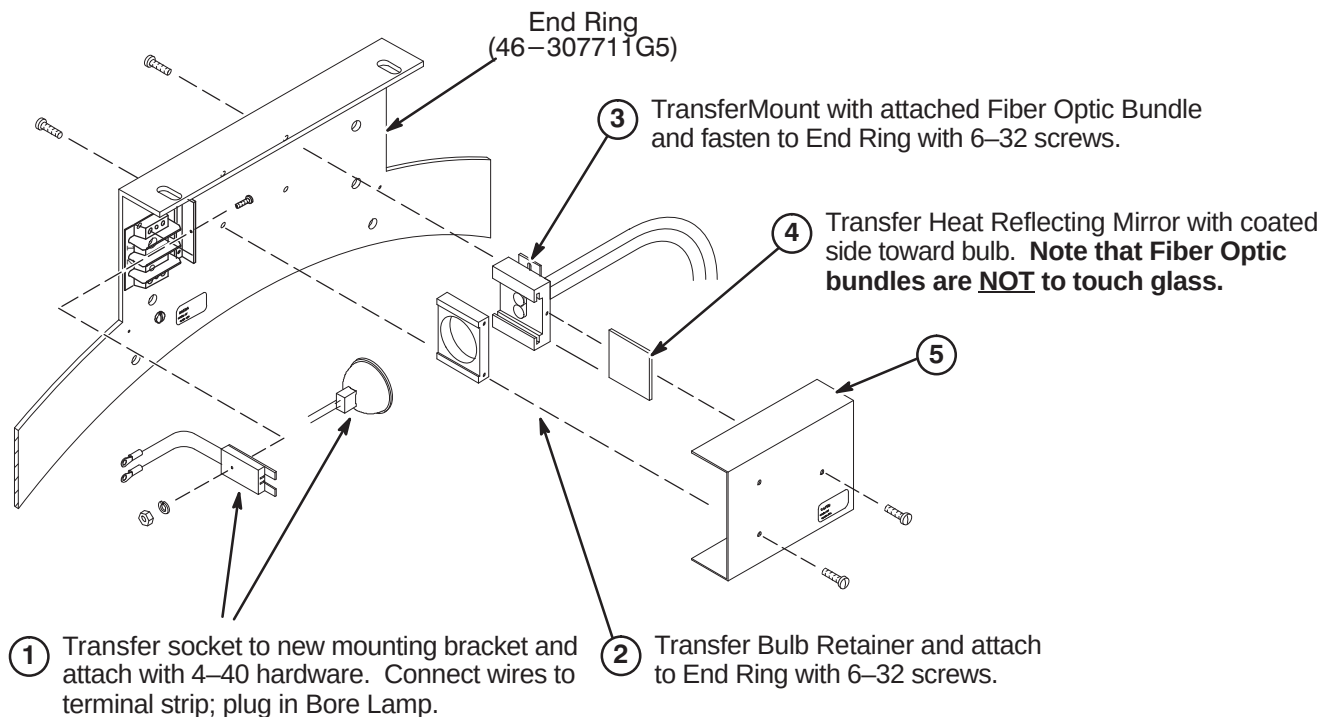
8-7-1-A Install Front Ring Assembly



8-7-1-B Route and Connect Runs 778 and 779

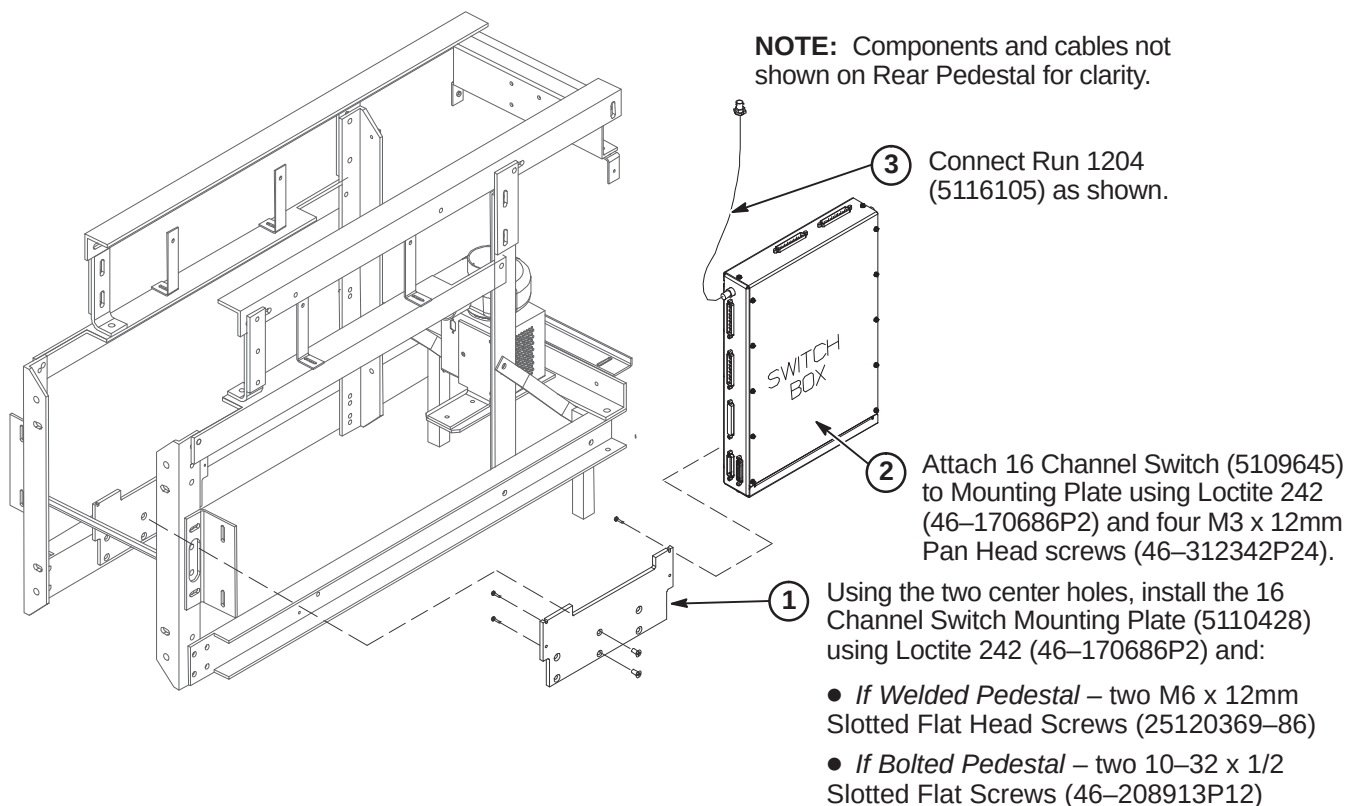


8-7-1-C Install Front Gasket and Mount End Bell**8-7-1-D Install and Connect Alignment Lights**

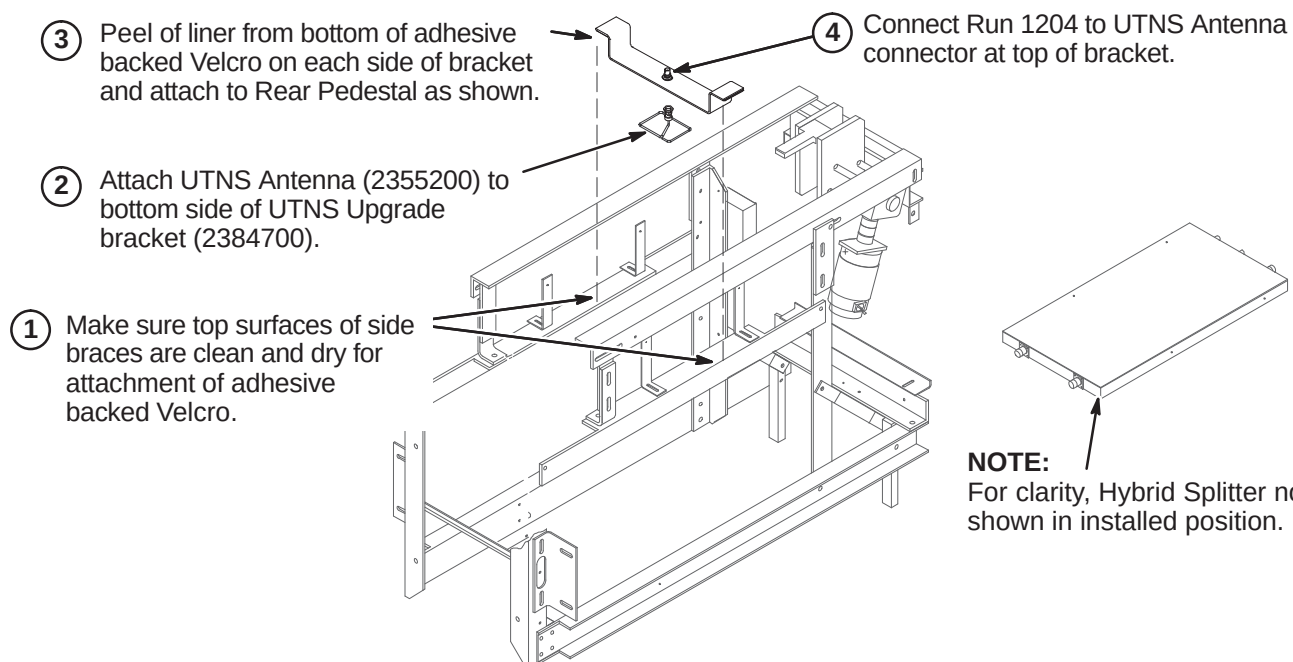
8-7-1-E Adjust Front Vertical Uprights & Installing Microphone Bracket**8-7-1-F Transfer & Assemble Bore Light to New Rear End Ring**

8-7-2 INSTALL NEW COMPONENTS ON REAR PEDESTAL

8-7-2-A Install 16 Channel Switch

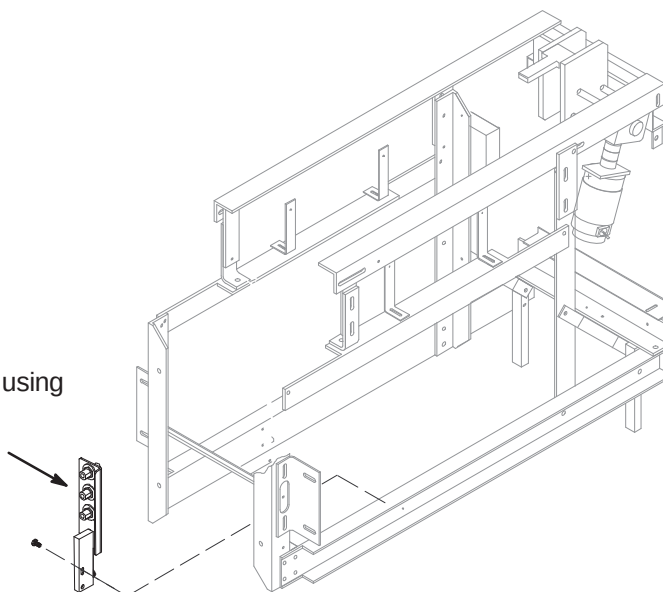


8-7-2-B Install UTNS Antenna



8-7-2-C Install Bulkhead Interface

- ① Install new Bulkhead Interface (5113250) using Loctite 242 (46-170686P2) and two M6 x 12mm pan Head Screws (2109873-24).



8-7-3 HYBRID SPLITTER UPGRADE

Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

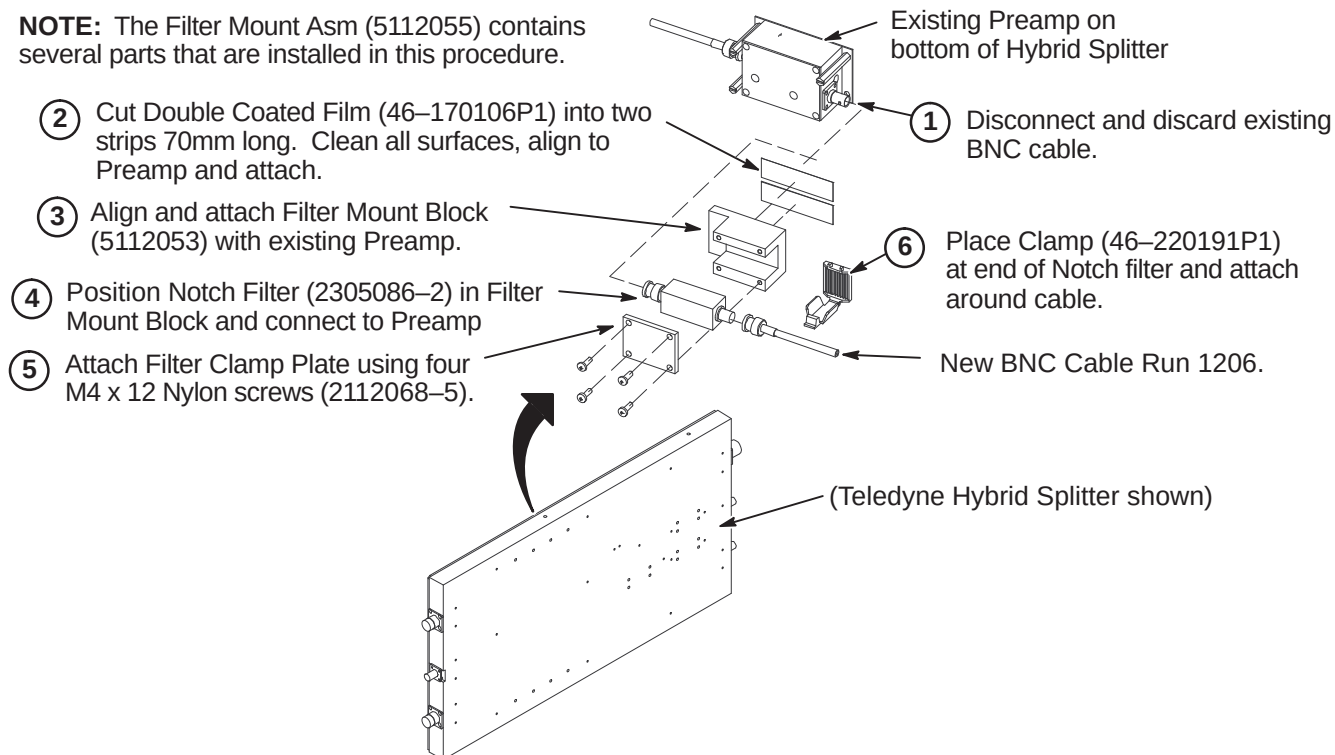
- Refer to Section 8-7-3-A for instructions on installing the Notch Filter and mounting block
- Refer to Section 8-7-3-B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 8-7-3-C for instructions on installing Isolation Filters on systems with Teledyne Splitter

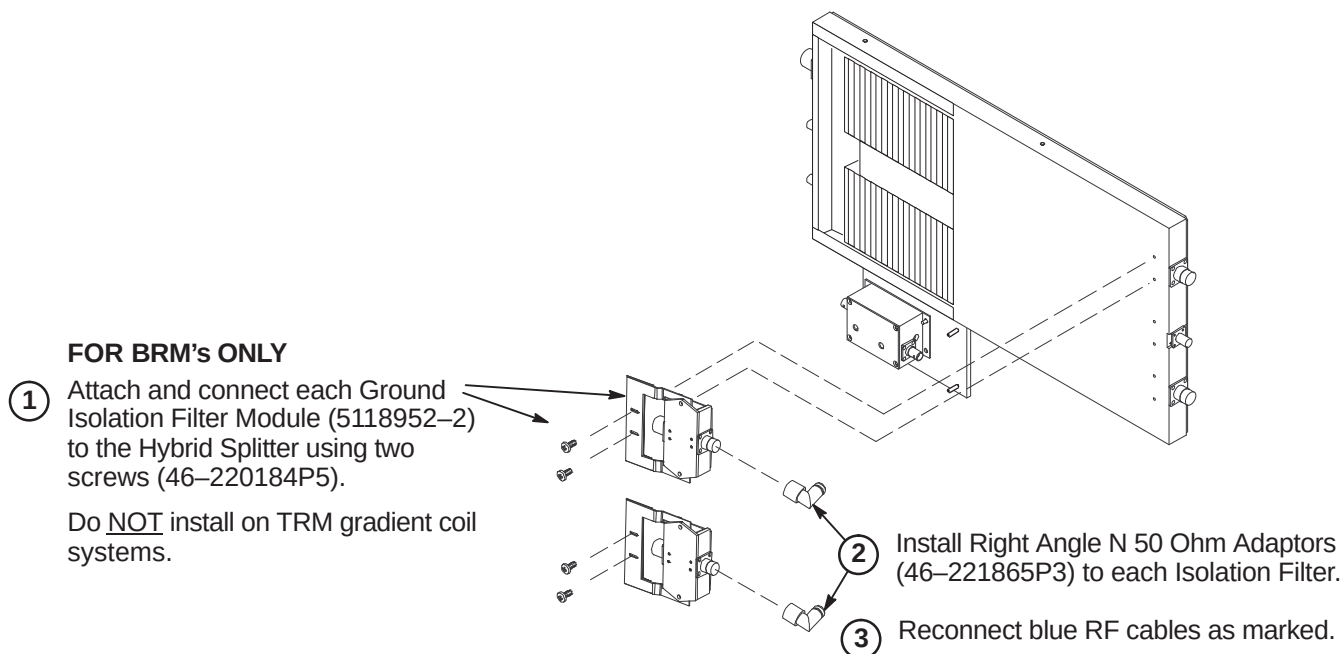
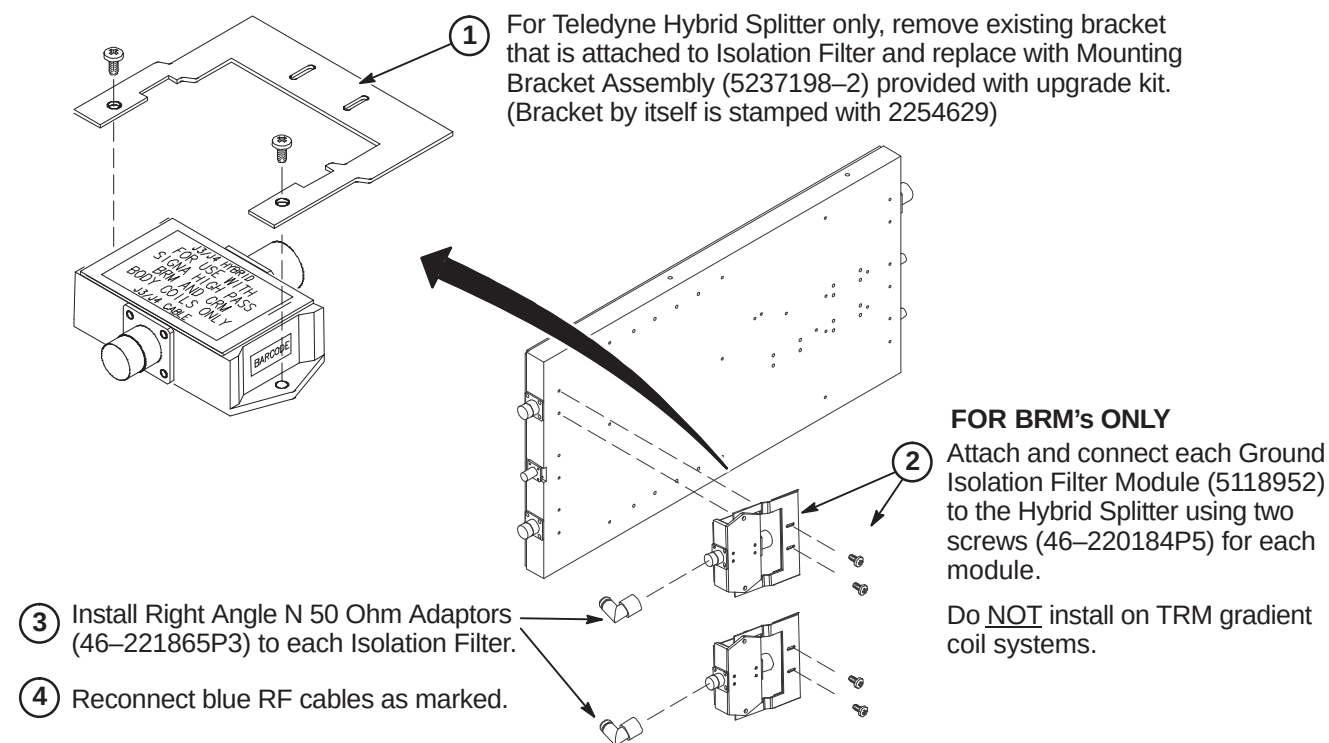
The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils.

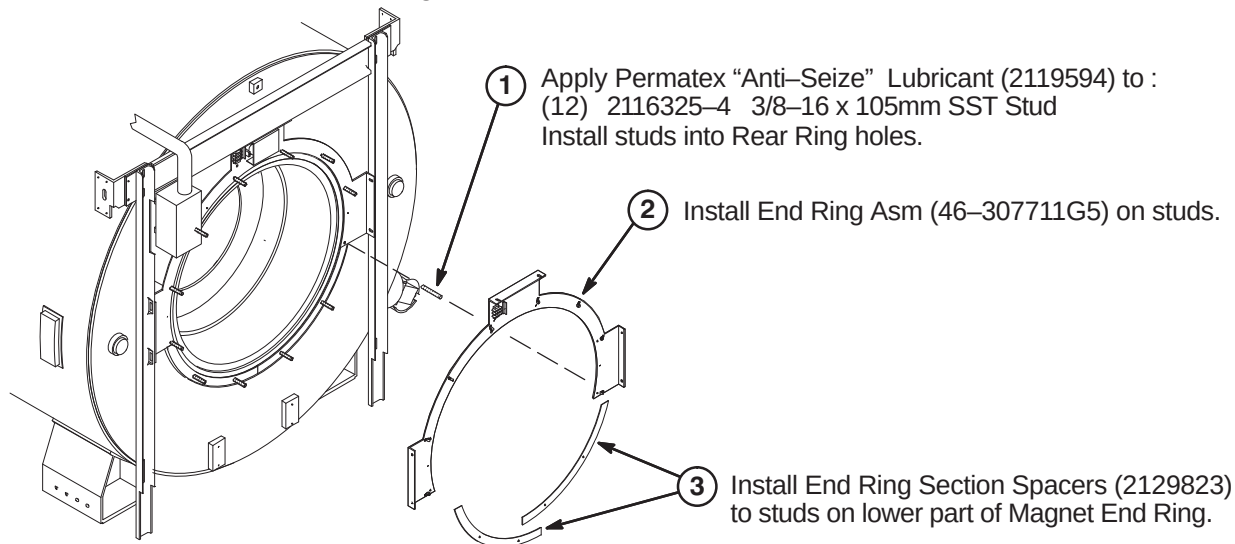
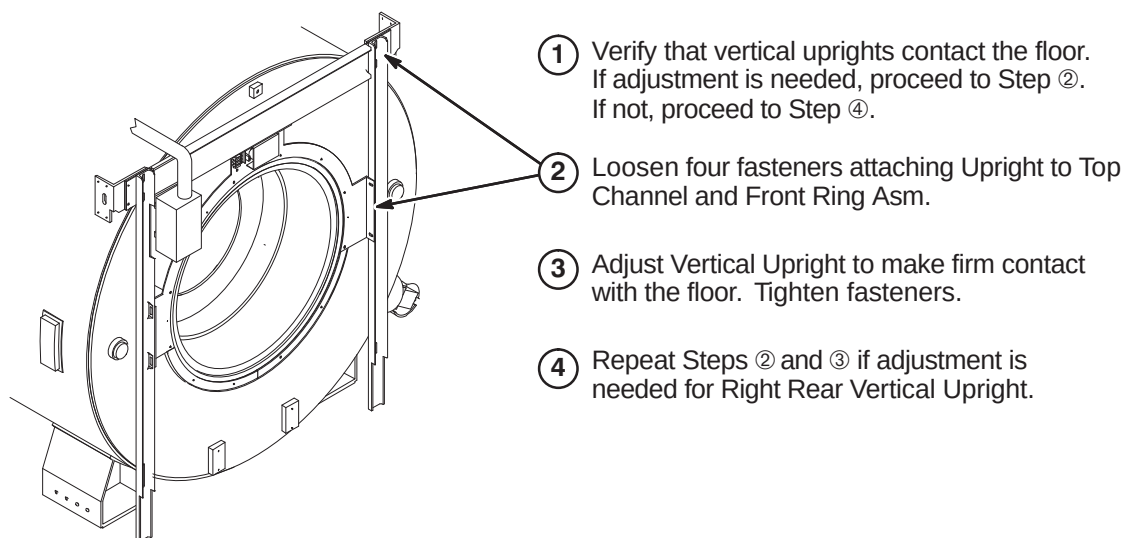
8-7-3-A Install Notch Filter and Filter Mount Block Asm

This Filter Mount assembly may not be required on newer Netcom Hybrid Splitter as it may already have a Notch Filter mounting clamp of a different style installed.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.

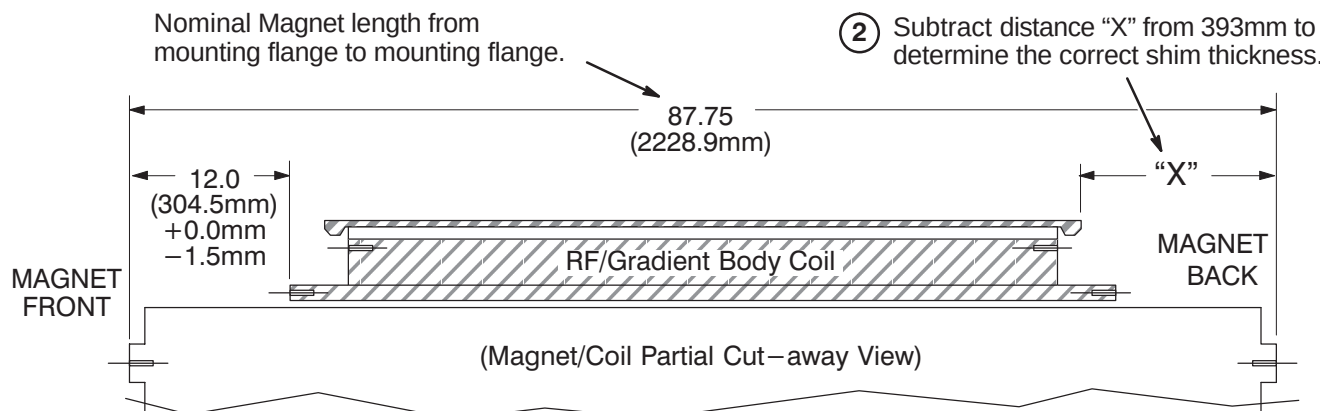


8-7-3-B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)**8-7-3-C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)**

8-7-4 INSTALL COMPONENTS AT REAR OF MAGNET**8-7-4-A Install Rear End Rings****8-7-4-B Adjusting Rear Vertical Uprights**

8-7-4-C Formula For Determining Rear Spacers

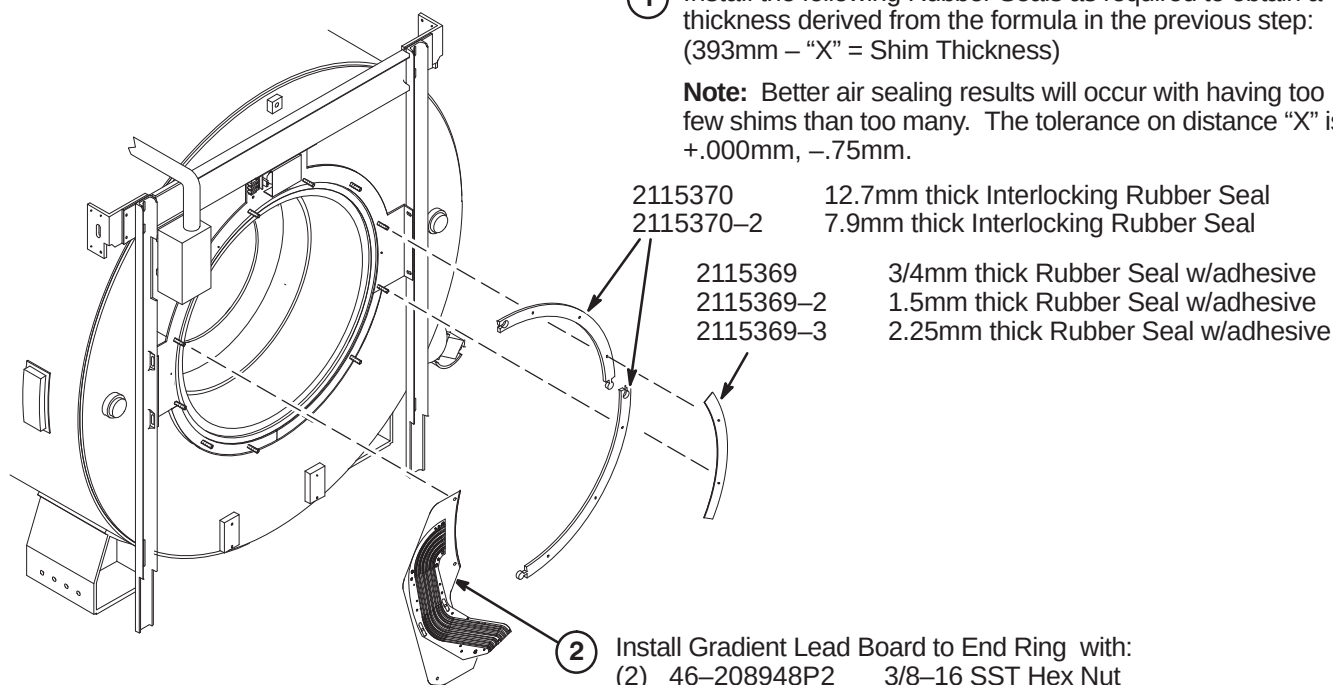
- ① With RF/Gradient Body Coil in place, measure in millimeters the distance from RF Tube to mounting flange on Magnet. This is distance "X".
Make sure the distance is measured to the Magnet mounting flange, not the Rear Frame Asm End Ring installed in Step D3.

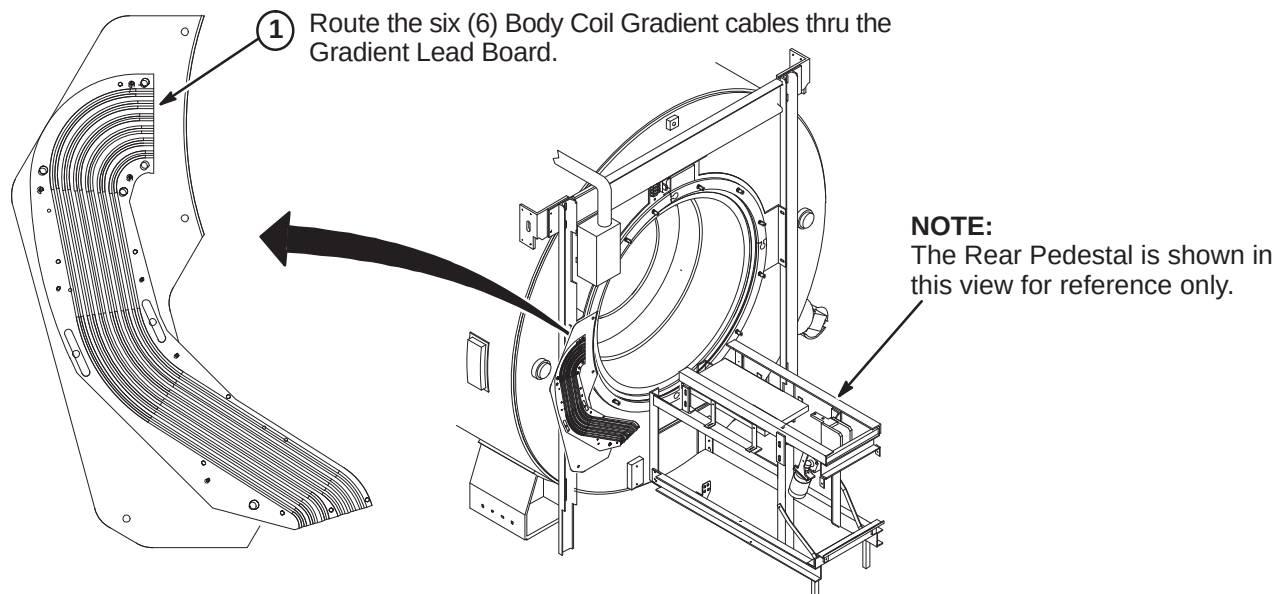
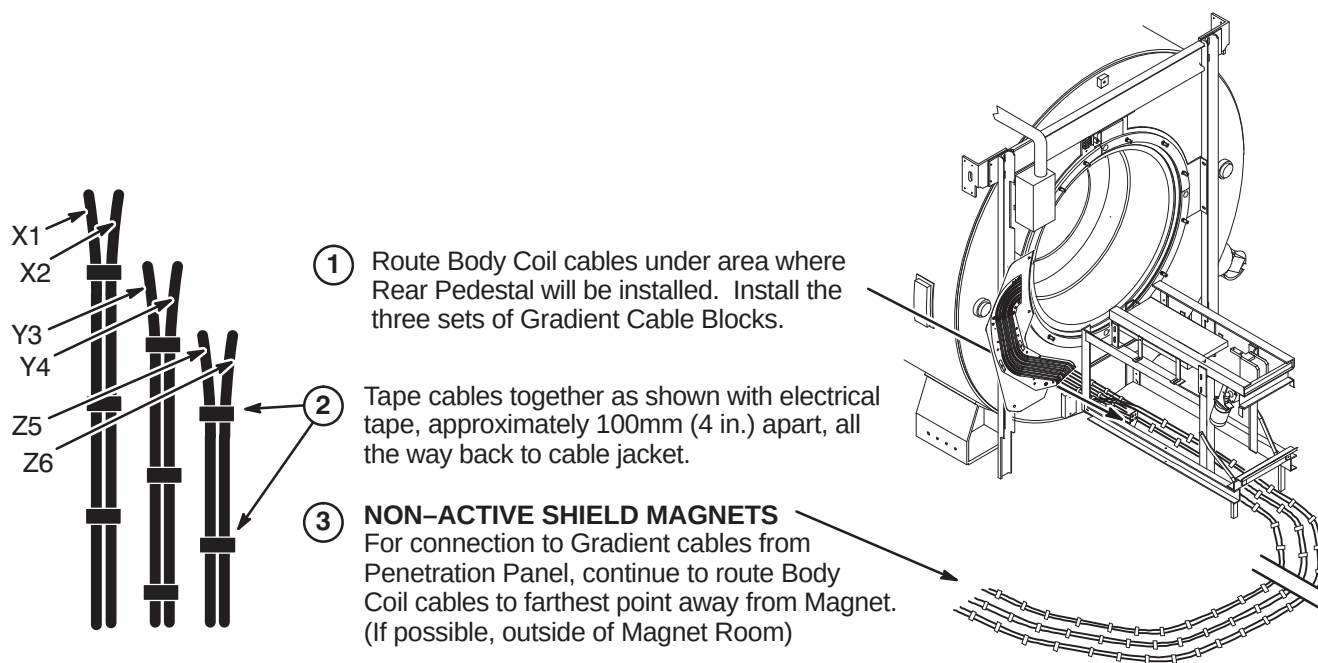


Note: Values shown here are for GE S-II, S-III, S-IV Magnets only.

8-7-4-D Install Rear Spacers And Gradient Terminal Block

- ① Install the following Rubber Seals as required to obtain a thickness derived from the formula in the previous step: $(393\text{mm} - "X" = \text{Shim Thickness})$
Note: Better air sealing results will occur with having too few shims than too many. The tolerance on distance "X" is +.000mm, -.75mm.



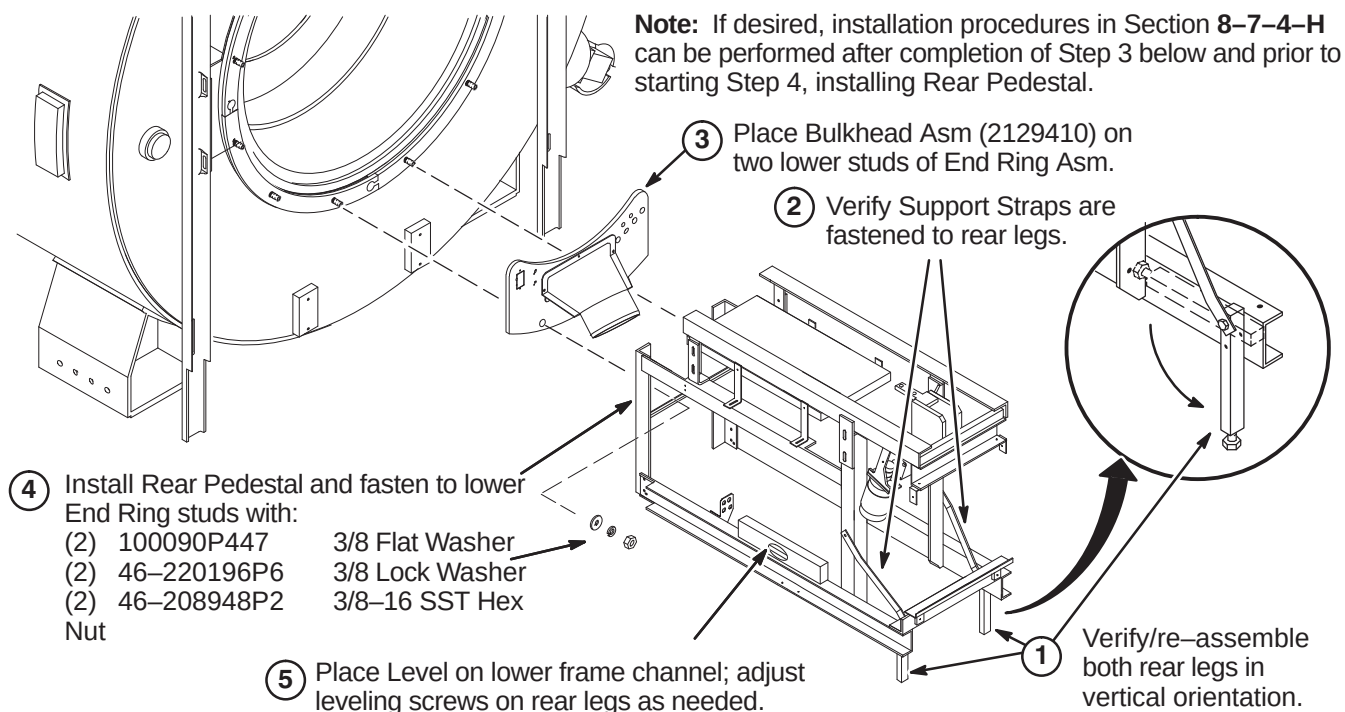
8-7-4-E Route and Install Gradient Cables in Lead Board**8-7-4-F Secure X, Y, and Z Gradient Cables**

8-7-4-G Install Rear Bulkhead And Rear Pedestal

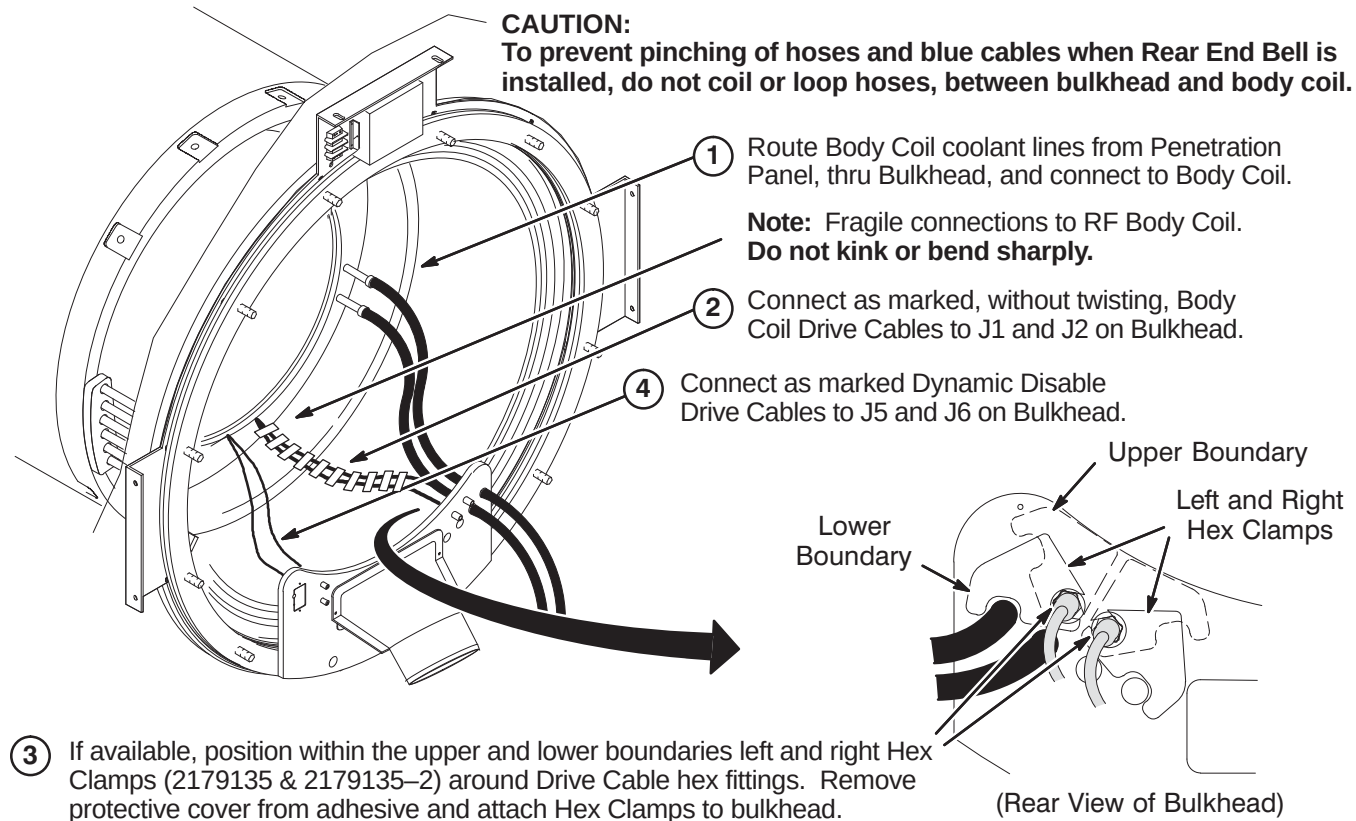
WARNING!

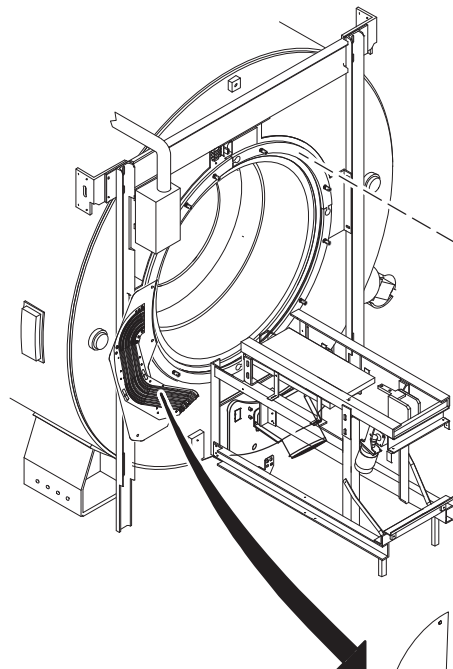
THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

USE PROPER SAFETY AND LIFTING TECHNIQUES.



8-7-4-H Route And Connect Bulkhead Cables And Hoses



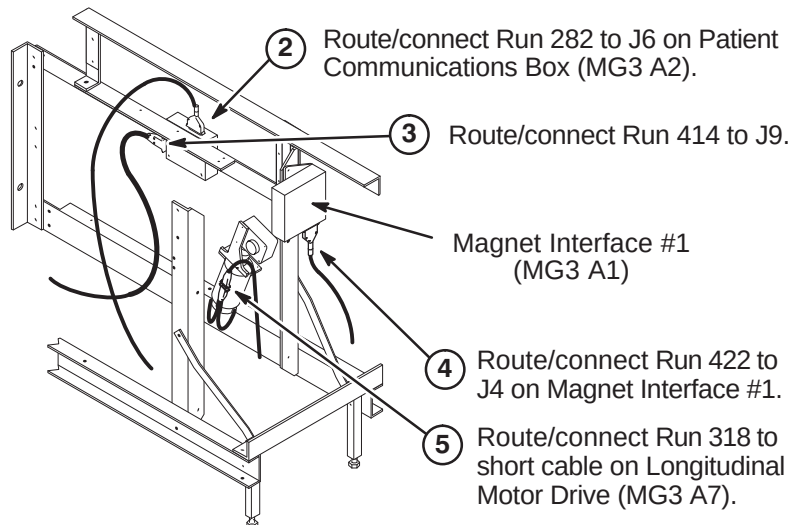
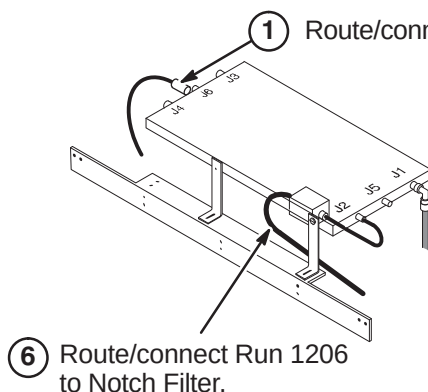
8-7-4-I Install Rear End Bell

Note: A good seal is required to prevent air leaks. Any air leaks between RF Tube (Body Coil) and End Bell joint can cause a loud humming noise during patient scanning. If possible, conduct an air leakage test prior to Bridge installation. This would require connecting power to the Blower Box and connecting 6 inch blower hose from Blower Box to Bulkhead.

- ① Install Rear End Bell (2113076) and fasten to studs with:
- | | | |
|-----|-------------|---------------------|
| (8) | 1000904P447 | 25/64 Flat Washer |
| (8) | 46-220196P6 | 3/8 SST Lock Washer |
| (8) | 46-208948P2 | 3/8-16 SST Hex Nut |

- ② Install Rear End Bell to Bulkhead Asm with:
- | | | |
|-----|------------|--------------------|
| (3) | 2109878-3 | M6 SST Flat Washer |
| (3) | 2109866-13 | M6 SST Hex Screw |

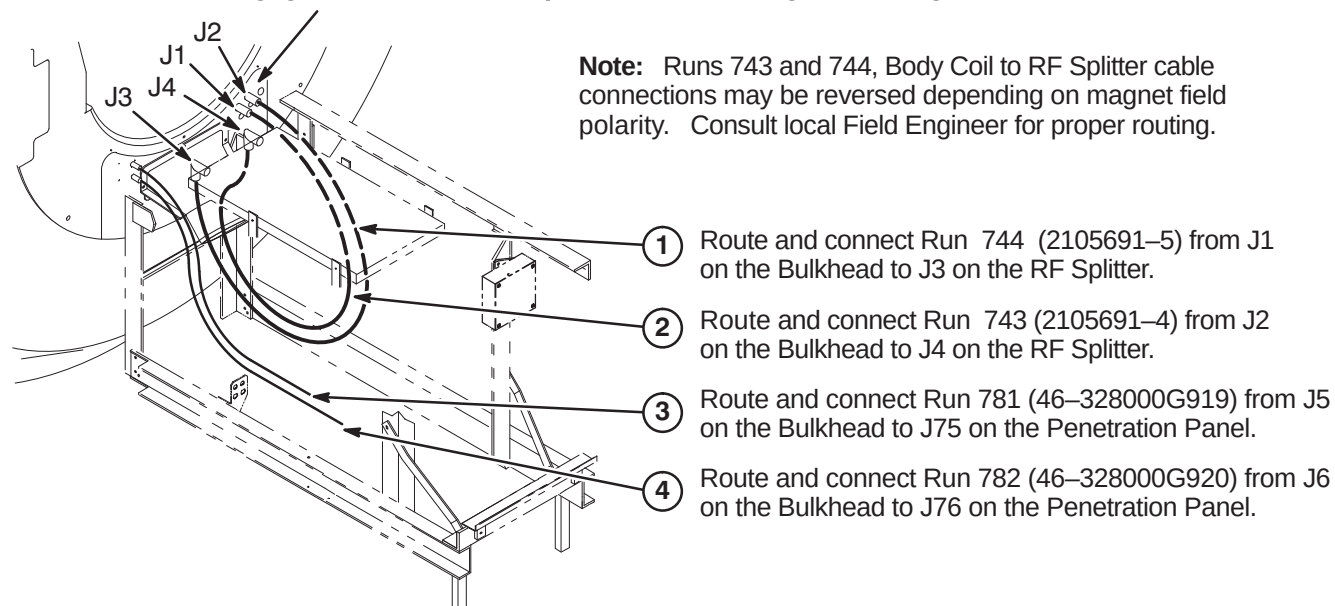
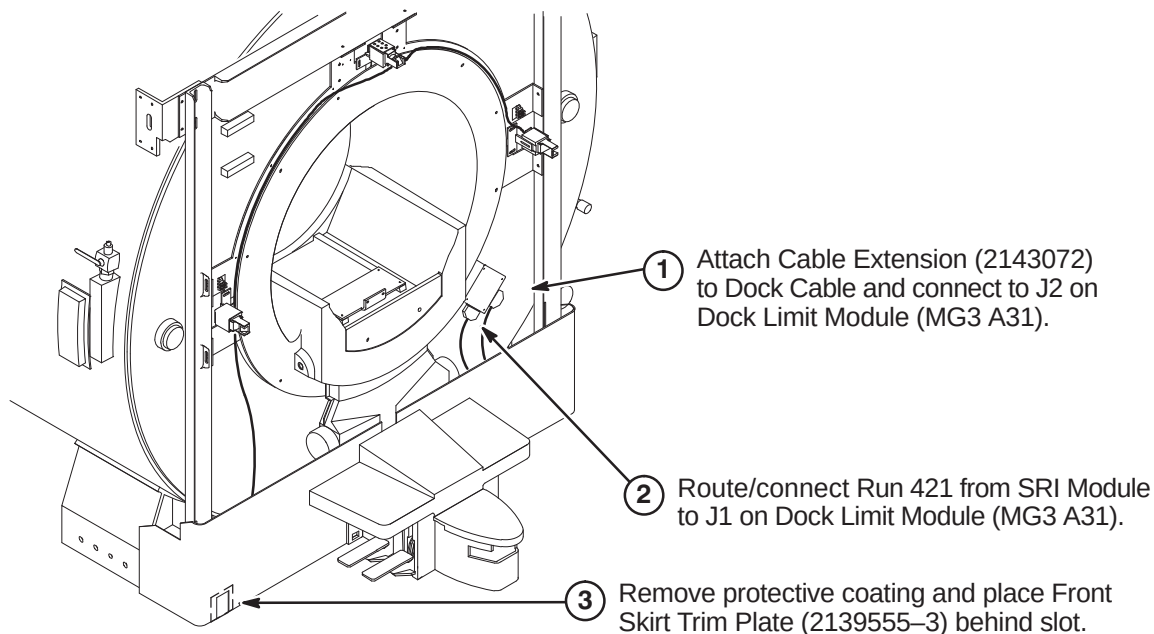
- ③ Install Lead Board Cover (2370909) to Gradient Lead Board using fasteners provided with assembly.

8-7-4-J Connect Runs 282, 318, 414, 422, 780, And 1206

8-7-4-K Route/Connect Runs 743, 744, 781, And 782**CAUTION**

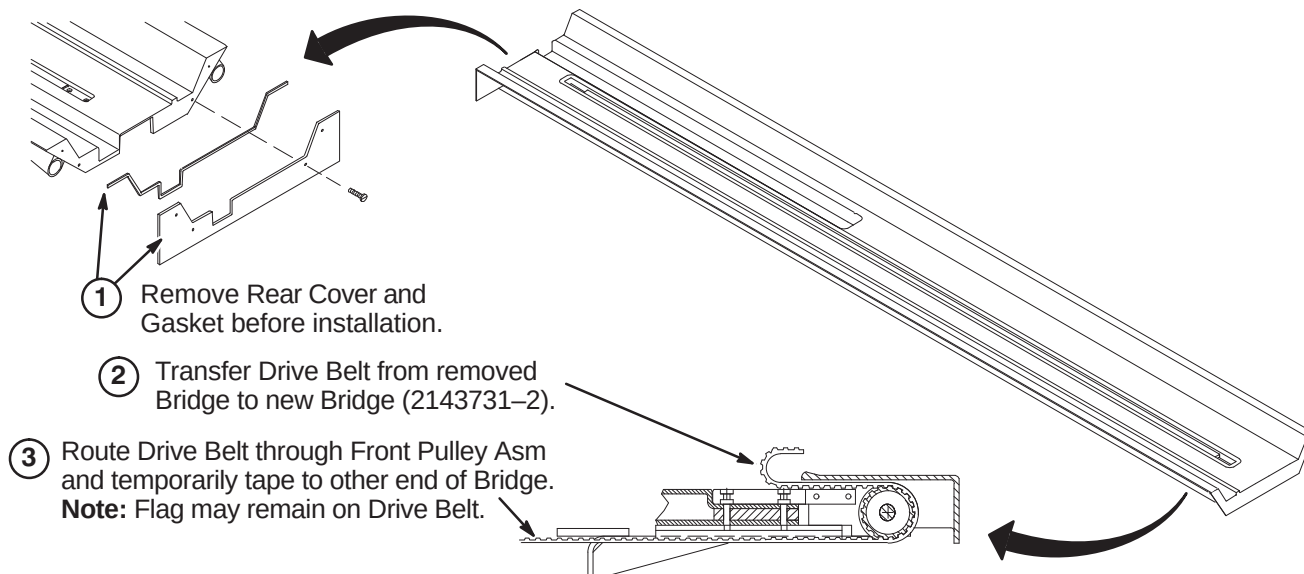
The RF Drive Input cables, Runs 743 and 744, utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

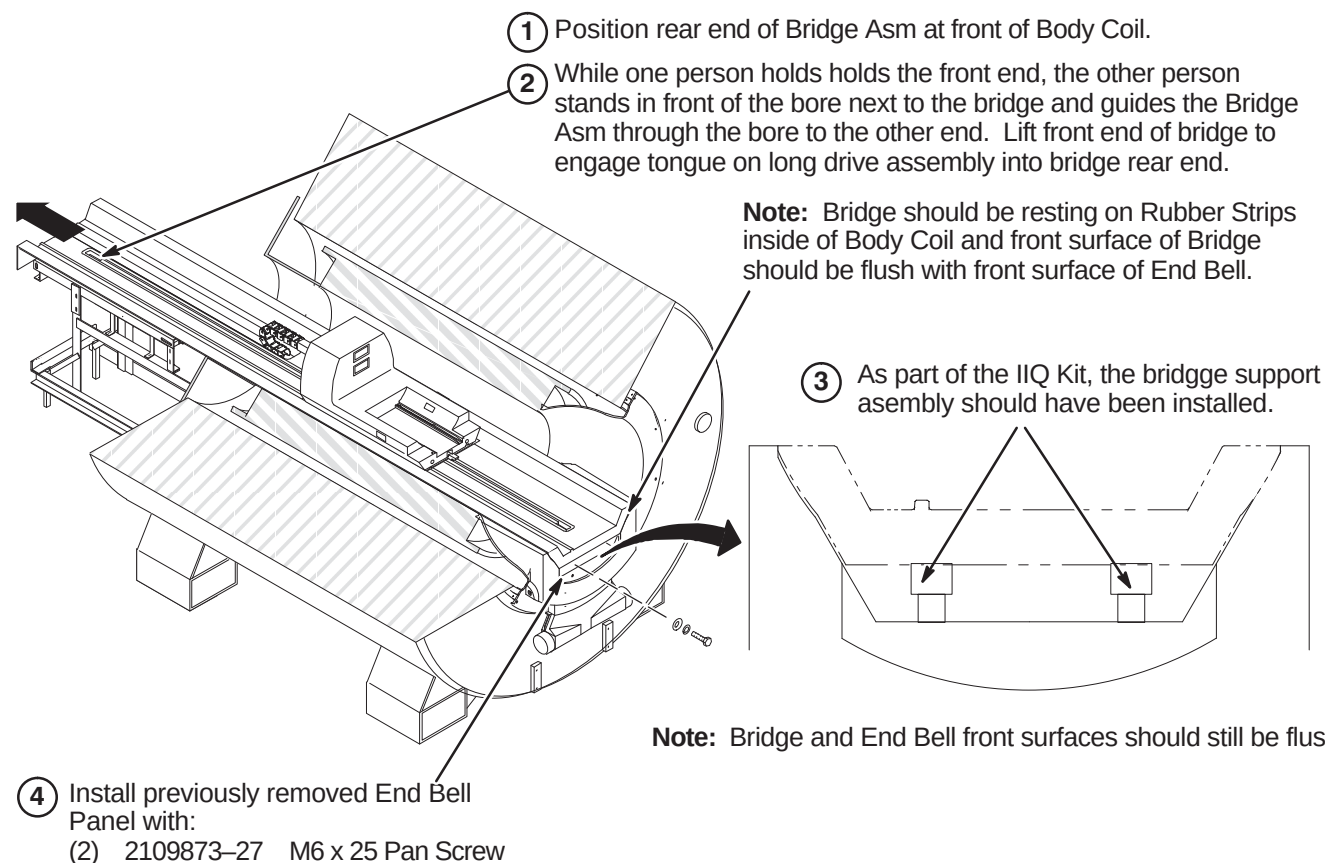
**8-7-4-L Route/Connect Runs 421 And Dock Cable**

8-7-5 INSTALL BRIDGE

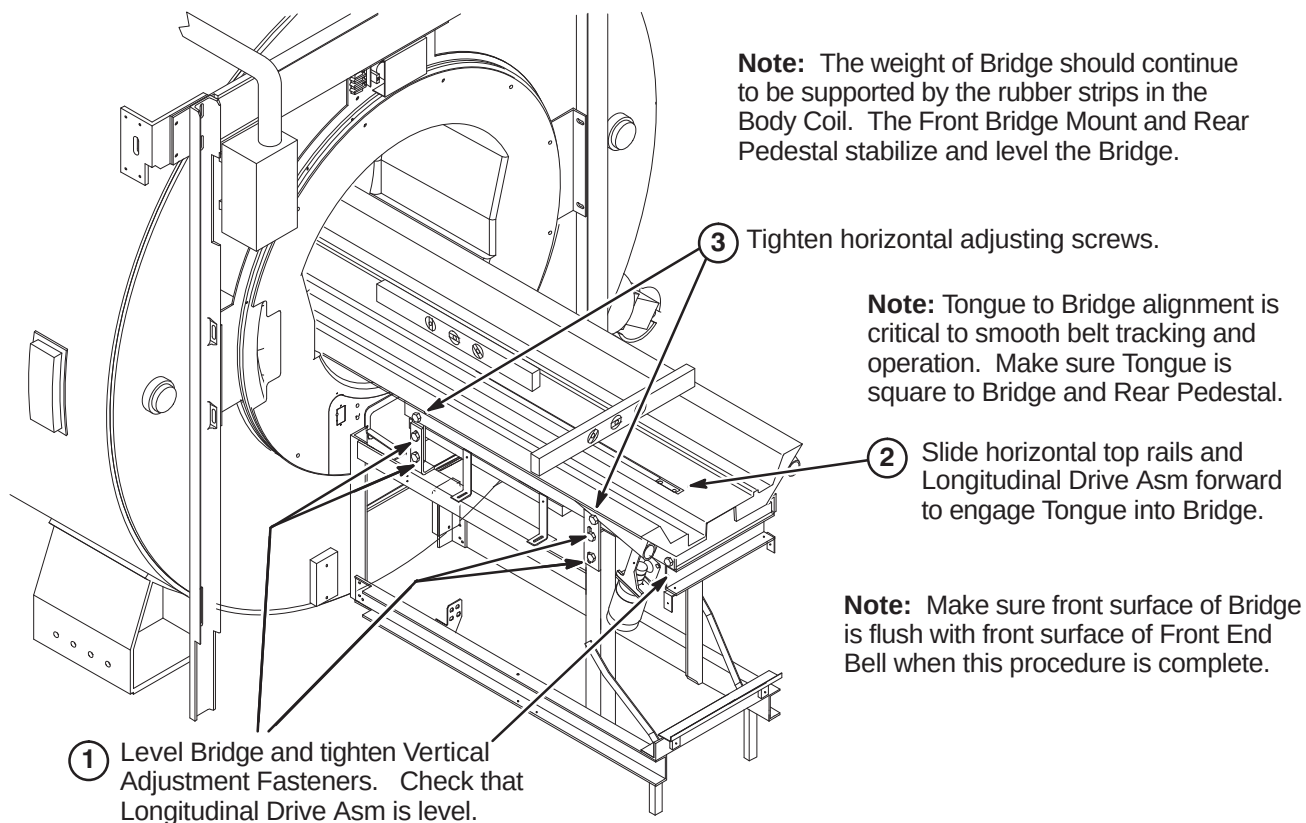
8-7-5-A Bridge Assembly Preparation Before Installing in Magnet



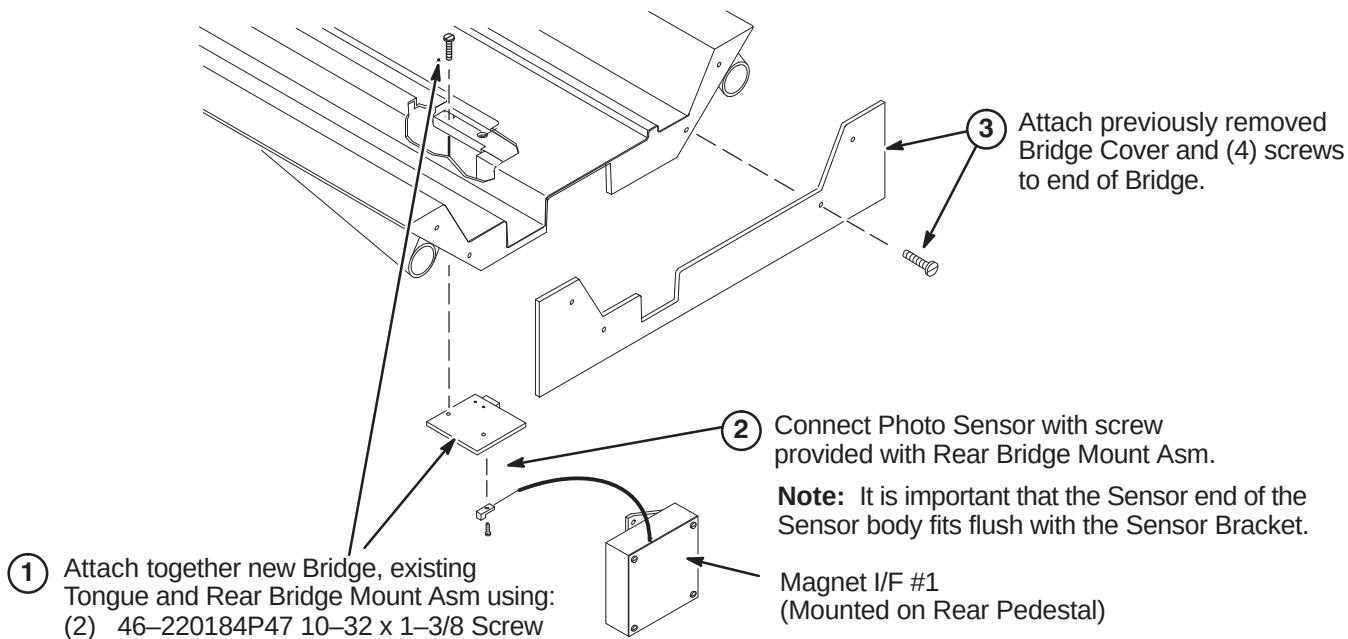
8-7-5-B Securing Front of Bridge

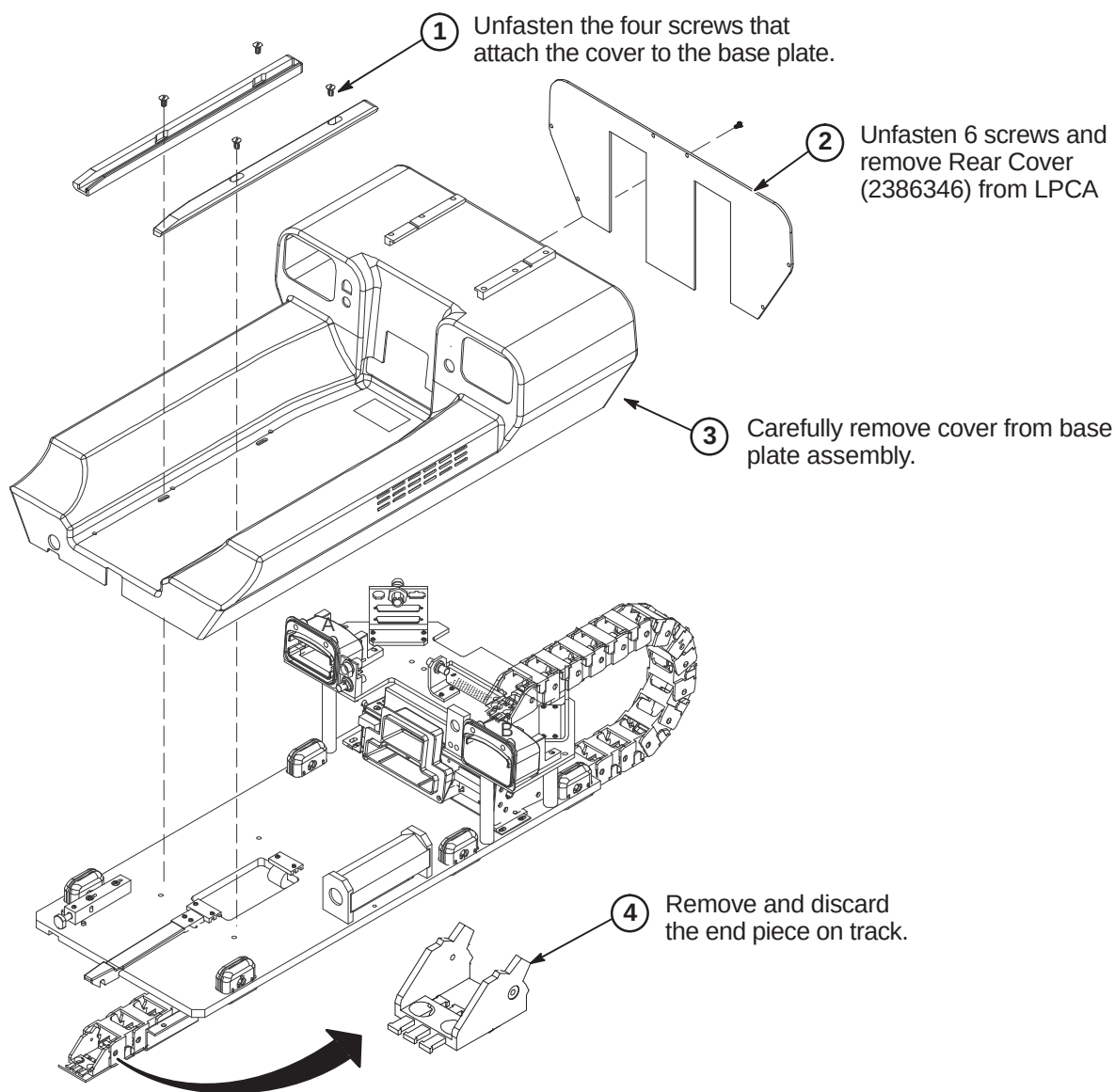


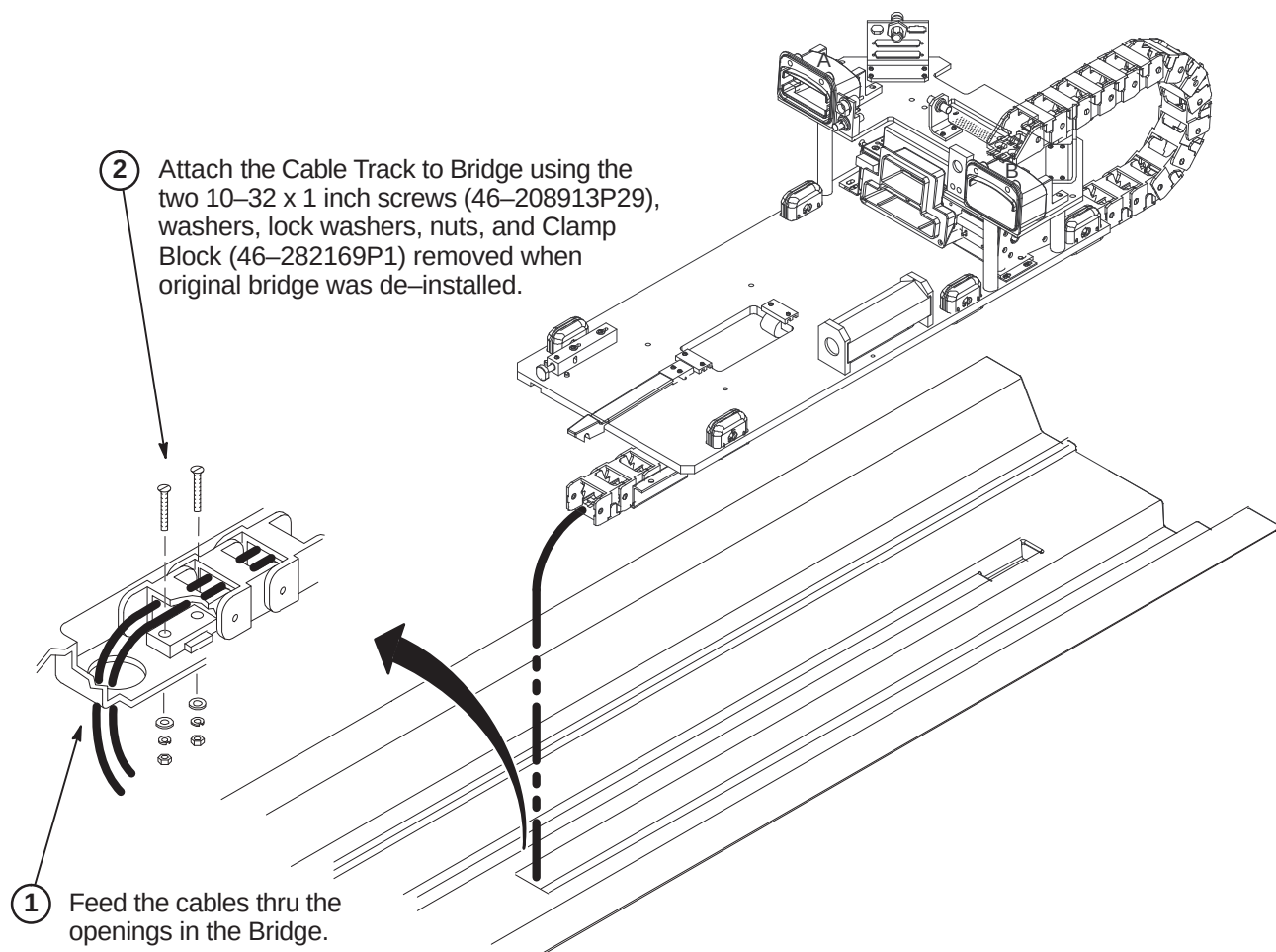
8-7-5-C Leveling Bridge

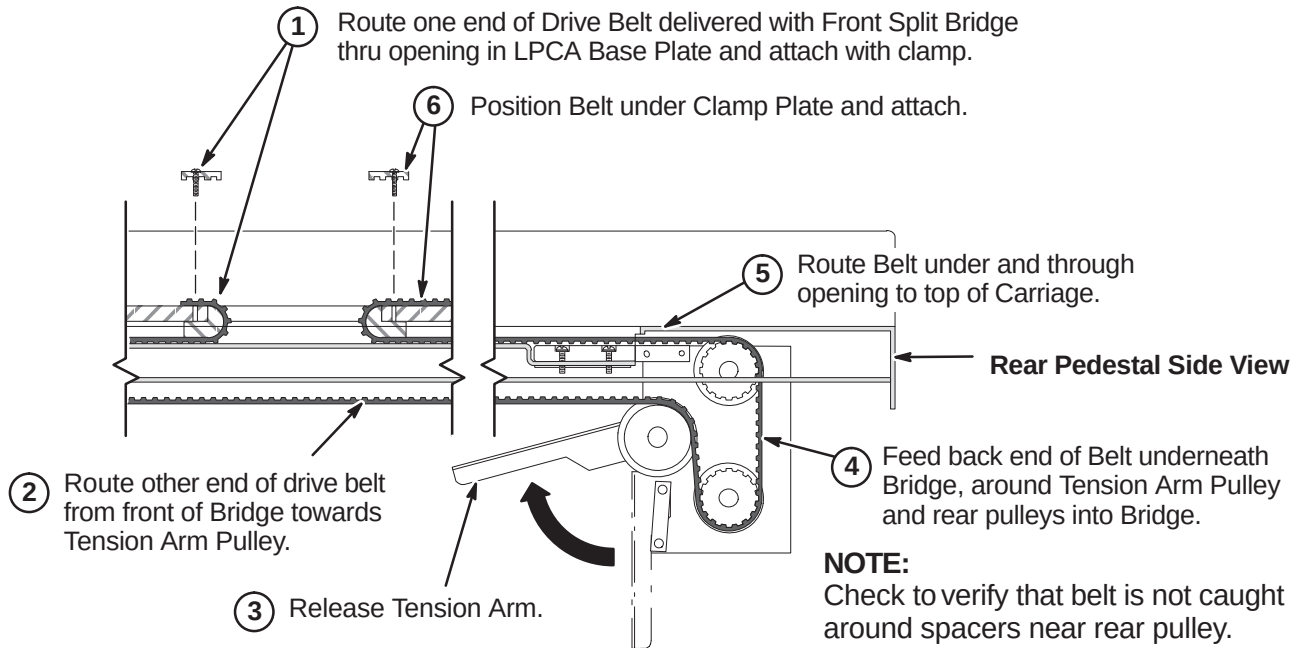
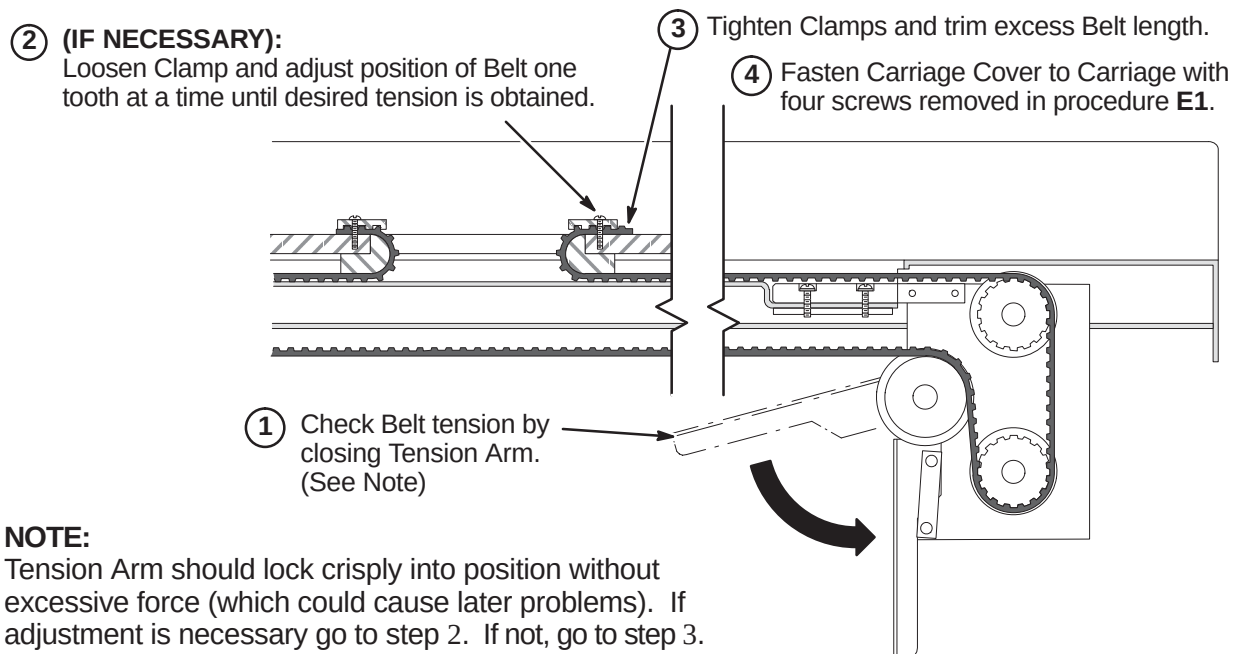


8-7-5-D Connect Photo Sensor & Bridge End Cover



8-7-6 INSTALL LPCA AND CABLE TRACK**8-7-6-A Remove LPCA Cover**

8-7-6-B Attach Cable Track to Bridge

8-7-6-C Route and Attach Drive Belt**8-7-6-D Final Adjustment of Drive Belt**

8-7-6-E Installation of Sensor Flag

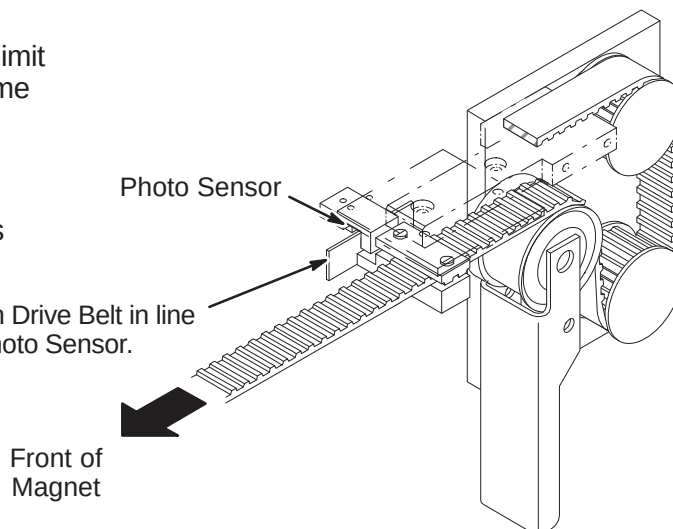
NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:

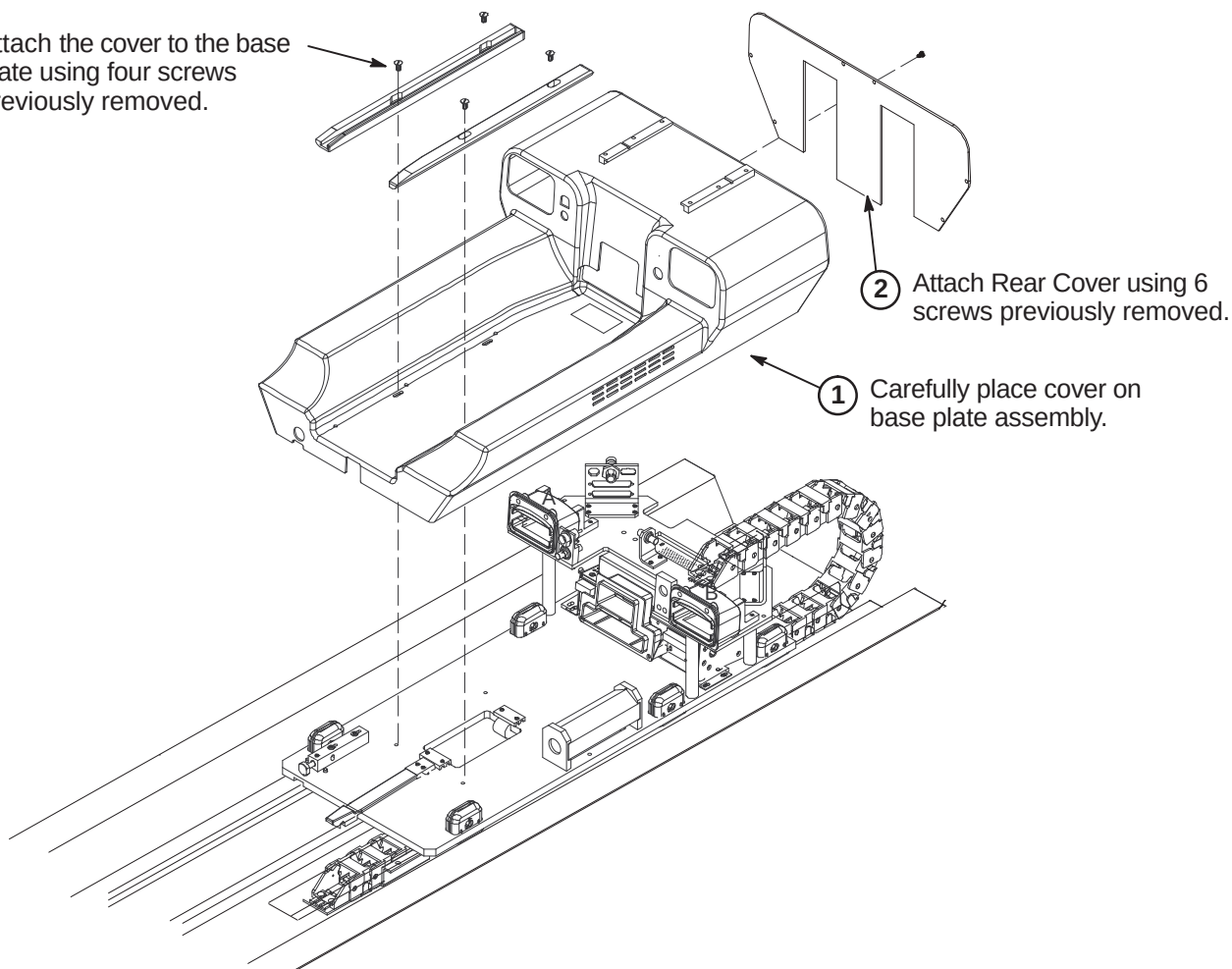
The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



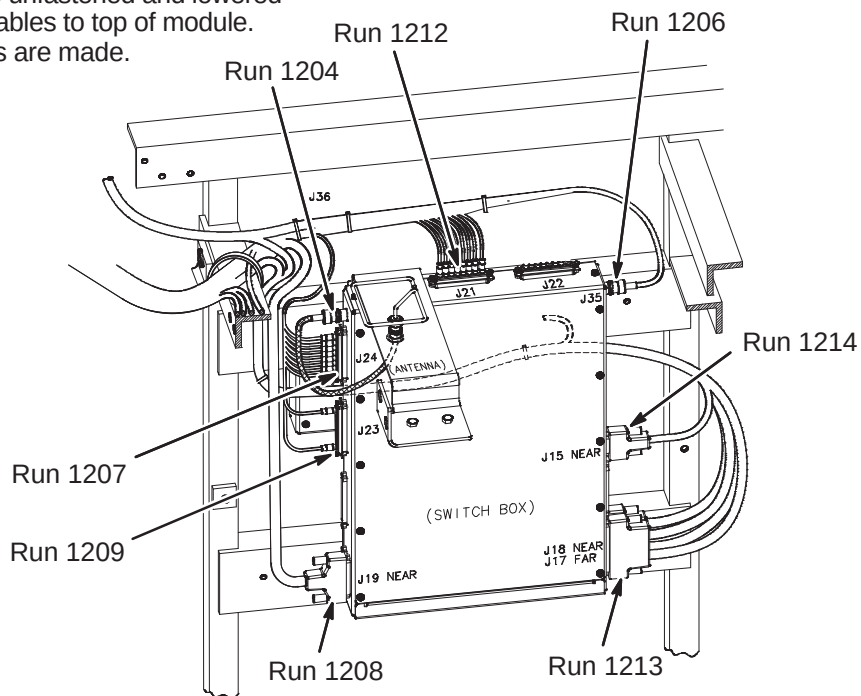
8-7-6-F Installation of LPCA Cover

- ③ Attach the cover to the base plate using four screws previously removed.



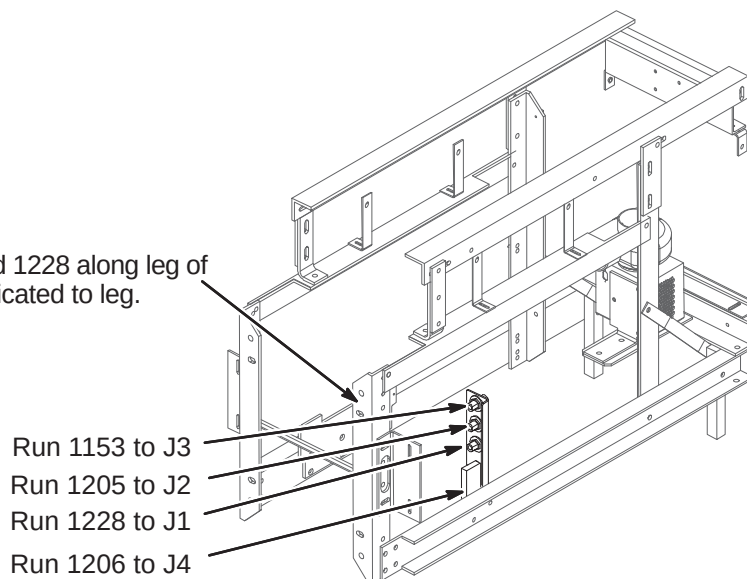
8-7-6-G Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**8-7-6-H Connect Cables from Cable Track to Bulkhead Interface**

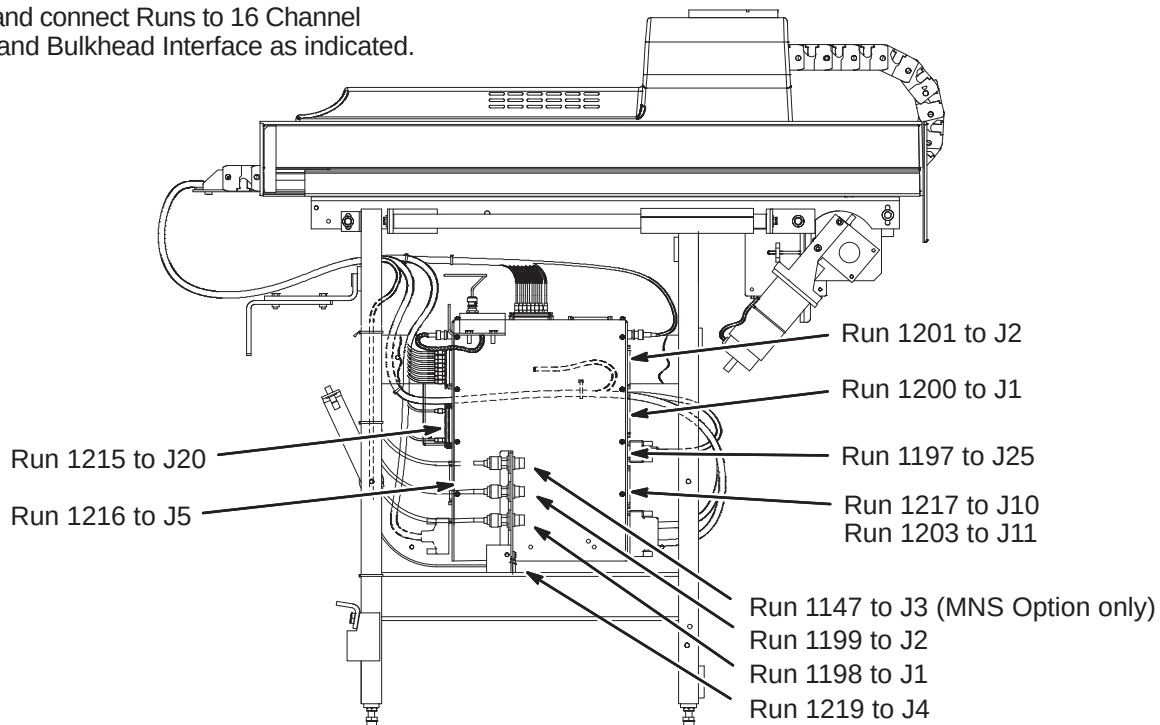
- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.

- ② Connect Runs to Bulkhead Interface as indicated.



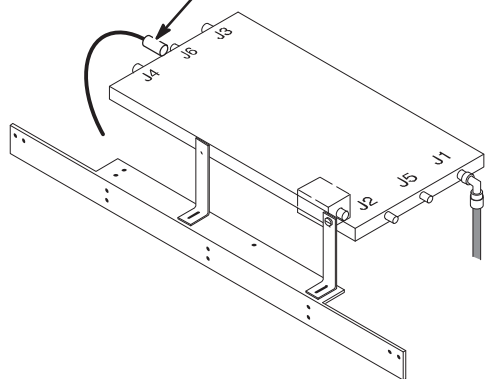
8-7-7 CONNECT REMAINING CABLES TO REAR PEDESTAL

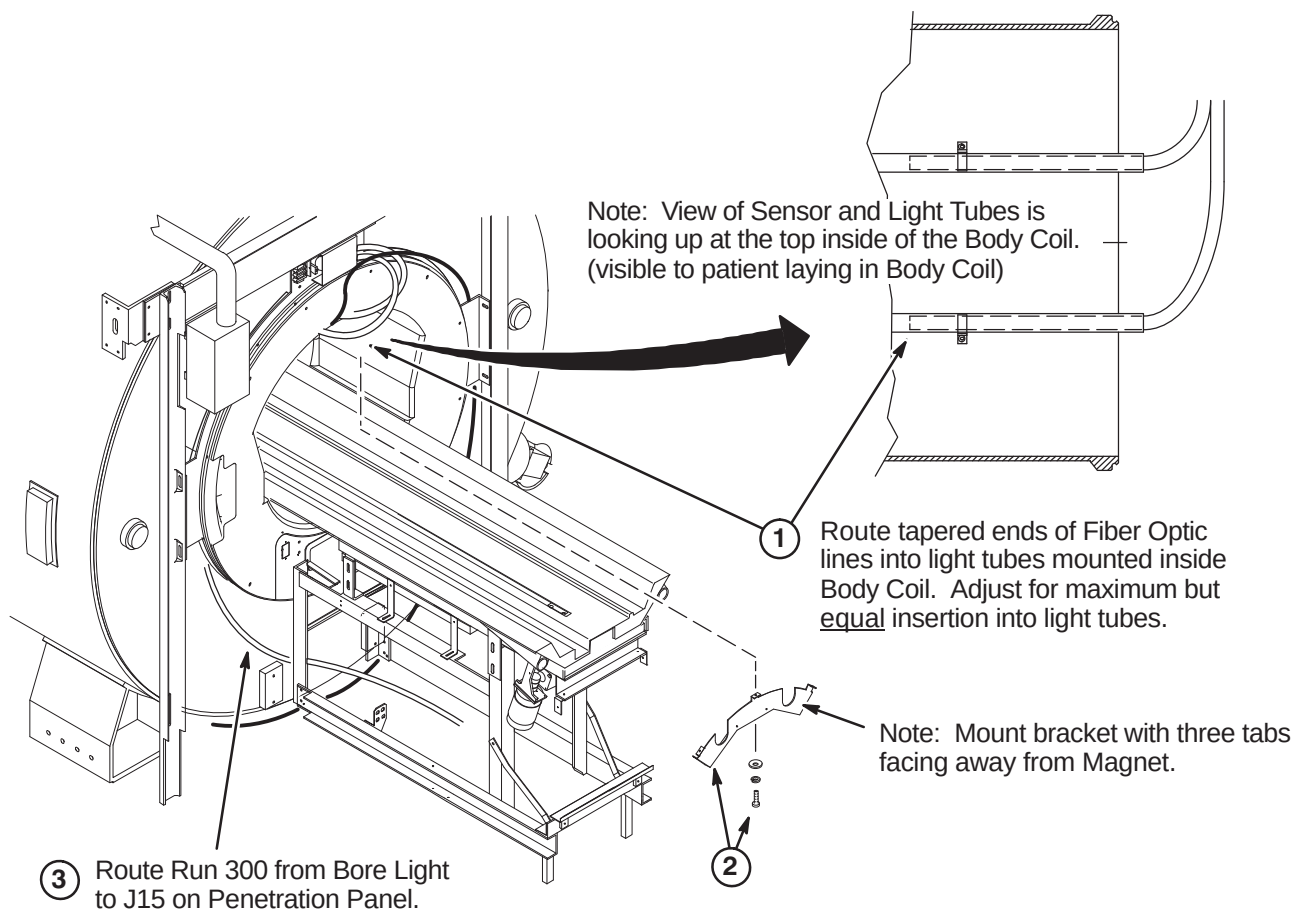
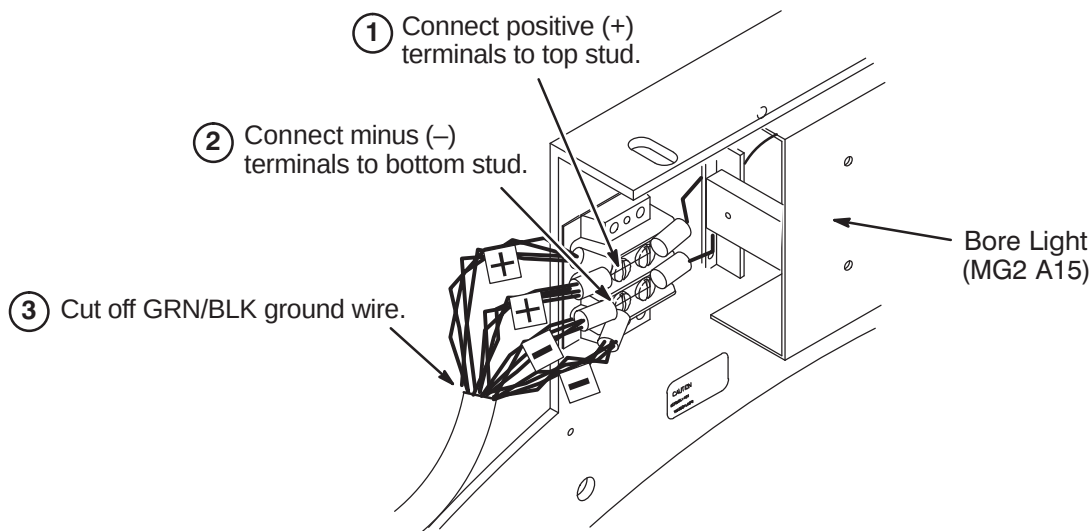
- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



- ② Route/connect Run 780 to J6 on RF Splitter.

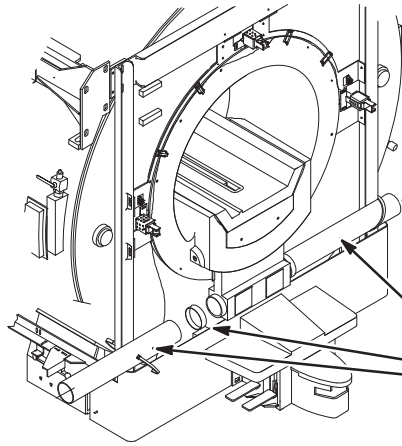
- ③ Connect Run 1250 to J1 on RF Splitter.



8-7-8 INSTALL BORE LIGHT**8-7-8-A Route Run 300 and Install Light Tube Fiber Optics****8-7-8-B Connect Run 300 To Bore Light**

8-7-9 INSTALL PATIENT COMFORT AIR SYSTEM

8-7-9-A Connecting Front Air Plenum Hoses

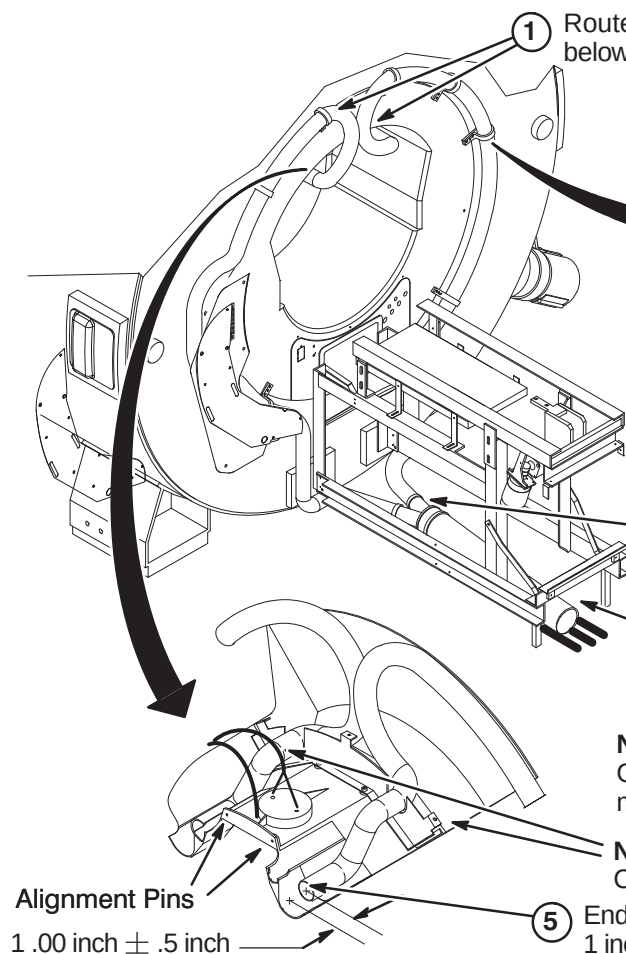


② Ty-wrap each hose to Vertical Uprights on each side.

Note: All the air hoses and clamps in the following steps are located in the Blower Hose Kit (2129412).

① Cut two (2) 26 inch lengths of 4 Inch Hose (46-282666P5) and attach one length to each side of Front Air Plenum using:
(2) 46-208765P34 4 inch Hose Clamp

8-7-9-B Routing/Connection Of Patient Comfort Air Hoses



① Route the (2) 2 inch diameter hoses (46-282668P18) as shown from below the Rear Pedestal, around and to the top of the End Ring.

② Secure 2 inch hose to End Ring in seven places by cutting Nylon Strap (2139049) into seven 12 inch long pieces and fastening with:

(7) 1000904P447 3/8 Flat Washer
(7) 46-208948P2 3/8-16 SST Hex Nut

③ Attach ends of 2 inch hoses to 2 Way Header (2120631) using:

(2) 46-208765P28 2 inch Hose Clamps

④ Attach remaining length of 4 inch Hose (46-282666P5) to 2-way Header using:
(1) 46-208765P34 4 inch Hose Clamp

Note: 4 Inch Hose should lie on top of previously routed Gradient Coil Cables. Six Inch Hose routed in Step L4 must lie on the floor next to Gradient Coil Cables.

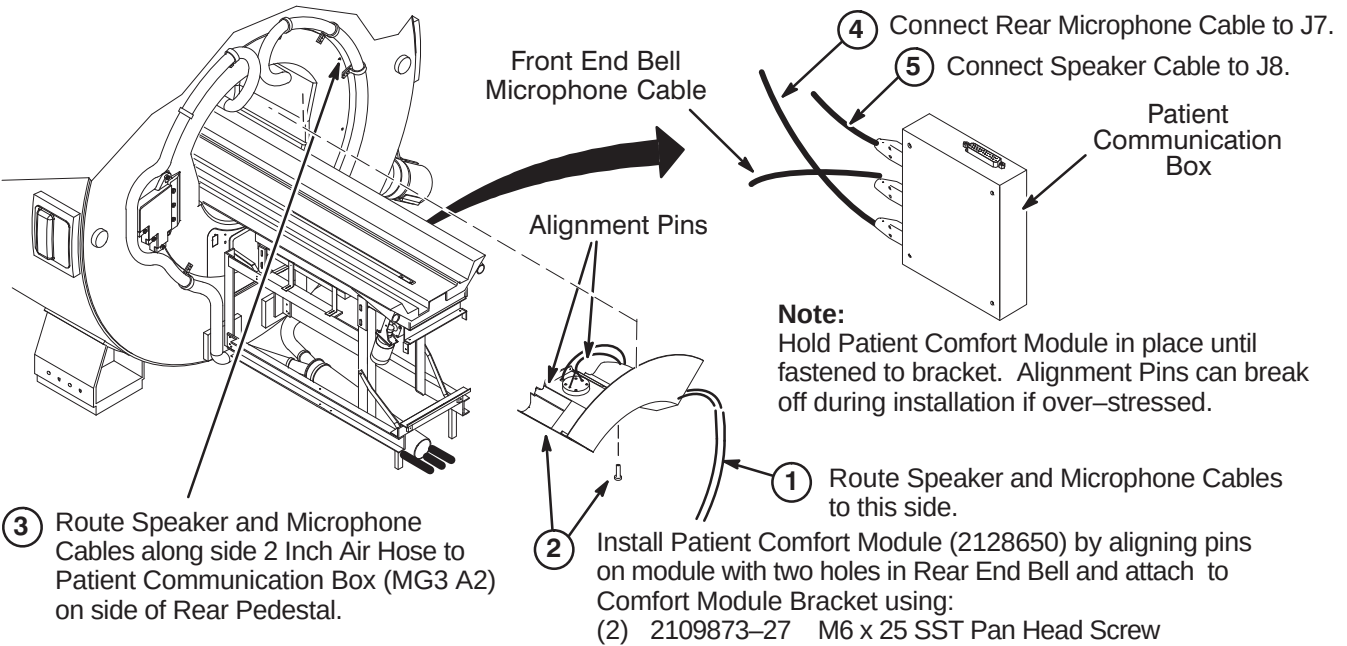
Note: For clarity, this view shows final assembly with Comfort Module Cover in place and cables routed to side.

⑤ Ends of 2 inch hoses should be routed to within 1 inch ($\pm .5$ inch) of edge of Comfort Module Cover.

Alignment Pins

1.00 inch $\pm .5$ inch

8-7-9-C Install Patient Comfort Module and Cables



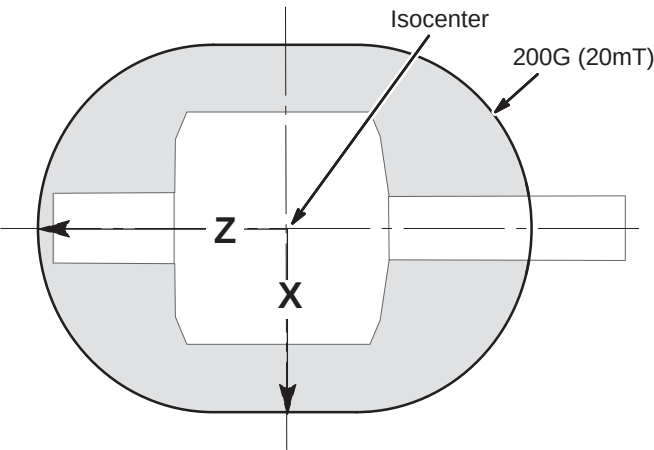
8-7-9-D 200 Gauss Zone Limit for Movement of Blower Box

Refer to the illustration and table below for applicable 200 Gauss zone for magnet at site.



THE BLOWER BOX IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.

Illustration is not to scale

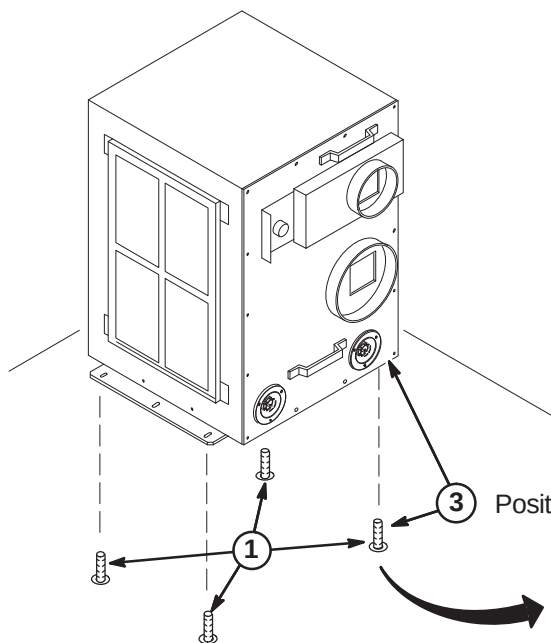


Approximate Distance from isocenter to 200G Line		
	X	Z
3.0T	180 cm (71 in) ¹	272 cm (107 in) ¹
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
1.0T & Lower	145 cm (57 in) ¹	195 cm (77 in) ¹
Unshielded Magnet	Measure ²	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

8-7-9-E Installation of Blower Box

WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

**IMPORTANT!**

Location of Blower Box and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural)/ Construction drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

CAUTION

***THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions.**

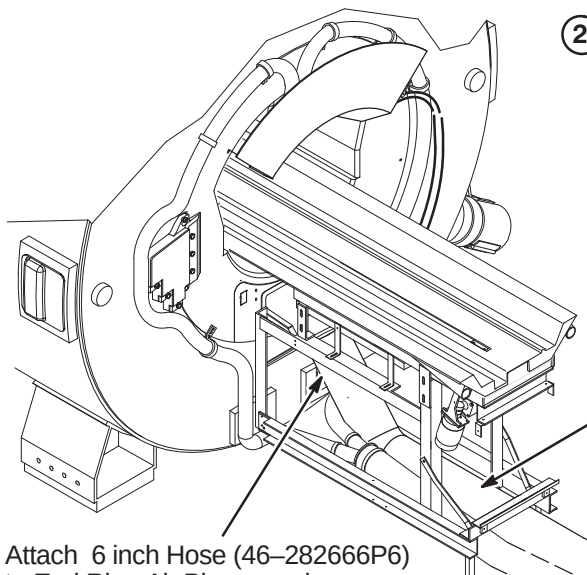
- ① Check for correct installation of the anchors:
 - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor. *
 - Confirm there are at least four (4) anchors.
 - If any of the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.

③ Position the blower box on top of anchors.

② Remove any mounting hardware from the anchors.

④ Secure the blower box using the hardware provided by the RF shield room vendor to attach.

8-7-9-F Install Blower Box Air Hoses from Rear of Magnet to Blower Box



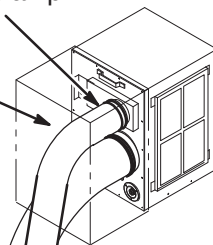
- ② Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:

- (1) 46-208765P34 4 inch Hose Clamp
- (1) 46-208765P29 6 inch Hose Clamp

Note: If needed, Blower Hose Cover can be cut and modified.

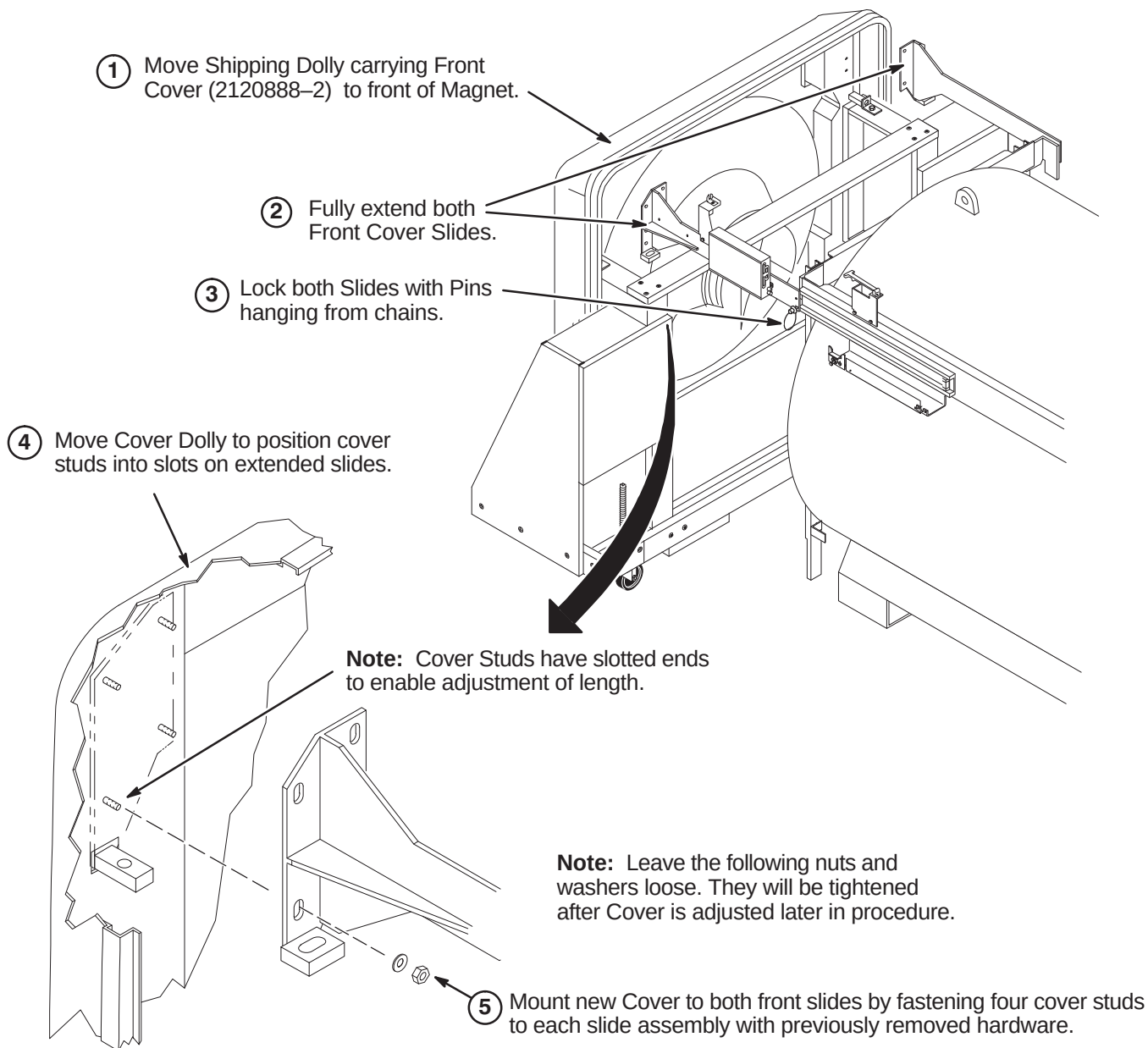
Note: 6 inch Hose must lie on the floor next to Gradient Coil Cables.

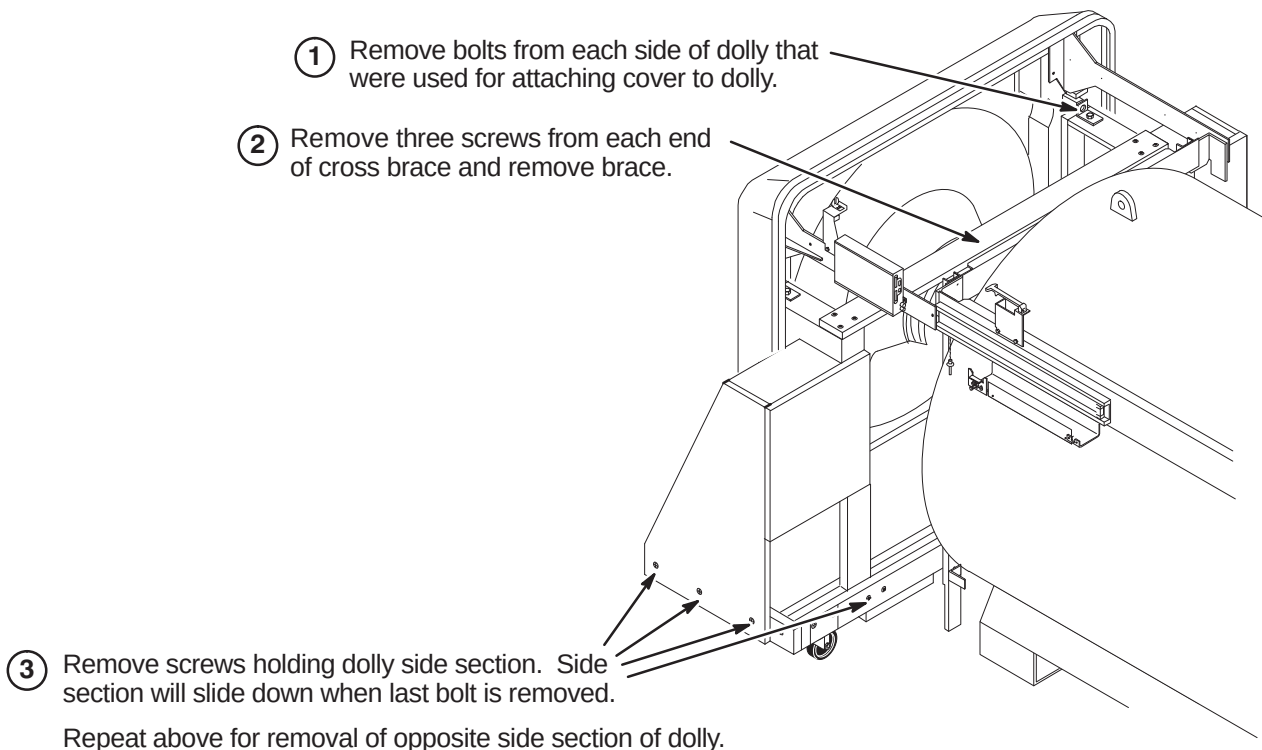
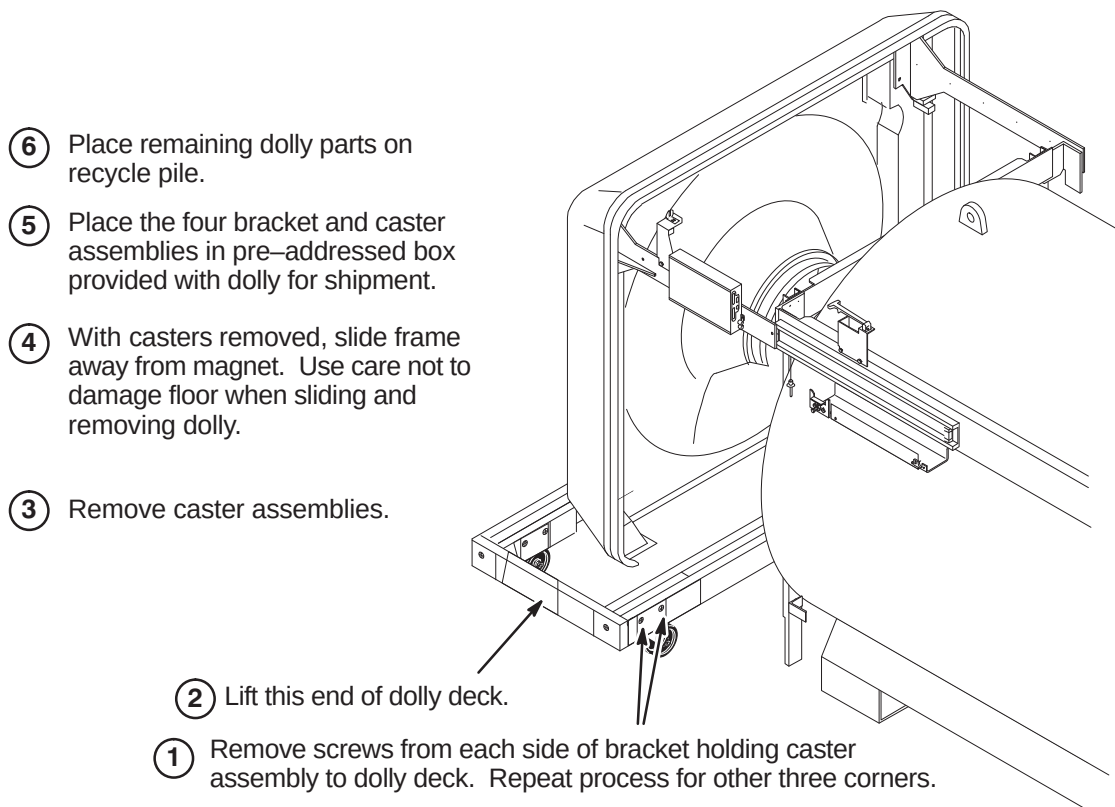
- ① Attach 6 inch Hose (46-282666P6) to End Ring Air Plenum using:
 - (1) 46-208765P29 6 inch Hose Clamp



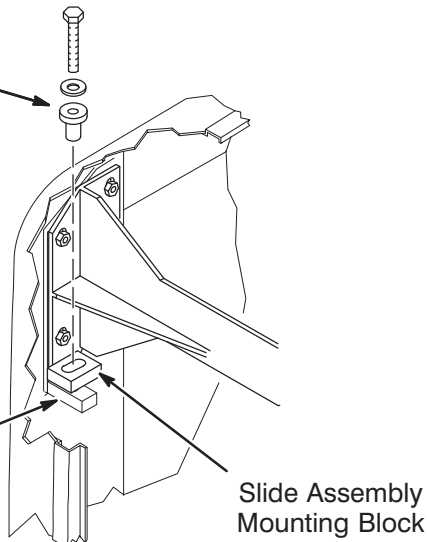
8-7-10 INSTALL FRONT COVER**8-7-10-A Mount Front Cover**

FERROUS MATERIAL HAZARD! THE COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE FRONT COVER FROM DOLLY AND CARRY COVER INTO POSITION AT FRONT OF MAGNET.

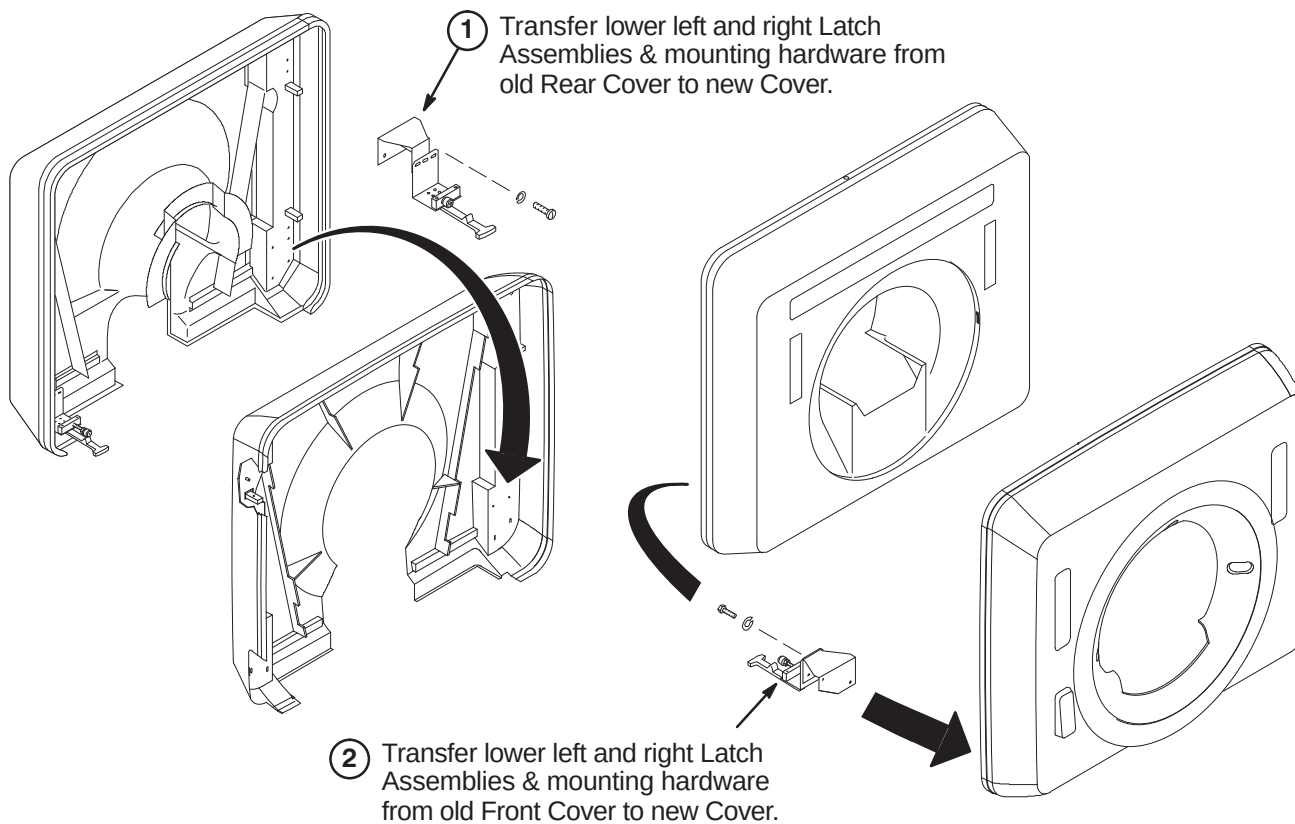


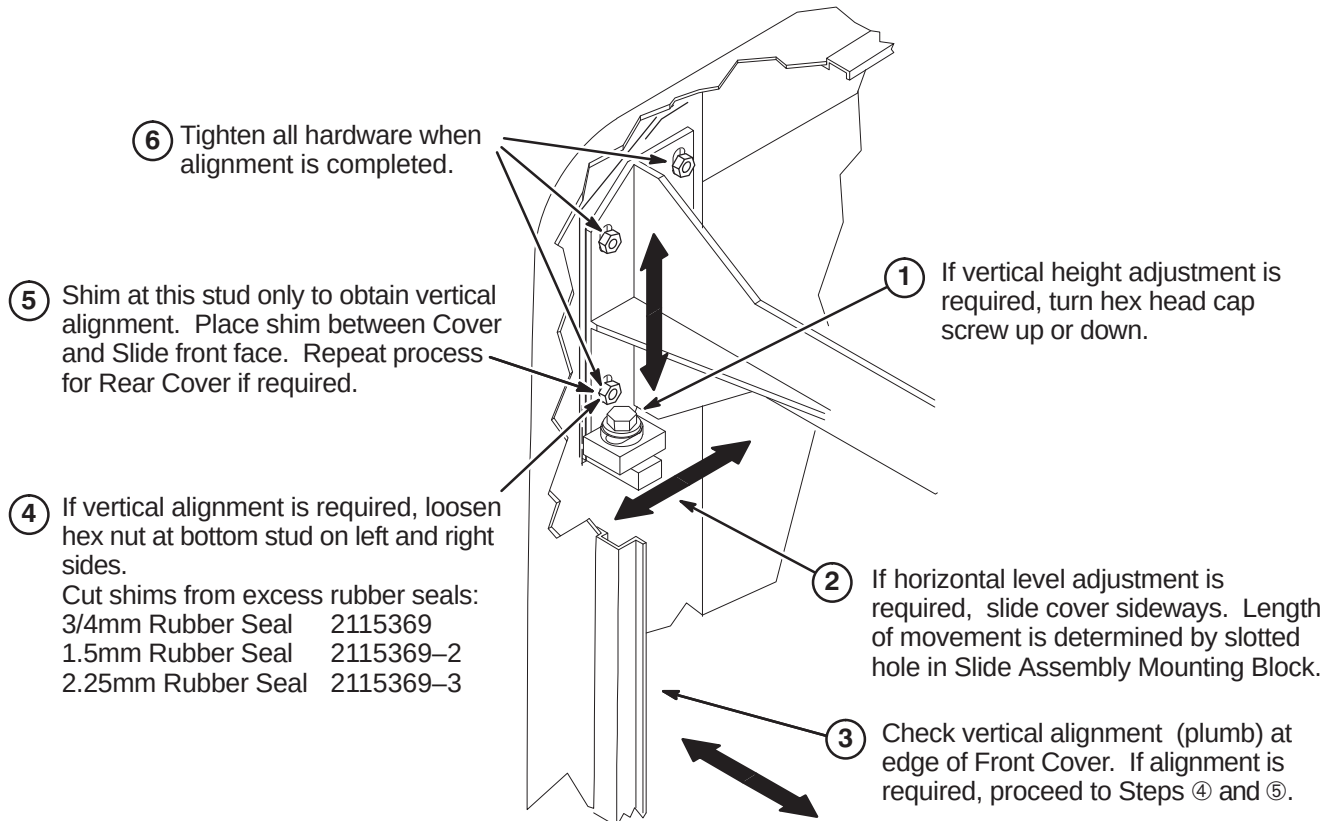
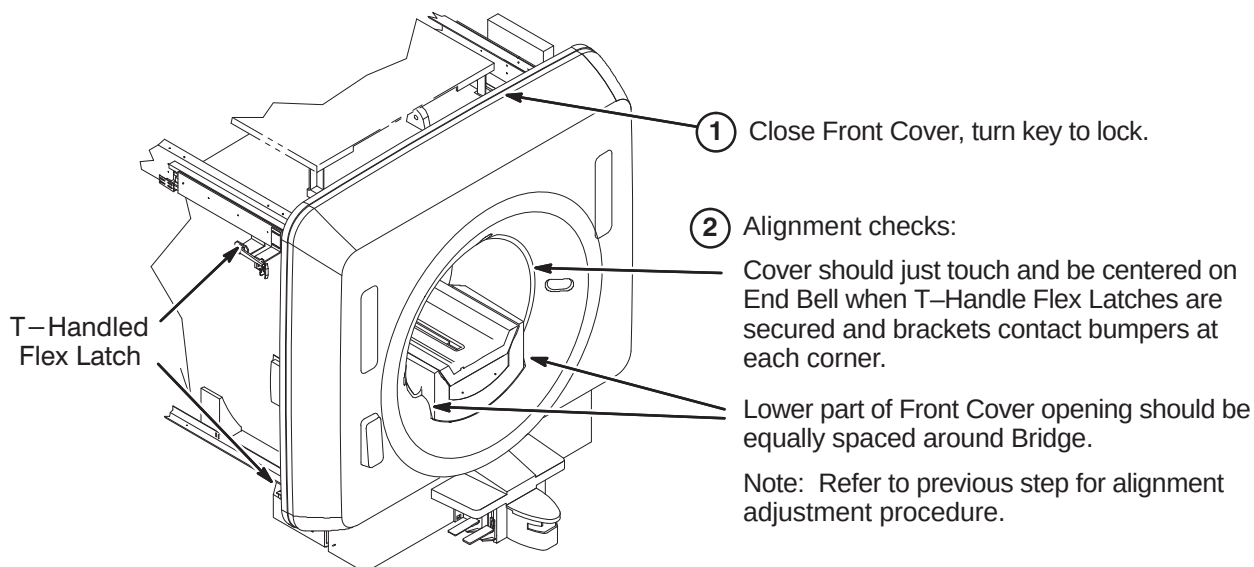
8-7-10-B Disassemble Cover Dolly**8-7-10-C Remove Cover Dolly**

8-7-10-D Install Front Cover Alignment Hardware

- ① Transfer and insert the Alignment Hardware from the old Front Cover into the oval hole on the Slide Assembly Mounting Block
 - ② Engage threads in Cover Mounting Block.
 - ③ Repeat process for other side of Front Cover.
- 

8-7-10-E Transfer Lower Latch Assemblies to New Front & Rear Covers

- ① Transfer lower left and right Latch Assemblies & mounting hardware from old Rear Cover to new Cover.
 - ② Transfer lower left and right Latch Assemblies & mounting hardware from old Front Cover to new Cover.
- 

8-7-10-F Adjust Front Cover Alignment**8-7-10-G Align Front Cover**

8-7-11 INSTALL REAR COVER



FERROUS MATERIAL HAZARD! THE REAR COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE COVER FROM DOLLY AND CARRY COVER INTO POSITION AT REAR OF MAGNET.

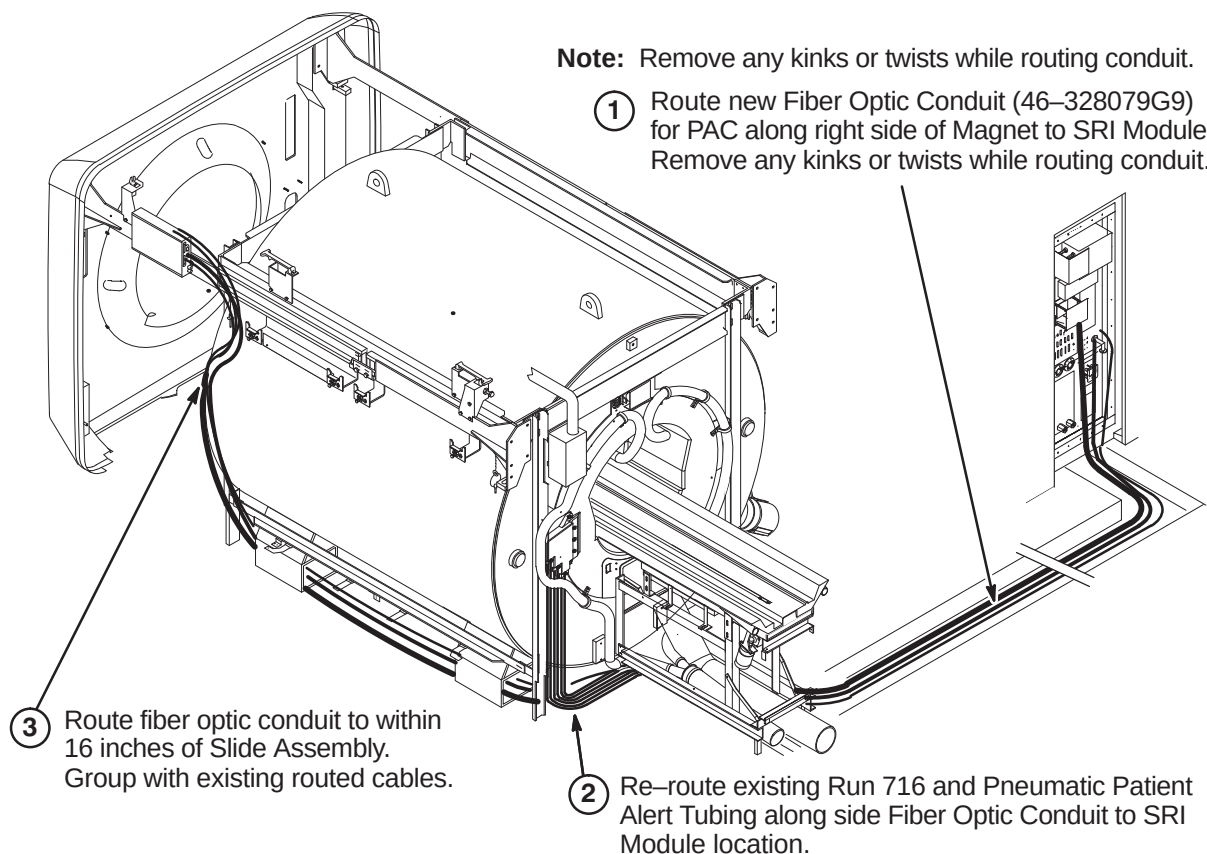
THE STEPS ARE THE SAME AS THE JUST COMPLETED FRONT COVER INSTALLATION PROCEDURE

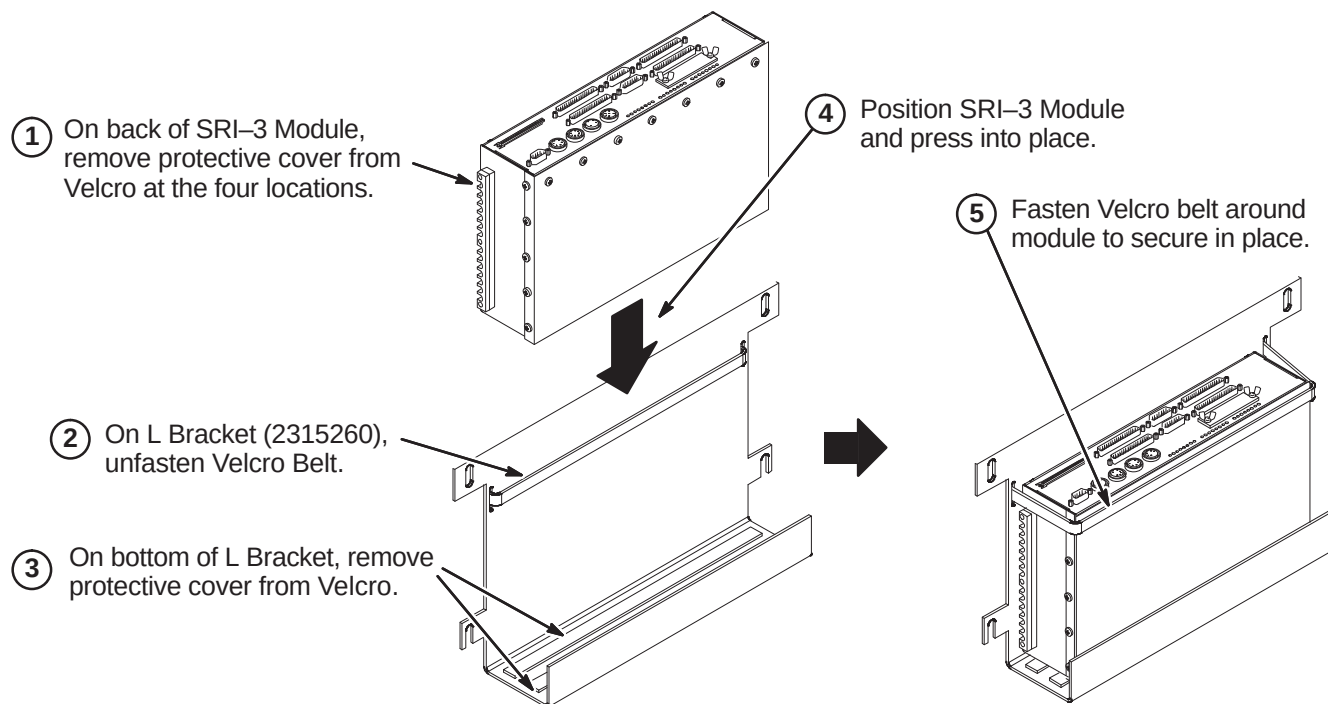
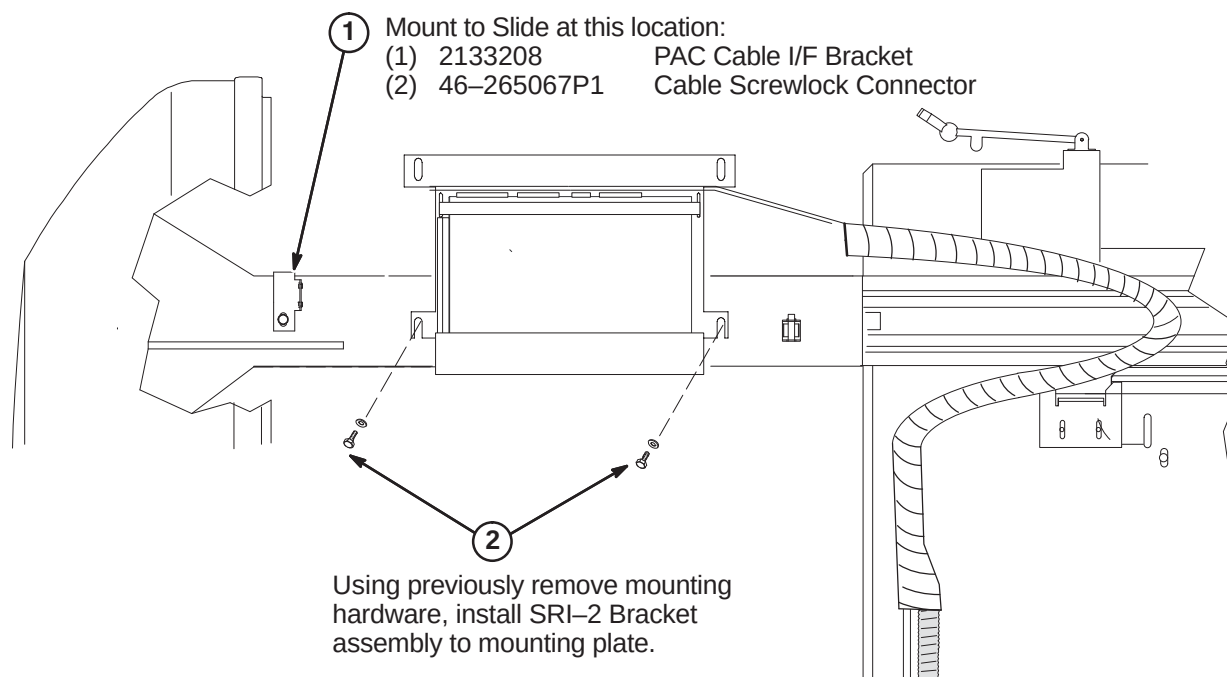
Note: Experience has shown that the Rear Cover does not require shimming for vertical levelness.

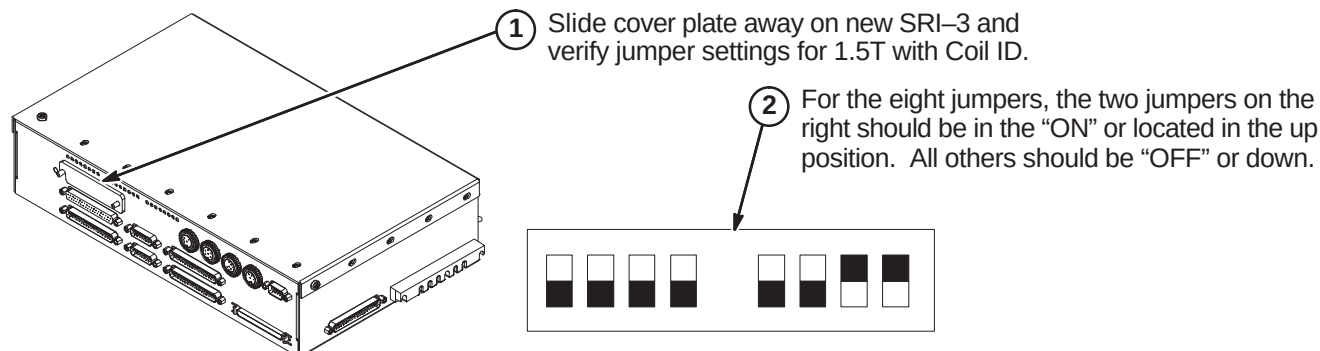
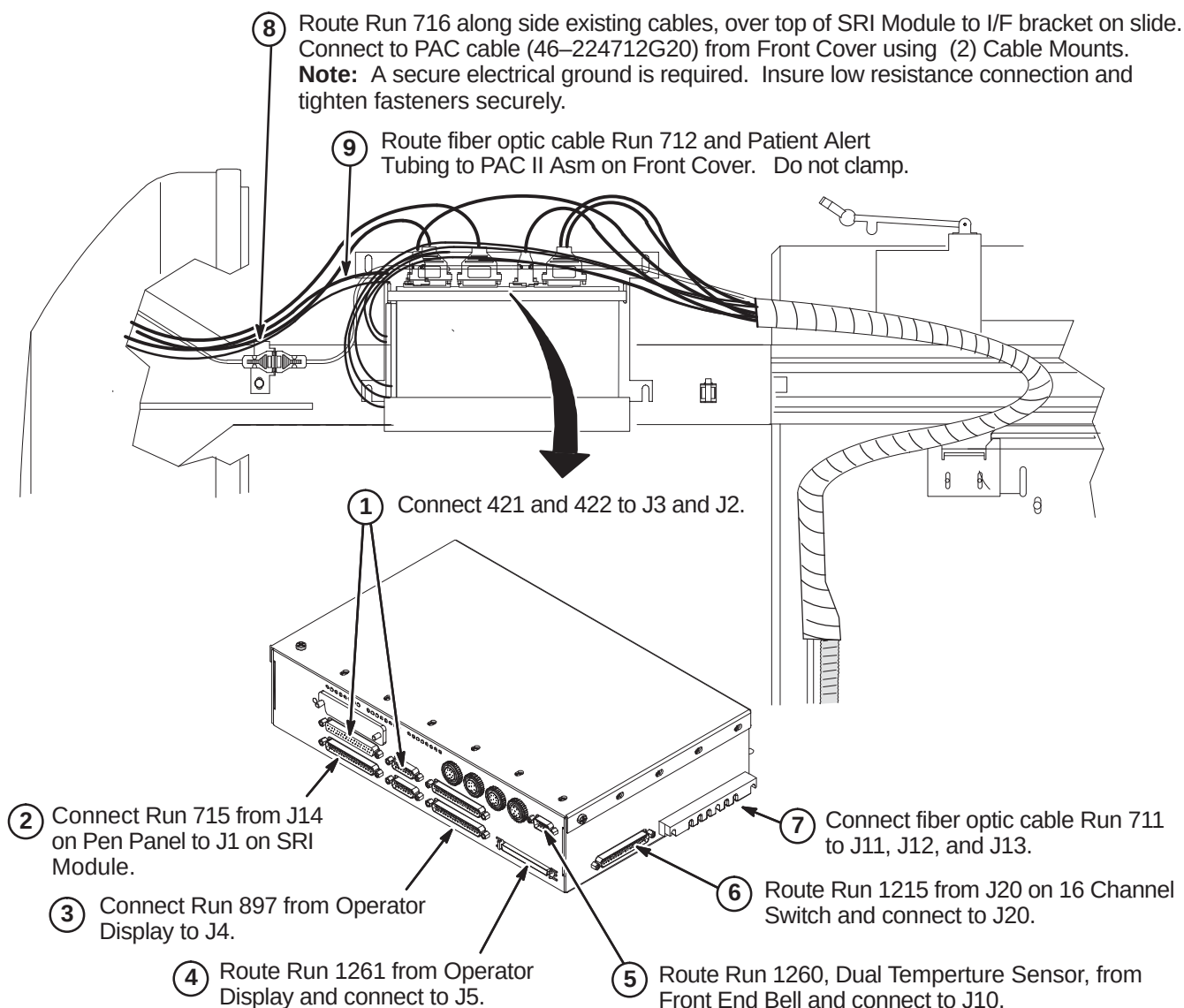
8-7-12 ROUTE/CONNECT RUNS 711/712, 716, AND PATIENT ALERT TUBING

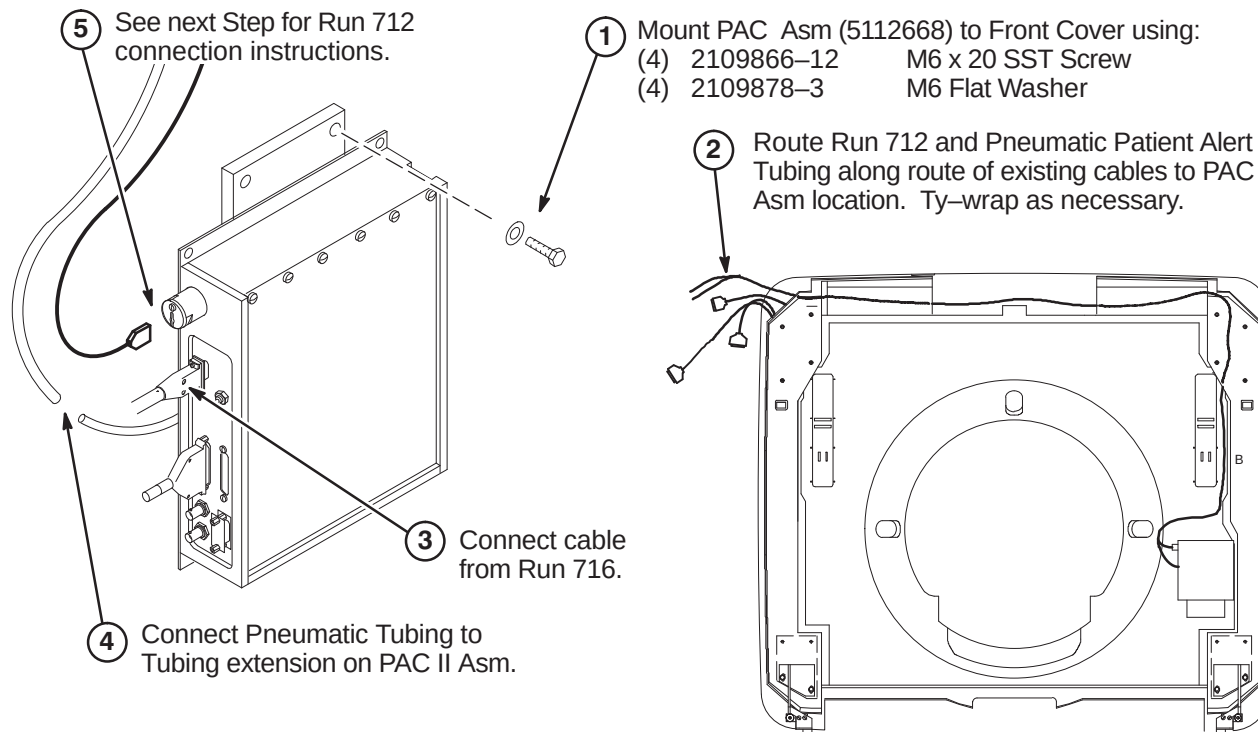
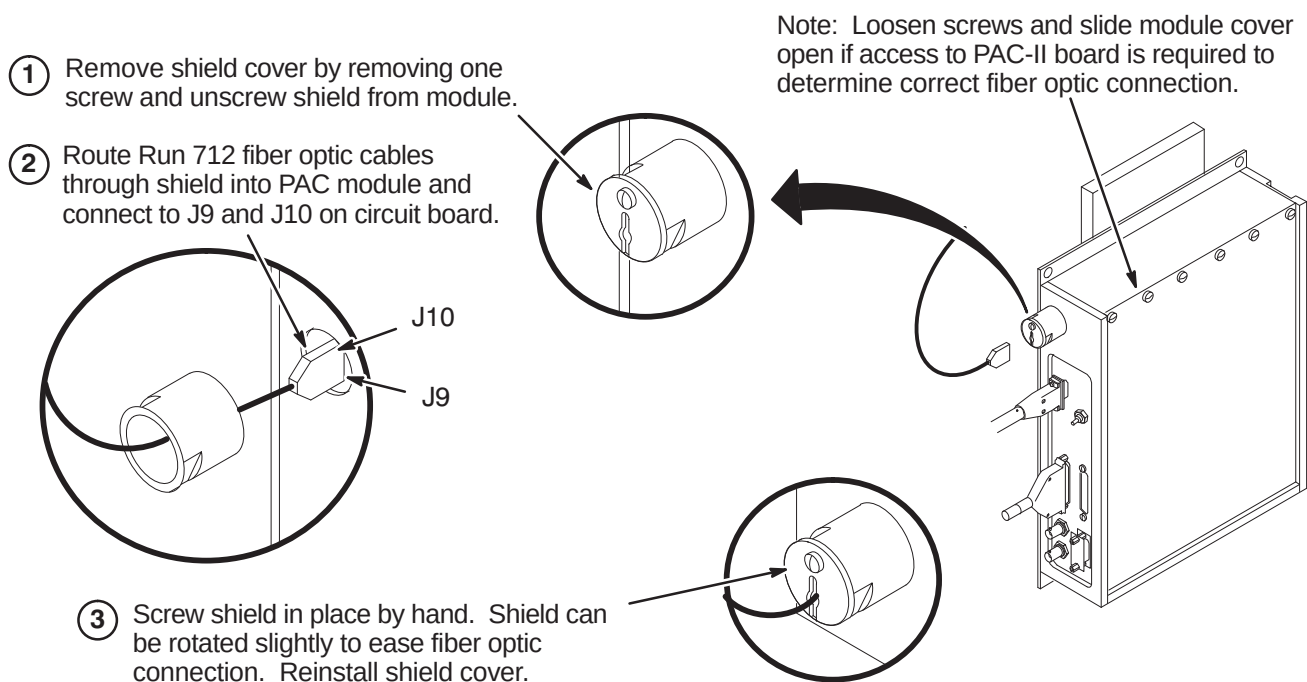


EQUIPMENT DAMAGE POSSIBILITY. Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend to radius smaller than two inches. Do not pull connector from cable. Avoid scratching connector ends. Keep ends protected until ready to connect.



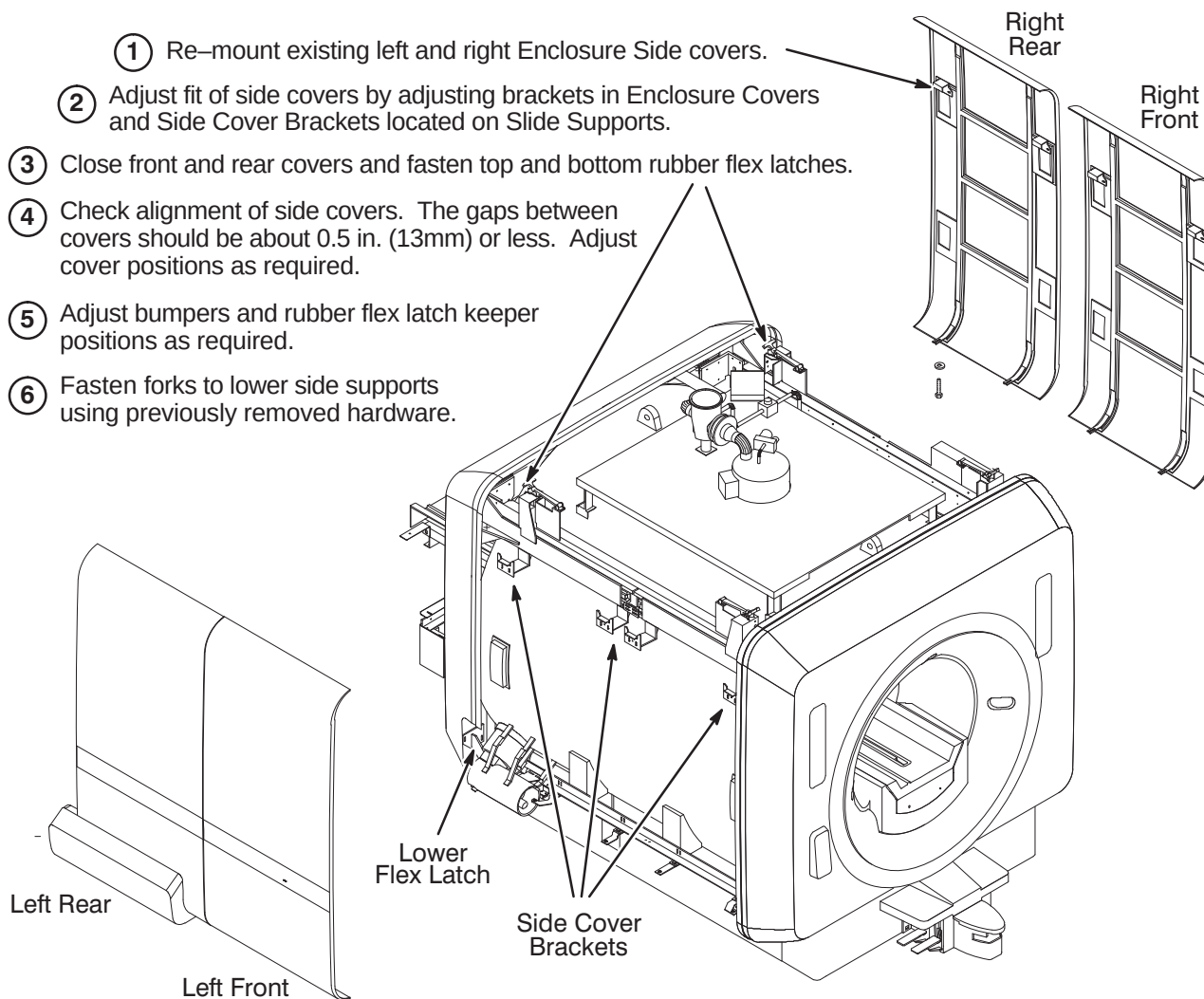
8-7-13 INSTALL SRI-3 MODULE**8-7-13-A Position SRI-3 in L Bracket****8-7-13-B Install SRI-3 on Magnet**

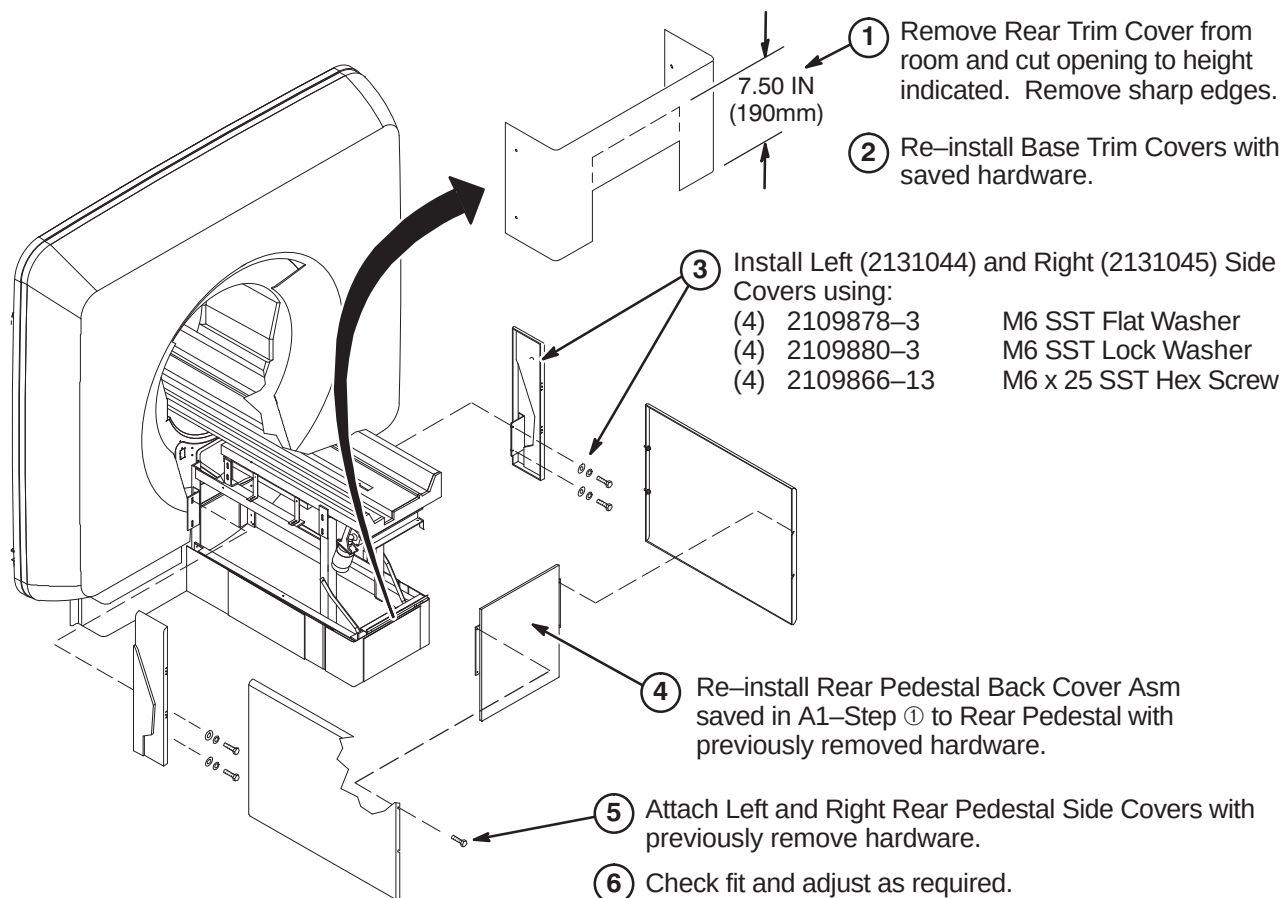
8-7-13-C Verify SRI-3 Jumper Settings**8-7-13-D Connect Cables at SRI-3 Module Location**

8-7-14 INSTALL NEW PAC MODULE**8-7-14-A Connect Run 712, 716, & Pneumatic Patient Alert Tubing****8-7-14-B Connect Run 712**

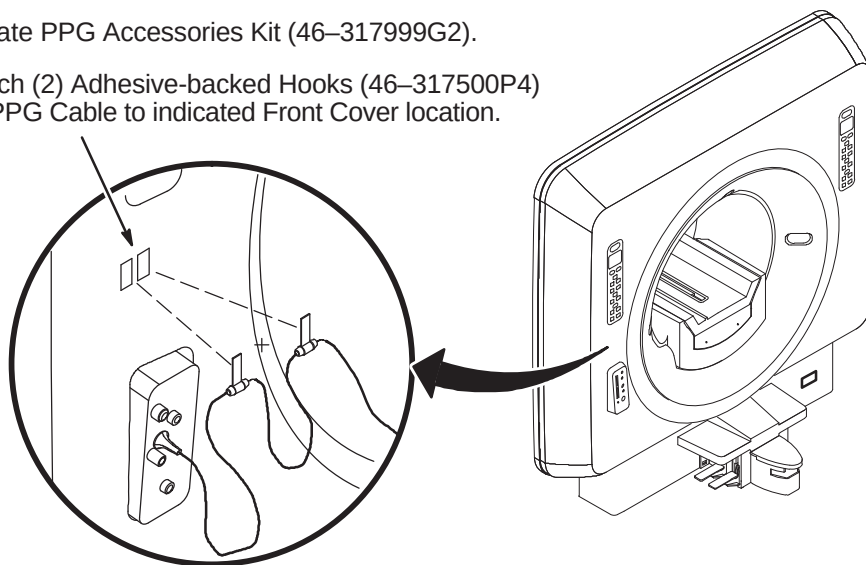
8-7-15 INSTALL ENCLOSURE SIDE COVERS

- ① Re-mount existing left and right Enclosure Side covers.
- ② Adjust fit of side covers by adjusting brackets in Enclosure Covers and Side Cover Brackets located on Slide Supports.
- ③ Close front and rear covers and fasten top and bottom rubber flex latches.
- ④ Check alignment of side covers. The gaps between covers should be about 0.5 in. (13mm) or less. Adjust cover positions as required.
- ⑤ Adjust bumpers and rubber flex latch keeper positions as required.
- ⑥ Fasten forks to lower side supports using previously removed hardware.

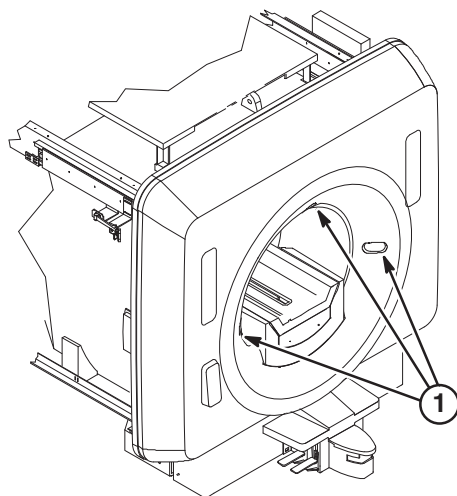


8-7-16 INSTALL REAR PEDESTAL COVERS**8-7-17 INSTALL PPG CABLE HOOKS ON FRONT COVER**

- 1** Locate PPG Accessories Kit (46-317999G2).
- 2** Attach (2) Adhesive-backed Hooks (46-317500P4) for PPG Cable to indicated Front Cover location.



8-8 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS



A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



Spanish



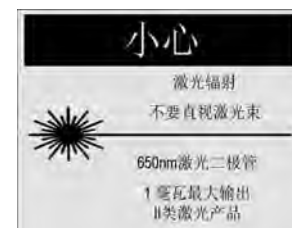
Italian



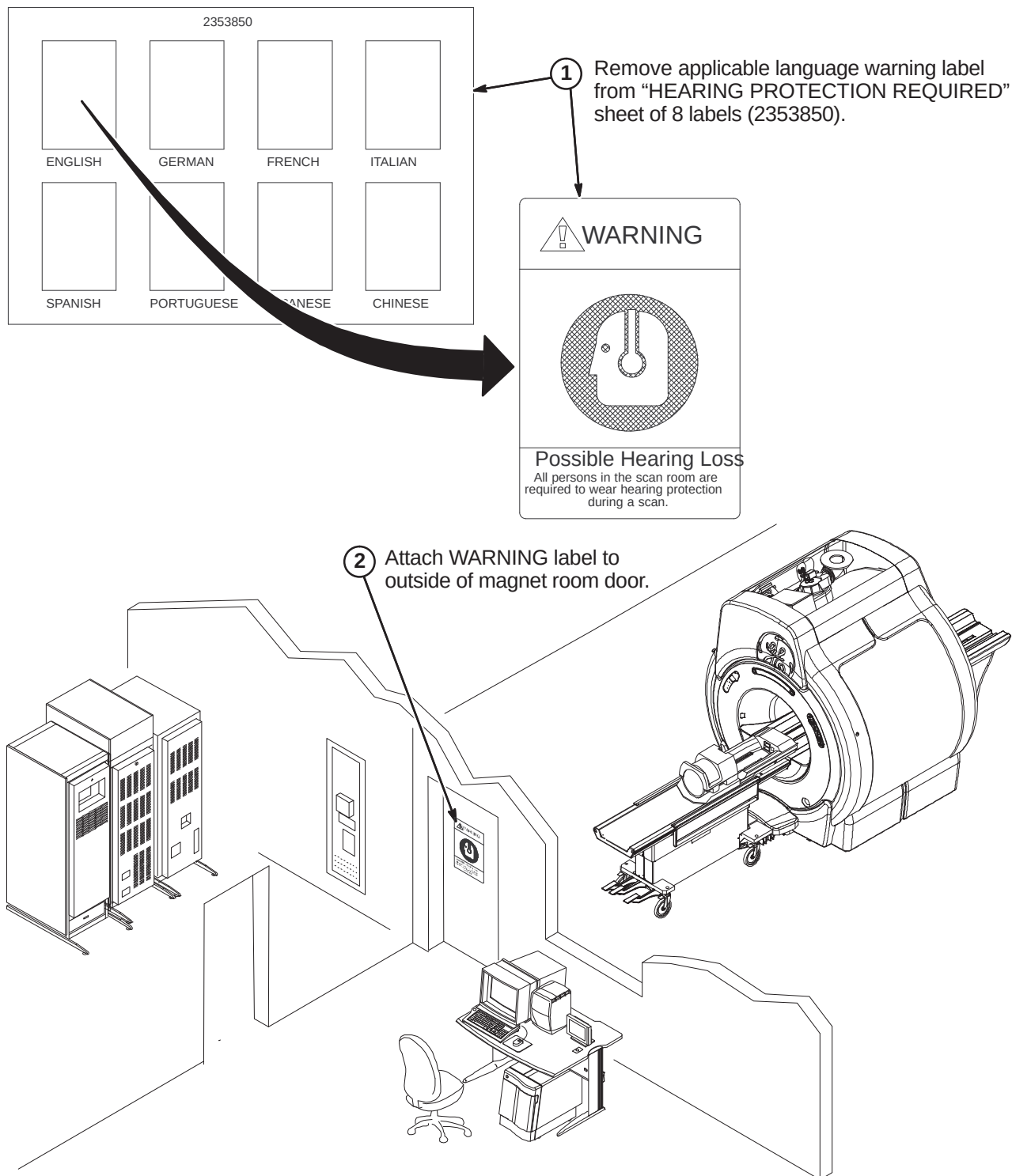
Swedish



Japanese



Chinese

8-9 INSTALL HEARING LOSS WARNING SIGN

SECTION 9 – MAGNISHIELD ADVANTAGE UPGRADE

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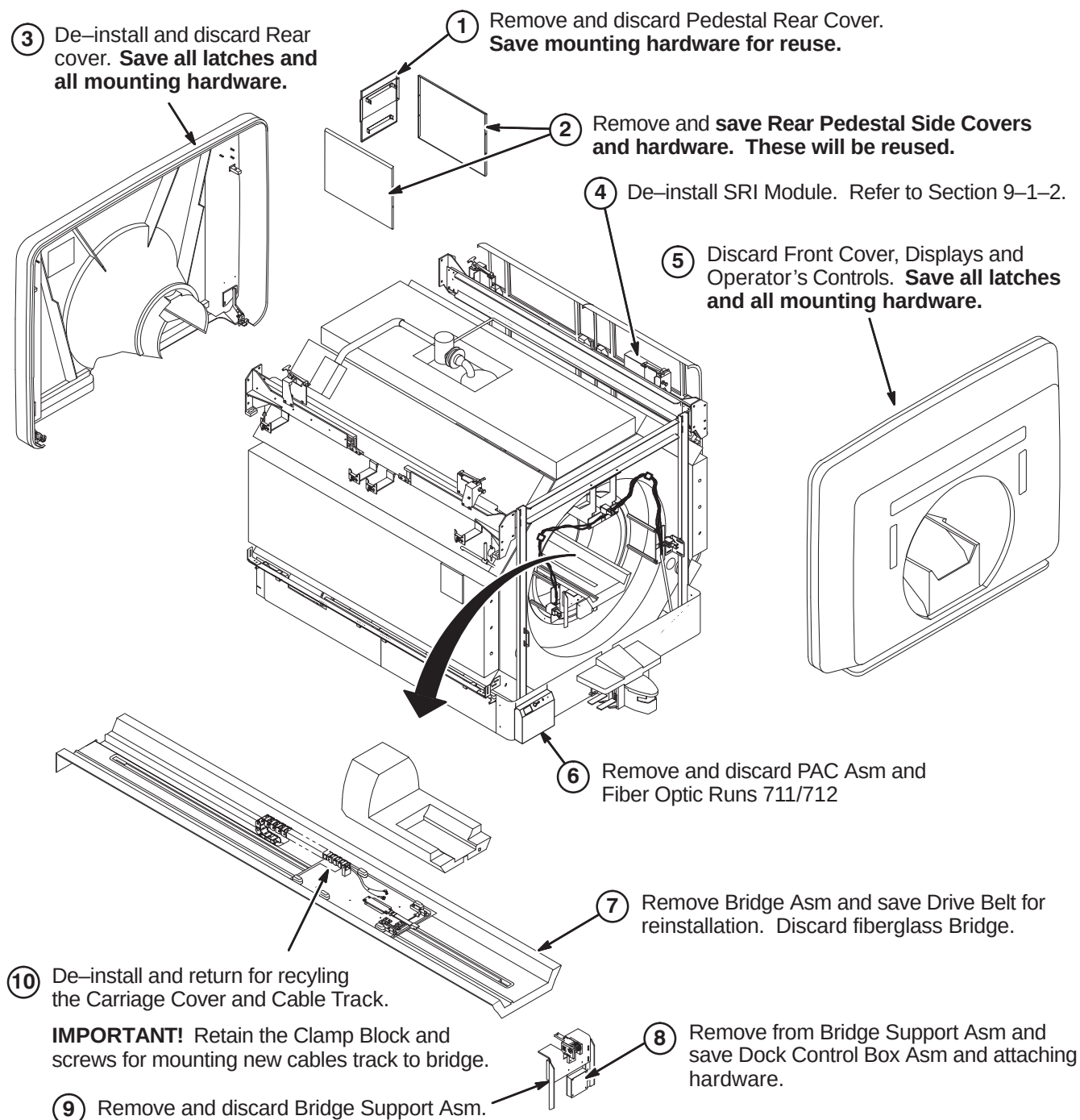
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9-8 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS	9-43
9-9 INSTALL HEARING LOSS WARNING SIGN	9-44

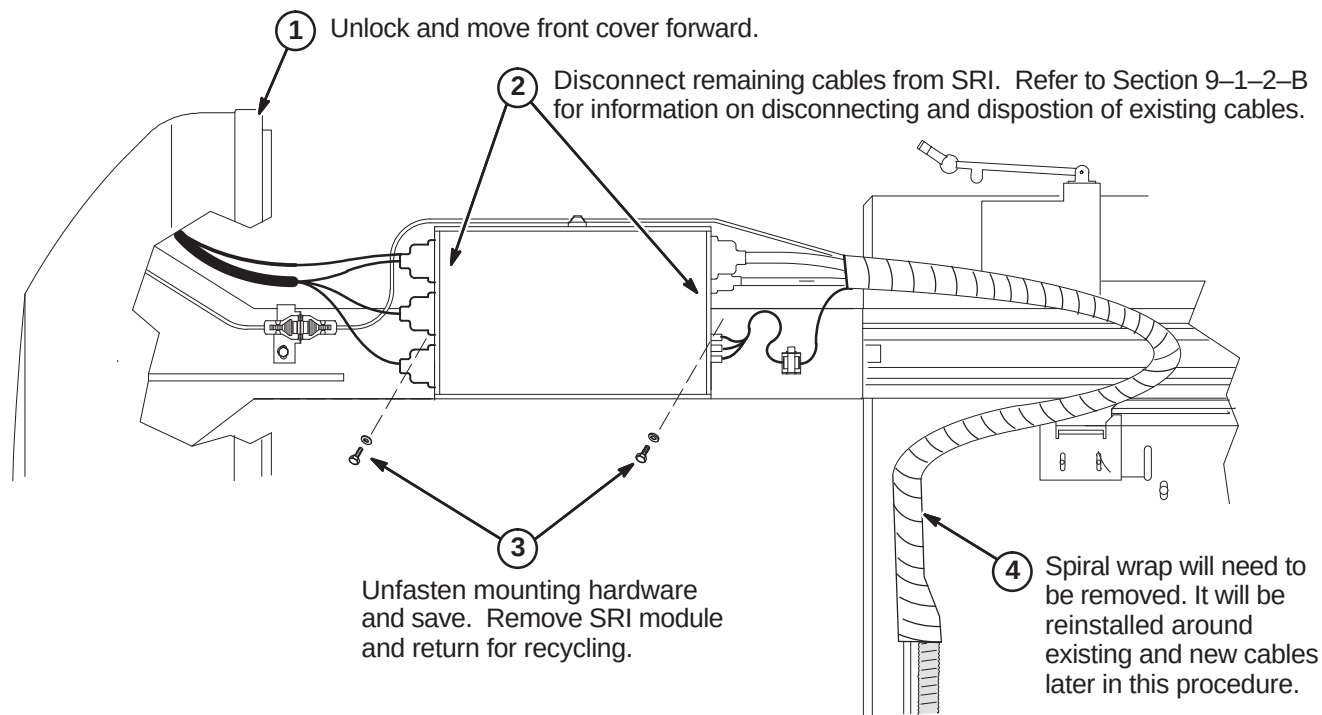
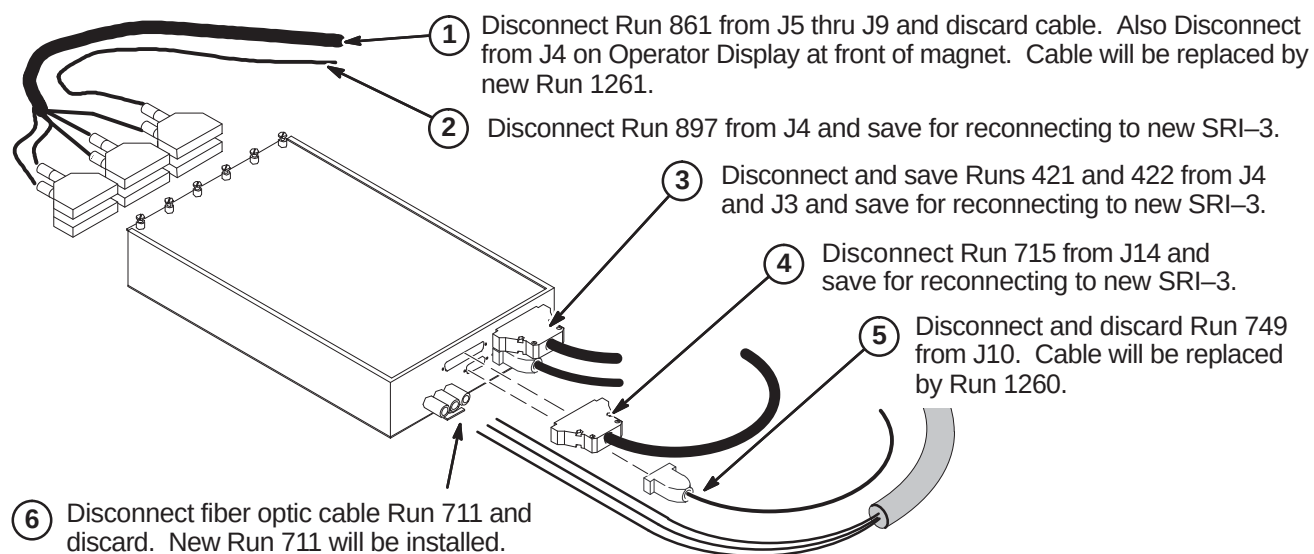
9-1 DE-INSTALL ADVANTAGE MAGNISHIELD MAGNET ENCLOSURE

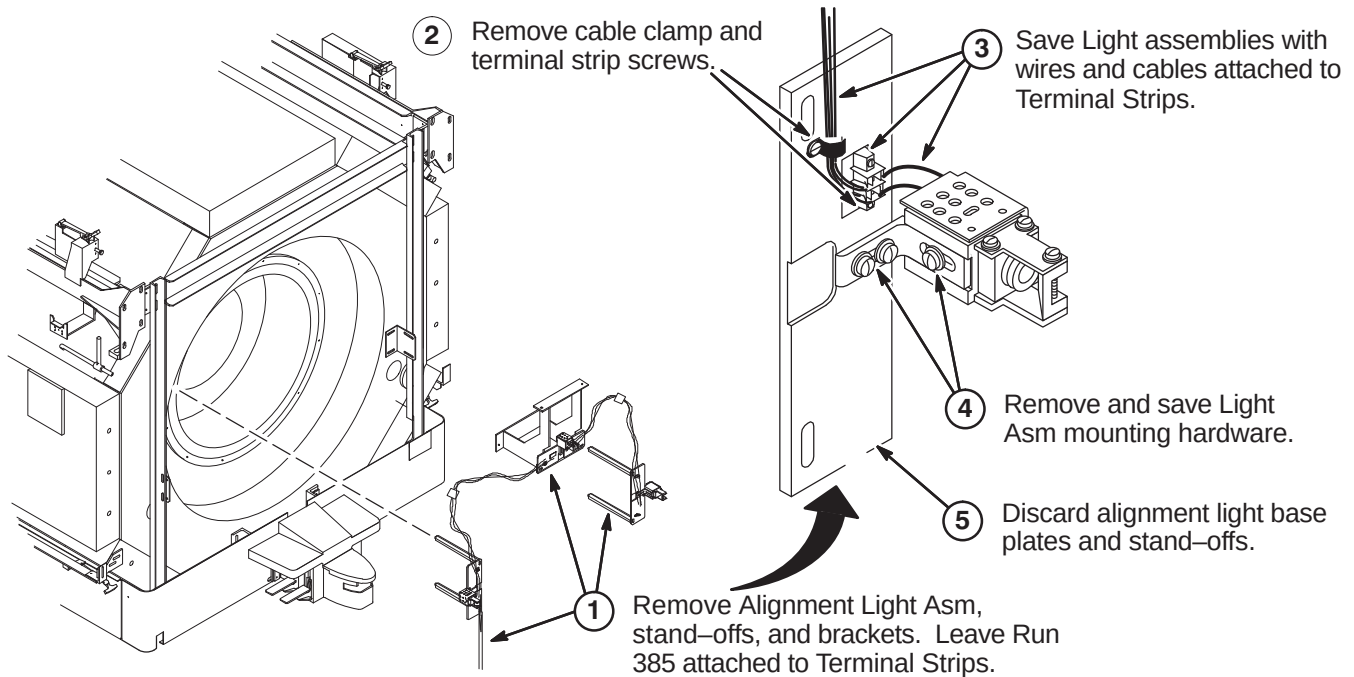
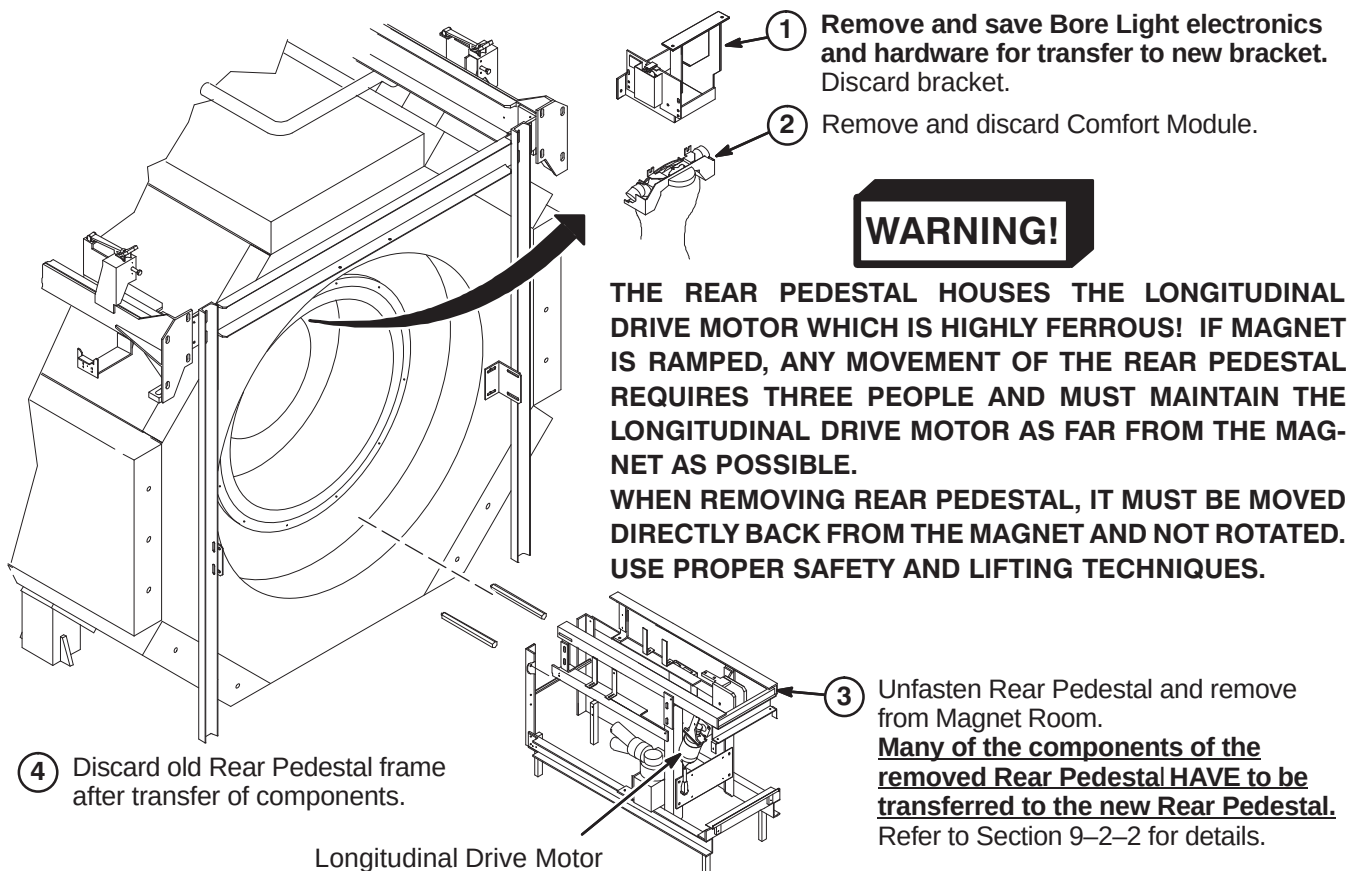
Refer to Section 4, Table 4-1 for information on cable removal. Also refer to applicable cable map for additional information on cable removal.

Discarded Material Return Policy – Refer to Section for procedures on how to handle all de-installed material that is designated as “discard” in the following procedures.

9-1-1 DE-INSTALL COVERS, BRIDGE, BRIDGE SUPPORT, DSIM/SRI AND PAC ASM

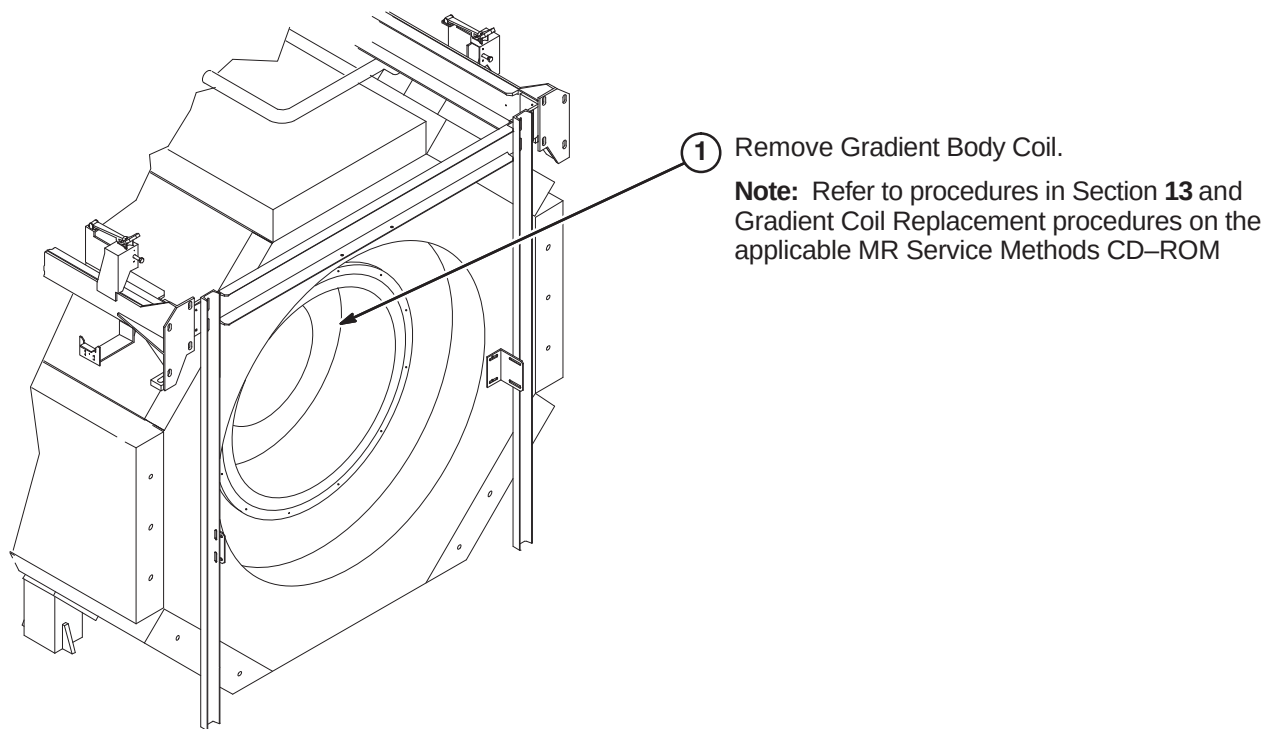
- 
- ③ De-install and discard Rear cover. **Save all latches and all mounting hardware.**
 - ① Remove and discard Pedestal Rear Cover. **Save mounting hardware for reuse.**
 - ② Remove and **save** Rear Pedestal Side Covers and hardware. **These will be reused.**
 - ④ De-install SRI Module. Refer to Section 9-1-2.
 - ⑤ Discard Front Cover, Displays and Operator's Controls. **Save all latches and all mounting hardware.**
 - ⑥ Remove and discard PAC Asm and Fiber Optic Runs 711/712
 - ⑦ Remove Bridge Asm and save Drive Belt for reinstallation. Discard fiberglass Bridge.
 - ⑧ Remove from Bridge Support Asm and save Dock Control Box Asm and attaching hardware.
 - ⑨ Remove and discard Bridge Support Asm.
 - ⑩ De-install and return for recycling the Carriage Cover and Cable Track.
IMPORTANT! Retain the Clamp Block and screws for mounting new cables track to bridge.

9-1-2 REMOVE EXISTING SRI MODULE**9-1-2-A De-Install SRI Module****9-1-2-B SRI Cable Disposition**

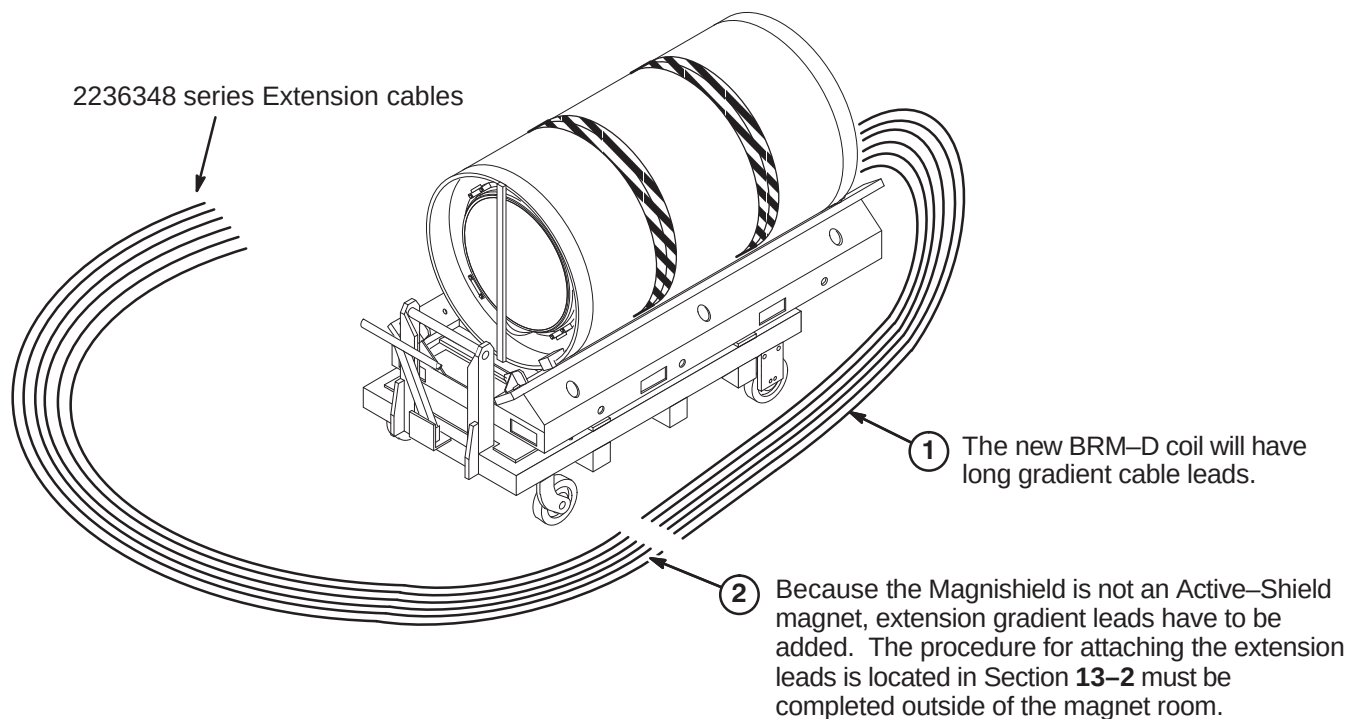
9-1-3 REMOVE ALIGNMENT LIGHTS**9-1-4 REMOVE REAR PEDESTAL, BORE LIGHT, & COMFORT MODULE**

9-1-5 BODY COIL REPLACEMENT

9-1-5-A Coil Removal Instructions

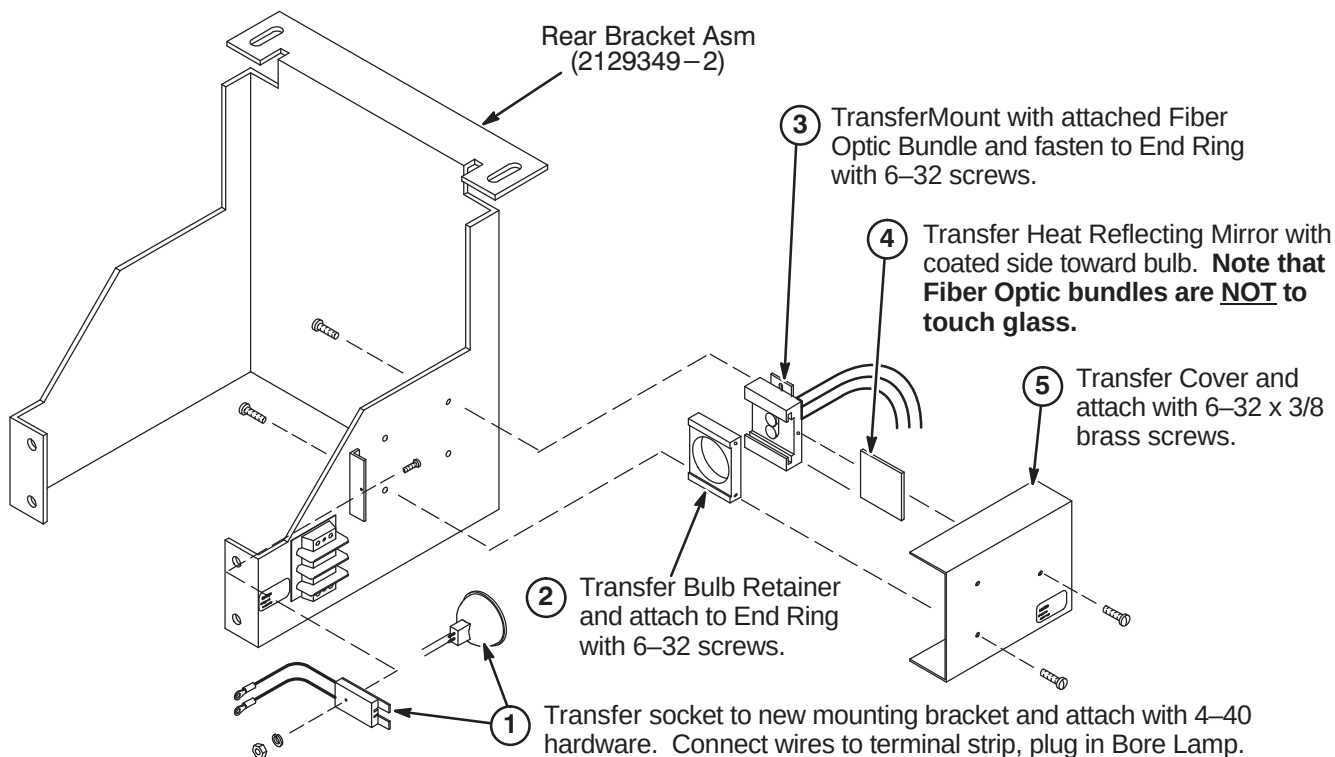


9-1-5-B BRM-D Gradient Coil Pre-Installation Instructions

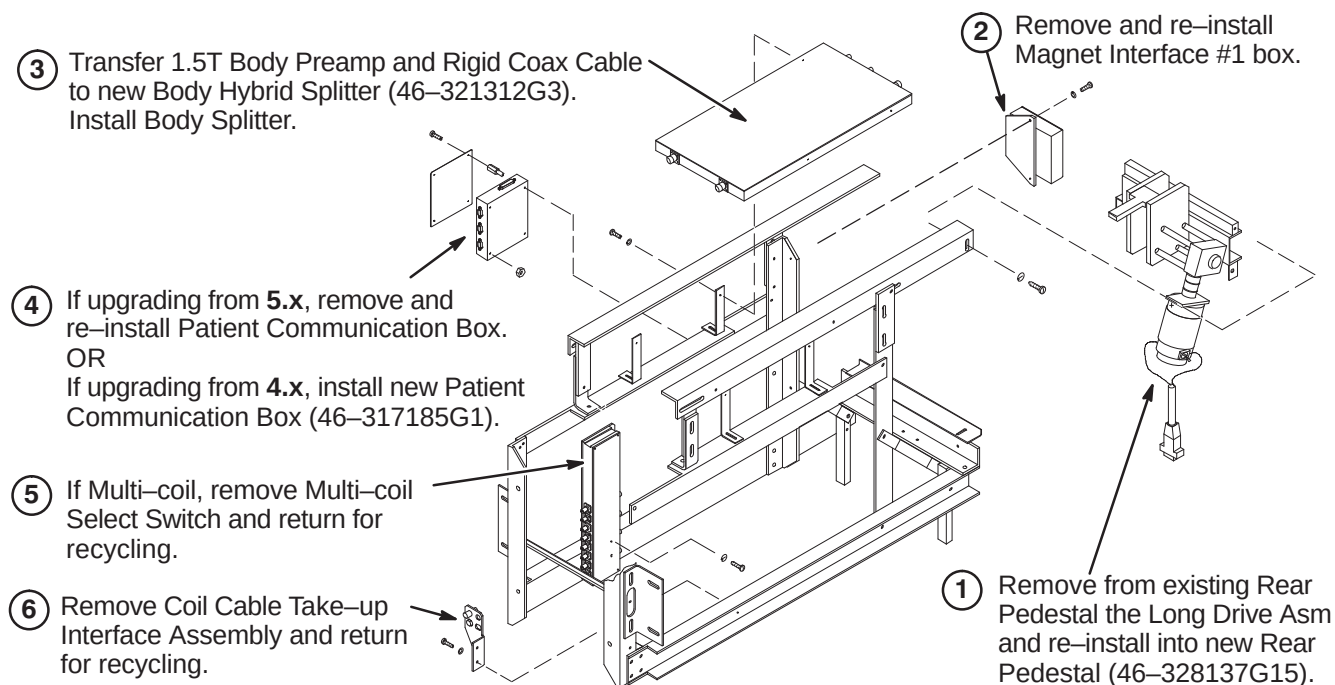


9-2 TRANSFER DE-INSTALLED PARTS TO NEW COMPONENTS

9-2-1 TRANSFER & ASSEMBLE BORE LIGHT TO NEW REAR BRACKET ASM



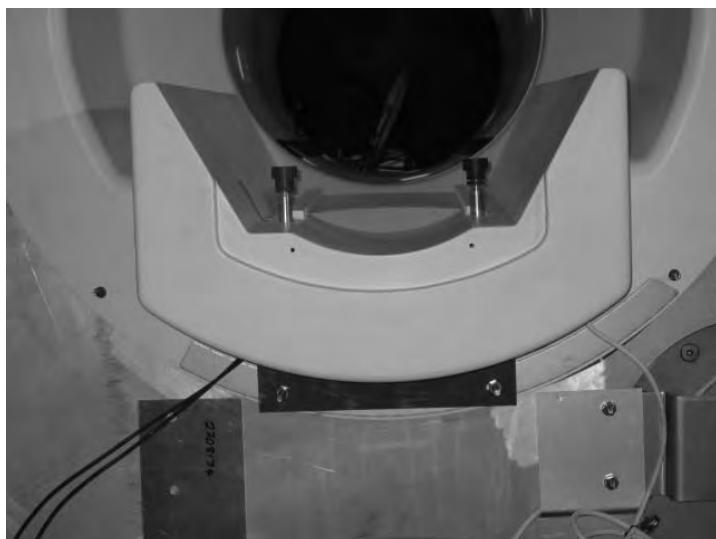
9-2-2 TRANSFER EXISTING COMPONENTS TO NEW REAR PEDESTAL FRAME



9-3 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for all magnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the EXCITE HD upgrade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S-Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade*, for instructions on installing this kit.

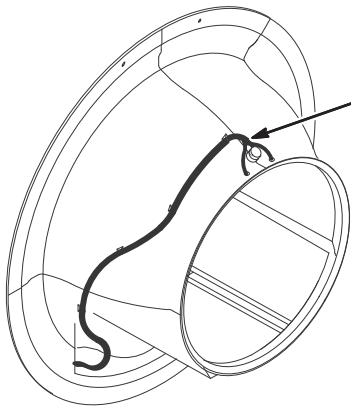
9-4 INSTALL GRADIENT LEAD BOARD AND RESTRAINT BLOCKS (If Required)

Prior to this upgrade, the site should have been audited to determine if the Gradient Lead Board Kit (5198802) has been installed. This kit includes parts and instructions for the installation of the Gradient Lead Board at the rear of the magnet.

Also, the gradient cable Restraint Blocks Kit (5186281-2) should have been installed.

These kits are required for all magnets with Horizon style enclosures. If these upgrades has not been performed on the existing enclosure, then M33335BA should have been ordered and it must be installed at this point of the HDx upgrade.

9-5 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL



The new Front End Bell is equipped with a Dual Temperature Sensor cable, Rune 749, that has to be replaced with a new cable, Run 1260 (5123812).

EXCITE HD requires Run 1260 to be installed.

Refer to Sections **10-3-7-B**, **10-3-8**, and **10-3-9** for instructions on removal of existing cable and installing the new cable.

9-6 INSTALL MAGNET MONITOR

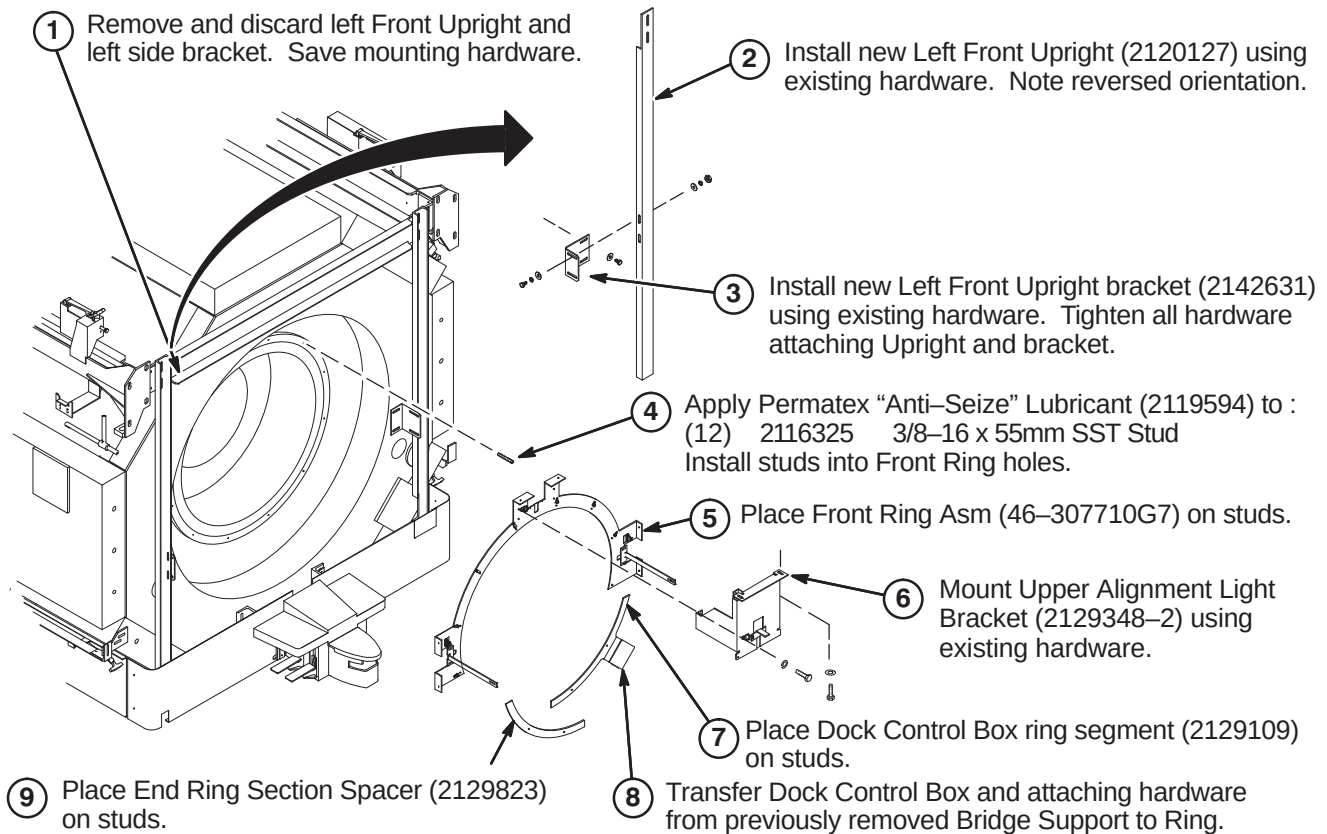
EXCITE HD requires Magnet Monitor on all systems. For this upgrade, all the parts and instructions are included in parts collector 5127838, Magnet Monitor Kit for Non-LCC Systems. Refer to Direction 2230681, included with collector, for instructions on installing magnet monitor on this magnet.

9-7 INSTALL NEW MAGNISHIELD COVERS, COMPONENTS, AND CABLES

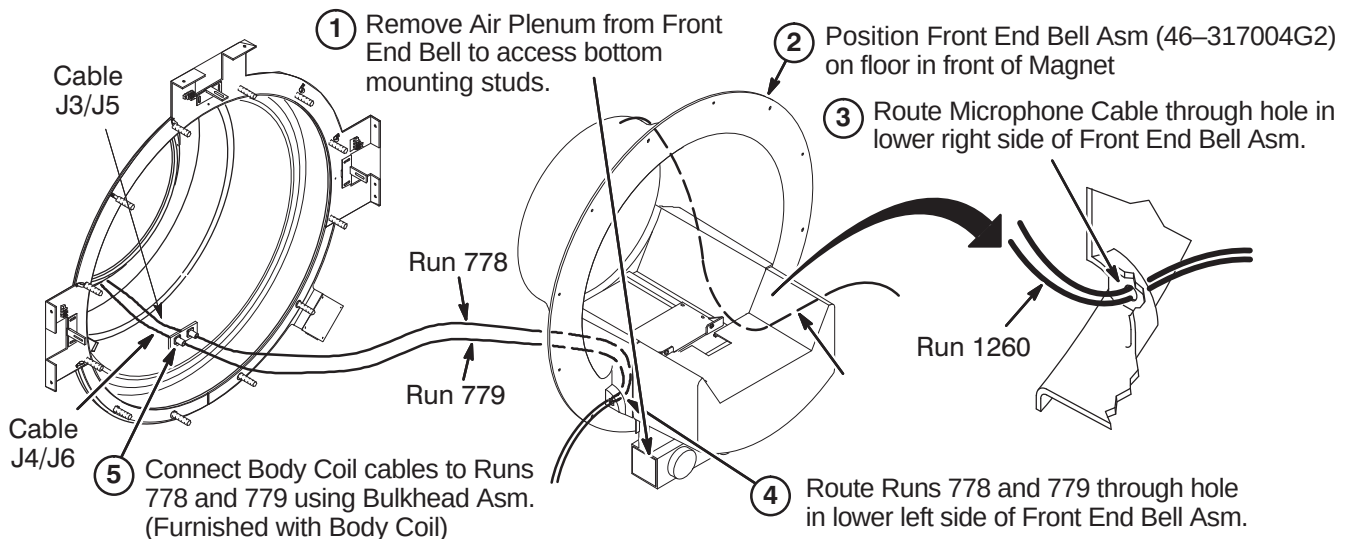
The new Gradient Coil must be installed before starting the installation of new parts referred to in the following steps. Refer to Section 13 for Gradient Coil installation instructions.

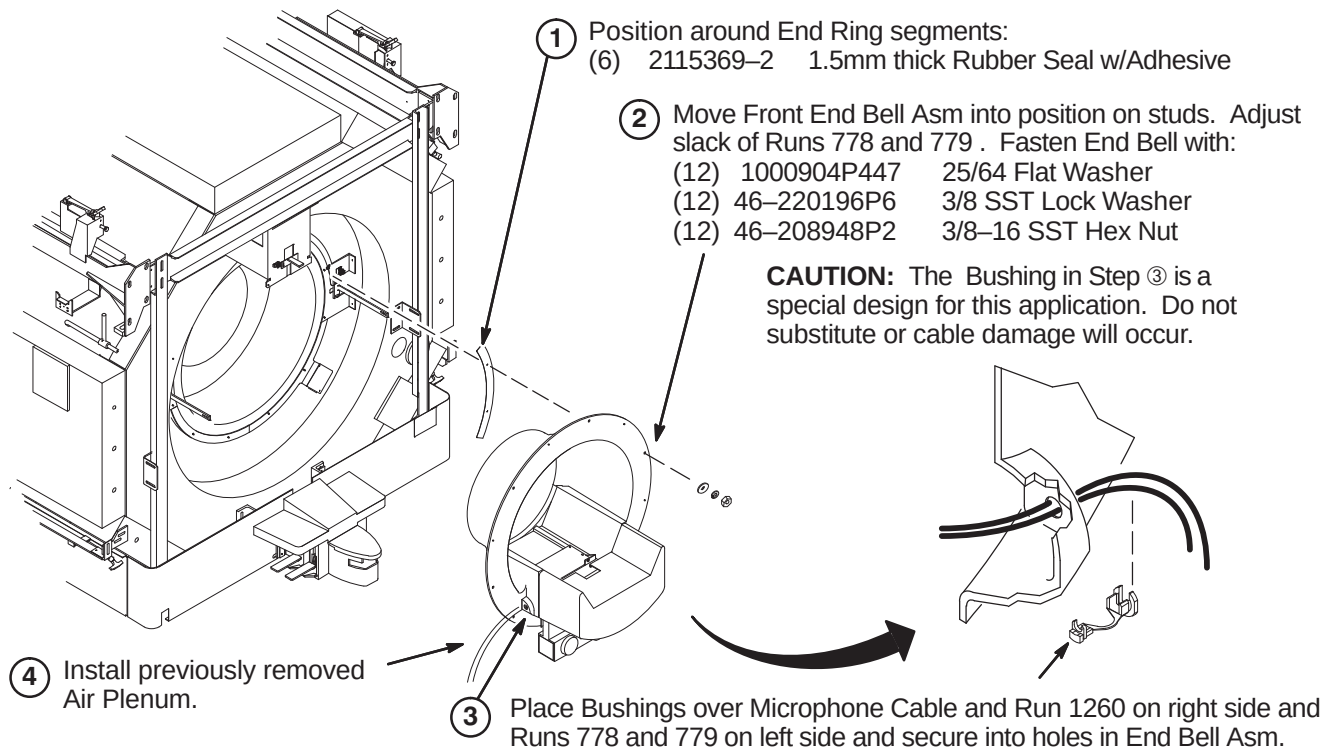
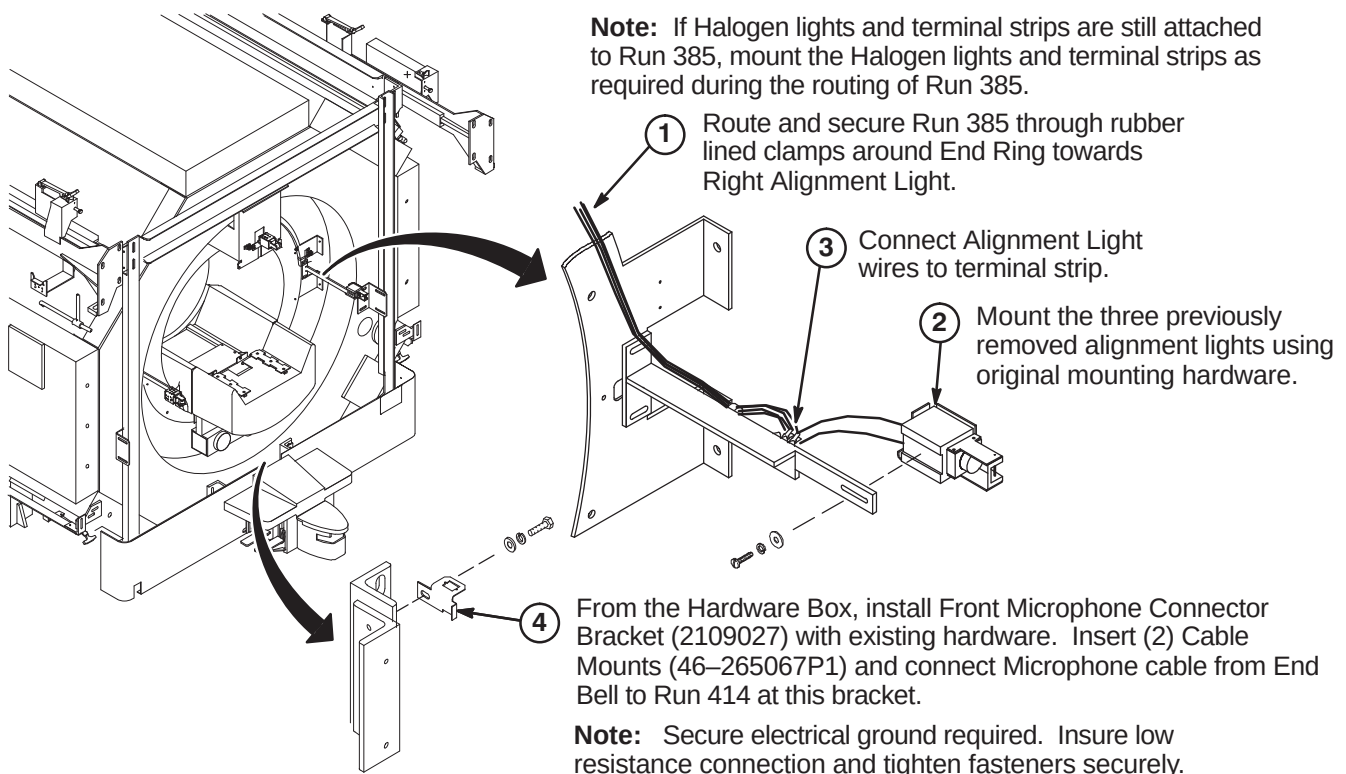
9-7-1 INSTALL COMPONENTS AT FRONT OF MAGNET

9-7-1-A Install Front Ring Asm and Vertical Upright



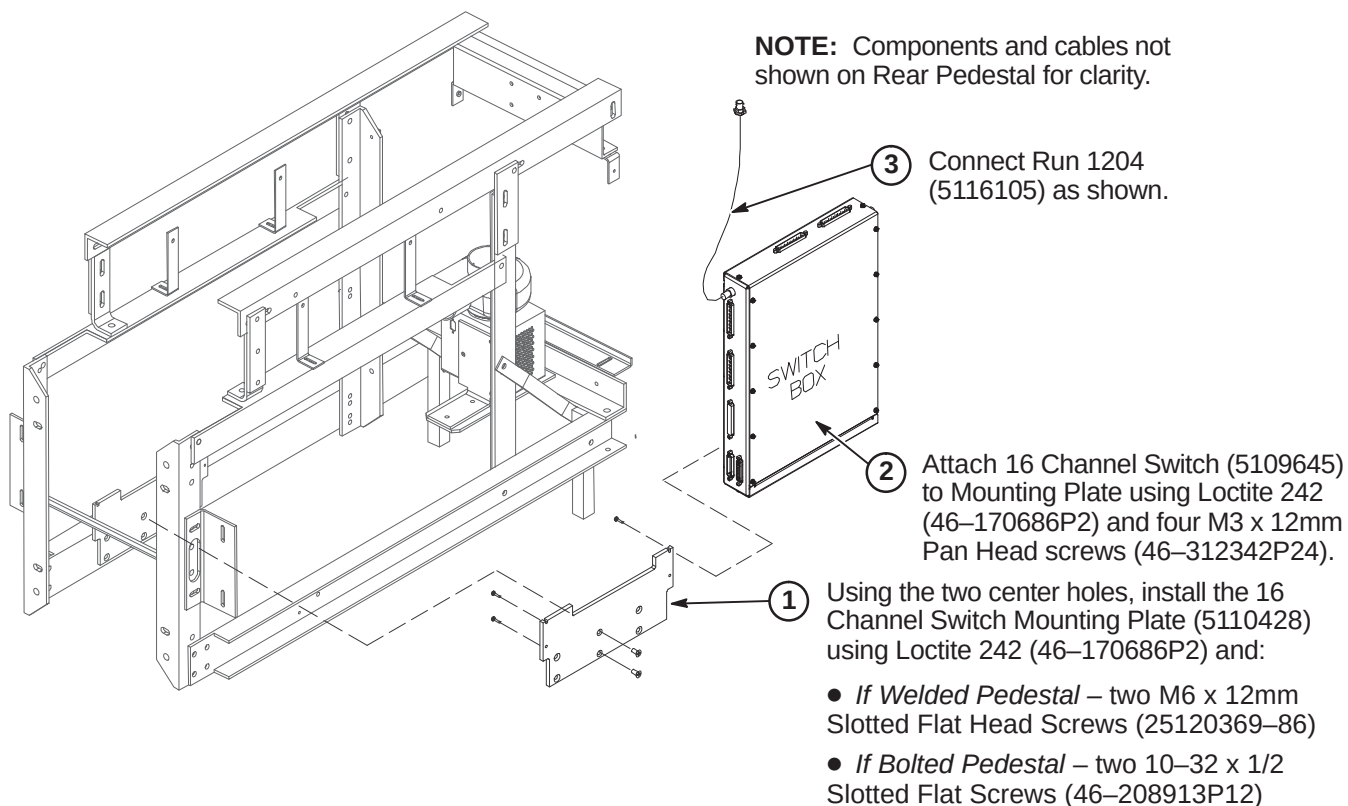
9-7-1-B Route and Connect Runs 778 and 779



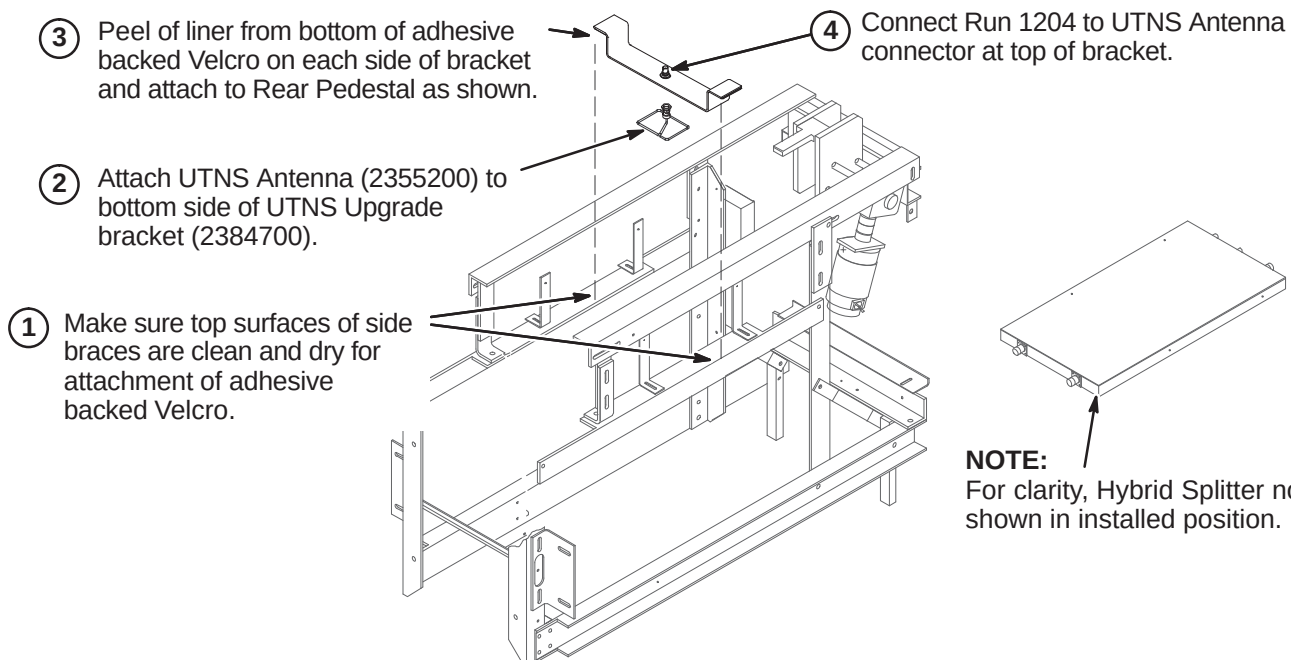
9-7-1-C Install Front Gasket and Mount End Bell**9-7-1-D Install/Connect Alignment Lights & Microphone Bracket**

9-7-2 INSTALL NEW COMPONENTS ON REAR PEDESTAL

9-7-2-A Install 16 Channel Switch

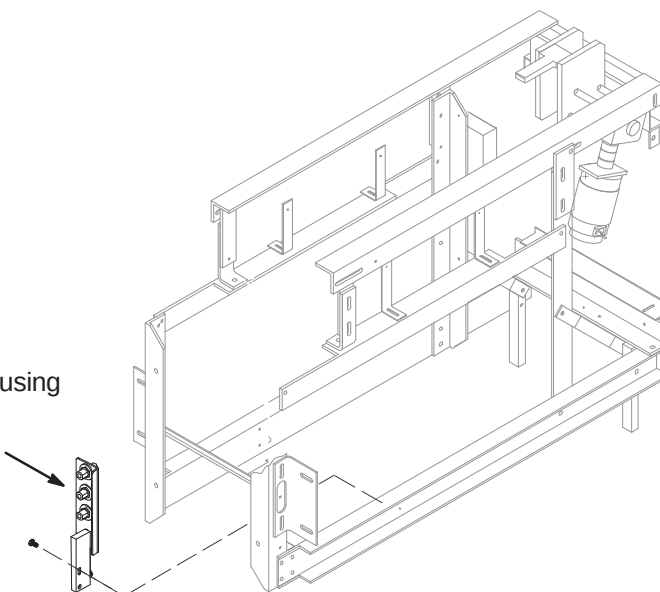


9-7-2-B Install UTNS Antenna



9-7-2-C Install Bulkhead Interface

- ① Install new Bulkhead Interface (5113250) using Loctite 242 (46-170686P2) and two M6 x 12mm Pan Head Screws (2109873-24).



9-7-3 HYBRID SPLITTER UPGRADE

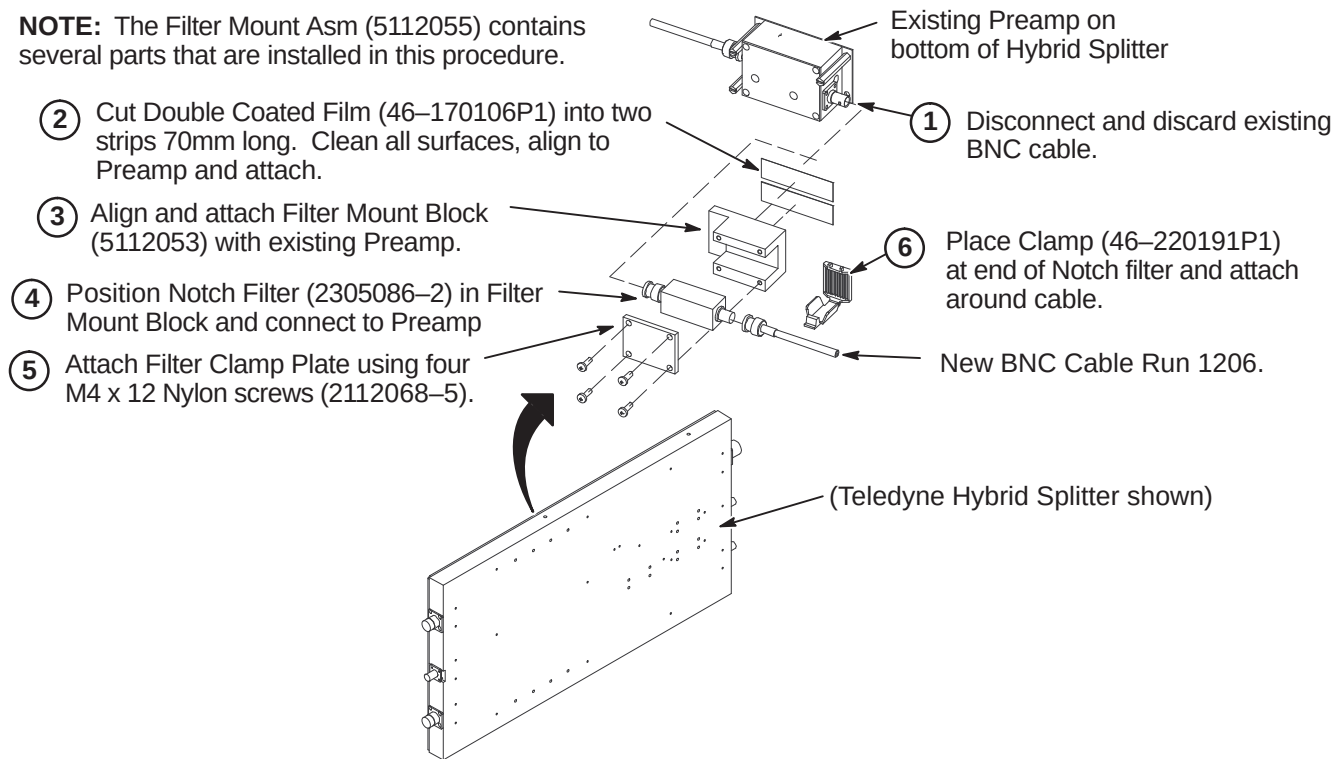
Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

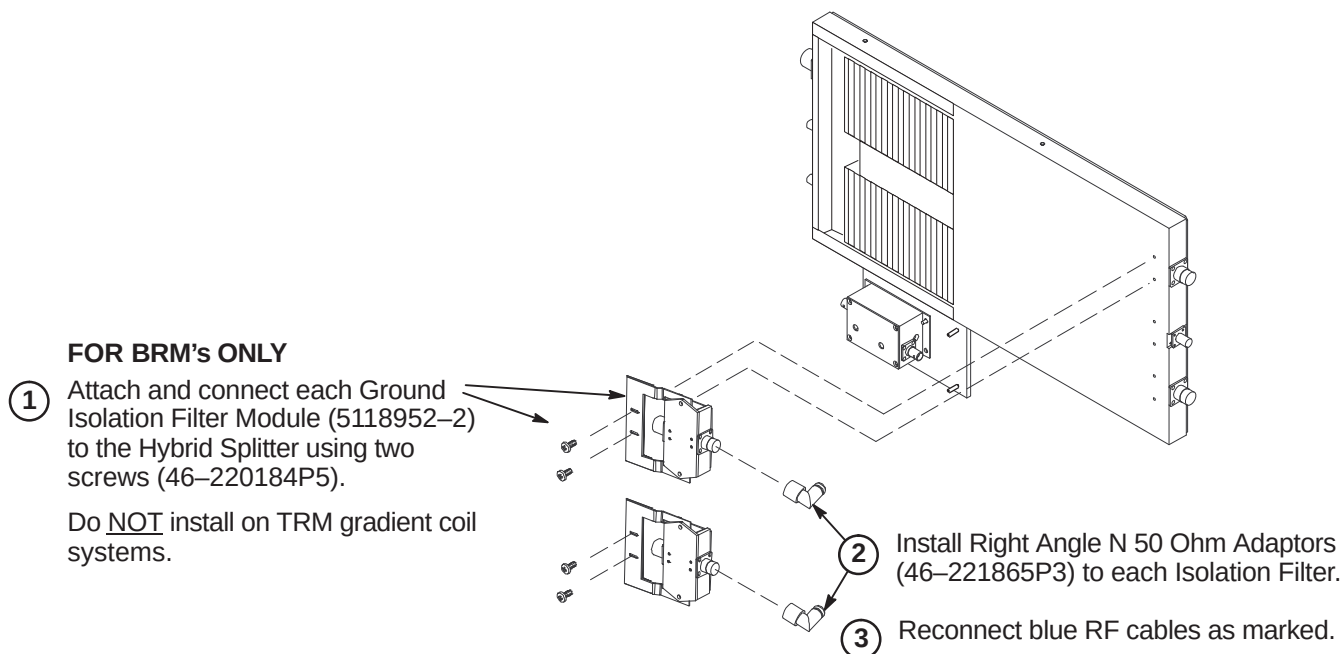
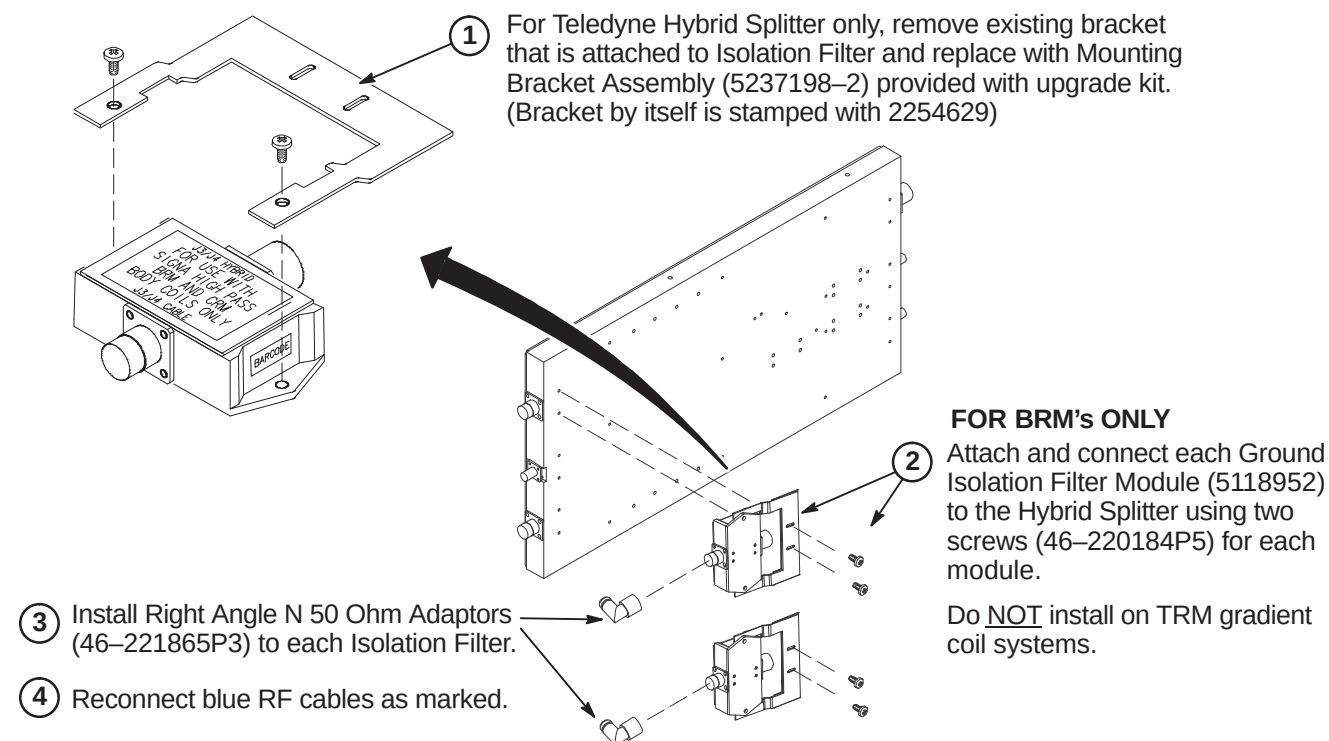
- Refer to Section 9-7-3-A for instructions on installing the Notch Filter and mounting block
- Refer to Section 9-7-3-B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 9-7-3-C for instructions on installing Isolation Filters on systems with Teledyne Splitter

The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils.

9-7-3-A Install Notch Filter and Filter Mount Block Asm

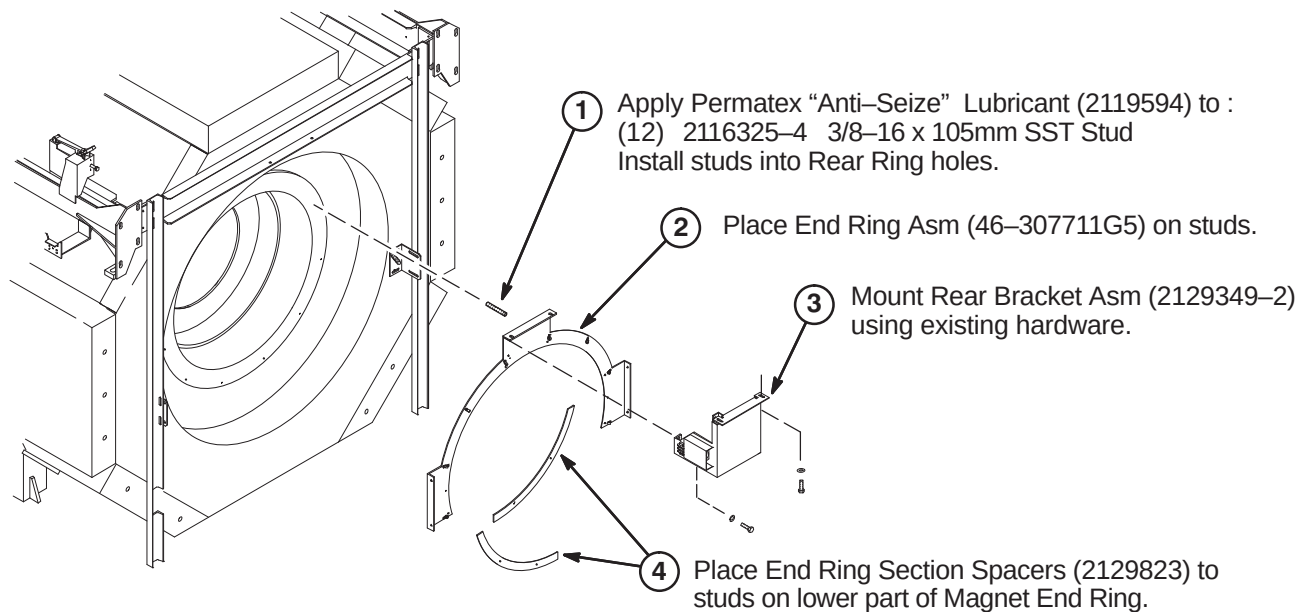
NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



9-7-3-B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)**9-7-3-C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)**

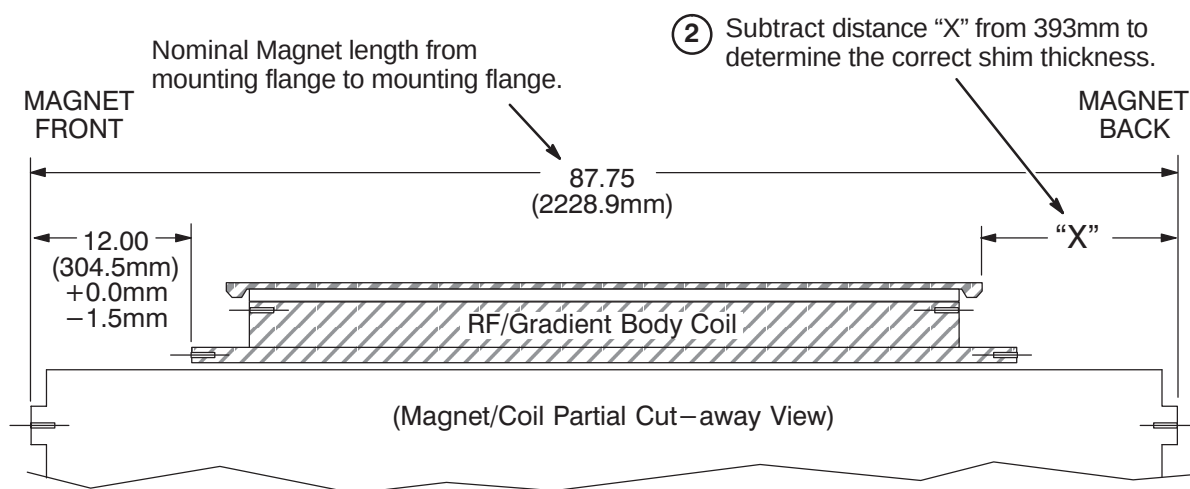
9-7-4 INSTALL COMPONENTS AT FRONT AND REAR OF MAGNET

9-7-4-A Install Rear End Rings and Bracket Asm

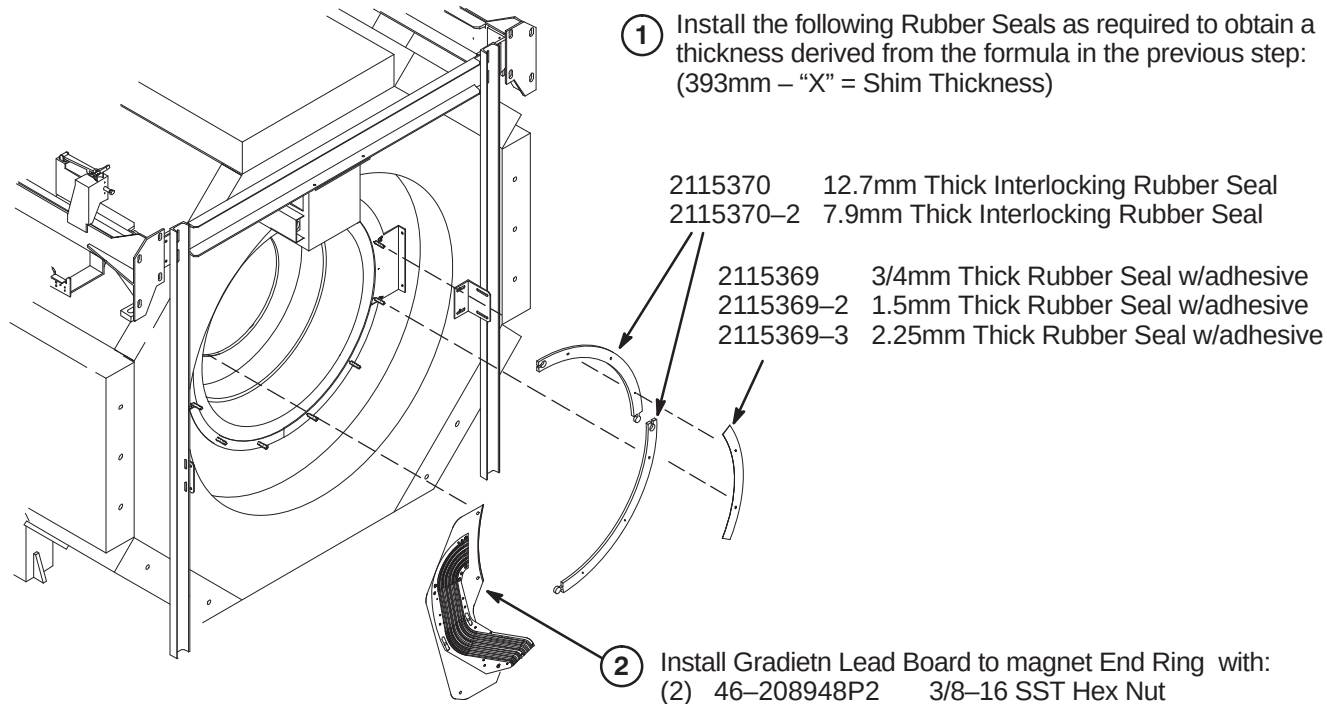
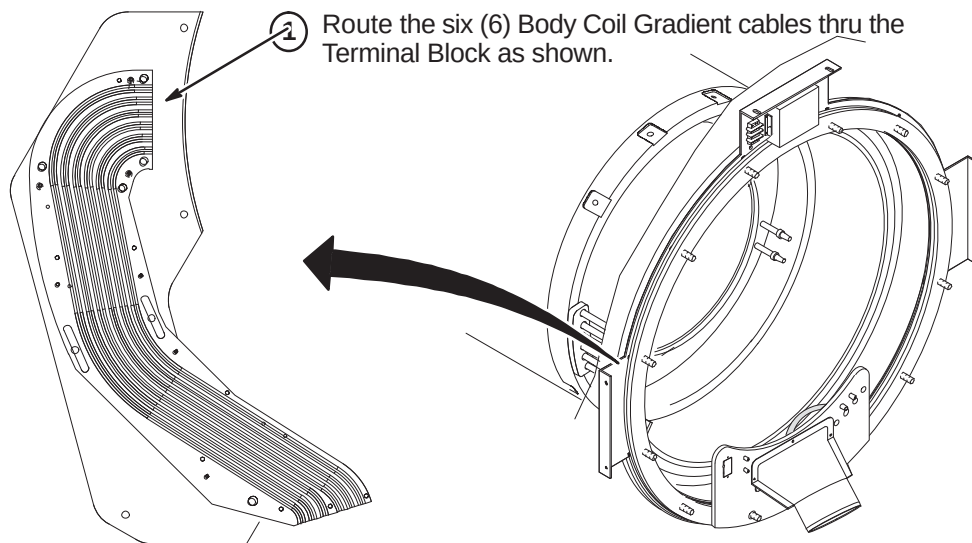


9-7-4-B Formula for Determining Rear Spacers

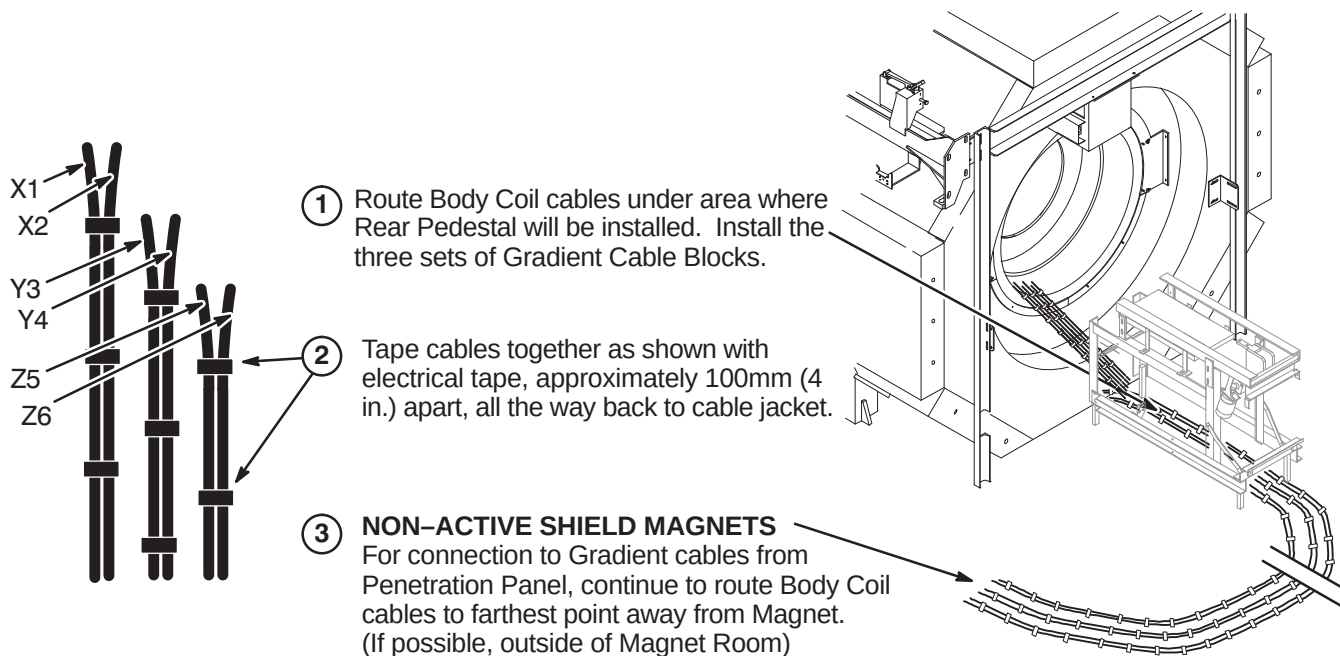
- ① With RF/Gradient Body Coil in place, measure in millimeters the distance from RF Tube to mounting flange on Magnet. This is distance "X".
Make sure the distance is measured to the Magnet mounting flange, not the Rear Frame Asm End Ring installed in Section 9-7-4-A.



Note: Values shown here are for GE S-II, S-III, S-IV, S-V and Magnishield Magnet only.

9-7-4-C Install Rear Spacers and Gradient Terminal Block**9-7-4-D Route and Install Gradient Cables in Terminal Block**

9-7-4-E Secure X, Y, and Z Gradient Cables

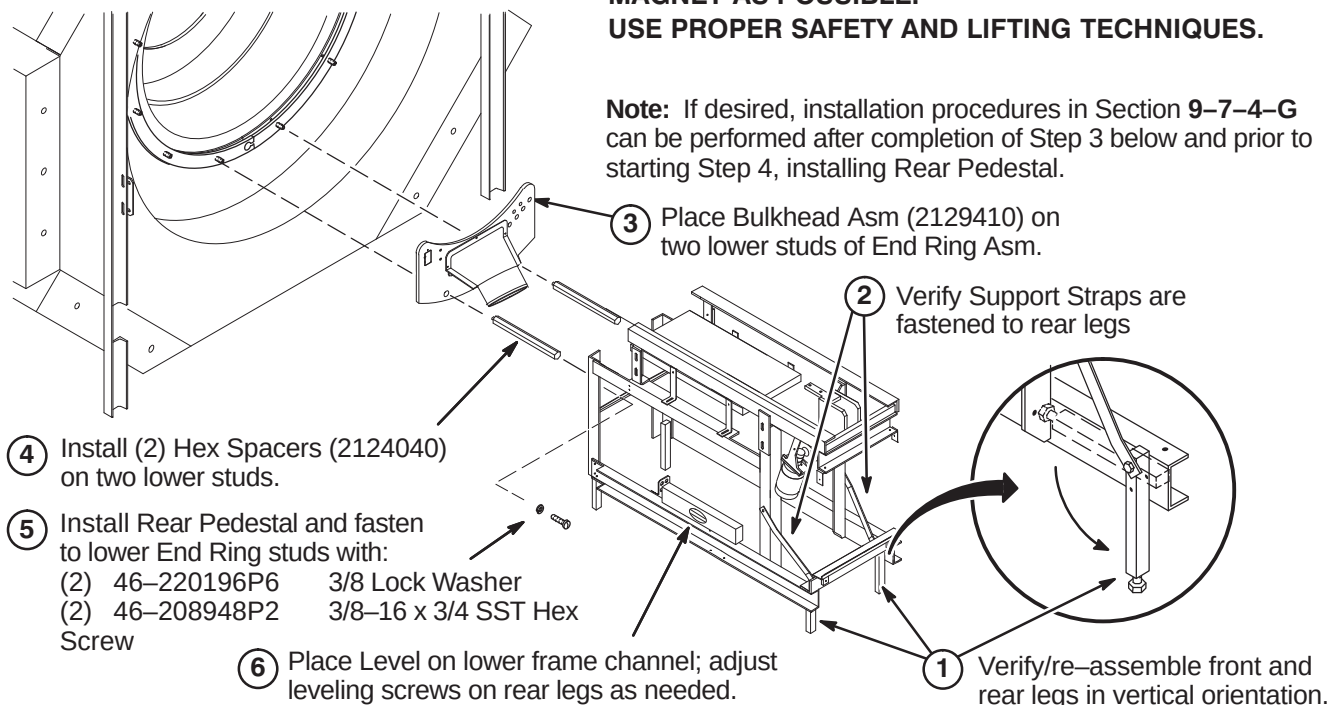


9-7-4-F Install Rear Bulkhead and Rear Pedestal

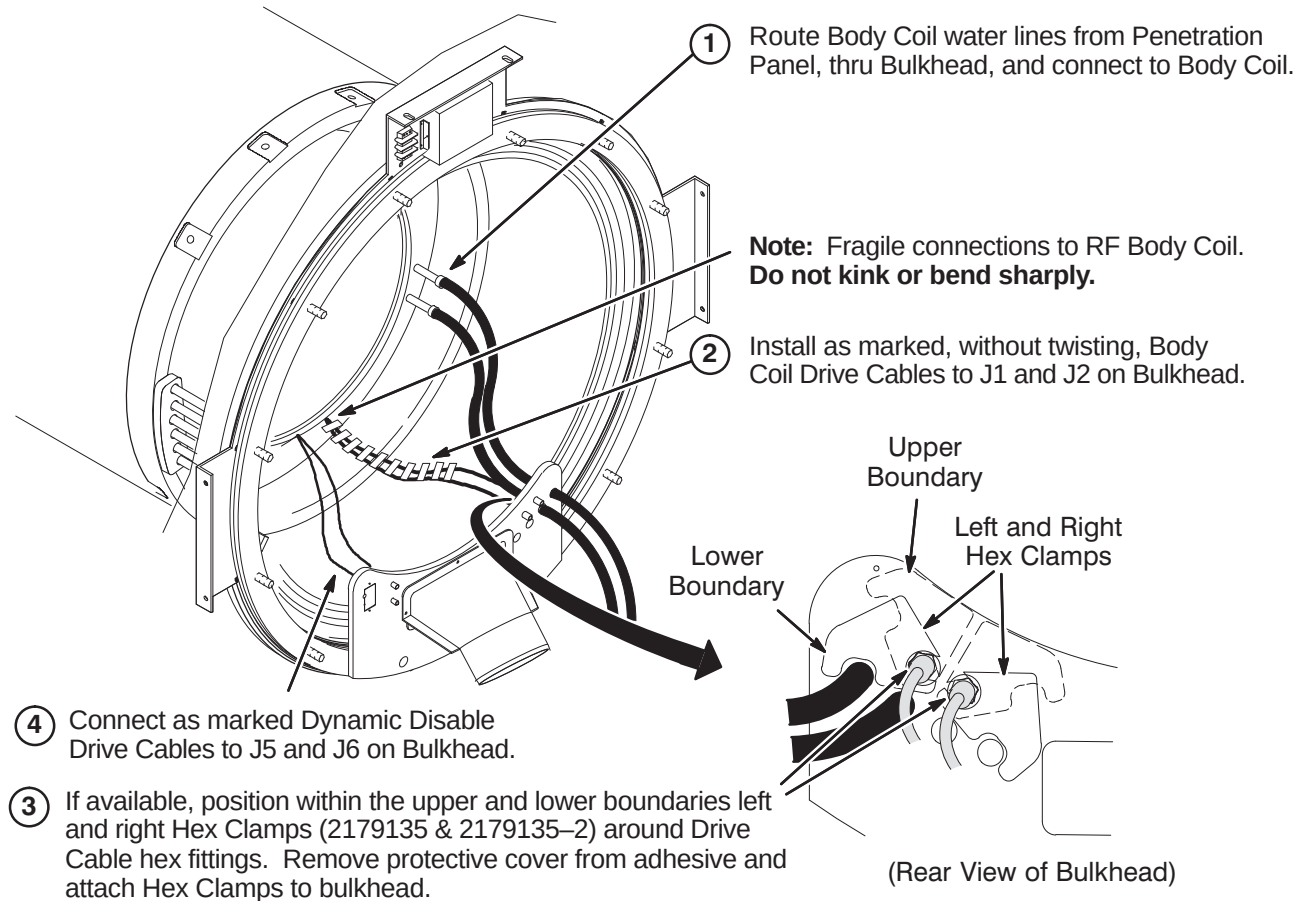
WARNING!

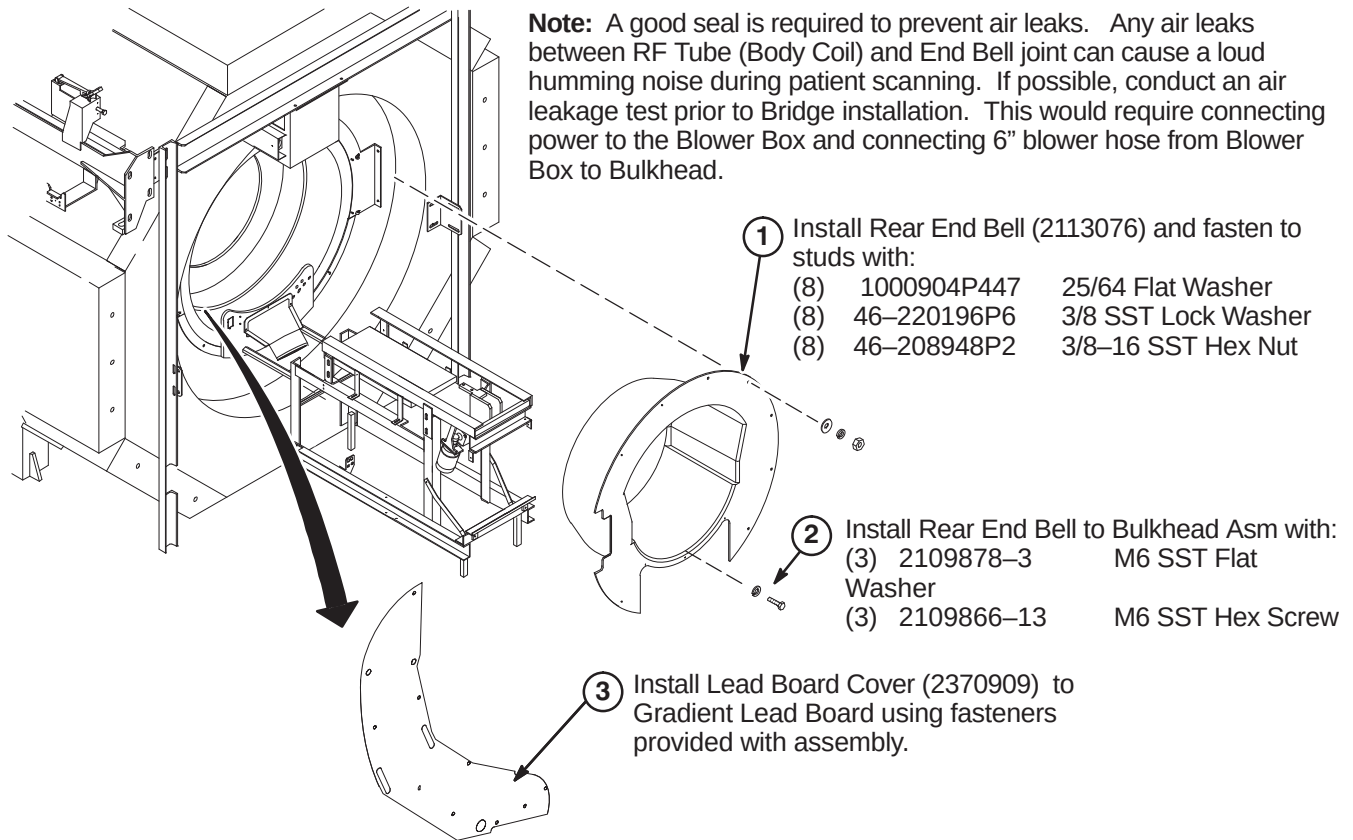
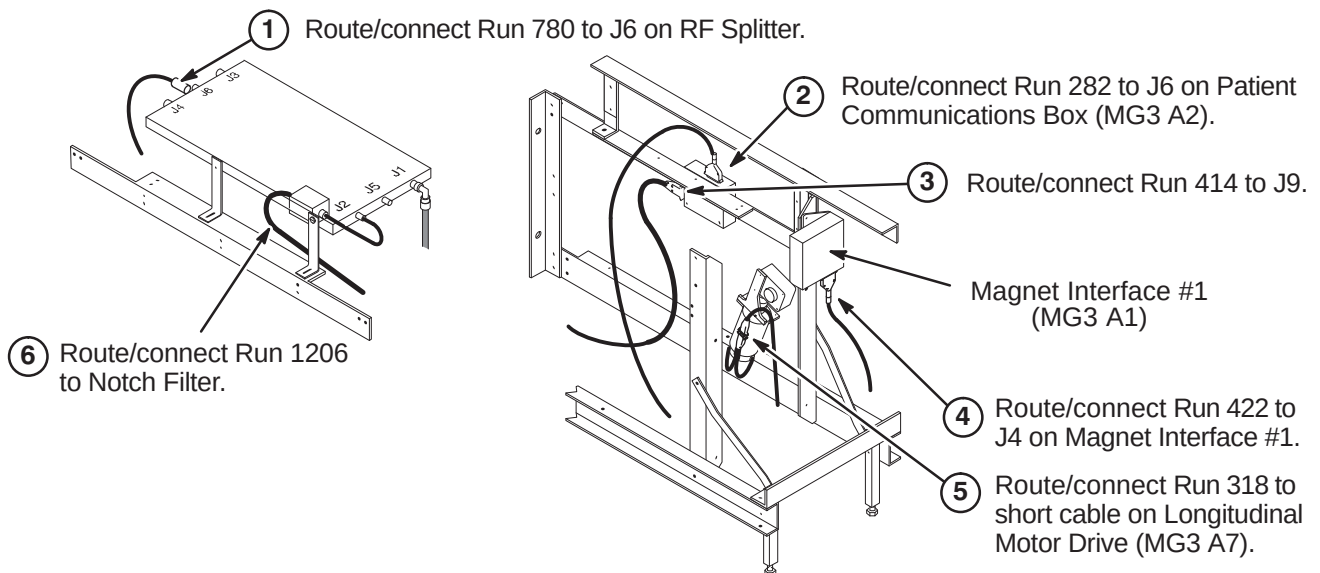
THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE. USE PROPER SAFETY AND LIFTING TECHNIQUES.

Note: If desired, installation procedures in Section 9-7-4-G can be performed after completion of Step 3 below and prior to starting Step 4, installing Rear Pedestal.



9-7-4-G Route and Connect Bulkhead Cables and Hoses



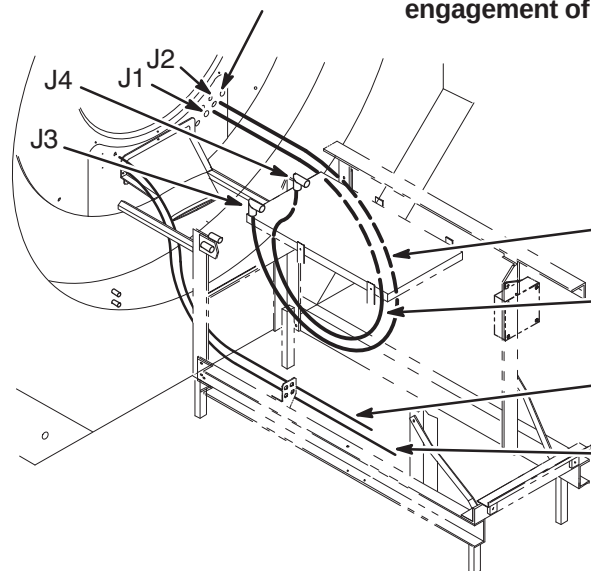
9-7-4-H Install Rear End Bell**9-7-4-I Connect Runs 282, 318, 414, 422, 780, and 1206**

9-7-4-J Route/Connect Runs 743, 744, 781, and 782

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

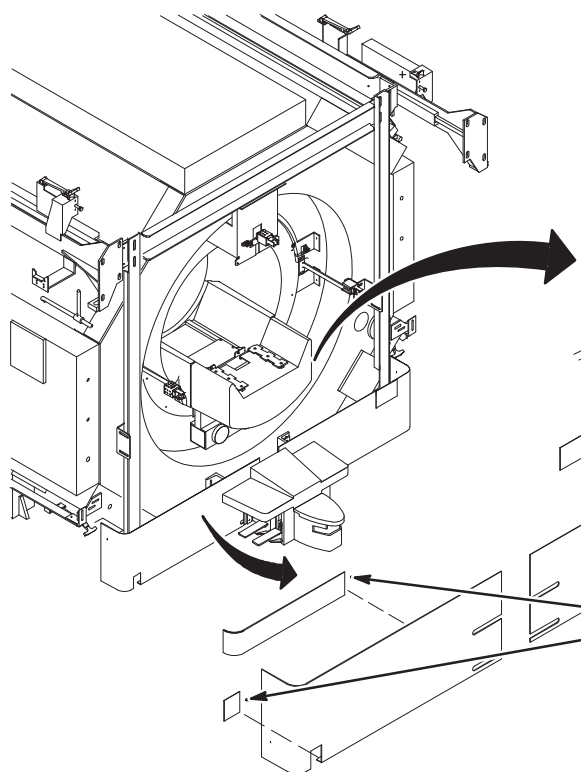
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

- ① Route and connect Run 744 (2105691-5) from J1 on the Bulkhead to J3 on the RF Splitter.
- ② Route and connect Run 743 (2105691-4) from J2 on the Bulkhead to J4 on the RF Splitter.
- ③ Route and connect Run 781 (46-328000G919) from J5 on the Bulkhead to J75 on the Penetration Panel.
- ④ Route and connect Run 782 (46-328000G920) from J6 on the Bulkhead to J76 on the Penetration Panel.

9-7-4-K Connect Runs 421 and Dock Cable & Install Skirt Upgrade Kit



- ① Attach Cable Extension (2143072) to Dock Cable and connect to J2 on Dock Limit Module (MG3 A31).

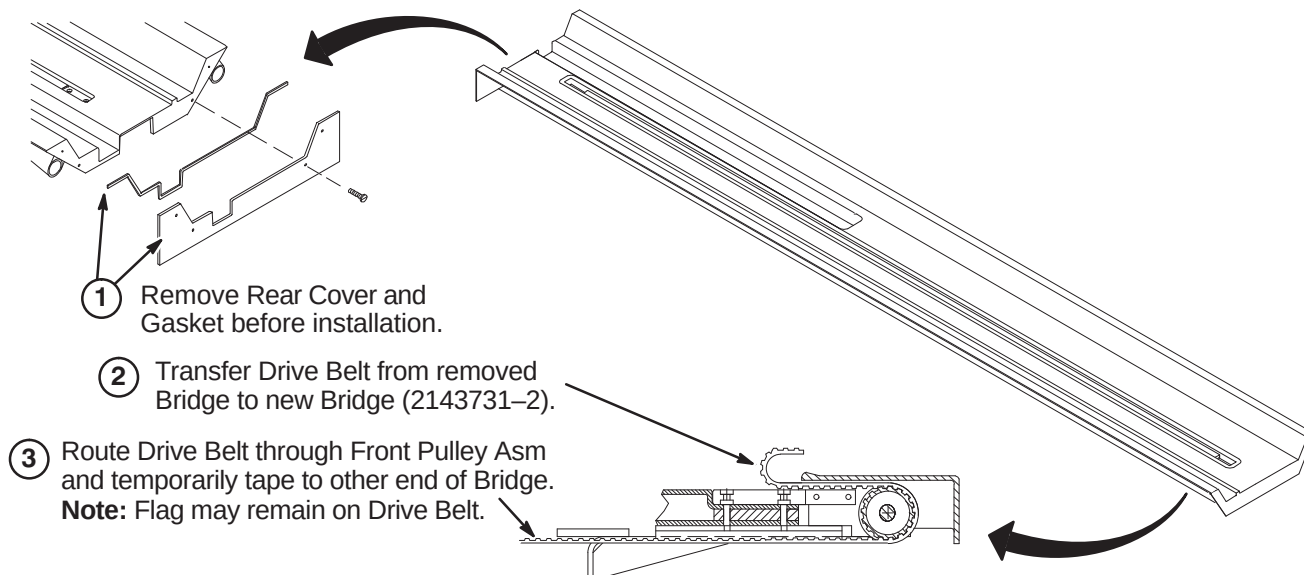
- ② Route/connect Run 421 from SRI Module to J1 on Dock Limit Module (MG3 A31).

Install Front Skirt Upgrade Kit (2139555). Remove protective coating from following parts and install as shown:

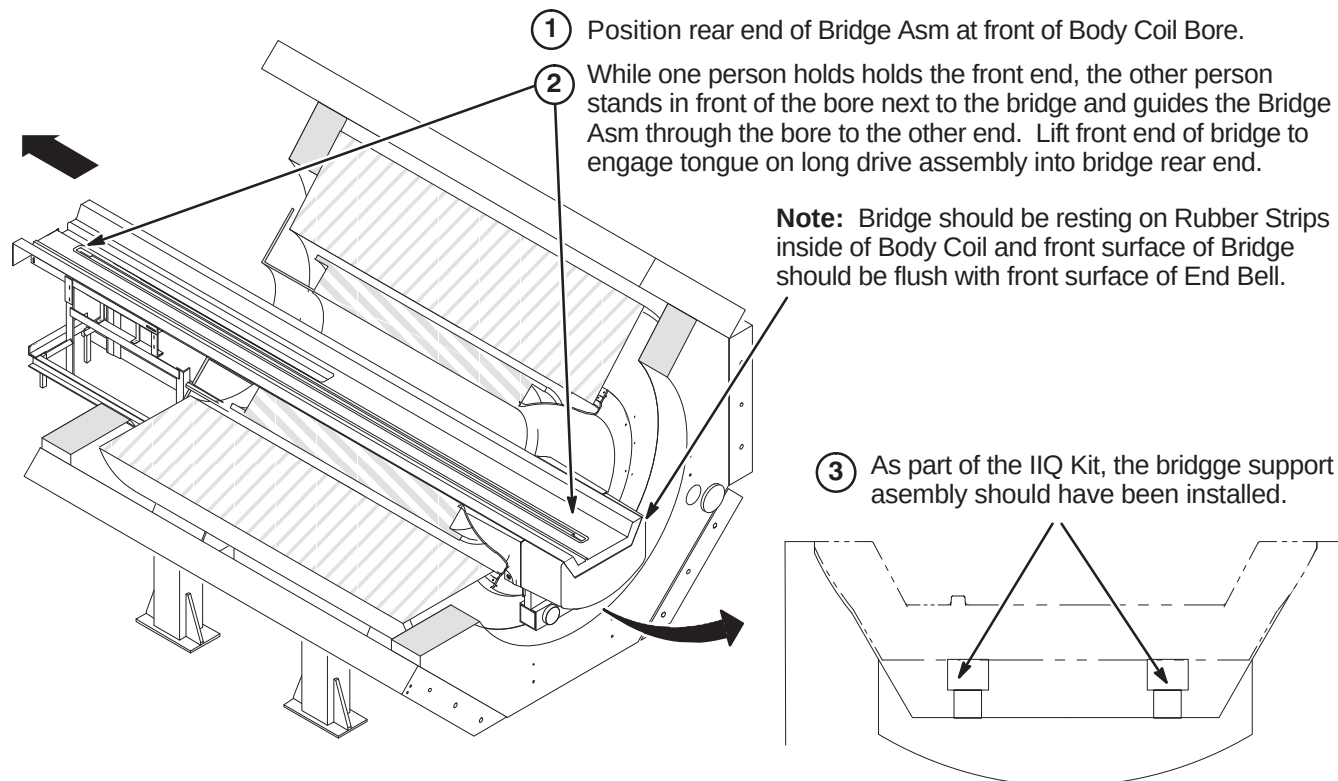
- | | |
|---------------|--------------------------|
| (1) 2139555-2 | Vacuum Break Port Access |
| (1) 2139555-3 | Trim for Front Skirt |
| (1) 2139555-4 | Front Skirt Wraparound |
| (1) 2139555-5 | Front Skirt Top Plate |
| (1) 2139555-6 | Front Skirt Corner Trim |

9-7-5 INSTALL BRIDGE

9-7-5-A Bridge Assembly Preparation Before Installing in Magnet

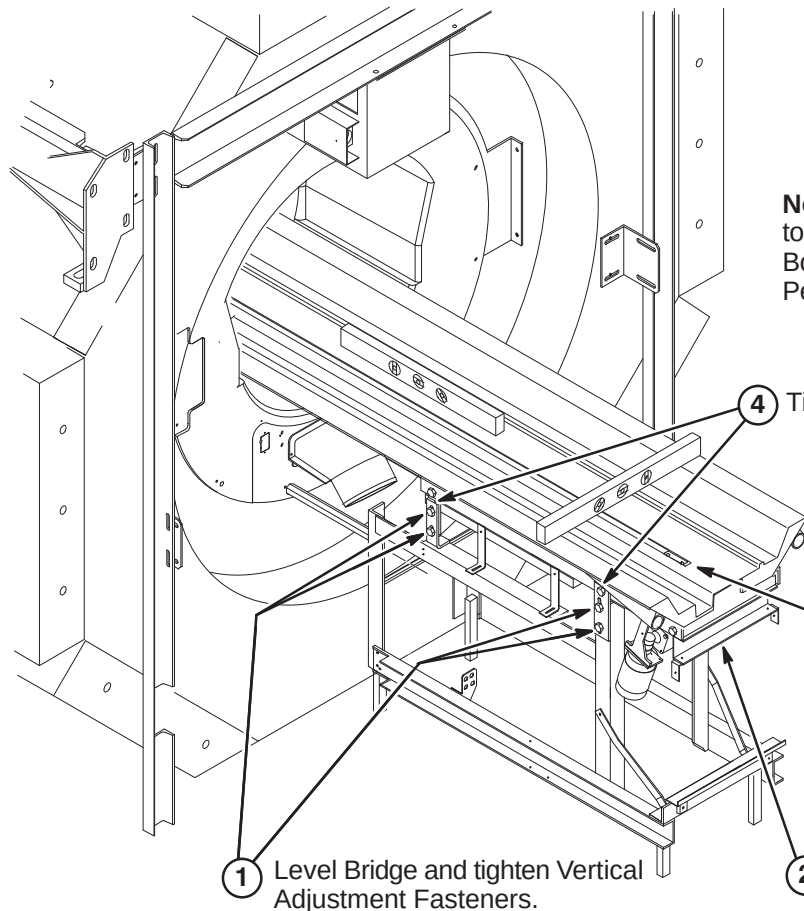


9-7-5-B Securing Front of Bridge



Note: Bridge and End Bell front surfaces should still be flush.

9-7-5-C Leveling Bridge



Note: The weight of Bridge should continue to be supported by the rubber strips in the Body Coil. The Front Bridge Mount and Rear Pedestal stabilize and level the Bridge.

④ Tighten horizontal adjusting screws.

Note: Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

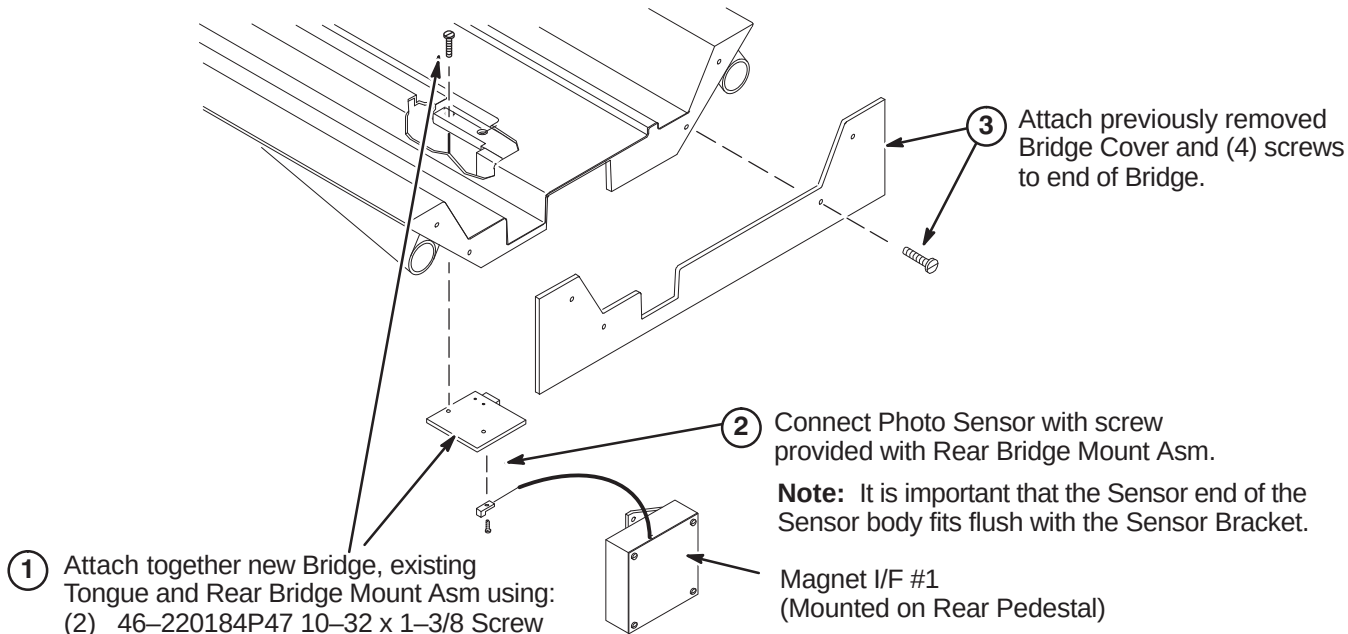
③ Slide horizontal top rails and Longitudinal Drive Asm forward to engage Tongue into Bridge.

Note: Make sure front surface of Bridge is flush with front surface of Front End Bell when this procedure is complete.

① Level Bridge and tighten Vertical Adjustment Fasteners.

② Check that Longitudinal Drive Asm is level.

9-7-5-D Connect Photo Sensor & Bridge End Cover



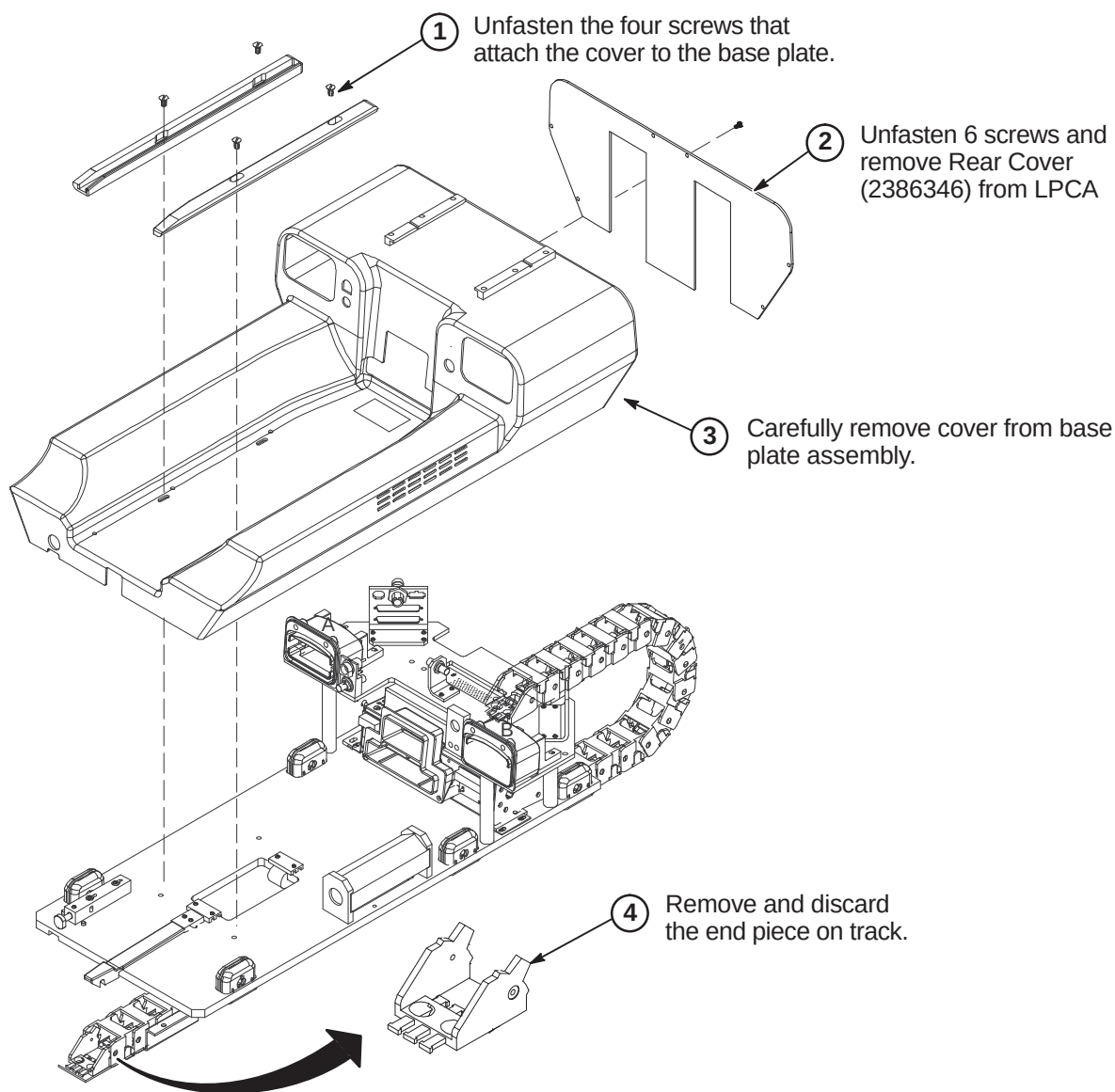
① Attach together new Bridge, existing Tongue and Rear Bridge Mount Asm using:
(2) 46-220184P47 10-32 x 1-3/8 Screw

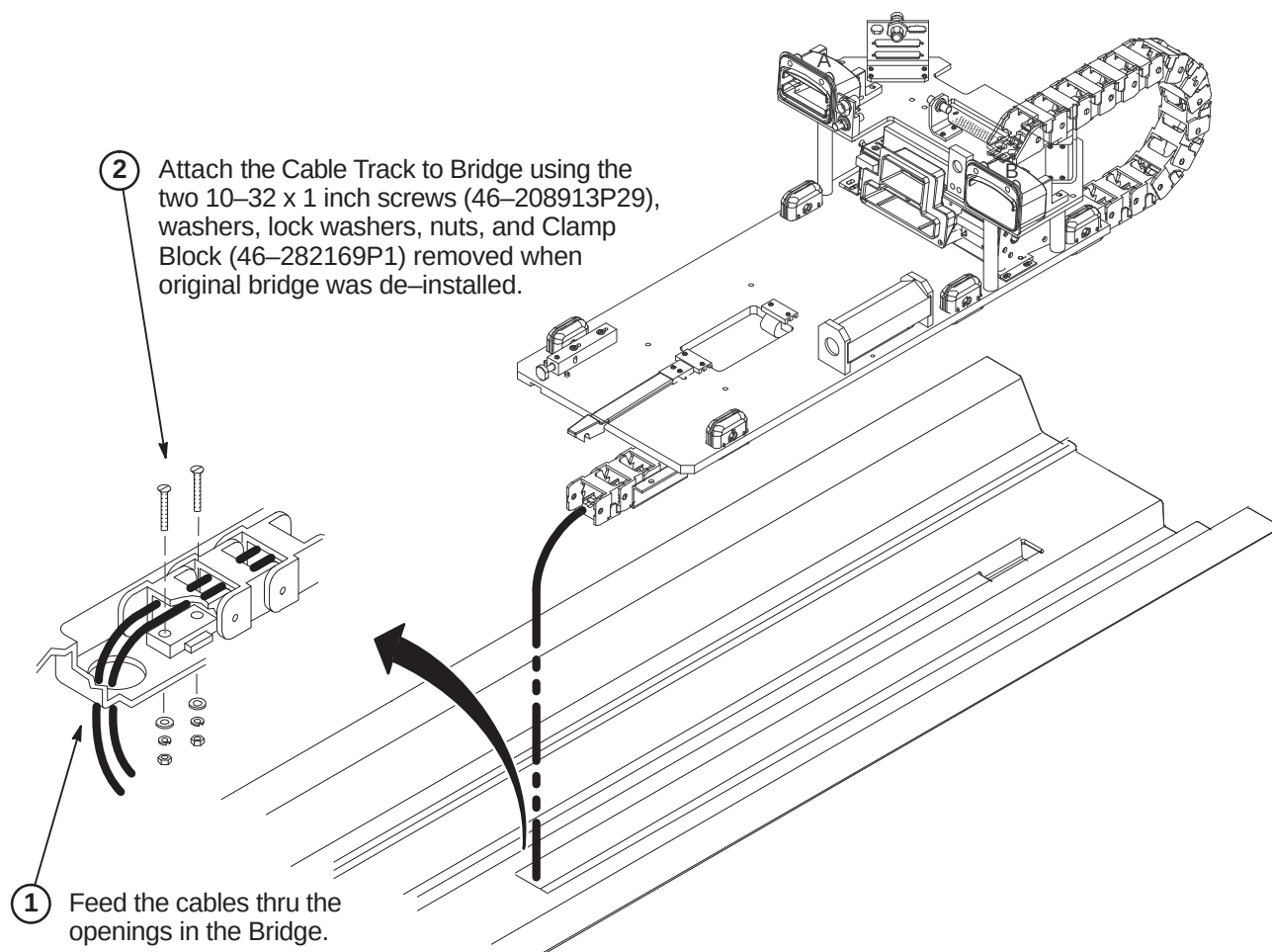
② Connect Photo Sensor with screw provided with Rear Bridge Mount Asm.

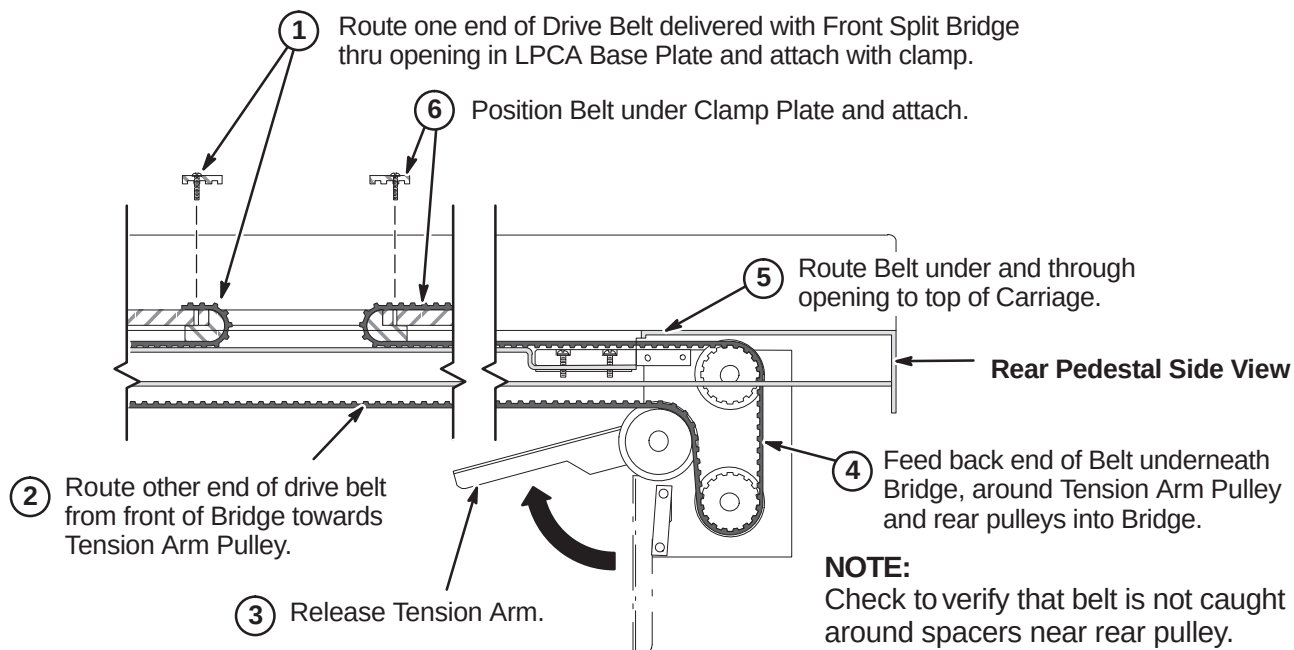
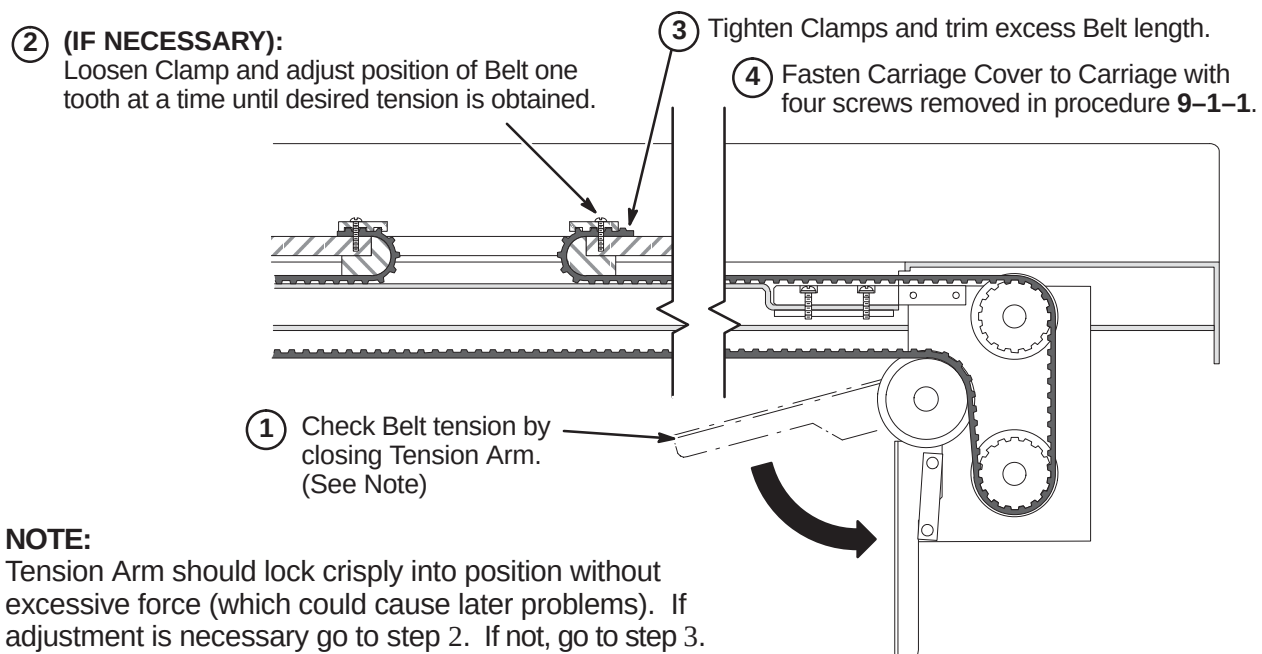
Note: It is important that the Sensor end of the Sensor body fits flush with the Sensor Bracket.

Magnet I/F #1
(Mounted on Rear Pedestal)

③ Attach previously removed Bridge Cover and (4) screws to end of Bridge.

9-7-6 INSTALL LPCA AND CABLE TRACK**9-7-6-A Remove LPCA Cover**

9-7-6-B Attach Cable Track to Bridge

9-7-6-C Route and Attach Drive Belt**9-7-6-D Final Adjustment of Drive Belt**

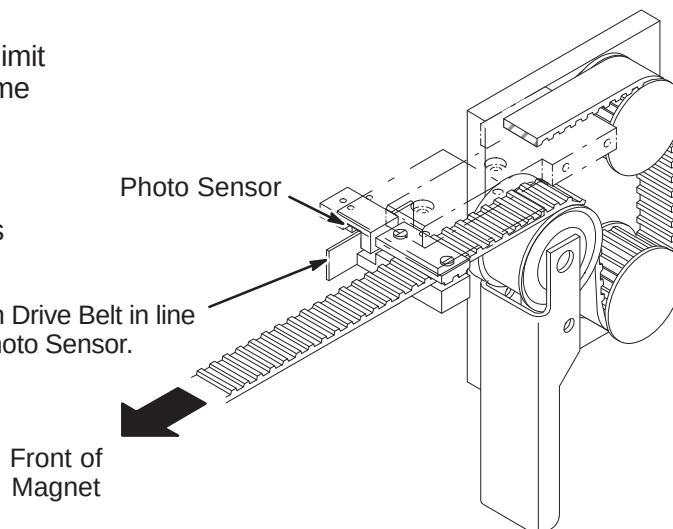
9-7-6-E Installation of Sensor Flag**NOTE:**

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

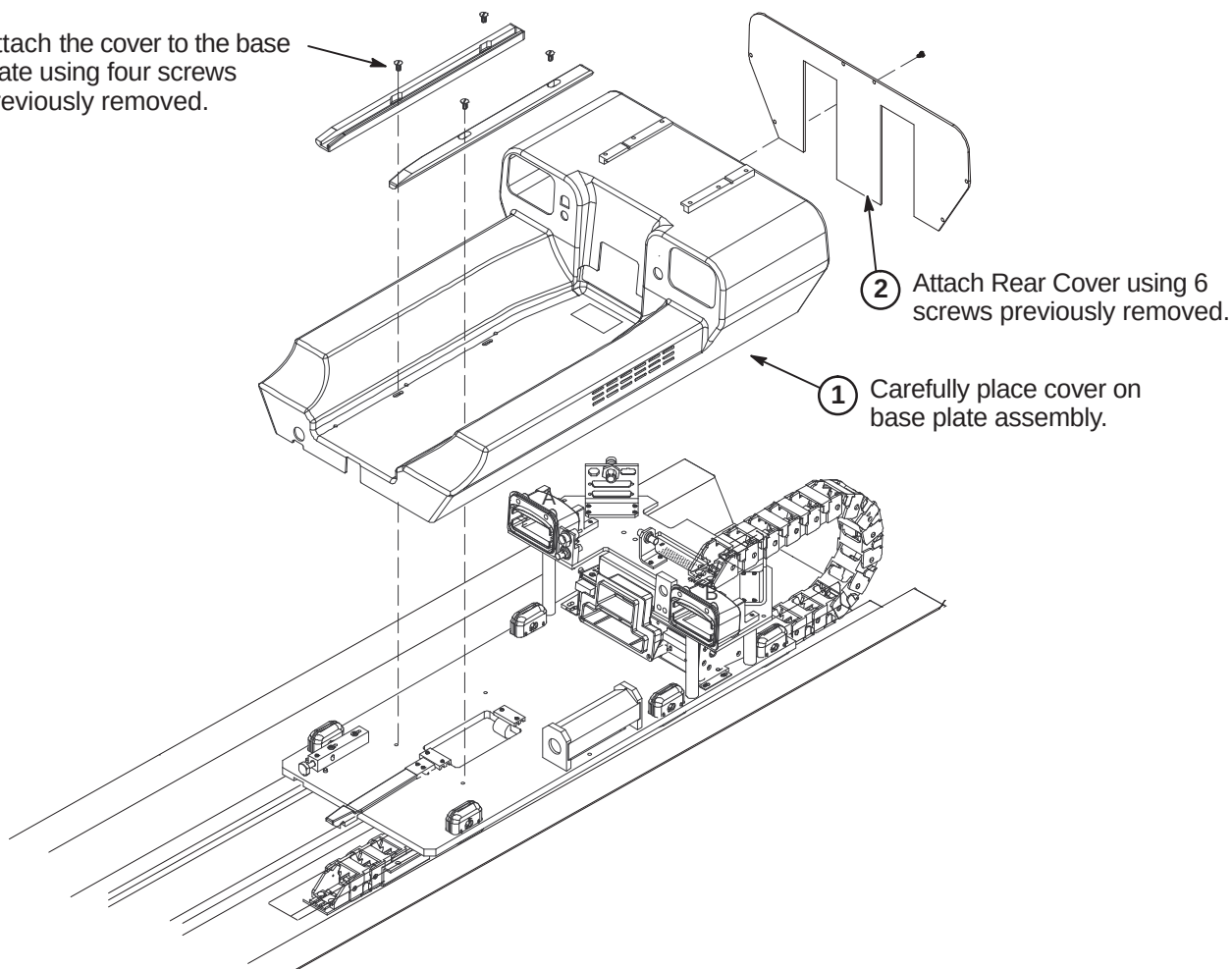
NOTE:

The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

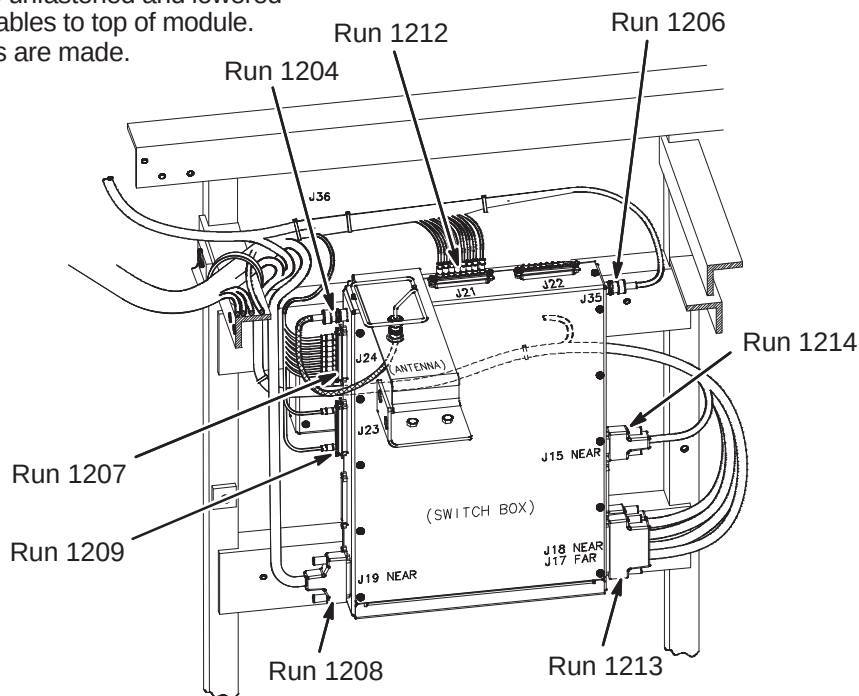
**9-7-6-F Installation of LPCA Cover**

- ③ Attach the cover to the base plate using four screws previously removed.



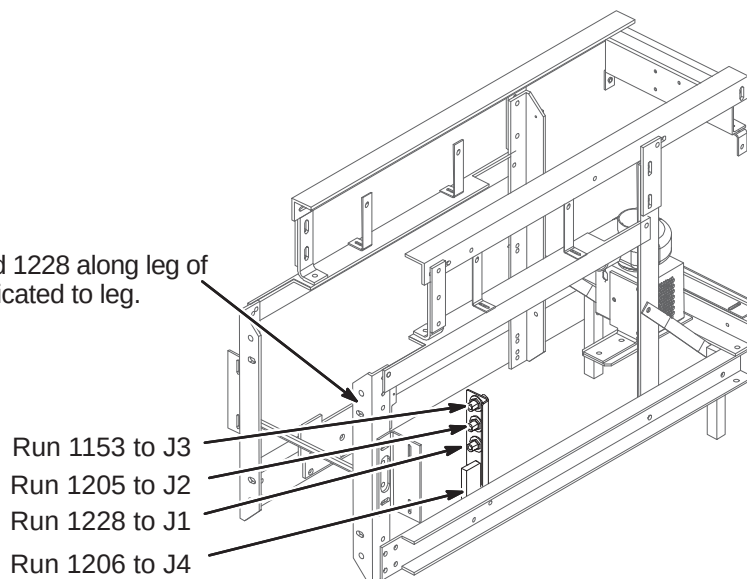
9-7-6-G Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**9-7-6-H Connect Cables from Cable Track to Bulkhead Interface**

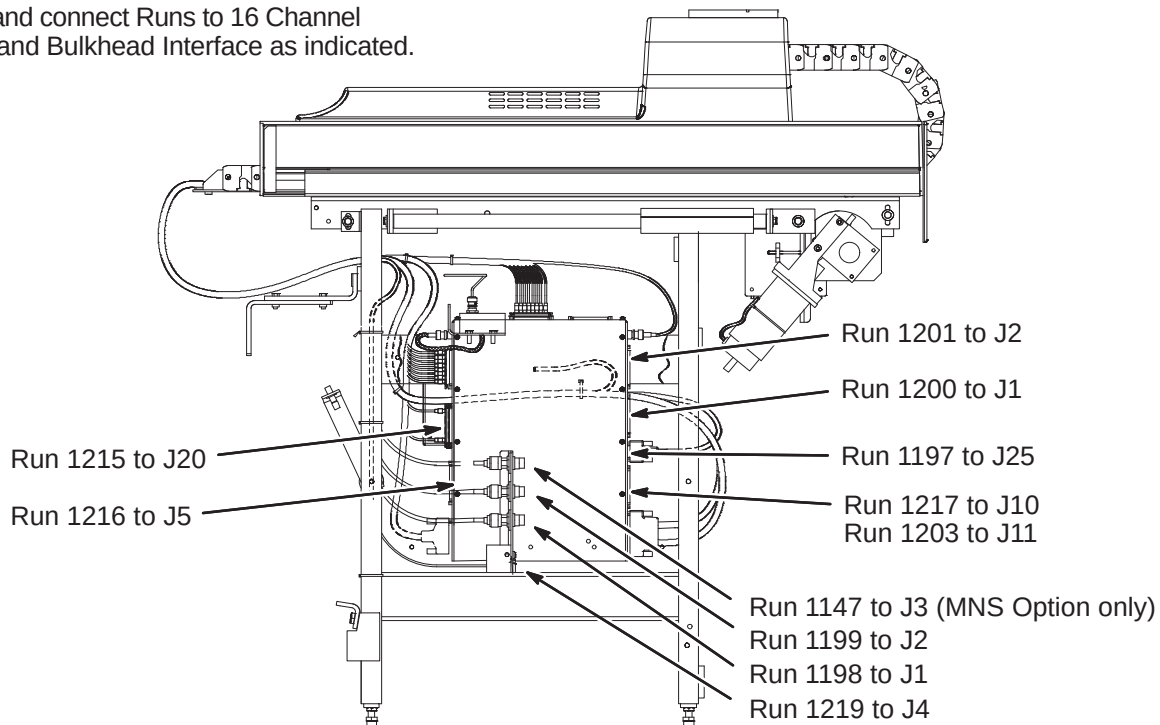
- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.

- ② Connect Runs to Bulkhead Interface as indicated.

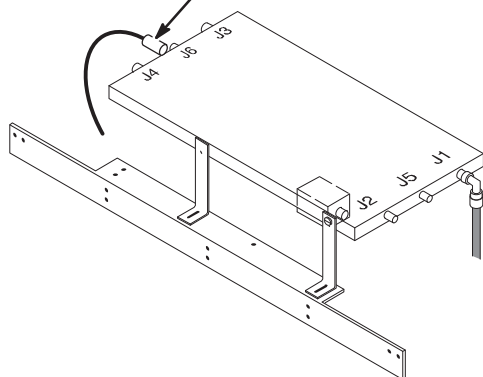


9-7-7 CONNECT REMAINING CABLES TO REAR PEDESTAL

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

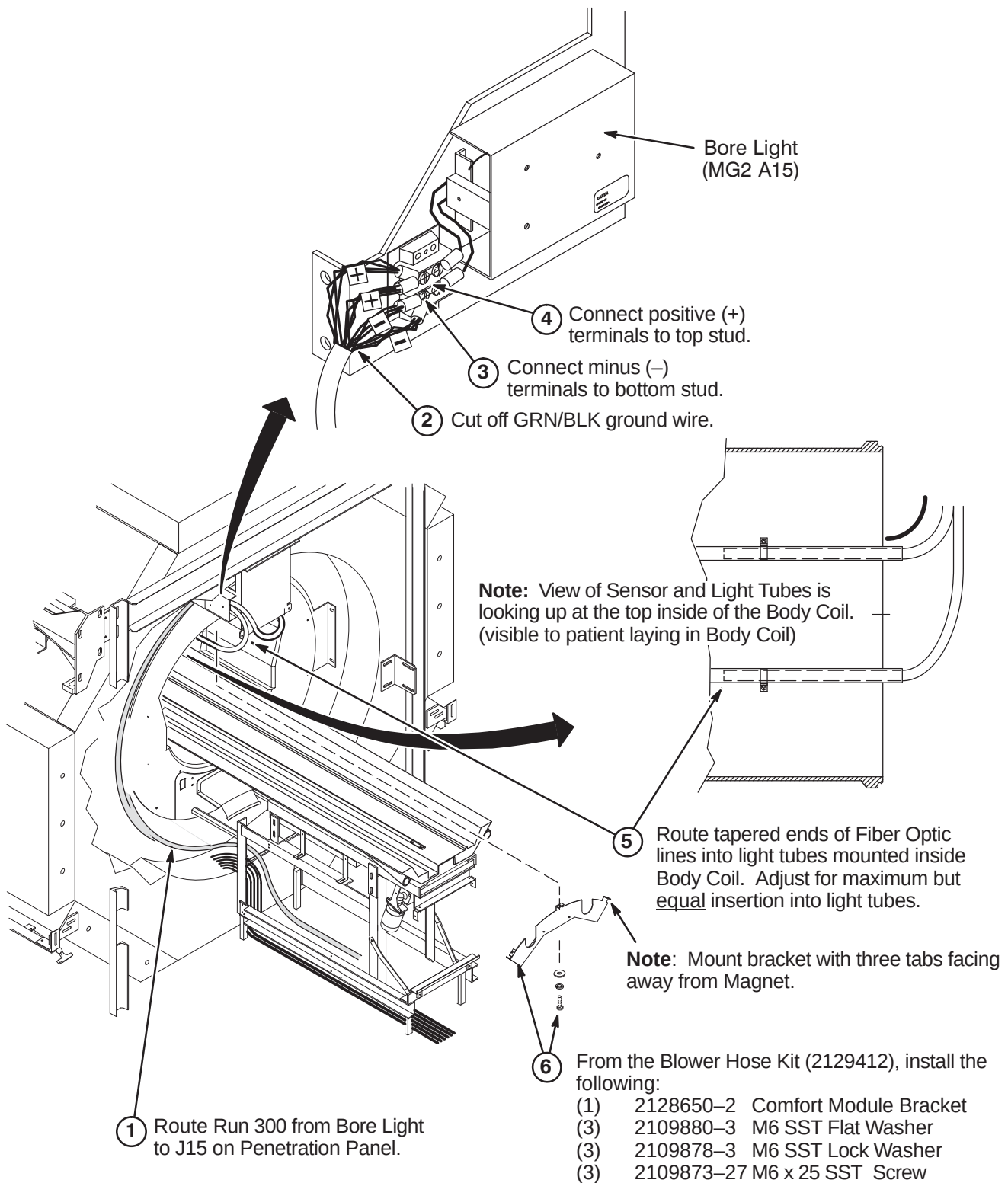


- ② Route/connect Run 780 to J6 on RF Splitter.



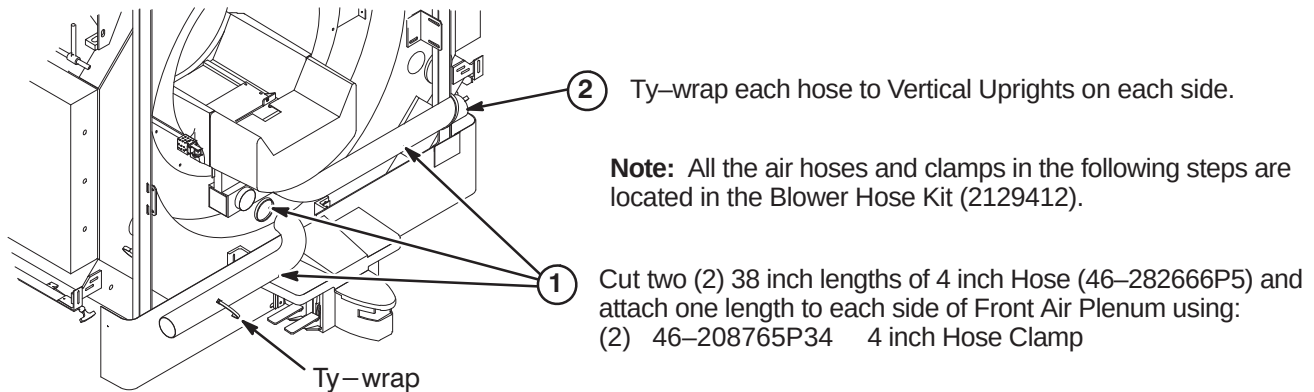
- ③ Connect Run 1250 to J1 on RF Splitter:

9-7-8 ROUTE AND CONNECT RUN 300 AND BORE LIGHT FIBER OPTIC LINES

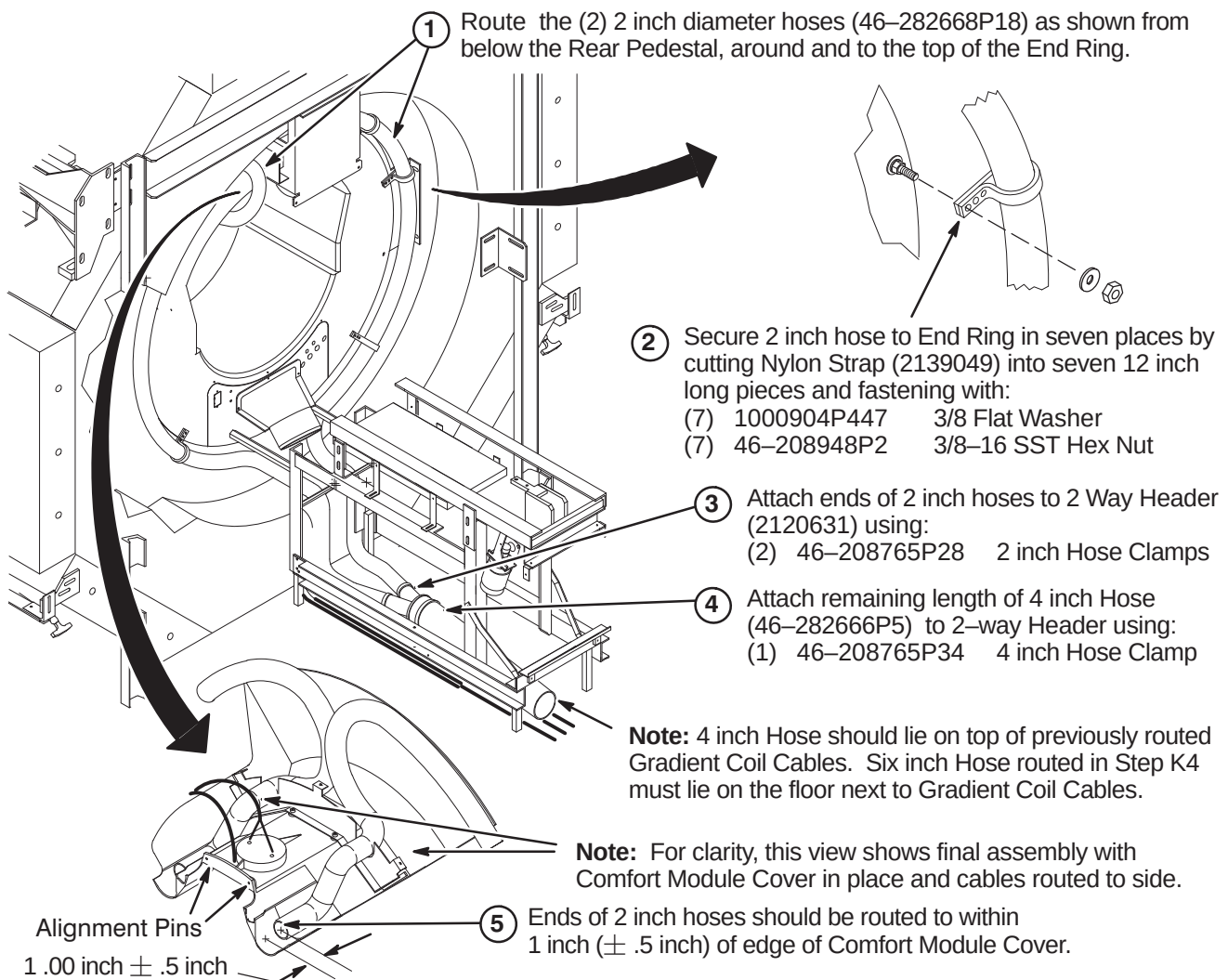


9-7-9 INSTALL PATIENT COMFORT AIR SYSTEM

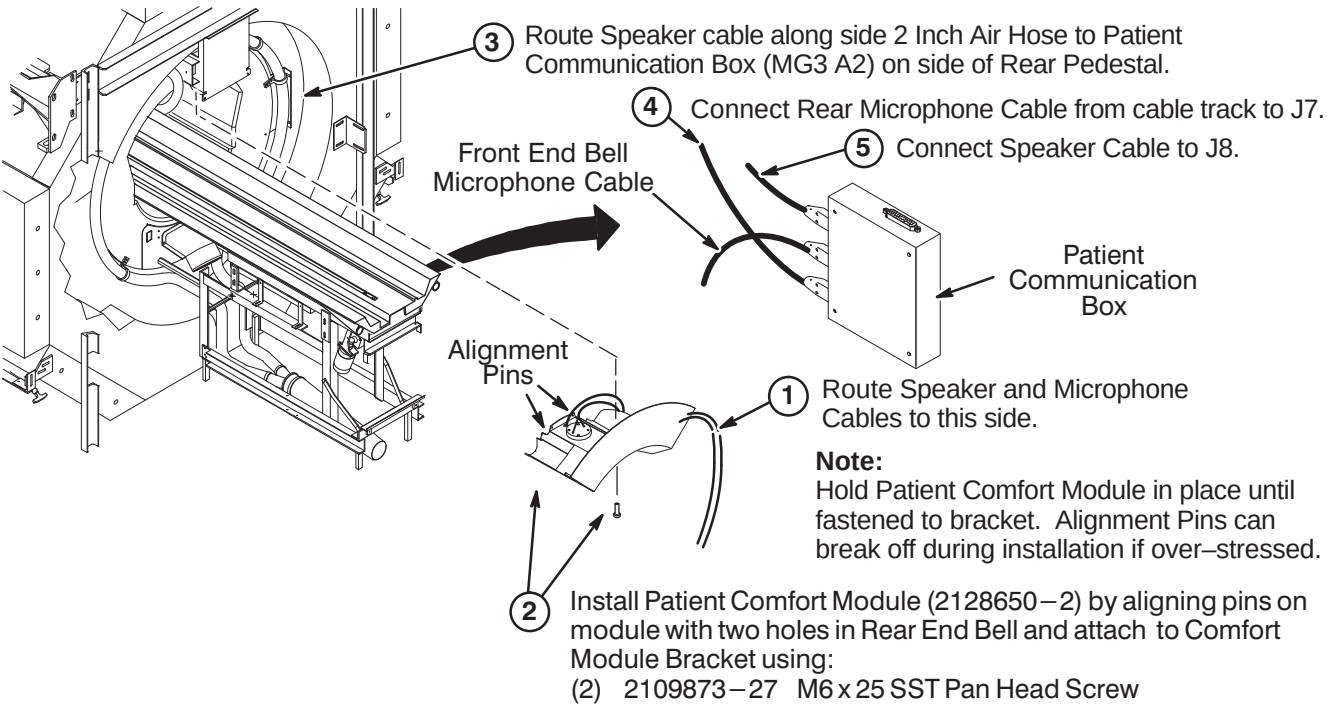
9-7-9-A Connecting Front Air Plenum Hoses (2129412)



9-7-9-B Routing/Connection of Patient Comfort Air Hoses



9-7-9-C Attaching Patient Comfort Module and Cables



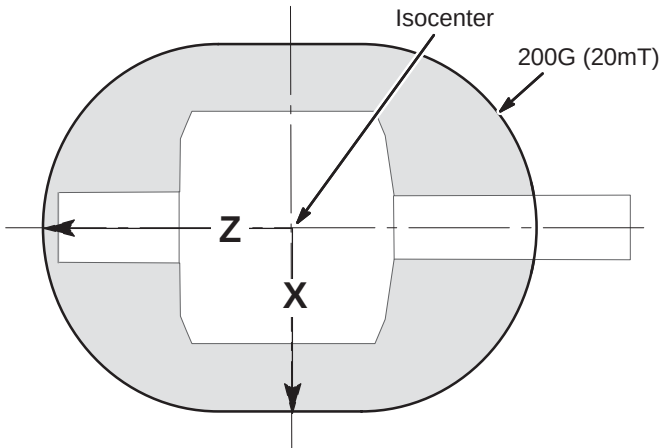
9-7-9-D 200 Gauss Zone Limit for Movement of Blower Box

Refer to the illustration and table below for applicable 200 Gauss zone for magnet at site.



THE BLOWER BOX IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.

Illustration is not to scale

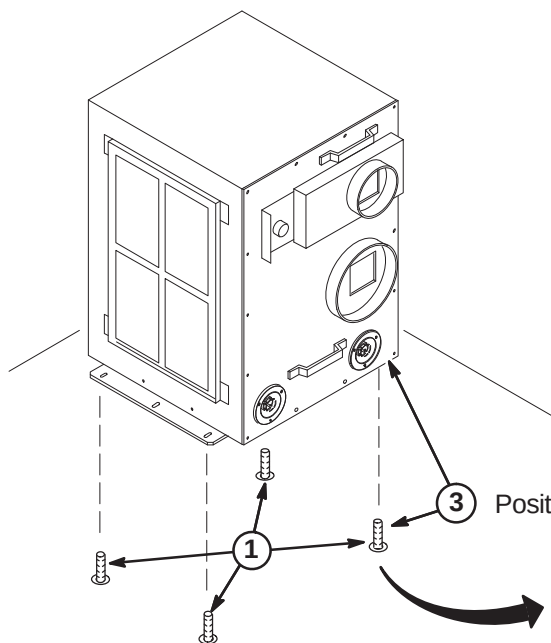


Approximate Distance from isocenter to 200G Line		
	X	Z
3.0T	180 cm (71 in) ¹	272 cm (107 in) ¹
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
Less than 1.0T	145 cm (57 in) ¹	195 cm (77 in) ¹
Unshielded Magnet	Measure ²	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

9-7-9-E Installation of Blower Box

WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

**IMPORTANT!**

Location of Blower Box and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural)/ Construction drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions.

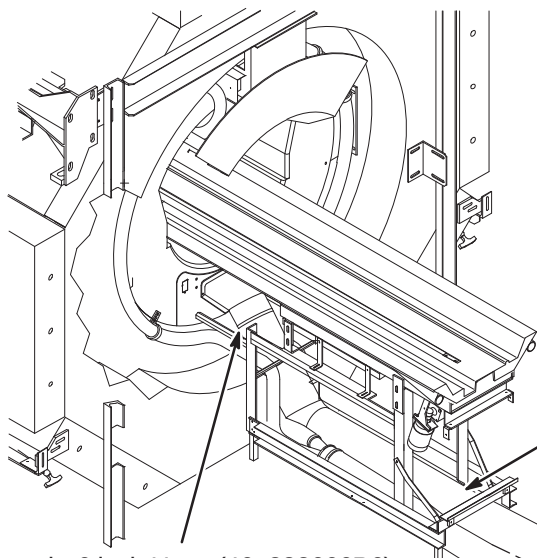
- ① Check for correct installation of the anchors:
 - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
 - Confirm there are at least four (4) anchors.
 - If any of the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.

③ Position the blower box on top of anchors.

② Remove any mounting hardware from the anchors.

④ Secure the blower box using the hardware provided by the RF shield room vendor to attach.

9-7-9-F Route/Connect Patient Comfort Cables and Air Hoses



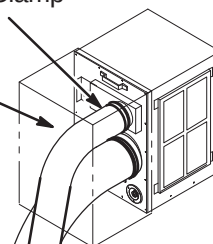
- ② Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:

- | | | |
|-----|--------------|-------------------|
| (1) | 46-208765P34 | 4 inch Hose Clamp |
| (1) | 46-208765P29 | 6 inch Hose Clamp |

Note: If needed, Blower Hose Cover can be cut and modified.

Note: 6 inch Hose must lie on the floor next to Gradient Coil Cables.

- ① Attach 6 inch Hose (46-28266P6) to End Ring Air Plenum using:
 - (1) 46-208765P29 6 inch Hose Clamp

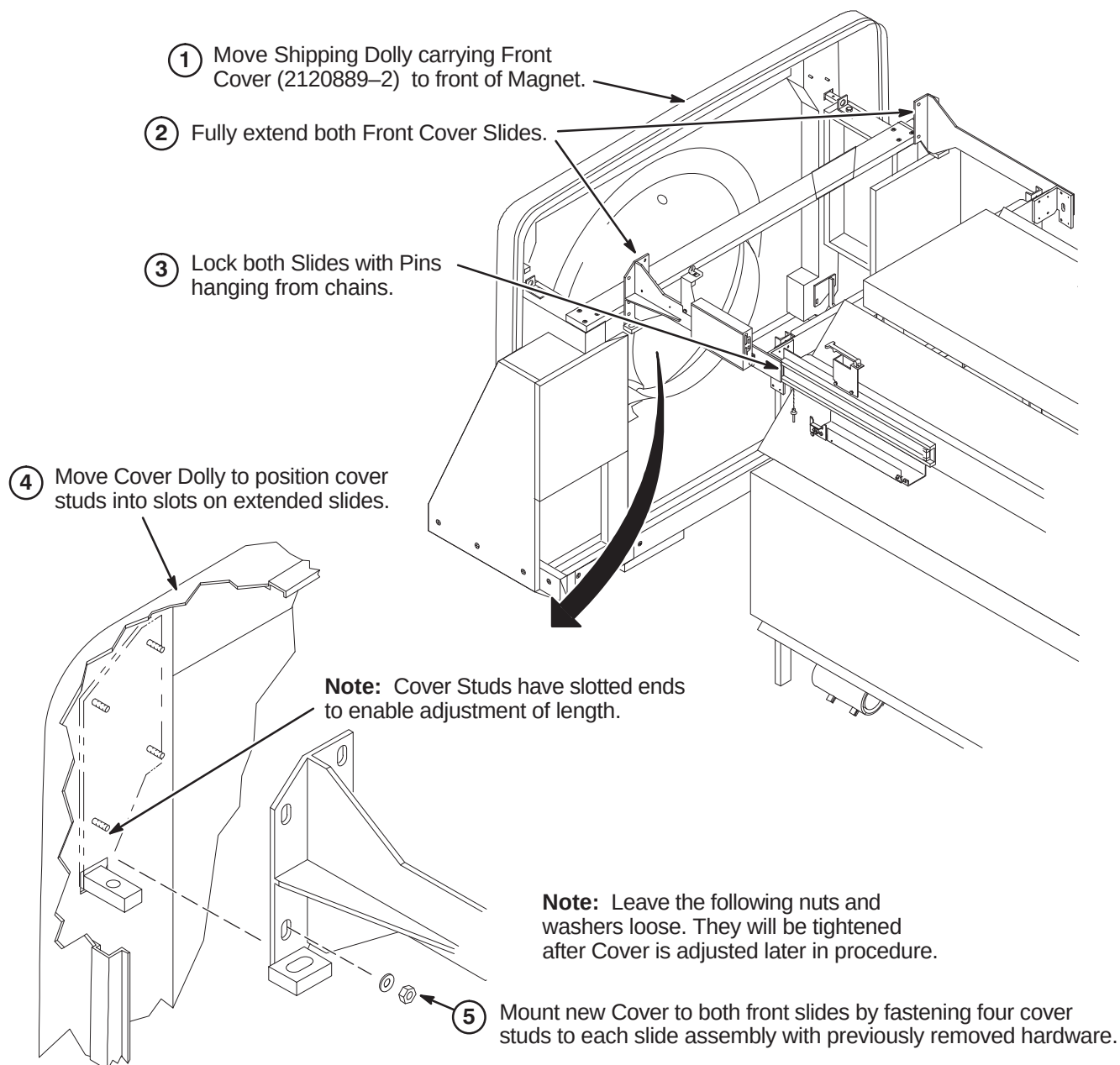


9-7-10 INSTALL FRONT COVER

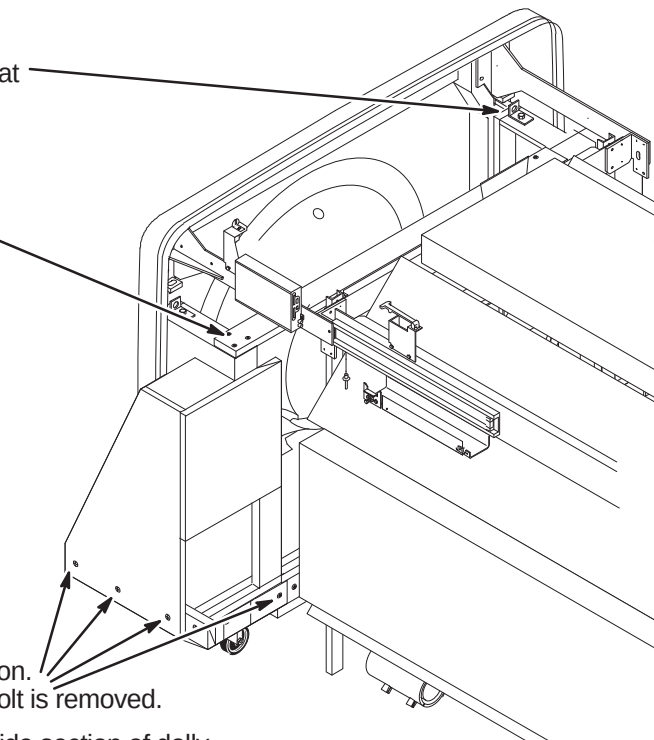
9-7-10-A Mount Front Cover



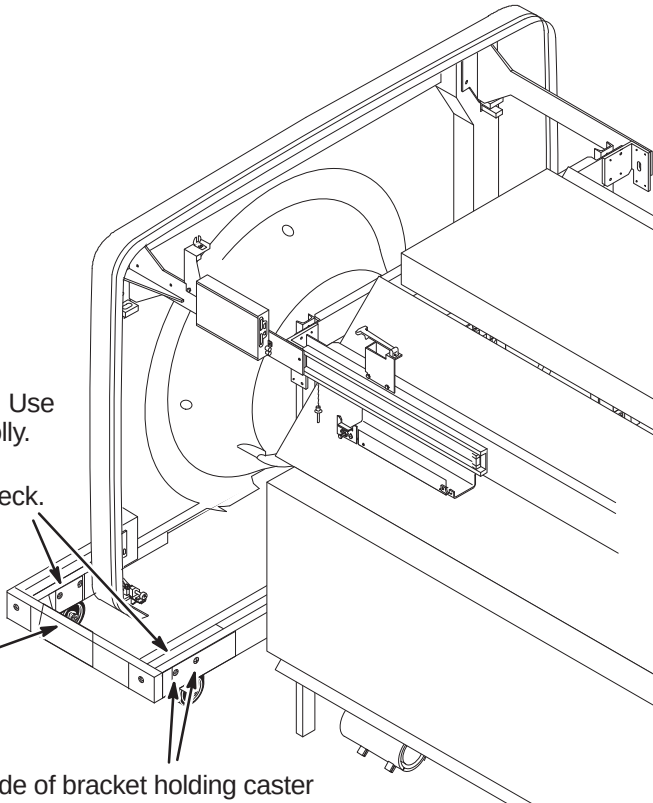
FERROUS MATERIAL HAZARD! THE COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE FRONT COVER FROM DOLLY AND CARRY COVER INTO POSITION AT FRONT OF MAGNET.



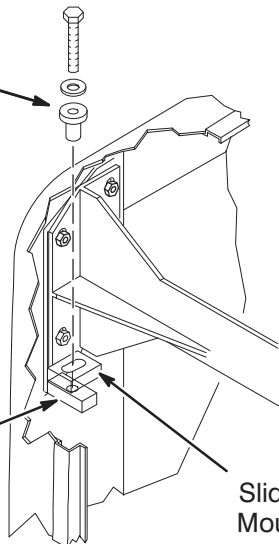
9-7-10-B Disassemble Cover Dolly

- 
- The diagram shows a side view of the magnet room enclosure with the dolly assembly. Arrows point from the numbered instructions to the specific components being removed: bolts on the side of the dolly, screws on the cross brace, and screws holding the side section.
- ① Remove bolts from each side of dolly that were used for attaching cover to dolly.
 - ② Remove three screws from each end of cross brace and remove brace.
 - ③ Remove screws holding dolly side section. Side section will slide down when last bolt is removed.
Repeat above for removal of opposite side section of dolly.

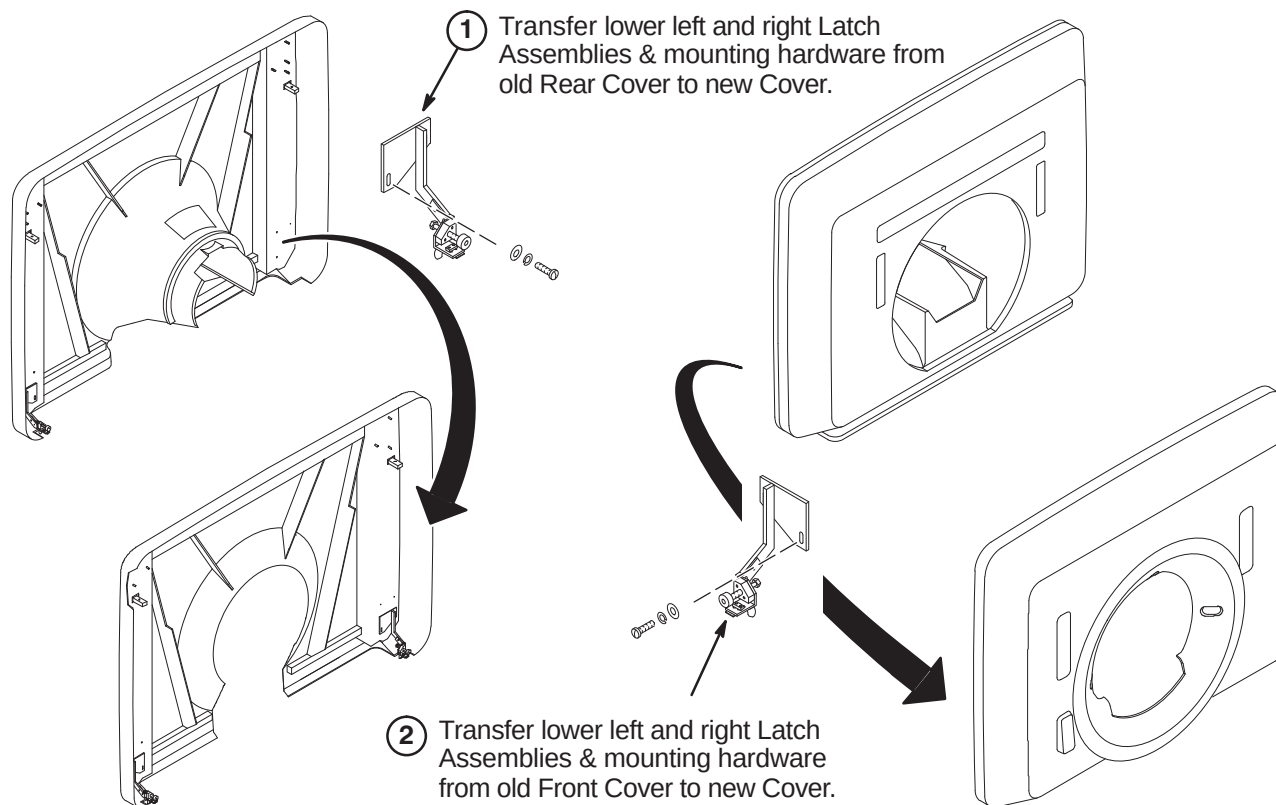
9-7-10-C Remove Cover Dolly

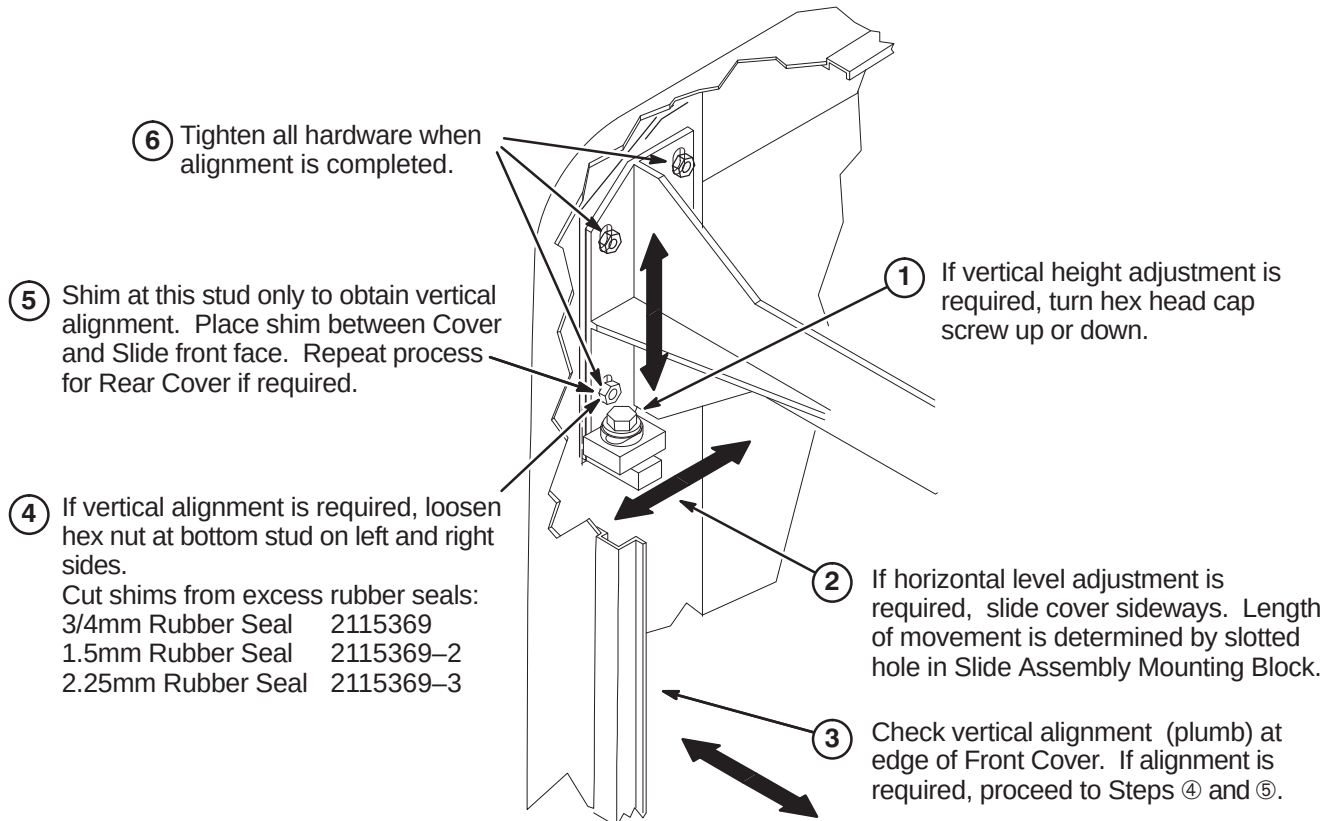
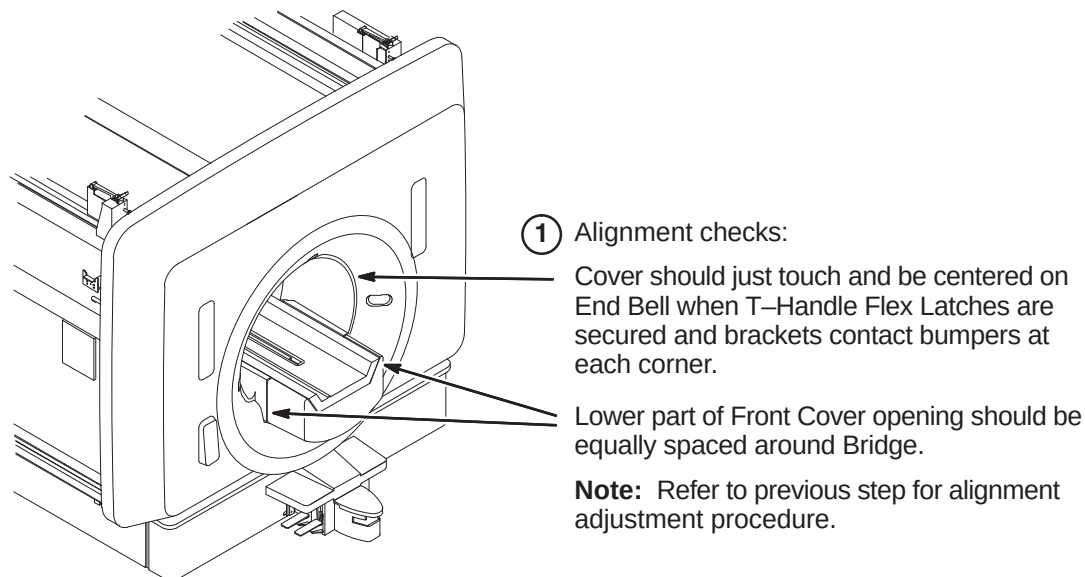
- 
- The diagram shows the dolly being slid away from the magnet. Arrows point from the numbered instructions to the remaining dolly parts, the bracket and caster assemblies, the dolly deck, and the screws holding the bracket.
- ⑥ Place remaining dolly parts on recycle pile.
 - ⑤ Place the four bracket and caster assemblies in pre-addressed box provided with dolly for shipment.
 - ④ With casters removed, slide frame away from magnet. Use care not to damage floor when sliding and removing dolly.
 - ③ With end lifted, remove caster assemblies from dolly deck.
Repeat process for other end of dolly deck.
 - ② Lift this end of dolly deck.
 - ① Remove screws from each side of bracket holding caster assembly to dolly deck. Repeat process for other three corners.

9-7-10-D Install Front Cover Alignment Hardware

- ① Transfer and insert the Alignment Hardware from the old Front Cover into the oval hole on the Slide Assembly Mounting Block
 - ② Engage threads in Cover Mounting Block.
 - ③ Repeat process for other side of Front Cover.
- 
- Slide Assembly Mounting Block

9-7-10-E Transfer Lower Latch Assemblies to New Front & Rear Covers

- ① Transfer lower left and right Latch Assemblies & mounting hardware from old Rear Cover to new Cover.
 - ② Transfer lower left and right Latch Assemblies & mounting hardware from old Front Cover to new Cover.
- 

9-7-10-F Adjust Front Cover Alignment**9-7-10-G Align Front Cover**

9-7-11 INSTALL REAR COVER



FERROUS MATERIAL HAZARD! THE REAR COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE COVER FROM DOLLY AND CARRY COVER INTO POSITION AT REAR OF MAGNET.

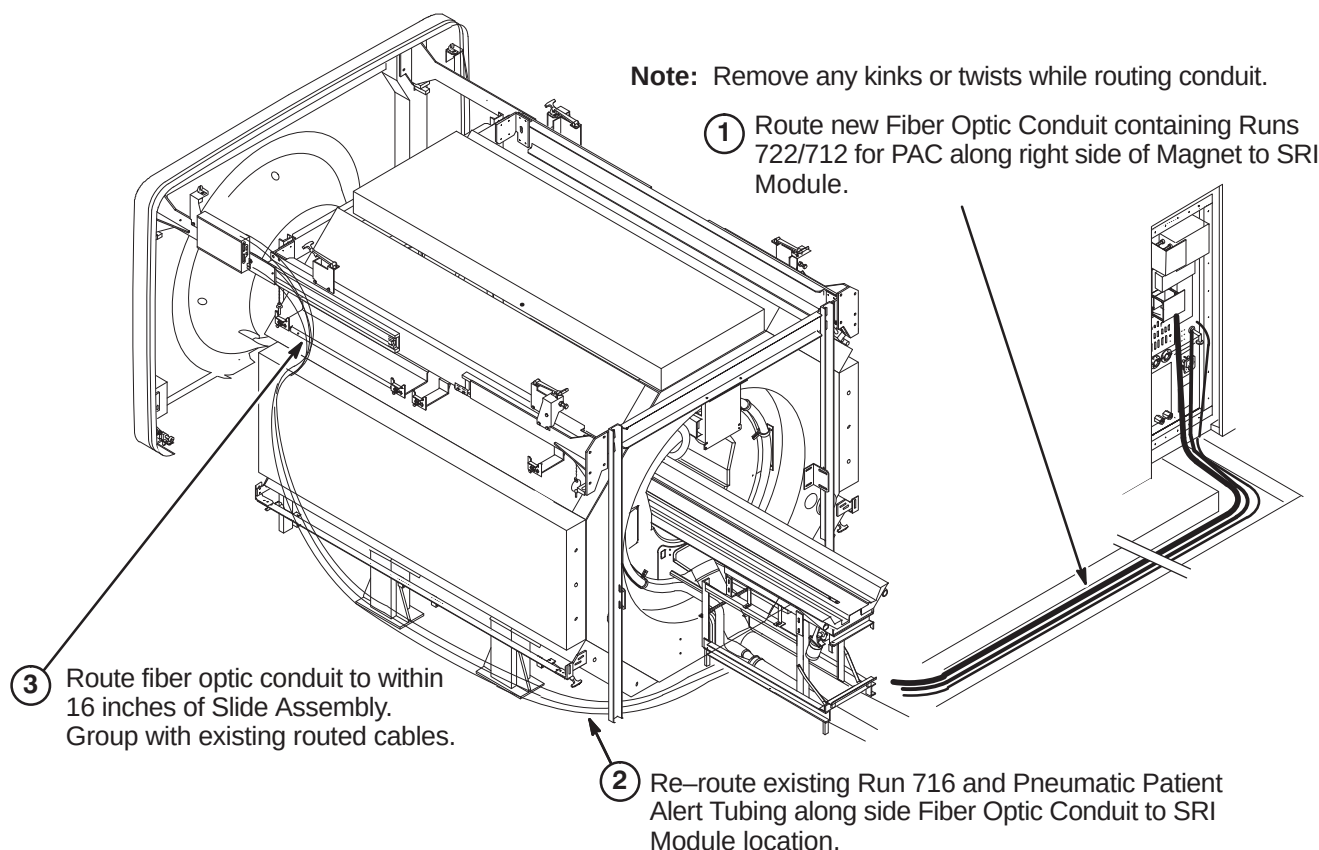
THE STEPS ARE THE SAME AS THE JUST COMPLETED FRONT COVER INSTALLATION PROCEDURE

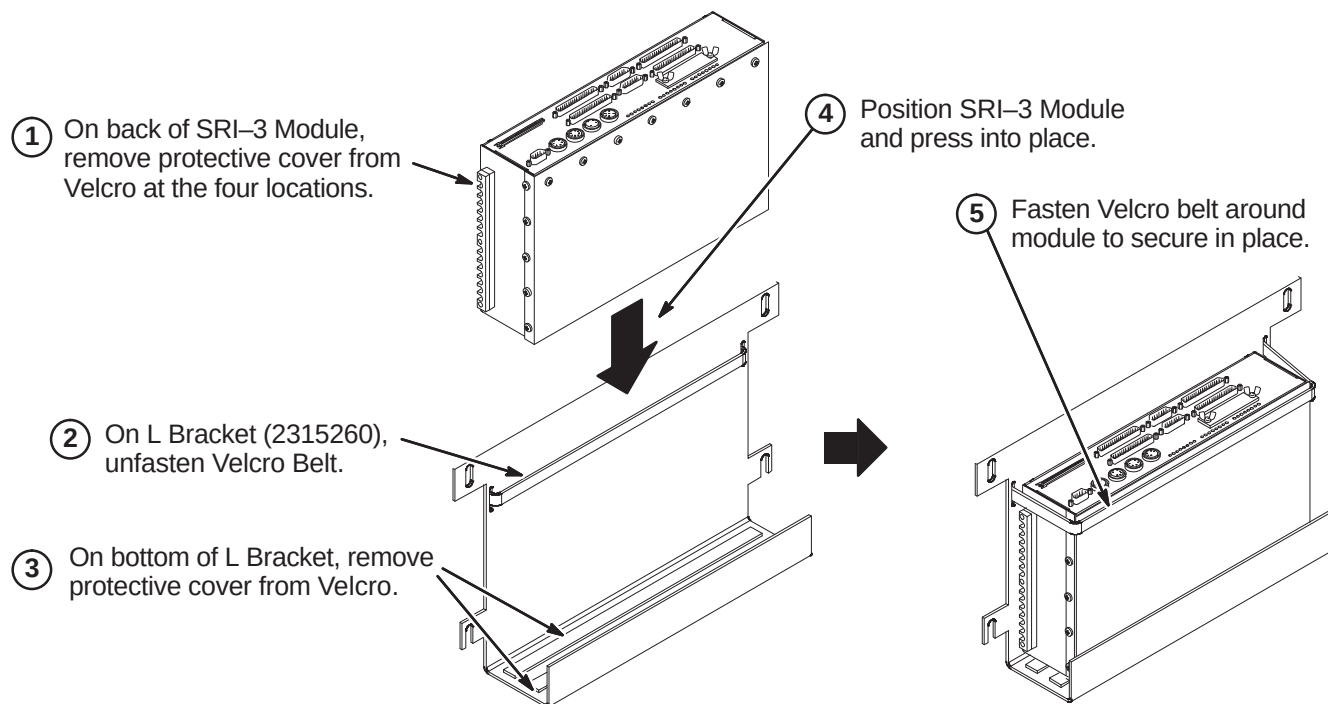
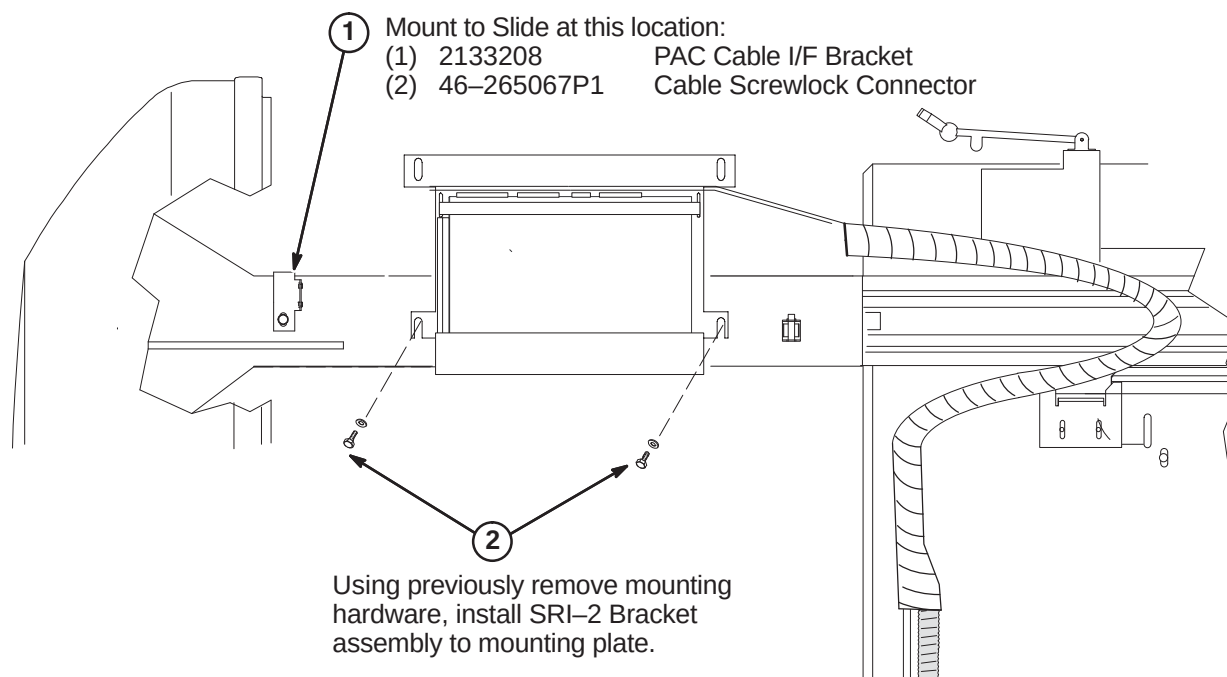
Note: Experience has shown that the Rear Cover does not require shimming for vertical levelness.

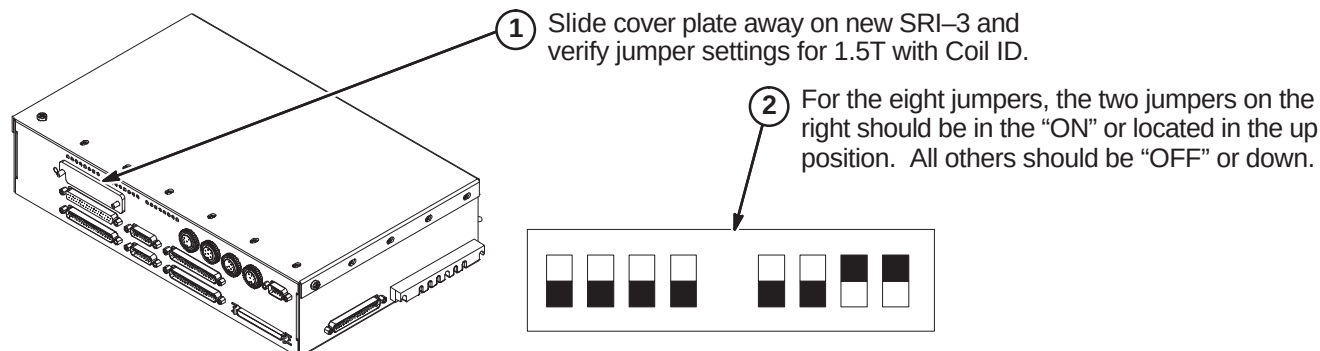
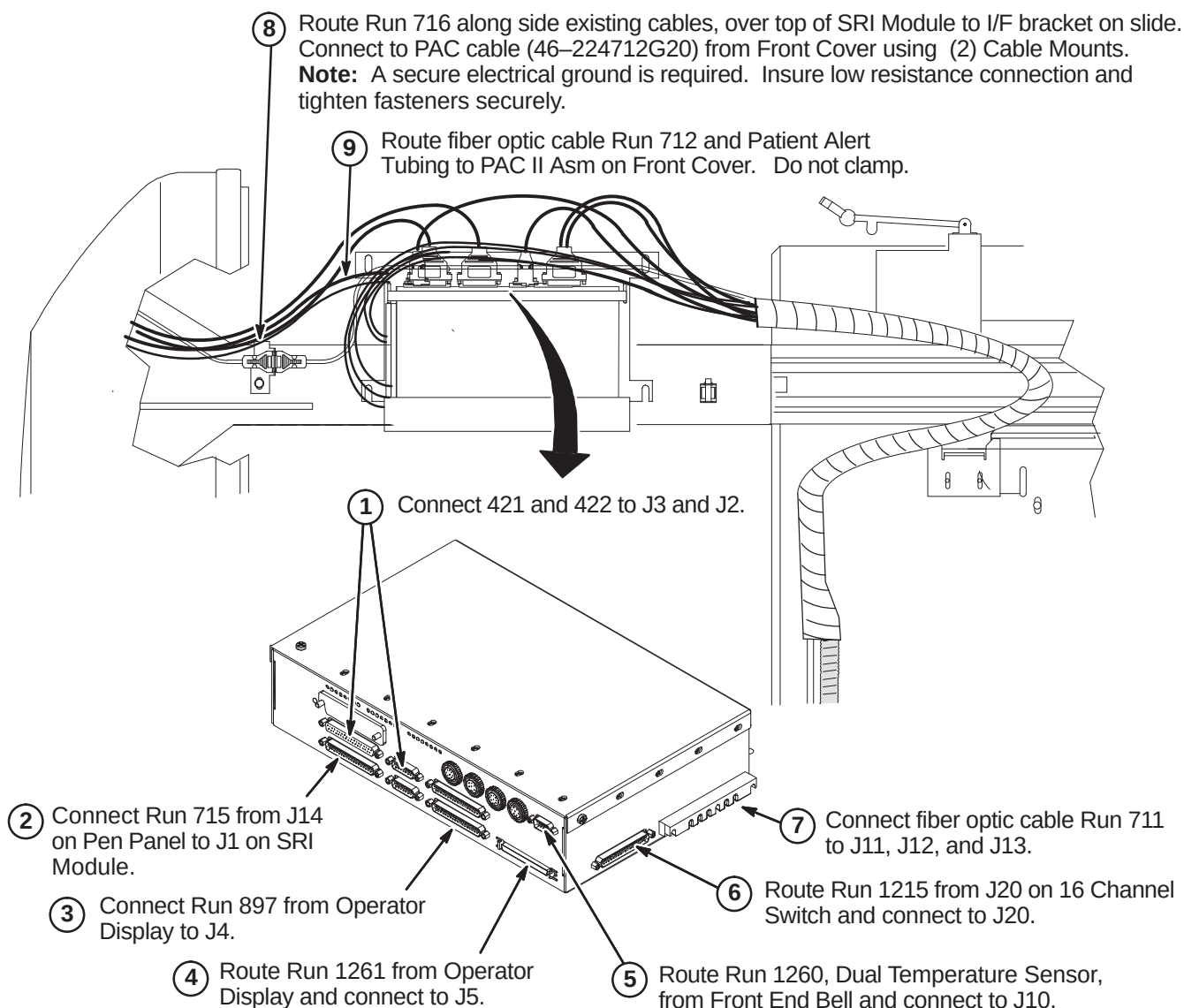
9-7-12 ROUTE AND CONNECT RUNS 711/712, 716, AND PATIENT ALERT TUBING



EQUIPMENT DAMAGE POSSIBILITY. Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend to radius smaller than two inches. Do not pull connector from cable. Avoid scratching connector ends. Keep ends protected until ready to connect.

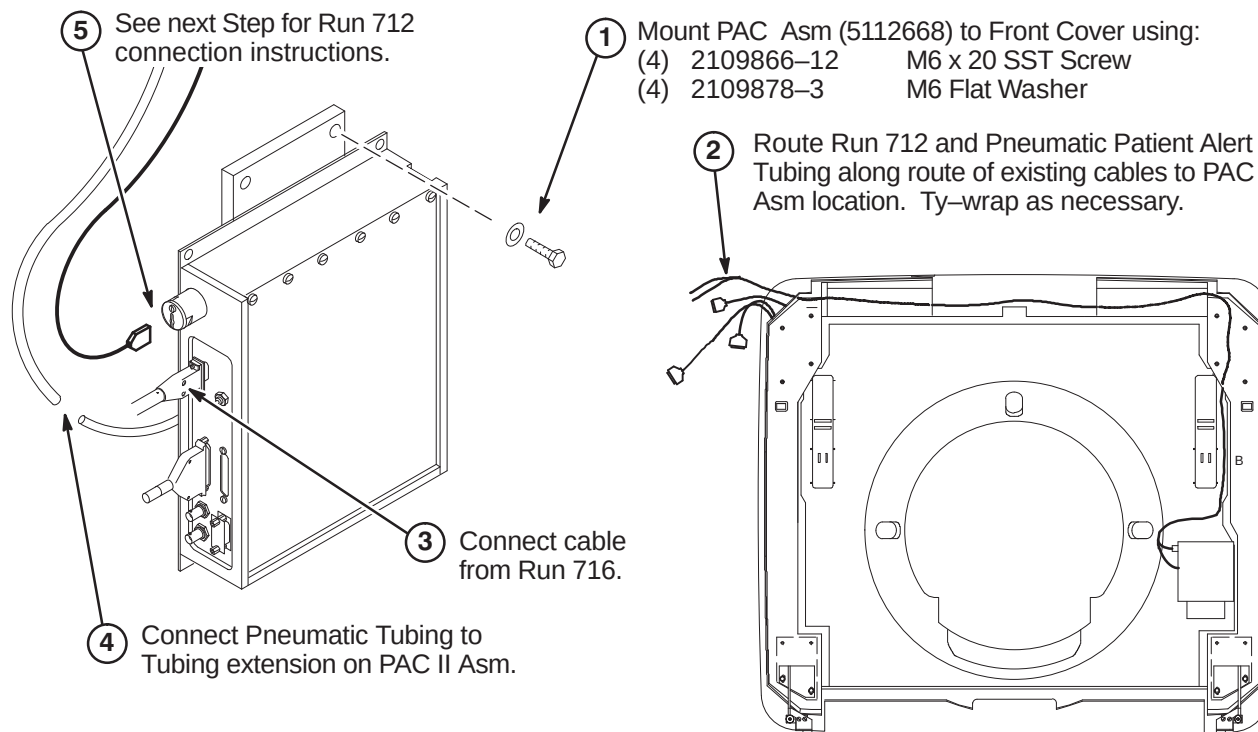


9-7-13 INSTALL SRI-3 MODULE**9-7-13-A Position SRI-3 in L Bracket****9-7-13-B Install SRI-3 on Magnet**

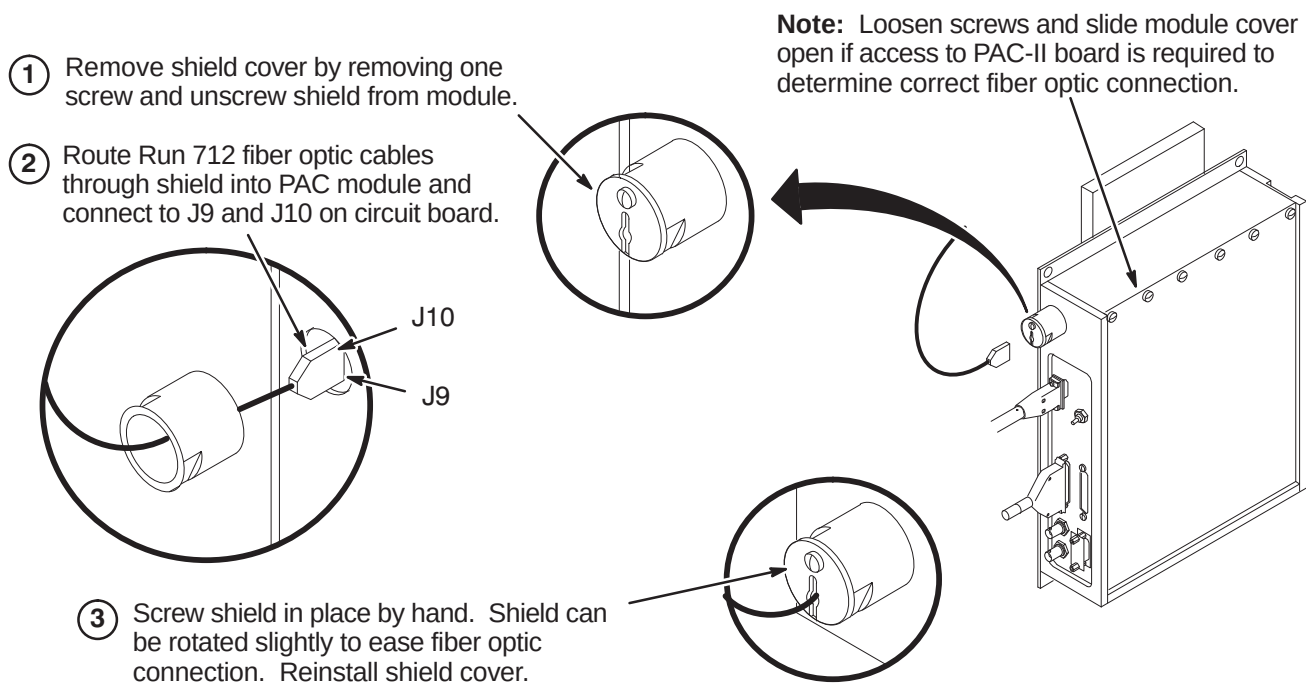
9-7-13-C Verify SRI-3 Jumper Settings**9-7-13-D Connect Cables at SRI-3 Module Location**

9-7-14 INSTALL NEW PAC MODULE

9-7-14-A Connect Run 712, 716, & Pneumatic Patient Alert Tubing

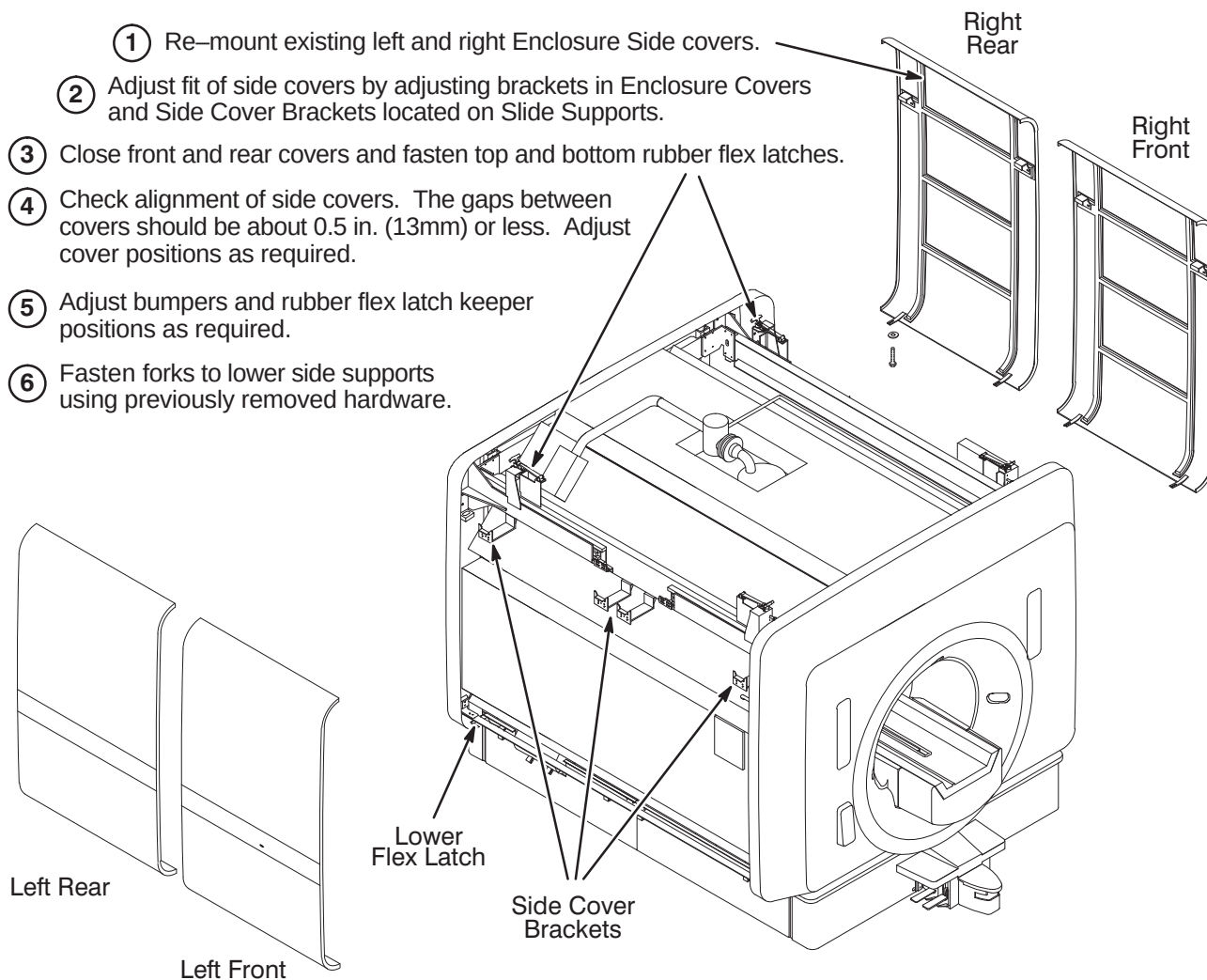


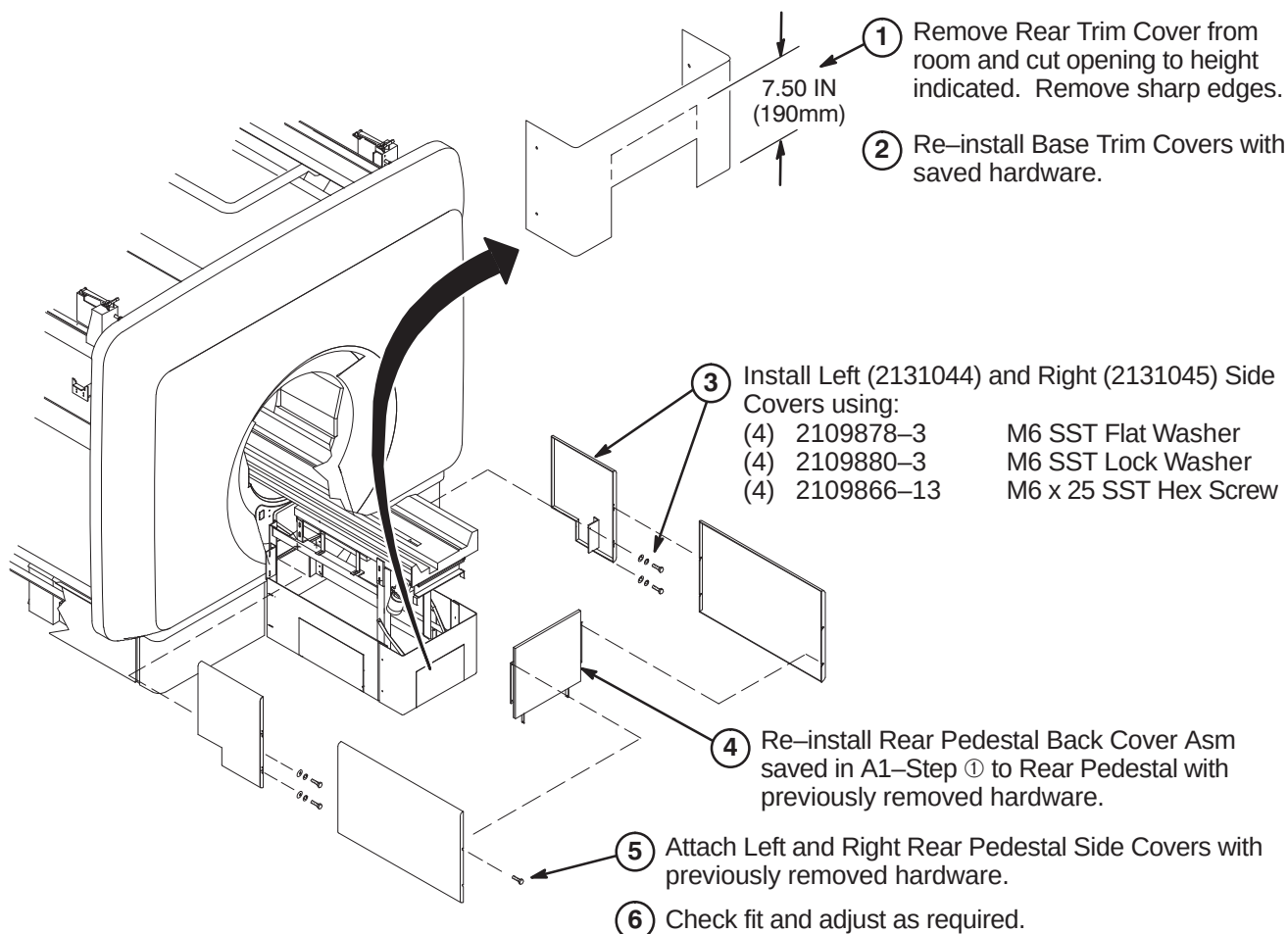
9-7-14-B Connect Run 712



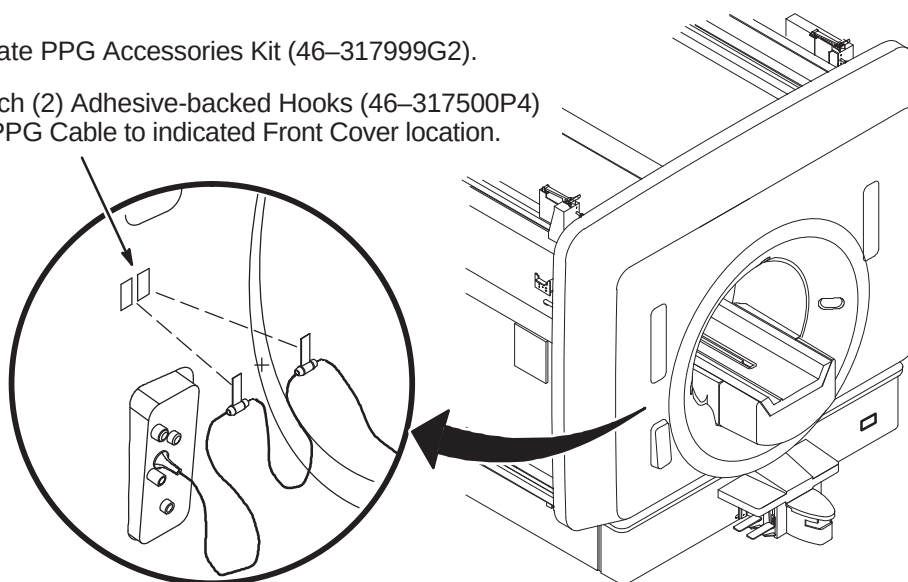
9-7-15 INSTALL ENCLOSURE SIDE COVERS

- ① Re-mount existing left and right Enclosure Side covers.
- ② Adjust fit of side covers by adjusting brackets in Enclosure Covers and Side Cover Brackets located on Slide Supports.
- ③ Close front and rear covers and fasten top and bottom rubber flex latches.
- ④ Check alignment of side covers. The gaps between covers should be about 0.5 in. (13mm) or less. Adjust cover positions as required.
- ⑤ Adjust bumpers and rubber flex latch keeper positions as required.
- ⑥ Fasten forks to lower side supports using previously removed hardware.

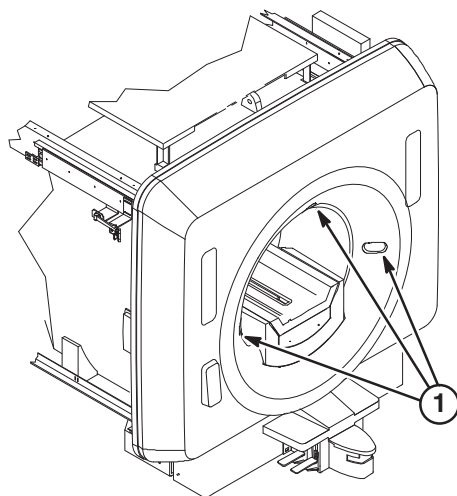


9-7-16 INSTALL REAR PEDESTAL COVERS**9-7-17 INSTALL PPG CABLE HOOKS ON FRONT COVER**

- 1** Locate PPG Accessories Kit (46-317999G2).
- 2** Attach (2) Adhesive-backed Hooks (46-317500P4) for PPG Cable to indicated Front Cover location.



9-8 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS



A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



Spanish



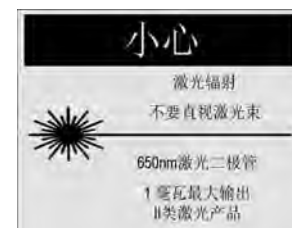
Italian



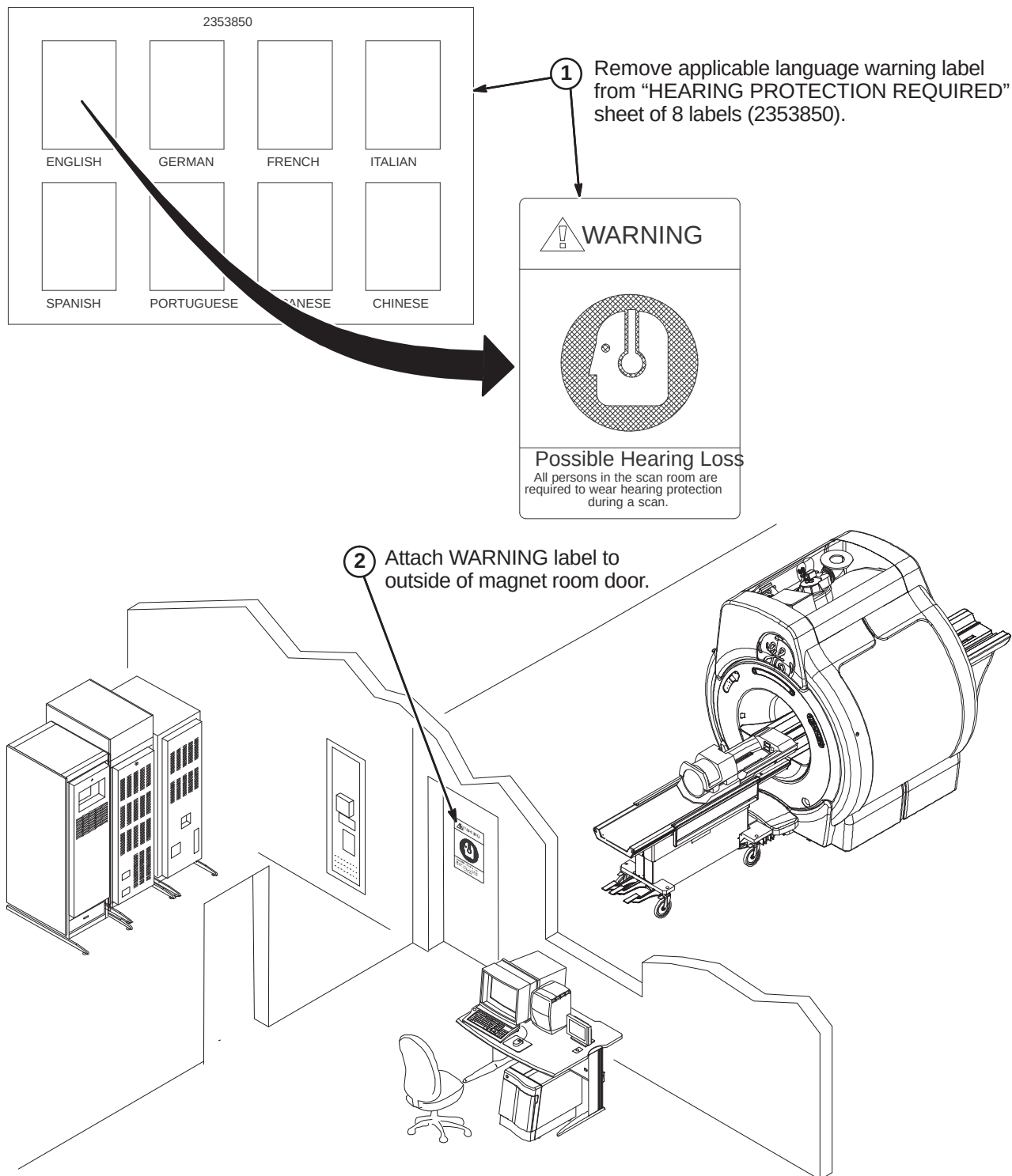
Swedish



Japanese



Chinese

9-9 INSTALL HEARING LOSS WARNING SIGN

SECTION 10 – DUAL TEMPERATURE SENSOR UPGRADE

TABLE OF CONTENTS

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10-1 INTRODUCTION

The Thermal Sensor Upgrade improves bore temperature sensing reliability by adding a second temperature sensor.

The following illustrations show an S3 magnet with a Horizon style enclosure. The instructions apply to any magnet enclosure. If needed, refer to the applicable magnet enclosure upgrade procedure in this manual for additional information on removal and installation of enclosure covers, bridge, and end bell for the installation on the new dual sensor cable, Run 1260.



WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

10-2 TOOLS AND EQUIPMENT

The following tools and equipment are required to perform this procedure:

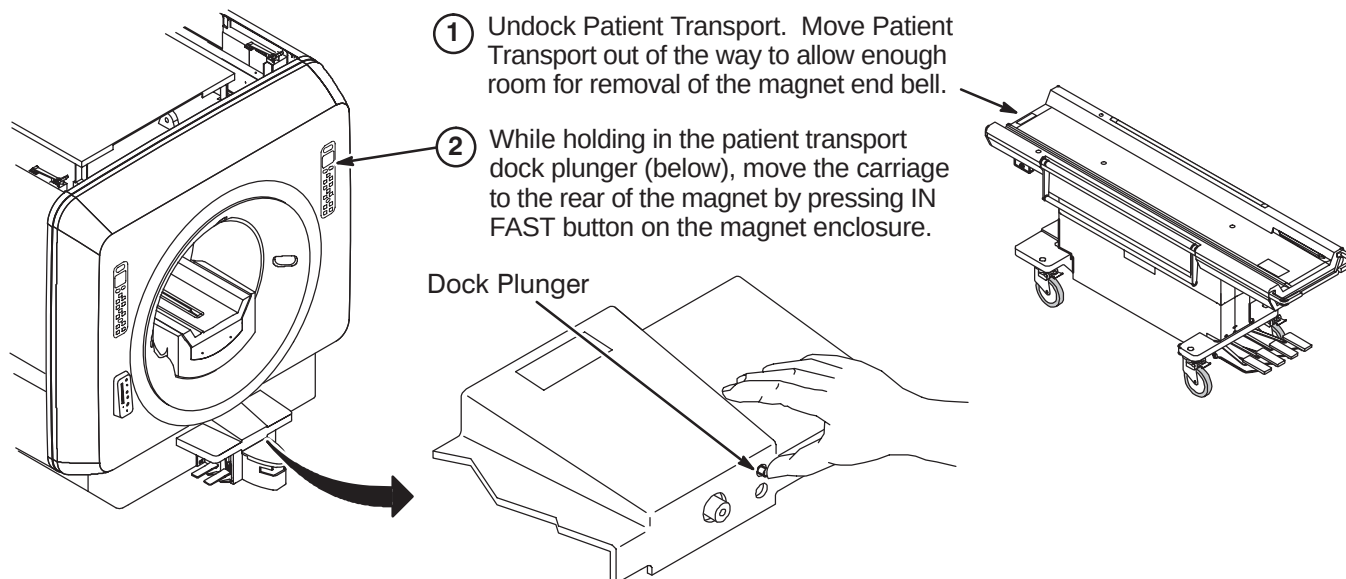
- 3mm Allen Wrench
- Drill bit, 7 mm or 5/16 inch
- Drill
- Dykes
- Digital Multimeter
- Lockout/Tagout equipment
- Tape, electrical or duct
- Permanent marker, fine-tip
- Slot-head screwdriver, 5/16" blade
- Nut driver, 5/16"
- Static-grounding wrist strap
- Copper tape (for replacement only)

The following items are provided in the Thermal Sensor Upgrade Kit, P/N 2287391

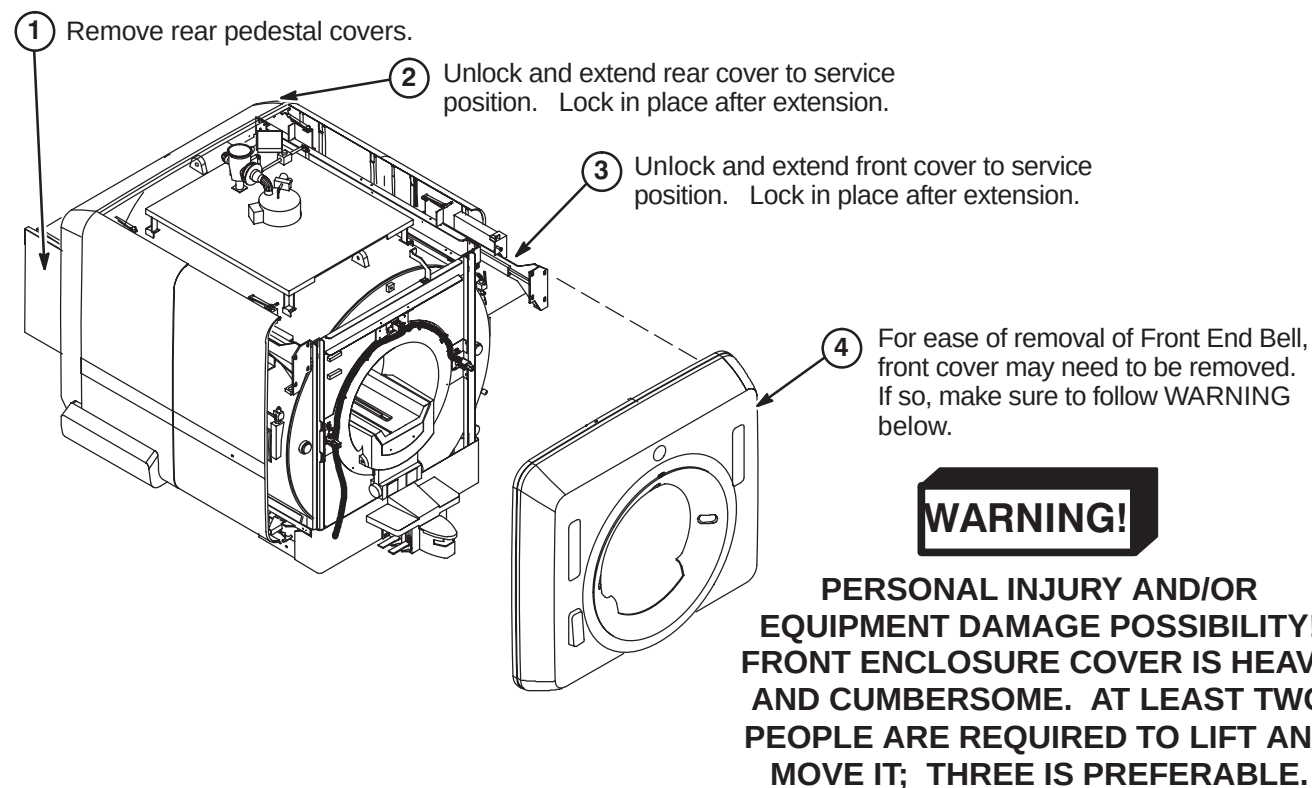
- 46-208747P1 Cable ties
- 46-282264P1 Epoxy adhesive
- 2298033 Applicator
- 46-320187P1 Nozzle

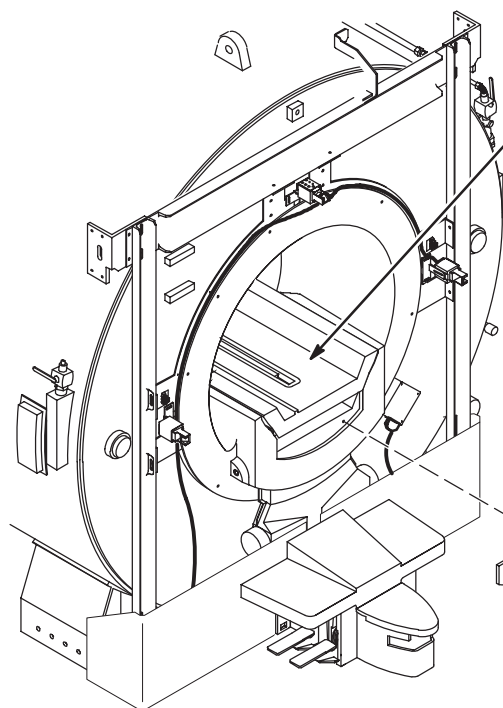
10-3 UPGRADE FOR HORIZON STYLE MAGNET ENCLOSURE

10-3-1 TABLE REMOVAL AND CARRIAGE POSITIONING



10-3-2 REMOVAL OF ENCLOSURE COVERS

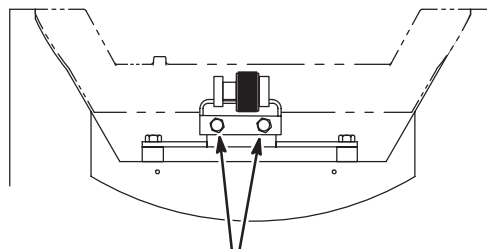


10-3-3 REMOVAL OF FRONT END BELL PANEL

③ For removal of Front End Bell, Bridge must be moved out of the way either by lifting the front end or complete removal.

Refer to Section **10-3-4-A** for procedure to lift the Bridge.

Refer to Section **10-3-4-B** for removal of Bridge.



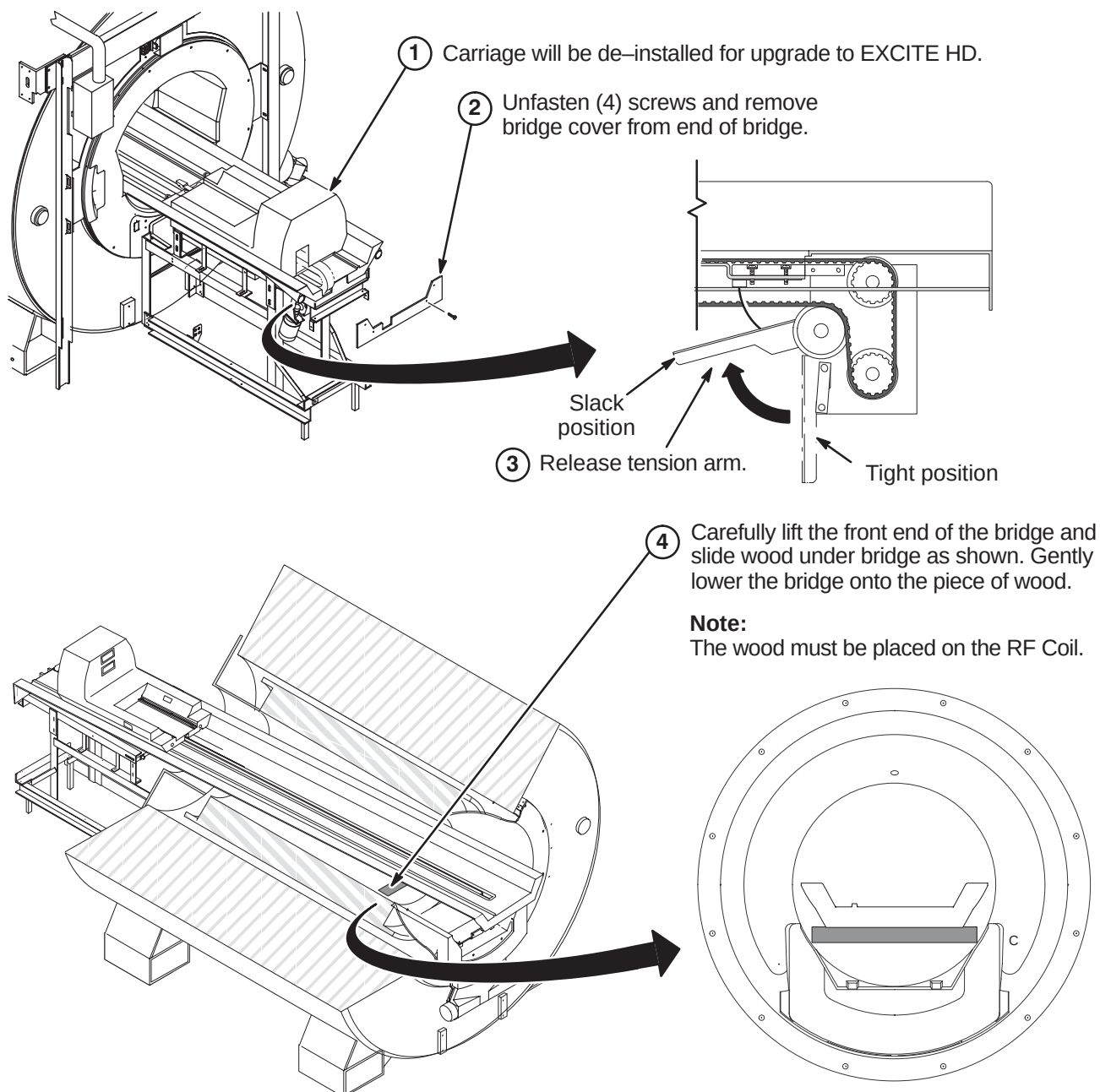
② Unfasten M10 flat washers, lock washers, and screws attaching front end of bridge asm to front end bell.

① Unfasten (2) M6 x 25 pan screws and remove end bell panel.

10-3-4 BRIDGE HANDLING OPTIONS PRIOR TO REMOVAL OF FRONT END BELL

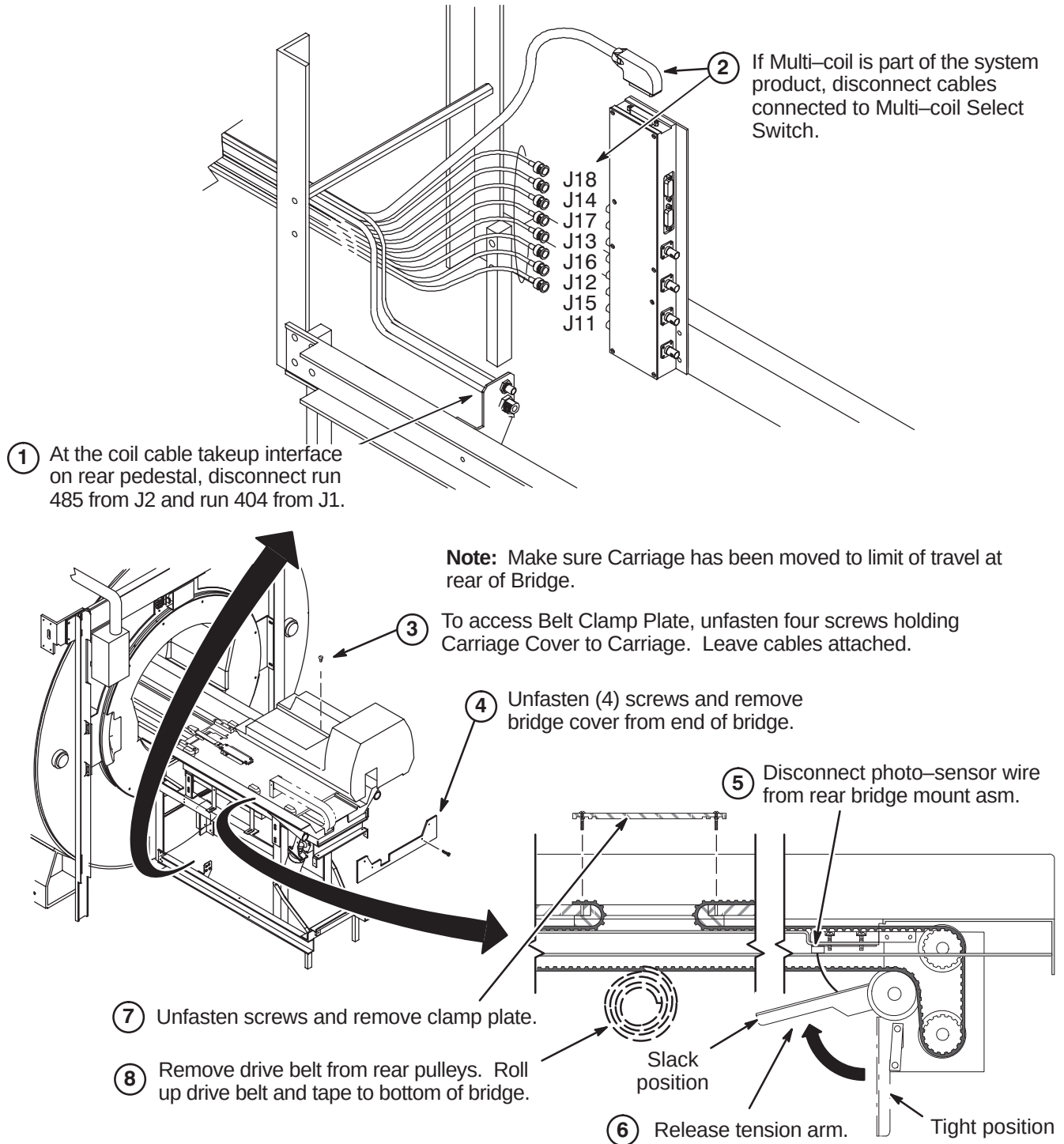
The Bridge needs to be repositioned or removed prior to removal of Front End Bell. Review sections **10-3-4-A** and **10-3-4-B**. Determine which method would best fit the needs of your site.

10-3-4-A Lifting Front End Of Bridge



10-3-4-B Removal of Bridge

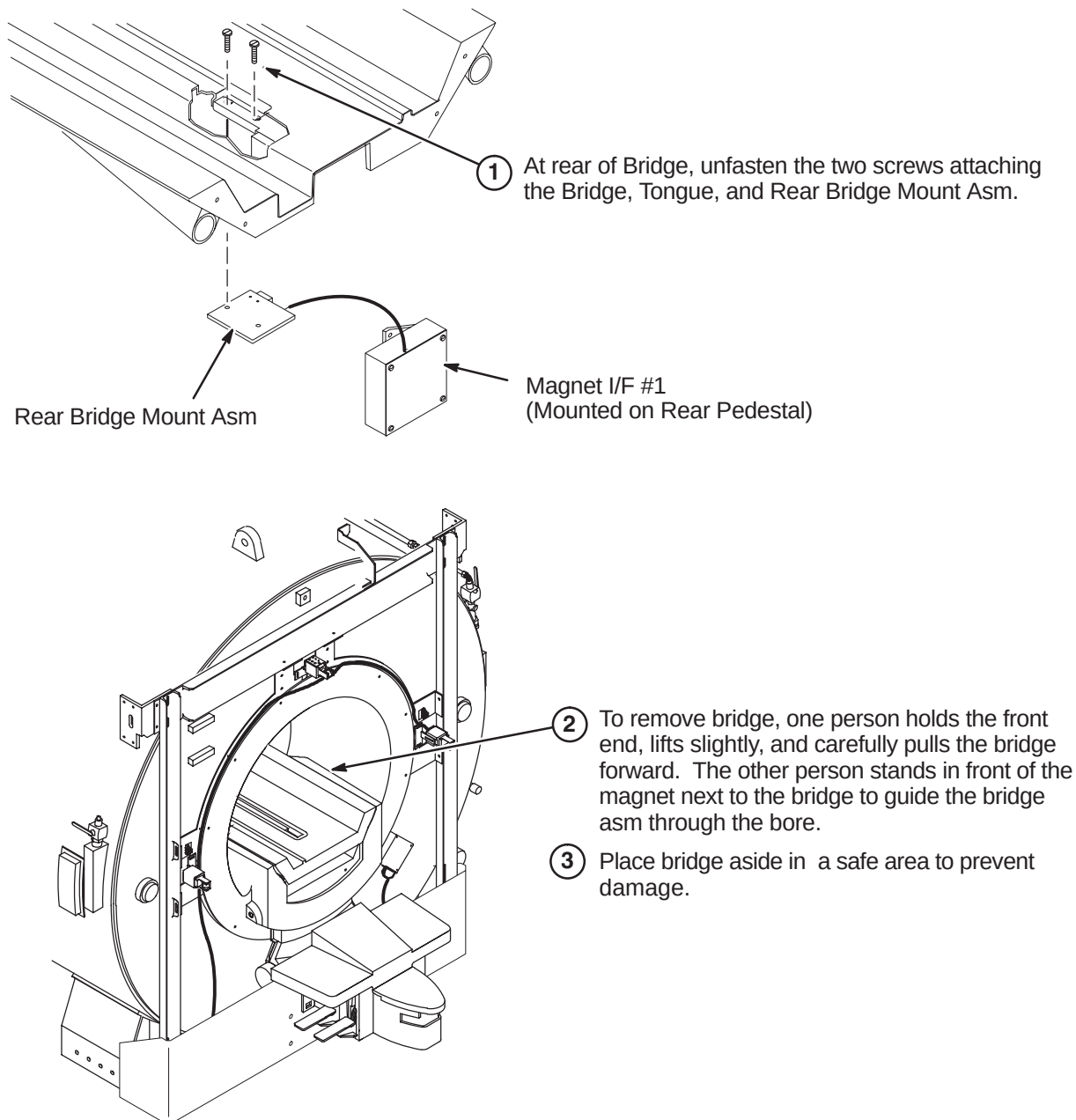
The upgrade to HDx requires a new LPCA with Cable Track and 16 Channel Switch. The de-installation instructions for existing components being replaced are located in the applicable magnet enclosure upgrade procedure in this manual. The steps below should be part of these de-installation instructions.



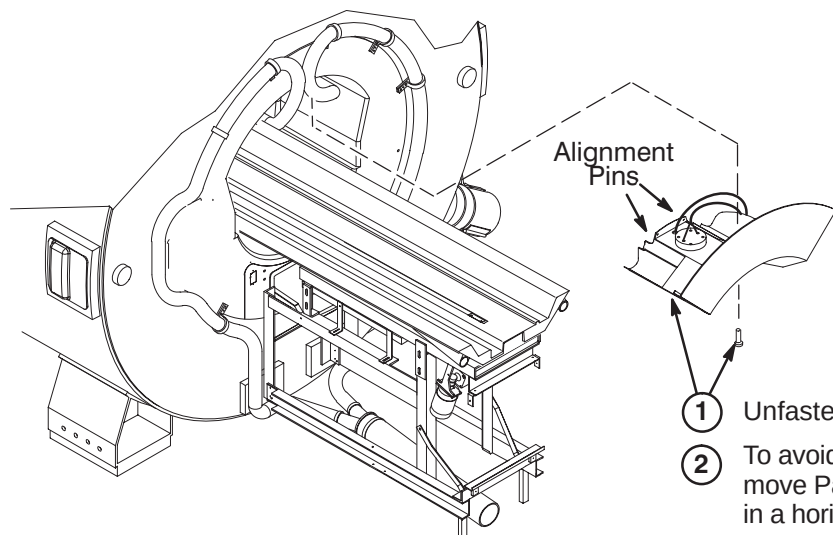
10-3-4-B Removal of Bridge (continued)

CAUTION

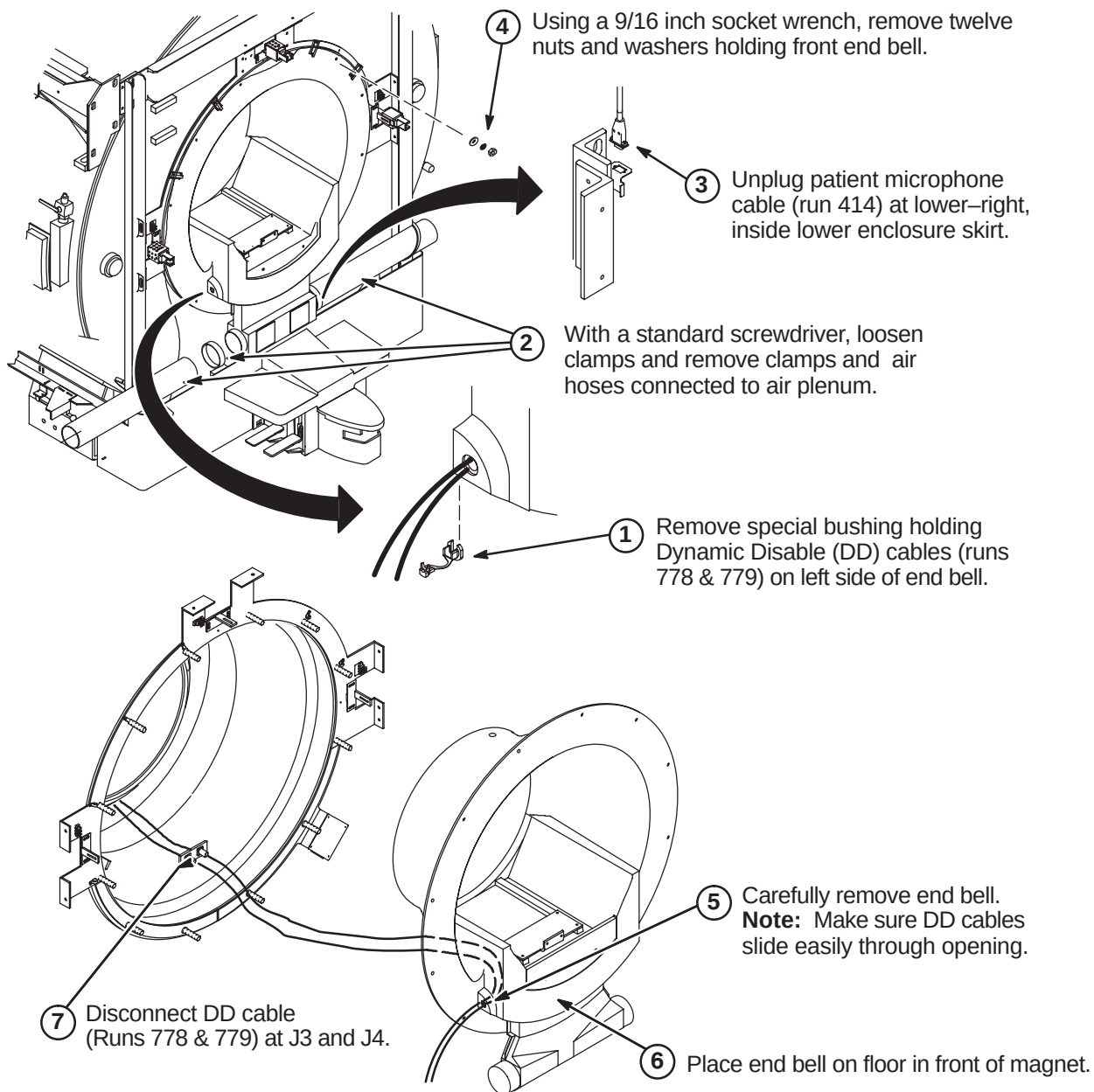
Possible equipment damage! Be careful not to damage the home flag on the drive belt when removing or replacing the bridge. The home flag is fragile and can easily be damaged.



L

10-3-5 REMOVE PATIENT COMFORT MODULE**Note:**

Hold Patient Comfort Module in place until mounting screws in Step 1 are removed. Alignment Pins can break off during removal if over-stressed.

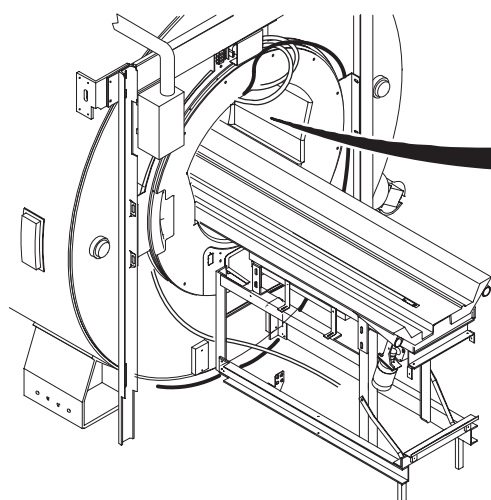
10-3-6 FRONT END BELL REMOVAL

10-3-7 TEMPERATURE SENSOR CABLE REMOVAL

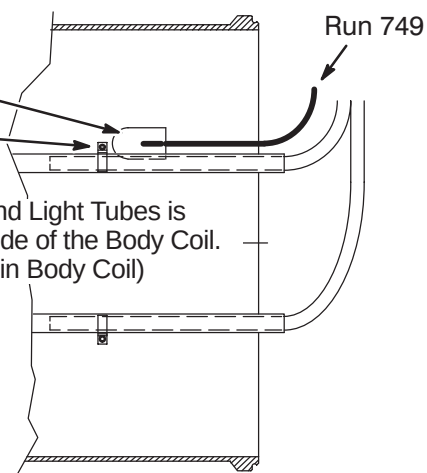
The site can have the existing Temperature Sensor Cable routed to the Rear end bell (earlier location) or Front end bell (later location). Either way, the cable needs to be removed and replaced with new cable, Run 1260. One of the two following procedures will be applicable to your site.

10-3-7-A Remove Existing Single Temperature Sensor Cable

- ① Unfasten existing Temperature Sensor Cable (Run 749) from top inside of Body Coil. It may be attached with:
adhesive fastener (as shown)
OR
with the Light and Cable clamp



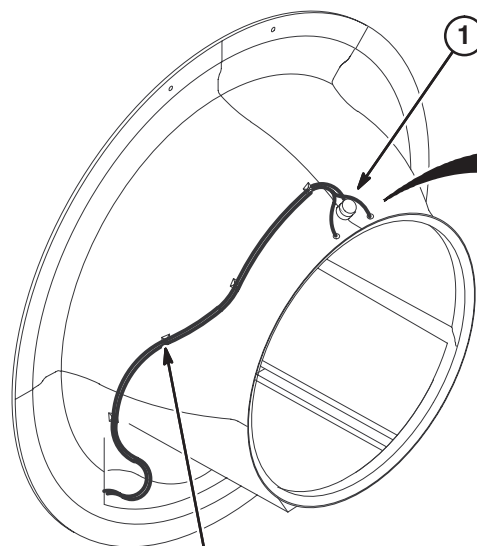
Note: View of Sensor and Light Tubes is looking up at the top inside of the Body Coil. (visible to patient laying in Body Coil)



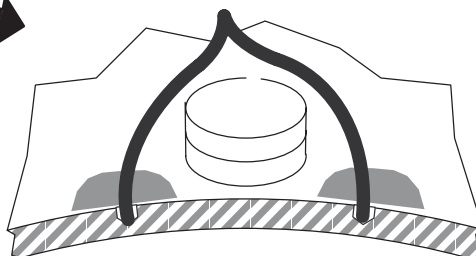
- ② Disconnect other end of Run 749 from J10 on SRI module.
- ③ Remove and discard cable.

10-3-7-B Remove Existing Dual Temperature Sensor Cable

The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



- ① Remove existing Dual Thermal Sensor Cable from end bell. Take care not to damage end bell.

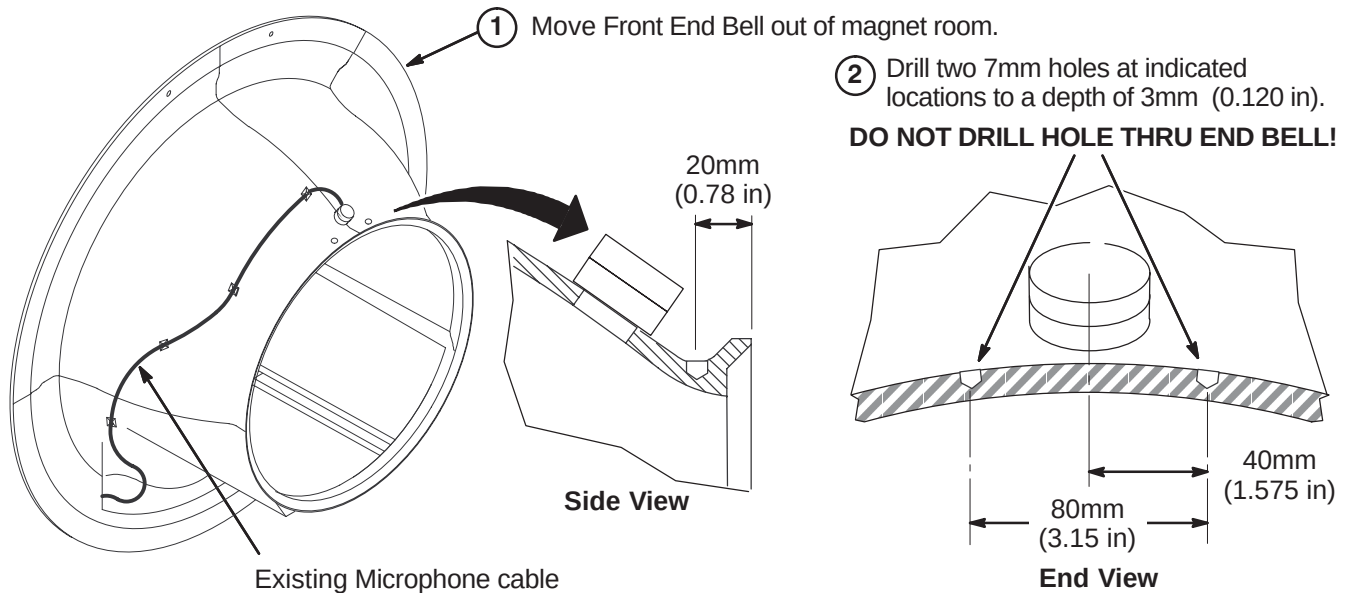


- ② Cut cable ties and remove Dual Thermal Sensor Cable from Front End Bell.

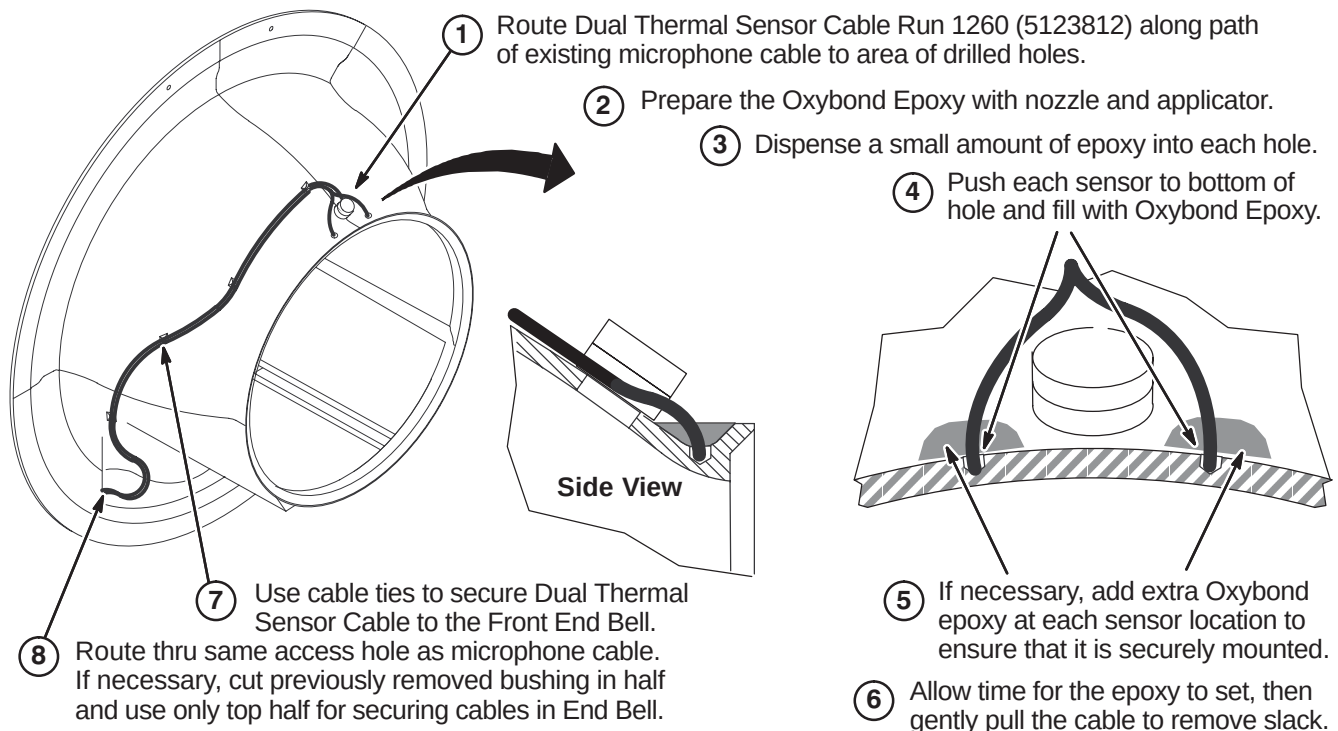
10-3-8 PREPARE END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!



10-3-9 ATTACH DUAL THERMAL SENSOR CABLE



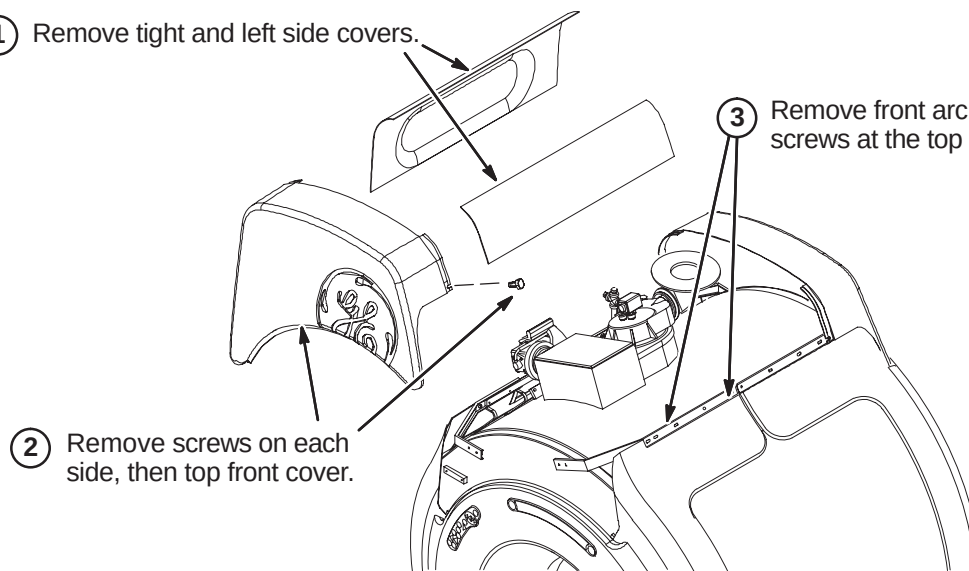
10-4 UPGRADE FOR WideOpen STYLE MAGNET ENCLOSURE

If a new Front End Bell has been delivered to the site, upgrade procedures in Sections 10-4-2, 10-4-3, and 10-4-4 will not be required. Proceed to Section 10-5.

10-4-1 FRONT END BELL REMOVAL

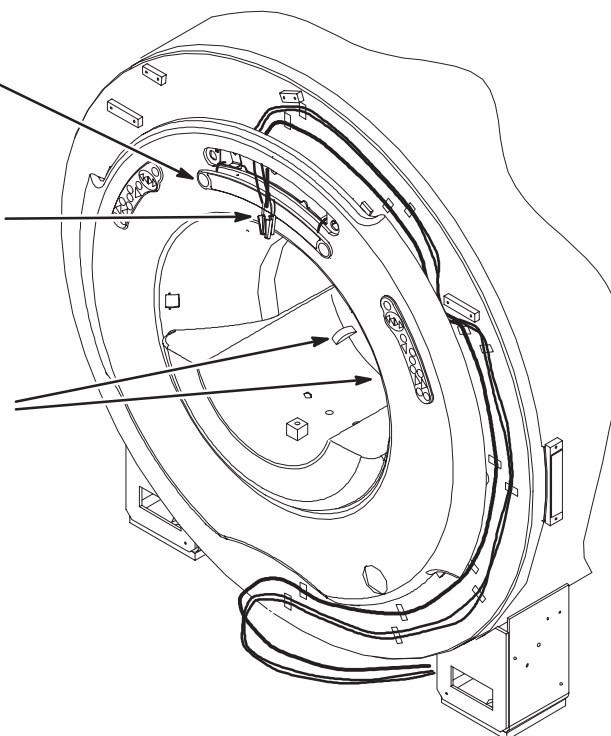
10-4-1-A Remove WideOpen Enclosure Covers

- 1 Remove tight and left side covers.



10-4-1-B Disconnect Operator Control Panel

- 1 Insert 3mm allen wrench into hole in Control Panel and turn counter clockwise 1/2 turn.
- 2 Carefully, using a small screwdriver, pry the Operator Display away from the front cover.
- 3 Disconnect the three cables attached to the Operator Display.
- 4 Lift open the two latches connecting the front end bell to the RF Coil.

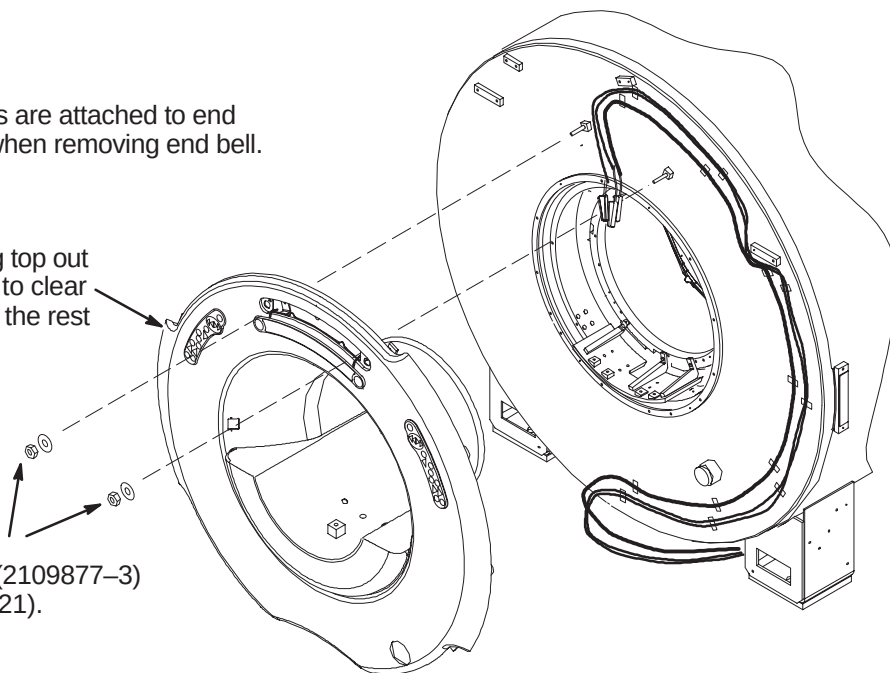


10-4-1-C Remove Front End Bell**CAUTION:**

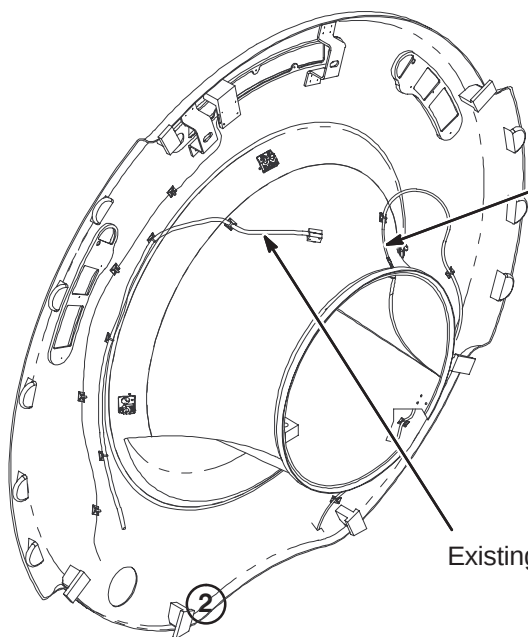
Bore Light inserts are attached to end bell. Take care when removing end bell.

- ② Remove front end bell by pulling top out past studs, then lifting vertically to clear end bell inserts, then pulling out the rest of the way.

- ① Unfasten the M10 Lock Nuts (2109877-3) and Rubber Washers (2167321).

**10-4-2 TEMPERATURE SENSOR CABLE REMOVAL**

The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



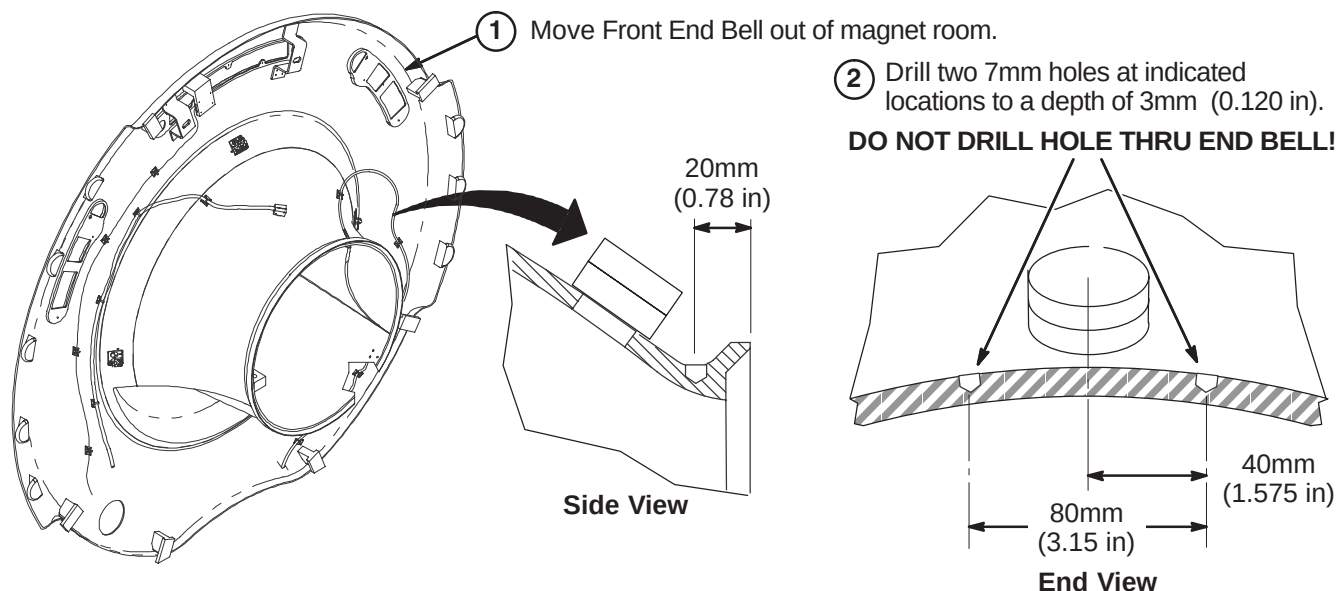
- ① Carefully remove tape that the sensor is located under.
- ② Remove the sensor by breaking the epoxy.
- ③ Pull sensor out by the cable from the felt lining.
- ④ Thread new sensor cable thru felt lining.

Existing Microphone cable

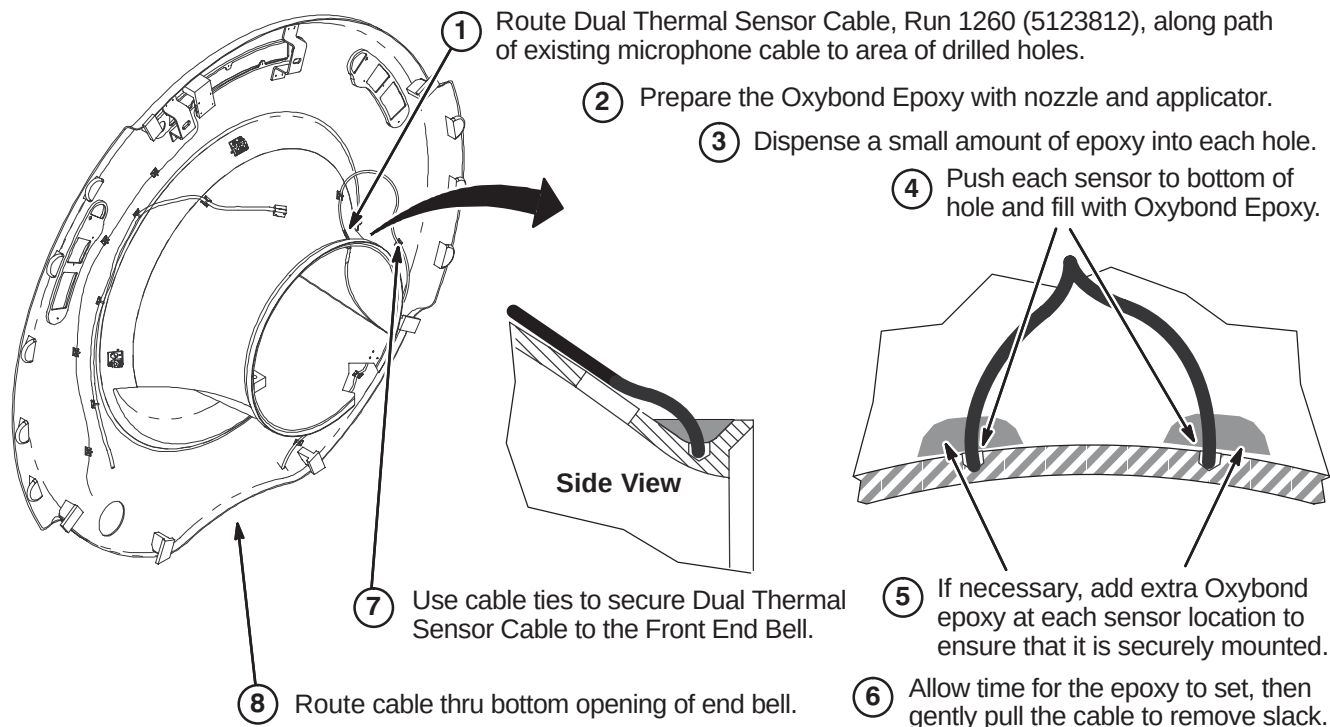
10-4-3 PREPARE END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!



10-4-4 ATTACH DUAL THERMAL SENSOR CABLE



10-5 REASSEMBLY OF MAGNET ENCLOSURE

For system replacing only the Temperature Sensor Cable:

10-5-1 HORIZON STYLE ENCLOSURE

Reverse the Front End Bell Removal procedure (Sections **10-3-6** to **10-3-2**) to reinstall the end bell, comfort module, and bridge.

10-5-2 WideOpen STYLE ENCLOSURE

Reverse the Front End Bell Removal procedure (Section **10-4-1**) to reinstall the end bell, comfort module, and bridge.

SECTION 11 – CRADLE RELEASE UPGRADE

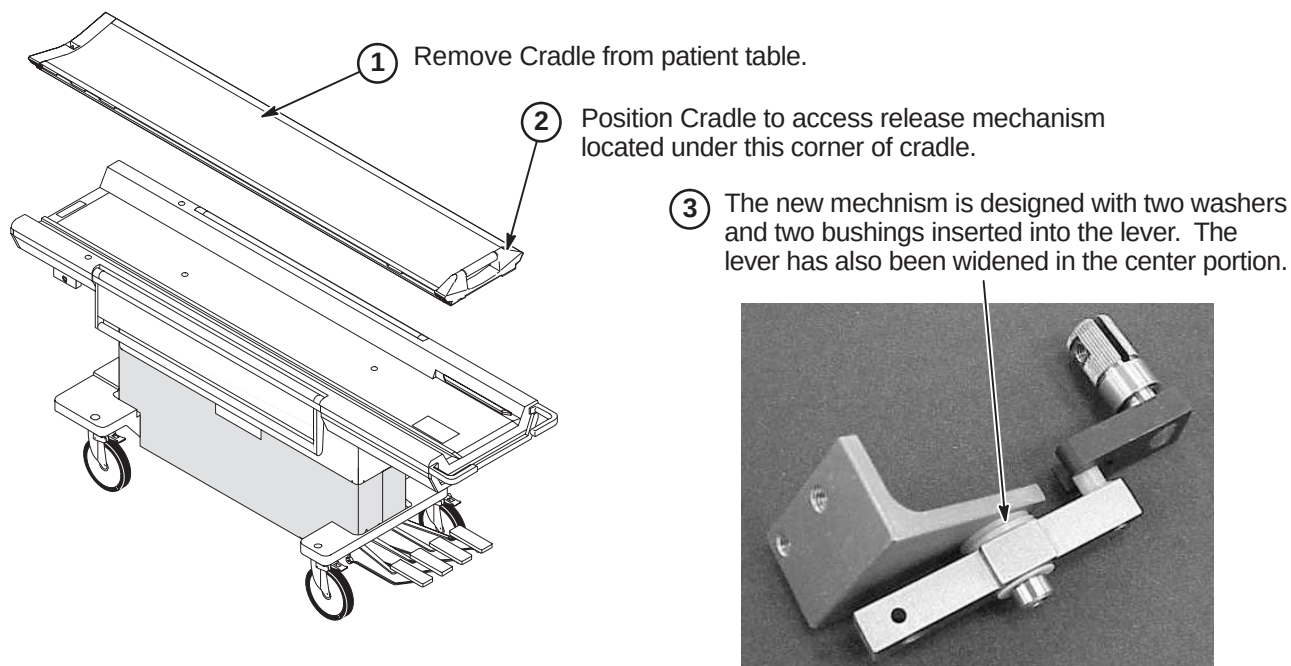
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11-1 PATIENT TABLE CRADLE RELEASE UPGRADE

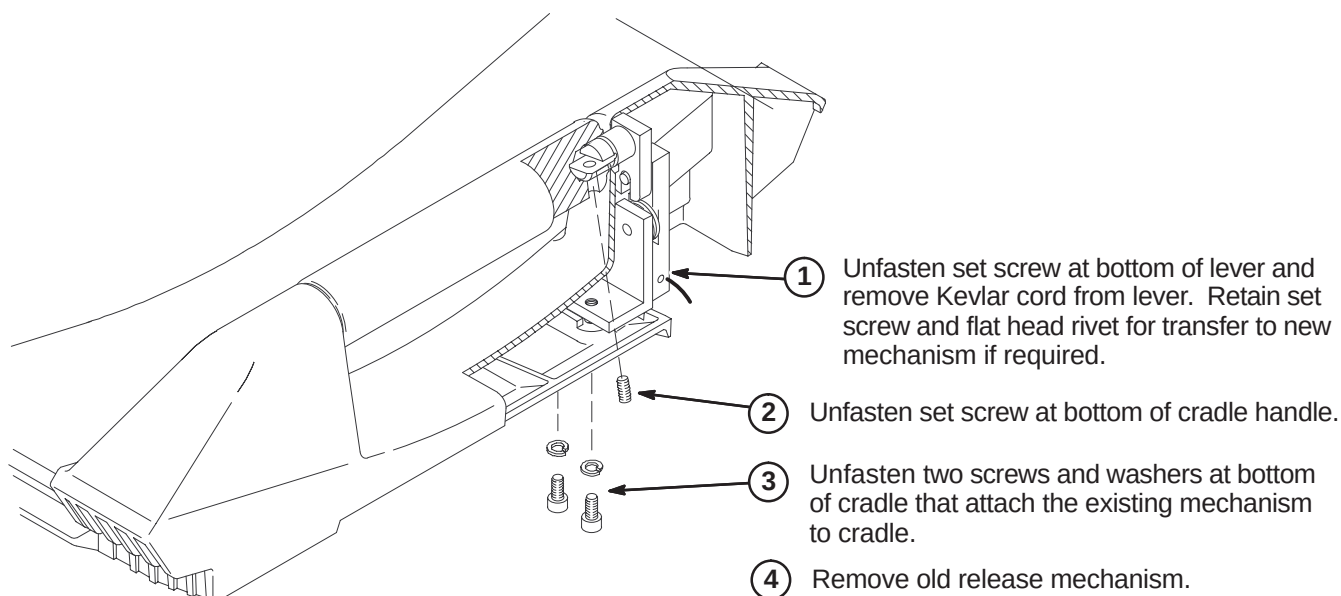
This upgrade is performed to improve the release mechanism and improve image quality. If this procedure is not completed by Mechanical Installer or Van Partner, make sure that the GE Field Engineer has been notified that this cradle release upgrade needs to be completed. For Van Partner, refer to the Design Review Checklist to report status of completion.

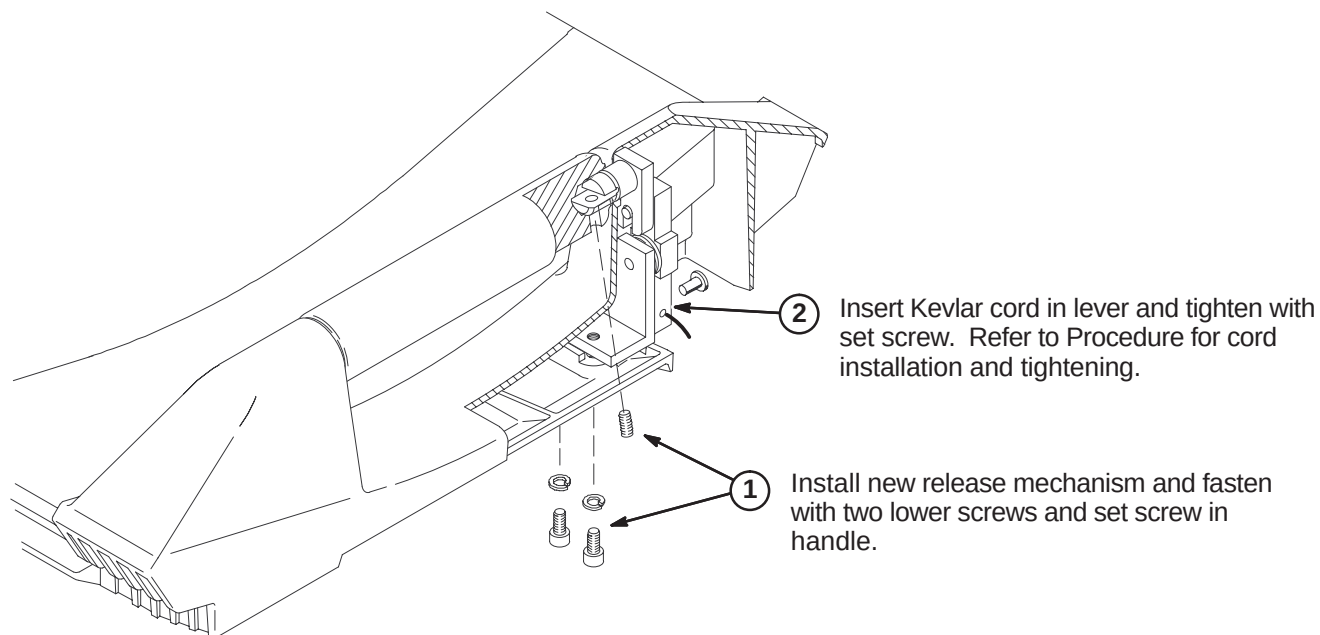
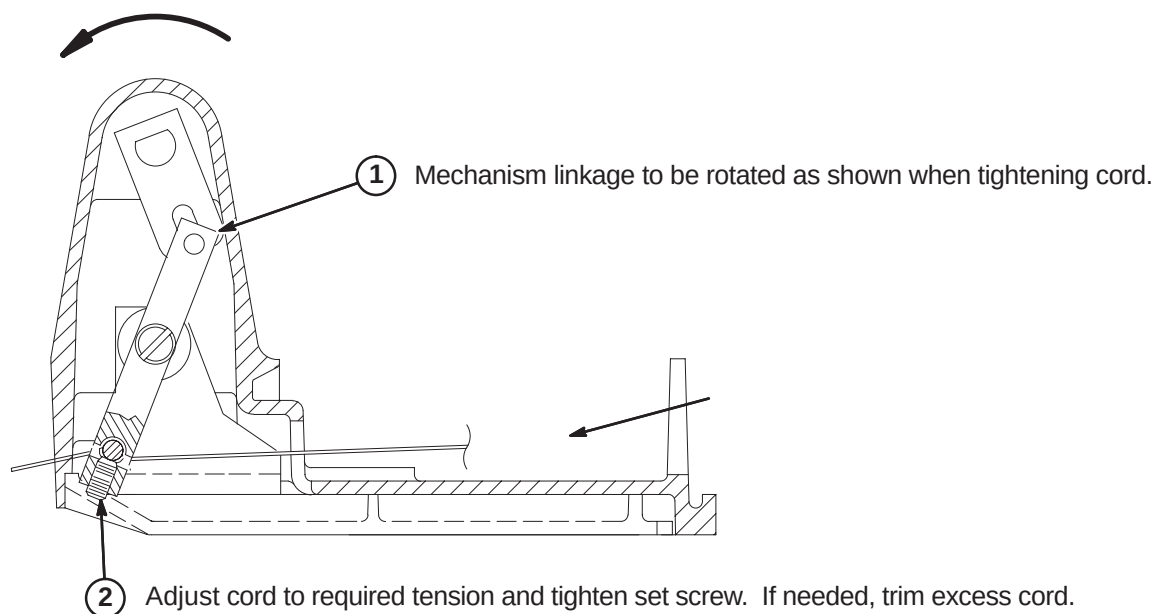
11-1-1 PREPARE CRADLE FOR UPGRADE



11-1-2 UPGRADE PROCEDURE

11-1-2-A Remove Existing Release Mechanism



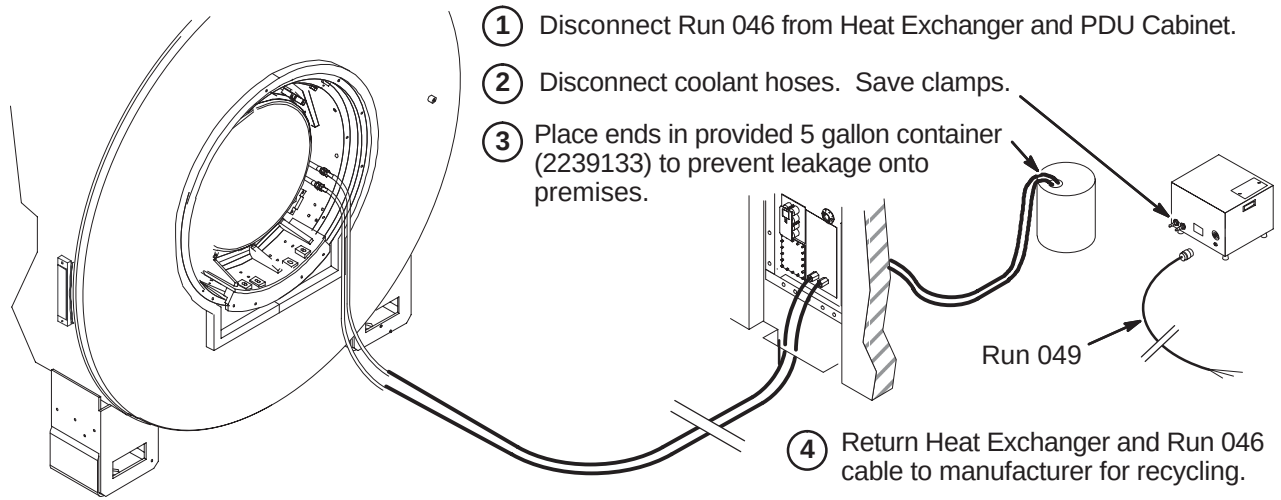
11-1-2-B Install New Release Mechanism**11-1-2-C Attach Cord**

SECTION 12 – WATER CHILLER INSTALLATION

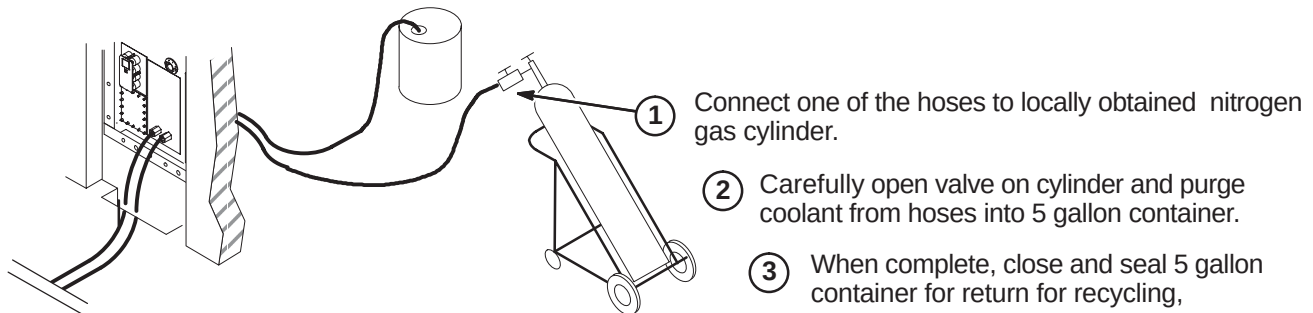
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12-1 DISCONNECT HEAT EXCHANGER



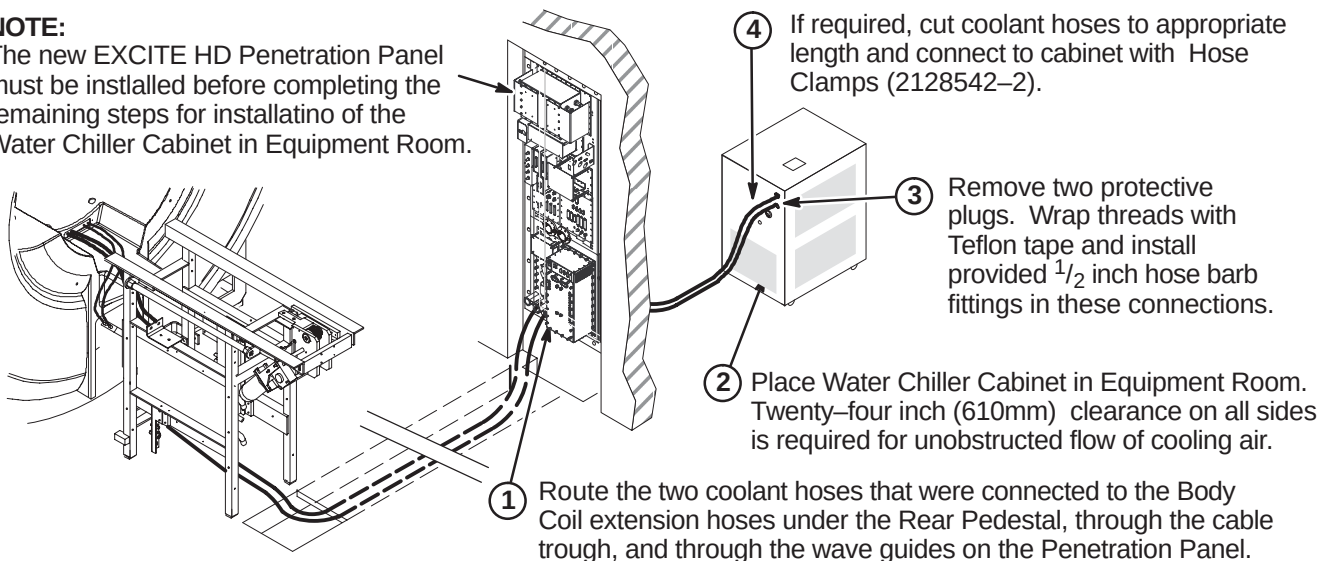
12-2 DRAIN COOLANT FROM COOLANT HOSES



12-3 POSITION WATER CHILLER CABINET AND CONNECT HOSES

NOTE:

The new EXCITE HD Penetration Panel must be installed before completing the remaining steps for installation of the Water Chiller Cabinet in Equipment Room.



12-4 INSTALL POWER CORD

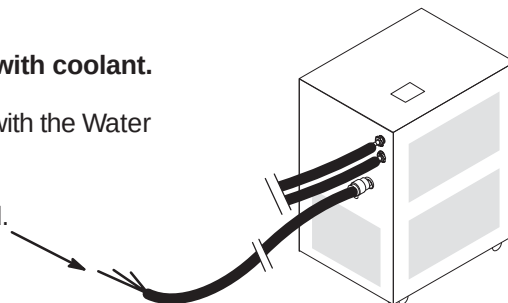
CAUTION:

Do not start Water Chiller pump before filling reservoir with coolant.

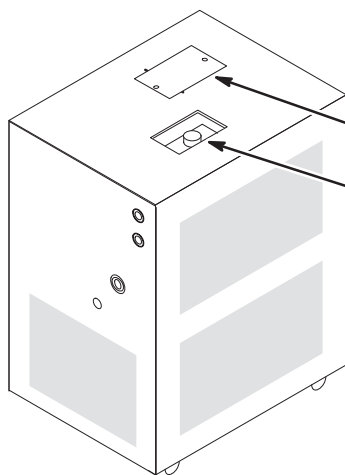
Consult vendor Water Chiller manual –

Installation, Operation, and Service Manual that is shipped with the Water Chiller.

- ① Connect power cable Run 970 (2280289) to PDU as marked.

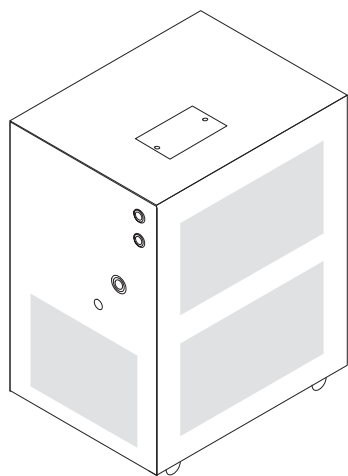


12-5 FILLING THE SYSTEM WITH COOLANT



- ① Locate the twelve (12) gallons of Body Coil Coolant (2138791) that was shipped with System
- ② Remove cover.
- ③ Consult vendor Water Chiller manual – *Installation: Filling Requirements*. Initial Filling of Reservoir, and the following steps for the proper procedure on priming and filling the system with coolant.

12-6 STARTING WATER CHILLER



- ① Consult vendor Water Chiller manual – *Operation: Start Up*. Follow instructions in manual for starting water chiller operation.
- ② Verify unit is set to control at 20° C.
- ③ If supplied, attach plastic sleeve to side of Water Chiller and insert vendor manual. This will allow easy access to any future service/maintenance procedures. If plastic sleeve is not present, attach vendor manual to Water Chiller or store it with other service manuals on site.

SECTION 13 – BRM-D & CRM BODY COIL INSTALLATION

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13-1 BODY COIL PRE-INSTALLATION INFORMATION

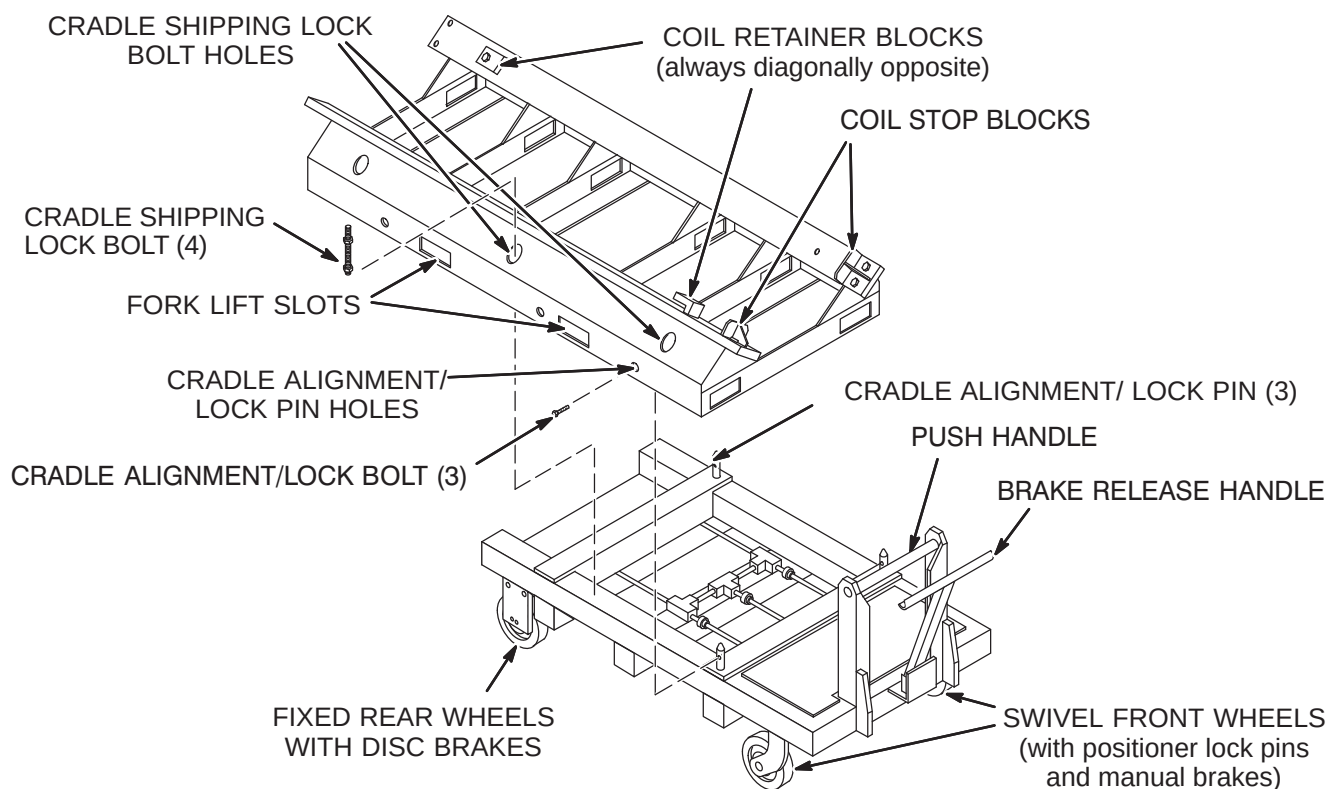
Refer to procedures in *Signa EXCITE HD 1.5T Service Methods CD-ROM*, Direction 5133315 Replacement>Coil Replacements>Gradient Coil Replacement for Non-Cx Magnets, for instructions on de-installing the existing Body Coil.

The upgrade body coils may be delivered on a cart/cradle or on a shipping pallet (cradle).

- **If body coil is delivered on a coil cart and cradle combination:**
The body coil is transported to the magnet room. Review Section 13-1-1, skip Section 13-1-2, and proceed to Section 13-1-3.
- **If body coil is delivered on a pallet (or cradle) for use with a coil cart:**
The body coil is loaded on the regional coil cart. Review Section 13-1-1 and proceed to Section 13-1-2.

13-1-1 BODY COIL CART AND CRADLE COMPONENTS

The Coil Cart and Cradle combination, as shown below is used to install the Body Coil in the magnet bore. The body coil may also be shipped on a disposable wood and metal pallet which performs the functions of the metal cradle shown below.



13-1-2 LOADING BODY COIL AND CRADLE OR PALLET ON COIL CART

- ① Have a fork lift truck raise the body coil and cradle (or pallet) combination. The gradient leads are at the end of the cradle away from the handle, and the Coil Stop Blocks are near the handle.

CAUTION

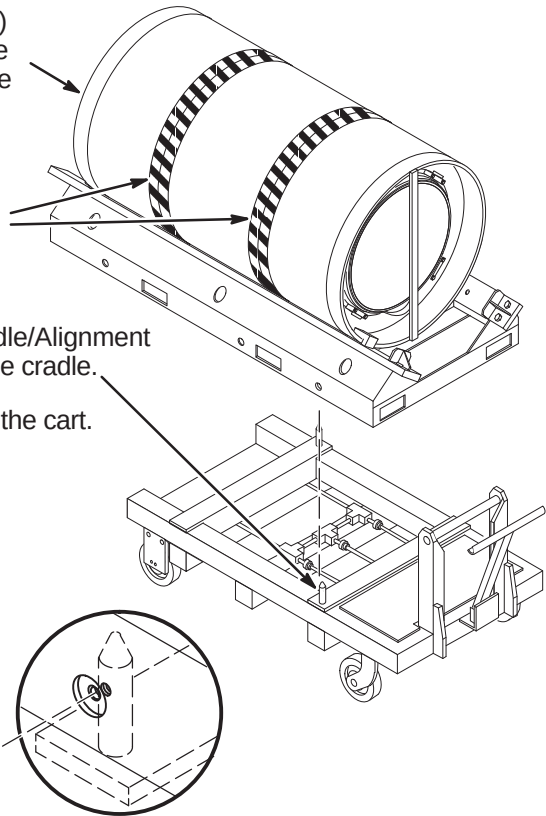
Do not lift or apply pressure to area of Body Coil marked with black/yellow tape labelled "NO LOAD ZONE". Doing so will damage the Body Coil.

- ② Position cart beneath cradle with Cradle/Alignment Lock Pins aligned with their holes in the cradle.
- ③ Lower body coil cradle (or pallet) onto the cart.

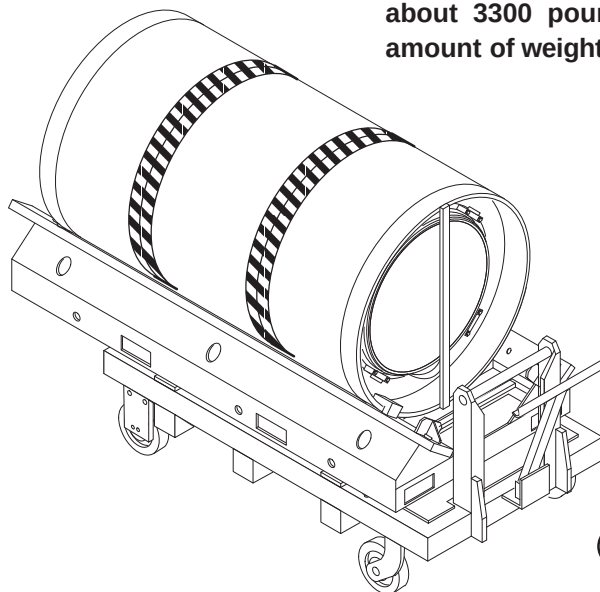
CAUTION

Not having the cradle anchored to the cart could cause the coil/cradle to tip. The three Cradle Alignment/Lock Bolts **MUST** be in place before attempting to move the coil cart.

- ④ Insert the three Cradle Alignment/Lock Bolts.

**13-1-3 USING CART TO TRANSPORT BODY COIL****CAUTION**

Avoid possible bodily injury. The combined coil/cradle/cart weigh about 3300 pounds. To avoid physical injury, and to control this amount of weight as it moves, a minimum of four people must be used.



- ① Check route to magnet room and verify route clearance and sufficient manpower to overcome inclines.
- ② Place plywood sheets or metal plates along route to distribute weight and protect floor surface from damage.

Note: The Brake Release Handle controls the disc brakes on the fixed rear wheels. It must be pushed against the Push Handle in order to move the cart.

- ③ Roll coil cart to outside of magnet room for connection of 10 foot (305cm) Gradient extension cables.

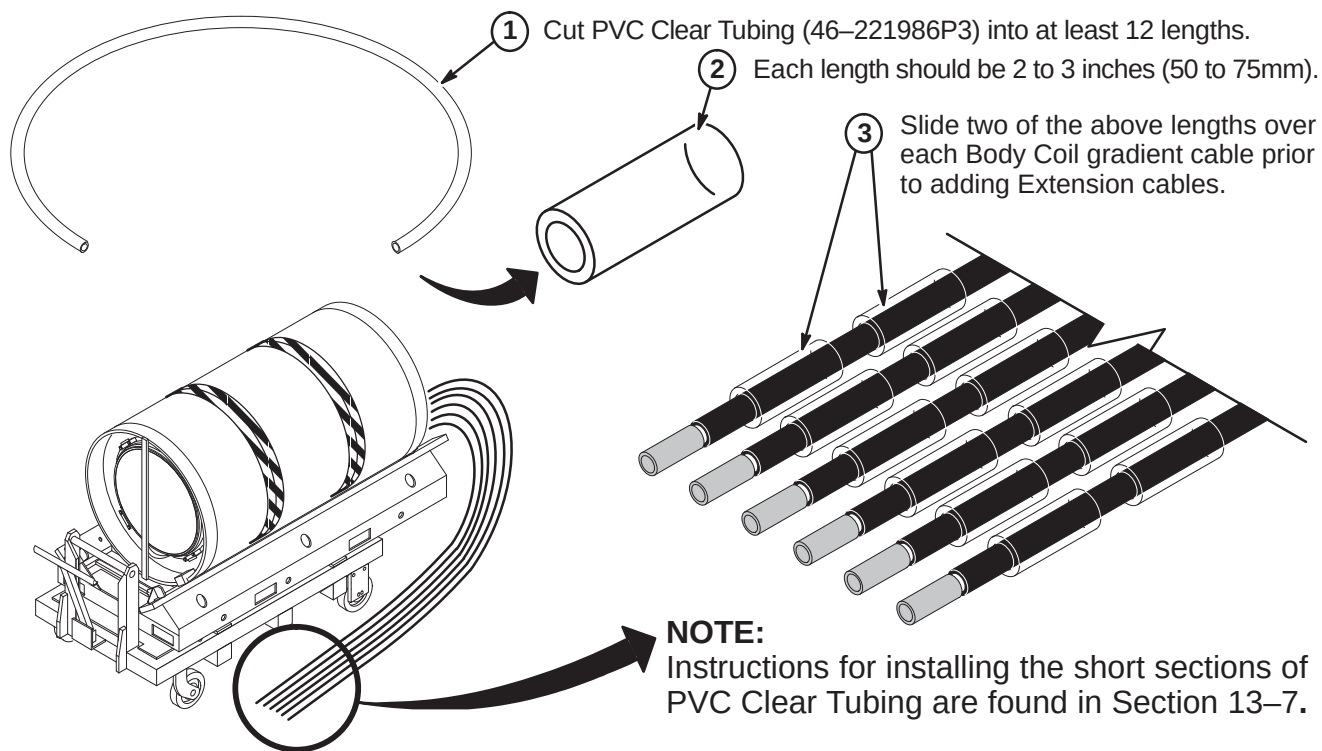
13-2 EXTENSION CABLE INSTALLATION FOR NON-ACTIVELY SHIELDED MAGNETS

WARNING!

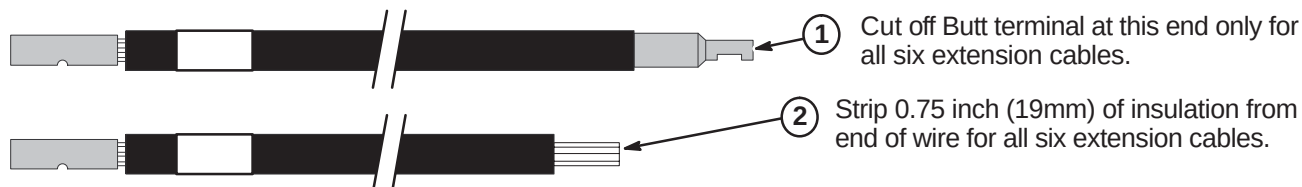
FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – ATTACHING THE 2119600 SERIES EXTENSION CABLES TO THE BODY COIL GRADIENT CABLES MUST BE COMPLETED OUTSIDE THE MAGNET ROOM.

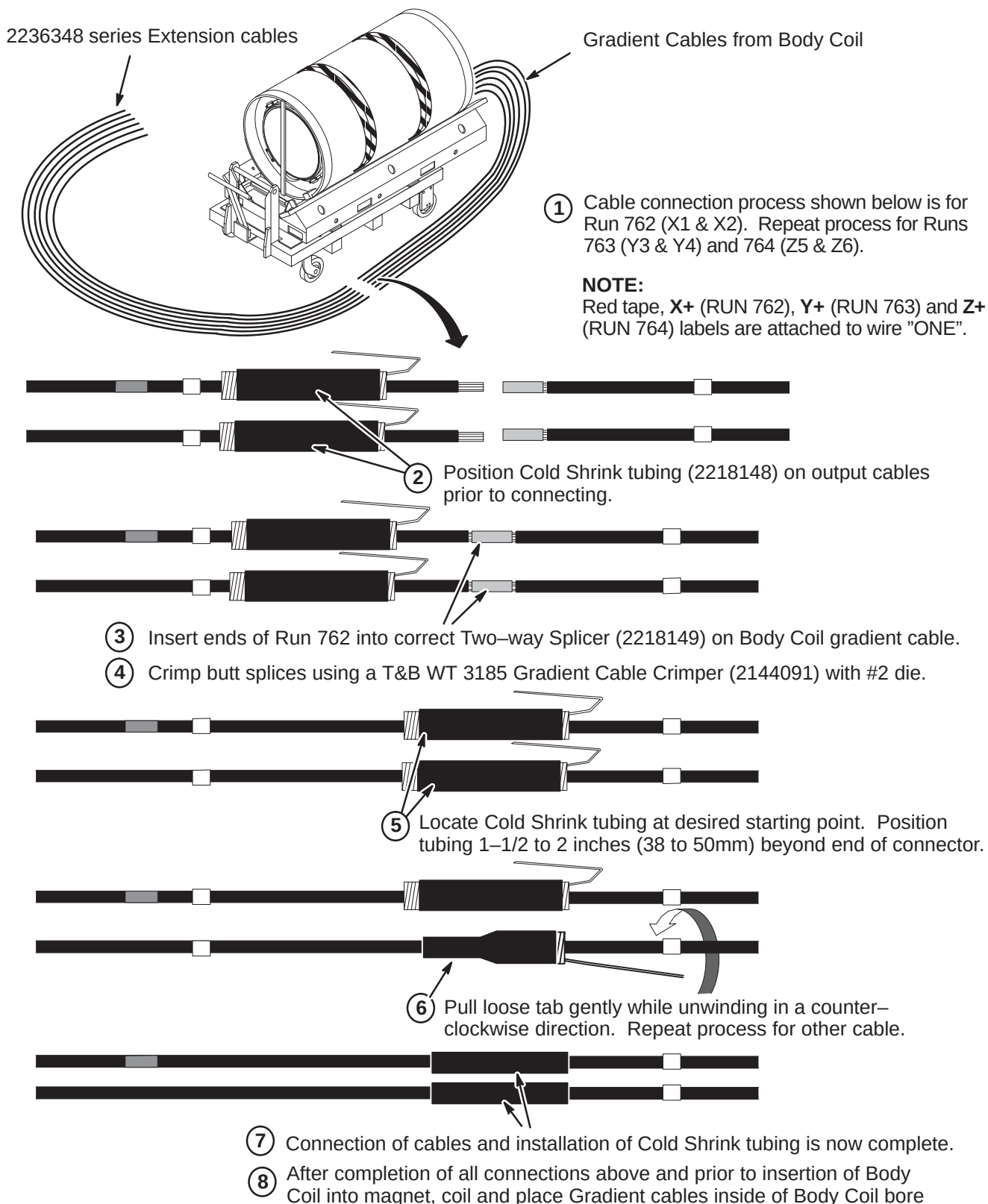
13-2-1 PRE-INSTALLATION OF PVC CLEAR TUBING



13-2-2 CABLE TERMINATIONS FOR GRADIENT EXTENSION CABLES

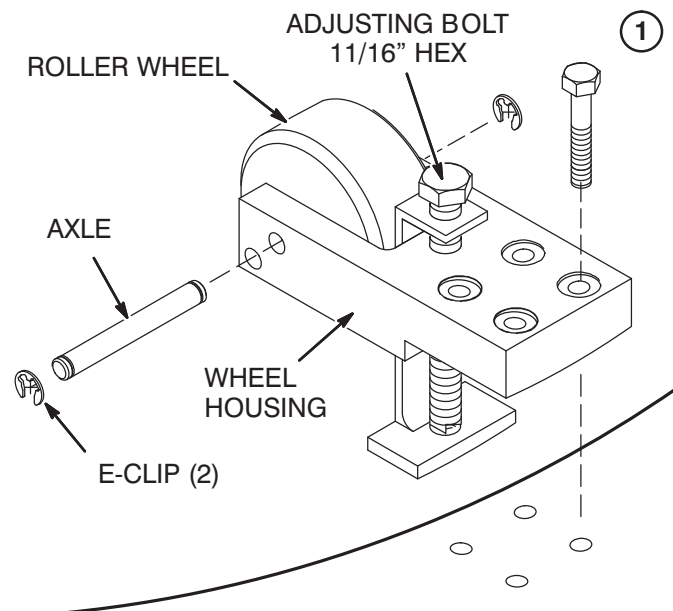


13-2-3 CONNECTION OF BODY COIL GRADIENT CABLES TO EXTENSION CABLES



13-3 INSTALL BODY COIL IN MAGNET BORE

13-3-1 INSTALLATION OF ROLLER WHEEL ASSEMBLIES



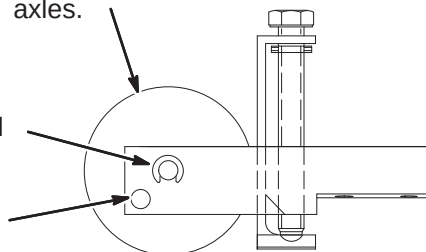
- ① Remove the two Coil Retainer Blocks (used to secure the coil to the cradle or pallet during shipment).

Note: The following steps are used to mount the four Roller Wheel Assemblies to the coil, two at each end.

- ② Back the adjusting bolt all the way out by turning it counterclockwise.
- ③ Remove one of the e-clips and remove wheel and axle.
- ④ Place Wheel Housing in position over the four threaded holes on the coil and fasten with four bolts.
- ⑤ Turn the adjusting bolt clockwise on all four Roller Wheel Assemblies to raise the coil to the point that the wheels and axles can be installed.
- ⑥ Note which position to use and install wheels and axles.

Use upper hole for **GE** magnets with passive or resistive shim and **Oxford** magnet.

Use lower hole for **GE** magnets without passive or resistive shim.

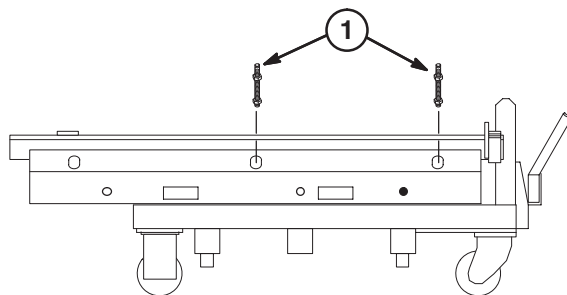
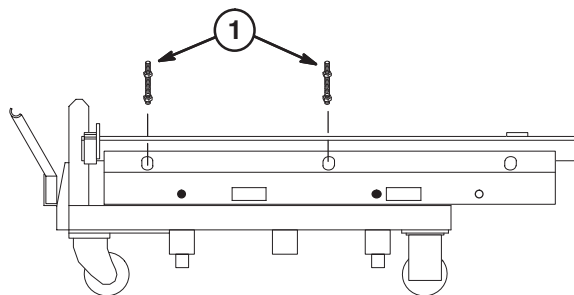


13-3-2 REMOVE CRADLE SHIPPING LOCKING BOLTS

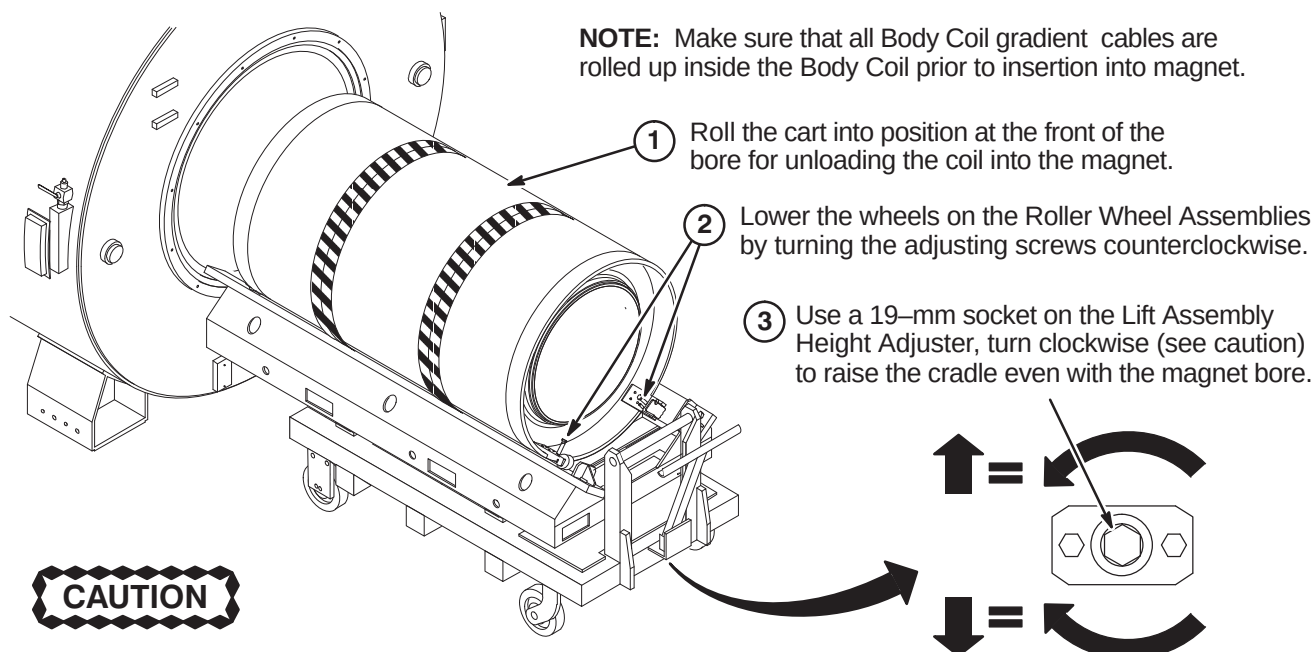
CAUTION

Attempting to raise the body coil with cradle shipping lock bolts installed will damage the elevating mechanism. The four Lock Bolts **MUST** be removed before attempting to raise the coil cart cradle.

- ① Remove the four Cradle Shipping Lock Bolts from the cart (two bolts on each side). This must be done before the Lift Assembly can be used to raise the cradle. These bolts belong with the cart; do not misplace them.



13-3-3 POSITION BODY COIL AT FRONT OF MAGNET BORE



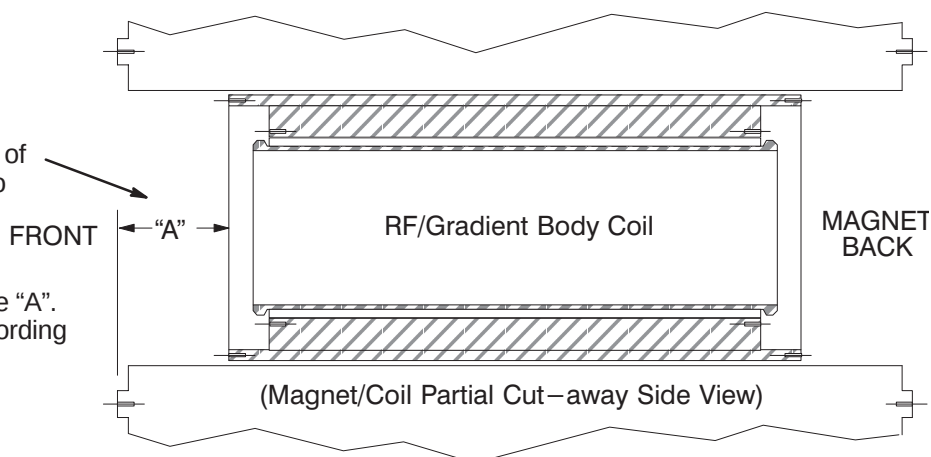
CAUTION

EQUIPMENT DAMAGE POSSIBILITY

Early carts had left-handed threads on the Lift Assembly Height Adjuster, i.e., clockwise rotation lowered the assembly, counterclockwise raised it. Gradually, these carts are being modified for right-handed threads. Be sure to check the cart that you are about to use. Damage to the mechanism can occur if the bolt is turned the wrong way with the full weight of a coil and cradle on it.

13-3-4 LOCATE POSITION OF BODY COIL IN MAGNET BORE

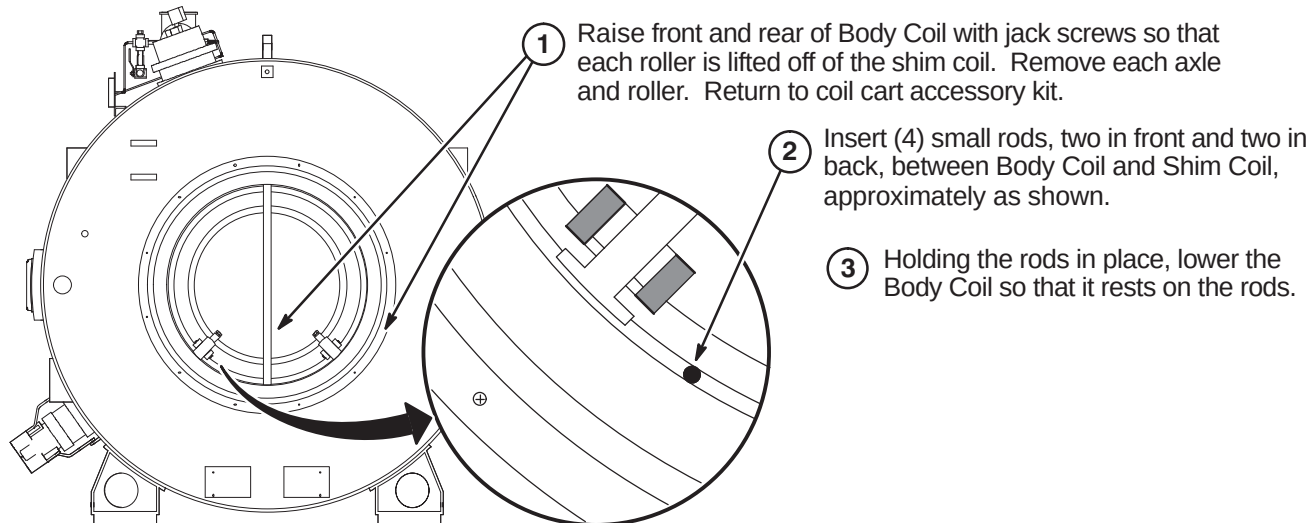
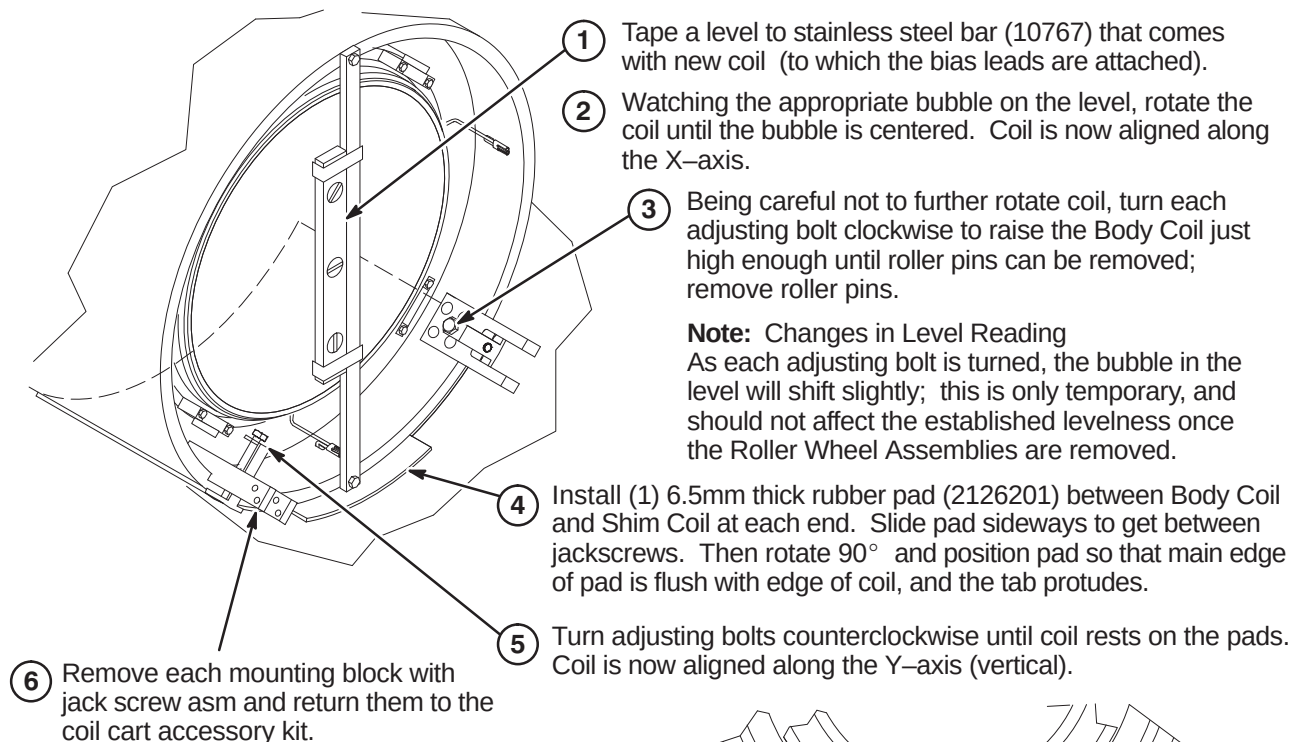
- 1 Slowly and carefully roll the Body Coil into the magnet bore.
Do not roll Body Coil past rear end of magnet bore.
- 2 Measure distance from surface of front Magnet mounting flange to edge of outer gradient coil.
(see Suggestion below)
- 3 Refer to table below for distance "A". Select the correct distance according to Magnet type.



Body Coil	BRM-D	CRM
"A" Distance	12.00 Inch (304.5mm)	10.43 Inch (264.5mm)
TOLERANCE: +0.0mm, -1.5mm		

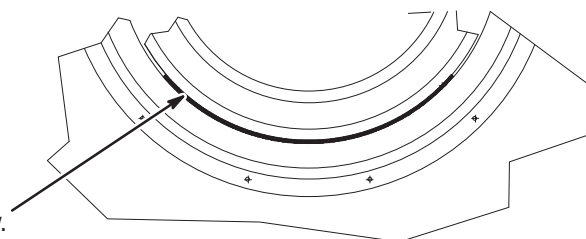
13-4 ALIGN BODY COIL IN MAGNET BORE WITH SHIM COIL

Perform this procedure if site has Shim Coil. If site does not have shim coil, go to Section **13-5**.

13-4-1 ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET WITH SHIM COIL**13-4-2 CENTERING BODY COIL IN GE MAGNET BORE WITH SHIM COIL**

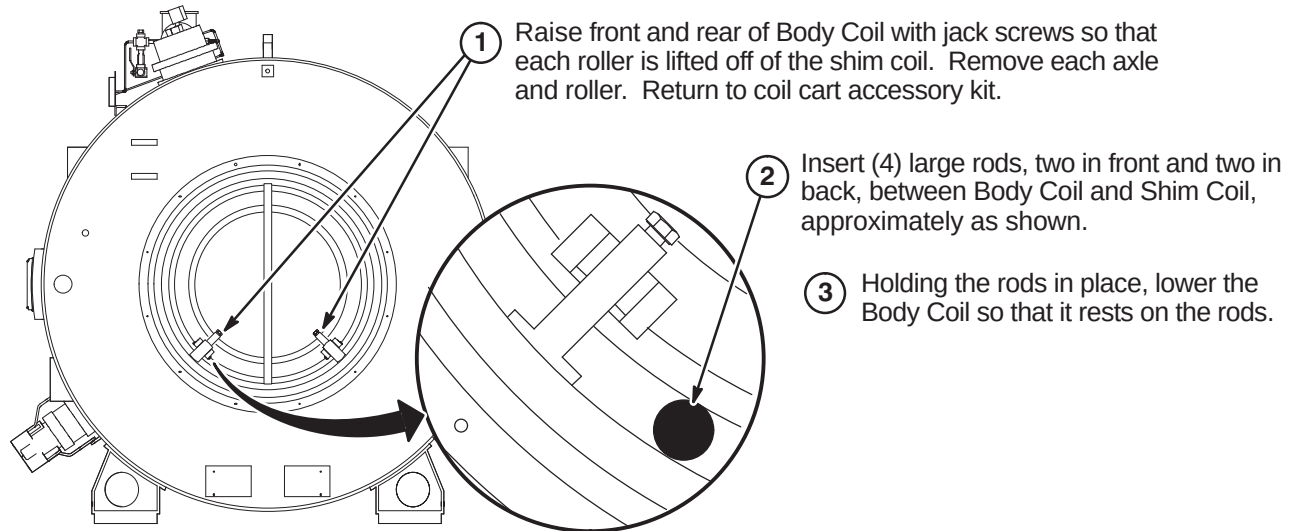
Note: Distance between top of Body Coil and top of Magnet bore should be equal to distance from bottom of Body Coil to bottom of Magnet bore.

Typical front and rear view.

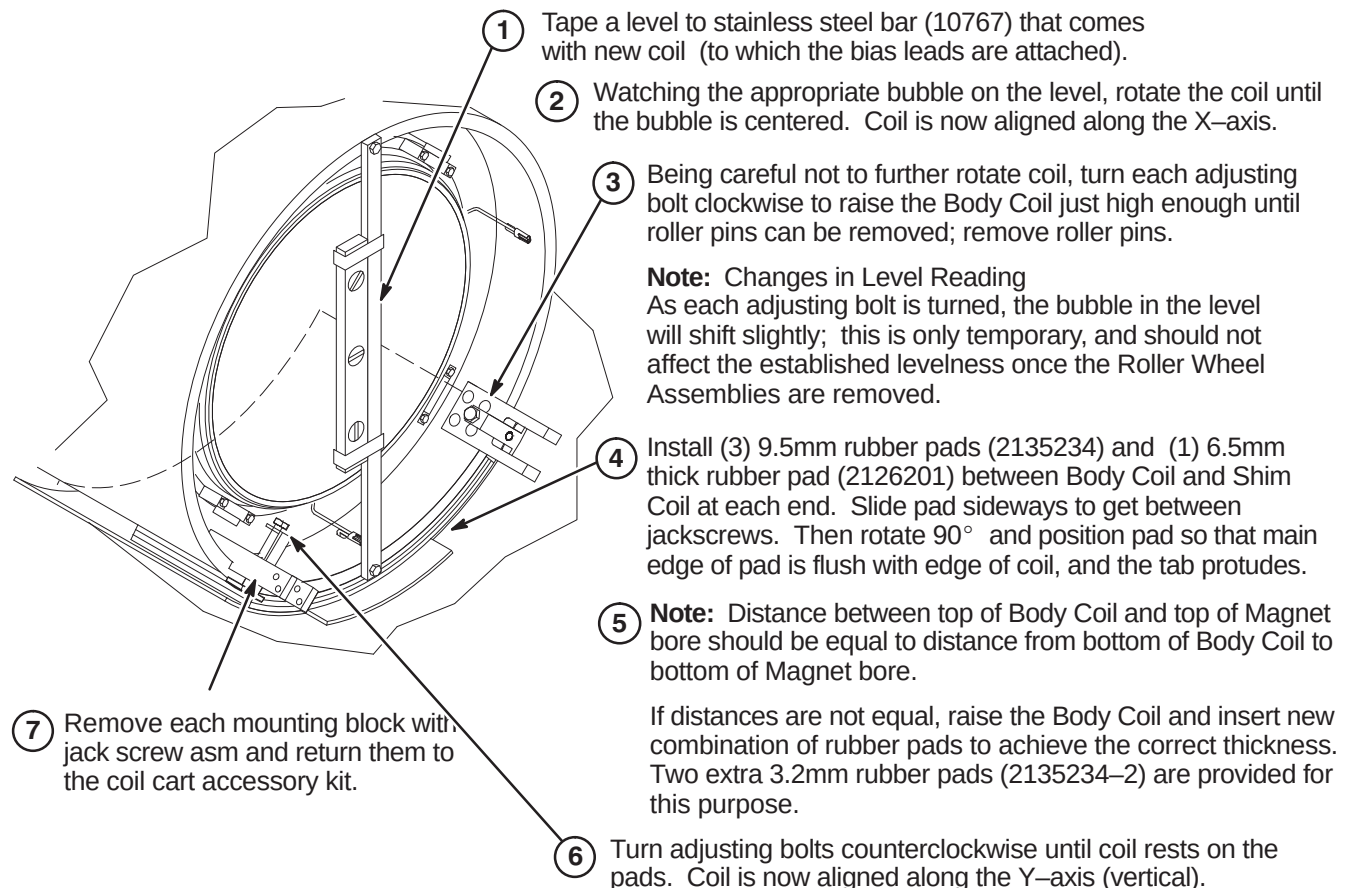


13-5 ALIGN BODY COIL IN MAGNET BORE WITHOUT SHIM COIL

13-5-1 ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET WITHOUT SHIM COIL



13-5-2 CENTERING BODY COIL IN GE MAGNET BORE WITHOUT SHIM COIL

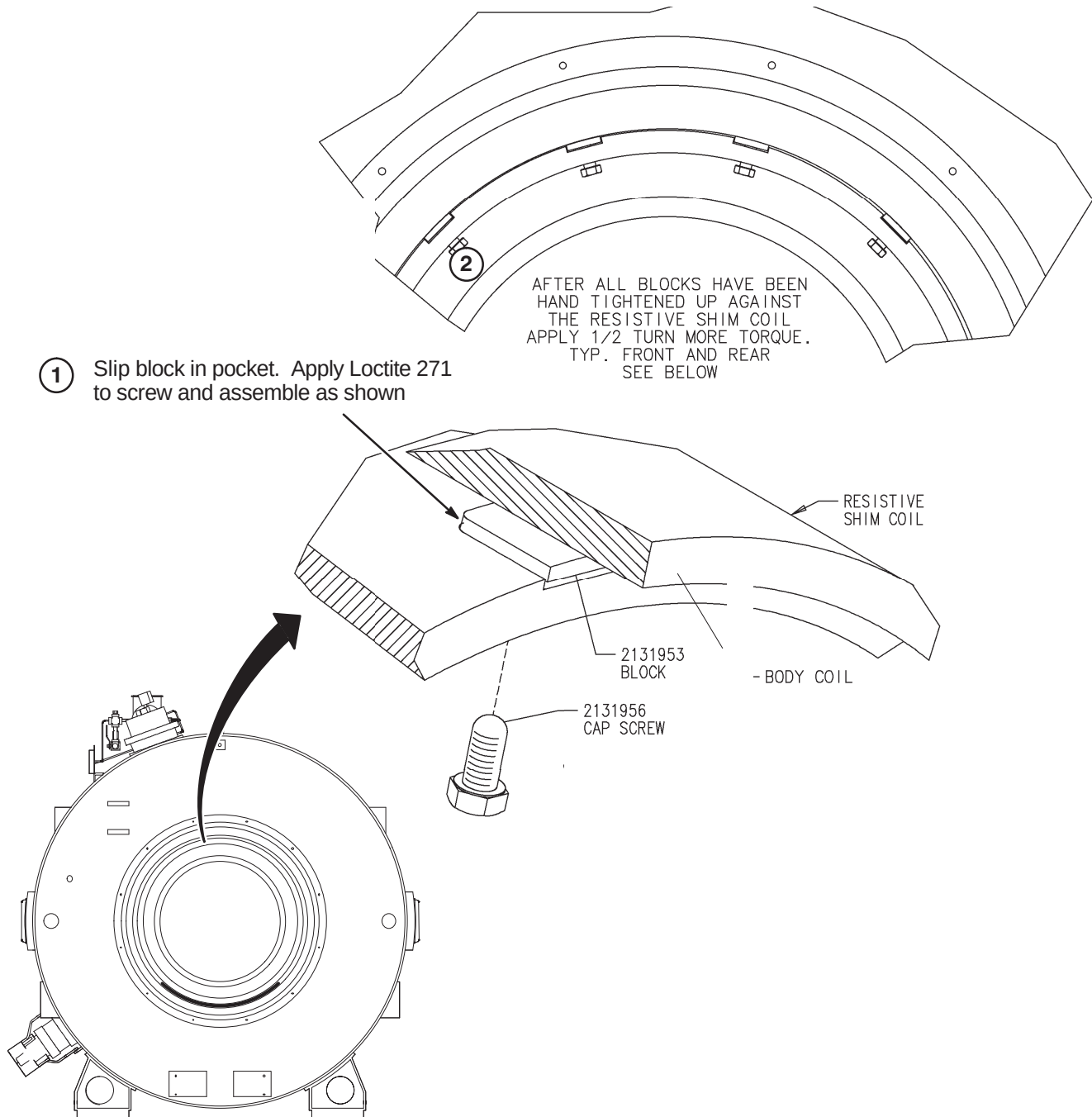


13-6 INSTALL RADIAL SUPPORT

Depending on site configuration, perform one of the following procedures.

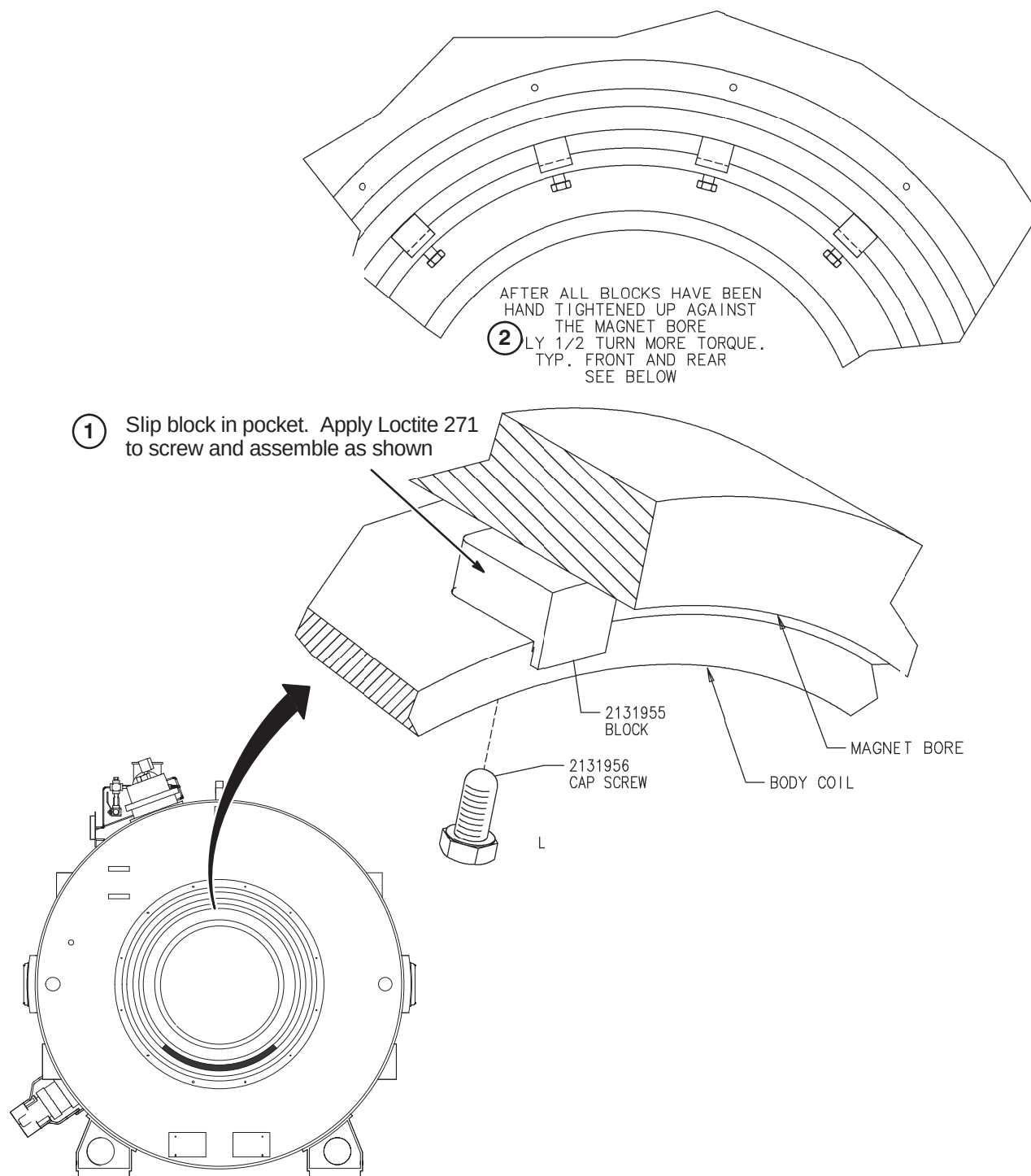
13-6-1 INSTALL RADIAL SUPPORT IN GE MAGNET WITH 900mm DIAMETER BORE

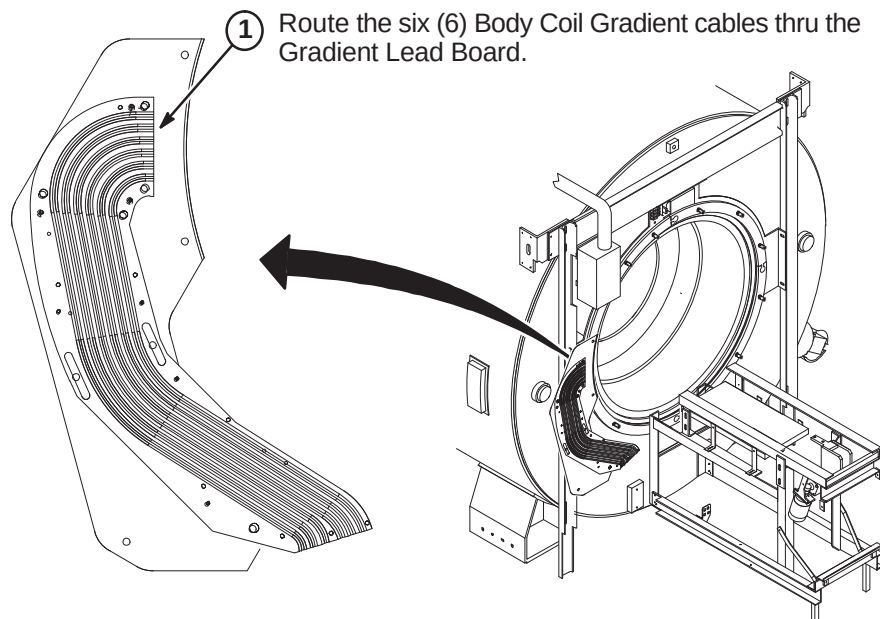
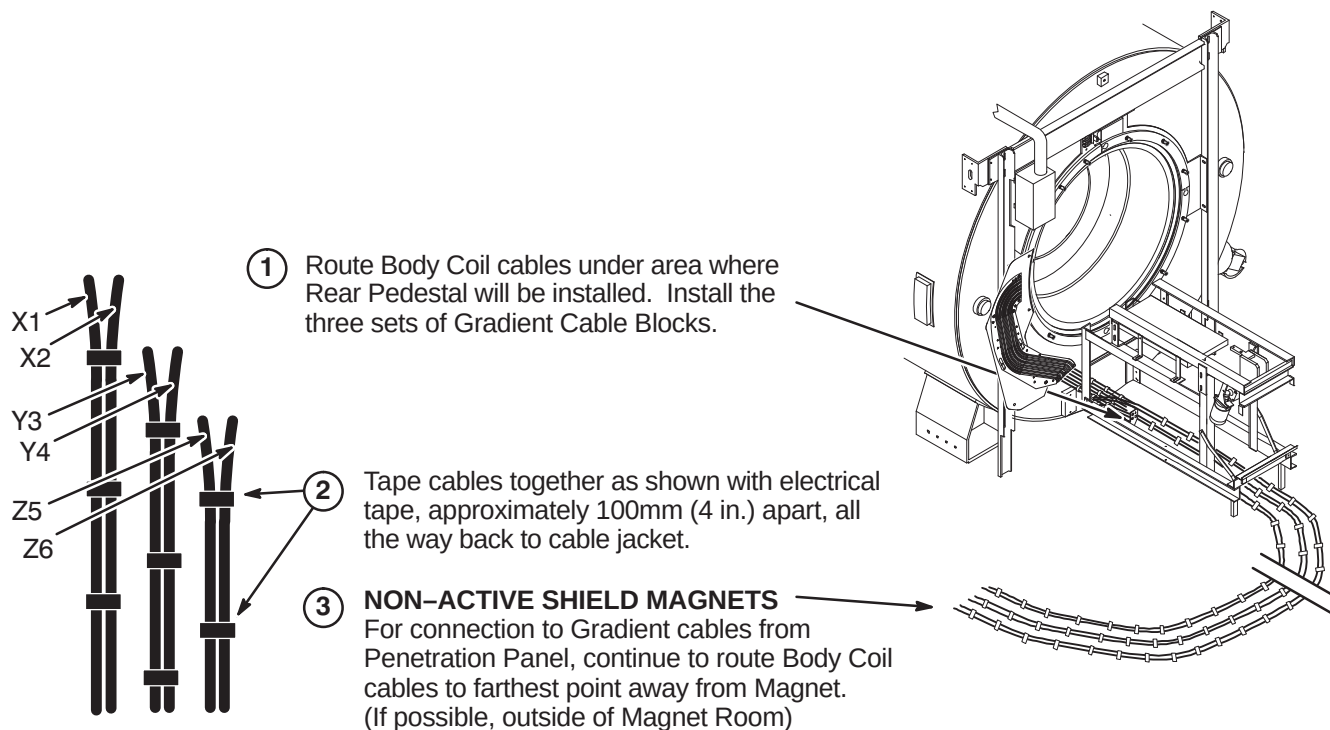
Refer to IIQ Kit (M3090NE) for parts and instructions for installing the radial supports.



13-6-2 INSTALL RADIAL SUPPORT IN GE MAGNET WITH 950mm DIAMETER BORE

Refer to IIQ Kit (M3090NE) for parts and instructions for installing the radial supports.



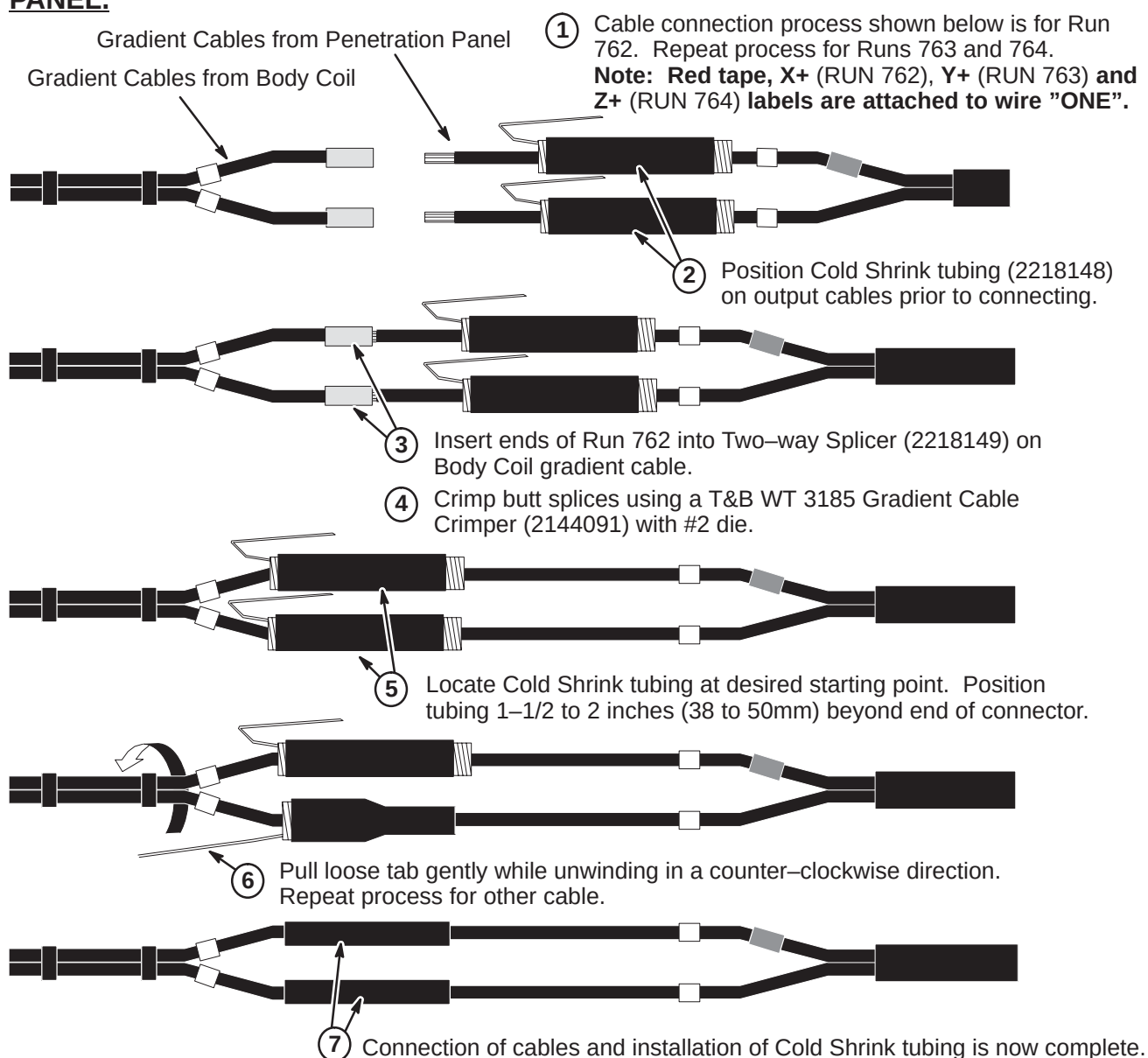
13-7 ROUTE AND INSTALL GRADIENT CABLES IN LEAD BOARD**13-8 SECURE X, Y, AND Z GRADIENT CABLES**

13-9 CONNECTION OF BODY COIL GRADIENT CABLES TO RUNS 762, 763, & 764

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

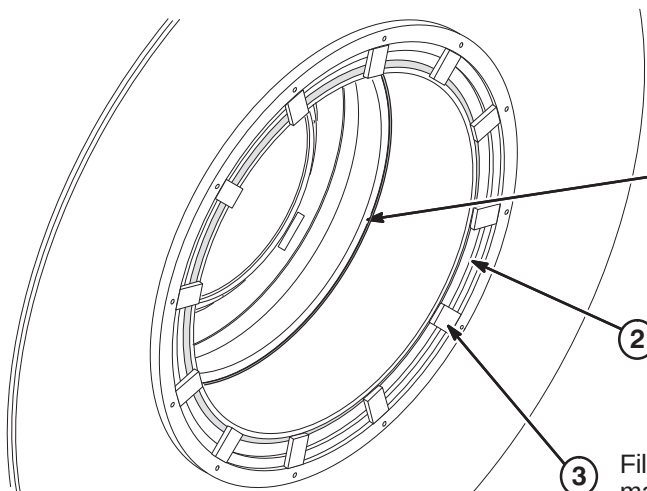
IF MAGNET IS NON-ACTIVE SHIELD AND IS AT FULL FIELD – MOVE END OF BODY COIL EXTENSION CABLES TO FARTHEST POINT AWAY FROM MAGNET TO PERFORM BUTT SPLICE CONNECTION TO RUNS 762, 763, AND 764 COMING FROM PENETRATION PANEL.



13-10 INSTALL BODY COIL INSULATION

Depending on site configuration, perform one of the following procedures.

13-10-1 INSTALLING INSULATION IN MAGNET WITH PASSIVE OR RESISTIVE SHIM

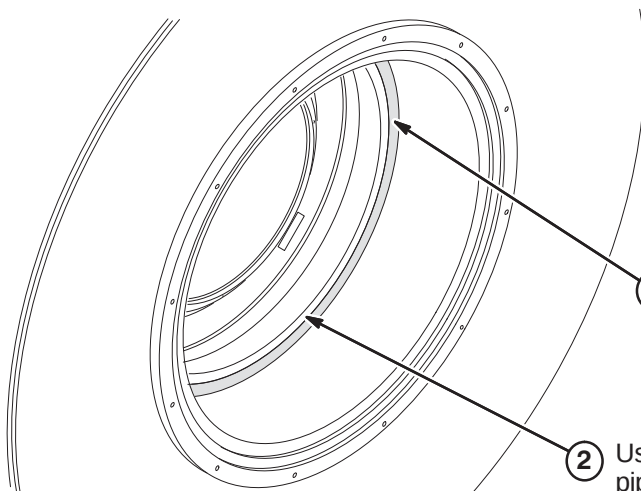


Note:

Rope caulk, pipe insulation and acoustical collar are installed after body coil centering.

- ① Fill the gap between the Passive Shim Coil and outer Gradient Coil at each end. Use one, two, or three strips (rolled together) of rope caulk (46-287118P1) around entire perimeter.
- ② Fill the gap between the Magnet bore and the Passive Shim Coil at each end. Use 8 in. (205 mm) sections of Pipe Insulation (46-320868P1) around entire perimeter.
- ③ Fill the gap between the Passive Shim drum stops and magnet bore. Use rope caulk (46-287118P1) to provide an air-tight seal around entire perimeter at the back of the magnet. This is to assure maximum air flow between the RF coil and the Gradient Coil.

13-10-2 INSTALLING INSULATION IN MAGNET WITHOUT PASSIVE OR RESISTIVE SHIM



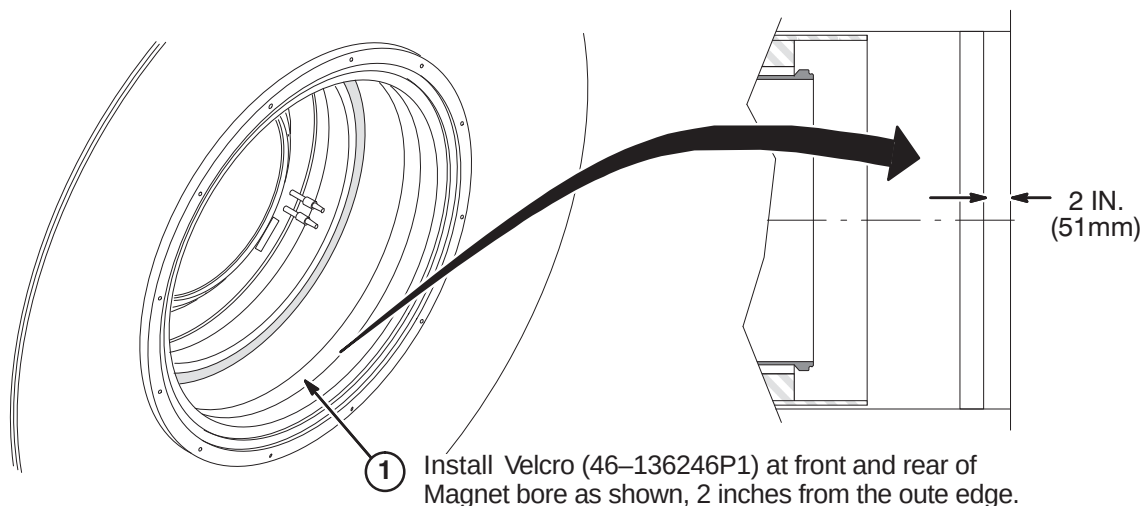
Note:

Rope caulk, pipe insulation, and acoustical collars are installed after body coil centering.

- ① Fill the gap between the Magnet bore and outside of Gradient Coil at each end. Use Pipe Insulation (46-320868P1) and rope caulk (46-287118P1) around entire perimeter.
- ② Use rope caulk (46-287118P1) as needed to supplement pipe insulation required for an air-tight seal between gradient coil and magnet bore at back of magnet. This is to assure maximum cooling air flow between the RF coil and the Gradient Coil.

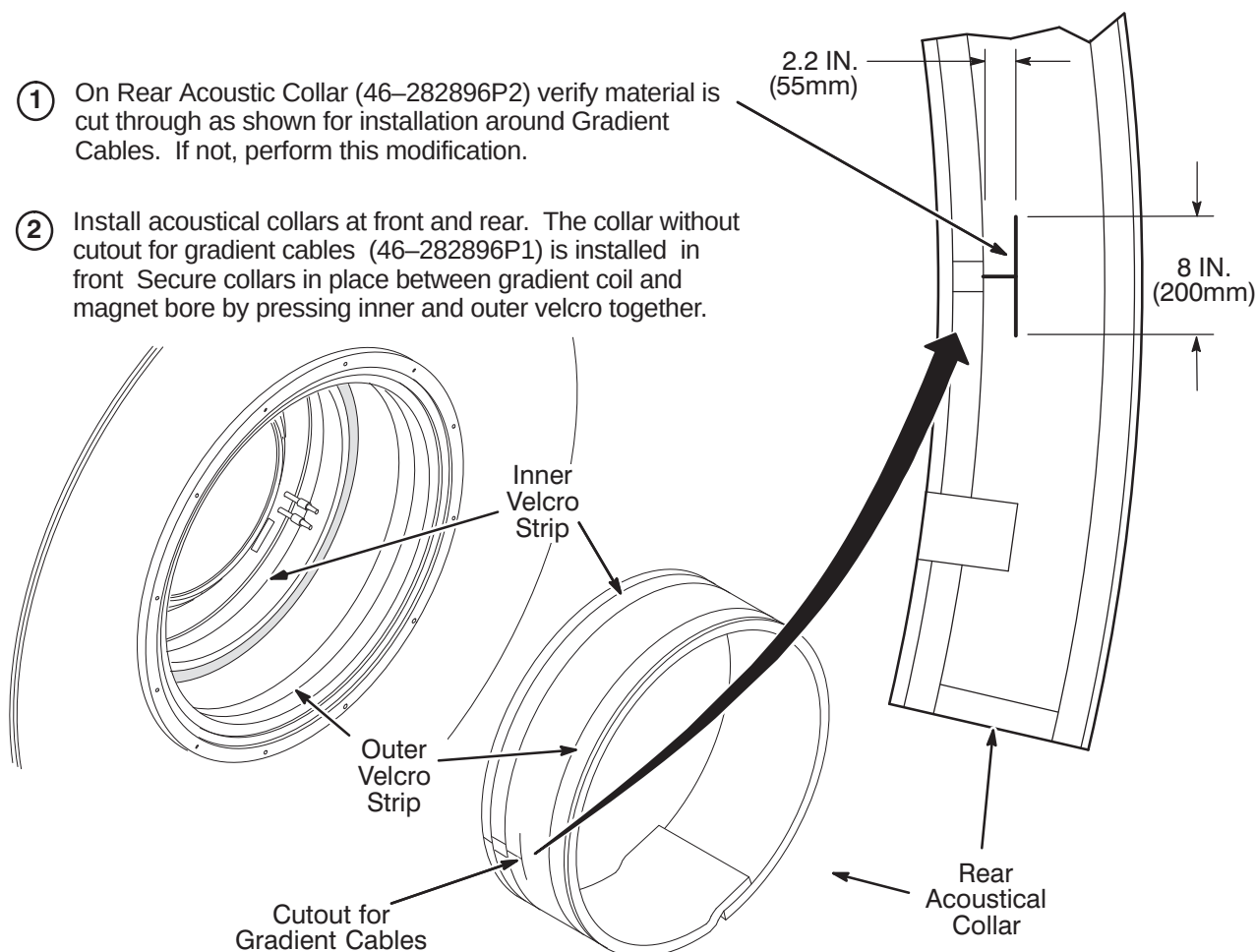
13-11 INSTALL ACOUSTIC COLLAR

13-11-1 INSTALLATION OF VELCRO IN MAGNET BORE



13-11-2 ATTACH ACOUSTIC COLLAR

- ① On Rear Acoustic Collar (46-282896P2) verify material is cut through as shown for installation around Gradient Cables. If not, perform this modification.
- ② Install acoustical collars at front and rear. The collar without cutout for gradient cables (46-282896P1) is installed in front. Secure collars in place between gradient coil and magnet bore by pressing inner and outer velcro together.



SECTION 14 – OPERATOR WORKSPACE

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14-1 INTRODUCTION



POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

14-1-1 DE-INSTALLATION INFORMATION

The equipment de-installed for the HDx upgrade depends on the existing site configuration. The general guideline for removal of equipment is:

- Advantage 4.x – De-install Operator Console and return for recycling
- Advantage 5.x or Horizon 5.x – De-install Operator Console and Computer.
- LX with Indigo – De-install modules on Operator Workspace table and Computer cabinet.
- LX with Octane – De-install indicated modules on Operator Workspace table and Computer cabinet.
- DAT and/or MOD drives are not supported in HDxt. Refer to Section 14-3 for legacy media support.

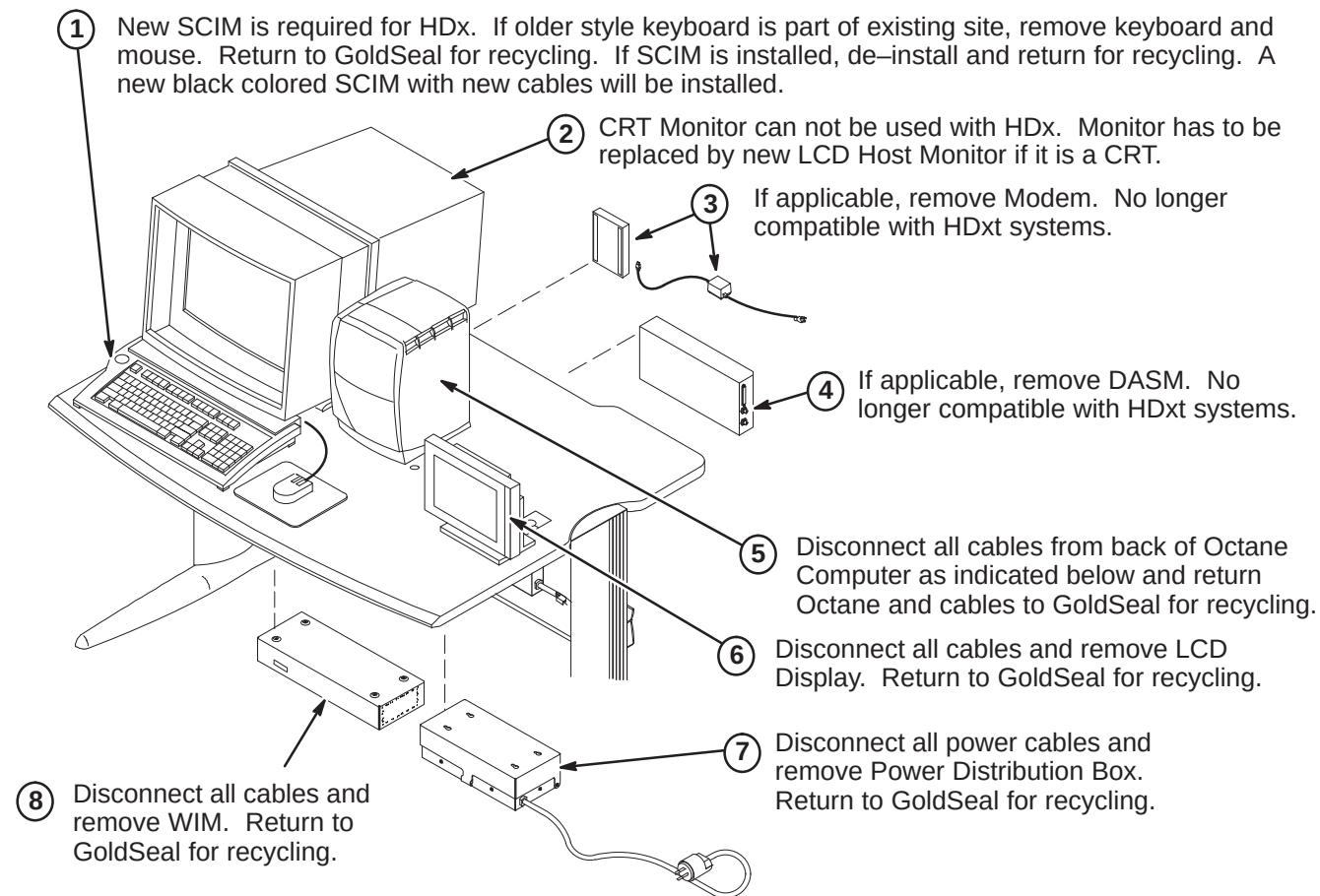
Refer to applicable de-installation instructions in next section for removal of obsolete equipment.

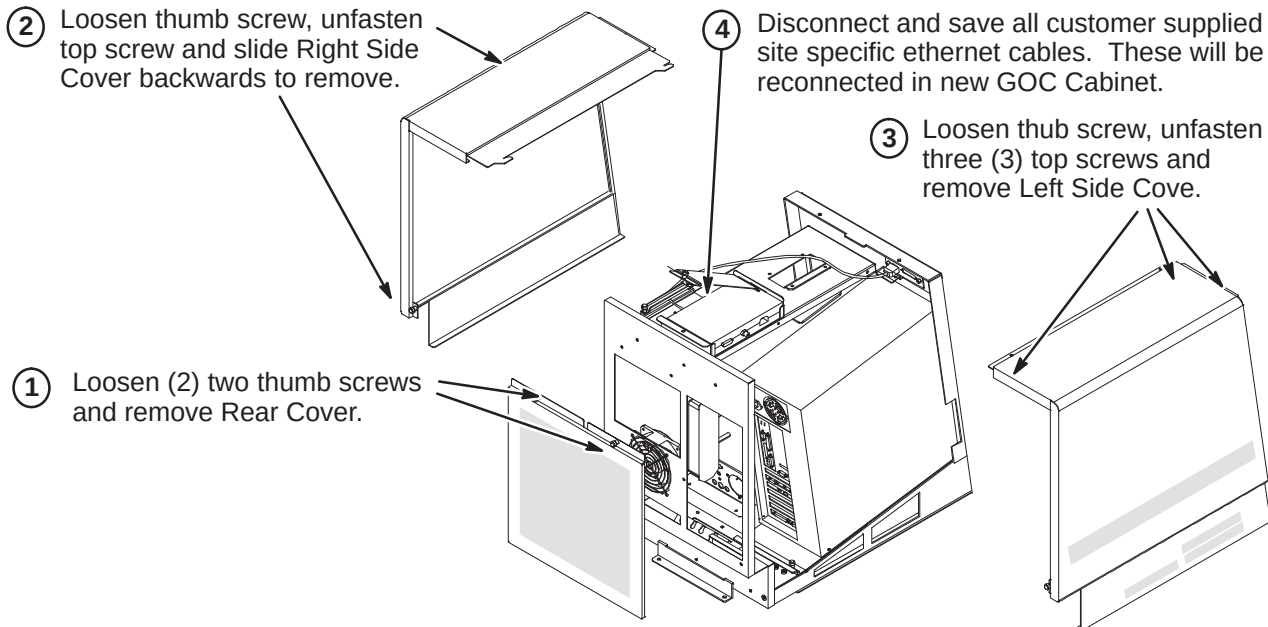
14-2 DE-INSTALLATION OF OPERATOR WORKSPACE

14-2-1 LX WITH OCTANE OPERATOR WORKSPACE

Confirm that all cables are disconnected from OperatorWorkspace and Computer. Perform the following procedure for return of the OW modules and Computer. Table will remain as part of upgrade.

14-2-1-1 De-Install Octane OW Table Modules



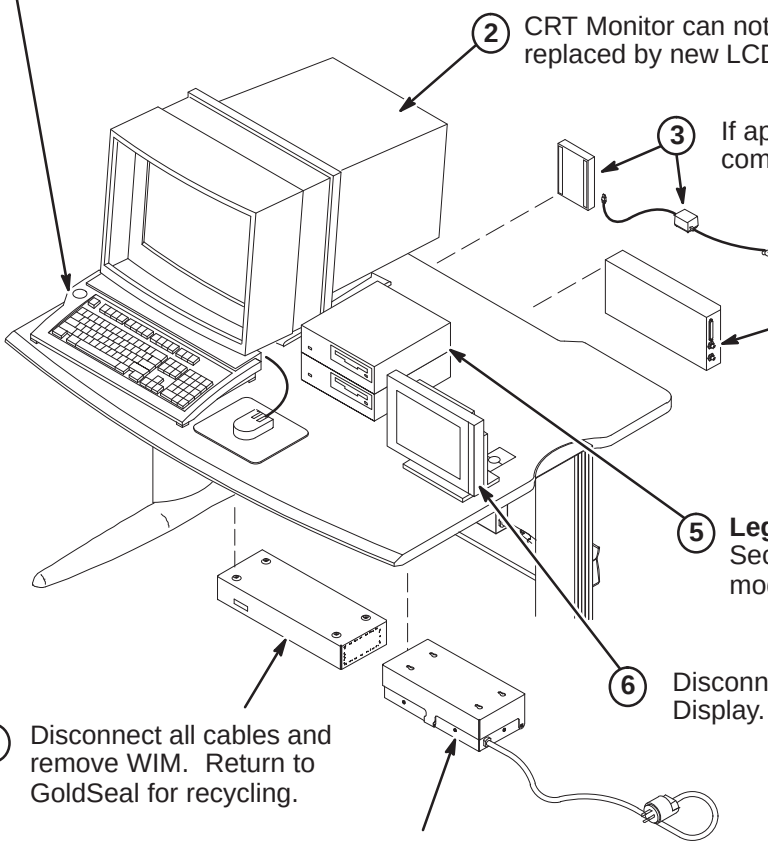
14-2-1-2 De-Install Octane Duplex Computer Cabinet

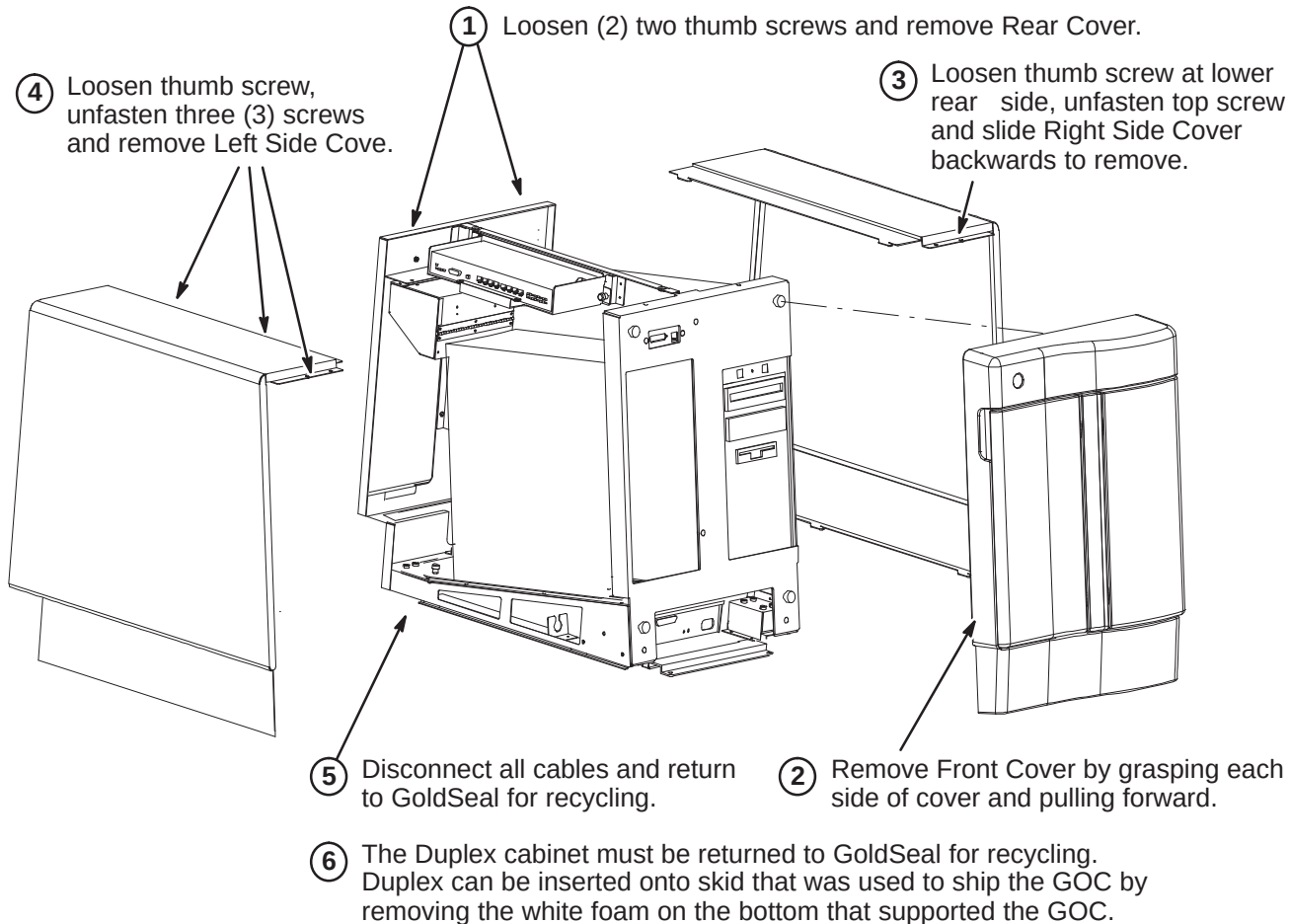
- ⑤ Refer to Section 14-3-1 for instructions on transferring Legacy Archive modules to HDx.
- ⑥ The remaining Duplex cables will not be used on the new GOC Cabinet and must be returned for recycling to Gold Seal.
- ⑦ The Duplex must be returned for recycling to Gold Seal. Duplex can be inserted onto skid that was used to ship the GOC by removing the white foam on the bottom that supported the GOC.

14-2-2 LX WITH INDIGO OPERATOR WORKSPACE

Confirm that all cables are disconnected from OperatorWorkspace and Computer. Perform the following procedure for return of the OW modules and Computer. Table will remain as part of upgrade.

14-2-2-1 De-Install indigo OW Table Modules

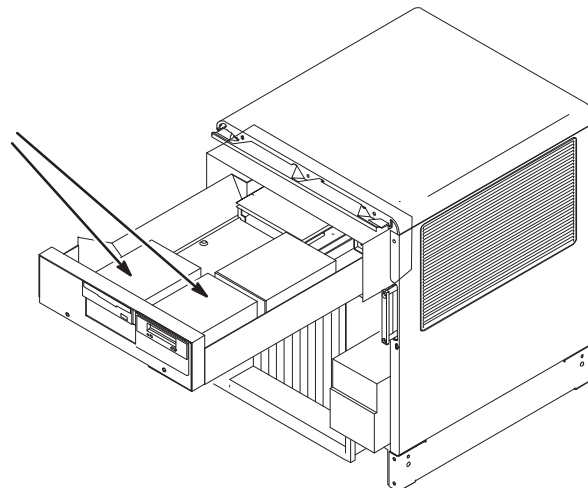
- ① New SCIM is required for HDx. If older style keyboard is part of existing site, remove keyboard and mouse. Return to GoldSeal for recycling. If SCIM is installed, de-install and return for recycling. A new black colored SCIM with new cables will be installed.
 - ② CRT Monitor can not be used with HDx. Monitor has to be replaced by new LCD Host Monitor if it is a CRT.
 - ③ If applicable, remove Modem. No longer compatible with HDxt systems.
 - ④ If applicable, remove DASM. No longer compatible with HDxt systems.
 - ⑤ **Legacy Modules:** Refer to instructions in Section 14-3-2 for tranfer of DAT and MOD modules to HDx system.
 - ⑥ Disconnect all cables and remove LCD Display. Return to GoldSeal for recycling.
 - ⑦ Disconnect all power cables and remove Power Distribution Box. Return to GoldSeal for recycling.
 - ⑧ Disconnect all cables and remove WIM. Return to GoldSeal for recycling.
- 
- The diagram illustrates the Indigo Operator Workspace setup. It includes a desk with a CRT monitor, a keyboard, and a mouse. Below the desk, there are various modules and components. Callout 1 points to the keyboard and mouse. Callout 2 points to the CRT monitor. Callout 3 points to a modem. Callout 4 points to a DASM module. Callout 5 points to legacy modules (DAT and MOD). Callout 6 points to the LCD display. Callout 7 points to the power distribution box. Callout 8 points to the WIM module.

14-2-2-2 De-Install Indigo Duplex Computer Cabinet

14-2-3 ADVANTAGE 5.x OR HORIZON 5.x OPERATOR CONSOLE AND COMPUTER

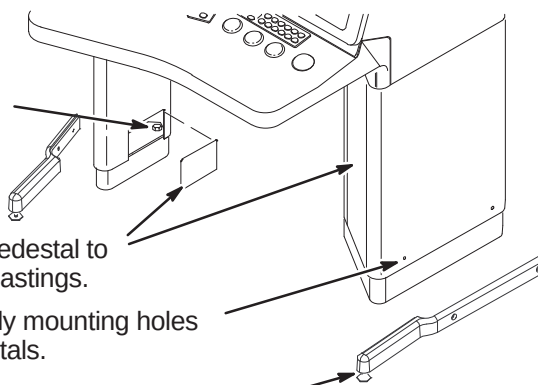
Confirm that all cables are disconnected from Operator Console and Computer. Complete the following procedure for return of the 5.x Operator Console and Computer.

Legacy Modules: Refer to instructions in Section **14-3-3** for transfer of DAT and MOD modules to HDx system.



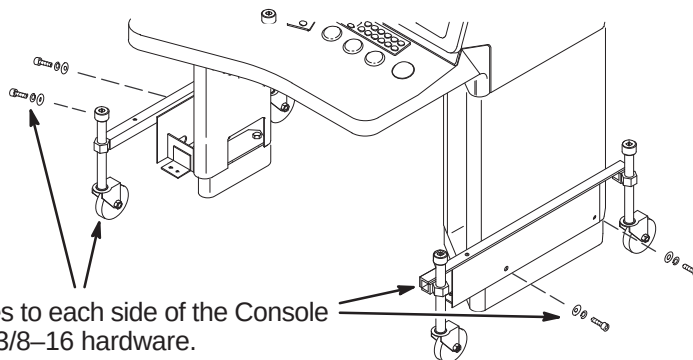
14-2-3-1 Prepare Operator Console Removal

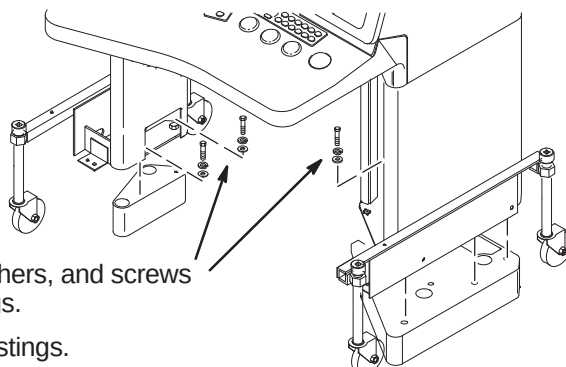
- ④ Loosen screws holding Console to pedestal risers.
- ③ Remove covers on sides of support pedestal to access mounting holes on the Base castings.
- ② Remove Plug Buttons from dolly mounting holes on each side of Console pedestals.
- ① If installed, unfasten and remove legs from console.



14-2-3-2 Attaching Dolly To Operator Console

- ① Attach shipping dollies to each side of the Console legs using furnished 3/8-16 hardware.

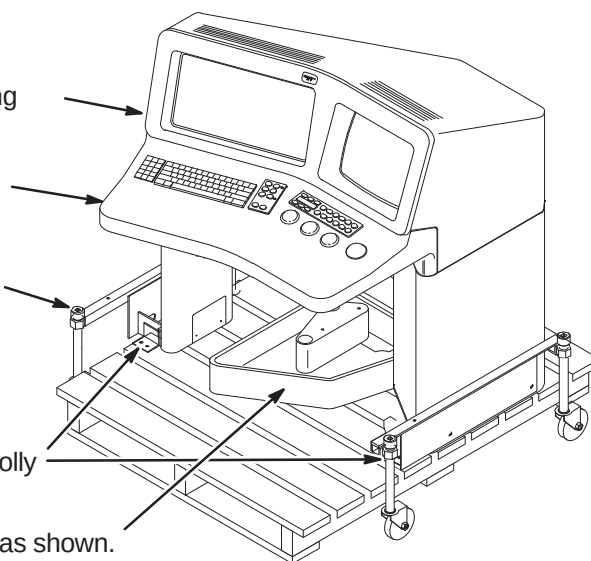


14-2-3-3 Removing Operator Console

- ① Remove lock washers, flat washers, and screws connecting legs to base castings.
- ② Raise console up over base castings.

14-2-3-4 Fastening Operator Console To Shipping Pallet

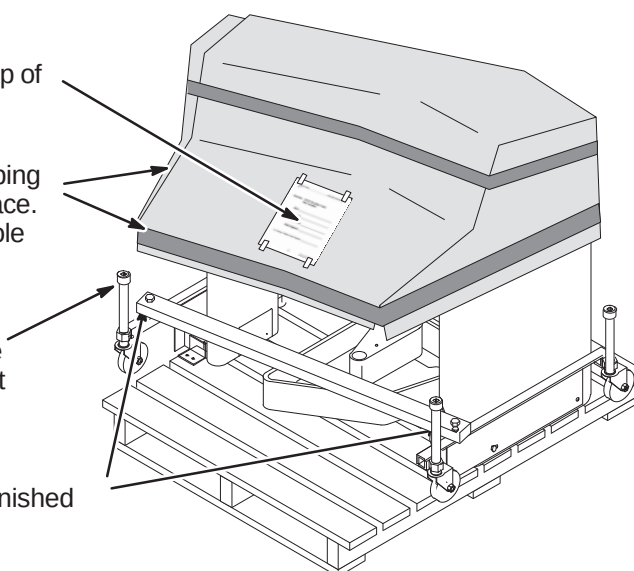
- ① Carefully roll Operator Console over an available shipping pallet.
- ② Position Console so that Keyboard Housing does not extend beyond shipping pallet.
- ③ Use a 1/2 inch socket wrench to turn the jack screws and lower the Operator Console onto pallet.
- ④ Locate (4) four provided 5/16 x 1-1/2 long lag bolts.



- ⑤ Install two of the lag bolts thru each shipping dolly bracket to fasten to shipping pallet.
- ⑥ Nest together Riser castings on pallet as shown.

14-2-3-5 Final Preparation Before Shipping

- ④ Fill out *GEMS Recycling Form* and attach to top of cardboard covering console.
- ③ Using any available cardboard (i.e. empty shipping boxes), cover Operator Console and tape in place. This is to prevent damage when shipping console back to manufacturer.
- ② Make sure that all four wheel assemblies are completely raised. If not, use 1/2 inch socket wrench to turn jack screws.
- ① Install provided dolly stiffener with furnished 1/4-20 x 1-1/2 long screws and nuts.



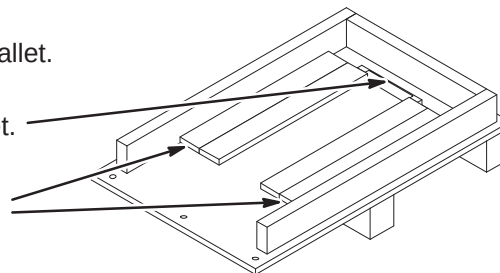
14-2-3-6 Preparing Shipping Pallet

Confirm that all cables are disconnected from Computer. Perform the following procedure for return of 5.x Computer reusing the shipping pallet that the new GOC Computer Cabinet arrived in.

- ① Locate the Operator Workspace Cabinet shipping pallet.

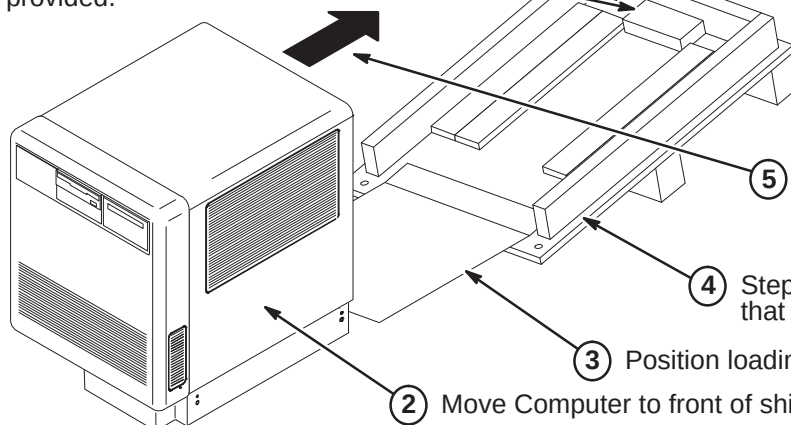
- ② Remove existing bracket.

- ③ Re-position the protective foam on each side from a vertical to a horizontal position between the side of the pallet and the existing horizontal foam.



14-2-3-7 Loading Computer Onto Shipping Pallet

- ① Attach 2 x 4 x 7 wood block with nails provided.



Note:

Pallet will return to horizontal position as Computer is rolled toward back of pallet.

- ⑤ With another person, carefully roll Computer onto shipping pallet so it will rest on the horizontal foam pieces.

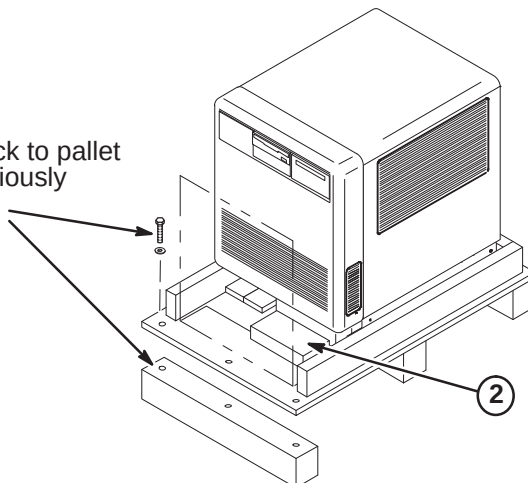
- ④ Step on front corner of shipping pallet so that bottom board of pallet touches floor.

- ③ Position loading board at front of pallet.

- ② Move Computer to front of shipping pallet.

14-2-3-8 Reassemble Shipping Pallet

- ① Re-attach wood block to pallet using the three previously removed fasteners.

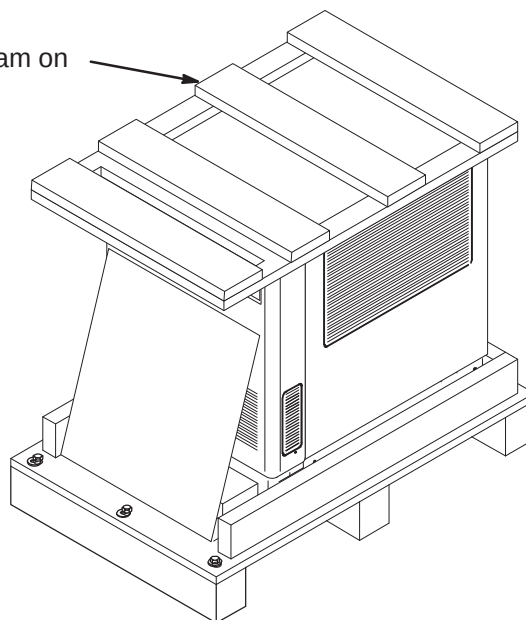


- ② Place 2 x 4 wood block against front right side of Computer and attach to pallet with nails provided.

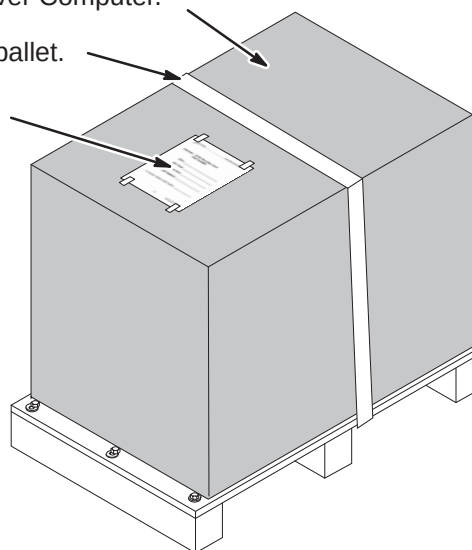
14-2-3-9 Installing Computer Top Shipping Foam

- ① Install the Operator Workspace Cabinet shipping foam on top of Computer.

Note: Shipping foam insert will not fit as well as it did with the new Operator Workspace Cabinet.

**14-2-3-10 Final Preparation Before Shipping**

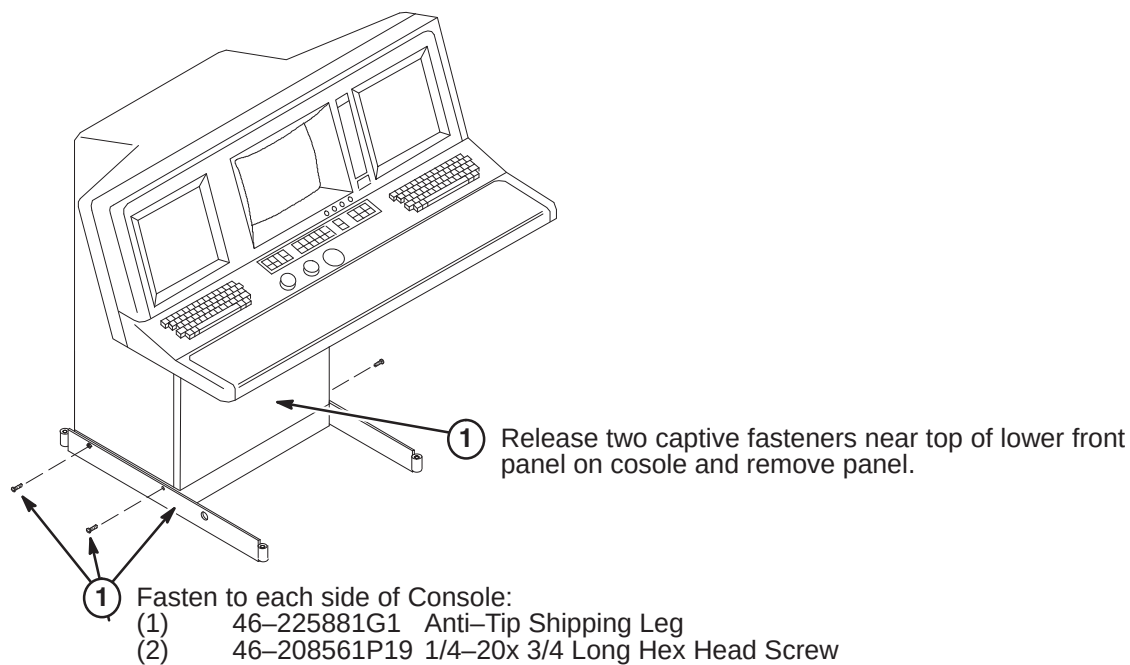
- ① Place Operator Workspace shipping box over Computer.
- ② Secure to shipping pallet.
- ③ Fill out *GEMS Recycling Form* and attach to top of shipping box.



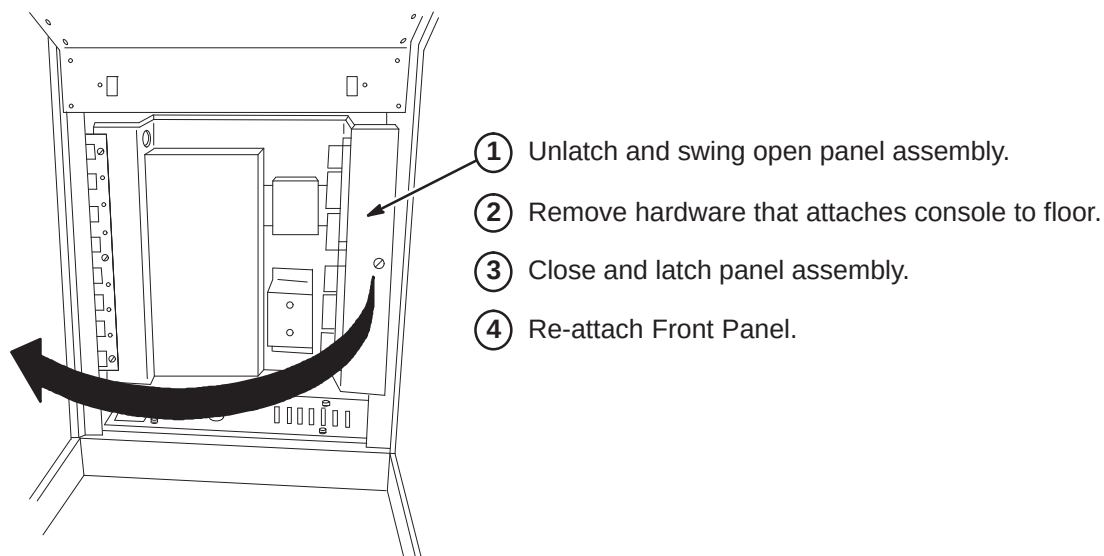
14-2-4 ADVANTAGE 4.x OPERATOR CONSOLE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of the 4.x Operator Console.

14-2-4-1 Install Operator Console Shipping Legs



14-2-4-2 Operator Console Removal



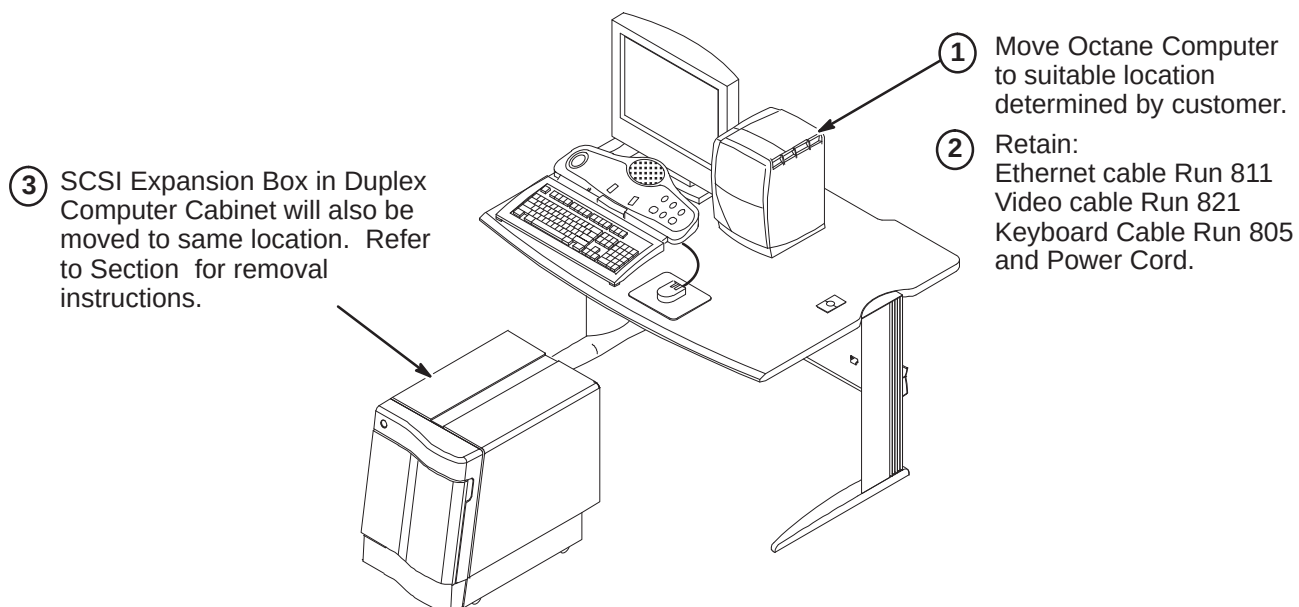
14-3 LEGACY ARCHIVE MEDIA SUPPORT

Signa HDx series does NOT support Legacy Archive Media, such as DAT and Pioneer MOD Archive Devices. Complete the applicable procedure for transfer of Legacy Archive Media for use with HDxt.

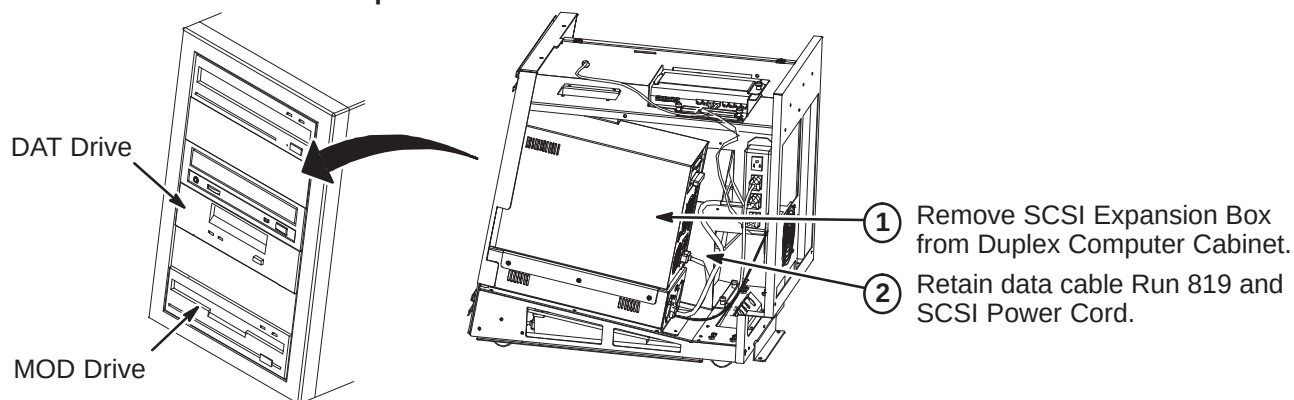
14-3-1 LEGACY MODULE UPGRADE FROM OCTANE

LX sites with Octane computers will retain their Octane computer and legacy Media Devices. This upgrade would require a Monitor (M1000NT) and Keyboard (M1000MN).

14-3-1-1 Deinstall Octane Computer

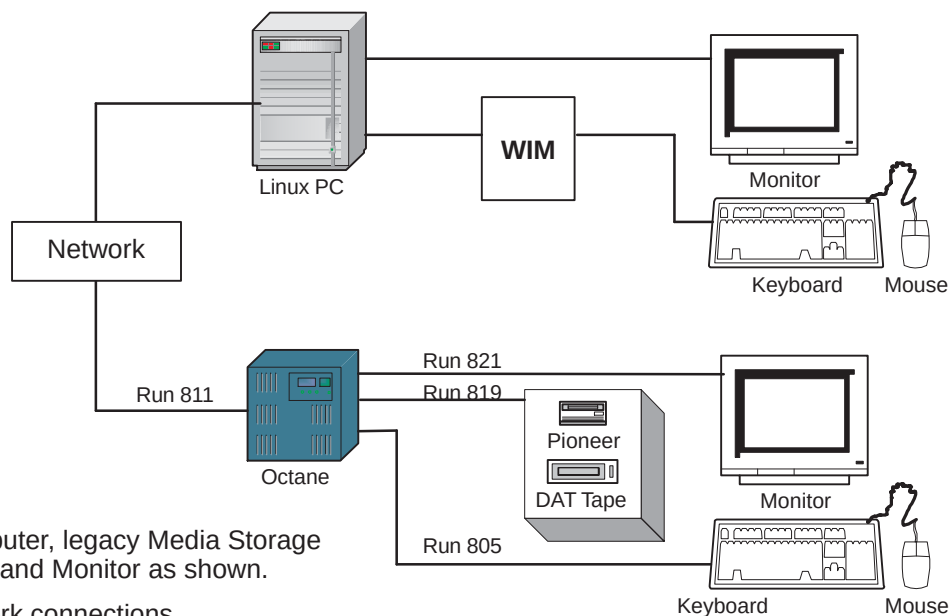


14-3-1-2 Deinstall SCSI Expansion Box and Cables



14-3-1-3 Hardware Configuration Interconnect

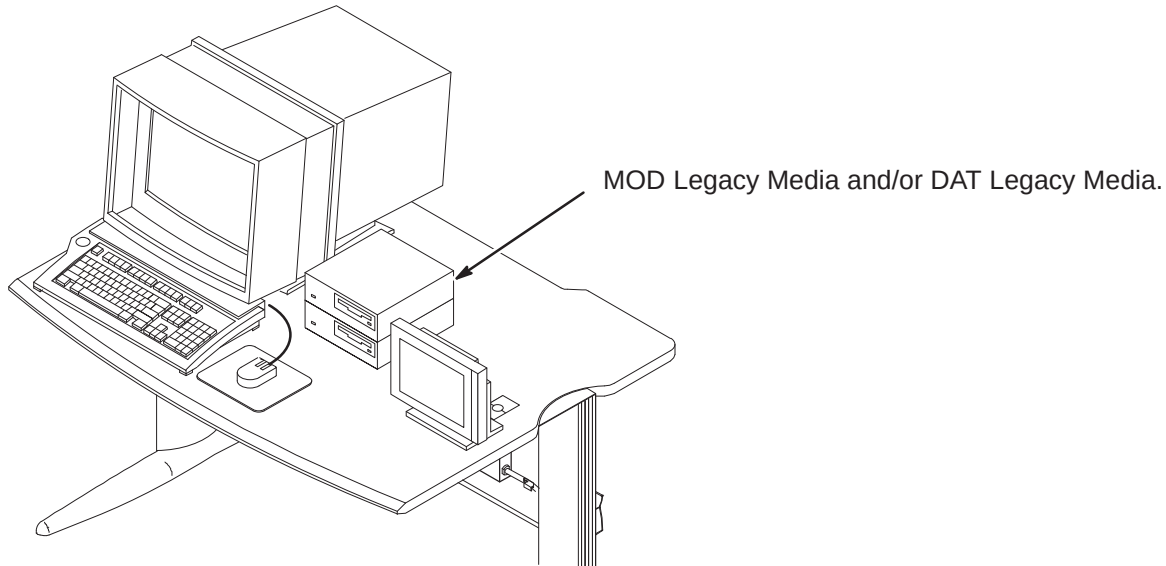
This procedure should be performed by FE. For Mechanical Installer or Van Partner, make sure the FE is notified that the steps below need to be completed.



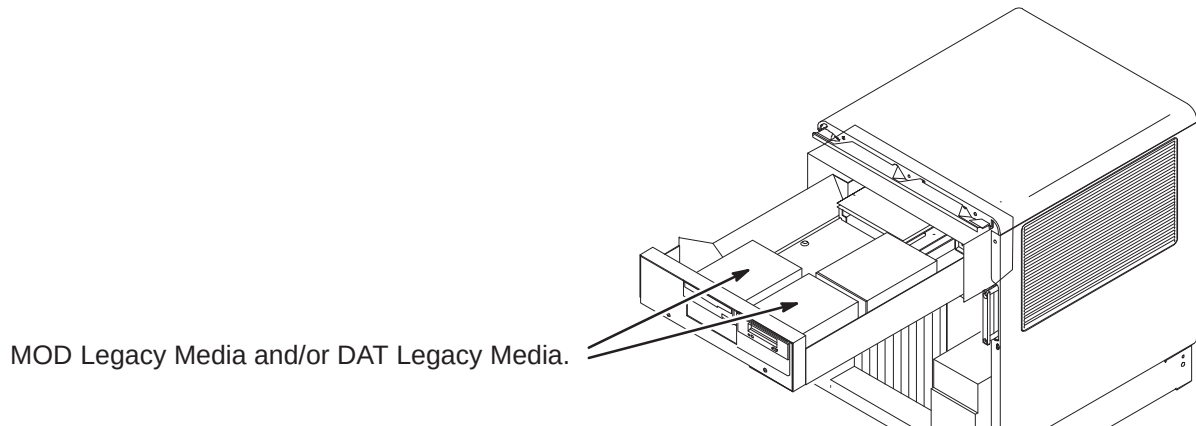
- ① Connect the Octane Computer, legacy Media Storage Device, Keyboard, Mouse and Monitor as shown.
- ② Connect power and Network connections.
- ③ After Octane computer is powered up and switched ON:
 1. Login as 'root'
 2. cd /export/home
 3. touch **.SHOW_MODE**
 4. Logout
 5. Login as 'signa'
- ④ The 'Octane' system will now boot up in **SHOW_MODE** which will enable to further setup the DICOM Network to the different hosts to transfer images.

14-3-2 LEGACY ARCHIVING FOR INDIGO

At this time there is no upgrade path for customers requesting to retain Legacy Archiving capability that own systems with host computers prior to the Octane platform. Instruct the Customer to contact sales to discuss possible options.

**14-3-3 LEGACY ARCHIVING FOR ADVANTAGE 5.x OR HORIZON 5.x**

At this time there is no upgrade path for customers requesting to retain Legacy Archiving capability that own systems with host computers prior to the Octane platform. Instruct the Customer to contact sales to discuss possible options.



14-4 OPERATOR WORKSPACE UPGRADE

14-4-1 OPERATOR WORKSPACE TABLE INSTALLATION (If Needed)

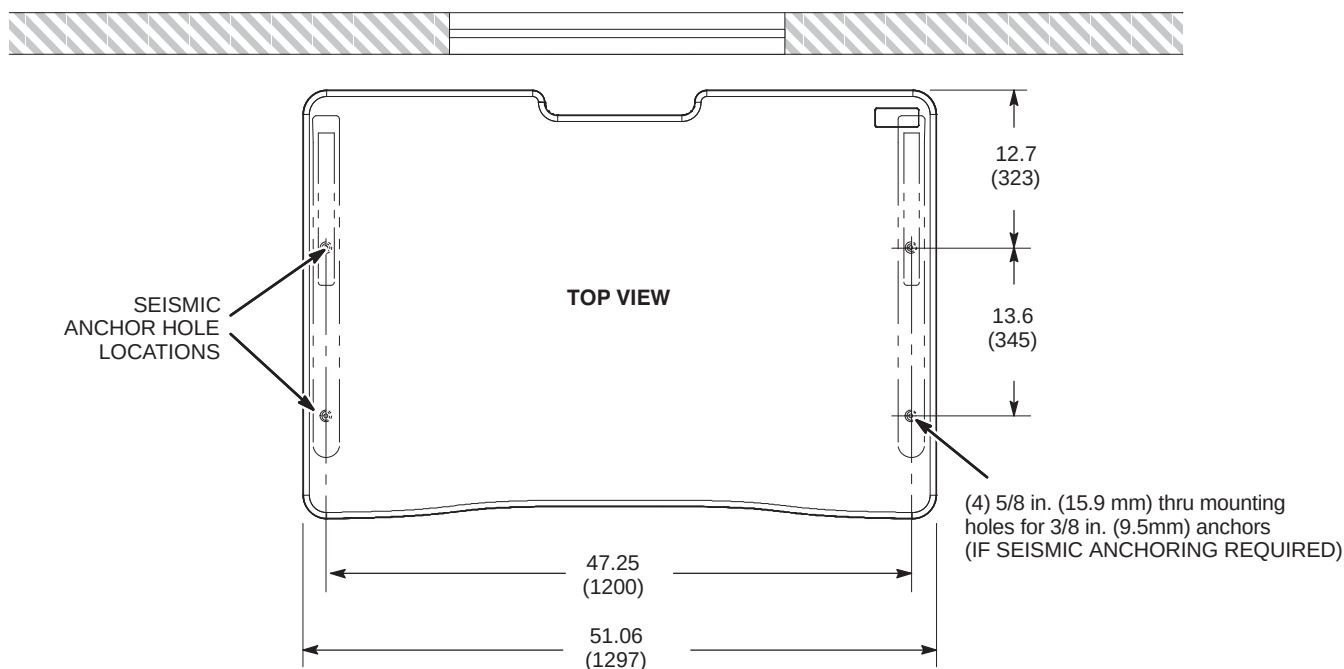
Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.

1. Open palletized M1000MW box that contains the Operator Workspace table. Use furnished *Operator's Workspace Table Assembly Instructions* (5189137–50) to assemble the model 5189137 table.
2. Move table into designated position. Refer to room layout plan and/or consult with Site customer for exact location of Operator Workspace Table. Typical location with table in front of Magnet Room viewing window is shown below.

Note

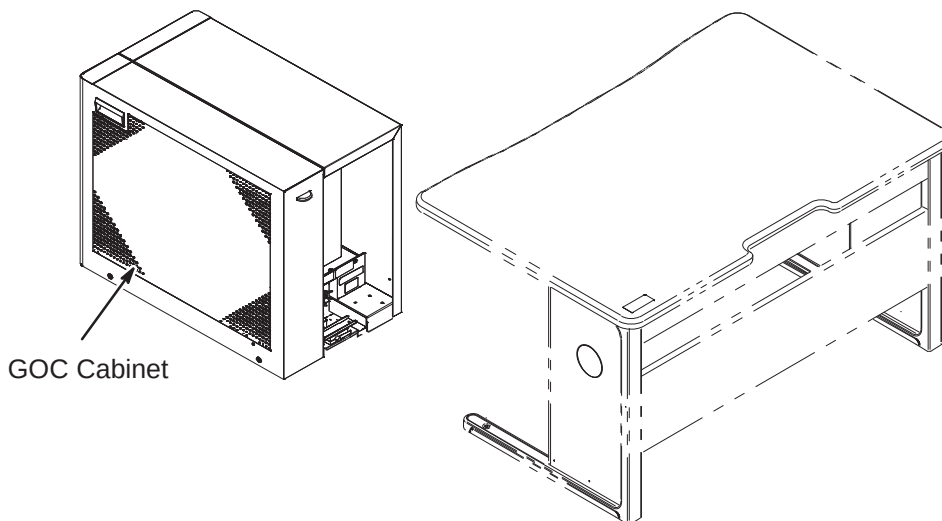
The design of the new model 5189137 table does not require anti-tip anchors. This table meets the ANSA/BIFMA anti-tip standards.

3. If seismic anchoring is required, refer to site architectural drawings and procure and install specified anchors.



14-4-2 POSITION COMPUTER CABINET

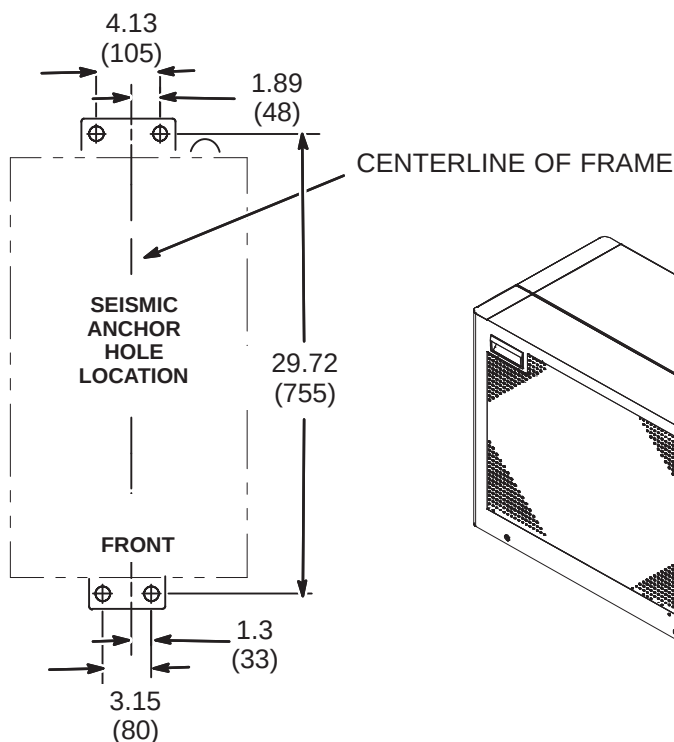
Consult with Site customer for exact location of Global Operator Console (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



14-4-3 SEISMIC ANCHORING (IF REQUIRED)

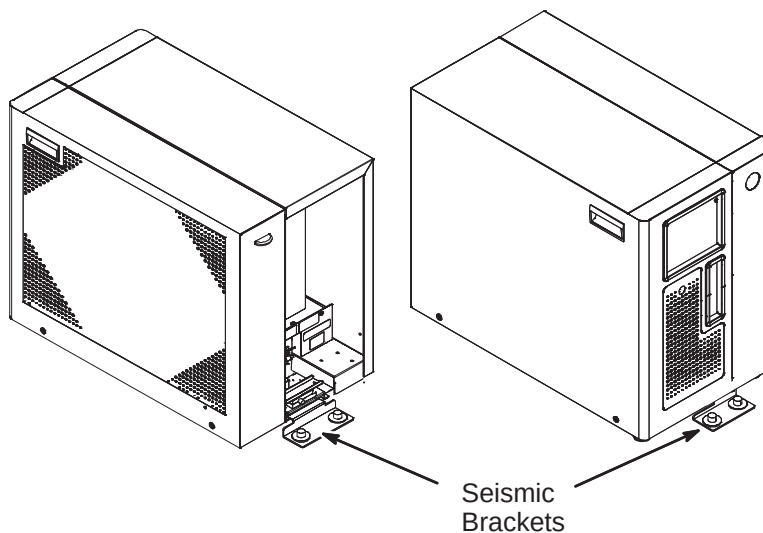
GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets.



NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



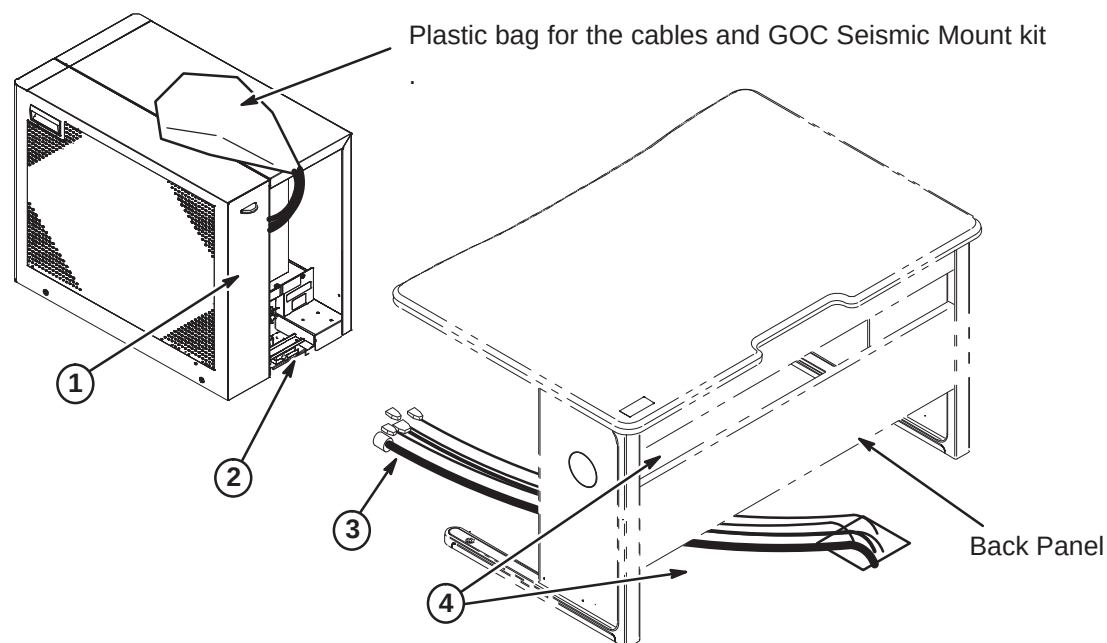
14-4-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Two power cords
- One video output cable
- Keyboard output cable
- SCIM output cable
- Modem output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1064, 1070, E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening.
(These cables are inside of plastic bag)
2. Runs 1381, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



14-4-5 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

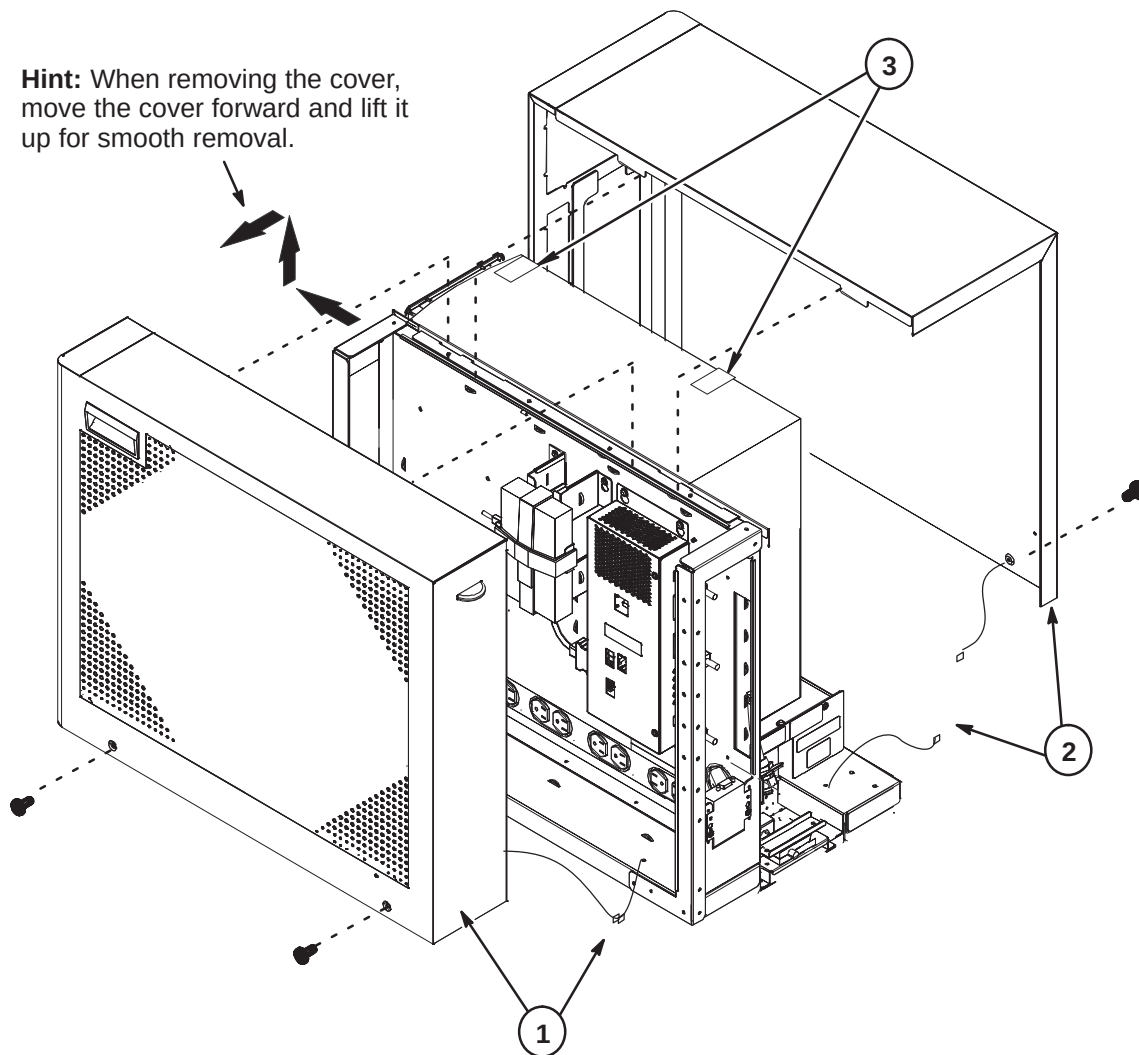
1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

3. Peel the two shipping tapes off from PC.

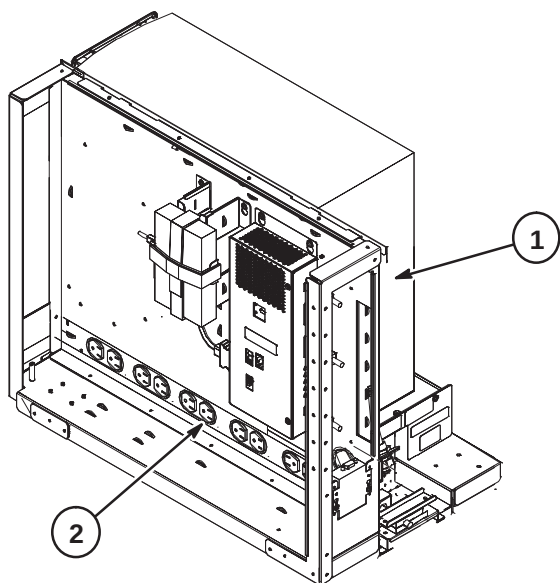
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



14-4-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

1. Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
2. Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



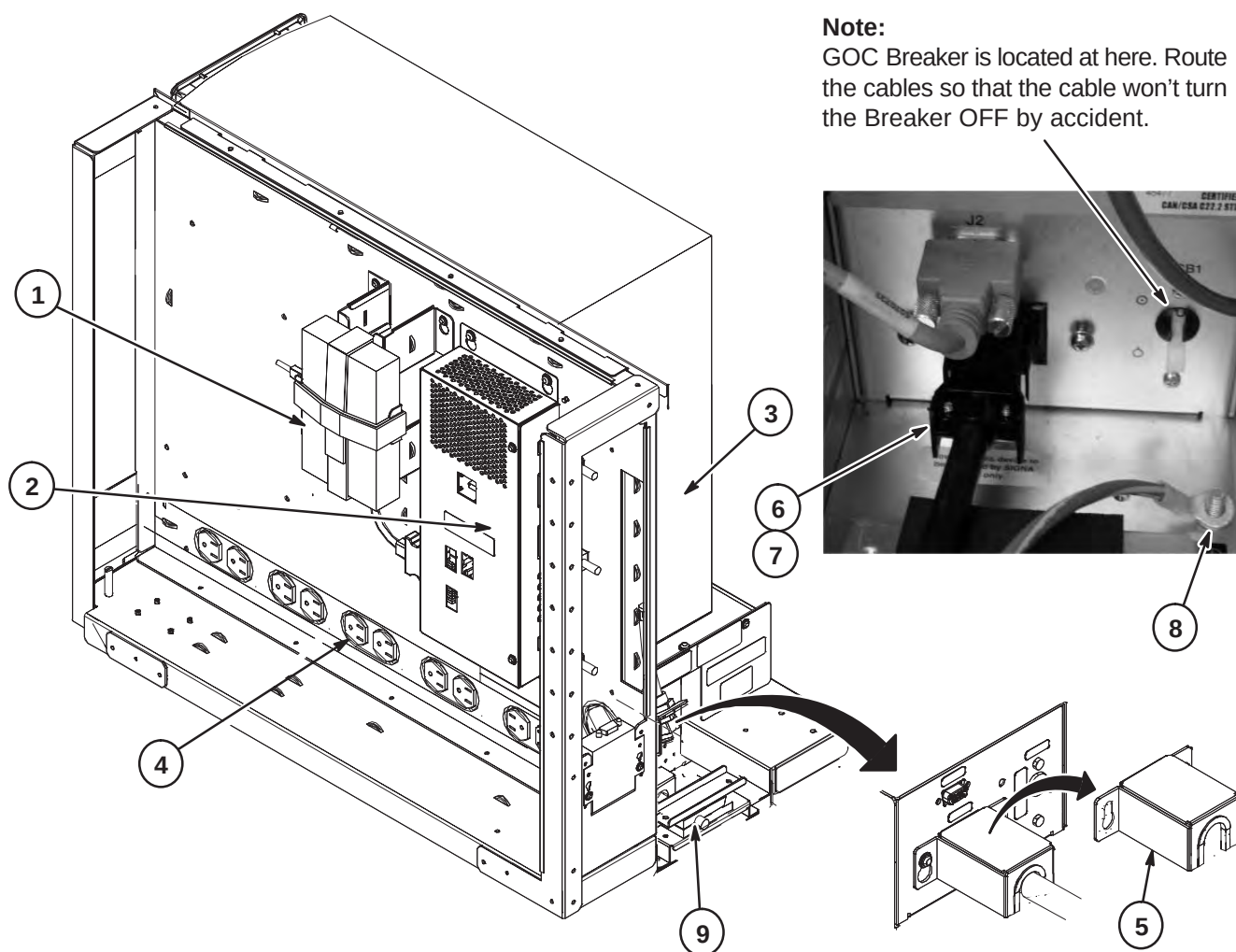
14-4-7 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

1. Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

2. Connect Run 1085 to J7 and Run 1255 to J18 on GOC AA.
3. Refer to Section 14-4-9 for all cables connected to Linux PC computer.
4. Refer to Section 14-4-8 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
5. Before connecting GOC power cable Run 1381 to J1, loosen screws and remove cover.
6. Route and connect GOC power cable Run 1381 to J1 on PDU.
7. Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover removed in Step 5. after connecting cable and tighten screws.
8. Connect Run 1382 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
9. Tighten clamp to secure cables in place.

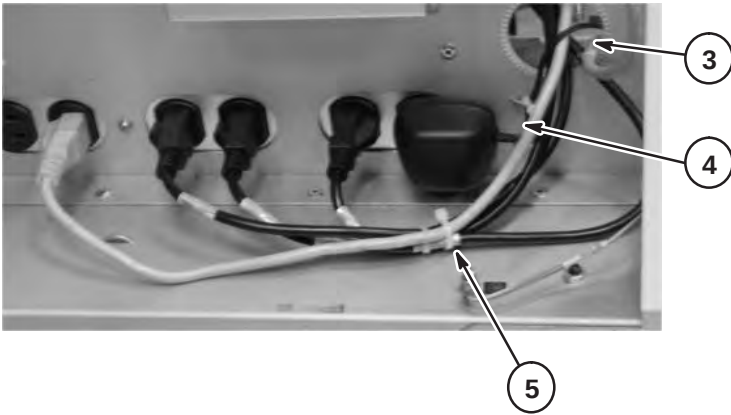
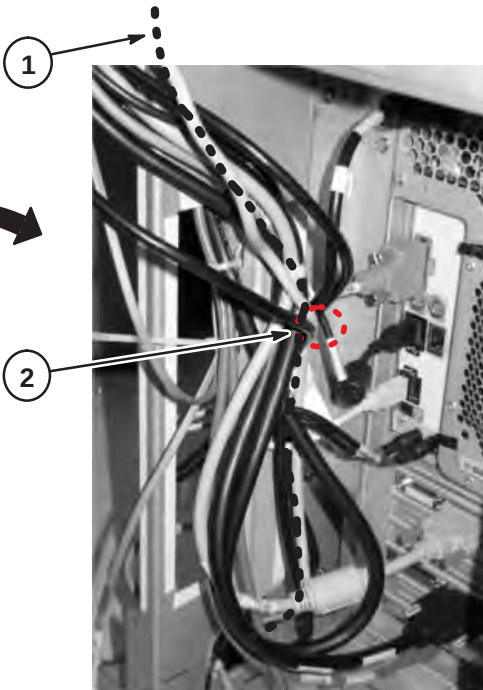
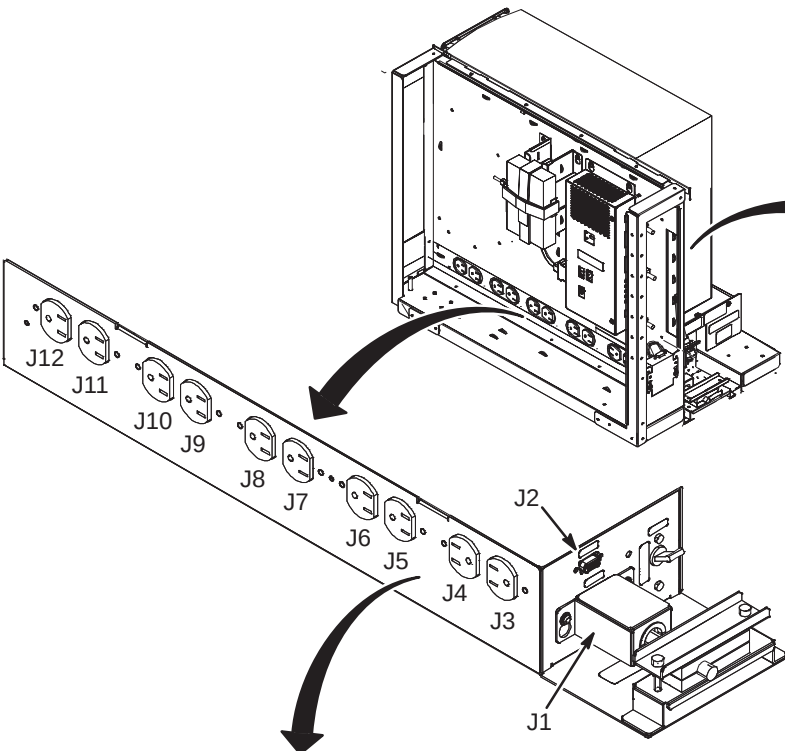


14-4-8 POWER CABLE CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1-PDM.

- 1. If Modem is installed, route power cable and connect to J7.
- 2. Tie-wrap point for Modem power cable.
- 3. Route power cables connected to J3 thru J7 through access hole.
- 4. route and connect Patient Alert power cable to J3 and tie-wrap at this point.
- 5. Tie-wrap power cables J4 to J7 at this point.

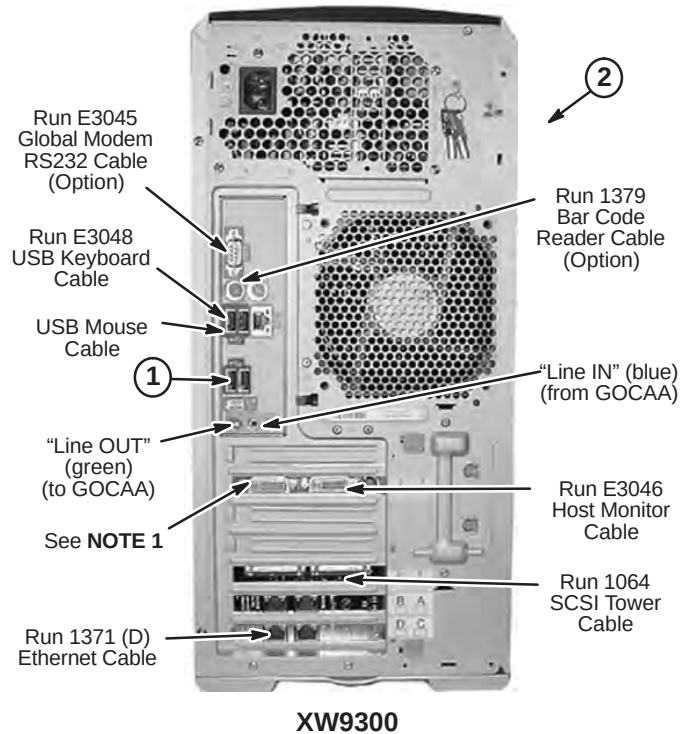
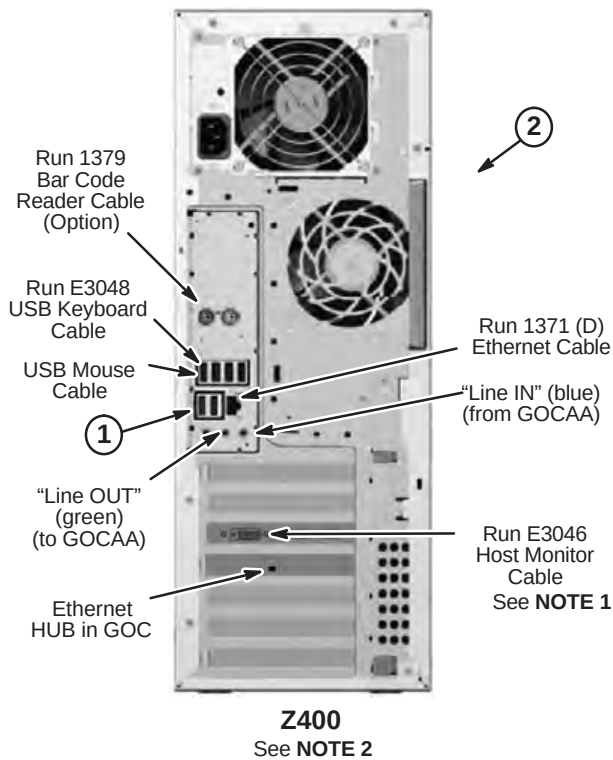


POSITION	NAME
J1	GOC Input
J2	GOCAA
J3	Patient Alert
J4	Host Computer
J5	Host LCD
J6	SCSI Tower
J7	Modem
J8	
J9	LAN Hub
J10	
J11	
J12	One Wire Hub

14-4-9 LINUX PC CABLE CONNECTIONS

The connection points of cables routed to the Linux PC are described in the following two sections. All cables shown are preconnected except for Run 1371 and USB cable for USB-Serial DB9 Adaptor which has to be routed and connected at site. Refer to the applicable PC for your site.

1. Refer to Section 14-4-10 for installing the USB to 1-Wire Adapter (5307784) to USB port.
2. Connect remaining cables as marked at rear of computer.

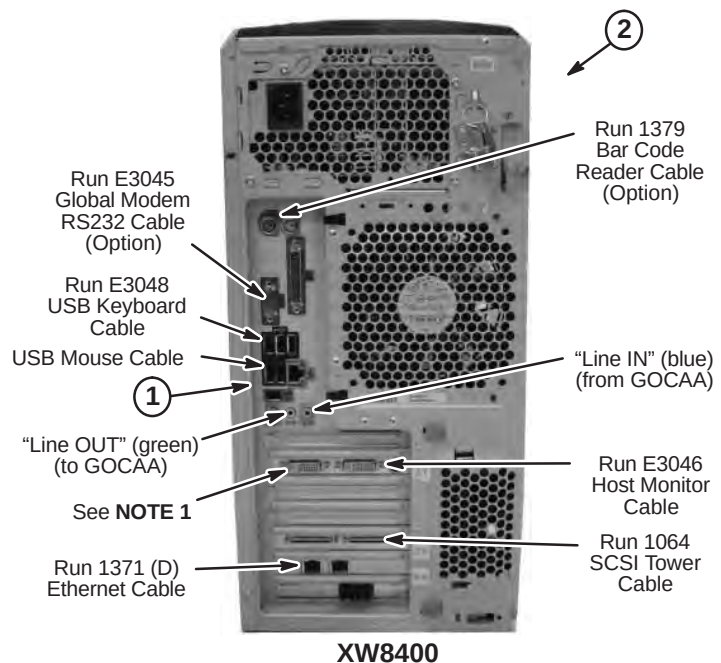


NOTE 1:

Do not connect any additional monitors prior to loading operating system software. Doing so will cause boot-up to stop.

NOTE 2:

SCSI Tower and DASM are not supported for Z400PC.

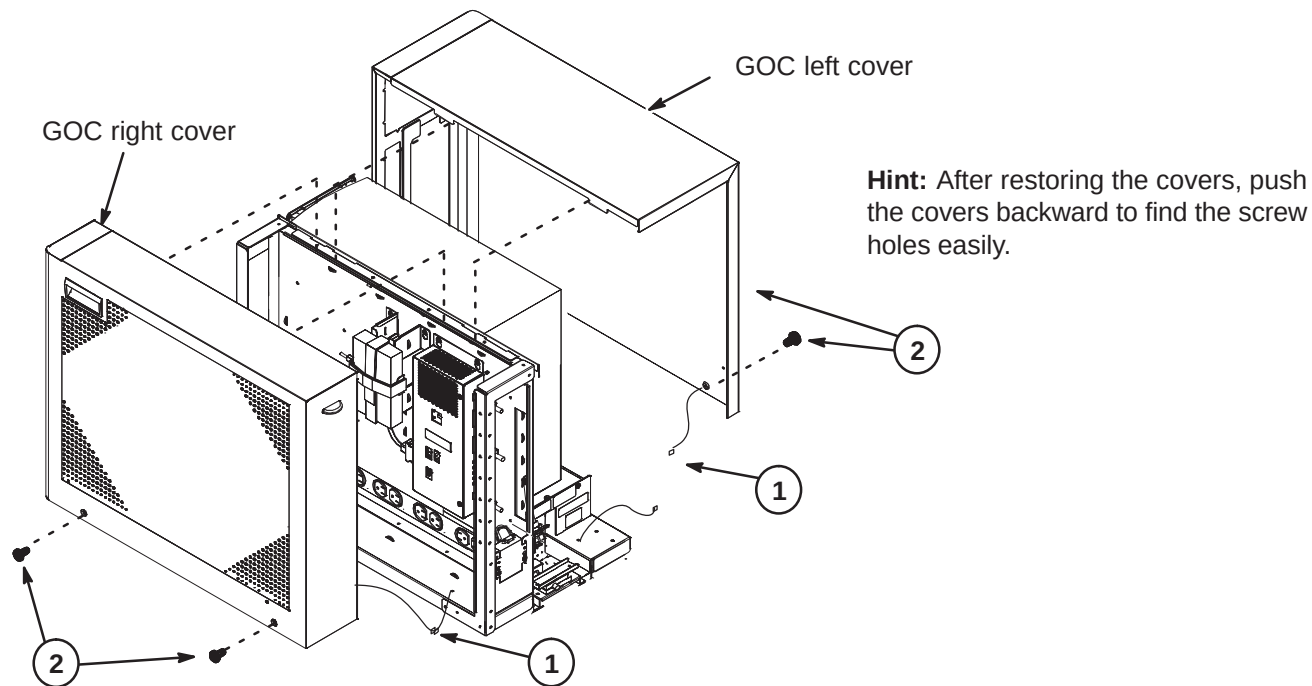


14-4-10 INSTALL TEMPERATURE AND HUMIDITY SENSOR COMPONENTS

The components of the Temperature and Humidity Sensor Kit (5147136-2) will be installed on the back of the GOC cabinet and on the back of the System Cabinet with one cable routed from the GOC and connected to it.

14-4-10-1 RESTORE GOC COVERS

1. Connect Ground cables and install both side covers by inserting tabs at top of covers into slots.
2. Install screws that were removed in earlier procedure

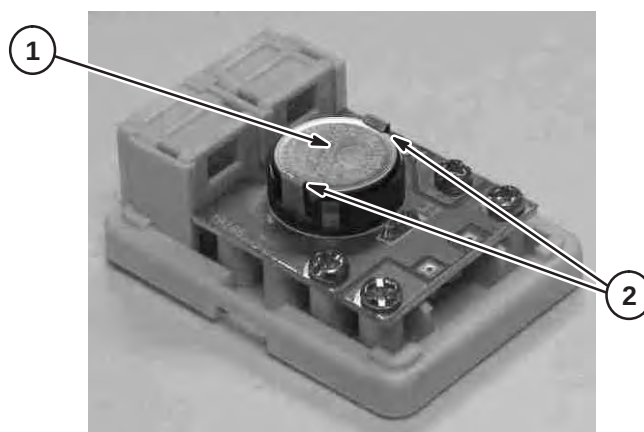
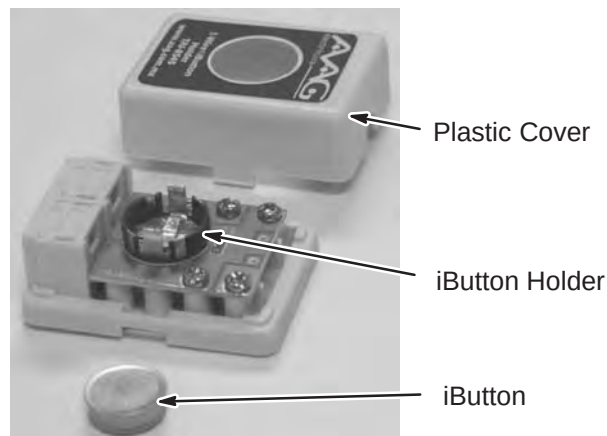


14-4-10-2 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton.

1. Install the iButton in the holder so that the solid part of the case is exposed.
2. Make sure the "fingers" latch over the top of the iButton and secure it in the holder.
3. If removed, reinstall the Plastic Cover.

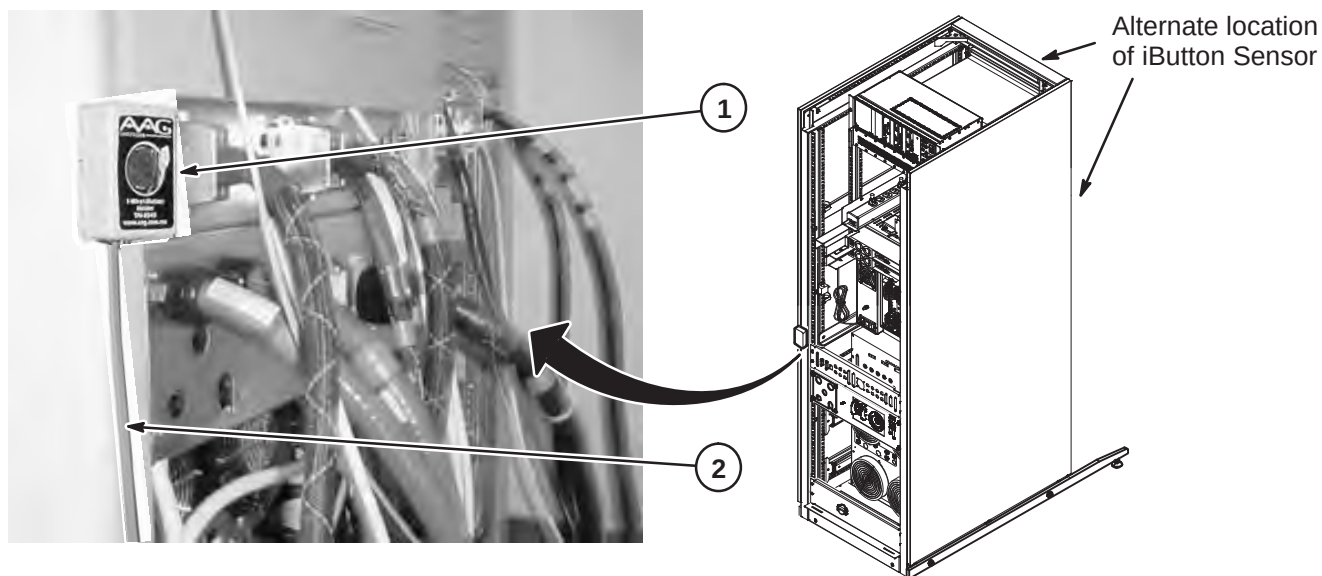


14-4-10-3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET

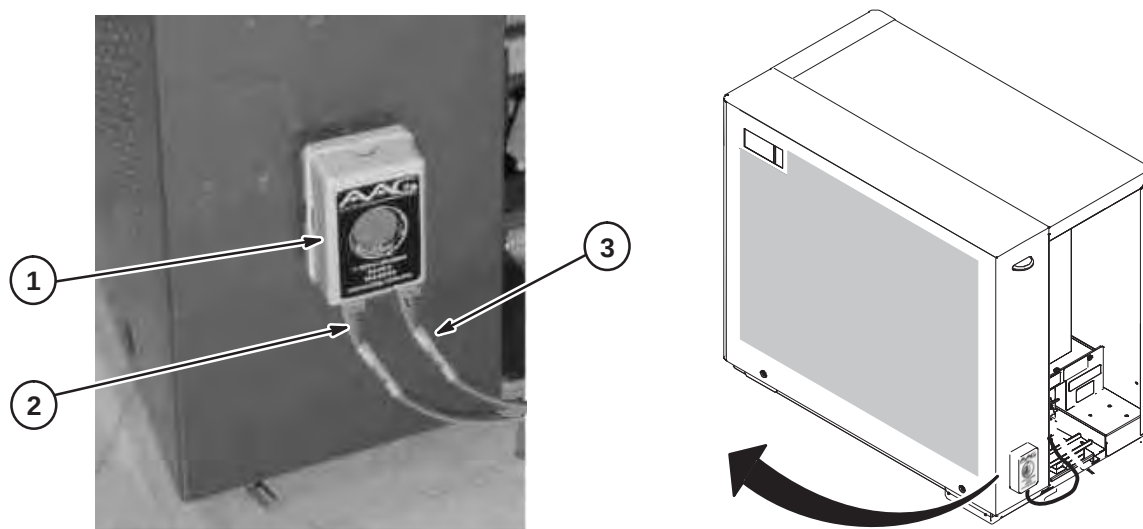
Note: iButton Holders are unique to each location. Make sure to install 5141857-4 to System Cabinet and 5141857-5 to GOC.

Alternate Location: For Mobile sites, or other sites where forced air may blow directly on the sensor which may cause false readings, the iButton Sensor can be located at the front or top of the System Cabinet.

1. Attach iButton Sensor assembly (Holder, 5141856, and Humidity iButton (5141857-4), to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Connect cable Run 1384 (5143794-2) to Sensor. Route and to iButton at rear of GOC.

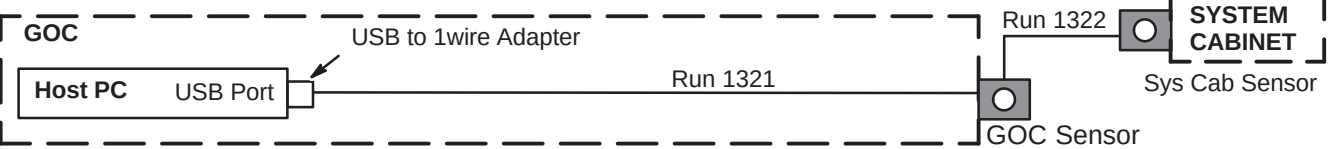
**14-4-10-4 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC**

1. Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857-5), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Route and connect Run 1322 (5143794) from System cabinet to Sensor at back of GOC.
3. Connect cable Run 1321 from Sensor to USB to 1-Wire Adapter (5307784) installed in any open USB port at rear of PC. Refer to step 1 in Section 14-4-9.

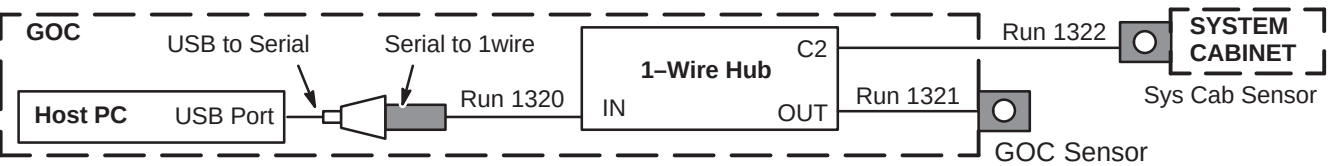


14-4-10-5 TEMPERATURE & HUMIDITY SENSOR WIRING SCHEMATICS

Environmental Sensors Block Diagram (Shipped after August 11, 2008) utilizing 1-wire 5147136-2



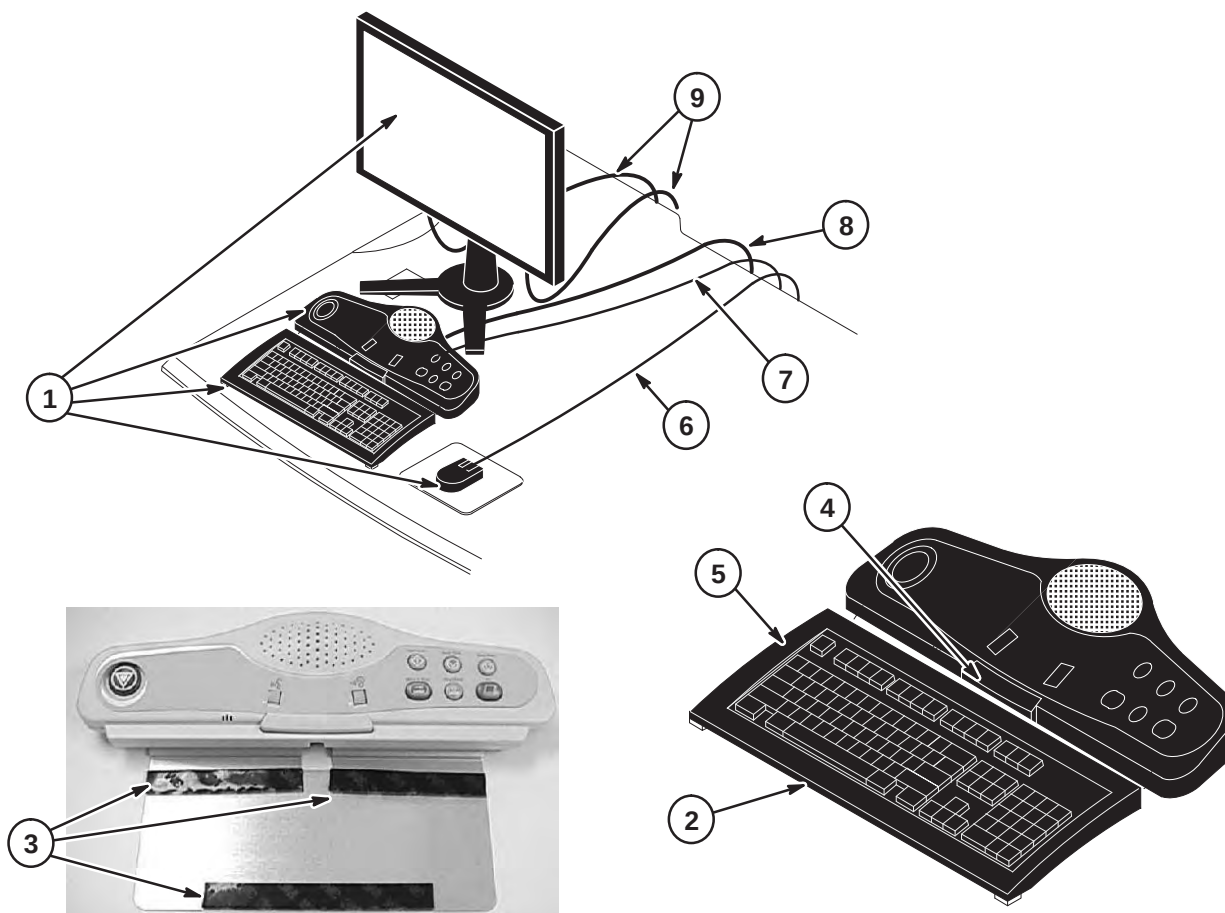
Environmental Sensors Block Diagram (Shipped before August 11, 2008) utilizing 1-wire 5147136



14-4-11 INSTALL MONITOR AND SCIM & KEYBOARD CABLES

Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

1. Position Host Display Monitor, SCIM, Mouse, and keyboard on table.
2. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
3. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface.
4. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
5. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.
6. Connect Mouse cable to USB port at rear of computer in GOC.
7. Connect Run E3048 to USB Keyboard cable. Route and connect other end to computer in GOC.
8. Connect Run E3047 to SCIM. Route and connect other end to computer in GOC.
9. Route power cord Run E0015 and Run E3046 from GOC and connect to Display Monitor.

**Note**

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

14-4-12 INSTALL SCSI TOWER and MODEM (Option)

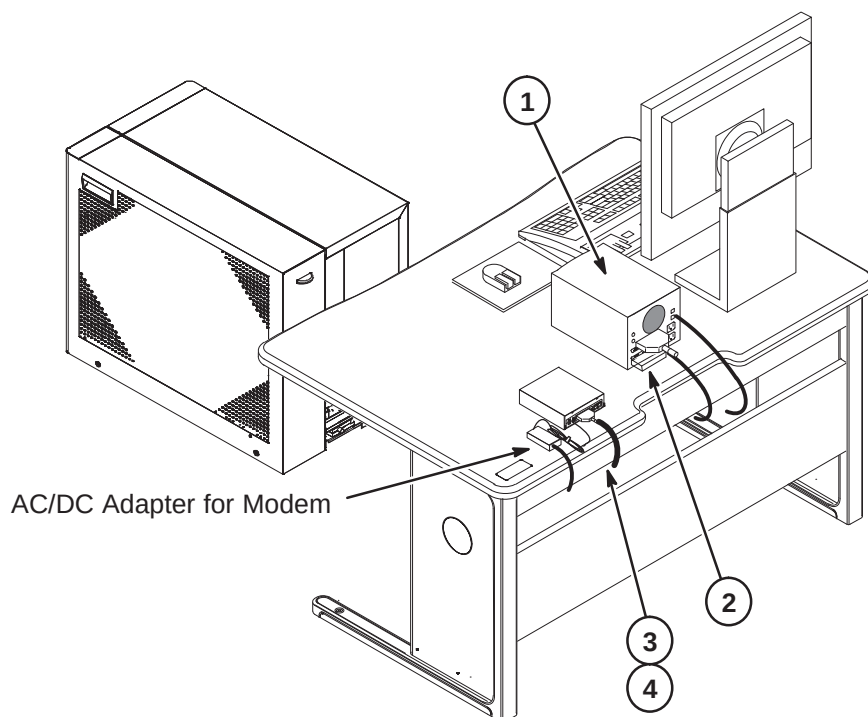
Note: SCSI Tower is an option with release 15.x or HDxt.

Note: SCSI Tower and Modem are not compatible with Z400 computer. The Z400 computer will be delivered and installed at sites around the 4th quarter of 2010.

1. 64 and 1070 from GOC Cabinet.
2. Install terminator on back of SCSI Tower.

The following steps are for the Modem option:

3. If applicable, place the Modem on the OW table and connect Runs E3045 and power cable from GOC Cabinet.
4. Connect customer supplied telephone line.



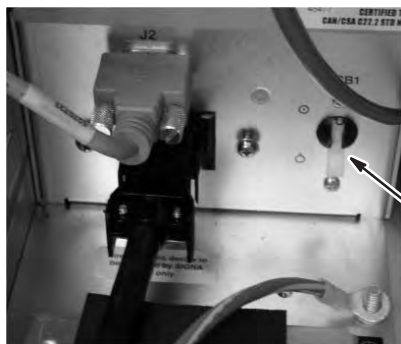
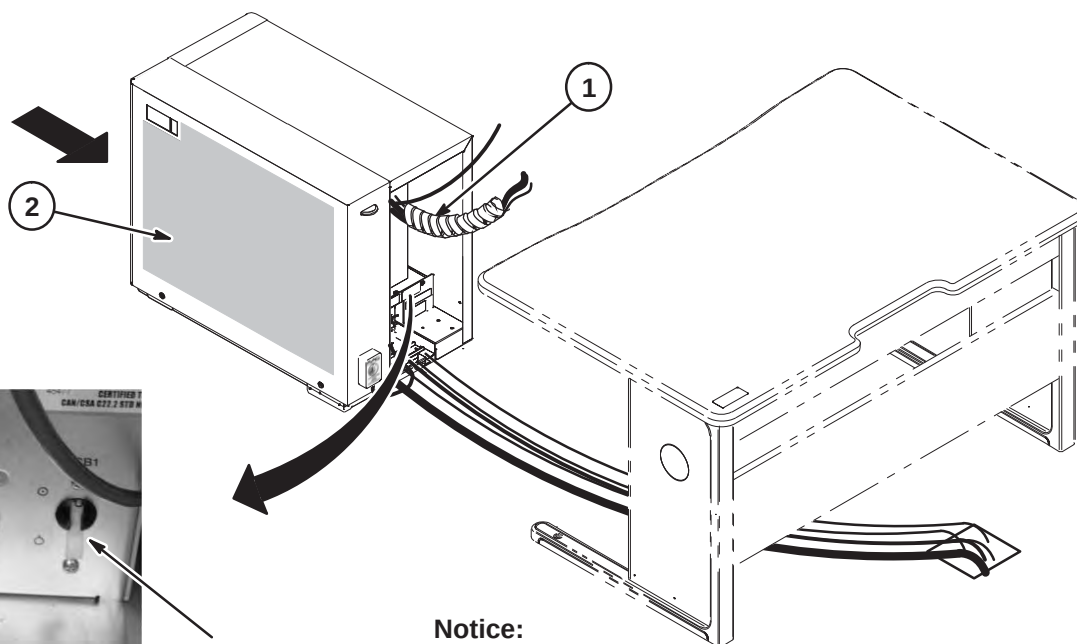
14-4-13 FINAL POSITIONING OF GOC CABINET

1. Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m) Spiral tube is included in the plastic bag which is shipped with GOC.

Notice

Do Not include Modem power cable in Spiral Tube.

- Three power cords (LCD Monitor ,SCSI Tower(if applicable))
 - One video cable
 - Keyboard cable
 - SCIM cable
 - SCSI Tower cable (if applicable)
 - Modem cable (if applicable)
2. Place cabinet in final position as determined by customer.

**Notice:**

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.

14-4-14 POWER ON INFORMATION**IMPORTANT**

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the rear of the GOC.

SECTION 15 – HFD/PDU CABINET

TABLE OF CONTENTS

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15-3-1 POSITION HFD/PDU CABINET	15-5
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15-3-5 CABLE CONNECTIONS TO HFD/PDU CABINET	15-8
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15-1 OVERVIEW

HDx systems require a new HFD/PDU cabinet or an HFD-S/PDU upgraded cabinet.

- Sites that have a pre-ACGD/PDU system will remove existing cabinets and install a new HFD/PDU cabinet. Refer to Sections **15-2** and **15-3**. Refer to Section **4-7-1** for connection of cables to HFD/PDU.
- Sites that have an ACGD/PDU installed will upgrade the cabinet to an HFD-S/PDU. To upgrade to HFD-S, M3000PT (1.5T non-TwinSpeed) or M3000PW (1.5T TwinSpeed), must be installed. Refer to Direction 5131778, *ACGD to HFD-S Upgrade Installation*, supplied with the catalogs for upgrade instructions. Also refer to Section **15-4** for connection of HDx cables to PDU.

All de-installed cabinets are to be returned for recycling. Refer to applicable cable removal map and applicable column in cable removal Table 4-2 for disconnection and removal of designated cables for system being de-installed.



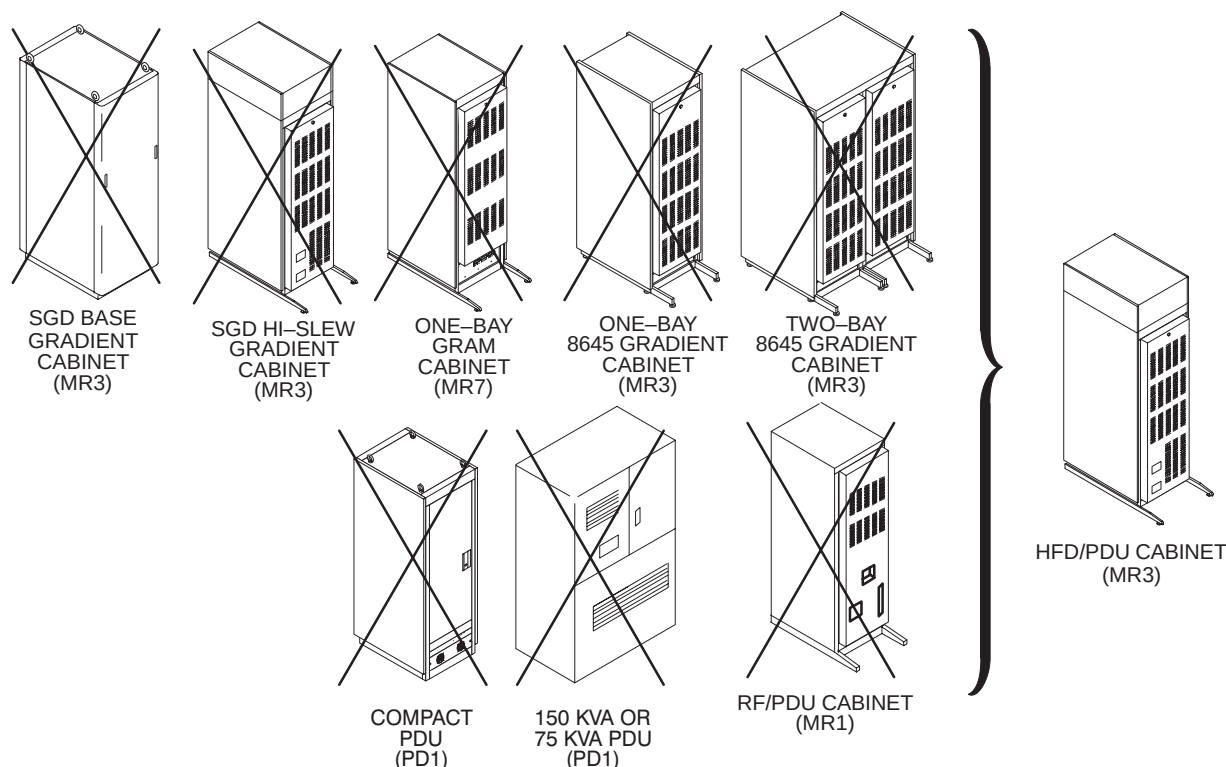
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

15-1-1 HFD/PDU CABINET REPLACEMENT

The HFD/PDU Cabinet will replace one or more of the cabinets shown below.

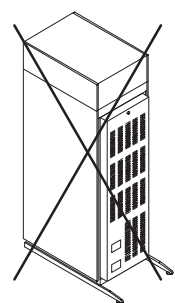
Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

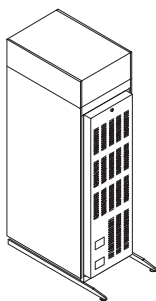


15-1-2 HFD-S/PDU CABINET REPLACEMENT

The ACGD/PDU Cabinet will be retained and upgraded to an HFD-S/PDU Cabinet.



ACGD/PDU CABINET
(MR3)



HFD-S/PDU
CABINET
(MR3)

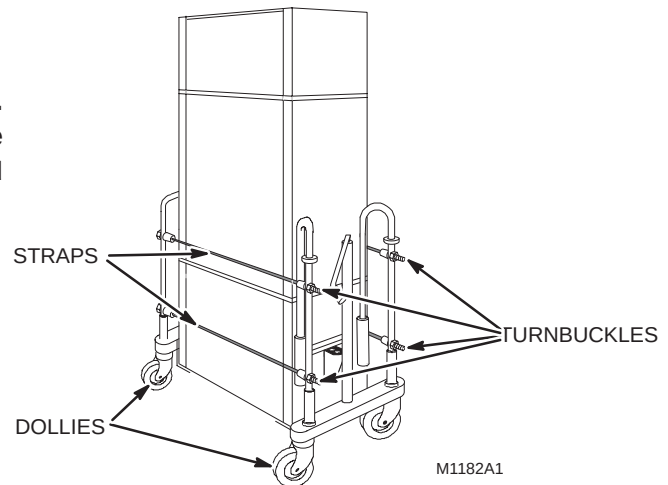
15-2 DE-INSTALL FULL-SIZE PDU (If installed on site)

15-2-1 SPECIAL HANDLING INSTRUCTIONS

All cabinets to be removed, except the Full-size PDU, have built-in casters for ease of movement. The full-size PDU does not have casters and will have to be moved with two shipping dollies placed on each side of the PDU and clamped in place using straps with turnbuckles as shown in illustration below. If dollies are not provided, the Full-size PDU will have to be removed by riggers or other locally supplied equipment moving personnel.

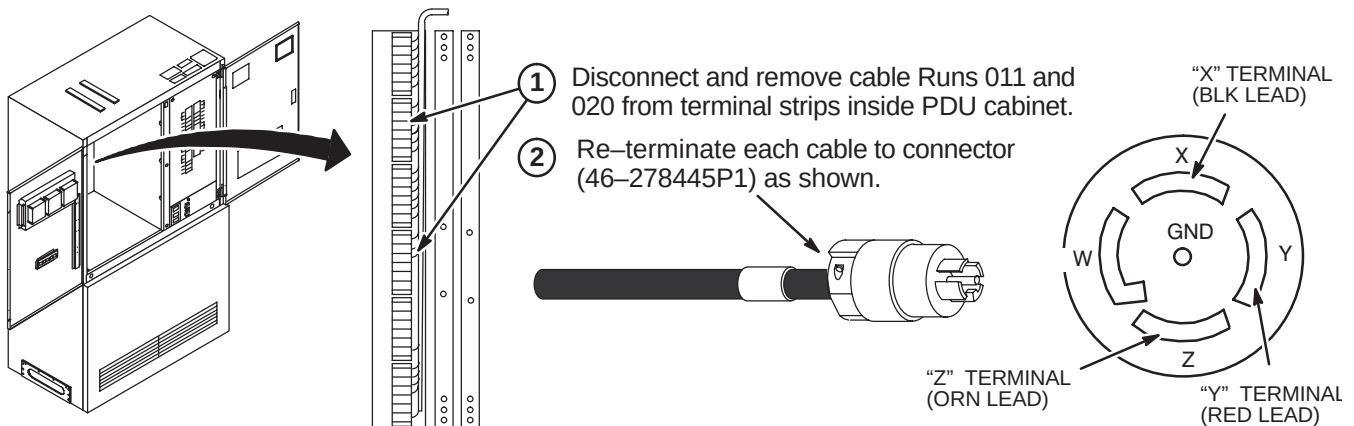
CAUTION

PDU weight is about 1000 pounds (455 kilograms). Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.



15-2-2 RETAIN RUNS 011 AND 020

If part of existing system, the shim and magnet power supply cables (Runs 011 and 020) need to be re-used for the new HFD/PDU cabinet. **DO NOT DISCARD RUNS 011 AND 020.** Refer to Illustration below for instructions on removing and reterminating Runs 011 and 020 for use with new HFD/PDU cabinet.



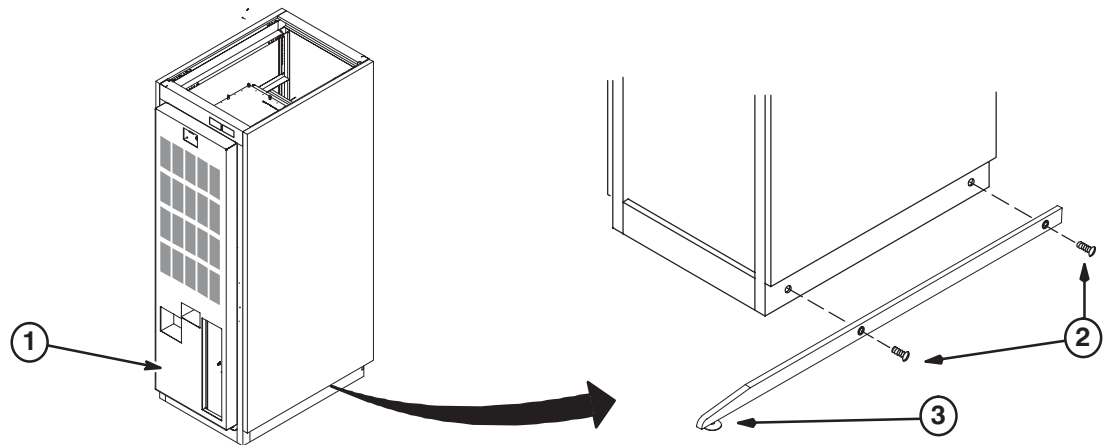
15-3 INSTALL HFD/PDU CABINET

15-3-1 POSITION HFD/PDU CABINET

WARNING!

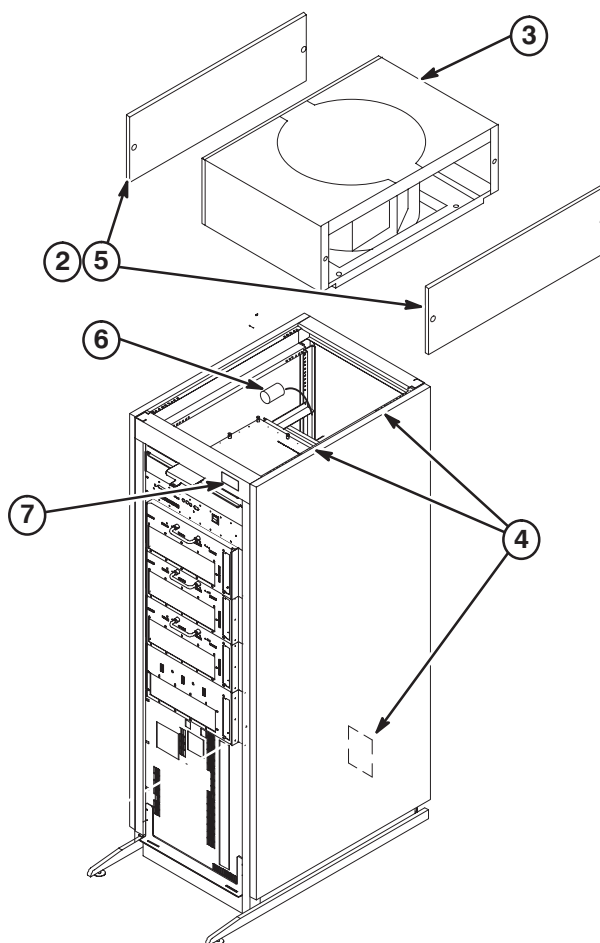
DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

1. Move cabinet to final position as indicated by site layout drawings.
2. Attach the anti-tip bars to cabinet. (Anti-tip bars are shipped separately)
3. Adjust leveling pads to level cabinet.



15-3-2 INSTALL FAN ASSEMBLY AND ATTACH CABINET IDENTIFICATION LABEL

1. Remove Fan Assembly from shipping container.
2. Remove two side trim panels (two screws from each).
3. Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
4. Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet. The fasteners are provided in a plastic bag located in PDU module. Rear PDU covers will need to be removed for access to plastic bags with fasteners.
5. Install two side trim panels with previously removed screws.
6. Connect cable from SGA Power Supply to the fan connector.
7. Make sure the "GRADIENT/PDU" cabinet magnet, part of the Cabinet ID Label Sheet (5126718), is attached to the front of cabinet as shown.

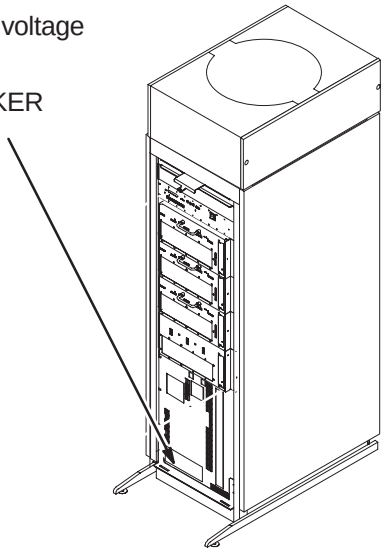
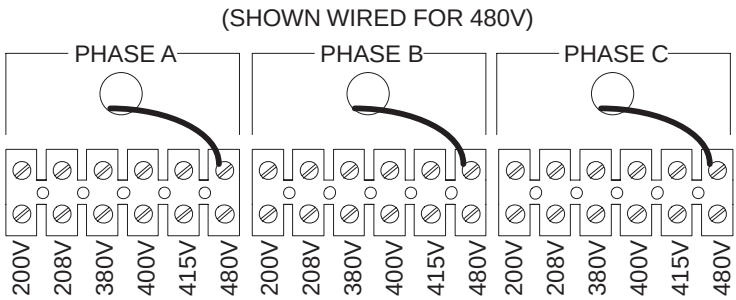




LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

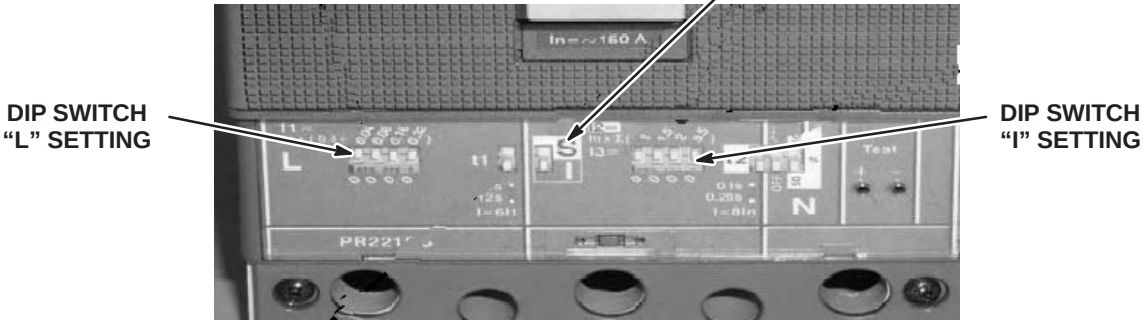
15-3-3 INPUT VOLTAGE SELECTION

- 1 Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
- 2 Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
- 3 For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



15-3-4 CIRCUIT BREAKER DIP SWITCH SETTINGS

- 1 Table below shows the position of the DIP switches for each input voltage selection. Set DIP switches according to input voltage. The white portion indicates the switch.
- 2 The "S/I" switch must be in the "I", down position



INPUT VOLTAGE	200	208	380	400	415	480
"L" SWITCH SETTINGS						
"I" SWITCH SETTINGS						

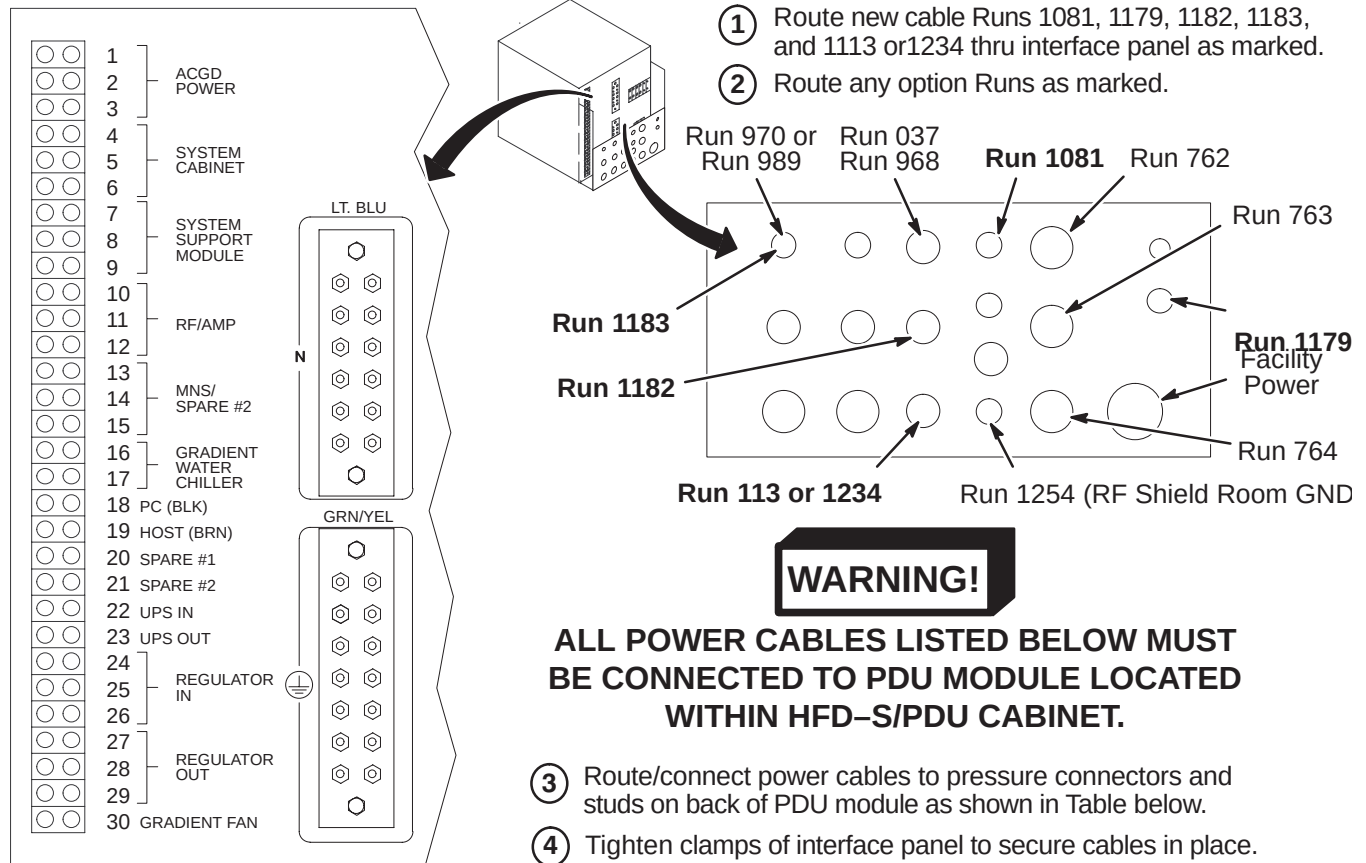
15-3-5 CABLE CONNECTIONS TO HFD/PDU CABINET

Refer to applicable cable “added” map for information on connecting new and retained cables to cabinet.

For gradient cables:

- Non-TwinSpeed sites will install new gradient cables in the equipment room.
- TwinSpeed sites will re-use the gradient cables that have not been disconnected from the retained TAC Cabinet. The other ends of the retained gradient cables will be connected to the new Penetration Panel and HFD/PDU Cabinet.

For all other cable connections, refer to applicable cable installation instructions in Section 4.

15-4 POWER CABLE CONNECTIONS TO HFD-S/PDU

	CONNECT TO	RUN	PHASE	DESTINATION	
NON-TwinSpeed	4,5,6	968	φ A, B, C	RFS Cabinet	3φ, Neutral, GND
	7	1182	φ A	PHPS Module	Single φ Neutral, GND
	10,11,12	1234	φ A, B, C	RF Amp in RFS Cabinet	3φ, Neutral, GND
	16,17	970	φ A,B	Water Chiller (Fixed sites)	Single φ GND
	18,19	1081	φ C,A	Operator Workspace	Single φ, Neutral, GND
	20	973	φ A	Heat Exchanger (Mobile only)	Single φ, Neutral, GND
	30	1183	φ A	Blower Box	Single φ, Neutral, GND
	GRN/YEL	037		System Cabinet Ground Wire	
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	
TwinSpeed	4,5,6	968	φ A, B, C	RFS Cabinet	3φ, Neutral, GND
	7	1182	φ A	PHPS Module	Single φ Neutral, GND
	10,11,12	1113	φ A, B, C	NB RF Amp Cabinet	3φ, Neutral, GND
	16,17	989	φ A,C	TAC Cabinet	Single φ, GND
	18,19	1081	φ C,A	Operator Workspace	Single φ, Neutral, GND
	(MDP)* or		φ A or	AirSys Water Cooled Cabinet	Single φ, Neutral, GND
	30 (PDU)		φ A	AirSys Heat Exchanger	Single φ, Neutral, GND
	GRN/YEL	037		System Cabinet Ground Wire	
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	

COLOR CODE: for remainder of cables:

3φ = Black, Red, Orange Neutral = Light blue

Single φ = Brown, Black

Gnd = Grn/Yel

* Connection to Main Disconnect Panel to be completed by customer electrician.

SECTION 16 – RFS CABINET

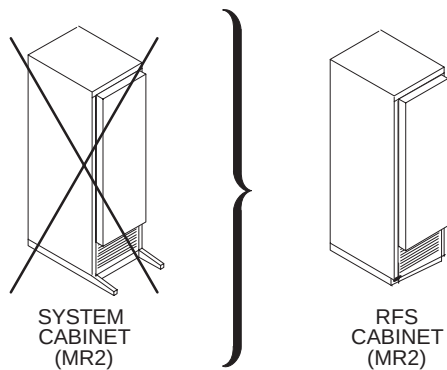
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16-2 POSITION RFS CABINET	16-3
16-3 CABLE CONNECTIONS TO RFS CABINET	16-3

16-1 OVERVIEW

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

The RFS Cabinet will replace the System cabinet shown below.



The de-installed System cabinet is to be returned for recycling. Refer to cable removal map in Section 4 for disconnection and removal of designated cables.

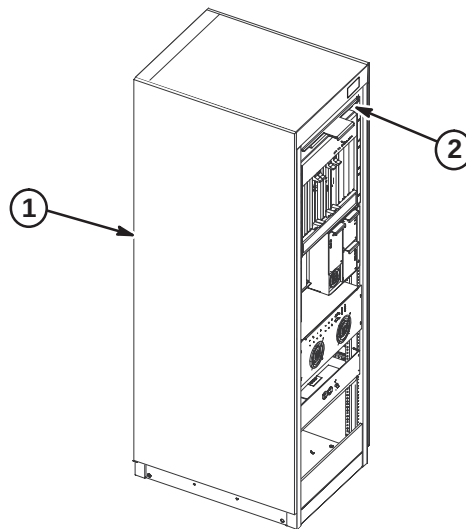
Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

16-2 POSITION RFS CABINET

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position RF/System cabinet to final position as indicated by site layout drawings.
2. From the Cabinet ID Label Sheet (5126718), locate the "RF/SYSTEMS CABINET" magnet and attach to front of cabinet as shown.



16-3 CABLE CONNECTIONS TO RFS CABINET

Refer to applicable cable "added" map in Section 4 for information on connecting new and retained cables to cabinet.

SECTION 17 – COMPLETION

17-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Have the on-site GE Healthcare Field Engineer, Service Engineer, or Project Manager of Installation (PMI) to confirm that the blower box has been anchored properly and installed outside the minimum service area.
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 17-2, 17-3, and 17-4.

17-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 17-3. Recycling Operation will route them to appropriate location.

17-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-8. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

17-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-9. To be performed by GE personnel.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

17-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.

SECTION 18 – APPENDIX

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18-1 PRE-EXCITE TO SIGNA HDx SYSTEM UPGRADE CONFIGURATIONS**18-1-1 SIGNA LX, HORIZON, 5.x, 4.x to HDx UPGRADES OVERVIEW**

The HDxt (Release 15.x) upgrade content for an existing 1.5T Signa LX, Signa Horizon, Signa 5.x, or Signa 4.x system is dependent on the configuration of the system being upgraded and the final product configuration. Table 18-1 lists illustrations which shows a high level summary of Release 15.x upgrade system equipment/cabinets.

TABLE 18-1
Pre-EXCITE to Release 12.x Upgrade Impact

System Configurations		Final System Major Equipment
Existing Pre-EXCITE	Upgraded Release 14.x	
Signa LX, Horizon, 5.x, or 4.x	Signa HDxt 1.5T	Illustration 18-4
Signa 1.5T TwinSpeed (9.x)	Signa HDxt 1.5T TwinSpeed	Illustration 18-5

Refer to the sub-section appropriate to the upgrade being planned:

- Section 18-1-2: Pre-EXCITE with LCC Magnet to HDxt Upgrade
- Section 18-1-3: Pre-EXCITE with Non-LCC Magnet to HDxt Upgrade

18-1-1 SIGNA LX, HORIZON, 5.x, 4.x to HDx UPGRADES OVERVIEW (Continued)

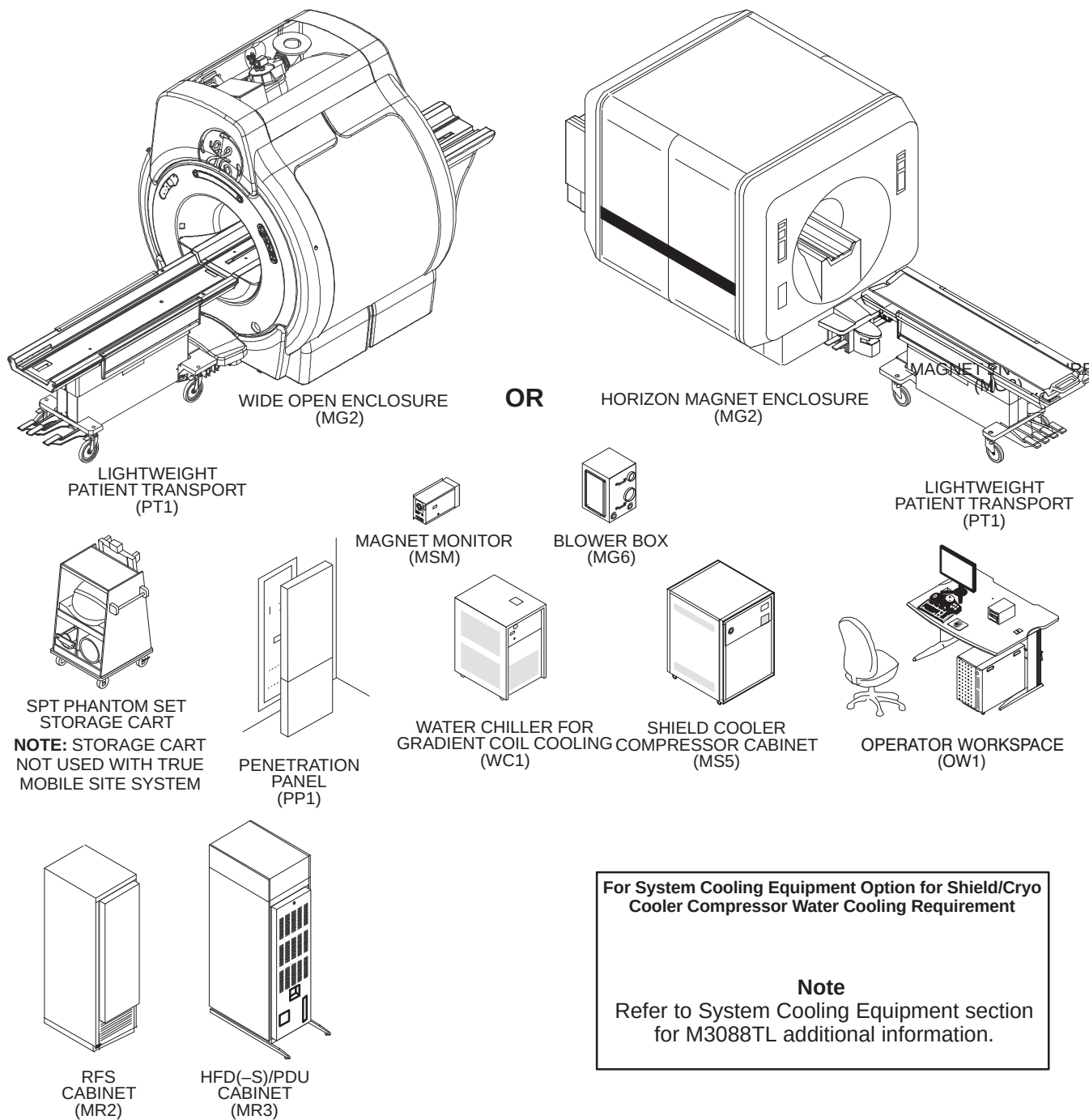


ILLUSTRATION 18-4
PRE-EXCITE UPGRADED TO Release 14.x NON-TWINSPEED SYSTEM MAJOR EQUIPMENT

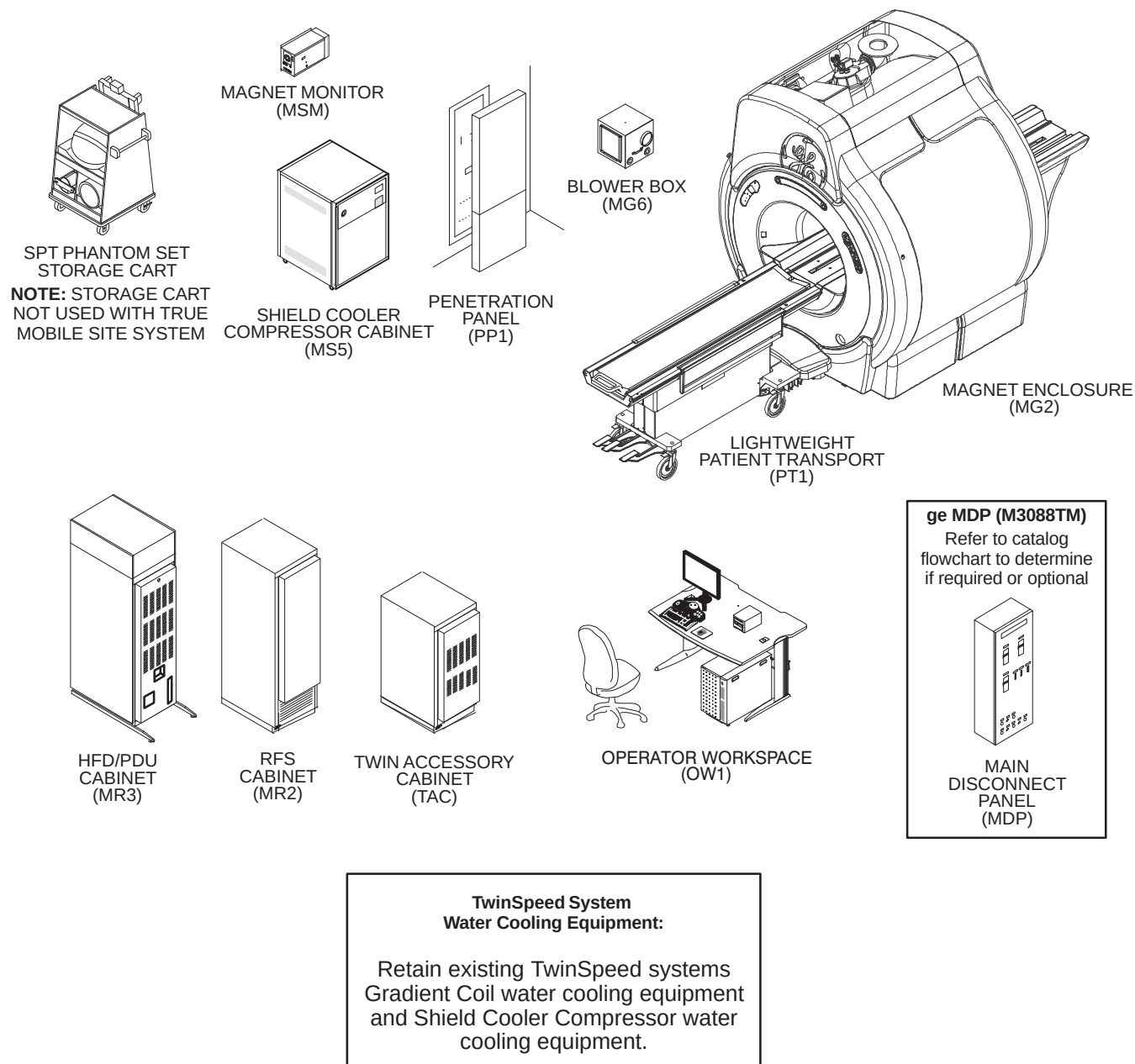
18-1-1 SIGNA LX, HORIZON, 5.x, 4.x to HDx UPGRADES OVERVIEW (Continued)

ILLUSTRATION 18-5

PRE-EXCITE TwinSpeed UPGRADED TO Signa HDx 1.5T TwinSpeed Release 14.x System

18-1-2 PRE-EXCITE WITH LCC MAGNET TO HDx UPGRADE

Signa LX with 1.5T LCC Magnet upgrade to Signa HDxt catalogs are listing in Illustration 18-6. Illustrations 18-7 and 18-18 shows the equipment/cabinets modified, replaced, and added by the listed catalogs.

The following Catalogs and collectors were current at time of publishing this manual and are only provided as a reference. They are subject to change at any time.

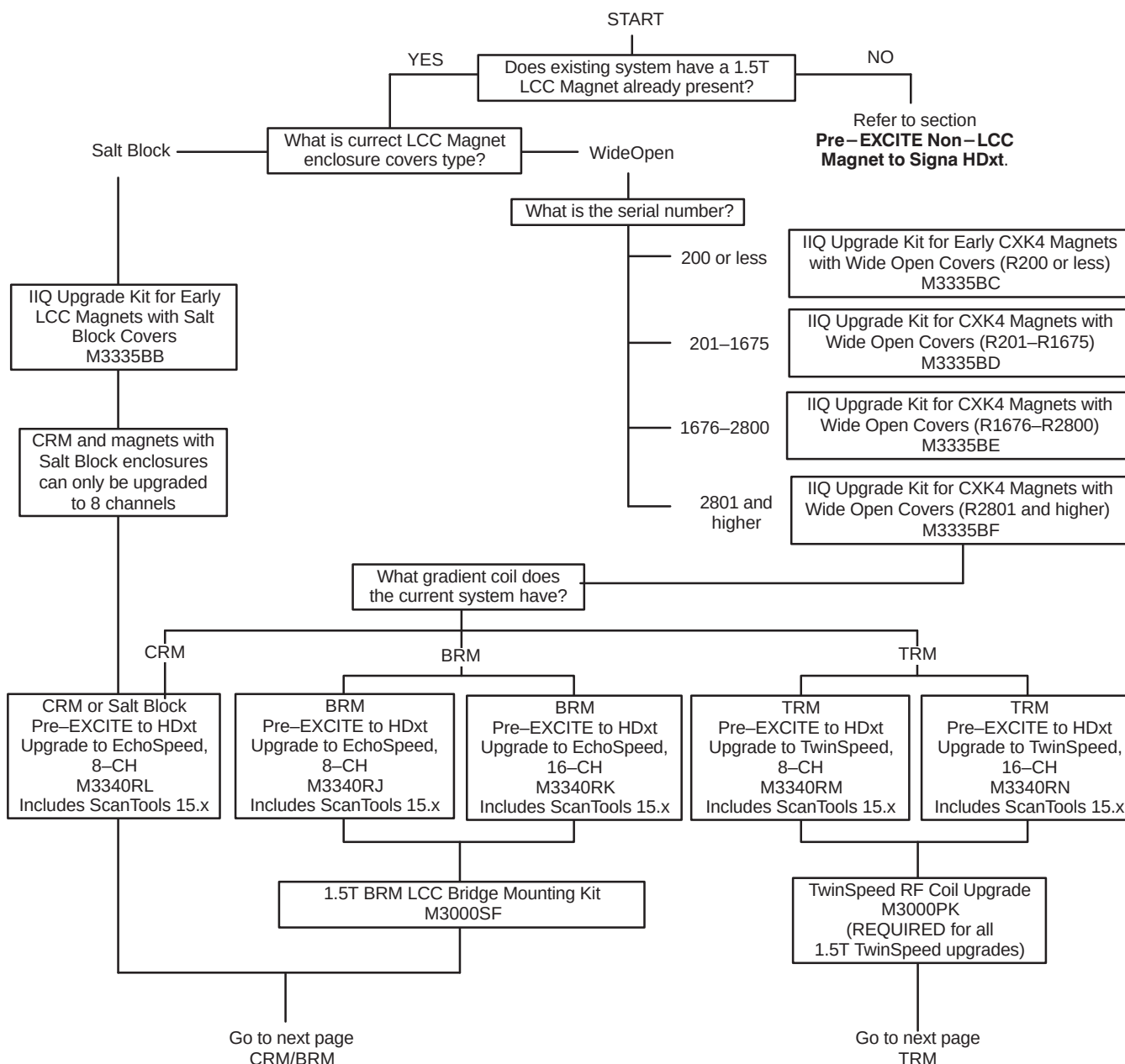


ILLUSTRATION 18-6
SIGNA LX WITH 1.5T LCC MAGNET TO SIGNA HDxt UPGRADE CATALOGS

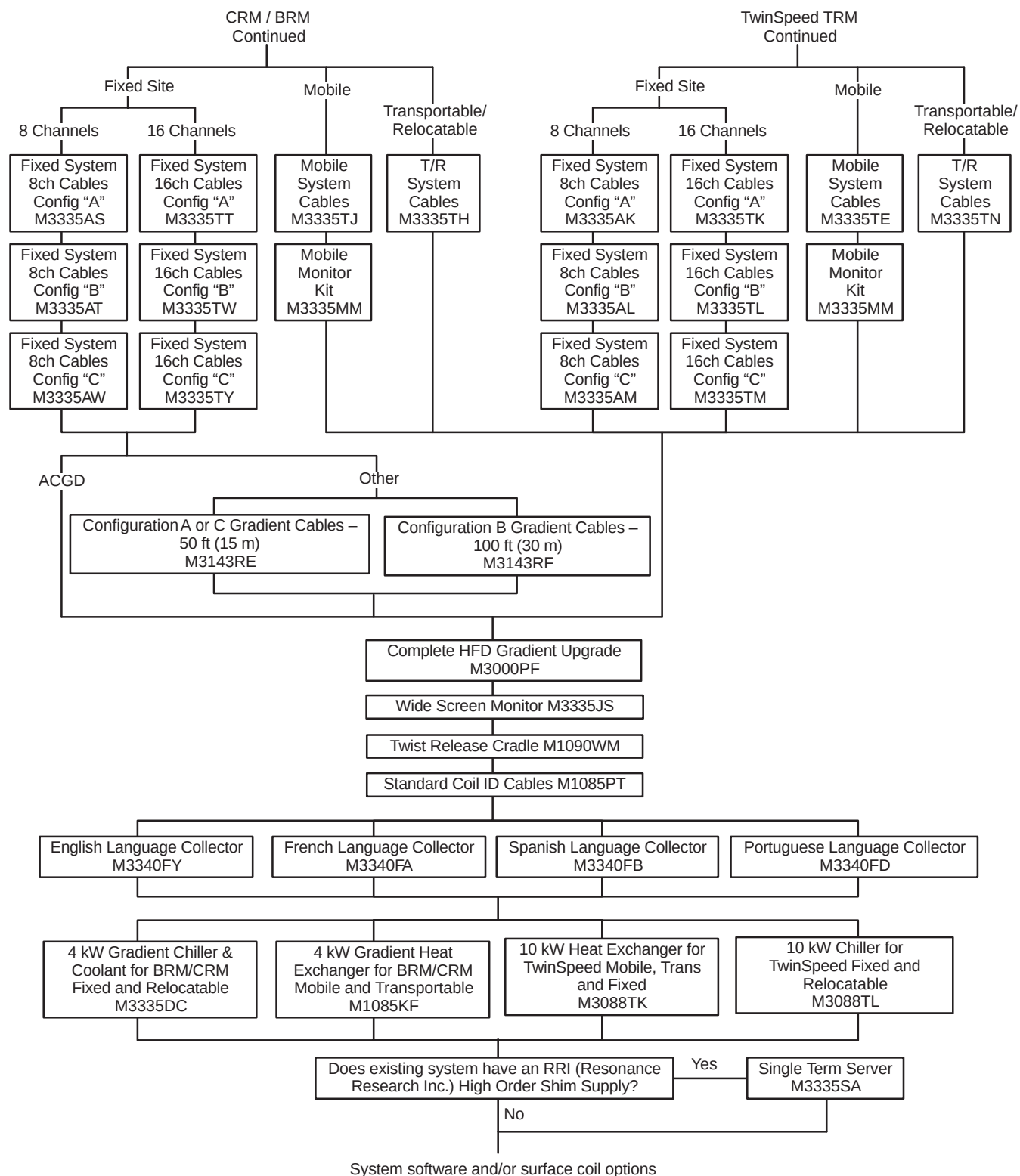
18-1-2 PRE-EXCITE WITH LCC MAGNET TO HDx UPGRADE (Continued)

ILLUSTRATION 18-6 (Continued)

SIGNA LX WITH 1.5T LCC MAGNET TO SIGNA HDxt UPGRADE CATALOGS

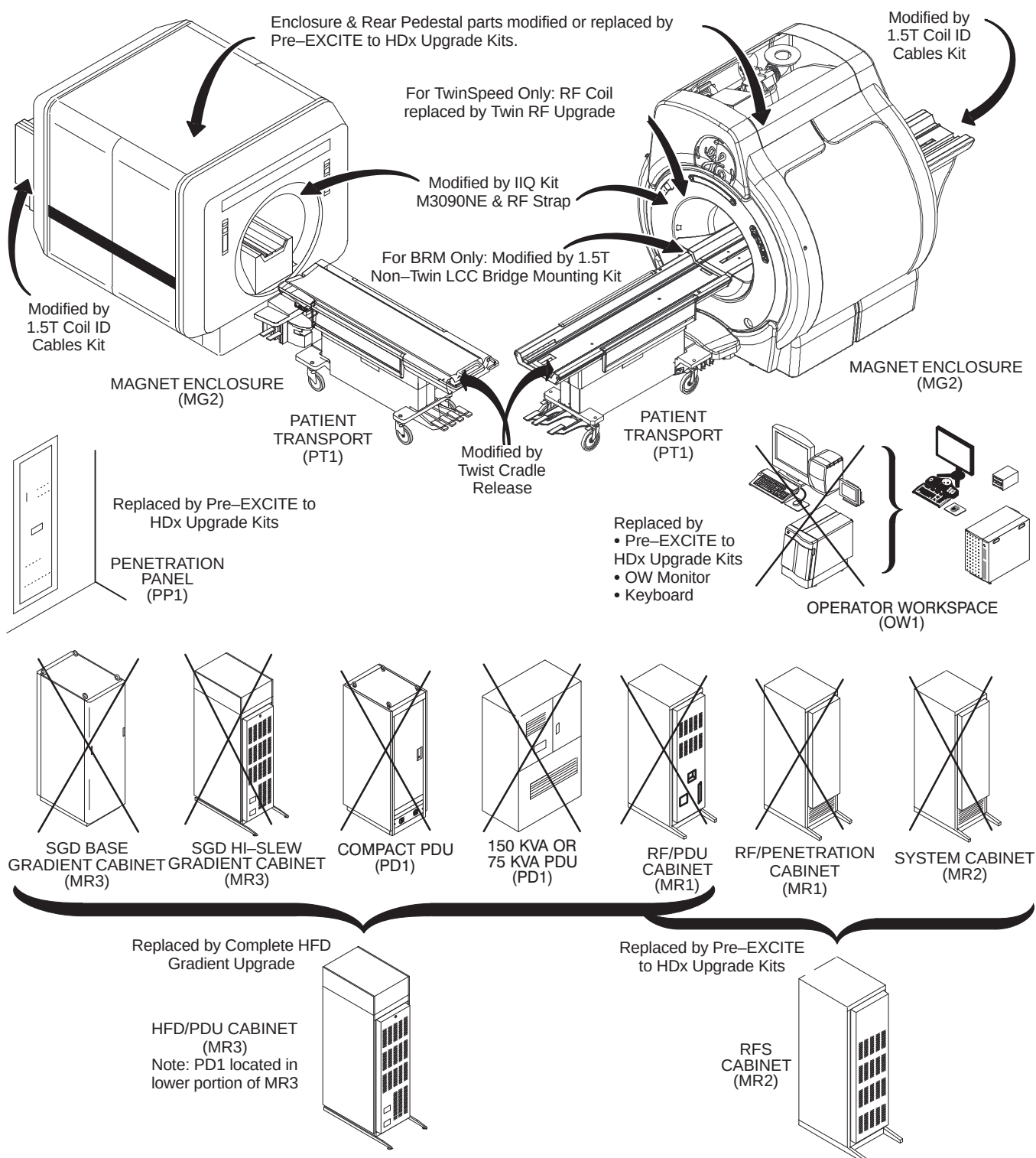
18-1-2 PRE-EXCITE WITH LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-7

SIGNA MR/I & SIGNA LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE

Signa LX, Signa Horizon, Signa Advantage 5.x, and Signa Advantage 4.x with non-LCC Magnet upgrade to Signa HDxt catalogs are listed Illustration 18-8. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as noted below:

- Signa LX see Illustration 18-9
- Signa Horizon (5.x) see Illustration 18-10
- Signa Advantage 5.x see Illustration 18-11
- Signa Advantage 4.x see Illustration 18-12

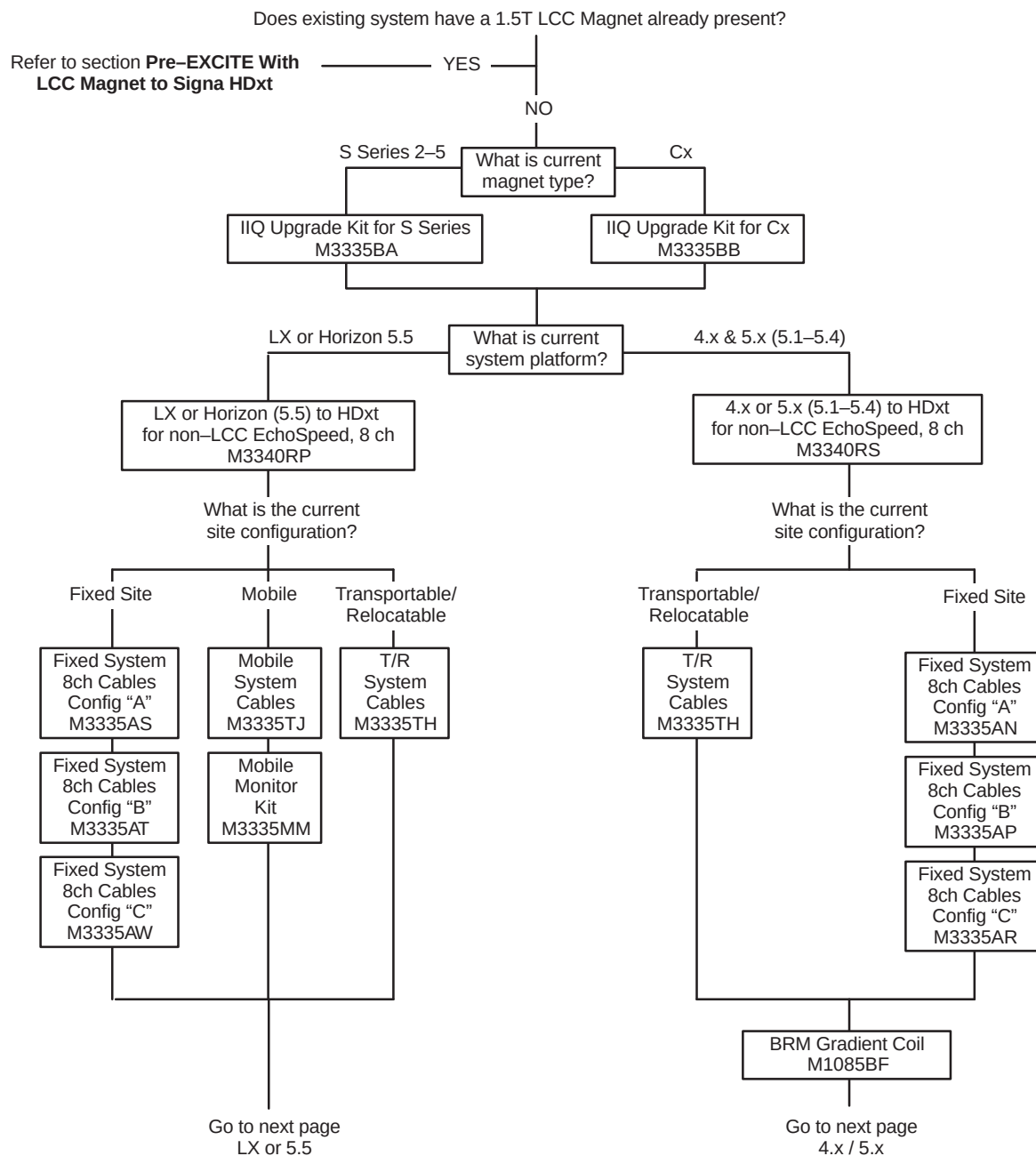
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-8
Pre-EXCITE WITH NON-LCC MAGNET TO SIGNA HDxt UPGRADE CATALOGS

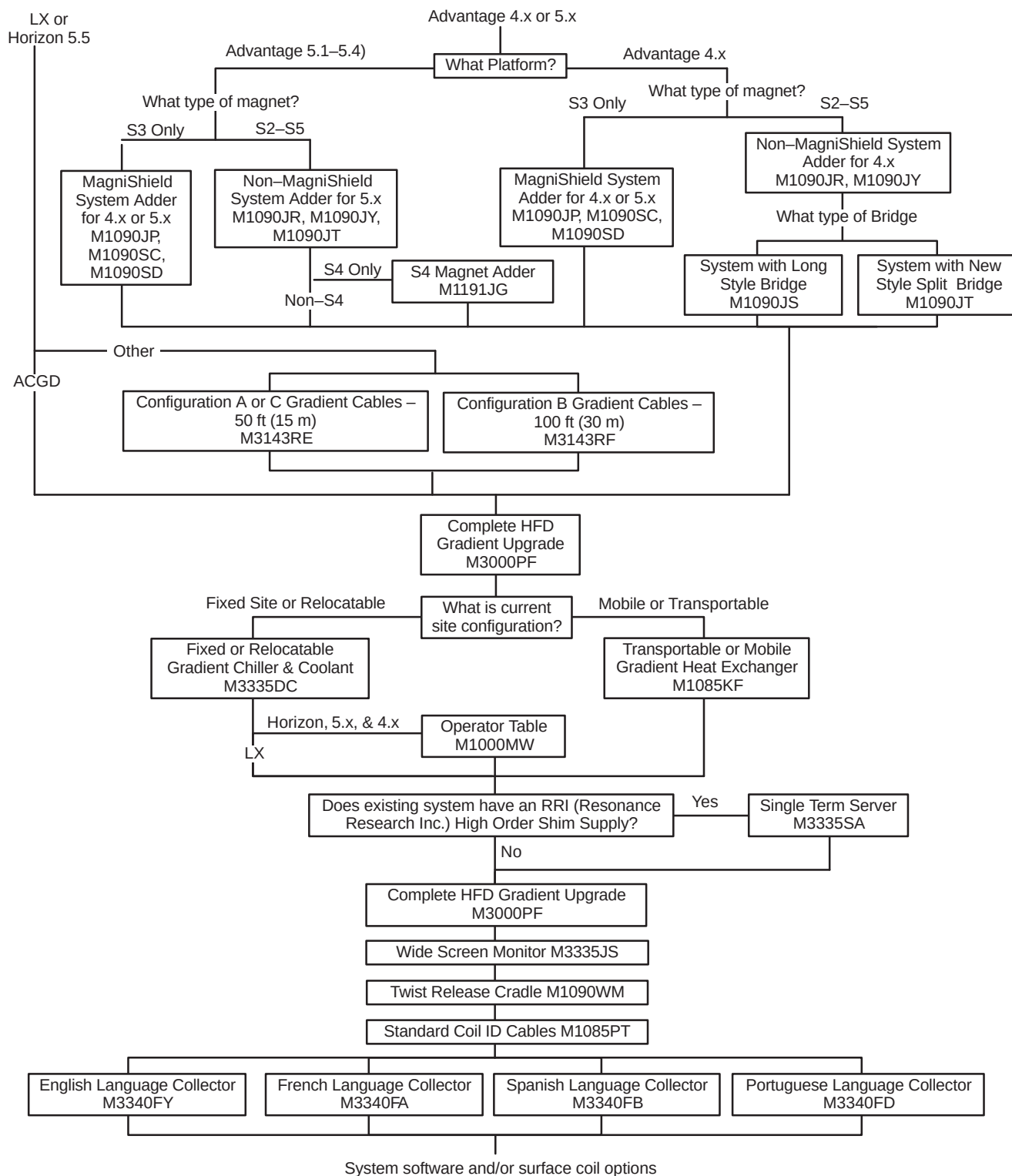
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-8 (Continued)
Pre-EXCITE WITH NON-LCC MAGNET TO SIGNA HDxt UPGRADE CATALOGS

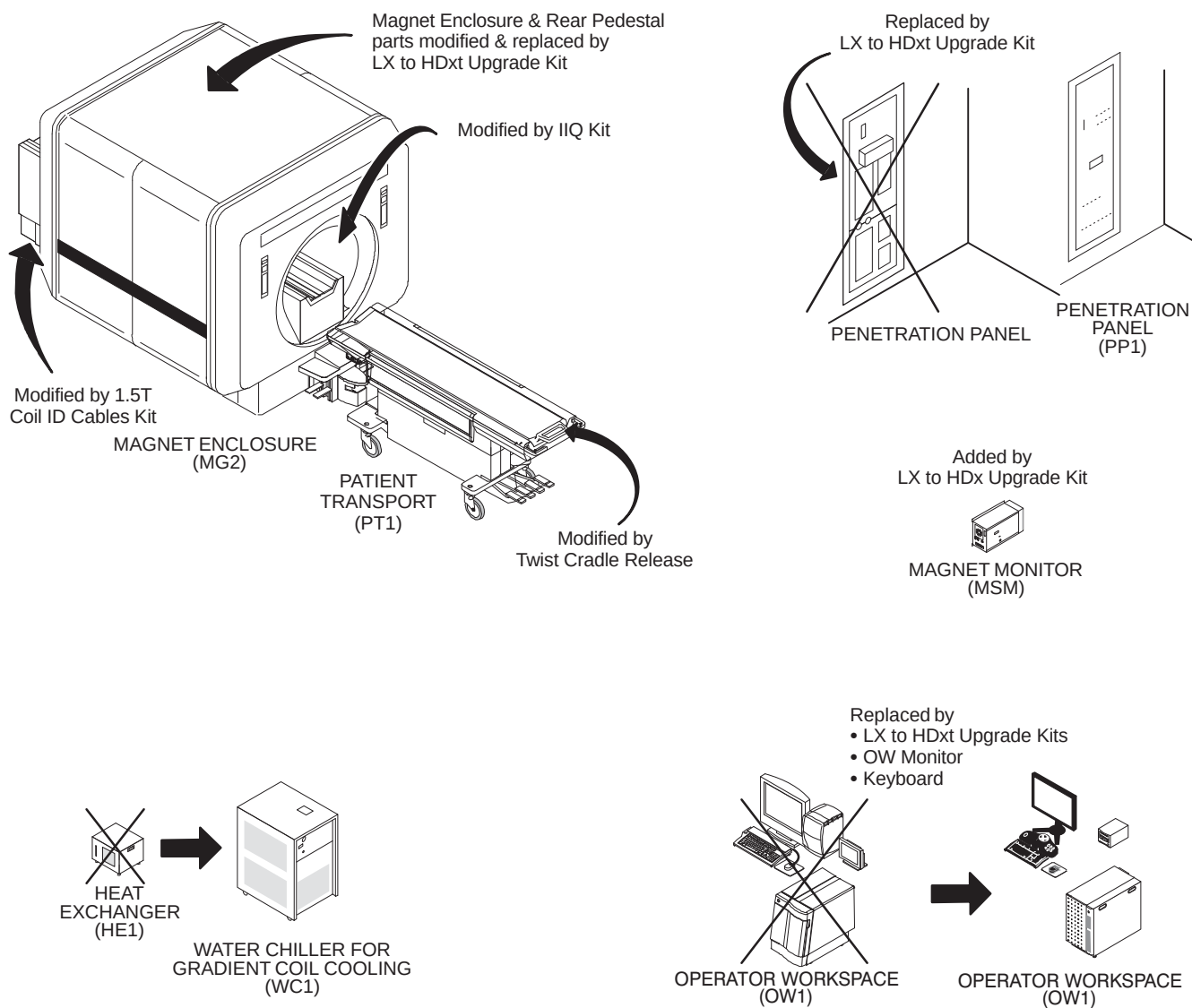
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-9
SIGNA LX EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

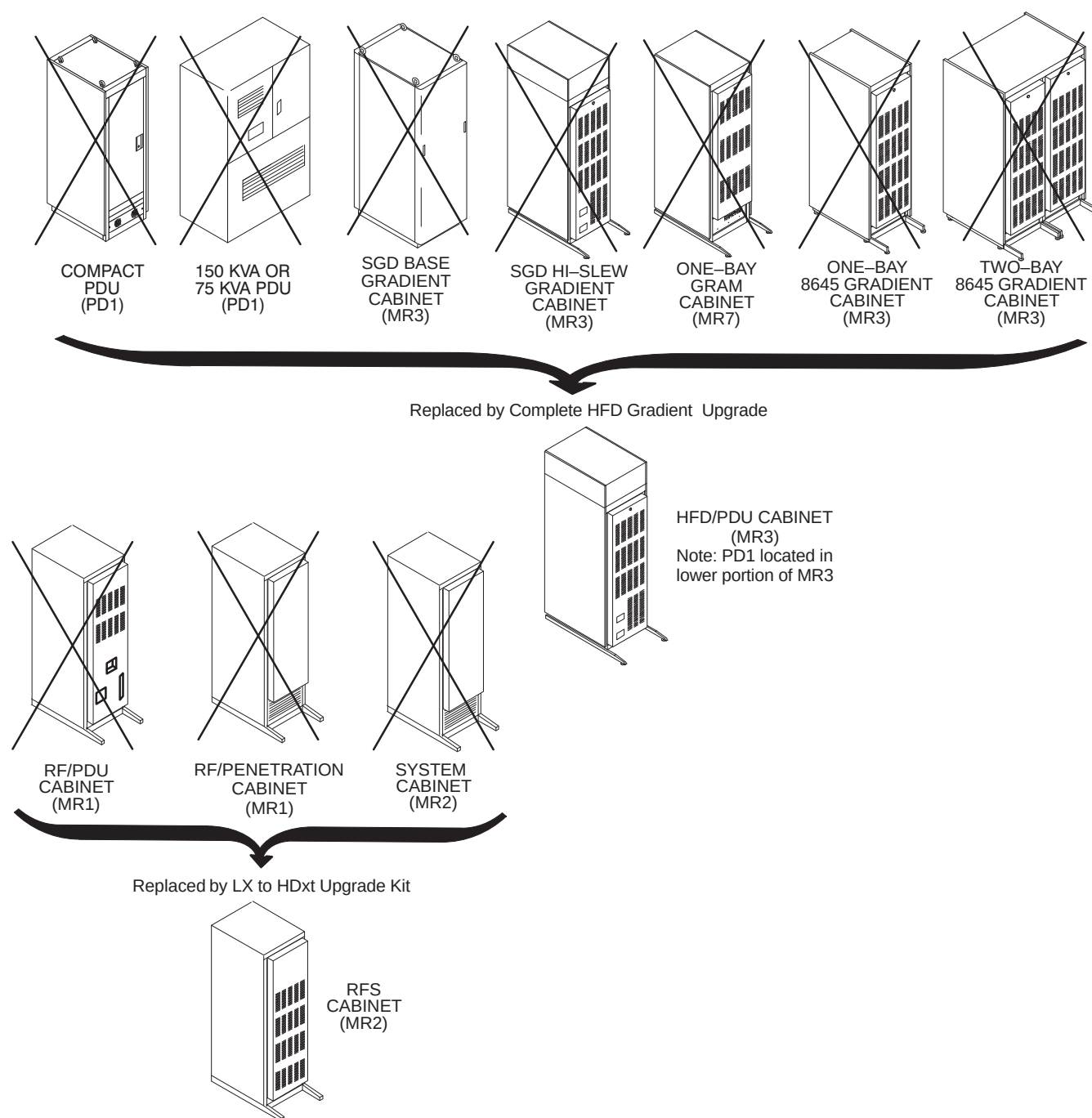
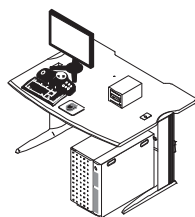
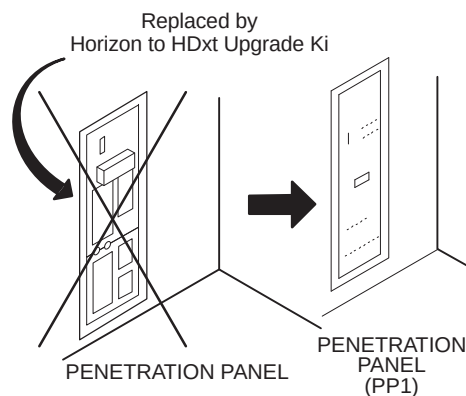
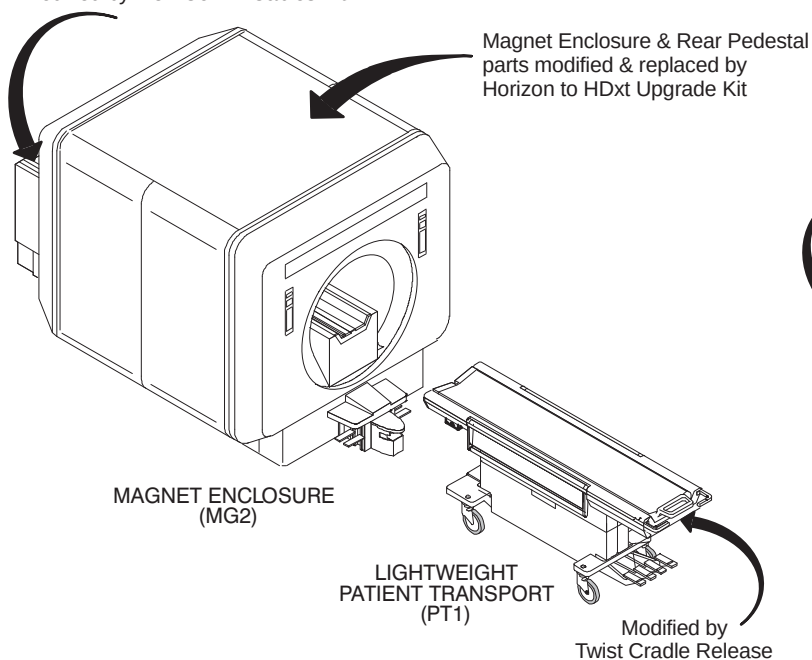
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-9 (Continued)
SIGNA LX EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

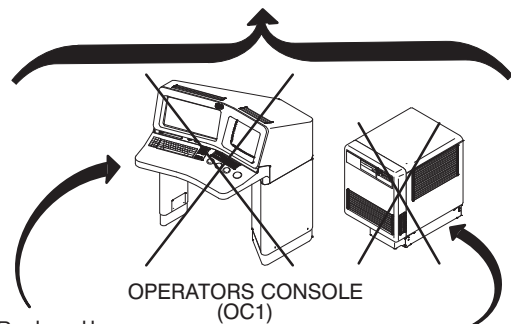
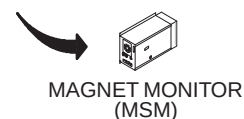
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

Modified by 1.5T Coil ID Cables Kit



OPERATOR WORKSPACE (OW1)

Added by Horizon to HDxt Upgrade Kit



Provided by Gradient Coil Water Chiller

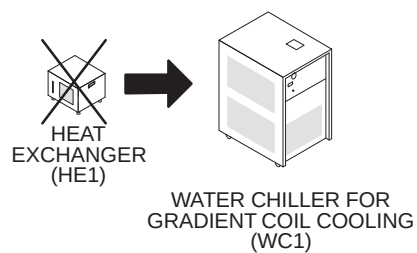


ILLUSTRATION 18-10
SIGNA HORIZON (5.X) EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

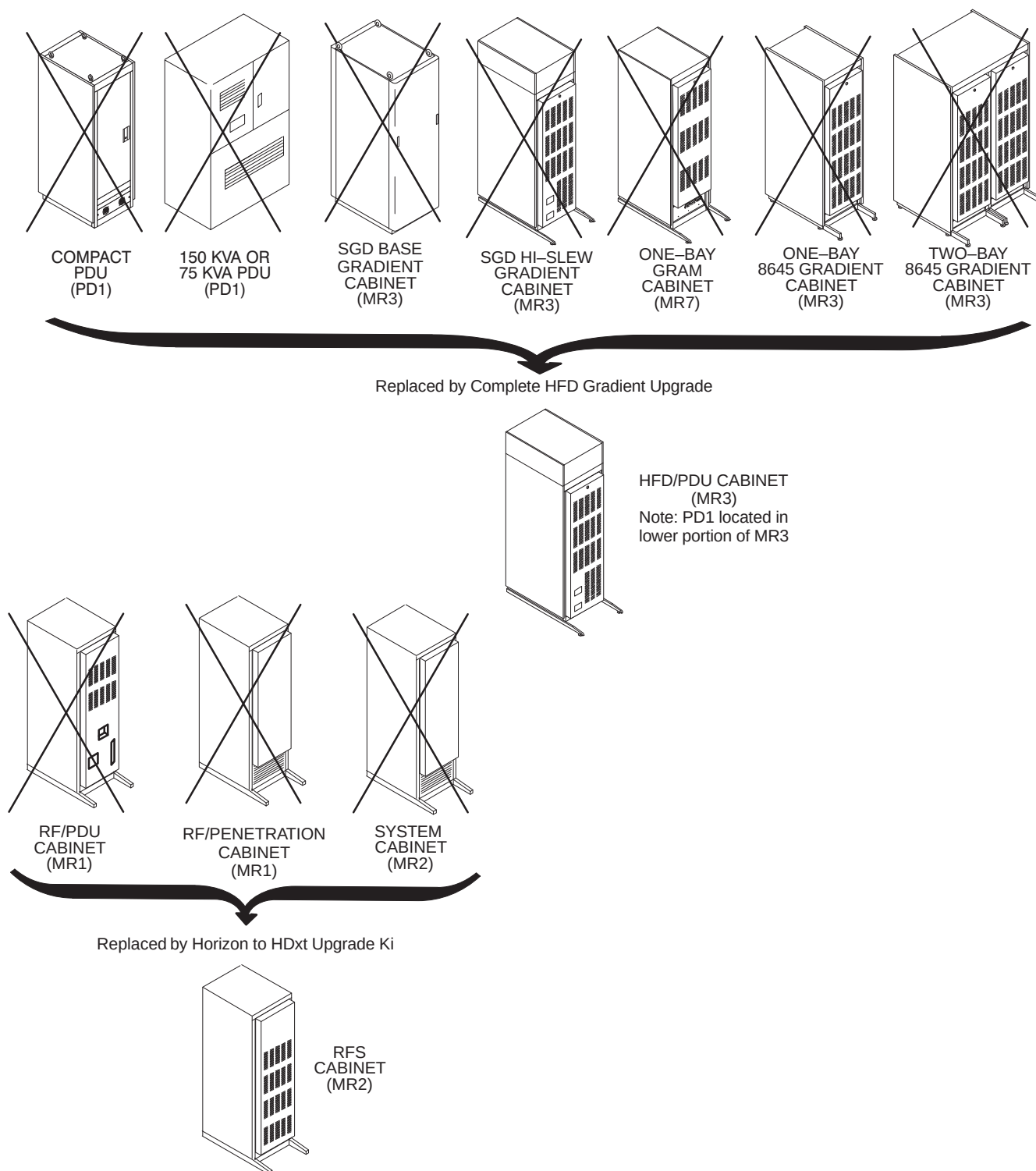
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-10 (Continued)

SIGNA HORIZON (5.X) EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

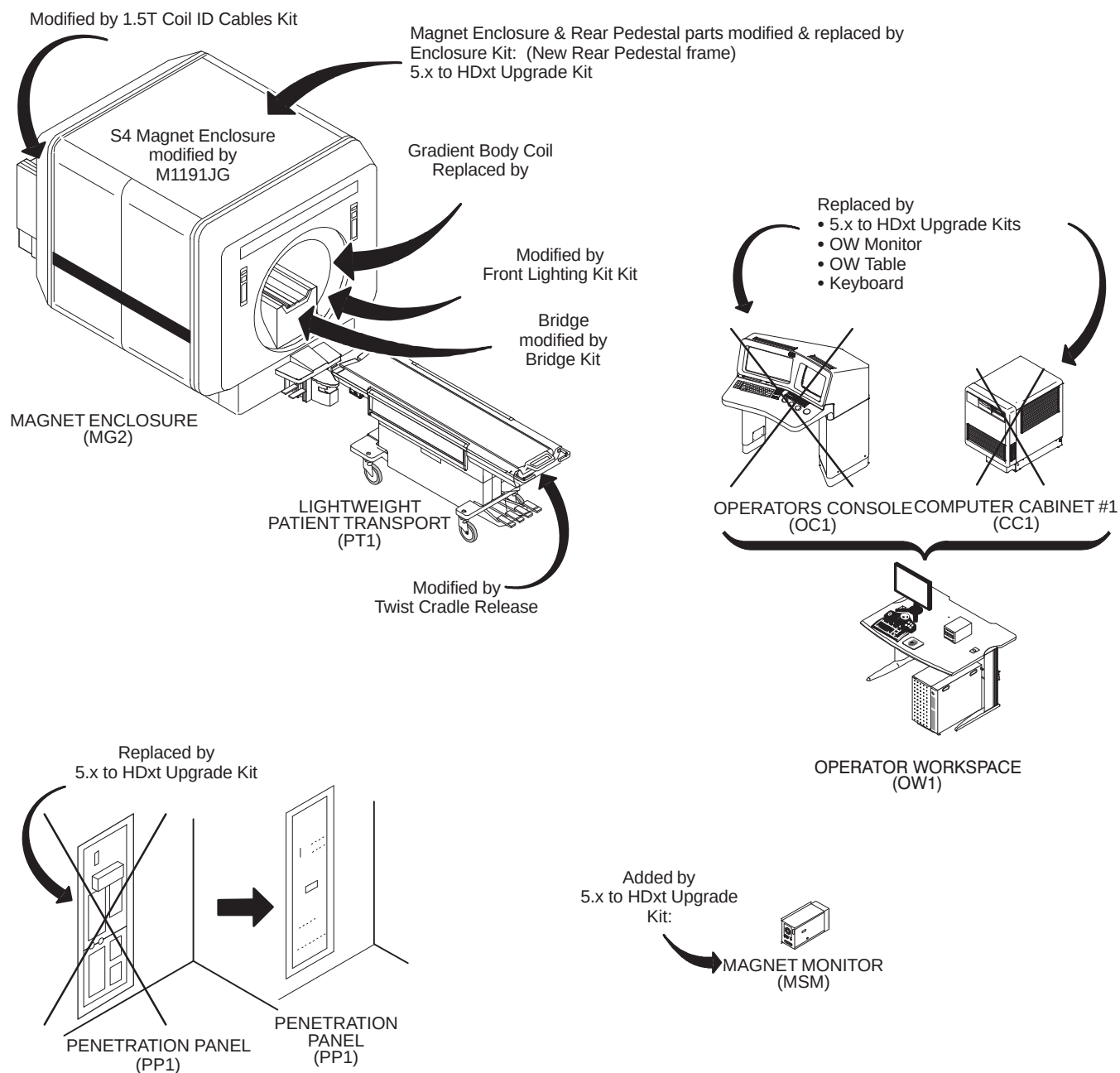
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-11

SIGNA ADVANTAGE 5.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

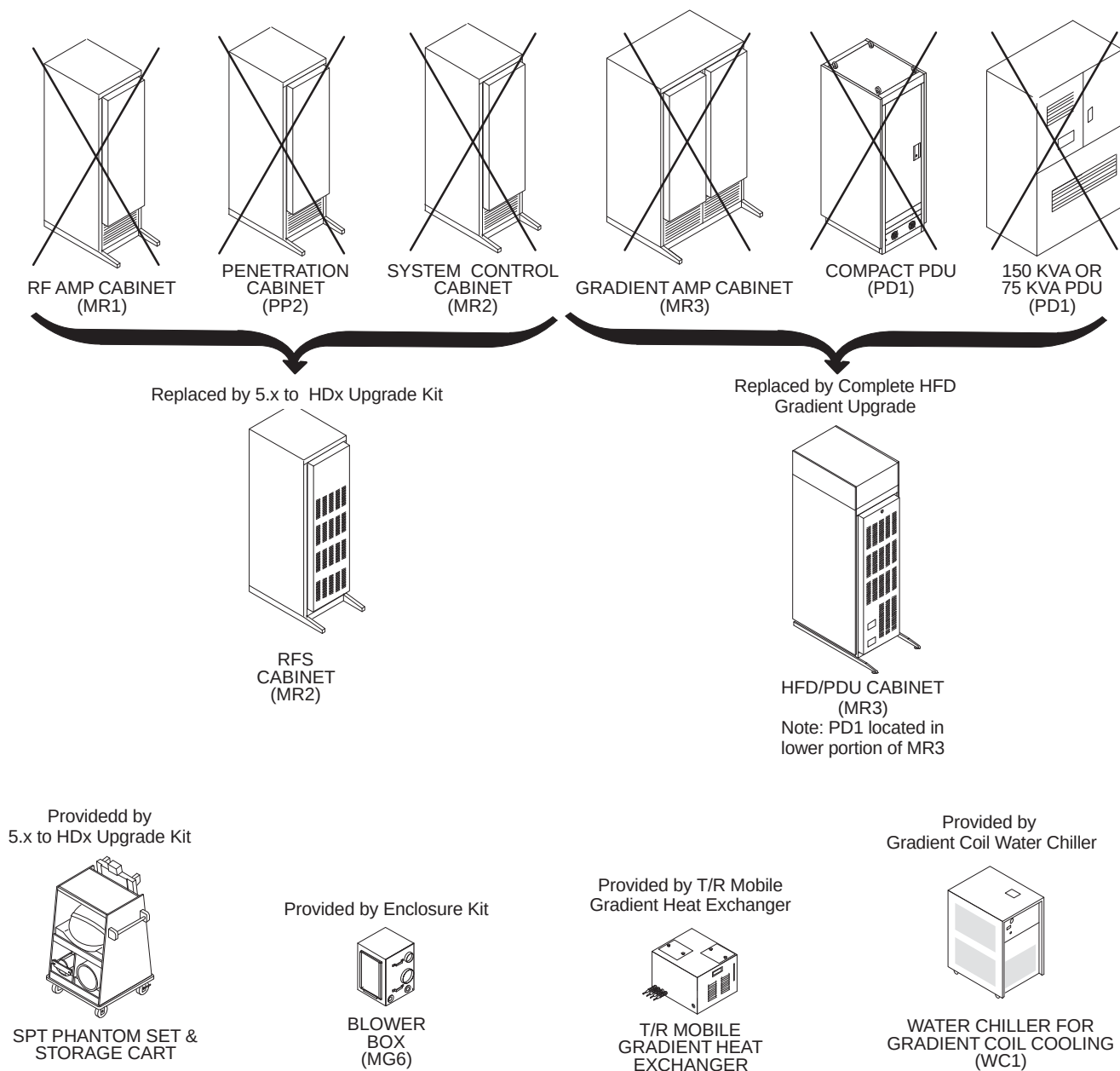
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-11 (Continued)

SIGNA ADVANTAGE 5.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

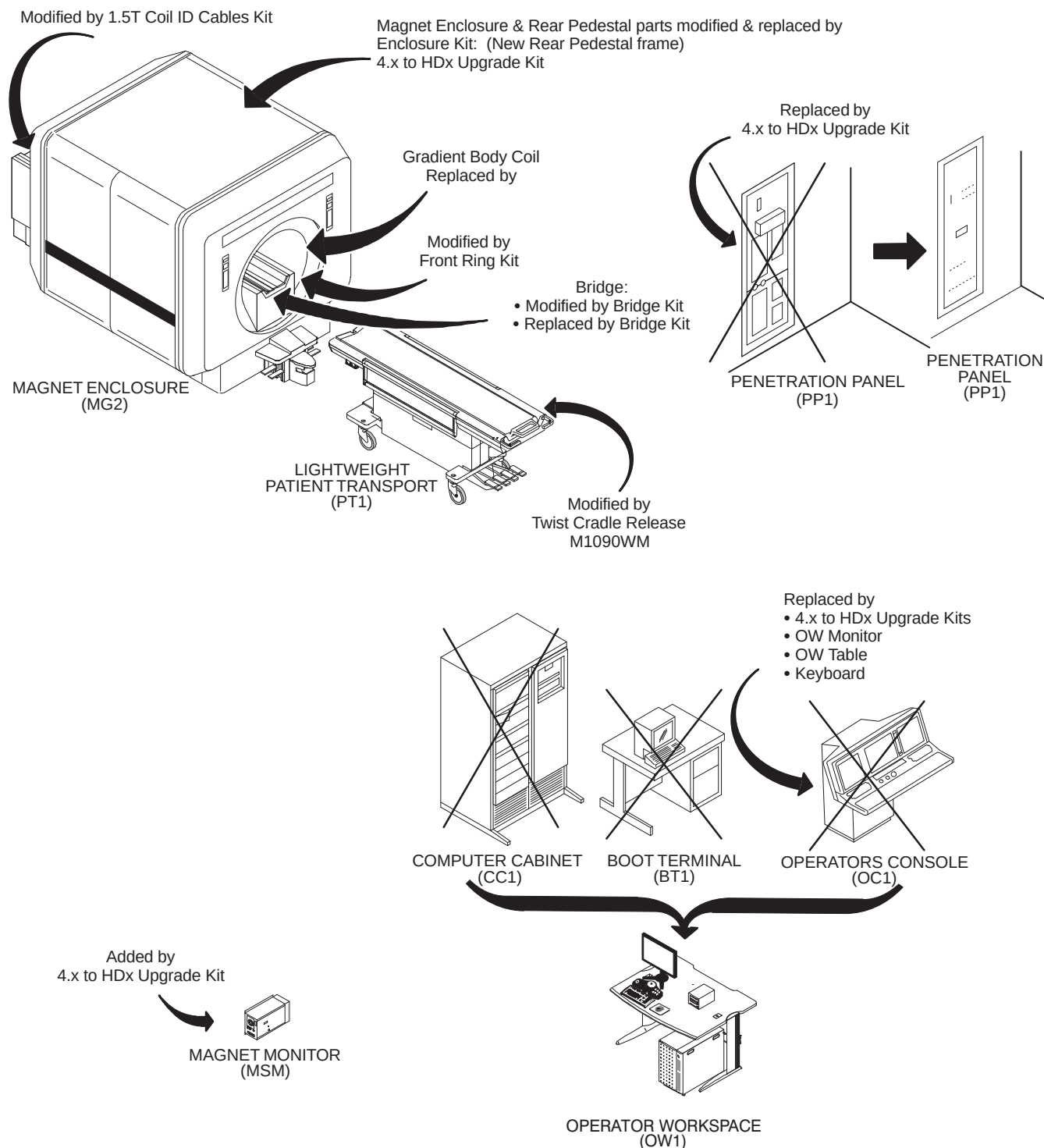
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-12
SIGNA ADVANTAGE 4.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

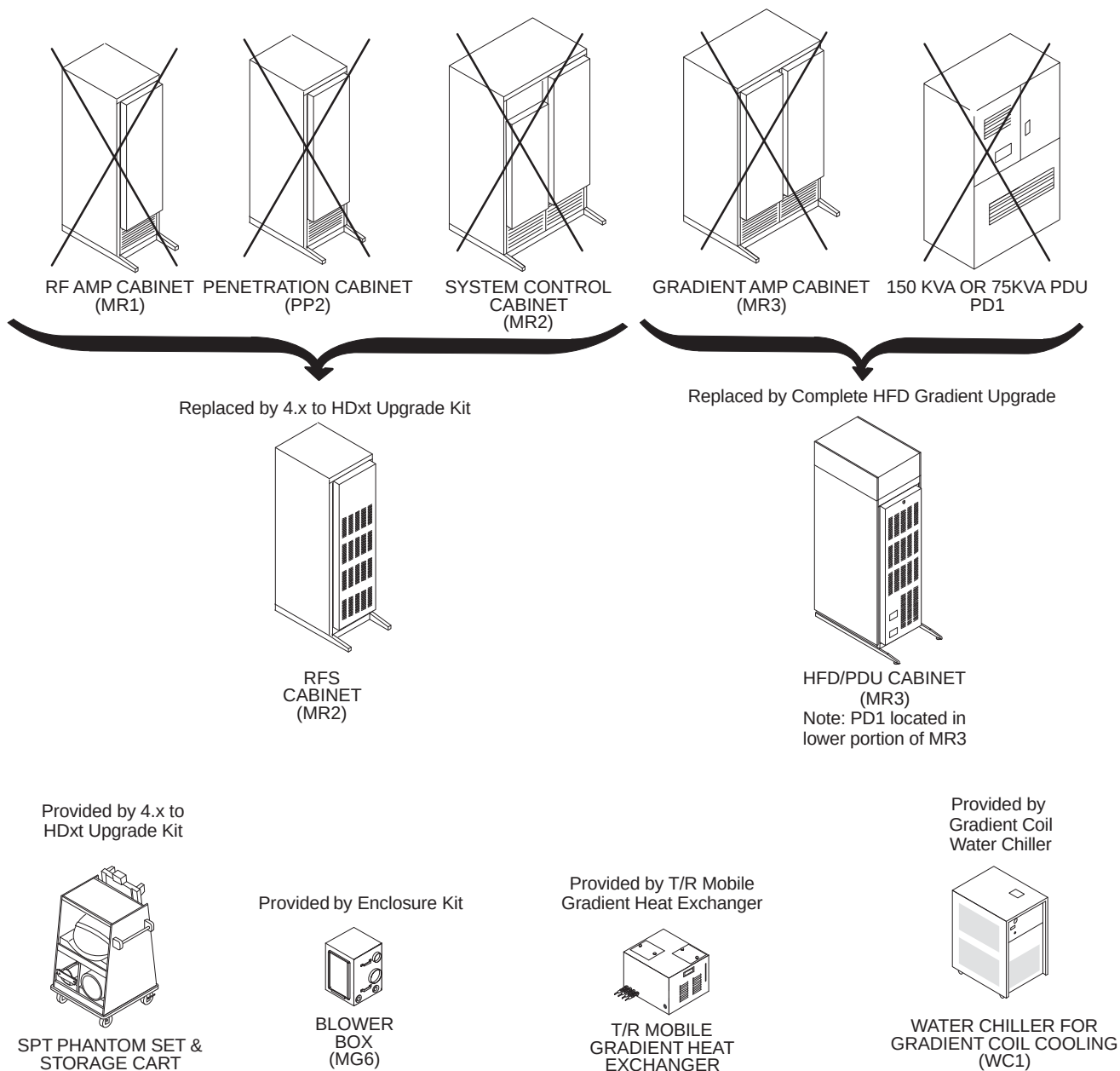
18-1-3 PRE-EXCITE NON-LCC MAGNET TO HDxt UPGRADE (Continued)

ILLUSTRATION 18-12 (Continued)

SIGNA ADVANTAGE 4.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18-2 SYSTEM COOLING EQUIPMENT

The following system cooling equipment is available with the MR system as indicated in the upgrade flowcharts:

- 50 and 60 Hz Water Chiller (WC1): The WC1 is a 4KW Water-to-Air Chiller for Gradient Coil cooling. Fixed site must order Water Chiller.
- Heat Exchanger (HE1): The HE1 is a Heat Exchanger for Gradient Coil cooling on T/R Mobile units
- MR Common Chiller (MRCC): The MRCC is a single-loop 10KW 50/60 Hz water chiller that can be used to cool either Shield/Cryo Cooler Compressor or the TwinSpeed Gradient Coil. The MRCC can be installed either indoors or outdoors. There is a Remote Panel to be installed in the MR system Equipment Room that can stop or restart the chiller and can display supply water temperature, chiller setpoints and alarms. The MRCC power is supplied by the MR system Main Disconnect Panel (MDP). This chiller will be serviced by a third party vendor.
- Gradient Water Heat Exchanger (GWHX): The GWHX is a 10KW 50/60 Hz water-to-water heat exchanger that is used for cooling the TwinSpeed Gradient Coil. The GWHX is for indoor use only. The GWHX requires facility provided chilled water. The GWHX power is supplied by the MR system Power Distribution Unit (PDU). This heat exchanger will be serviced by GE Service. When the GWHX is used then the customer must supply water cooling for the Shield/Cryo Cooler Compressor.

Note

TRM Gradient Coil water cooling must be supplied by cooling equipment (MRCC or GWHX) supplied with the MR system to prevent contamination/damage to the coil and for proper image quality.

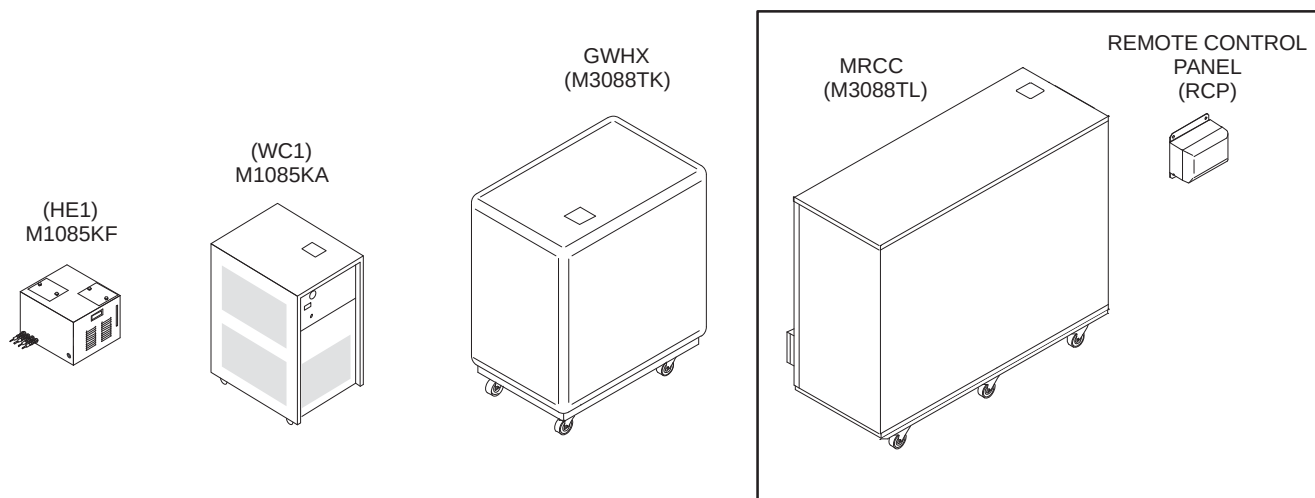


ILLUSTRATION 18-13
SYSTEM COOLING EQUIPMENT SELECTIONS

18-3 CABLE IDENTIFICATION

Cables delivered to site for installation have specifications dependent on its usage. The following tables contain information on cable runs for this system as it was originally installed.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
"FROM" & "TO"	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continuously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material may be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material's ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable's use in the MR system	

18-3-1 PDU POWER AND GROUND CABLES

The cables listed in Table 18-2 are connected at one end to the PDU module in the HFD/PDU Cabinet.

TABLE 18-2
PDU POWER AND GROUND CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW-1	System Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW-1	Magnet Ground Cable
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW-1	Blower Cabinet Separate ground
968	MR1 A12 J1	MR3 PD1	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	RFS Cabinet Power
970	WC1	MR3 PD1	82 ft Power (3-#10)	STO	600	208	90	FT-4	HFD Water Chiller Power
988	PP1 A7 GND	TAC GND	60 ft. Ground (#8)	SO	600	0	80	VW-1	TAC to PP1 ground (3)
989	MR3 PD1	TAC J8	25 ft. Power (3-#10)	SO	600	120	90	FT-4	TAC Power
1081	OW1 A19 J1	MR3 PD1	65.5 ft Power (4-#10)	SO	600	120	75	FT-4	Operator Workspace Power
1182	PP1 PHPS PWR	MR3 PD1	61 ft. Power (3-#10)	AWM	600	120	90	FT-4	PHPS Module Power
1183	PP1 F7, F8	MR3 PD1	41 ft Power (3-#10)	SO	600	0	105	VW-1	Blower Cabinet Power
1234	MR2 RF AMP	MR3 PD1	35.5 ft Power (4-#8)	SO	600	208	90	FT-4	RF Amplifier Power
1254	MR3 PD1	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW-1	Magnet Ground Cable
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5-#8) With 30A Plug and Receptacle						Cable for PDU to GE Magnet Power Supply or SC Shim Supply Power

18-3-2 MAGNET ROOM CABLES NOT CUT TO LENGTH

The cables listed in Table 18-3 are routed and connected in the Magnet Room and are not cut to length.

TABLE 18-3
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Table Elevation motor power
414	MG3 A35	MG3 A2 J9	30 ft. 9-pin Sub-D	CI3	300	<30	80	FT-4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Long Drive Encoder/Limit Sw
458	PP1 J17	OM3	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional Remote O2 Sensor
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	PAC Power
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT-4	Patient Cooling Blower power
786	PP1 F7, F8	MG6 J2	40 ft. 2 cond power cable	SO	600	120	80	FT-4	Gradient cooling blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub-D/rt. angl	CI3	300	<30	80	FT-4	Operator Display to PAC II
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Operator Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Operator Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW-1	Splitter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub-D/26 pin	CI3	300	<30	80	FT-4	Operator Display to SRI

18-3-2 MAGNET ROOM CABLES NOT CUT TO LENGTH (Continued)

TABLE 18-3 (CONT'D)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1152	PP1 A5 J1	MG2 A12 A3 P1	29.5 ft. 37 pin CPC to 16 pin	PLTC	300		105	FT-4	Shim Filter Cable
1197	PP1 RFTXSW J150	MG3 16CHSW J25	49 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	
1200	PP1 J200	MG3 16CHSW J1	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1201	PP1 J201	MG3 16CHSW J2	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1202	PP1 J202	MG3 A17 J3	2-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1203	PP1 MC SW FLTR J203	MG3 16CHSW J11	49 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	
1215	MG3 16CHSW J20	MG2 SRI J20	37 pin Sub-D	CI3	300	<30	80	FT-4	
1216	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
1217	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	SRI Power
1250	MG3 A3 J1	CONNECT TO RUN 257							
1260	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9-Sub D	CI3	300	<30	105	FT-4	Bore Temperature Sensor
1261	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80-#28)	CI2	300	<30	80	VW-1	SRI Splitter to Oper Display

18-3-3 MAGNET MONITORING CABLES

The Magnet Monitoring cables listed in Table 18-4 are delivered with the Magnet.

TABLE 18-4
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8-#22) & Data Cable (4-#22)	CI3	300	<30	80	FT-4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT-4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8-#24)	CI3	300	<30	80	FT-4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8-#22)	CM	300	<30	80	FT-4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6-#24)	CM	300	<30	80	FT-4	Compressor to Compressor I/F
939	MDP UPS	MSM4	6 ft. Power (5-#15)	SO	600	120	85	VW-1	MDP to UPS
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor to Modem
1077	MSM1 J5	OW1 A2 A4 PRT7	80 ft. Cat 5 Ethernet	CI3	300	<30	80	FT-4	Monitor to OW Ethernet
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

18-3-4 GE MAGNET CABLES

The Magnet cables listed in Table 18-5 arrive on site with the magnet and are installed at the time of Magnet delivery.

**TABLE 18-5
GE MAGNET CABLES**

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12-#22)	CI3	150	60 VDC	80	FT-4	Switch Heaters
606	MS4	MS1 A3 A1	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Emergency Run Down Unit
609	MS7 OR MS8	PDU	Power Cable (5-#16)	SO	300	208Y	105	FT-4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Cold Head Control Cable

18-3-5 FIBER OPTIC CABLES

The Fiber Optic cables listed in Table 18-6 are routed and connected in the Magnet Room and the Equipment Room.

**TABLE 18-6
FIBER OPTIC CABLES**

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
711/ 712	PP1-J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC-II
1045	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1083	MR2 A11 J22	OW1 A15 XMT1	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1179	MR2 A11 J17	MR3 A11 J2	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic cables

18-3-6 EQUIPMENT ROOM AND MAGNET ROOM CABLES AND HOSES CUT TO LENGTH

The Gradient cables and the RF Head and Body cables listed in Table 18-7 will be routed, cut, and terminated at designated cable ends according to instructions found in the CABLE INSTALLATION section of this manual.

TABLE 18-7
EQUIPMENT ROOM AND MAGNET ROOM CABLES & HOSES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
235	CONNECT TO RUN 1259	PP1 J44	Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	1.5T Body Coil Run created from kit
257	PP1 J44	CONNECT TO RUN 1250	Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	1.5T Body Coil Transmit
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
762	MR3 CABI/F J3	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT-4	X Gradient output
762	MG3 A32 1/2	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT-4	X Gradient output
763	MR3 CABI/F J4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT-4	Y Gradient output
763	MG3 A32 3/4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT-4	Y Gradient output
764	MR3 CABI/F J5	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT-4	Z Gradient output
764	MG3 A32 5/6	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT-4	Z Gradient output
976	MR3 CABI/F J3	TAC $\pm$ X	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Power
977	MR3 CABI/F J4	TAC $\pm$ Y	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Power
978	MR3 CABI/F J5	TAC $\pm$ Z	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Power
979	TAC X+,WB-,ZM-	PP1X+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Filter
980	TAC Y+,WB-,ZM-	PP1Y+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Filter
981	TAC Z+,WB-,ZM-	PP1Z+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Filter
982	PP1 ZZM+,ZM-	MG2 A12 A1 7/8	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Zzm Gradient Output
983	PP1 YZM+,ZM-	MG2 A12 A1 9/10	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Yzm Gradient Output
984	PP1 XZM+,ZM-	MG2 A12 A1 11/12	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Xzm Gradient Output
985	PP1 ZWB+,WB-	MG2 A12 A1 1/2	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin ZWB Gradient Output
986	PP1 YWB+,WB-	MG2 A12 A1 3/4	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin YWB Gradient Output
987	PP1 XWB+,WB-	MG2 A12 A1 5/6	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin XWB Gradient Output
1198	PP1 RFTXSW J151	MG3 PEDI/F J1	Times Microwave LMR-600FR	CATV R	4000 VDC		85	FT-4	RF Head Transmit
1199	PP1 RX TX SW J153	MG3 PEDI/F J2	Times Microwave LMR-600FR	CATV R	4000 VDC		85	FT-4	RF Head Transmit
1238	PP1 J4	CONNECT TO RUN 1249	Times Microwave LMR-600FR	CATV R	6000 VDC		85	FT-4	RF Head Transmit

18-3-7 EQUIPMENT AND OPERATOR ROOM CABLES

The cables listed in Table 18-8 are routed and connected between Penetration Pnael, cabinets in equipment room and Operator Workspace.

**TABLE 18-8
EQUIPMENT & OPERATOR ROOM RUNS**

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional O2 Monitor cable
611	TAC J22	PP1 SHIM							Resistive Shim P.S. Option
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	RF Door switch
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	PDU Status (ACGD)
974	MR3 PD1 J5	TAC J12	49 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Twin GIP/GP
1056	OW1 A17 PRT6	OW1-A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Laptop Access cable
1059	OW1 A17 PRT2	OW1-A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Lan Hub cable
1060	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	GOCAA Power cord
1061	OW1 A15 KYBD/MS	OW1 A21 J8	2.9 ft. 25 pin Sub-D 2 PS/2s	CI3	300	<30	90	CL-2	Keyboard/Mouse cable
1063	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW-1	Host LCD Display Video cable
1064	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW-1	SCSI Tower Data cable
1065	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host LCD Display Power cable
1066	OW1 A15	OW1 A13	3.9 ft. 25/9 pin Sub-D	CM	300	<30	80	FT-4	Modem I/O cable
1067	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW-1	DASM I/O cable
1068	OW1 A15 Video2	OW1 A3 LCD	9.9 ft. Data (15-#28)	CI3	300	<30	80	VW-1	LCD Display Video cable
1069	OW1 A11 J8	OW1 A3 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	LCD Display Power cable
1070	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	SCSI Tower Power cable
1071	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host Computer Power cable
1074	OW1 A19 J2	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1075	OW1 A19 J4	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1082	OW1 A15 ETH SLOT 2	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT-4	EXCITE Ethernet C'cable
1085	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Intercom
1180	MR3 J7	MR2 A11 J56	26 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1181	MR3 J8	PP1 PHPS J154	65 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1184	MR2 A11 J125	PP1 PHPS J32	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	
1185	MR2 A11 J200	PP1 J200	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1186	MR2 A11 J201	PP1 J201	8-Coax 8W8 Combo D	CATV E	1500	<30	85	FT-4	
1187	MR2 A11 J202	PP1 J202	2-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1188	MR2 DRV J3	PP1 A1 J77	65 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	16 Channel SW Cntrl Power
1189	MR2 DRV J5	PP1 A17 J78	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1190	MR2 DRV J10	PP1 J203	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive

18-3-7 EQUIPMENT AND OPERATOR ROOM CABLES (Continued)

TABLE 18-8 (CONT'D)
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1191	MR2 DRV J21	PP1 A11 J92	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1192	MR2 DRV J22	PP1 A11 J93	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1193	MR2 DRV J23	PP1 A11 J94	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1194	MR2 DRV J24	PP1 A11 J95	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1195	MR2 DRV J25	PP1 A11 J96	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1196	PP1 A17 J1	PP1 PHPS J35	2 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	Repeater Board Power
1233	TAC J116 (J5)	PP1 PHPS J155	30 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1249	MR8 J3	CONNECT TO RUN 1238	5 ft. Times Microwave LMR	CATV R	6000 VDC		85	FT-4	RF Extension cable
1255	OW1 A21 J18	MR2 A11 J26	60 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	E-stop
1256	OW1 A20, A7, A8	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub-D	CI3	300	<30	80	VW-1	SCIM, Mouse, Keyboard cable
1259	MR2 RF AMP J4	CONNECT TO RUN 235							
1320	OW1 A15 USB	HUB IN	2000mm RJ11						Temp/Humidity Sensor
1321	OW1 A2 USB	HUB OUT	2000mm RJ11						Temp/Humidity Sensor
1322	MR2 A11 Cabinet Sensor	HUB C2	100 ft. RJ11						Temp/Humidity Sensor
1329	MR2 DRV J10	PP1 A17 J78	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1330	MR2 DRV J5	PP1 J203	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1371	OW1 A15 ETH D	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT-4	EXCITE Ethernet C'able
1379	Bar Code Reader	OW1 A15 KYBRD	10 ft. Mini Din 6 Male to Female extender						Bar Code Reader Extension Cable (option)

18-3-8 CABLE SPARES (FRU'S)

TABLE 18-9
EQUIPMENT & OPERATOR ROOM RUNS

CABLE PART NUMBER	CABLE RUN
46-244710G30	Run 1227 Run 1236
46-244712G30	Run 1188
46-244713G30	Run 1181
46-244714G30	Run 1180 Run 1181 Run 1233
46-258877G30	Run 1189 Run 1190

18-3-9 EXCITE 4KW AND 8KW MNS CABLES (Optional)

TABLE 18-10
EXCITE 4KW AND 8KW MNS RUNS (optional)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1129	CPC J8	MR3 PD1	35 ft. Power (3-#10)	STO	600	208	90	FT-4	4KW CPC Amp Power cable
1130	CPC J3	MR2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	4KW MNS cable
1136	CPC J9	MR2 A11 J4	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	4KW MNS cable
1137	RUN 1158	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	4KW MNS cable
1138	MR6 J12	MR3 PD1	65 ft. Power (5-#8)	SO	600	208Y	75	FT-4	BB RF Amp Power Cable
1139	AMT J3	Mr2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1140	MR1 J123	MR6 J1	30 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	8KW MNS cable
1141	MR1 J102	AMT J5	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1142	MR1 J103	AMT J6	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1143	MR1 J104	AMT J7	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1144	MR1 J105	AMT J8	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1145	MR2 A11 J4	AMT J11	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1146	RUN 1159	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	8KW MNS cable
1147	MG3 A17 J3	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	4KW & 8KW MNS cable
1158	RUN 1137	MR1 CPC J2	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT-4	4KW MNS Extension cable
1248	RUN 1146	MR9 J4	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT-4	8KW MNS Extension cable

18-4 FIBER OPTIC TERMINATION INSTRUCTIONS

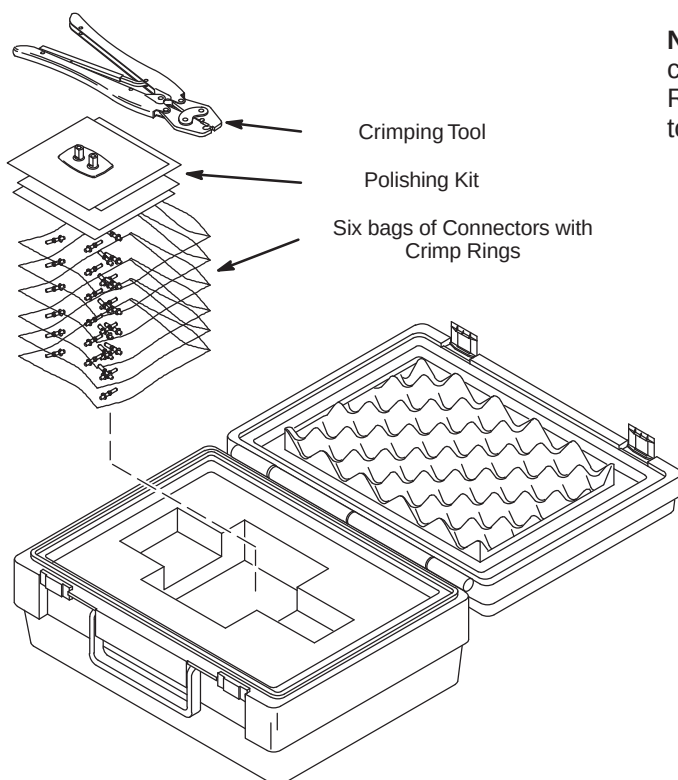
WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

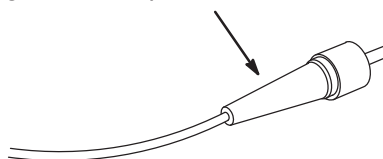
18-4-1 FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.

1. The Fiber Optic Repair Kit, 46-301450G1 shown below, contains all the necessary materials to replace a damaged connector.
2. The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.
3. Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.



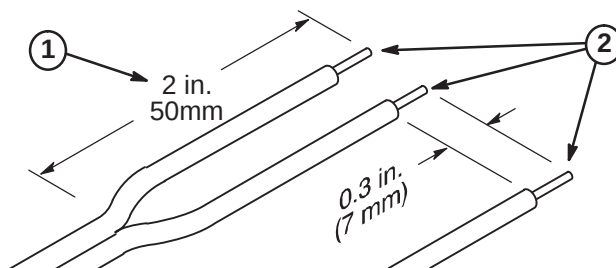
Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



18-4-2 STRIPPING FIBER OPTIC OUTER JACKET

The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

1. For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



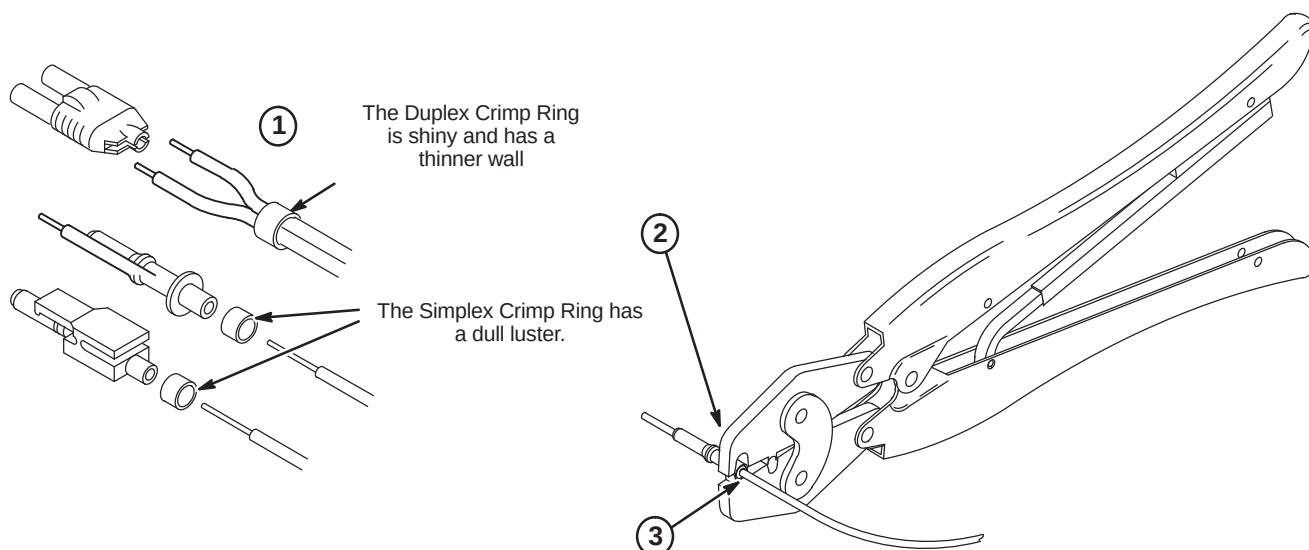
18-4-3 CRIMPING CONNECTOR TO FIBER OPTIC CABLE

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

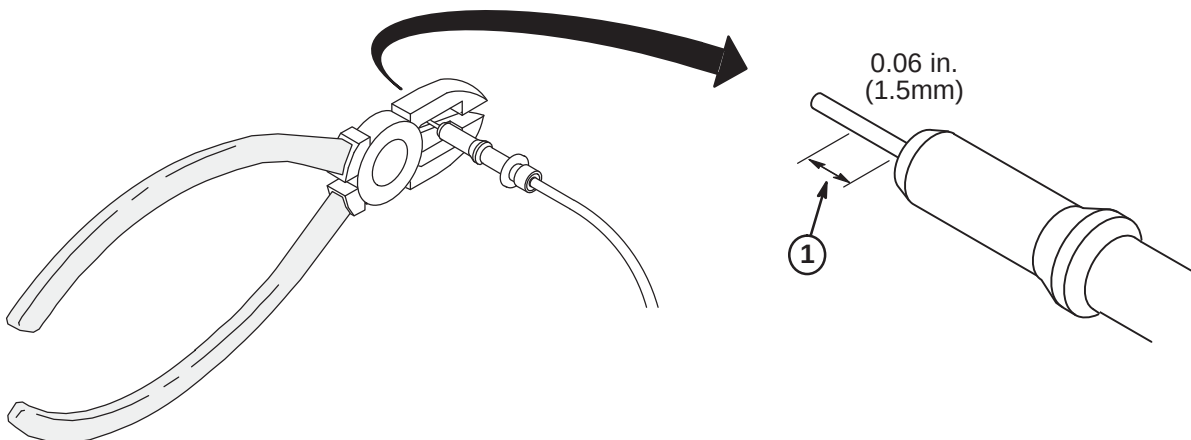
The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

1. Place the Crimp Ring and Connector over the end of the cable.
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.
3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.



18-4-4 TRIMMING EXCESS FIBER FROM CONNECTOR

1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 inch (1-1/2mm).

**18-4-5 POLISHING FIBER OPTIC END**

The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

